Testing isomorphism of finite groups in time polynomial in the group order

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Abstract

An algorithm that decides if two finite groups of order N are isomorphic in time $\mathsf{poly}(N)$ is presented. An important case is to show that testing isomorphism of $n \times n \times n$ tensors over $\mathbb{F}_q$ can be solved in time $q^{O(n)}$.

Before this work, the $N^{\log N + O(1)}$ generator-enumeration algorithm for group isomorphism was known since the 1970's (Tarjan, see Miller, *STOC*'78), and an $N^{(\frac{1}{4}+o(1))\log N}$-time was devised by Rosenbaum (*arXiv*'13). Testing isomorphism of $n \times n \times n$ tensors over $\mathbb{F}_q$ was known to be solvable in $q^{\tilde{O}(n^{1.5})}$ time (Ivanyos–Mendoza–Qiao–Sun–Zhang, *FOCS*'24) following Sun's breakthrough (*STOC*'23).

The group isomorphism result removes a well-recognised obstacle for putting graph isomorphism in P (Babai, *STOC*'16). By the Tensor Isomorphism complexity class (Grochow–Qiao, *SIAM J. Comput.*'23), the tensor isomorphism result improves the worst-case time complexities for polynomial isomorphism and algebra isomorphism.

A key innovation is to combine matrix groups with linear algebraic individualising techniques. The results build on techniques and results including the Luks–Miyazaki matrix group divide-and-conquer framework (Luks, *FOCS*'92), results on linearly primitive groups (Jaikin-Zapirain–Pyber, *Ann. Math.*'11), algorithms for testing isomorphism of groups without abelian normal subgroups (Babai–Codenotti–Qiao, *ICALP*'12), algorithms based on $*$-algebra (Brooksbank–Wilson, *Trans. Amer. Math. Soc.*'12; Ivanyos–Qiao, *SIAM J. Comput.*'19)), linear-algebraic individualising techniques (Li–Qiao, *FOCS*'17; Sun, *STOC*'23), and group extensions and cohomology (Grochow–Qiao, *SIAM J. Comput.*'17).

Brief AI methodology statement. The main results in this paper were generated by ChatGPT during sustained conversations between the authors and ChatGPT. The AI-generated results have clear traces back to previous results on group isomorphism, including several in the past 20 years. The authors put in their best efforts in a limited time to understand and rewrite the results in the presentation in the current form, and take full responsibility for the results presented in this paper. A more detailed AI methodology statement can be found at the end of the paper.

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1 Introduction

1.1 Results on group isomorphism

In this paper, we study the finite group isomorphism problem. The goal is to present an algorithm testing whether two finite groups are isomorphic in time polynomial in their orders. For this purpose, one can assume that the groups are given as Cayley (multiplication) tables. Indeed, if groups are given as permutation groups or matrix groups, their Cayley tables can be computed from such compact representations in time polynomial in the group order. In the following, we use Gpl-Table to denote the finite group isomorphism problem when Cayley tables are given.

Finite group isomorphism problem attracts attention from theoretical computer scientist and group theorists. As a group of order N has a generating set of size $\lceil \log N \rceil$, Gpl-Table admits a natural generator-enumeration algorithm in time $N^{\log N + O(1)}$. Methods in computational group theory in the 1960's [FN68, FN70] can be shown to imply this time bound, and this algorithm with an explicit time bound was recorded in theoretical computer science [Mil78] in the 1970's. (In [Mil78], Gary Miller attributed the algorithm for Gpl-Table to Robert Tarjan.) The $N^{\log N + O(1)}$-time bound was barely improved since the 1970's; the record before this work is an algorithm in time $N^{(\frac{1}{4}+o(1))\log N}$ by Rosenbaum [Ros13] (see Section 1.5 for more details on this). The main result of this work brings this to $N^{O(1)}$.

Theorem 1.1 (Group isomorphism in time polynomial in group order)**.** *Let G and H be two finite groups of order N, given by their Cayley (multiplication) tables. There is an algorithm that decides if G and H are isomorphic in time polynomial in N.*

A key step towards Theorem 1.1 is a new algorithm for tensor isomorphism (TI) over finite fields. The bridge between TI and Gpl-Table is through a proper generalisation of TI and the group extension and cohomology through a characteristic subgroup chain in the solvable radical as in [CH03, GQ17]. We now turn to tensor isomorphism.

1.2 Results on 3-tensor isomorphism over finite fields

Let U, V, W be three vector spaces. The 3-Tensor Isomorphism (3TI) problem asks whether two tensors in $U \otimes V \otimes W$ are the same up to the natural action of $\mathrm{GL}(U) \times \mathrm{GL}(V) \times \mathrm{GL}(W)$ on $U \otimes V \otimes W$.

In algorithms, after fixing bases of U, V and W, we record 3-tensors by 3-way arrays; recall that a 3-way array over $\mathbb{F}$ is an array with 3 indices whose elements are from $\mathbb{F}$.

Definition 1. Let $\mathtt{A} = (a_{i,j,k})$, $\mathtt{B} = (b_{i,j,k})$, where $i \in [\ell]$, $j \in [m]$, $k \in [n]$, $a_{i,j,k}, b_{i,j,k} \in \mathbb{F}$, be 3-way arrays over $\mathbb{F}$. We say that $\mathtt{A}$ and $\mathtt{B}$ are isomorphic as 3-tensors, if there exist $R = (r_{i,i'}) \in \mathrm{GL}(\ell, \mathbb{F})$, $S = (s_{j,j'}) \in \mathrm{GL}(m, \mathbb{F})$, and $T = (t_{k,k'}) \in \mathrm{GL}(n, \mathbb{F})$ such that for any i, j, k, $a_{i,j,k} = \sum_{i',j',k'} r_{i,i'} s_{j,j'} t_{k,k'} b_{i',j',k'}$.

3-Tensor Isomorphism over finite fields is closely related to Cayley Table Group Isomorphism. The connection is via a class of groups called *p-groups of class 2 and exponent p*. Recall that a p-group G is of class 2 if its commutator subgroup $[G, G]$ is contained in the centre $\mathrm{Z}(G)$; a group G is of exponent p if for any $g \in G$, $g^p = \mathrm{id}$. Among various group families, p-groups of class 2 and exponent p is usually considered to be a bottleneck. We shall review some reasons for supporting this family as the bottleneck in Section 1.5.

In [GQ23a, GQ24, CGQ+24, GQ23b, GQSS25], the complexity class TI was introduced, defined as the class of problems polynomial-time reducible to the tensor isomorphism problem. While the

definition of TI does not specify tensors of degree 3, a result in [GQ23a] shows that k-TI reduces to 3TI in polynomial time. In [GQ23b], it is shown that testing isomorphism of p-groups of class 2 and exponent p in time polynomial in group order is equivalent to solving 3TI for $\ell \times m \times n$ tensors over $\mathbb{F}_q$ in time $q^{O(\ell+m+n)}$.

Let us consider 3TI for $n \times n \times n$ tensors over $\mathbb{F}_q$. A brute-force algorithm runs in time $q^{O(n^2)}$. Recently there was some notable progress. The first breakthrough came from Sun [Sun23], who presented an algorithm in time $q^{O(n^{1.8} \cdot \log q)}$ over fields of odd characteristic. This was improved to $q^{\tilde{O}(n^{1.5})}$ in [IMQ+26]; a recent work extended this time bound to fields of characteristic 2 [GIQS26]. Our second main result brings this to $q^{O(n)}$ (regardless of q is even or odd).

Theorem 1.2 (Tensor isomorphism in singly exponential time)**.** *Let* $\mathtt{A}, \mathtt{B}$ *be two* $\ell \times m \times n$ 3*-way arrays over* $\mathbb{F}_q$*. There is an algorithm that decides if* $\mathtt{A}$ *and* $\mathtt{B}$ *are isomorphic as tensors in time* $q^{O(\ell+m+n)}$.

The strategy in previous works on 3TI [Sun23, IMQ+26, GIQS26] has some inherent limitation to go beyond $q^{O(n^{1.5})}$. There is also a relevant matrix group divide-and-conquer framework by Luks and Miyazaki [Luk92, Miy99, LM11], which also meets difficulties to go beyond solvable groups. Theorem 1.2 is obtained by further both matrix groups and linear algebraic individualisation techniques [LQ17, Sun23], together with several recent advances on linearly primitive groups [JZP11, HMPQ19] and algebras with involutions [Wil09a, BW12, IQ19], to obtain an algorithm with the desired running time.

One technical step towards Theorem 1.2 is a new algorithm for matrix tuple congruence (MTC) over fields of characteristic 2. We now turn to matrix tuple congruence.

1.3 Results on matrix tuple congruence over finite fields of characteristic 2

Let $\mathbf{A} = (A_1, \ldots, A_m), \mathbf{B} = (B_1, \ldots, B_m) \in \mathrm{M}(n, q)^m$ be two m-tuples of $n \times n$ matrices over $\mathbb{F}_q$. We say that $\mathbf{A}$ and $\mathbf{B}$ are congruent if there exists $L \in \mathrm{GL}(n, q)$, such that for any $i \in [m]$, $A_i = LB_iL^{\mathrm{t}}$. The Matrix Tuple Congruence (MTC) problem asks to decide whether two matrix tuples are congruent.

The worst-case complexity of MTC depends on whether the field characteristic is even or odd. Over finite fields of odd characteristics, MTC can be solved in randomised $\mathrm{poly}(n, \log q)$-time, and deterministic $\mathrm{poly}(n, q)$ time [IQ19]. Over $\mathbb{F}_2$, MTC has a sibling problem, called the Quadratic Form Tuple Isometry (QFTI) problem, which asks whether two tuples of quadratic forms are isometric, defined as follows.

Definition 2. Let $(f_1, \ldots, f_m), (g_1, \ldots, g_m) \in \mathbb{F}_2[x_1, \ldots, x_n]^m$ be two tuples of quadratic forms (homogeneous quadratic polynomials). We say that they are isometric if there exists $A = (a_{i,j}) \in \mathrm{GL}(n, 2)$ such that for any $i \in [m]$, $f_i = g_i \circ A$, where

$$(g_i \circ A)(x_1, \ldots, x_n) = g_i(\sum_j a_{1,j}x_j, \ldots, \sum_j a_{n,j}x_n). \tag{1.1}$$

In a recent work [GIQS26], Quadratic Form Tuple Isometry was shown to admit an algorithm in time $2^{O((m+n)^{1.5})}$, and this is the key technical contribution to the $N^{O(\sqrt{\log N})}$-time algorithm for testing isomorphism of 2-groups of class 2 in [GIQS26].

We present a new algorithm for MTC and QFTI as follows.

Theorem 1.3 (Matrix Tuple Congruence over $\mathbb{F}_2$)**.** *Consider* Matrix Tuple Congruence *for inputs in* $\mathrm{M}(n, 2)^m$*, and* Quadratic Form Tuple Isometry *for* m*-tuples of* n*-variate quadratic forms. Both problems can be solved in time* $2^{O(n)} \cdot \mathrm{poly}(n, m)$.

1.4 Implications to other isomorphism problems

Graph isomorphism. Gpl-Table reduces to the celebrated graph isomorphism problem (Gfl) [KST93]. Babai's breathrough result gives a quasipolynomial-time algorithm for Gfl [Bab16b], where Babai listed putting Gpl-Table in P as a bottleneck for putting Gfl in P. Theorem 1.1 removes this bottleneck, and its techniques should be helpful in dealing with primitive groups of almost simple type (see Section 2).

Polynomial and algebra isomorphism. In [GQ23a, GQ23b], isomorphism problems for several algebraic structures are shown to be Tl-complete. We mention two problem families here.

The first family is polynomial isomorphism, introduced in [Pat96] as a security basis for some digital signature schemes. Two problems from this family are as follows. The first is called cubic polynomial equivalence: given two cubic polynomials $f, g \in \mathbb{F}_q[x_1, \ldots, x_n]$, decide if there exists $A \in \mathrm{GL}(n, q)$, such that $f = g \circ A$, where $g \circ A$ is defined as in Eq. (1.1). The second is polynomial isomorphism with two secrets, which asks whether two quadratic form tuples $(f_1, \ldots, f_m), (g_1, \ldots, g_m) \in \mathbb{F}_q[x_1, \ldots, x_n]^m$, when viewed as quadratic maps $\mathbb{F}_q^n \to \mathbb{F}_q^m$, are equivalent under the natural action of $\mathrm{GL}(n, q) \times \mathrm{GL}(m, q)$.

The second problem is algebra isomorphism, which encompasses associative and Lie algebra isomorphism. Let $\alpha, \beta : \mathbb{F}_q^n \times \mathbb{F}_q^n \to \mathbb{F}_q^n$ be two bilinear maps. The algebra isomorphism asks whether α, β are equivalent under the natural action of $\mathrm{GL}(n, q)$. The algebra isomorphism was studied in [AS06].

By the linear-length reductions in [GQ23b], Theorem 1.2 implies the following.

Corollary 1.4. *1. Let $\mathbb{F}_q$ be a finite field of characteristic not 2 or 3. There is an algorithm in time $q^{O(n)}$ testing whether two cubic forms in $\mathbb{F}_q[x_1, \ldots, x_n]$ are equivanet.*

2. Let $\mathbb{F}_q$ be a finite field of characteristic not 2. Let $(f_1, \ldots, f_m), (g_1, \ldots, g_m) \in \mathbb{F}_q[x_1, \ldots, x_n]$ be two quadratic form tuples. Then there is an algorithm in time $q^{O(n+m)}$ testing whether the two quadratic form tuples are isomorphic as quadratic maps from $\mathbb{F}_q^n$ to $\mathbb{F}_q^m$.

3. Let $\alpha, \beta : \mathbb{F}_q^n \times \mathbb{F}_q^n \to \mathbb{F}_q^n$ be two bilinear maps. There is an algorithm in time $q^{O(n)}$ testing whether α and β are isomorphic.

Before this work, these problems were known to be solvable in time $q^{\tilde{O}(n^{1.5})}$ as a result of [IMQ+26, GQ23b] (see [GQ23b, Corollary 1.6]).

1.5 A review of previous works

In this section we review works on group isomorphism, tensor isomorphism, and matrix tuple congruence related problems. We shall mostly focus on algorithms with rigorous analyses. We recognise a strong body of works on practical group isomorphism; we refer the readers to [GQ15b, Appendix C] for a discussion and references to some of these results. In the following, when we say polynomial-time, we mean polynomial in the group order.

General group isomorphism. For groups of order N, the natural generator-enumeration algorithm with the explicit time bound $N^{\log N + O(1)}$ was observed by Tarjan, as recorded by Miller [Mil78]. Rosenbaum showed that the coefficient of $\log N$ on the exponent can be improved from 1 to $\frac{1}{2} + o(1)$ [Ros13]. This could be further improved to 1/2 by methods of Babai or Luks (see [LGR16]).

Restricted families of groups. As general Gpl-Table seems difficult, a natural strategy is to make progress on special group families.

The abelian group isomorphism problem can be solved in linear time in the group order [Ili85, Kav07]).

In [BCGQ11, BCQ12], a polynomial-time algorithm is presented for groups without abelian normal subgroups. This result will be used as the base case of the recursive algorithm for group isomorphism in this paper Theorem 9.1.

In [Gal09, QST11, BQ12], coprime extensions were studied, leading to a polynomial-time solution for testing isomorphism of groups with abelian Sylow towers. In [GQ15b], groups that are the so-called tame extensions were studied.

On p-group of class 2 and exponent p of order N, notable classes with polynomial-time algorithms include quotients of generalised Heisenberg groups [LW12], genus-2 groups [BMW17], and groups whose commutator subgroup order bounded by $p^{O(\sqrt{N})}$ [IQ19]. We shall leave progress on general p-groups of class 2 and exponent p to later when we discuss tensor isomorphism.

In [DW21], it is shown that for most group orders, group isomorphism can be solved in linear time. Note that the orders covered there do not include groups of prime power order.

Recently, Skresanov [Skr26a] solved testing isomorphism of solvable groups with abelian Sylow subgroups, also known as solvable A-groups. Another work of Skresanov [Skr26b] solves testing isomorphism of groups that are extensions of abelian by bounded generated groups.

Complexity and logic based on group isomorphism. While Gpl-Table reduces to Gfl, there is no AC^0 reduction from Gfl to Gpl-Table [CTW13]. Recently, there has been a nice line of works on parallel complexities on Gpl-Table; we refer the readers to [Lev26, GL26] and references there in. An important technique in this direction is the Weisfeiler–Leman technique for group isomorphism, see also [BS20, BS22].

Previous works on tensor isomorphism over finite fields. By [GQ23b], testing isomorphism of p-groups of class 2 and exponent p in polynomial time is polynomial-time equivalent to testing isomorphism of $n \times n \times n$ tensors over $\mathbb{F}_q$ in time $q^{O(n)}$.

Tensor isomorphism over finite fields can be studied from the worst-case, average-case, and heuristic, algorithm viewpoints.

For worst-case algorithms, we've mentioned Sun's breakthrough [Sun23] as well as the improvements in [IMQ$^+$26], where an algorithm in time $q^{\tilde{O}(n^{1.5})}$ was established.

For average-case algorithms, an algorithm in time $q^{O(n)}$ was presented in [LQ17], with better average-case analysis in [BLQW20]. A recent work [LLQ$^+$26] exhibits a rich landscape of average-case complexities of TI-complete problems.

For heuristic algorithms, which are closely related to cryptography , there was an algorithm in time $q^{\frac{n}{2}} \cdot \mathrm{poly}(n, \log q)$ [NQT24]. A later result achieves the same running time but extends to non-cube settings [CL25].

Tensor isomorphism: complexity and cryptography. We've mentioned a series of works establishing the complexity class TI [GQ23a, GQ24, CGQ$^+$24, GQ23b, GQSS25, NQT24]. 3TI has been studied in proof complexity [GGPS23].

TI can serve as the security basis of some cryptographic schemes. There were two submissions to NIST's call for post-quantum digital signatures based on TI-complete problems [CNP$^+$23, BDN$^+$23], built on the zero-knowledge protocol for graph isomorphism and the Fiat–Shamir transformation [GMW91, FS86].

TI over $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Q}$ and $\mathbb{Z}$ demonstrate quite different complexities, as studied in e.g. [CGQ+24, GQSS25]. This also holds for TI under different group actions, as discussed in [CGQ+24, CEMQ26].

1.6 Outlooks and open problems

We list some problems for future study.

- An algorithm for Gpl-Table in time polynomial in the group order removes one well-recognised bottleneck for putting graph isomorphism in P. This makes another bottleneck to put Gfl in P, the tournament isomorphism, more prominent (see [RS26]).
- An important question is to examine how low the complexity can go. Works on Weisfeiler–Leman on group isomorphism, some of which were mentioned in Section 1.5, should be of key interest.
- Can one improve the complexity of linear code equivalence using matrix group computation? This is supported by a result of Luks [Luk92, Theorem 3.2 (6)] which can be viewed as solving linear code equivalence in a restricted setting.
- Is it possible to get an algorithm for $n \times n \times n$ tensor isomorphism over $\mathbb{F}_q$ in time $q^{o(n)}$? This may be viewed as a Exponential Time Hypothesis (ETH) for tensor isomorphism. Note that ETH for k-SAT was recently claimed to be refuted [Ope26].
- The complexity of TI over $\mathbb{C}$ is only known to be in AM. Is it in $\mathrm{AM} \cap \mathrm{coAM}$?

2 Overview of algorithmic techniques

As discussed in Section 1, the group isomorphism result (Theorem 1.1) resorts to the tensor isomorphism result (Theorem 1.2), while the tensor isomorphism result relies on the matrix tuple congruence result over $\mathbb{F}_2$ (Theorem 1.3). We first provide some high-level remarks on the technical difficulties and the approaches addressing them, and then walk through some key points in the following subsections.

2.1 High-level remarks on Matrix Tuple Congruence

Consider two m-tuples of $n \times n$ matrices over $\mathbb{F}_q$. Previous works [BW12, IQ19] handles Matrix Tuple Congruence (MTC) over fields of odd characteristics. They introduce the $*$-algebra framework, which combines mathematical structures of algebras with involutions together with computational methods for algebras [Rón90, Iva00, BL08]. The strategy is to solve a so-called $*$-symmetric decomposition problem by filtering through the algebra structure: first solve for the semisimple algebra case, and then lift the solutions through the radical layers. The main difficulty to tackle Matrix Tuple Congruence over characteristic-2 fields lies in the lifting through radical step. Over fields of odd characteristics, a solution in the semisimple quotient can always be lifted through the radical by a clever divide-by-2 technique. Over fields of characteristic 2, this is no longer the case. This difficulty was partly addressed in [GIQS26], which uses canonical forms for left-right actions on matrix tuples [QS24] and matrix group algorithms [Luk92]. However, it falls short of getting an algorithm in $q^{O(n)}$.

The main lead to Theorem 1.3 is a structural understanding of the obstruction to lifting, that is the obstruction space defined in Eq. (4.6), which has traces back to the works of Bak, Ranicki, and Serre [Bak79, Ran80, Ser14]. This distillation of an obstruction space allows for the use of matrix

group algorithms in a more efficient approach that leads to an algorithm with the desired running time.

2.2 High-level remarks on 3-Tensor Isomorphism

Consider two $n \times n \times n$ tensors over $\mathbb{F}_q$. By slicing along one direction, we obtain two n-tuples of matrices $\mathbf{A} = (A_1, \ldots, A_n), \mathbf{B} = (B_1, \ldots, B_n) \in \mathrm{M}(n, q)^n$. We shall refer $\mathtt{A}$ and $\mathtt{B}$ as the tensors, and A_i and B_i as slices of the tensors.

Unlike matrix tuple congruence, for tensor isomorphism, we have $(L, R, T) \in \mathrm{GL}(n, q)^{\times 3}$ sending $\mathbf{A}$ to $(L, R, T) \circ \mathbf{A} := (A'_1, \ldots, A'_n)$, where for $T = (t_{i,j})_{i,j\in[n]}$, $A'_i := L^{\mathrm{t}}(\sum_{j\in[n]} t_{i,j} A_j)R$. We shall refer to the L, R, or T sides of the tensors.

Recent progress on TI. The recent progress on 3-Tensor Isomorphism [LQ17, BLQW20, Sun23, IMQ+26, GIQS26] could be viewed as developing combinatorial-type techniques for Graph Isomorphism. We review some techniques there.

1. In [LQ17], the basic idea is to fix the images of $A_1, \ldots, A_c$ for some constant c, which amounts to enumerating the first c columns of T, and examine the consequence of this fixing on the choices of L and R. This is similar to individualising c vertices in the graph isomorphism setting.

   Whether consequence of this fixing is substantial depends on $A_1, \ldots, A_c$. It can be shown that this is the case for random $A_1, \ldots, A_c$, leading to an average-case algorithm in [LQ17].

2. In [Sun23], the individualisation step consists of enumerating the first r columns of both L and R. This has two consequences on the T side. The first is that, up to a change of basis, there exist several slices that are fixed by eligible T. The second is that, the rest slices span a matrix space of low rank, which in turn could be put in a format where a large of zero block is present, thanks to the connection between commutative and non-commutative rank of matrix spaces [IQS18]. A delicate manipulation then collects the structural restrictions to a Matrix Tuple Congruence instance, which can then be solved in polynomial time over fields of odd characteristics [IQ19].

   In [IMQ+26], this strategy is sharpened, and it was suggested there that this strategy meets a bottleneck for going below $q^{O(n^{1.5})}$ as r above is best set as $r = \Theta(n^{0.5})$.

Matrix group algorithms. Most notable progress on graph isomorphism relies on an interaction between group-theoretic and combinatorial techniques [Bab16b]. The group-theoretic techniques for GfI were pioneered by Babai [Bab79] and Luks [Luk82]. In particular, Luks's divide-and-conquer framework for permutation groups is a milestone for graph isomorphism.

TI admits a natural matrix group formulation as follows. Consider the group $G = \mathrm{GL}(n, q) \times \mathrm{GL}(n, q) \times \mathrm{GL}(n, q)$. The natural module of G is $\mathbb{F}_q^n \oplus \mathbb{F}_q^n \oplus \mathbb{F}_q^n$, while G has a natural induced action on $V = \mathbb{F}_q^n \otimes \mathbb{F}_q^n \otimes \mathbb{F}_q^n$. Then TI becomes an instance of the point transporter problem: given two 3-tenosrs $\mathbf{A}, \mathbf{B} \in V$, compute $\mathrm{Trans}_G(\mathbf{A}, \mathbf{B}) = \{(L, R, T) \in G \mid (L, R, T) \circ \mathbf{A} = \mathbf{B}\}$, which is either empty (for non-isomorphic $\mathbf{A}$ and $\mathbf{B}$) or a coset of the group $\mathrm{Stab}_G(\mathbf{A})$.

For the purpose of an $q^{O(n)}$-time algorithm for TI, several tasks for matrix groups can be performed within this time bound, such as order, membership, and point transporter in the natural module. Even some more sophisticated tasks, such as invariant subspaces, can also be computed thanks to the computer algebra algorithms [Rón90, Iva00].

For TI, as the dimension of $\mathbb{F}_q^n \otimes \mathbb{F}_q^n \otimes \mathbb{F}_q^n$ is n^3, TI as the transporter problem is not easily seen to be solvable in $q^{O(n)}$. To make progress, it is natural to resort to a matrix group divide-and-conquer framework, which was announced in Luks [Luk92] and further developed by Luks and Miyazaki [Miy99, LM11].

We give a brief review of the group divide-and-conquer framework in Section 2.4. As in the permutation group setting, the bottlenck for matrix group divide-and-conquer lies on large linearly primitive groups (see Section 2.4 for a definition), and 3TI falls into this bottleneck case.

The approach forward. To overcome the difficulties met in both the recent progress and the matrix group algorithms, the approach forward is to develop the divide-and-conquer framework further, and also to use some ideas in the recent progress to address the large linearly primitive groups.

An algorithm in the divide-and-conquer framework is perhaps easiest described as a recursive algorithm. The algorithm for 3TI in Section 8 is also of a recursive nature. To accommodate recursion, extra data needs to be tracked, including:

1. Matrix groups and cosets: we enlarge on the group side to accommodate any matrix groups, even though we start with $\mathrm{GL}(n,q) \times \mathrm{GL}(n,q) \times \mathrm{GL}(n,q)$.

2. Tensor modules (Definition 3), which is essentially a direct sum of tensors supported by possibly more than 3 vector spaces.

3. A 1-cocyle as a translation, to accomodate recursion into submodules. This is called a crossed homomorphism by Luks [Luk92].

There are more but these are the major ones (Section 8).

We also need a library of matrix group algorithms to compute the matrix group structure (see Section 5), including invariant subspaces, linear imprimitive systems, local components, and diagonal links between components. Roughly speaking, local components are those linearly primitive actions, and diagonal links record their correlations. We distinguish local components into two types: large classical groups, and alternating dominant ones. As we are allowed $q^{O(n)}$ time, the algorithmic parts of matrix group computation in our setting are usually straightforward, and more crucial is the structural result on linearly primitive groups [JZP11] (see also [HMPQ19]).

The algorithm for 3TI in Section 8 then has two interacting major components, the group component and the tensor component. It starts by exploiting the input matrix group structure while keeping track of the tensors, until it hits the large classical groups as the linearly primitive action, when it turns to the tensor component. On the tensor side, the basic idea is to individualise a constant number of slices as in [LQ17], which we call contraction. Such a contraction leads to restriction of groups (which goes back to the group component), or a low-rank situation as in [Sun23, IMQ$^+$26], or a controlled situation which may be understood as a generalisation of Matrix Tuple Congruence and can be solved (Section 7). A progress measure is formulated which ensures that this recursive algorithm runs in the desired time bound.

2.3 High-level remarks on Cayley Table Group Isomorphism

To solve Cayley Table Group Isomorphism, we use the characteristic chains and the group extension methodology in [CH03, GQ17].

Reducing to group extensions of elementary abelian groups. Let G be a finite group. Let $\mathrm{srad}(G)$ be the solvable radical of G, that is the largest solvable normal subgroup of G.

Suppose we want to test if two groups G and H are isomorphic. If one of $\mathrm{srad}(G)$ and $\mathrm{srad}(H)$ is trivial while the other is not, G and H cannot be isomorphic.

We then compute the derived series of $\mathrm{srad}(G)$

$$1 = S_0 \trianglelefteq S_1 \trianglelefteq S_2 \trianglelefteq \ldots \trianglelefteq S_\ell = \mathrm{srad}(G),$$

where $S_{i-1} = [S_i, S_i]$ for $i \in [\ell]$. If the derived series lengths of $\mathrm{srad}(G)$ and $\mathrm{srad}(H)$ are different, reject.

The algorithm then performs the following induction on $\mathrm{srad}(G)$ and $\mathrm{srad}(H)$.

If $\mathrm{srad}(G)$ and $\mathrm{srad}(H)$ are trivial, then G and H are groups without nontrivial abelian normal subgroups. For such groups, the algorithm in [BCQ12, BCGQ11] decides if G and H are isomorphic, and in the case of isomorphism, returns the coset of isomorphisms from G to H.

If $\mathrm{srad}(G)$ and $\mathrm{srad}(H)$ are not trivial, we construct a characteristic elementary abelian subgroup as follows. Recall that S_1 is the last non-identity subgroup in the derived series. Choose the least prime p that divides $|S_1|$, let D be the unique p-Sylow subgroup of S_1, and set $\mathrm{AC}(G) := \{x \in D \mid x^p = \mathrm{id}\}$. Then $\mathrm{AC}(G)$ is an elementary abelian characteristic subgroup.

To test isomorphism of two groups G and H, compute $\mathrm{AC}(G)$ and $\mathrm{AC}(H)$. For $G \cong H$, it is necessary that $\mathrm{AC}(G) \cong \mathrm{AC}(H)$, in which case we identify both as $\mathbb{Z}_p^a$. By inductive hypothesis, we can assume that $G/\mathrm{AC}(G) \cong H/\mathrm{AC}(H)$ and identify them as Q.

Group extension data. Let G and H be identified as extensions of $\mathbb{Z}_p^a$ by Q, with $\mathrm{Aut}(Q)$ given by a generating set. We need to decide if G and H are isomorphic, and if so, compute the coset of isomorphisms.

Group extensions are governed by group actions and cohomology, a classical topic in group theory [EM47] that found use in computational group theory [HEO05]. Here we follow the notations and conventions in [GQ17], which studies group isomorphism via group extensions in the Cayley table model.

Let a group G be an extension of $A = \mathbb{Z}_p^n$ by Q. We use $+$ for group operations in A. That is, $A \trianglelefteq G$ and $Q \cong G/A$. The first function associated with this extension is the homomorphism $\theta' : G \to \mathrm{Aut}(A)$ by $\theta'(g)$ sending a to gag^{-1}. As A is abelian, this induces a homomorphism $\theta : Q \to \mathrm{Aut}(A)$. This is called the *group action* associated with this extension.

Let $\pi : G \to G/A \cong Q$ be the natural projection. A function $s : Q \to G$ is a section if s represents a selection of coset representatives, that is $\pi(s(x)) = x$ for any $x \in Q$. From a section s we can define a function $f_s \colon Q \times Q \to A$ defined by $f_s(x, y) := s(x)s(y)s(xy)^{-1}$. This function is called a 2-cycle as it satisfies

$$f_s(x, y) + f_s(xy, z) = \theta_x(f_s(y, z)) + f_s(x, yz) \qquad \text{(the 2-cocycle identity)}.$$

Without loss of generality, we can choose $s(1) = 1$ so that $f_s(1, x) = f_s(x, 1) = 0$ for all $x \in Q$, called a normalised 2-cocycle. In the following 2-cocycles are normalized unless stated otherwise.

A 2-cocycle $f : Q \times Q \to A$ is called a 2-coboundary, if it is derived from a function $u : Q \to A$ through $f_u(x, y) := u(x) + \theta_x(u(y)) - u(xy)$. Two 2-cocycles are cohomologous if they differ by a 2-coboundary.

The action θ and the 2-cocycle f_s govern the isomorphic type of G as an extension of A by Q in the following sense. Suppose $|Q| = s$; we can assume that the elements in Q are labelled by $[s]$.

First, θ as a function $Q \to \mathrm{Aut}(A) \cong \mathrm{GL}(n, p)$ can be expressed as a matrix tuple $(B_1, \ldots, B_s) \in \mathrm{M}(n, p)^s$. Then $\mathrm{Aut}(Q)$ as a subgroup of $\mathrm{Sym}([s])$ permutes B_i's, and $L \in \mathrm{Aut}(A) \cong \mathrm{GL}(n, p)$ acts on B_i's by simultaneous conjugation.

Second, f as a function $Q \times Q \to A$ can be recorded as a 3-way array $\mathtt{A} = (a_{i,j,k})$ of size $s \times s \times n$, where $a_{i,j,k} = f(i,j)_k \in \mathbb{Z}_p$. Then $\mathrm{Aut}(Q)$ as a subgroup of $\mathrm{Sym}([s])$ permutes on the first two indices of $\mathtt{A}$, and $\mathrm{Aut}(A) \cong \mathrm{GL}(n,p)$ acts on the third index of $\mathtt{A}$ by linear combinations.

A key criterion is that the isomorphic type of G is determined by the equivalence class of (θ, f_s) under the $\mathrm{Aut}(Q) \times \mathrm{Aut}(A)$ action. Note that two 2-cocycles should be considered as same up to 2-coboundaries. This is the classical extension-isomorphism criterion (see Holt, Eick, and O'Brien [HEO05, Section 2.7.4, pp. 54–55], and Grochow and Qiao [GQ17, Lemma 2.3]).

The group divide-and-conquer framework. Let G be an extension of $A \cong \mathbb{Z}_p^n$ by Q as before. As described above, the action $\theta : Q \to \mathrm{Aut}(A)$ and the 2-cocycle $f_s : Q \times Q \to A$ can be recorded as a matrix tuple and a 3-way array, respectively. This gives a first indication of why 3-tensor isomorphism may suffice for group isomorphism. However, the 3-way array from f_s has size $|Q| \times |Q| \times n$, so applying the tensor running-time bound directly would give an exponential dependence on $|Q|$, whereas our goal is an algorithm in time polynomial in $|G| = p^n|Q|$. Therefore, we need to exploit the structure of Q.

Just as we make use of a filtration of a module in the 3TI setting, in the group setting we resort to a characteristic subgroup chain of Q (Section 9.2.1), such that the quotients of consecutive subgroups are either elementary abelian or products of non-abelian simple groups. For the elementary abelian case, we are quite close to the tensor isomorphism setting. For products of non-abelian simple groups, the local cohomology data are represented as invariant subquotients of direct sums of tensor modules. The key thing to note is that these terminal restriction comparisons are instances of the degree-3 quotient tensor-module transporter (Section 8.3). The overall framework and controls can be delicate (Section 9), but this gives a brief idea of how divide-and-conquer may work.

2.4 Divide-and-conquer for group computation

In [Luk82], Luks presented his celebrated result of a polynomial-time algorithm for testing isomorphism of graphs with bounded degrees. This algorithm for graph isomorphism depends on what is now known as Luks' divide-and-conquer strategy for permutation groups. This strategy also underpins Babai's quasipolynomial-time algorithm for graph isomorphism [Bab16b].

In [Luk92], Luks extended this divide-and-conquer strategy from permutation groups to matrix groups over finite fields. The main results in [Luk92] concern computing with solvable matrix groups. In Miyazaki's thesis [Miy99], this framework was developed further, and several results inc[Luk92] were extended to matrix groups in Γ_d.[1] Applications also include computing normalisers in permutation groups within an ambient permutation group in Γ_d [LM11].

We briefly explain some basic structures of the divide-and-conquer strategies for permutation and matrix groups.

Divide and conquer in permutation groups. Let $P \leq \mathrm{Sym}_n$ be a permutation group. Recall that $S \subseteq [n]$ is P-invariant, if for any $\sigma \in P$, $\sigma(S) = S$. If the only P-invariant sets are $\emptyset$ and $[n]$, then P is transitive. For a transitive P, $B \subseteq [n]$ is a P-block, if for any $\sigma \in P$, either $\sigma(B) = B$ or $\sigma(B) \cap B = \emptyset$. If the only P-blocks are singleton sets or $[n]$, then P is called primitive.

Luks's divide-and-conquer strategy for permutation groups exploits (non-trivial) P-invariant sets and P-blocks to reduce to primitive actions. The terminal cases are then P being primitive; in the case of P in Γ_d where d is a constant, this can be handled as primitive Γ_d permutation groups are of polynomial order [BCP82].

[1] A group is in Γ_d if non-abelian composition factors are isomorphic to subgroups of Sym_d. This group class was introduced in [Luk82].

Divide and conquer in matrix groups. Let $G \leq \mathrm{GL}(n, q)$ be a matrix group over $\mathbb{F}_q$. Recall that $U \leq \mathbb{F}_q^n$ is G-invariant, if for any $A \in G$, $A(U) = U$. If the only G-invariant subspaces are 0 and $\mathbb{F}_q^n$, then G is called irreducible. For an irreducible G, a G-imprimitive system is a direct sum decomposition $\mathbb{F}_q^n = V_1 \oplus \cdots \oplus V_m$, $m \geq 2$, such that for any $A \in G$ and $i \in [m]$, $A(V_i)$ is one of V_j's. In this case, $\dim(V_i)$'s are the same ($=n/m$), and G has an induced permutation action on the set of subspaces $\{V_1, \ldots, V_m\}$.

Luks's divide-and-conquer for matrix groups similarly exploits (non-trivial) G-invariant subspaces and G-imprimitive systems [Luk92, Theorem 6.1]. In the case of solvable matrix groups considered in [Luk92], the terminal cases are when there is an abelian subgroup with polynomially bounded index; the polynomial bound on the order of primitive solvable permutation groups [Pál82] is also needed. In the case of Γ_d groups in [Miy99], the corresponding bound in [BCP82] is also required.

The bottleneck: large primitive groups. The bottlenecks in Luks's divide-and-conquer are primitive groups of large order. In the case of permutation groups, building on the O'Nan–Scott theorem [Sco80, LPS88], Cameron [Cam81] (see also Maróti [Mar02]) proved important structural constraints for primitive permutation groups of orders larger than quasipolynomial. These structural constraints underpin an important group-theoretic reduction in Babai's quasipolynomial-time algorithm for graph isomorphism [Bab16b].

For matrix groups, the barriers to Luks's divide-and-conquer strategy lie in primitive linear groups. Note though that compared to Luks's original setting [Luk92], we are allowed a rather generous time bound, namely polynomial in the underlying vector space size. While this generous time bound makes some tasks easier, it does not help with working with large linear primitive groups for our purpose. To handle such groups is one main technical contributions of this work.

2.5 An organising problem: quotient tensor module transporter

We introduce the following problem as an organising problem in this paper, called the quotient tensor module transporter problem, short for the point transporter problem in quotient tensor modules under 1-cocyle induced affine actions.

We first define the notion of tensor modules. It uses the notion of Frobenius twists (Definition 7).

Definition 3 (Tensor modules)**.** Let $V_1, \ldots, V_s$ be vector spaces over $\mathbb{F}_q$. A *tensor module* over $(V_1, \ldots, V_s)$ is a finite direct sum of tensor products of V_i, the Frobenius twist $V_i^{[r]}$, and their dual spaces, with repetitions allowed.

Each tensor product appearing as a summand is called a *component* of the tensor module. The *degree* of a component is its number of tensor factors. The trivial module of dimension 1 is of degree 0. The *degree* of a tensor module is the maximum degree of its components.

We now introduce a key algorithmic problem in this paper.

Definition 4 (Quotient tensor module transporter)**.** An instance of the quotient tensor module transporter problem over $\mathbb{F}_q$ consists of the following data.

1. Vector spaces $V_1, \ldots, V_s$, where $V_i \cong \mathbb{F}_q^{d_i}$. Set $V = \bigoplus_{i \in [s]} V_i$ and $d = \sum_i d_i$.
2. A subgroup
$$H = \langle S \rangle \leq \prod_{i=1}^{s} \mathrm{GL}(V_i) \leq \mathrm{GL}(V),$$

given by explicit generator matrices, where the product is embedded block-diagonally with respect to $V = \bigoplus_{i=1}^{s} V_i$.

3. An H-tensor module Z.
4. Explicitly given H-invariant subspaces $P \leq N \leq Z$. Let $Q = N/P$, and let ϕ be the induced action of H on Q.
5. For each $g \in S$, a vector $c_g \in Q$ is given, with the promise that the maps $\psi_g : Q \to Q$ defined by $\psi_g(z) = \phi_g(z) + c_g$ $(g \in S)$ extend to an affine action of H on Q.
6. Two points $x, y \in Q$.

The task is to compute the coset $\mathrm{Trans}_H(x, y) = \{h \in H : \psi_h(x) = y\}$.

Quotient tensor module transporter and divide-and-conquer. Quotient tensor module transporter has a natural origin in Luks's divide-and-conquer for matrix groups. Consider $G \leq \mathrm{GL}(n, q)$ acting on $V = \mathbb{F}_q^n$, and let $U \leq V$ be G-invariant. Suppose $x', y' \in V$ and we wish to compute $\mathrm{Trans}_G(x', y') = \{g \in G : g(x') = y'\}$. There is an induced action of G on $Q = V/U$. A natural divide-and-conquer strategy first computes the transporter of $x = x' + U$ and $y = y' + U$ in Q. If this transporter is empty, then $\mathrm{Trans}_G(x', y')$ is empty. Otherwise, write $\mathrm{Trans}_G(x, y) = tH$, where $H = G_x$ is the stabilizer of x in the quotient action.

To recover $\mathrm{Trans}_G(x', y')$, put $z = t^{-1}y' - x' \in U$. For $h \in H$, the vector $c_h = hx' - x'$ lies in U because h fixes $x' + U$. Moreover, $c_{hk} = c_h + hc_k$, so $\psi_h(u) = hu + c_h$ defines an affine action of H on U. Now $thx' = y'$ holds exactly when $\psi_h(0) = z$. Consequently, $\mathrm{Trans}_G(x', y') = t\ \mathrm{Trans}_{H,\psi}(0, z)$. Thus the remaining problem is an affine transporter problem on U. If its answer is aK, where $K = \{h \in H : \psi_h(0) = 0\}$, then the original transporter is taK; if its answer is empty, so is the original transporter. This quotient–fibre construction, as already observed by Luks [Luk92] (Lemma 3.5) explains why affine actions arise naturally even when the original action is linear.

In this paper we mostly use the quotient tensor module transporter problem arising from the setting above, except in the lifting step for matrix tuple congruence in characteristic two (Section 4.4), where the affine action instead describes changes in the obstruction to lifting a solution of $x^*x = h$.

Quotient tensor module transporter and our main problems. We use quotient tensor module transporter when the ambient module Z is a tensor module (Definition 3). The degree of the tensor module Z is a key parameter for the difficulty of this problem. Depending on the degree, Definition 4 has a close relation with Matrix Tuple Congruence, 3-Tensor Isomorphism, and Cayley Table Group Isomorphism.

- Degree-1 quotient tensor module transporter: this is a relatively straightforward application of the techniques already in Luks and Miyazaki [Luk92, Miy99] (Theorem 4.9). It is a subroutine used in degree-2 quotient tensor module transporter.
- Degree-2 quotient tensor module transporter: this subsumes and uses Matrix Tuple Congruence as a subroutine (Theorem 7.1). It cannot be simply solved by Luks and Miyazaki's techniques anymore. It is also closely related to matrix group intersection (Theorem 7.3). It is a subroutine used in tensor isomorphism.
- Degree-3 quotient tensor module transporter: it subsumes and uses 3-Tensor Isomorphism as a subroutine (Theorem 8.10). It is a subroutine used in group isomorphism.

2.6 An exposition on some technical aspects of Matrix Tuple Congruence

The reduction in Section 4.2 converts congruence of two m-tuples of $n \times n$ matrices over $\mathbb{F}_q$, where $q = 2^f$, into solving $x^*x = h$ in the finite-dimensional unital $*$-algebra $F \leq \mathrm{M}(n,q) \times \mathrm{M}(n,q)$ defined in Eq. (4.2). Here $*$ is an $\mathbb{F}_q$-linear involution, satisfying $(ab)^* = b^*a^*$ and $(a^*)^* = a$, and $h \in F^\times$ is $*$-*symmetric*, meaning $h^* = h$. We explain how the obstruction to lifting solutions through the radical leads to a point affine transporter problem.

The obstruction space. For an $\mathbb{F}_q$-vector space M with a linear involution s, define

$$Q(M) := \frac{\ker(I_M + s)}{\mathrm{im}(I_M + s)},$$

where I_M is the identity map on M. Since $(I_M + s)^2 = 0$ in characteristic two, its image lies in its kernel. Thus $Q(M)$ consists of classes of elements fixed by s, modulo those of the form $m + s(m)$. For $J = \operatorname{rad} F$, the powers of J are preserved by $*$. Hence each radical layer $M_i := J^i/J^{i+1}$ has an induced involution $s_i(y + J^{i+1}) = y^* + J^{i+1}$. The obstruction space used in Section 4.4 is precisely $Q_i = Q(M_i)$; see Eq. (4.6).

Suppose $x_i \in F^\times$ satisfies $x_i^* x_i \equiv h \pmod{J^i}$, with $i \geq 1$. A lift of this candidate has the form $x_{i+1} = (1_F + y)x_i$, with $y \in J^i$, where 1_F is the identity element of F. Since $y^*y \in J^{2i} \subseteq J^{i+1}$, the next lifting step is exactly the linear correction equation in Eq. (4.5):

$$(I_{M_i} + s_i)\overline{y} = \delta_i, \qquad \delta_i := \left((x_i^{-1})^* h x_i^{-1} - 1_F\right) + J^{i+1}, \qquad \overline{y} = y + J^{i+1}.$$

The error δ_i is fixed by s_i, so the *obstruction class* $\mathrm{obs}_i(x_i) := \delta_i + \mathrm{im}(I_{M_i} + s_i)$ belongs to Q_i. The solution $x_i \bmod J^i$ lifts to a solution modulo J^{i+1} if and only if $\mathrm{obs}_i(x_i) = 0$.

Classical interpretations. Let the generator of the two-element group C_2 act on M by s. Then $Q(M) = \widehat{H}^0(C_2, M)$, the degree-zero *Tate cohomology* group, namely the fixed subspace modulo $\mathrm{im}(I_M + s)$. For $M = F$ and $s = *$, Bak's maximal and minimal form parameters (with parameter $\lambda = 1$) are precisely $\ker(I_F + *)$ and $\mathrm{im}(I_F + *)$. Bak [Bak79, p. 3] equips their quotient with the action $a \cdot [b] = [aba^*]$, where brackets denote quotient classes. Ranicki [Ran80, p. 112] explicitly describes both the Tate quotient and this module action. The same formula defines an action on Q_i that factors through F/J. Additivity in a holds because, for $b \in M_i$ fixed by s_i, the cross terms $abc^* + cba^* = (I_{M_i} + s_i)(abc^*)$ vanish in Q_i. This is a module action of the underlying ring: a scalar $c \in \mathbb{F}_q$ acts as c^2, which explains the appearance of Frobenius twists (Definition 7).

Serre [Ser14, §3.7, Proposition 3.7.4] gives a closely related *lifting application* for a square-zero radical extension of a symplectic matrix algebra. His equations (3.7.9)–(3.7.10) express liftability by a congruence modulo elements $z + z^*$, equivalently by preservation of an associated quadratic form.

Connection to point transporters. For a group H acting affinely on a vector space Q by $\alpha_g(u) = L_g(u) + c_g$, with L_g invertible and linear and $c_g \in Q$, the *point affine transporter problem* is to compute $\mathrm{Trans}_{H,\alpha}(u, v) := \{g \in H : \alpha_g(u) = v\}$; see Definition 12 for the formal input. Here let $H_i := \{g \in F^\times : g^*g \equiv 1_F \pmod{J^i}\}$. All solutions modulo J^i are represented by $H_i x_i$. Lemma 4.4 constructs an affine action on Q_i satisfying $\mathrm{obs}_i(gx_i) = \alpha_g(\mathrm{obs}_i(x_i))$. The translation term matters: a linear action alone cannot send a nonzero obstruction class to zero. Thus $\mathrm{Trans}_{H_i,\alpha}(\mathrm{obs}_i(x_i), 0)$ consists exactly of the changes g for which gx_i can be lifted. For each such g, solving Eq. (4.5) with x_i replaced by gx_i produces a lift modulo J^{i+1}.

The decomposition in Lemma 4.6 is explained by the Bak–Ranicki module structure above: semisimplicity of F/J splits Q_i into simple modules, with the scalar twist described above, and these summands are invariant under the linear parts L_g. Every simple F/J-module occurs as a composition factor of the faithful F-module $\mathbb{F}_q^{2n}$, so the summands have $\mathbb{F}_q$-dimension at most $2n$. A concrete model is symmetric matrices modulo alternating matrices, where only the diagonal survives and transforms by entrywise-squared matrices; the symplectic version appears explicitly in [AHS11, §3, equation (3.2)].

Computing Q_i and correcting a fixed liftable candidate take polynomial time, but finding a suitable $x \in H_i$ remains an orbit search. Treating Q_i as an unstructured vector space gives a search over up to $q^{\dim_{\mathbb{F}_q} Q_i}$ points, hence only a $q^{O(n^2)}$ bound since $\dim_{\mathbb{F}_q} Q_i \le \dim_{\mathbb{F}_q} F \le 2n^2$. The key contribution of Lemma 4.6 is an explicit decomposition into invariant summands of dimension at most $d = 2n$. Proposition 4.5 uses the resulting flag to perform orbit and stabilizer computations on at most q^d points at a time; maintaining transporter cosets avoids multiplying the search spaces across the flag. This yields $q^{O(d)}\mathrm{poly}(\mathrm{InSize}) = q^{O(n)}\mathrm{poly}(\mathrm{InSize})$ overall, where InSize is the input bit length; the number of radical layers is polynomial.

3 Preliminary

3.1 Notations and conventions

For $n \in \mathbb{N}$, $[n]$ denotes $\{1, 2, \ldots, n\}$. We write $\bigsqcup_{i \in I} X_i$ for the *disjoint union* of the sets X_i.

We use InSize to denote the input size to an algorithmic problem. In asymptotic bounds, a subscript such as r in $O_r(\cdot)$ allows the hidden constant to depend on r, but not on the other input parameters.

We use $\mathrm{M}(\ell \times m, q)$ to denote the linear space of $\ell \times m$ matrices over $\mathbb{F}_q$, and $\mathrm{GL}(n, q)$ the *general linear group* of degree n over $\mathbb{F}_q$. Thus $\mathrm{GL}(n, q)$ consists of the invertible $n \times n$ matrices over $\mathbb{F}_q$, with matrix multiplication as its group operation. We abbreviate $\mathrm{M}(n \times n, q)$ to $\mathrm{M}(n, q)$. For an explicitly named field $\mathbb{K}$, the notations $\mathrm{Mat}(n, \mathbb{K})$ and $\mathrm{Mat}_n(\mathbb{K})$ denote this same square-matrix space over $\mathbb{K}$; likewise $\mathrm{GL}(n, \mathbb{K}) = \mathrm{GL}_n(\mathbb{K})$, and $\mathrm{GL}(n, q) = \mathrm{GL}(n, \mathbb{F}_q)$. We write X^{t} for transpose and $X^{-\mathrm{t}} = (X^{-1})^{\mathrm{t}}$ for inverse transpose.

3.2 Linear and multilinear algebra

Let $\mathbb{F}$ be a field. We use $\mathbb{F}^n$ to denote the linear space of length-n column vectors over $\mathbb{F}$.

Let U, V be vector spaces over $\mathbb{F}$. A map $\phi : U \to V$ is $\mathbb{F}$-*linear* if

$$\phi(au + bv) = a\phi(u) + b\phi(v) \text{ for } a, b \in \mathbb{F},\ u, v \in U.$$

If, in addition, every element of V has exactly one preimage under ϕ, we call ϕ an $\mathbb{F}$-*linear isomorphism* and write $\phi : U \xrightarrow{\sim} V$.

All vector spaces are assumed to be finite dimensional.

We write $\mathrm{Hom}_{\mathbb{F}}(U, V)$ for the vector space of $\mathbb{F}$-linear maps from U to V, and $\mathrm{End}_{\mathbb{F}}(U) = \mathrm{Hom}_{\mathbb{F}}(U, U)$, with multiplication given by composition. Its invertible elements form $\mathrm{GL}(U)$. The *dual space* is $U^* = \mathrm{Hom}_{\mathbb{F}}(U, \mathbb{F})$. For a basis $e_1, \ldots, e_n$ of U, the *dual basis* $e_1^*, \ldots, e_n^*$ is specified by $e_i^*(e_j) = 1$ if $i = j$ and $e_i^*(e_j) = 0$ otherwise. For $S \subseteq U$, $\mathrm{span}_{\mathbb{F}} S$ is the set of finite $\mathbb{F}$-linear combinations of elements of S; for vectors, angle brackets also denote their linear span when the field is clear.

For a linear map $f : U \to V$, we use

$$\ker f = \{u \in U : f(u) = 0\}, \qquad \mathrm{im}\, f = \{f(u) : u \in U\}, \qquad \mathrm{rank}\, f = \dim_{\mathbb{F}}(\mathrm{im}\, f).$$

For any map $f : U \to V$ and subset $S \subseteq V$, its *inverse image* is $f^{-1}(S) = \{u \in U : f(u) \in S\}$; this notation does not require f to be invertible. We also write $f^{-1}(v)$ for the fiber $f^{-1}(\{v\})$ when the context is clear. If f is an isomorphism, $f^{-1} : V \to U$ denotes its inverse map instead. For $W \subseteq U$, the *restriction* $f|_W$ is given by $(f|_W)(w) = f(w)$ for $w \in W$. The arrows $\hookrightarrow$ and $\twoheadrightarrow$ denote, respectively, an injective map (equal images imply equal inputs) and a surjective map (every target element has a preimage).

For subspaces $P \leq N$, the *quotient space* is $N/P = \{x+P : x \in N\}$, where $x+P = \{x+p : p \in P\}$ and $x+P = y+P$ exactly when $x-y \in P$. Its operations are $(x+P)+(y+P) = (x+y)+P$ and $a(x+P) = ax+P$; the *quotient map* is $N \twoheadrightarrow N/P$, $x \mapsto x+P$. The *direct sum* $U \oplus V$ consists of pairs (u, v) with componentwise operations, and $U^{\oplus m}$ denotes m copies of U. For subspaces of one space, $U+V = U \oplus V$ means that every vector in $U+V$ has a unique expression $u+v$, equivalently $U \cap V = 0$.

The *tensor product* $U \otimes_{\mathbb{F}} V$ is the vector space generated by symbols $u \otimes v$, subject to the bilinearity relations

$$\begin{aligned}(au + bu') \otimes v &= a(u \otimes v) + b(u' \otimes v),\\ u \otimes (av + bv') &= a(u \otimes v) + b(u \otimes v') \qquad (a, b \in \mathbb{F}).\end{aligned}$$

If (e_i) and (f_j) are bases of U and V, then $(e_i \otimes f_j)_{i,j}$ is a basis of $U \otimes_{\mathbb{F}} V$. For linear maps $A : U \to U'$ and $B : V \to V'$, the induced map is $(A \otimes B)(u \otimes v) = A(u) \otimes B(v)$, extended linearly. We omit the field subscript when it is clear.

For a finite field extension $K/\mathbb{F}$, its *degree* is $[K : \mathbb{F}] = \dim_{\mathbb{F}} K$. *Restriction of scalars* views a K-space as an $\mathbb{F}$-space with the same vectors and addition, allowing only scalars from $\mathbb{F}$; its dimension is multiplied by $[K : \mathbb{F}]$. For any field extension $K/\mathbb{F}$, *extension of scalars* sends an $\mathbb{F}$-space U to the K-space $K \otimes_{\mathbb{F}} U$. An *algebraic closure* $\overline{\mathbb{F}}$ is an algebraic extension of $\mathbb{F}$ that is algebraically closed, so every nonconstant polynomial over $\overline{\mathbb{F}}$ has a root in $\overline{\mathbb{F}}$. For K-spaces and $\sigma \in \mathrm{Aut}(K)$, a map f is *σ-semilinear* if it is additive and $f(av) = \sigma(a)f(v)$ for all $a \in K$ and v in its domain; here $\mathrm{Aut}(K)$ denotes the field automorphism group of K.

3.3 Groups and algebras

Group theory basic notions. Let G be a finite group. For $S \subseteq G$, $\langle S \rangle$ is the smallest subgroup of G containing S, equivalently the set of products of elements of S and their inverses, including the empty product 1. For $H \leq G$, a *left coset* is $gH = \{gh : h \in H\}$, and the *index* $[G : H]$ is the number of left cosets, equal to $|G|/|H|$. We write $H \trianglelefteq G$ if $gHg^{-1} = H$ for every $g \in G$; then G/H is the quotient group, with $(gH)(hH) = ghH$. A *group homomorphism* $f : G \to H$ satisfies $f(gh) = f(g)f(h)$; its kernel is $\ker f = \{g \in G : f(g) = 1_H\}$. A bijective homomorphism is an *isomorphism*, and $\mathrm{Aut}(G)$ is the group of isomorphisms $G \to G$ under composition.

We use $\mathrm{Z}(G)$ and G' to denote the *centre* and *commutator subgroup* of G. Explicitly,

$$\mathrm{Z}(G) = \{z \in G : zg = gz \text{ for all } g \in G\}, \qquad G' = [G, G] = \langle g^{-1}h^{-1}gh : g, h \in G \rangle.$$

A group is *simple* if it is nontrivial and its only normal subgroups are 1 and itself. It is *solvable* if the sequence $G^{(0)} = G$, $G^{(i+1)} = (G^{(i)})'$ reaches 1. An *elementary abelian p-group* is an abelian group in which every nonidentity element has order p; it is the additive group of an $\mathbb{F}_p$-vector space. A subgroup is *characteristic* if every automorphism of G preserves it. An *inner automorphism* has the form $x \mapsto gxg^{-1}$ for some $g \in G$. A group G is *perfect* if $G = G'$. It is *quasisimple* if it is perfect, and $G/\mathrm{Z}(G)$ nonabelian simple.

We use $G \cong H$ to denote that G and H are isomorphic. If two finite groups G and H are isomorphic, the set of isomorphisms from G to H form a coset of $\mathrm{Aut}(G)$. When we say that we

compute the coset of isomorphisms from G to H, we mean the following: if $G \cong H$, the output is an isomorphism from G to H and a generating set of $\mathrm{Aut}(G)$; if $G \ncong H$, the output is an empty set.

The *solvable radical* of G, $\mathrm{srad}(G)$, is the largest solvable normal subgroups. Following [BCGQ11, BCQ12], a group with trivial solvable radical is called *semisimple*. The *p-core* of G, $\mathrm{O}_p(G)$, is the largest normal subgroup of p-power order.

A *subnormal subgroup* H of G is a subgroup of G such that there exists a normal subgroup chain $H \trianglelefteq H_1 \trianglelefteq \ldots \trianglelefteq H_k \trianglelefteq G$. A *component* of G is a quasisimple subnormal subgroup of G. We say that G is a *central product* of $H_1, H_2 \leq G$, if G is generated by $H_1 \cup H_2$ and H_1 and H_2 commute elementwise; note that in this case we have $H_1 \cap H_2 \subseteq \mathrm{Z}(H_1) \cap \mathrm{Z}(H_2)$.

For $D \leq G$, its *normalizer* and *centralizer* are

$$N_G(D) = \{g \in G : gDg^{-1} = D\},$$
$$C_G(D) = \mathrm{Cent}_G(D) = \{g \in G : gd = dg \text{ for every } d \in D\}.$$

For a finite set X, $\mathrm{Sym}(X)$ is the group of bijections $X \to X$ under composition. We write $\mathrm{Sym}_m = S_m = \mathrm{Sym}([m])$, and Alt_m for its subgroup of even permutations, those expressible as a product of an even number of transpositions. The *special linear group* is $\mathrm{SL}(n, q) = \{g \in \mathrm{GL}(n, q) : \det g = 1\}$.

Actions and representations. A (left) *action* of a group G on a set X assigns a bijection $\alpha_g : X \to X$ to each $g \in G$, with $\alpha_1 = \mathrm{id}_X$ and $\alpha_{gh} = \alpha_g \circ \alpha_h$. A *representation* on an $\mathbb{F}$-space U is a homomorphism $\rho : G \to \mathrm{GL}(U)$; thus $\rho(gh) = \rho(g)\rho(h)$ and $\rho(1) = I_U$. It defines a linear action $g \cdot u = \rho(g)u$. An action is *faithful* if only the identity of G acts as the identity map (for a representation, $\ker \rho = \{1\}$), and *trivial* if every group element acts as the identity map.

Algebras. Let $\mathbb{F}$ be a field, let V be a vector space over $\mathbb{F}$, and let $E \leq \mathrm{End}_{\mathbb{F}}(V)$ be a unital associative $\mathbb{F}$-algebra given by a matrix basis.

A *unital subalgebra* of E is an $\mathbb{F}$-linear subspace closed under multiplication and containing the identity 1_E.

Write $E^\times$ for its *unit group*. Thus $E^\times = \{a \in E : ab = ba = 1_E \text{ for some } b \in E\}$. An *ideal* is a subspace $I \leq E$ with $EI + IE \subseteq I$.

An ideal I is *nilpotent* if $I^t = 0$ for some integer $t \geq 1$, where I^t is the $\mathbb{F}$-span of all products of t elements of I.

Modules are finite-dimensional left modules. A left E-module is an $\mathbb{F}$-space M with an $\mathbb{F}$-bilinear action $E \times M \to M$, $(a, m) \mapsto am$, satisfying $(ab)m = a(bm)$ and $1_E m = m$. A submodule is a linear subspace stable under this action. An E-module is *simple* if it is nonzero and has no nonzero proper submodule, and *semisimple* if it is a direct sum of simple modules. A *composition series* is a flag with simple successive quotients, called the *composition factors*. The *Jacobson radical* $J = \mathrm{rad}\, E$ is the nilpotent ideal annihilating every simple E-module [Ben91, Sections 1.2–1.5].

A finite-dimensional algebra E is *semisimple* if $\mathrm{rad}\, E = 0$, equivalently if it is semisimple as a left module over itself.

For an E-module M, write a_M for the action of $a \in E$. For E-modules M, N, solve $fa_M = a_N f$ for the unknown $f \in \mathrm{Hom}_{\mathbb{F}}(M, N)$, with a ranging over a matrix basis of E, to compute $\mathrm{Hom}_E(M, N)$. In particular, the *commutant* on M is $\mathrm{End}_E(M)$. Let $H = \langle T \rangle$ be a finite group, let W be a vector space over $\mathbb{F}$, and let $\rho : H \to \mathrm{GL}(W)$ be a *representation*. The *enveloping algebra*

$$A_\rho = \mathrm{span}_{\mathbb{F}}\{\rho(h) : h \in H\} \leq \mathrm{End}_{\mathbb{F}}(W)$$

is the image of the linear extension of ρ to the *group algebra* $\mathbb{F}[H]$. Here $\mathbb{F}[H]$, also written $\mathbb{F}H$, consists of formal sums $\sum_{h\in H} a_h h$ with $a_h \in \mathbb{F}$; addition is coefficientwise and multiplication extends the group law bilinearly. The extension sends $\sum_h a_h h$ to $\sum_h a_h \rho(h)$, making W an $\mathbb{F}[H]$-module. Matrix-span closure of $\{\rho(t) : t \in T\}$ and the identity computes a basis of A_ρ.

An *idempotent* is an element $e \in E$ satisfying $e^2 = e$. It is *central* if it commutes with every element of E, and *primitive* if it is nonzero and cannot be written as $e = e_1 + e_2$ for nonzero idempotents with $e_1 e_2 = e_2 e_1 = 0$. A *primitive central idempotent* is a nonzero central idempotent admitting no such decomposition into central idempotents.

The quotient E/J is a product of full matrix algebras over finite-dimensional division $\mathbb{F}$-algebras; when $\mathbb{F}$ is finite, these division algebras are finite fields. Its *primitive central idempotents* are the projections onto the factors; within a matrix factor, a diagonal matrix unit is a *primitive idempotent*. For an idempotent $e \in E$, write $eE = \{ea : a \in E\}$ for the corresponding *right ideal*. A *Wedderburn–Malcev complement* is a semisimple subalgebra $S \le E$ such that

$$E = S \oplus J \quad \text{as } \mathbb{F}\text{-vector spaces}, \quad S \cong E/J.$$

Such a complement exists whenever E/J is separable over $\mathbb{F}$, in particular when $\mathbb{F}$ is perfect (and hence when $\mathbb{F}$ is finite) [dGIKR97, Theorem 1.2].

Algebras with involutions. Let E be a unital $\mathbb{F}$-algebra. An *involution* on E is an $\mathbb{F}$-linear map $* : E \to E$ satisfying, for $a, b \in E$,

$$(ab)^* = b^* a^*, \quad (a^*)^* = a.$$

The pair $(E, *)$ is a *$*$-algebra*. An element $h \in E$ is *$*$-symmetric* if $h^* = h$; the *unitary group* is

$$\mathrm{U}(E) = \{u \in E^\times : u^* u = 1\}.$$

Given a $*$-algebra $(E, *)$ and an element $h \in E^\times$ satisfying $h^* = h$, the *$*$-symmetric decomposition problem* is to find $x \in E^\times$ satisfying $x^* x = h$, or determine that no such element exists [IQ19, Problem 17].

A *$*$-subalgebra* is a subalgebra invariant under $*$. An *E-bimodule* B has commuting left and right E-actions; $\mathrm{U}(E)$ acts on B by $b \mapsto ubu^{-1}$ for $u \in \mathrm{U}(E)$ and $b \in B$.

For any field $\mathbb{K}$ and $n \ge 1$, the *exchange involution* on the $\mathbb{K}$-algebra $E = \mathrm{Mat}(n, \mathbb{K}) \times \mathrm{Mat}(n, \mathbb{K})$ is

$$(a, b)^* = (b^{\mathrm{t}}, a^{\mathrm{t}}), \quad \mathrm{U}(E) = \{(g, g^{-\mathrm{t}}) : g \in \mathrm{GL}(n, \mathbb{K})\}.$$

Here a^{t} denotes transpose and $g^{-\mathrm{t}} = (g^{-1})^{\mathrm{t}}$. For the *adjoint involution* on $E = \mathrm{End}_{\mathbb{K}}(V)$ associated with a nondegenerate symmetric or alternating bilinear form, or a nondegenerate Hermitian form, on a $\mathbb{K}$-space V, $\mathrm{U}(E)$ is the isometry group of that form. In the Hermitian case, E is viewed as an algebra over the fixed field of the involution on $\mathbb{K}$. Writing the form as b, the adjoint is characterized by $b(av, w) = b(v, a^* w)$ for all $a \in E$ and $v, w \in V$. An isometry is a $g \in \mathrm{GL}(V)$ satisfying $b(gv, gw) = b(v, w)$ for all $v, w \in V$.

3.4 Representations of groups and algebras

Let U be a vector space over $\mathbb{F}$ and G be a group with a representation $\rho : G \to \mathrm{GL}(U)$. Then U is called a *G-module*. A module is *simple* if it is nonzero and its only submodules are 0 and itself, *semisimple* if it is a direct sum of simple modules, and *indecomposable* if it is not a direct sum of two nonzero submodules. It is *absolutely simple* if it remains simple over an algebraic closure,

meaning that $\overline{\mathbb{F}} \otimes_{\mathbb{F}} U$ is simple under the scalar-extended action. For a module M over an algebra A (in particular, $A = \mathbb{F}[G]$ for a G-module), its *socle* $\mathrm{soc}_A(M)$ is the sum of its simple submodules. A *constituent* means a simple composition factor, considered up to isomorphism. For a semisimple module, the *X-isotypic summand* is the sum of all simple submodules isomorphic to a given simple module X; it is a direct sum of copies of X.

A sequence of module homomorphisms is *exact* if the image of each map equals the kernel of the next. In a *short exact sequence*

$$0 \longrightarrow U \xrightarrow{\iota} M \xrightarrow{\pi} W \longrightarrow 0,$$

ι is injective, π is surjective, and $\operatorname{im}\iota = \ker\pi$. It is an extension of W by U, identifying U with $\iota(U)$. The extension is *split* if this copy of U has an A-invariant complement in M.

Definition 5 (Natural representation and natural module)**.** Let $\mathbb{K}$ be a field and $G \le \mathrm{GL}(n,\mathbb{K})$ be a matrix group. For this matrix realization, the *natural representation* is the inclusion $G \hookrightarrow \mathrm{GL}(n,\mathbb{K})$. The corresponding *natural module* is the space of column vectors $\mathbb{K}^n$ with the action

$$g \cdot v := gv \qquad (g \in G,\ v \in \mathbb{K}^n),$$

where gv is matrix multiplication. For a classical group, this refers to its usual defining matrix representation; for example, the natural module of $\mathrm{SL}(n,\mathbb{K})$ is $\mathbb{K}^n$ with this action. If $\mathbb{K}/\mathbb{F}_p$ is finite, restriction of scalars gives the same action on an $\mathbb{F}_p$-space of dimension $n[\mathbb{K}:\mathbb{F}_p]$.

Definition 6 (Contragredient action)**.** Let U be a finite-dimensional vector space over $\mathbb{F}$, and let $\rho : G \to \mathrm{GL}(U)$ be a representation. The *contragredient action* of G on $U^* = \mathrm{Hom}_{\mathbb{F}}(U,\mathbb{F})$ is defined by

$$(g \cdot a)(u) := a\big(\rho(g)^{-1}u\big) \quad (g \in G,\ a \in U^*,\ u \in U).$$

In the corresponding dual basis, its matrix is $\rho(g)^{-\mathrm{t}}$, where $-\mathrm{t}$ denotes inverse transpose.

Definition 7 (Frobenius twists)**.** Let $q = p^f$, where p is prime, and let $\rho : G \to \mathrm{GL}(U)$ be a representation of G on $U = \mathbb{F}_q^n$. For an integer $r \ge 0$, the *r-th Frobenius twist* $U^{[r]}$ is the representation of G on U defined by $\rho^{[r]} : G \to \mathrm{GL}(U)$ such that the (i,j)th entry of $\rho^{[r]}(g)$ is $\rho(g)_{i,j}^{p^r}$.

The isomorphism class of a Frobenius representation is independent of the chosen basis. When a module is subsequently viewed over $\mathbb{F}_p$ by restriction of scalars, its specified defining field $\mathbb{F}_q$, representation, and twist exponent are retained: the twist is formed over $\mathbb{F}_q$ before restriction of scalars. If the representation is given only over $\mathbb{F}_p$, entrywise p^r-th powers leave its matrices unchanged, so its Frobenius twists introduce no new action.

Definition 8 (Induced actions on tensor modules and their quotients)**.** Let $q = p^f$, where p is prime. Let H be a finite group, and let $V_1,\dots,V_s$ be finite-dimensional $\mathbb{F}_q$-vector spaces with fixed bases and given representations $\rho_i : H \to \mathrm{GL}(V_i)$ for $i \in [s]$. Consider a tensor module

$$Z = \bigoplus_{a \in I} \bigotimes_{j=1}^{k_a} W_{a,j},$$

where I is a finite set of summand labels fixed by H, each k_a is a nonnegative integer, and each factor $W_{a,j}$ is a specified $V_i^{[r]}$ or $(V_i^{[r]})^*$ for some $i \in [s]$ and $r \ge 0$. Here $r = 0$ includes the untwisted modules. All tensor products are over $\mathbb{F}_q$.

For each specified factor W, define its representation by

$$\rho_W(h) := \begin{cases} \rho_i^{[r]}(h), & W = V_i^{[r]}, \\ \rho_i^{[r]}(h)^{-\mathrm{t}}, & W = (V_i^{[r]})^*, \end{cases} \qquad (h \in H),$$

where $-\mathrm{t}$ denotes *inverse transpose*, using the corresponding dual basis in the second case.

For $z \in Z$, write $z = (z_a)_{a \in I}$ according to the given direct-sum decomposition, with $z_a \in \bigotimes_{j=1}^{k_a} W_{a,j}$. Define the action of $h \in H$ on Z by

$$h \cdot z := \left(\left(\bigotimes_{j=1}^{k_a} \rho_{W_{a,j}}(h) \right) z_a \right)_{a \in I}.$$

When $k_a = 0$, the corresponding summand is $\mathbb{F}_q$ and the empty tensor product of matrices is the identity on $\mathbb{F}_q$.

Let $P \leq N \leq Z$ be invariant under this action, and set $Q := N/P$. For each $h \in H$, the *induced linear map* $L_h : Q \to Q$ is defined by

$$L_h(z + P) := h \cdot z + P \quad (z \in N).$$

The invariance of N and P makes these maps well defined. They satisfy $L_{1_H} = \mathrm{id}_Q$ and $L_{gh} = L_g \circ L_h$ for $g, h \in H$, and hence define a linear action of H on Q.

Definition 9 (Enveloping algebra)**.** Let U be a vector space over $\mathbb{F}$, and let $\rho : G \to \mathrm{GL}(U)$ be a representation. The *enveloping algebra* of ρ over $\mathbb{F}$ is

$$A_\rho = \mathrm{Span}_{\mathbb{F}}\{\rho(g) : g \in G\} \leq \mathrm{End}_{\mathbb{F}}(U).$$

Equivalently, A_ρ is the smallest unital associative $\mathbb{F}$-subalgebra of $\mathrm{End}_{\mathbb{F}}(U)$ containing $\rho(G)$.

A subspace of U is G-invariant if and only if it is invariant under A_ρ.

3.5 Computing with groups and algebras

Group encodings and input/output conventions. All fields, vector spaces, and linear maps are given in coordinates. An input group $H = \langle \mathcal{S} \rangle$ is given by generators in an explicit encoding supporting multiplication, inversion, and equality testing in polynomial time. Its matrix and permutation coordinates define a homomorphism

$$\begin{aligned} \pi : H &\longrightarrow \mathrm{GL}(c, q) \times \mathrm{Sym}(\Omega), \\ h &\longmapsto (M_h, \sigma_h), \end{aligned}$$

where $\mathbb{F}_q$ has characteristic p and Ω is an explicitly listed finite set. The matrix M_h combines the encoding matrices block-diagonally with a specified block decomposition, so c is the sum of their representation dimensions. Absent matrix or permutation coordinates are represented by $c = 0$ or $\Omega = \varnothing$, with the corresponding group interpreted as trivial.

Write $A := \ker \pi$. The encoding allows A to be nontrivial, in which case it is an elementary abelian p-group with additive coordinates given by an injective homomorphism

$$\eta : A \longrightarrow (\mathbb{F}_p^a, +).$$

The case $A = 1$ is included by taking $a = 0$. For any word w in $\mathcal{S}$ with $\pi(w) = 1$, its coordinate vector $\eta(w)$ can be recovered in polynomial time in the encoded data and word representation. Conjugation is supplied in these coordinates: for each $s \in \mathcal{S}$, a matrix $C_s \in \mathrm{GL}(a, p)$ is given such that

$$\eta(sbs^{-1}) = C_s\eta(b) \text{ for } b \in A.$$

The coordinate formulas for the original group operations include any corrections required by the group law. In particular, if $\pi(h) = \pi(g)$, then

$$h = g \quad \Longleftrightarrow \quad \eta(hg^{-1}) = 0.$$

Thus the matrix and permutation images, together with the additive kernel coordinates, distinguish the original group elements.

Representations and affine actions are specified on the given generators. Thus a representation $\rho : H \to \mathrm{GL}(V)$ is given by the matrices $\rho(s)$ for $s \in \mathcal{S}$, and an *affine action* $\alpha_h(u) = L_h u + c_h$ is given by L_s and c_s for $s \in \mathcal{S}$. These data are required to define the stated representation or action. Notation indexed by all $h \in H$ does not require listing its values on every group element.

Each original matrix representation used in an algorithmic instance, including those used only for the group encoding, is counted once in its matrix dimension parameter d. In particular,

$$c \le d.$$

This includes encoding spaces that occur in no tensor summand. Repeated tensor summands, the permutation degree $|\Omega|$, the additive coordinate dimension a, and all explicit maps contribute to the total input length InSize.

All subgroup generators, witnesses, and kernel elements are returned in the original group encoding, with straight-line programs in $\mathcal{S}$ as defined below. Products act from right to left: $\alpha_{gh} = \alpha_g \circ \alpha_h$. For a nonempty transporter under a left action α, the output consists of a witness g_0 and generators of the *source stabilizer* H_x, so that

$$\mathrm{Trans}_{H,\alpha}(x, y) = g_0 H_x, \quad H_x := \{h \in H : \alpha_h(x) = x\}.$$

Permutation group algorithms: when they are useful for matrix groups. A matrix group $G \le \mathrm{GL}(c, \mathbb{F}_p)$, with $c \ge 1$, acts faithfully on the p^c vectors of $\mathbb{F}_p^c$. More generally, jointly faithful linear actions of G on spaces $V_1, \ldots, V_s$ give a faithful permutation action on $\bigsqcup_i V_i$, of degree $\sum_i p^{\dim V_i}$. Thus, for $d = \sum_i \dim V_i$, polynomial-time permutation-group algorithms run in $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$ time, within our allowed bound. An additional explicit permutation action is included by disjoint union. Symbolic additive kernels are treated separately below.

The permutation group model. Let X be a finite set. Permutation groups $G \le \mathrm{Sym}(X)$ are given by lists of generators; write $G = \langle S \rangle$ and $D = |X|$. A *straight-line program* of length ℓ in S is a sequence $g_1, \ldots, g_\ell$ of group elements, with instructions specifying for each i either

$$g_i \in S, \quad g_i = g_j g_k \ (j, k < i), \quad \text{or} \quad g_i = g_j^{-1} \ (j < i).$$

The program specifies output indices; the elements at those positions are its outputs [Bab16a, Definition 3.1.9]. Computing a subgroup means producing generators; computing a coset aK, where $a \in G$ and $K \le G$, means producing a and generators of K. Group elements produced by our algorithms have straight-line programs in the input generators. The same instructions evaluate any additional matrix or affine actions.

The following tasks can be solved in deterministic polynomial time in D and the input length. Computing with permutation groups is a major area in computational group theory, tracing back to the foundational work of Sims in the 1970s [Sim71]; see [Luk93, Section 3] and [Ser03, Sections 2.1.1, 3.1, 4.1–4.2, and 5.1].

(i) *Constructive membership:* given $\sigma \in \mathrm{Sym}(X)$, decide whether $\sigma \in G$; if so, produce a straight-line program in S computing σ.

(ii) *Order:* compute $|G|$.

(iii) *Orbits and stabilizers:* given $u \in X$ and $A \subseteq X$, compute $Gu = \{gu : g \in G\}$, $G_u = \{g \in G : gu = u\}$, and $G_{(A)} = \{g \in G : ga = a \text{ for all } a \in A\}$.

(iv) *Point transporter:* given $u, v \in X$, compute $\mathrm{Trans}_G(u, v) = \{g \in G : gu = v\}$ as either $\varnothing$ or aG_u, where $a \in G$ and $au = v$.

(v) *Normal closure:* given $T \subseteq G$, compute $\langle T \rangle^G$, the least normal subgroup of G containing T.

(vi) *Succinct generating set:* given $K = \langle T \rangle \leq G$, find $T' \subseteq T$ with $\langle T' \rangle = K$ and $|T'| \leq \lfloor \log_2 |K| \rfloor$.

The quotient group model. In the Kantor–Luks model [KL90b, Sections 3–4], the input consists of generating sets S for $G \leq \mathrm{Sym}(X)$ and T for $K \trianglelefteq G$. A coset gK is represented by $g \in G$; two cosets are equal if and only if their representatives differ by an element of K. A subgroup $U = \langle g_1K, \ldots, g_rK \rangle \leq G/K$ is specified by representatives $g_1, \ldots, g_r \in G$ of its generators. Its preimage in G is $\langle T \cup \{g_1, \ldots, g_r\} \rangle$. Polynomial time is measured in $|X|$ and the input length.

Write $Q = G/K$. The following tasks can be solved in deterministic polynomial time [KL90b, Sections 4 and 12].

(i) *Order:* compute $|Q|$.

(ii) *Presentation:* compute generators $x_1, \ldots, x_t \in Q$ and a finite set R of defining relator words such that $Q = \langle x_1, \ldots, x_t \mid R \rangle$.

(iii) *Normal closure:* given $H \leq Q$, compute H^Q, the least normal subgroup of Q containing H.

(iv) *Centralizer of a normal subgroup:* given $N \trianglelefteq Q$, compute $C_Q(N) = \{q \in Q : qn = nq \text{ for all } n \in N\}$.

(v) *Composition series:* compute $Q = Q_0 \triangleright Q_1 \triangleright \cdots \triangleright Q_t = 1$ with each Q_i/Q_{i+1} simple.

(vi) *Socle:* compute $\mathrm{Soc}(Q)$, the subgroup generated by the minimal nontrivial normal subgroups of Q.

(vii) *Solvable radical:* compute $\mathrm{srad}(Q)$, the largest solvable normal subgroup of Q.

(viii) *ℓ-core:* given a prime ℓ, compute $O_\ell(Q)$, the largest normal ℓ-subgroup of Q.

Representatives of output elements have straight-line programs in S. The case $K = 1$ gives presentations of permutation groups.

Computing kernels of homomorphisms. Let E and Q be finite groups, and let T be a generating set of E. A surjective homomorphism $\pi : E \to Q$ is specified by the images $\pi(u)$ for $u \in T$. The task is to compute generators of $\ker \pi$. Suppose generators $x_1, \ldots, x_t \in Q$ and a finite set R of defining relator words give a presentation $Q = \langle x_1, \ldots, x_t \mid R \rangle$. Assume also that *lifts* $e_i \in E$ with

$\pi(e_i) = x_i$ and words w_u satisfying $w_u(x_1, \dots, x_t) = \pi(u)$, for $u \in T$, are known. For $r \in R$, write $r(e_1, \dots, e_t)$ for the evaluation of the relator on these lifts. Then [Luk92, Lemma 4.2]

$$\ker \pi = \left\langle \{r(e_1, \dots, e_t) : r \in R\} \cup \{u\, w_u(e_1, \dots, e_t)^{-1} : u \in T\} \right\rangle^E.$$

Thus normal closure converts the displayed normal generators into subgroup generators. The identity requires no solvability assumption. When E is given as a permutation group, normal closure takes polynomial time. For an elementary abelian kernel given in vector coordinates, it is closure of a linear span under conjugation.

Applying this identity to the elementary abelian image, together with Gaussian elimination on the generator images, gives the following lemma.

Lemma 3.1 (Kernels of additive homomorphisms)**.** *Let $H = \langle S \rangle \le \mathrm{Sym}(X)$, let $m \ge 0$ be an integer, and put $B = \mathbb{F}_p^m$. Let $\eta : S \to B$ be a function extending to a homomorphism $H \to (B, +)$, also denoted by η. A basis $b_1, \dots, b_t$ of $\eta(H)$, lifts $h_i \in H$ with $\eta(h_i) = b_i$, and generators of $\ker \eta$ can be computed deterministically in time polynomial in $|X|, |S|, m$, and $\log p$. The output, including straight-line programs in S, has the same polynomial bound.*

For the symbolic additive coordinates in the group encoding specified above, Lemma 7.7 gives subgroup membership tests and succinct generating sets in $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$ time. Normal closure then uses successive conjugation and membership tests. That lemma computes additive kernels directly from the presentation identity and linear algebra, without enumerating their vector spaces.

Transporters for group actions. For an action α of a group H on a set Ω and $x, y \in \Omega$, write

$$\mathrm{Trans}_{H,\alpha}(x, y) = \{h \in H : \alpha_h(x) = y\}, \quad \mathrm{Stab}_{H,\alpha}(x) = \mathrm{Trans}_{H,\alpha}(x, x).$$

We omit α when the action is clear. For example, if α is an affine action on a vector space Ω and $V \le \Omega$ is invariant under its linear parts, then the induced action on Ω/V is $h \cdot (v + V) = \alpha_h(v) + V$, and we write

$$\mathrm{Trans}_H(x + V, y + V) = \{h \in H : \alpha_h(x) + V = y + V\}.$$

Combining transporter cosets. Let H act on a set Ω, and let $x, y \in \Omega$. Suppose the nonempty transporter is given as a finite union $\mathrm{Trans}_H(x, y) = \bigcup_{j \in J} g_j S_j$, where $g_j \in H$ and $S_j \le H$ for every $j \in J$. For any $g_0 \in \mathrm{Trans}_H(x, y)$,

$$\begin{aligned} \mathrm{Trans}_H(x, y) &= g_0\, \mathrm{Stab}_H(x), \\ \mathrm{Stab}_H(x) &= \langle S_j,\ g_0^{-1} g_j : j \in J \rangle. \end{aligned} \tag{3.1}$$

This is the standard combination of coset answers in divide-and-conquer algorithms; see [Luk82, Sections 2.2 and 3.4].

Successive equations. Let H act on sets $\Omega_1, \dots, \Omega_t$. For each $i = 1, \dots, t$, fix $x_i, y_i \in \Omega_i$. The task is to compute

$$T = \{h \in H : hx_i = y_i \text{ for all } i = 1, \dots, t\}.$$

Suppose point transporters can be computed in each subgroup of H arising below. Start with $a_0 = 1$ and $H_0 = H$. For $i = 1, \dots, t$, compute $T_i = \mathrm{Trans}_{H_{i-1}}(x_i, a_{i-1}^{-1} y_i)$. If $T_i = \varnothing$, then $T = \varnothing$. Otherwise, write

$$T_i = h_i H_i, \quad H_i = \mathrm{Stab}_{H_{i-1}}(x_i), \quad a_i = a_{i-1} h_i. \tag{3.2}$$

At each step, $a_i H_i$ consists exactly of the elements satisfying the first i equations. Thus, if all comparisons are nonempty, $T = a_t H_t$. The cost is the sum of the comparison costs and group operations. A polynomial number of comparisons with uniform bound $p^{O_r(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$, for fixed r, in the original parameters therefore has the same bound.

Computing algebra structures. A *Las Vegas* algorithm always returns a correct answer; its running time is random, and the stated bounds are for its expected running time. For the algorithms in the following lemma, see [Rón90, Iva00], [CIW97, Section 4, Algorithm 1, and Section 4.1], [IQ19, Theorem 18], [dGIKR97, Theorem 3.1], and [Sho92, Theorem 1].

Lemma 3.2 (Structure of a represented finite-dimensional algebra)**.** *Let V be a vector space of dimension m, and let $E \leq \mathrm{End}_{\mathbb{F}_p}(V)$ be a unital algebra given by a matrix basis. A basis of $J = \operatorname{rad} E$ can be computed deterministically in* $\mathrm{poly}(\mathrm{InSize}, m, \log p)$ *time. The following are computable by a Las Vegas algorithm of expected time* $\mathrm{poly}(\mathrm{InSize}, m, \log p)$*, or deterministically in* $\mathrm{poly}(\mathrm{InSize}, m, p)$ *time: positive integers $a_1, \ldots, a_t$, finite extension fields $\mathbb{K}_1, \ldots, \mathbb{K}_t$ of $\mathbb{F}_p$ in explicit coordinates, a quotient map*

$$\pi : E \longrightarrow \prod_{j=1}^{t} \mathrm{Mat}(a_j, \mathbb{K}_j), \quad \ker \pi = J,$$

and a Wedderburn–Malcev complement to J. A generating set for $E^\times$ is computable deterministically in $\mathrm{poly}(\mathrm{InSize}, m, p)$ *time. For input given by algebra generators, matrix-span closure first computes a basis in deterministic* $\mathrm{poly}(\mathrm{InSize}, m, \log p)$ *time.*

Over $\mathbb{F}_q$, where $q = p^e$ with $e \geq 1$, these bounds are Las Vegas $\mathrm{poly}(\mathrm{InSize}, m, \log q)$ and deterministic $\mathrm{poly}(\mathrm{InSize}, m, \log q, p)$, hence polynomial in InSize, m, q. Module isomorphism is deterministic polynomial time in the explicit input length [CIK97, Corollary 3].

3.6 Group extensions and cohomology

Let D be a finite group and E a finite-dimensional $\mathbb{F}_p$-space with an action $\theta : D \to \mathrm{GL}(E)$. We write $ge = \theta(g)e$ and suppress θ in the cohomology notation. For $i = 1, 2$, an *i-cochain* is a map $D^i \to E$. These maps form an $\mathbb{F}_p$-space $C^i(D, E)$ under pointwise operations.

A 1-cochain $z : D \to E$ is a 1-*cocycle* if

$$z(gh) = z(g) + gz(h) \quad (g, h \in D).$$

It is a 1-*coboundary* if $z(g) = gb - b$ for some fixed $b \in E$. A 1-cocycle defines an affine action $\alpha_g(x) = L_g(x) + c_g$, where $L_g = \theta(g)$ and $c_g = z(g)$; this action can be made linear by a change of origin if and only if z is a 1-coboundary.

A 2-cochain $f : D \times D \to E$ is a 2-*cocycle* if

$$f(g, h) + f(gh, k) = gf(h, k) + f(g, hk) \quad (g, h, k \in D).$$

It is a 2-*coboundary* if there is a 1-cochain $u : D \to E$ such that

$$f(g, h) = (\delta u)(g, h) := u(g) + gu(h) - u(gh).$$

A 2-cocycle is *normalised* if $f(1, g) = f(g, 1) = 0$ for all $g \in D$.

For $i = 1, 2$, write $Z^i(D, E)$ and $B^i(D, E)$ for the spaces of i-cocycles and i-coboundaries, respectively. Then $B^i(D, E) \leq Z^i(D, E) \leq C^i(D, E)$, and the *$i$-th cohomology group* is

$$H^i(D, E) = Z^i(D, E)/B^i(D, E).$$

Two i-cocycles are *cohomologous* if their difference is an i-coboundary. We write $[f] = f + B^i(D, E)$ for the *cohomology class* of f, and call f a *representative* of $[f]$. Every class in $H^2(D, E)$ has a normalised representative. We also write $H^0(D, E) = E^D$ for the *fixed-point space*. Explicitly,

$$E^D = \{e \in E : \theta(d)e = e \text{ for every } d \in D\}.$$

More generally, V^D denotes the subspace fixed by every element of D in a representation on V.

Following [GQ17], an *extension* $E \hookrightarrow G \overset{\pi}{\twoheadrightarrow} D$ identifies the additive group of E with $\ker \pi$. Its *action* is the action of D on E induced by conjugation in G. A *section* is a set map $s : D \to G$ with $\pi \circ s = \mathrm{id}_D$. A section with $s(1) = 1$ determines a normalised 2-cocycle

$$f_s(g, h) = s(g)s(h)s(gh)^{-1} \in E,$$

whose cohomology class is independent of the section. For a fixed action θ, $H^2(D, E)$ classifies extensions up to *equivalence*, meaning an isomorphism inducing the identity on both E and D. An extension *splits* if π admits a homomorphic section, equivalently, if $[f_s] = 0$. It is *central* if $E \leq \mathrm{Z}(G)$, equivalently, if its action is trivial.

3.7 Module (representation) extensions and cohomology

We recall the basic facts about module extensions used below, with examples for readers with relatively little background in module (representation) extensions. For monograph treatments, see Benson [Ben91, Sections 2.5–2.6 and 3.7] and Weibel [Wei94, Sections 3.4, 6.4, and 6.6]; Etingof et al. [EGH$^+$11, Problem 3.9.1] give an introductory treatment in terms of representations.

The common setting. Throughout this and the next subsection, G is a finite group, U is a finite-dimensional $\mathbb{F}_pG$-module, and $V \leq U$ is a submodule. Put $W = U/V$ and let $\pi : U \twoheadrightarrow W$ be the quotient map. Write $\rho : G \to \mathrm{GL}(U)$ for the action on U, and $\rho_h = \rho(h)$ for $h \in G$. The *restricted action* $\check{\rho} : G \to \mathrm{GL}(V)$ is given by $\check{\rho}_h = \rho_h|_V$. The induced action $\hat{\rho} : G \to \mathrm{GL}(W)$ is given by $\hat{\rho}_h(u + V) = \rho_h(u) + V$. Thus $\pi \circ \rho_h = \hat{\rho}_h \circ \pi$. A linear map between G-modules is *equivariant*, or an *G-module homomorphism*, if it commutes with the action of G. For a vector space X, write $X^+ = (X, +)$ for its additive group. In the quotient tensor-module problems, this local notation is used with $G = H$, $U = N$, $V = P$, and $W = Q = N/P$. The maps and group elements in the lemmas below retain their original-group meanings under this identification.

Module extensions. The *short exact sequence*

$$0 \longrightarrow V \longrightarrow U \xrightarrow{\pi} W \longrightarrow 0$$

is an *extension of W by V*. Equivalently, it is an extension of representations of G. Two such extensions are *equivalent* if there is a G-module isomorphism between their middle terms inducing the identity on V and W. A *linear section* of π is an $\mathbb{F}_p$-linear map $s : W \to U$ satisfying $\pi \circ s = \mathrm{id}_W$. Such a section always exists: choose representatives in U of a basis of W and extend linearly. It gives the vector-space decomposition $U = V \oplus s(W)$. The extension *splits* if an equivariant section exists, equivalently, if V has a G-invariant complement in U.

Sections and 1-cocycles. Fix a linear section $s : W \to U$. For $h \in G$ and $w \in W$, both $s(w)$ and $\rho_h(s(\hat{\rho}_h^{-1}(w)))$ are representatives of w in U. Their difference therefore lies in V. Define the linear map $\gamma_h^s : W \to V$ by

$$\gamma_h^s(w) = \rho_h\big(s(\hat{\rho}_h^{-1}(w))\big) - s(w).$$

The superscript records the choice of section. The section s is equivariant exactly when $\gamma_h^s = 0$ for every $h \in G$. Give $\mathrm{Hom}_{\mathbb{F}_p}(W, V)$ the G-action $h \cdot \phi = \check{\rho}_h \circ \phi \circ \hat{\rho}_h^{-1}$. The map $\gamma^s : G \to \mathrm{Hom}_{\mathbb{F}_p}(W, V)$, $h \mapsto \gamma_h^s$, satisfies the 1-cocycle identity

$$\gamma_{gh}^s = \gamma_g^s + g \cdot \gamma_h^s \quad (g, h \in G),$$

as follows from $\rho_{gh} = \rho_g \circ \rho_h$.

Any other linear section has the form $s' = s + t$, where $t : W \to V$ is linear. Substituting $s + t$ in the definition gives $\gamma_h^{s'} = \gamma_h^s + h \cdot t - t$. For each $t \in \mathrm{Hom}_{\mathbb{F}_p}(W, V)$, the map $b_t : G \to \mathrm{Hom}_{\mathbb{F}_p}(W, V)$ defined by $b_t(h) = h \cdot t - t$ is a 1-*coboundary*. Thus changing the section changes its 1-cocycle by a 1-coboundary. In particular, if s_0 is equivariant, then $\gamma^{s_0} = 0$, and every section $s_0 + t$ gives the coboundary b_t. Conversely, if $\gamma^s = b_t$, then $s - t$ is equivariant. This explains why coboundaries correspond to split extensions.

The class $[\gamma^s] \in H^1(G, \mathrm{Hom}_{\mathbb{F}_p}(W, V))$ is independent of the section. Every class arises from an extension, and two extensions give the same class exactly when they are equivalent. The *space of extension classes*, denoted by $\mathrm{Ext}^1_{\mathbb{F}_p G}(W, V)$, is therefore naturally isomorphic to $H^1(G, \mathrm{Hom}_{\mathbb{F}_p}(W, V))$; the zero class corresponds to split extensions [Ben91, Proposition 3.1.8(iii)].

Affine actions and 1-cocycles. For a vector space X, let $\mathrm{AGL}(X)$ denote the group of *invertible affine maps* of X. For affine actions on W, write $L : G \to \mathrm{GL}(W)$ for the induced linear representation, with $L_h := \hat{\rho}_h$. Given a map $c : G \to W$, write $c_h = c(h)$. The maps $\alpha_h : W \to W$, $w \mapsto L_h(w) + c_h$, form an affine action of G precisely when c satisfies the 1-cocycle identity

$$c_{gh} = c_g + L_g(c_h) \quad (g, h \in G).$$

Indeed, this is the condition $\alpha_{gh} = \alpha_g \circ \alpha_h$. Changing the origin from 0 to $b \in W$ replaces c_h by $c_h + L_h(b) - b$. The map $h \mapsto L_h(b) - b$ is a W-valued 1-coboundary. Consequently the affine action has a fixed point exactly when c is a coboundary, or equivalently, when $[c] = 0$ in $H^1(G, W)$.

This is also a special case of the module-extension construction. Given such a 1-cocycle c, let $M_c = W \oplus \mathbb{F}_p$ as a vector space, with G-action $h \cdot (w, t) = (L_h(w) + tc_h, t)$. The inclusion $w \mapsto (w, 0)$ and the map $\varepsilon : M_c \to \mathbb{F}_p$, $(w, t) \mapsto t$, give an extension $0 \to W \to M_c \to \mathbb{F}_p \to 0$, where G acts trivially on $\mathbb{F}_p$. The inverse image of 1 is the invariant set $\varepsilon^{-1}(1) = \{(w, 1) : w \in W\}$. Using $(0, 1)$ as its origin identifies this set with W and its action with α. An equivariant section of ε chooses a fixed point in this set. Hence this extension splits exactly when $[c] = 0$ [Ben91, Proposition 3.7.2].

Example 1 (A nonsplit extension and an affine line)**.** Let $G = \langle g \rangle$ be cyclic of order p, let $U = \mathbb{F}_p e_1 \oplus \mathbb{F}_p e_2$, and set $V = \mathbb{F}_p e_1$. Define $ge_1 = e_1$ and $ge_2 = e_2 + e_1$. Both V and $W = U/V$ are trivial modules. The section $s(e_2 + V) = e_2$ has $\gamma_g^s(e_2 + V) = e_1$. No representative $e_2 + te_1$ of $e_2 + V$ is fixed by g, so the extension does not split. The invariant coset $e_2 + V = \{e_2 + te_1 : t \in \mathbb{F}_p\}$ is an affine line modelled on V. In its coordinate t, the action of g is $t \mapsto t + 1$. Thus a linear action on U becomes an affine action on an invariant coset, and the lack of an invariant complement appears as the lack of a fixed point on that coset.

Quotients and changes of presentation. Return to the common setting $V \leq U$ and $W = U/V$. Fix a 1-cocycle $c : G \to W$, the associated affine action $\alpha_h(w) = L_h(w) + c_h$, and points $x, y \in W$. We use this notation for the rest of this and the next subsection. The following facts allow us to describe submodules and quotient modules in convenient coordinates. The first follows from the isomorphism theorems for modules [AB95, Section 12, pp. 110–111]; it is also a special case of the snake lemma [Wei94, Lemma 1.3.2].

Fact 3.3 (The kernel of an equivariant map)**.** *Let U' be an $\mathbb{F}_pG$-module, and let $\phi : U \to U'$ be equivariant. Put $U_\phi = \ker \phi$ and $V_\phi = V \cap \ker \phi$. There is a short exact sequence of $\mathbb{F}_pG$-modules*

$$0 \longrightarrow U_\phi/V_\phi \xrightarrow{\iota} U/V \xrightarrow{\bar{\phi}} \phi(U)/\phi(V) \longrightarrow 0,$$

where $\iota(u + V_\phi) = u + V$ for $u \in U_\phi$ and $\bar{\phi}(u + V) = \phi(u) + \phi(V)$ for $u \in U$. The maps and the inverse of ι on $\ker \bar{\phi}$ are computable by linear algebra.

Equivariant bijections preserve transporter sets and stabilizers, with the witnessing group elements unchanged [AB95, Section 3, Lemma 6]. Together with the isomorphism theorem for modules [AB95, Section 12, p. 110], this gives the next fact.

Fact 3.4 (Pulling back a quotient presentation)**.** *Let E be an $\mathbb{F}_pG$-module, and let $\psi : E \twoheadrightarrow U$ be equivariant. Put $\widetilde{V} = \psi^{-1}(V)$ and $\widetilde{W} = E/\widetilde{V}$. The map $\theta : \widetilde{W} \to W$ defined by $\theta(e+\widetilde{V}) = \psi(e)+V$ is an equivariant linear isomorphism. Define $\widetilde{x} = \theta^{-1}(x)$, $\widetilde{y} = \theta^{-1}(y)$, and $\widetilde{\alpha}_h = \theta^{-1} \circ \alpha_h \circ \theta$. Then $\mathrm{Trans}_{G,\widetilde{\alpha}}(\widetilde{x}, \widetilde{y}) = \mathrm{Trans}_{G,\alpha}(x, y)$ and $\mathrm{Stab}_{G,\widetilde{\alpha}}(\widetilde{x}) = \mathrm{Stab}_{G,\alpha}(x)$. The quotient bases, θ, and θ^{-1} are computable by linear algebra.*

When a surjection is given onto a module containing U, apply the fact to its restriction to the inverse image of U.

Example 2 (A quotient of an invariant subspace)**.** Let Y be a G-module, let $F \leq Y$ be a submodule, and suppose $V \leq U \leq Y/F$. Write $\psi : Y \to Y/F$ for the quotient map. The submodules $E = \psi^{-1}(U)$ and $\widetilde{V} = \psi^{-1}(V)$ satisfy $E/\widetilde{V} \cong U/V$, by applying Fact 3.4 to $\psi|_E : E \to U$. Thus the quotient U/V can be represented using subspaces of Y. Tensoring ψ with identity maps on other factors gives further equivariant surjections to which the same fact applies.

3.8 Affine transport on module quotients

Continue with the setting of Subsection 3.7: $V \leq U$, $W = U/V$, and $\alpha_h(w) = L_h(w) + c_h$. We wish to compute the complete transporter from x to y under this action. Luks [Luk92, Section 6.2] and Miyazaki [Miy99, Chapter IV, Section 2, pp. 49–51] formulate this problem in terms of *crossed homomorphisms*, another name for 1-cocycles. After changing the source to 0, the problem is to compute $\{h \in G : c_h = y\}$. The *quotient–fiber reduction* below first solves the problem on a quotient module and then solves the remaining problem on a coset of the submodule. Two further lemmas describe the translation subgroup that arises when the linear action factors through a quotient of G.

Quotients and fibers. The *fiber* of a map of sets $\varpi : A \to B$ over $b \in B$ is $\varpi^{-1}(\{b\})$. For a subspace $T \leq W$, the fiber of $W \to W/T$ over $x + T$ is the coset $x + T \subseteq W$. The following is Luks's quotient–fiber reduction [Luk92, Section 6.2], written for left affine actions.

Lemma 3.5 (An affine quotient and its fiber)**.** *Let $T \leq W$ be invariant under $L(G)$. Suppose*

$$\mathrm{Trans}_G(x + T, y + T) = aG_{\mathrm{quot}}, \quad G_{\mathrm{quot}} = \mathrm{Stab}_G(x + T), \quad a \in G.$$

Put

$$y_T = \alpha_{a^{-1}}(y) - x \in T, \quad \beta_h(t) = L_h(t) + \alpha_h(x) - x \quad (h \in G_{\mathrm{quot}},\ t \in T). \tag{3.3}$$

Then $\beta : G_{\mathrm{quot}} \to \mathrm{AGL}(T)$ is an affine action, and

$$\mathrm{Trans}_G(x, y) = a\,\mathrm{Trans}_{G_{\mathrm{quot}},\beta}(0, y_T), \quad \mathrm{Stab}_G(x) = \mathrm{Stab}_{G_{\mathrm{quot}},\beta}(0).$$

The stabilizers include the kernel of the original action.

Translations from a linear kernel. For the next two lemmas, take $x = 0$, so that $\mathrm{Trans}_{G,\alpha}(0, y) = \{h \in G : c_h = y\}$. A general source x reduces to this case by replacing c_h with $L_h(x) + c_h - x$ and the target y with $y - x$.

Let J be a finite group and $\lambda : G \to J$ a surjective homomorphism. Set $K_{\mathrm{lin}} = \ker \lambda$, and assume $L_u = 1_W$ for every $u \in K_{\mathrm{lin}}$, where 1_W is the identity map on W. Define $\tau : K_{\mathrm{lin}} \to W^+$ by $\tau(u) = c_u$, and put $E = \tau(K_{\mathrm{lin}})$. The computational input consists of the actions and λ on generators, generators of K_{lin}, and lifts in G of generators of J. We use the group encodings and operations of Subsection 3.5, including its implementation for symbolic additive coordinates. The descent to W/E uses the cocycle argument in [Sha, Theorem 1.8.10].

The proof of the following lemma is given in Appendix A.1.

Lemma 3.6 (Removing translations of a linear kernel)**.** *The map τ is a homomorphism, and E is a $L(G)$-invariant $\mathbb{F}_p$-subspace of W. The affine action on W/E factors through J. Given the complete transporter from $0+E$ to $y+E$ in J, linear algebra and the group operations of Lemma 3.1 compute* $\mathrm{Trans}_{G,\alpha}(0, y)$ *with its full source stabilizer.*

Translations from a flag kernel. Suppose $G \leq \prod_{a=1}^{s} \mathrm{GL}(Y_a)$, where the Y_a are finite-dimensional $\mathbb{F}_p$-spaces with specified G-invariant flags. Put

$$0 = Y_{a,0} < Y_{a,1} < \cdots < Y_{a,\ell_a} = Y_a, \quad X_{a,i} = Y_{a,i}/Y_{a,i-1}, \quad d = \sum_{a=1}^{s} \dim_{\mathbb{F}_p} Y_a.$$

Write $h_{a,i}$ for the induced action of h on $X_{a,i}$, and define

$$\lambda_{\mathrm{gr}} : G \longrightarrow \prod_{a=1}^{s} \prod_{i=1}^{\ell_a} \mathrm{GL}(X_{a,i}), \quad \lambda_{\mathrm{gr}}(h) = (h_{a,i})_{a,i}, \quad R = \ker \lambda_{\mathrm{gr}}.$$

Take $\lambda = \lambda_{\mathrm{gr}}$ with codomain $J = \mathrm{im}\, \lambda_{\mathrm{gr}} \cong G/R$, so $K_{\mathrm{lin}} = R$, and assume $L_r = 1_W$ for $r \in R$. Kernel generators and lifts are computed by the group operations above. No relation between W and the spaces $Y_a, X_{a,i}$ is assumed. The action of G/R on $\mathrm{Hom}_{\mathbb{F}_p}(X_{a,j}, X_{a,i})$ is $f \mapsto h_{a,i} f h_{a,j}^{-1}$. The following uses the standard off-diagonal filtration of a flag kernel; see [ABS90, Glu91, GM19].

The proof of the following lemma is given in Appendix A.2.

Lemma 3.7 (Translations from a flag kernel)**.** *The space $E = \tau(R)$ is an $\mathbb{F}_p[G/R]$-module. Every composition factor of E occurs among those of*

$$\bigoplus_{a=1}^{s} \bigoplus_{1 \leq i < j \leq \ell_a} \mathrm{Hom}_{\mathbb{F}_p}(X_{a,j}, X_{a,i}).$$

A basis of E, word lifts in R of its basis vectors, and generators of $\ker \tau$ *can be computed in* $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$ *time, where* InSize *is the total explicit input length.*

4 Matrix tuple congruence over fields of characteristic 2

4.1 Problem and main results

For preliminaries, we mainly follow Sec 3.3.

Definition 10 (Matrix tuple equivalence, conjugacy, and congruence)**.** Let $\mathbf{A} = (A_1, \ldots, A_m), \mathbf{B} = (B_1, \ldots, B_m) \in \mathrm{M}(n, q)^m$ be two tuples of matrices.

1. $\mathbf{A}$ and $\mathbf{B}$ are equivalent if there exist $L, R \in \mathrm{GL}(n, q)$, such that for any $i \in [m]$, $A_i = LB_iR^{-1}$.
2. $\mathbf{A}$ and $\mathbf{B}$ are conjugate if there exists $L \in \mathrm{GL}(n, q)$, such that for any $i \in [m]$, $A_i = LB_iL^{-1}$.
3. $\mathbf{A}$ and $\mathbf{B}$ are congruent if there exists $L \in \mathrm{GL}(n, q)$, such that for any $i \in [m]$, $A_i = LB_iL^{\mathrm{t}}$.

Definition 11 (Quadratic form isomorphism problem)**.** Let $(f_1, \ldots, f_m)$ and $(g_1, \ldots, g_m)$ be two tuples of quadratic forms (homogeneous quadratic polynomials) in $\mathbb{F}_q[x_1, \ldots, x_n]$. We say that they are isometric if there exists $A = (a_{i,j}) \in \mathrm{GL}(n, q)$ such that for any $i \in [m]$, $f_i = g_i \circ A$, where $(g_i \circ A)(x_1, \ldots, x_n) = g_i(\sum_j a_{1,j}x_j, \ldots, \sum_j a_{n,j}x_n)$.

We will prove the following theorems in this section.

Theorem 4.1 (Matrix tuple congruence algorithm over $\mathbb{F}_{2^f}$)**.** *Let $\mathbf{A} = (A_1, \ldots, A_m), \mathbf{B} = (B_1, \ldots, B_m) \in \mathrm{M}(n, 2^f)^m$ be two tuples of matrices. There is a $2^{O(fn)} \cdot \mathrm{poly}(n, m)$-time algorithm that decides whether the two tuples are congruent.*

Corollary 4.2 (Quadratic-form tuple isometry algorithm over $\mathbb{F}_{2^f}$)**.** *Let $(f_1, \ldots, f_m), (g_1, \ldots, g_m) \in \mathbb{F}_{2^f}[x_1, \ldots, x_n]^m$ be two tuples of quadratic forms. There is a $2^{O(fn)} \cdot \mathrm{poly}(n, m)$-time algorithm that decides whether the two tuples are isometric.*

4.2 Reducing to problems on $*$-algebras

For the matrix tuple congruence algorithm, we follow the approach of Ivanyos–Qiao [IQ19, Sections 3.3–3.5]: first test equivalence of matrix tuples, and then solve the $*$-symmetric decomposition problem [IQ19, Problem 17]. We will give the modification needed to handle arbitrary matrix tuples in characteristic 2.

Let $q = 2^f$ and $\mathbf{A}, \mathbf{B} \in \mathrm{M}(n, q)^m$ be the input tuples. We form two matrix tuples:

$$\widehat{\mathbf{A}} = (A_1, \ldots, A_m, A_1^{\mathrm{t}}, \ldots, A_m^{\mathrm{t}}),$$
$$\widehat{\mathbf{B}} = (B_1, \ldots, B_m, B_1^{\mathrm{t}}, \ldots, B_m^{\mathrm{t}}).$$

Apply [IQ19, Proposition 14], which works over arbitrary fields, to find $L, R \in \mathrm{GL}(n, q)$ such that

$$LB_i = A_iR, \quad LB_i^{\mathrm{t}} = A_i^{\mathrm{t}}R \quad \text{for } i \in [m]. \tag{4.1}$$

If no such pair exists, we can return emptiness for the matrix tuple congruence algorithm. This is because any $G \in \mathrm{GL}(n, q)$ satisfying $A_i = GB_iG^{\mathrm{t}}$ for every $i \in [m]$ would give a solution $(L, R) = (G, G^{-\mathrm{t}})$.

Suppose that a solution $c = (L, R)$ has been found. Let $E := \mathrm{M}(n, q) \times \mathrm{M}(n, q)$, with component-wise multiplication and involution $(U, V)^* := (V^{\mathrm{t}}, U^{\mathrm{t}})$, and define

$$F := \{(U, V) \in E : UB_i = B_iV,\ UB_i^{\mathrm{t}} = B_i^{\mathrm{t}}V \text{ for every } i \in [m]\}. \tag{4.2}$$

The space F contains $1_F := (I_n, I_n)$ and is closed under component-wise multiplication, so it is a unital subalgebra of E. An $\mathbb{F}_q$-basis of F can be efficiently computed by Gaussian elimination.

Let $F^* := \{(V^{\mathrm{t}}, U^{\mathrm{t}}) : (U, V) \in F\}$. Transposing the equations defining F shows that $F^* = F$, so $(F, *)$ is a $*$-algebra. Left multiplication by $c^{-1} := (L^{-1}, R^{-1})$ identifies the invertible solutions of (4.1) with $F^\times := \{(U, V) \in F : U, V \in \mathrm{GL}(n, q)\}$. Thus the set of all invertible matrix pairs satisfying (4.1) is exactly $cF^\times$.

Since $c = (L, R)$, we have $h = c^*c = (R^{\mathrm{t}}L, L^{\mathrm{t}}R)$. The equations in (4.1) and their transposes give

$$(R^{\mathrm{t}}L)B_i = R^{\mathrm{t}}A_iR = B_i(L^{\mathrm{t}}R),$$
$$(R^{\mathrm{t}}L)B_i^{\mathrm{t}} = R^{\mathrm{t}}A_i^{\mathrm{t}}R = B_i^{\mathrm{t}}(L^{\mathrm{t}}R).$$

Thus $h \in F^\times$ and $h^* = h$. We use $\mathrm{U}(E) := \{(G, G^{-\mathrm{t}}) : G \in \mathrm{GL}(n, q)\}$ to denote the set of $*$-unitary elements of E. Thus the matrices G satisfying $A_i = GB_iG^{\mathrm{t}}$ for every $i \in [m]$ correspond to the $*$-unitary elements of $cF^\times$. For $a \in F^\times$,

$$ca \in \mathrm{U}(E) \quad \iff \quad a^*ha = 1_F \quad \iff \quad (a^{-1})^*a^{-1} = h.$$

We have therefore reduced the problem to finding $x \in F^\times$ satisfying $x^*x = h$, which is the $*$-symmetric decomposition problem as stated in [IQ19, Problem 17]. Indeed, suppose that $x_0 = (U_0, V_0) \in F^\times$ satisfies $x_0^*x_0 = h$. Then $cx_0^{-1} = (LU_0^{-1}, RV_0^{-1})$ satisfies (4.1) and belongs to $\mathrm{U}(E)$. It follows that $L_0 = LU_0^{-1}$ satisfies $A_i = L_0B_iL_0^{\mathrm{t}}$ for every $i \in [m]$.

Therefore, the set of all such congruence solutions for matrix tuples corresponds to the set of all solutions for $x^*x = h$, which is

$$\mathrm{U}(F)x_0 := \{ux_0 : u \in F^\times,\ u^*u = 1_F\}.$$

To compute this solution set, we first consider the induced equation in the semisimple quotient F/J, where $J = \operatorname{rad} F$. For the semisimple step, we follow the reductions of [IQ19, Section 3.5], which solves the odd characteristic case. The next subsection addresses how to find $\overline{x}_0 \in (F/J)^\times$ satisfying $\overline{x}_0^*\overline{x}_0 = h + J$, and how to compute generators of $\mathrm{U}(F/J)$ in characteristic 2.

4.3 Computing the semisimple quotient solutions

Lemma 4.3 (Computations in the semisimple quotient)**.** *Let $q = 2^f$. Recall that F is defined in* (4.2)*, $F^\times := \{(U, V) \in F : U, V \in \mathrm{GL}(n, q)\}$, and $J = \operatorname{rad} F$. The following tasks can be performed deterministically in time* poly(InSize)*, where* InSize *is the total bit length of the input, including an $\mathbb{F}_q$-basis of F and an element $h \in F^\times$ satisfying $h^* = h$.*

1. *Find $\overline{x}_0 \in (F/J)^\times$ satisfying*
$$\overline{x}_0^*\overline{x}_0 = h + J,$$
or determine that no such element exists.

2. *Compute generators of*
$$\mathrm{U}(F/J) := \{\overline{u} \in (F/J)^\times : \overline{u}^*\overline{u} = 1_F + J\}.$$

Here $*$ also denotes the induced involution on F/J, given by $((U, V) + J)^* := (V^{\mathrm{t}}, U^{\mathrm{t}}) + J$ for $(U, V) \in F$.

Proof. Put $\overline{F} := F/J$ and $\overline{h} := h + J$. Embed F block-diagonally in $\mathrm{M}(2n, q)$ and restrict scalars to $\mathbb{F}_2$. Lemma 3.2, applied with $p = 2$, computes J and explicit coordinates

$$\overline{F} \cong \prod_{j=1}^{t} \mathrm{Mat}(r_j, K_j)$$

in deterministic polynomial time, where the K_j are finite fields of characteristic 2. Since J is $*$-stable, we can compute the induced involution in these coordinates. It permutes the simple factors

in orbits of size one or two. Both tasks can therefore be performed separately on these orbits, following [IQ19, Proposition 24].

Suppose first that $*$ exchanges two simple factors S_1 and S_2, and let $\varphi : S_1 \to S_2$ be its restriction. This is an anti-isomorphism. If (h_1, h_2) is the component of $\overline{h}$, then $h_2 = \varphi(h_1)$, and $(h_1, 1)$ solves the required equation on this pair. Moreover,

$$\mathrm{U}(S_1 \times S_2) = \{(a, \varphi(a^{-1})) : a \in S_1^\times\}.$$

Thus generators of $S_1^\times$, computed by Lemma 3.2, give generators of the unitary group on this pair.

Now consider a fixed factor $S = \mathrm{Mat}(r, K)$, and write h_S for the component of $\overline{h}$. The restriction of $*$ to the center gives an automorphism σ of K of order at most two. The intertwining equations used in [IQ19, Proposition 21],

$$DY^* = Y^{\sigma\mathrm{t}}D,$$

for a matrix basis of S, give a nonzero $D \in \mathrm{Mat}(r, K)$ by linear algebra. The existence of such a solution follows because composing $*$ with σ-transpose gives an inner K-algebra automorphism of S. Every nonzero solution is invertible, and hence $Y^* = D^{-1}Y^{\sigma\mathrm{t}}D$. Here $Y^{\sigma\mathrm{t}}$ means transpose after applying σ entrywise. We may arrange that $D^{\sigma\mathrm{t}} = D$ as follows. Initially $D^{\sigma\mathrm{t}} = \epsilon D$ for a scalar ϵ with $\epsilon^\sigma\epsilon = 1$. If $\sigma = 1$, characteristic 2 gives $\epsilon = 1$. Otherwise choose $z \in K$ such that $\lambda := z + \epsilon z^\sigma \neq 0$, and replace D by λD. Testing an $\mathbb{F}_2$-basis of K finds such a z, since this linear map is nonzero, and $\lambda^\sigma\epsilon = \lambda$ gives the desired normalization.

The reduction of [IQ19, Proposition 25] now gives

$$x^*x = h_S \quad \Longleftrightarrow \quad x^{\sigma\mathrm{t}}Dx = Dh_S.$$

The matrices D and Dh_S are Gram matrices of nonsingular symmetric bilinear forms when $\sigma = 1$, and of nonsingular Hermitian forms otherwise. Their isometry can be tested, and an isometry constructed, by the Gram–Schmidt procedure recalled in [IQ19, Theorem 26], with the following characteristic-2 normalizations. Alternating forms reduce to the standard symplectic form. A nonalternating symmetric form reduces to I_r: split off vectors of nonzero self-pairing and normalize using square roots. If an alternating remainder occurs, combine a normalized vector e with a hyperbolic pair u, v; the vectors $e+u, e+v, e+u+v$ are orthonormal, so the reduction continues. For a Hermitian form, a nonisotropic vector can be found using a linear trace equation, and orthogonal elimination gives diagonal entries in the fixed field K^σ. An entry $a \neq 0$ is normalized using $\sqrt{a} \in K^\sigma$, since its norm is a. Square roots and trace equations in these finite fields are computable deterministically in polynomial time. In particular, when $\sigma = 1$, we reject exactly when one of D, Dh_S is alternating and the other is nonalternating. Otherwise these reductions give the required x.

Independently of the outcome of this test, we compute the full isometry group of D. For alternating and Hermitian forms, use the standard root-subgroup, torus, and Weyl-group generators in symplectic and unitary coordinates; see [Tay92, Chapters 8 and 10]. There are $O(r^2)$ root subgroups, whose additive parameters are generated using field bases. For a nonabelian unitary root subgroup, take lifts of an additive basis of its noncentral parameter, obtained by solving linear trace equations, together with a basis of its trace-zero central parameter. Multiplicative parameters are supplied by Lemma 3.2 applied to the relevant fields. These constructions give the symplectic or special unitary group; the exceptional case of dimension 3 over $\mathbb{F}_4$ can also be handled directly in constant size. After transferring the special unitary generators to orthonormal coordinates using the computed form isometries, adjoin

$$\mathrm{diag}(t/\sigma(t), 1, \ldots, 1)$$

for generators t of $K^\times$. Their determinants generate $\{a \in K^\times : a\sigma(a) = 1\}$, so together with the special unitary generators they give the full group. In dimension one these diagonal elements alone suffice. All these constructions have polynomial cost in the matrix dimension and the field bit length.

For completeness, the nonalternating symmetric bilinear case requires a different group description. Work in the normalized coordinates $b(v, w) = v^{\mathrm{t}}w$ and put $v_0 := (1, \ldots, 1)^{\mathrm{t}}$. Every isometry fixes v_0, since

$$b(v_0, v) = \sqrt{b(v, v)} \quad (v \in K^r).$$

If r is odd, then Kv_0 is nondegenerate and $v_0^\perp$ is nondegenerate alternating. The full group is therefore obtained by extending the generators of $\mathrm{Sp}(v_0^\perp)$ to fix v_0.

If r is even, take v_1 to be the first coordinate vector, so $b(v_0, v_1) = b(v_1, v_1) = 1$, and put $W := \langle v_0, v_1\rangle^\perp$. The restriction of b to W is nondegenerate alternating. Preservation of these pairings shows that every isometry has a unique expression

$$\begin{aligned} v_0 &\longmapsto v_0, \\ w &\longmapsto gw + b(gw, z)v_0 \quad \text{for } w \in W, \quad g \in \mathrm{Sp}(W),\ z \in W,\ a \in K. \\ v_1 &\longmapsto v_1 + z + av_0, \end{aligned}$$

Take the generators with g ranging over generators of $\mathrm{Sp}(W)$ and $z = a = 0$. Also take $g = 1$, first with z ranging over additive generators of W and $a = 0$, and then with $z = 0$ and a ranging over additive generators of K. The generators with $g = 1$ generate the entire kernel of projection onto $\mathrm{Sp}(W)$: its quotient by the subgroup $z = 0$ is $(W, +)$, while that subgroup is $(K, +)$. Together they therefore generate the full isometry group. For each fixed factor, conjugating back through the computed changes of basis gives the generators in its original coordinates.

Finally, combine the solutions from all factor orbits to obtain $\overline{x}_0$, unless a fixed factor has no solution. Embed each orbit's group generators with identity components on the other orbits to obtain generators of $\mathrm{U}(\overline{F})$. All coordinate conversions are supplied by Lemma 3.2. The number and sizes of the factors, the required linear algebra, and all generating lists are polynomial in InSize, proving both assertions. □

Lemma 4.3 gives either an empty quotient solution set or a description of all quotient solutions by one representative $\overline{x}_0$ and generators of $\mathrm{U}(F/J)$. An empty quotient solution set rules out a solution in F. Otherwise, since J is nilpotent, we lift the solutions successively through the quotients $F/J, F/J^2, \ldots, F/J^t = F$, until $J^t = 0$, so that the final output is either emptiness or a solution x_0 together with generators of $\mathrm{U}(F)$.

The computations in the semisimple quotient F/J provide the first stage, namely the solution set modulo J. Each subsequent stage uses the solution set modulo J^i to compute the solution set modulo J^{i+1}. In odd characteristic, [IQ19, Proposition 23] guarantees that every quotient solution can be lifted from J^i to J^{i+1}, using division by 2. In characteristic 2, a particular solution modulo J^i may fail to lift modulo J^{i+1}, but this does not rule out lifts of other solutions modulo J^i. We must therefore search for which solutions can be lifted. The next subsection addresses this lifting problem.

4.4 Finding all solutions that can be lifted

Recall that $J = \operatorname{rad} F$. Since J is nilpotent, we can solve $x^*x = h$ by successively lifting solutions from modulo J^i to modulo J^{i+1}. Fix $i \geq 1$. We first consider a fixed solution $x_i \in F^\times$ satisfying

$$x_i^* x_i \equiv h \pmod{J^i}.$$

Our immediate goal is to determine whether there exists $x_{i+1} \in F^\times$ such that

$$x_{i+1} \equiv x_i \pmod{J^i}, \quad x_{i+1}^* x_{i+1} \equiv h \pmod{J^{i+1}}. \tag{4.3}$$

Every element congruent to x_i modulo J^i has the form $x_{i+1} = (1_F + y)x_i$ with $y := (x_{i+1} - x_i)x_i^{-1} \in J^i$, and $1_F + y$ is invertible because J is nilpotent. Substituting this expression into the second equation and multiplying by $(x_i^{-1})^*$ on the left and x_i^{-1} on the right gives

$$(1_F + y)^*(1_F + y) \equiv (x_i^{-1})^* h x_i^{-1} \pmod{J^{i+1}}.$$

Since $y^* y \in J^{2i} \subseteq J^{i+1}$, this is equivalent to the following equation:

$$y + y^* \equiv (x_i^{-1})^* h x_i^{-1} - 1_F \pmod{J^{i+1}}. \tag{4.4}$$

For the chosen x_i, finding $y \in J^i$ satisfying (4.4) is equivalent to finding x_{i+1} satisfying both equations in (4.3). In odd characteristic, every solution x_i modulo J^i can be lifted in this way, which means that (4.4) has a solution $y \in J^i$ for every choice of x_i satisfying $x_i^* x_i \equiv h \pmod{J^i}$. In characteristic 2, the nonexistence of y for the chosen x_i does not rule out the existence of a solution modulo J^{i+1}.

To describe precisely when (4.4) is solvable, let $M_i := J^i/J^{i+1} = \{y + J^{i+1} : y \in J^i\}$. Since J and its powers are preserved by $*$, the involution on F induces an involution s_i on M_i, given by $s_i(y + J^{i+1}) = y^* + J^{i+1}$.

For the chosen candidate x_i, define its error in M_i by

$$\delta_i := \left((x_i^{-1})^* h x_i^{-1} - 1_F\right) + J^{i+1}.$$

It follows that $(x_i^{-1})^* h x_i^{-1} - 1_F \in J^i$ since $x_i^* x_i \equiv h \pmod{J^i}$, and it is fixed by $*$ since $h^* = h$. Hence $\delta_i \in M_i$ and $s_i(\delta_i) = \delta_i$.

Writing $\overline{y} = y + J^{i+1}$, (4.4) becomes

$$(I_{M_i} + s_i)\overline{y} = \delta_i, \tag{4.5}$$

where I_{M_i} denotes the identity map on M_i. Thus the errors that can be removed by a correction $y \in J^i$ are exactly the elements of $\operatorname{im}(I_{M_i} + s_i)$.

In characteristic 2, the identity $s_i^2 = I_{M_i}$ gives $(I_{M_i} + s_i)^2 = 0$, so $\operatorname{im}(I_{M_i} + s_i) \subseteq \ker(I_{M_i} + s_i)$. We therefore define the obstruction space

$$Q_i := \ker(I_{M_i} + s_i)/\operatorname{im}(I_{M_i} + s_i). \tag{4.6}$$

Since $s_i(\delta_i) = \delta_i$ and the characteristic is 2, we have $(I_{M_i} + s_i)(\delta_i) = 0$, so $\delta_i \in \ker(I_{M_i} + s_i)$. The obstruction class of x_i is defined as

$$\operatorname{obs}_i(x_i) := \delta_i + \operatorname{im}(I_{M_i} + s_i) \in Q_i.$$

We say the obstruction class $\operatorname{obs}_i(x_i)$ is zero if and only if δ_i lies in $\operatorname{im}(I_{M_i} + s_i)$, which is equivalent to the existence of a solution $\overline{y}$ to (4.5). Choosing any representative $y \in J^i$ of such a solution gives $x_{i+1} = (1_F + y)x_i$ satisfying both conditions in (4.3). Conversely, any x_{i+1} satisfying those conditions determines such a correction y and therefore makes the obstruction class zero. Therefore, we can conclude that $x_i \bmod J^i$ lifts to a solution modulo J^{i+1} if and only if $\operatorname{obs}_i(x_i) = 0$.

Let $H_i := \{g \in F^\times : g^* g \equiv 1_F \pmod{J^i}\}$. Since $x_i^* x_i \equiv h \pmod{J^i}$, an element $x \in F^\times$ satisfies $x^* x \equiv h \pmod{J^i}$ if and only if $x x_i^{-1} \in H_i$, i.e., $x \in H_i x_i := \{g x_i : g \in H_i\}$. It therefore suffices to search for which $g \in H_i$ make $g x_i$ lift to a solution modulo J^{i+1}, or equivalently satisfy $\operatorname{obs}_i(g x_i) = 0$.

We express this search as the following affine transporter problem for two points in a given vector space.

Definition 12 (Point affine transporter problem)**.** Let q be any prime power. Given a finite group H, a finite-dimensional $\mathbb{F}_q$-vector space Q, and an affine group action $\alpha : H \times Q \to Q$ such that, for each $g \in H$,

$$\alpha_g(u) = L_g(u) + c_g,$$

where L_g is an invertible linear map on Q and $c_g \in Q$. For $u, v \in Q$, the problem is to compute

$$\mathrm{Trans}_{H,\alpha}(u, v) := \{g \in H : \alpha_g(u) = v\}.$$

Lemma 4.4 (Reduction of lifting to point affine transporter)**.** *Let $q = 2^f$ and $i \geq 1$. Given $x_i \in F^\times$ satisfying $x_i^* x_i \equiv h \pmod{J^i}$ and generators of $H_i = \{g \in F^\times : g^* g \equiv 1_F \pmod{J^i}\}$, represented in $\mathrm{GL}(2n, q)$ by $(U, V) \mapsto \mathrm{diag}(U, V)$, one can construct in polynomial time an affine group action $\alpha : H_i \times Q_i \to Q_i$ such that*

$$\mathrm{obs}_i(gx_i) = \alpha_g\big(\mathrm{obs}_i(x_i)\big) \text{ for } g \in H_i.$$

Consequently,

$$\mathrm{Trans}_{H_i,\alpha}\big(\mathrm{obs}_i(x_i), 0\big)$$

consists exactly of the elements $g \in H_i$ for which $gx_i \bmod J^i$ lifts to a solution modulo J^{i+1}.

Proof. For $z \in J^i$ with $z + J^{i+1} \in \ker(I_{M_i} + s_i)$, write

$$[z] := (z + J^{i+1}) + \mathrm{im}(I_{M_i} + s_i) \in Q_i.$$

For $g \in H_i$, define

$$L_g([z]) := [(g^{-1})^* z g^{-1}], \quad c_g := [(g^{-1})^* g^{-1} - 1_F].$$

These expressions are well-defined. Indeed, the map $z + J^{i+1} \mapsto (g^{-1})^* z g^{-1} + J^{i+1}$ is an $\mathbb{F}_q$-linear automorphism of M_i commuting with s_i. It therefore preserves both $\ker(I_{M_i} + s_i)$ and $\mathrm{im}(I_{M_i} + s_i)$ and induces the invertible linear map L_g on Q_i. Moreover,

$$(g^{-1})^* g^{-1} - 1_F = (g^{-1})^*(1_F - g^* g) g^{-1} \in J^i,$$

and this element is fixed by $*$, so its class c_g belongs to Q_i.

The defining formulas give, for $g, k \in H_i$,

$$L_{gk} = L_g L_k, \quad c_{gk} = c_g + L_g(c_k), \quad L_{1_F} = I_{Q_i}, \quad c_{1_F} = 0.$$

Thus c is a 1-cocycle for the linear action L. By the affine-action construction in Section 3.7, the maps

$$\alpha_g(u) := L_g(u) + c_g \text{ for } g \in H_i,\ u \in Q_i$$

define an affine group action of H_i on Q_i.

To check its relation to lifting, expand the normalized error of gx_i:

$$\begin{aligned}\big((gx_i)^{-1}\big)^* h (gx_i)^{-1} - 1_F = (g^{-1})^*\big((x_i^{-1})^* h x_i^{-1} - 1_F\big) g^{-1} \\ + (g^{-1})^* g^{-1} - 1_F.\end{aligned}$$

Taking classes in Q_i gives $\mathrm{obs}_i(gx_i) = L_g(\mathrm{obs}_i(x_i)) + c_g = \alpha_g(\mathrm{obs}_i(x_i))$. The characterization of zero obstruction established after (4.6) now gives the claimed description of the transporter.

Finally, a basis of J is available from Lemma 3.2. Products of basis elements and Gaussian elimination compute its powers, M_i, the kernel and image of $I_{M_i} + s_i$, and quotient coordinates for Q_i. Evaluating the displayed formulas on a basis of Q_i computes L_g and c_g for each given generator g of H_i. Under the conventions of Section 3.5, these generator images specify the action. All these operations take deterministic polynomial time in the explicit input length. □

We present an algorithm solving Definition 12 for the case where the linear part of the action preserves a flag with small-dimensional quotients.

Proposition 4.5 (Solve point affine transporter along an invariant flag)**.** *Let q be any prime power and consider an instance of Definition 12 over $\mathbb{F}_q$. Assume that H is given in the encoding of Section 3.5 with matrix coordinates $M_h \in \mathrm{GL}(c, q)$ and $c \le d$, and that we are given a chain of subspaces*

$$0 = Q_0 < Q_1 < \cdots < Q_r = Q$$

such that

$$L_g(Q_j) = Q_j \text{ for every } g \in H, \quad \dim_{\mathbb{F}_q}(Q_j/Q_{j-1}) \le d \quad \text{for } j \in [r].$$

Then $\mathrm{Trans}_{H,\alpha}(u, v)$ can be computed deterministically in time $q^{O(d)}\mathrm{poly}(\mathrm{InSize})$, where InSize *is the total bit length of the input, including the generators of H, affine action α, subspace bases of Q, and points u, v.*

Proof. We apply Lemma 3.5 successively along the given flag, starting with its top quotient. If $Q = 0$, the transporter is H, so suppose $r \ge 1$ and put $T := Q_{r-1}$. Since T is preserved by every L_g, the affine action induces an action on Q/T. This quotient has at most q^d elements. The orbit and stabilizer algorithms of Section 3.5 compute

$$\mathrm{Trans}_{H,\alpha}(u + T, v + T)$$

as either empty or aK, where $a \in H$ and $K = \mathrm{Stab}_{H,\alpha}(u + T)$. Here α also denotes the induced quotient action. We evaluate the Schreier generators in the original group H, so K includes the entire kernel of its action on Q/T.

If the quotient transporter is empty, so is the original transporter. Otherwise, Lemma 3.5 gives

$$w := \alpha_{a^{-1}}(v) - u \in T,$$
$$\beta_g(t) := L_g(t) + \alpha_g(u) - u \text{ for } g \in K,\ t \in T,$$

and reduces the remaining problem to

$$\mathrm{Trans}_{H,\alpha}(u, v) = a\,\mathrm{Trans}_{K,\beta}(0, w).$$

The linear part of β_g is $L_g|_T$, so the remaining flag $0 = Q_0 < \cdots < Q_{r-1} = T$ is still invariant and has the same quotient-dimension bound. Apply the same procedure recursively to $(K, T, \beta, 0, w)$. If it returns a coset bS, return abS; if it returns emptiness, return emptiness. By Lemma 3.5, S is the full stabilizer of u in H. Thus the procedure returns the complete transporter in the original group encoding.

There is only one recursive call after each quotient comparison, hence at most $r \le \dim_{\mathbb{F}_q} Q$ comparisons. Each orbit has size at most q^d. Use the subgroup membership and succinct-generating-set routines of Section 3.5, including its treatment of additive kernel coordinates, to compress each returned subgroup to at most $\lfloor \log_2 |H| \rfloor$ generators. This number is polynomial in InSize under the original encoding conventions. Write $q = p^f$, where p is prime. Restriction of scalars represents the matrix coordinates of H in $\mathrm{GL}(cf, p)$, so these group operations take $p^{O(cf)}\mathrm{poly}(\mathrm{InSize}) = q^{O(c)}\mathrm{poly}(\mathrm{InSize})$ time, which is bounded by $q^{O(d)}\mathrm{poly}(\mathrm{InSize})$ since $c \le d$; permutation degrees and additive dimensions enter only the polynomial factor.

All recursive groups remain subgroups of the original H. Store their words as shared straight-line programs and evaluate their affine maps in the current coordinates, as in Section 3.5. Quotient and fiber coordinates are computed by linear algebra, and the affine data have polynomial size in InSize. Consequently each comparison has the same bound in the original parameters, and summing over the flag gives $q^{O(d)}\mathrm{poly}(\mathrm{InSize})$ time. □

To apply Proposition 4.5 to the affine action constructed in Lemma 4.4, we need an invariant flag in Q_i with bounded quotient dimensions. The following lemma provides such a flag.

Lemma 4.6 (Structure of the obstruction space)**.** *Recall that the obstruction space Q_i is defined in* (4.6) *for the $*$-algebra $F \le \mathrm{M}(n,q) \times \mathrm{M}(n,q)$. Let $q = 2^f$ and $d = 2n$, with d measured over $\mathbb{F}_q$. Denote the affine action $\alpha : H_i \times Q_i \to Q_i$ constructed in Lemma 4.4 by*

$$\alpha_g(u) = L_g(u) + c_g \text{ for } g \in H_i,\ u \in Q_i.$$

For every $i \ge 1$, Q_i admits a decomposition

$$Q_i = \bigoplus_{j=1}^{r_i} W_{i,j}$$

into nonzero $\mathbb{F}_q$-subspaces satisfying

$$L_g(W_{i,j}) = W_{i,j} \text{ for every } g \in H_i, \quad \dim_{\mathbb{F}_q} W_{i,j} \le d \text{ for } j \in [r_i].$$

The subspaces $W_{i,j}$ and their bases can be computed in polynomial time, with $r_i = 0$ when $Q_i = 0$.

In particular, setting

$$Q_{i,0} := 0, \quad Q_{i,j} := \bigoplus_{k=1}^{j} W_{i,k} \quad \text{for } j \in [r_i]$$

gives an explicit flag

$$0 = Q_{i,0} < Q_{i,1} < \cdots < Q_{i,r_i} = Q_i$$

preserved by every L_g, with

$$\dim_{\mathbb{F}_q}(Q_{i,j}/Q_{i,j-1}) \le d = 2n.$$

Proof. Put $\overline{F} := F/J$. Since J annihilates $M_i = J^i/J^{i+1}$ on both sides, M_i is an $\overline{F}$-bimodule. For $g \in H_i$, write $\overline{g} := g + J \in \mathrm{U}(\overline{F})$. The formula for L_g in Lemma 4.4, together with $(g^{-1})^* \equiv g \pmod J$, shows that L_g is induced by conjugation $z \mapsto \overline{g} z \overline{g}^{-1}$ on M_i. It therefore suffices to construct the required decomposition invariant under $\mathrm{U}(\overline{F})$.

Use Lemma 3.2 to compute matrix-factor coordinates

$$\overline{F} \cong \prod_{\nu=1}^{t} S_\nu, \quad S_\nu = \mathrm{Mat}(a_\nu, K_\nu),$$

and let e_ν denote the corresponding central idempotents. To incorporate the involution into these coordinates, regard M_i as a module for $\mathcal{R} := \overline{F} \otimes_{\mathbb{F}_q} \overline{F}$ by

$$(a \otimes b)z := azb^* \text{ for } a, b \in \overline{F},\ z \in M_i.$$

The defining compatibility of s_i with the bimodule actions gives

$$s_i((a \otimes b)z) = (b \otimes a)s_i(z).$$

Moreover, for $u \in \mathrm{U}(\overline{F})$, the element $u \otimes u$ acts as conjugation by u. Thus every $\mathcal{R}$-submodule is invariant under this group action.

First decompose M_i into the subspaces $(e_\nu \otimes e_\lambda)M_i$. The involution s_i exchanges the subspaces indexed by (ν, λ) and (λ, ν). Whenever it exchanges two distinct summands C, C', every fixed vector

in $C \oplus C'$ is $z + s_i(z)$ for some $z \in C$, so this pair contributes zero to Q_i. It remains to consider the summands with $\nu = \lambda$.

Fix such a summand and write $S_\nu = \mathrm{Mat}(a, K)$ and $e = [K : \mathbb{F}_q]$. Its two central K-actions decompose it further using the explicit isomorphism

$$K \otimes_{\mathbb{F}_q} K \longrightarrow \prod_{k=0}^{e-1} K, \quad \lambda \otimes \mu \longmapsto (\lambda \mu^{q^k})_{k=0}^{e-1}.$$

Interchanging the tensor factors sends the component indexed by k to that indexed by $-k$ modulo e. Components exchanged in this way again contribute zero. A fixed component other than $k = 0$ occurs only when e is even and $k = e/2$. On that component, s_i is semilinear for the nontrivial field involution $\theta \mapsto \theta^{q^{e/2}}$. Choose $\theta \in K$ with $\theta + \theta^{q^{e/2}} = 1$ by solving a linear trace equation. Every fixed vector z then satisfies $(I_{M_i} + s_i)(\theta z) = z$, so this component also contributes zero.

Let C be the remaining component, indexed by $k = 0$. Its acting algebra is $\mathrm{Mat}(a, K) \otimes_K \mathrm{Mat}(a, K)$. Let E_{rs} be the standard matrix units of $\mathrm{Mat}(a, K)$ and put $D := (E_{11} \otimes E_{11})C$. The idempotent $E_{11} \otimes E_{11}$ is fixed by the tensor interchange, so D is s_i-stable and $\eta := s_i|_D$ is a K-linear involution. The matrix units give an isomorphism

$$\mathrm{Mat}(a, K) \otimes_K D \longrightarrow C, \quad E_{rs} \otimes z \longmapsto (E_{r1} \otimes E_{s1})z.$$

In these coordinates, $s_i(X \otimes z) = X^{\mathrm{t}} \otimes \eta(z)$. If g is the S_ν-component of $u \in \mathrm{U}(\overline{F})$, its action in these tensor coordinates is $X \otimes z \mapsto (gXg^{\mathrm{t}}) \otimes z$.

Since $(\eta - I_D)^2 = 0$, the Jordan decomposition of η consists of one-dimensional fixed blocks and nontrivial two-dimensional blocks. Each two-dimensional block admits a basis pair exchanged by η. For a two-dimensional block, s_i sends (X, Y) to $(Y^{\mathrm{t}}, X^{\mathrm{t}})$. Every fixed pair (X, X^{t}) is $(I_{M_i} + s_i)(X, 0)$, so this block contributes zero. A one-dimensional block contributes

$$\{X \in \mathrm{Mat}(a, K) : X^{\mathrm{t}} = X\} / \{Y + Y^{\mathrm{t}} : Y \in \mathrm{Mat}(a, K)\}.$$

The diagonal map identifies this quotient with K^a. It respects the group action through the identity

$$\mathrm{diag}(gXg^{\mathrm{t}}) = g^{(2)} \,\mathrm{diag}(X) \quad (X^{\mathrm{t}} = X),$$

where $g^{(2)}$ is obtained by squaring each entry of g. Thus each surviving block is an invariant $\mathbb{F}_q$-subspace of dimension $a[K : \mathbb{F}_q]$. Taking the obstruction quotient commutes with the decompositions above into s_i-stable pieces. The resulting nonzero summands in Q_i are the required $W_{i,j}$.

For the dimension bound, Lemma 3.2 supplies a Wedderburn–Malcev complement to J in F. Through the block-diagonal embedding $F \leq \mathrm{M}(2n, q)$, this complement gives a faithful $2n$-dimensional $\mathbb{F}_q$-representation of $\overline{F}$. Every factor $\mathrm{Mat}(a, K)$ must therefore occur on this module, and its simple module has dimension $a[K : \mathbb{F}_q]$. Consequently $\dim_{\mathbb{F}_q} W_{i,j} = a[K : \mathbb{F}_q] \leq 2n$.

All constructions are deterministic polynomial time. The matrix-factor coordinates are supplied by Lemma 3.2 in characteristic 2. The central projections, the inverse of the displayed field-tensor isomorphism, matrix-unit coordinates, trace equations, Jordan bases, and quotient maps are computed by linear algebra and finite-field arithmetic. Passing the surviving blocks to Q_i gives their $\mathbb{F}_q$-bases. If there are no surviving blocks, then $Q_i = 0$ and $r_i = 0$. Taking partial sums gives the claimed invariant flag. □

4.5 Solving matrix tuple congruence over characteristic 2

Proof of Theorem 4.1. Let $q = 2^f$. Applying the reduction in Section 4.2, we either determine that the matrix tuples are not congruent, or construct $F \leq \mathrm{M}(n,q) \times \mathrm{M}(n,q)$ as in (4.2) and $h \in F^\times$ such that matrix tuple congruence is equivalent to the existence of $x \in F^\times$ satisfying $x^*x = h$. Compute $J = \operatorname{rad} F$ using Lemma 3.2 and its powers by linear algebra. Let t be the least positive integer with $J^t = 0$.

At each stage i, we maintain a solution $x_i \in F^\times$ satisfying $x_i^* x_i \equiv h \pmod{J^i}$ and generators of $H_i := \{g \in F^\times : g^* g \equiv 1_F \pmod{J^i}\}$, so that all solutions modulo J^i are represented by $H_i x_i$.

First of all, applying Lemma 4.3, if the quotient equation has no solution, we return that the tuples are not congruent; otherwise, let $\overline{x}_0 \in (F/J)^\times$ be the returned solution and choose any $x_1 \in F$ such that $x_1 + J = \overline{x}_0$. Since J is nilpotent, this representative is invertible. Then $x_1^* x_1 \equiv h \pmod{J}$. Reduction modulo J gives $H_1/(1_F + J) \cong \mathrm{U}(F/J)$. Therefore, generators of H_1 are efficiently obtained by lifting the generators of $\mathrm{U}(F/J)$ and adjoining generators of $1_F + J$. For $1 \leq k < t$, generators of $1_F + J^k$ can be computed in polynomial time using

$$(1_F + J^a)/(1_F + J^{a+1}) \cong (J^a/J^{a+1}, +) \qquad (k \leq a < t).$$

For each layer, lift additive generators of J^a/J^{a+1} to elements $z \in J^a$ and include $1_F + z$. The union of these lists generates $1_F + J^k$; for $k = t$ the group is trivial. Additive generators of an $\mathbb{F}_q$-space are obtained by multiplying its basis vectors by additive generators of $\mathbb{F}_q$.

For $1 \leq i < t$, apply Lemmas 4.4 and 4.6, together with Proposition 4.5 for $d = 2n$, to compute

$$\mathrm{Trans}_{H_i,\alpha}(\mathrm{obs}_i(x_i), 0).$$

If this transporter is empty, no solution exists modulo J^{i+1}, so we reject. Otherwise, choose a returned transporter witness $u \in H_i$. By Lemma 4.4, equation (4.4) with x_i replaced by ux_i is solvable. Solve it for $y \in J^i$ by Gaussian elimination and set

$$x_{i+1} := (1_F + y)ux_i.$$

Then $x_{i+1}^* x_{i+1} \equiv h \pmod{J^{i+1}}$.

To compute generators of H_{i+1}, apply the same two lemmas and proposition with $x_i = 1_F$ and $h = 1_F$. This computes the stabilizer of $0 \in Q_i$ in H_i. By Lemma 4.4, an element $u \in H_i$ belongs to this stabilizer exactly when $(1_F + y)u \in H_{i+1}$ for some $y \in J^i$. Hence the stabilizer is $(1_F + J^i)H_{i+1}$. Let $u_1, \ldots, u_r$ be its returned generators. For each $j \in [r]$, solve (4.4) with $h = 1_F$ and $x_i = u_j$ for $y_j \in J^i$, and set $v_j := (1_F + y_j)u_j \in H_{i+1}$. We claim that

$$H_{i+1} = \langle v_1, \ldots, v_r,\ (1_F + J^i) \cap H_{i+1} \rangle.$$

Indeed, for any $g \in H_{i+1}$, write $g = u_{j_1}^{\varepsilon_1} \cdots u_{j_k}^{\varepsilon_k}$ where $\varepsilon_a \in \{1, -1\}$ for $a \in [k]$. Define $v := v_{j_1}^{\varepsilon_1} \cdots v_{j_k}^{\varepsilon_k}$. Since $v_j \equiv u_j \pmod{J^i}$ for every j, we have $v \equiv g \pmod{J^i}$, which implies $gv^{-1} \in 1_F + J^i$. Both g and v belong to H_{i+1}, so $gv^{-1} \in (1_F + J^i) \cap H_{i+1}$. The identity $g = (gv^{-1})v$ proves the claim. Thus, having obtained $v_1, \ldots, v_r$, it remains to compute generators of $(1_F + J^i) \cap H_{i+1}$.

Recall that $M_i = J^i/J^{i+1}$ and $s_i(y + J^{i+1}) = y^* + J^{i+1}$. For $y \in J^i$, one can verify that

$$1_F + y \in H_{i+1} \iff y + y^* \in J^{i+1} \iff (I_{M_i} + s_i)(y + J^{i+1}) = 0.$$

Compute additive generators $\overline{z}_1, \ldots, \overline{z}_s$ of $\ker(I_{M_i} + s_i)$ by linear algebra. For each $j \in [s]$, choose $z_j \in J^i$ satisfying

$$z_j + J^{i+1} = \overline{z}_j.$$

The above equivalence shows that $1_F + z_j \in (1_F + J^i) \cap H_{i+1}$. We claim these elements, together with generators of $1_F + J^{i+1}$, generate the required intersection:

$$(1_F + J^i) \cap H_{i+1} = \langle 1_F + z_1, \ldots, 1_F + z_s,\ 1_F + J^{i+1} \rangle.$$

Since $\overline{z}_1, \ldots, \overline{z}_s$ generate $\ker(I_{M_i} + s_i)$ additively and $J^{2i} \subseteq J^{i+1}$, every $1_F + y \in (1_F + J^i) \cap H_{i+1}$ is congruent modulo J^{i+1} to some $w \in \langle 1_F + z_1, \ldots, 1_F + z_s \rangle$. Hence $(1_F + y)w^{-1} \in 1_F + J^{i+1}$, which proves the claimed generation of $(1_F + J^i) \cap H_{i+1}$.

If all stages succeed, we reach $J^t = 0$ and obtain $x_t^* x_t = h$, proving that the tuples are congruent. At this stage $H_t = \mathrm{U}(F)$, so its generators are also available. When $Q_i \neq 0$, Proposition 4.5 compresses each returned stabilizer to $O(n^2 f)$ generators, since $H_i \leq \mathrm{GL}(2n, q)$; when $Q_i = 0$, it retains the current generating list. Each lifting stage adds only $O(n^2 f)$ radical-kernel generators, so all lists remain polynomially bounded over the $O(n^2)$ stages. There are at most $\dim_{\mathbb{F}_q} F \leq 2n^2$ lifting stages. Each lifting stage takes $q^{O(n)}\mathrm{poly}(n, m, f)$ time. Thus the total running time is $q^{O(n)}\mathrm{poly}(n, m, f) = 2^{O(fn)}\mathrm{poly}(n, m)$. □

The represented-algebra outputs. The construction above also computes unitary generators and solves norm equations for a unital represented $\mathbb{F}_p$-algebra $\mathcal{A} \leq \mathrm{End}_{\mathbb{F}_p}(V)$ with an explicitly given $\mathbb{F}_p$-linear involution $*$. Put $m = \dim_{\mathbb{F}_p} V$. The map

$$a \longmapsto \left(a, (a^*)^{\mathrm{t}}\right)$$

identifies $\mathcal{A}$ with a $*$-stable subalgebra of $\mathrm{M}(m, p) \times \mathrm{M}(m, p)$ under the exchange-transpose involution, with a faithful representation of dimension $2m$. In characteristic 2, the proofs of Lemmas 4.3, 4.4, and 4.6 use only the represented algebra, its involution, its radical filtration, and this faithful dimension bound. They therefore apply to this image of $\mathcal{A}$. The lifting construction just proved computes generators of $\mathrm{U}(\mathcal{A})$ and, for any given $h \in \mathcal{A}^\times$ with $h^* = h$, either reports that $x^* x = h$ has no solution or returns the complete solution coset $\mathrm{U}(\mathcal{A})x_0$. Generators of $\mathrm{U}(\mathcal{A})$ can always be computed by running the construction with $h = 1$, independently of the other norm equation.

In odd characteristic, the semisimple reduction and norm solver of [IQ19, Section 3.5] apply; use the deterministic finite-field versions of its algebra and form computations, whose polynomial cost may depend on p. We also compute the full unitary group of the semisimple quotient. On an exchanged pair of simple factors, use unit-group generators from Lemma 3.2. On a fixed factor $\mathrm{Mat}(a, K)$, the involution is the adjoint of a nonsingular symmetric, alternating, or Hermitian form. Generate its full isometry group by enumerating the isometric images of a fixed nondegenerate one- or two-dimensional frame. Witt extension and reduction to standard form produce an isometry for each such image. The pointwise stabilizer is the isometry group on the orthogonal complement; compute its generators recursively once and reuse them for all frame images [Tay92, Chapters 7–11]. These transporters and stabilizer generators generate the full group. The cost satisfies $T(a) \leq |K|^{O(a)} + T(a-1)$, with a two-dimensional decrease in the alternating case, and is therefore $|K|^{O(a)}$. Here $a[K : \mathbb{F}_p] \leq 2m$ by the faithful representation bound above. Put $J = \operatorname{rad} \mathcal{A}$. The radical correction of [IQ19, Proposition 23] lifts each quotient-unitary generator to a unitary element with the same residue modulo J. For the kernel, put $K_i = \mathrm{U}(\mathcal{A}) \cap (1 + J^i)$. The map $1 + z \mapsto z + J^{i+1}$ identifies K_i/K_{i+1} with the additive space of classes satisfying $z + z^* = 0$ in J^i/J^{i+1}. Indeed, lift each additive basis class to $y \in J^i$, replace it by $z = (y - y^*)/2$, and use the unitary lift $(1 - z/2)^{-1}(1 + z/2)$, which is congruent to $1 + z$ modulo J^{i+1}. These lifts over all radical layers generate K_1. Together with the lifted quotient generators they generate $\mathrm{U}(\mathcal{A})$. All lists can be compressed in the faithful $2m$-dimensional representation using Section 3.5. The running time is $p^{O(m)}\mathrm{poly}(\mathrm{InSize}, \log p)$.

We also prove the following corollary, which clearly covers Corollary 4.2.

Corollary 4.7 (Simultaneous isometry of bilinear and quadratic forms in characteristic 2)**.** *Let $q = 2^f$ and $\mathbf{A}, \mathbf{B} \in \mathrm{M}(n,q)^m$. Let $(f_1, \dots, f_r), (g_1, \dots, g_r) \in \mathbb{F}_q[x_1, \dots, x_n]^r$ be two tuples of homogeneous quadratic forms. There is a deterministic $2^{O(fn)}\mathrm{poly}(n,m,r)$-time algorithm that computes the complete set*

$$\{L \in \mathrm{GL}(n,q) : A_i = LB_iL^{\mathrm{t}} \ \text{ for } i \in [m], \quad f_j = g_j \circ L^{\mathrm{t}} \ \text{ for } j \in [r]\}.$$

The output is either emptiness or a coset L_0K, specified by one solution $L_0 \in \mathrm{GL}(n,q)$ and generators of the automorphism group $K := \{U \in \mathrm{GL}(n,q) : UB_iU^{\mathrm{t}} = B_i \ \text{ for } i \in [m], \quad g_j \circ U^{\mathrm{t}} = g_j \ \text{ for } j \in [r]\}$.

Proof. For a quadratic form φ, let P_φ be its polar matrix, defined by

$$\varphi(v+w) - \varphi(v) - \varphi(w) = v^{\mathrm{t}} P_\varphi w \text{ for } v, w \in \mathbb{F}_q^n.$$

Use the left action $U \cdot \varphi := \varphi \circ U^{\mathrm{t}}$. Its polar matrix is $UP_\varphi U^{\mathrm{t}}$. Apply the algorithm in the proof of Theorem 4.1 to the augmented tuples

$$(A_1, \dots, A_m, P_{f_1}, \dots, P_{f_r}), \quad (B_1, \dots, B_m, P_{g_1}, \dots, P_{g_r}).$$

For the algebra F associated with these augmented tuples, that algorithm computes a norm-equation solution and generators of $\mathrm{U}(F)$. Projection onto the first matrix component identifies $\mathrm{U}(F)$ with the full automorphism group of the source matrix tuple, and Section 4.2 recovers a congruence witness from the norm solution. Thus we obtain either emptiness or the complete matrix transporter aH, where $H \le \mathrm{GL}(n,q)$ preserves every B_i and P_{g_j}. In the empty case we return emptiness.

The kernel of polarization on homogeneous quadratic forms is

$$\mathcal{D} := \left\{ \sum_{k=1}^{n} b_k x_k^2 : b_k \in \mathbb{F}_q \right\}, \quad \dim_{\mathbb{F}_q} \mathcal{D} = n.$$

Apply the quotient–fiber reduction of Lemma 3.5 to the map that retains the bilinear coordinates and polarizes the quadratic coordinates. Its kernel is $Q := \mathcal{D}^{\oplus r}$. With the quotient transporter aH found above, the remaining target and affine action on Q are

$$t := \left(f_j \circ (a^{-1})^{\mathrm{t}} - g_j\right)_{j=1}^{r},$$
$$\beta_U(v) := \left((g_j + v_j) \circ U^{\mathrm{t}} - g_j\right)_{j=1}^{r},$$

where $U \in H$ and $v = (v_1, \dots, v_r) \in Q$. The matched polar matrices ensure that $t \in Q$ and that $\beta_U(v) \in Q$. By that lemma, the required set of simultaneous isometries is

$$a\,\mathrm{Trans}_{H,\beta}(0,t).$$

Identify each copy of $\mathcal{D}$ with its column vector of coefficients $(b_1, \dots, b_n)^{\mathrm{t}}$. The linear part of β_U acts on this vector by $U^{(2)}$, obtained by squaring every entry of U. Hence the partial sums of the r copies of $\mathcal{D}$ form an explicit flag in Q preserved by the linear parts of β, with every successive quotient of dimension n. Since H is encoded in $\mathrm{GL}(n,q)$, Proposition 4.5 applies with $d = n$. It returns emptiness or a coset bK, where

$$K = \mathrm{Stab}_{H,\beta}(0) = \{U \in H : g_j \circ U^{\mathrm{t}} = g_j \text{ for every } j \in [r]\}.$$

Preserving a quadratic form also preserves its polar matrix, so this is exactly the automorphism group K in the statement. In the nonempty case, return $L_0 := ab$ and the computed generators of K. If $r = 0$, the affine space is zero and the matrix transporter already gives the answer.

Both the augmented matrix computation and the flag computation take $q^{O(n)}\mathrm{poly}(n,m,r,f)$ time. All polar matrices, coefficient vectors, and affine generator images are computed in polynomial time. Thus the total cost is $2^{O(fn)}\mathrm{poly}(n,m,r)$, as claimed. □

4.6 A simple extension: solving degree-1 quotient tensor module transporter

We solve the degree-1 quotient tensor module transporter problem by constructing an invariant flag with small-dimensional quotients, similarly to how we solve the matrix tuple congruence problem.

We follow the group encoding and input/output conventions of Section 3.5. Here all original matrix representations and the matrix coordinates of H are over the base field $\mathbb{F}_q$, and d is measured over $\mathbb{F}_q$. The list $V_1, \dots, V_s$ below includes every original matrix representation used to encode H, even if it occurs in no tensor summand. Set $d := \sum_{i=1}^{s} \dim_{\mathbb{F}_q} V_i$, counting each original representation once; repeated tensor summands do not increase d.

Definition 13 (Degree-1 quotient tensor module transporter)**.** An instance over $\mathbb{F}_q$, where $q = p^f$ with p prime and $f \geq 1$, consists of the following data.

(i) Vector spaces $V_1, \dots, V_s$ over $\mathbb{F}_q$.

(ii) A finite group H with representations $\rho_i : H \to \mathrm{GL}(V_i)$ for $i \in [s]$.

(iii) A tensor module of degree at most 1,

$$Z = \bigoplus_{a \in I} \bigotimes_{j=1}^{k_a} W_{a,j}, \quad k_a \in \{0, 1\},$$

where I is a finite set of summand labels fixed by H, and each $W_{a,j}$ is a specified V_i, its dual, or a Frobenius twist of either. Repetitions are allowed, and an empty tensor product means the trivial module $\mathbb{F}_q$. All tensor products are over $\mathbb{F}_q$. A Frobenius twist is specified by $r \in \{0, \dots, f-1\}$ and applies $a \mapsto a^{p^r}$ entrywise to the $\mathbb{F}_q$-action matrices; for $f = 1$ this is the identity twist.

(iv) Two H-invariant subspaces $P \leq N \leq Z$, with quotient coordinates for $Q := N/P$. For each $h \in H$, let $L_h : Q \to Q$ be the linear map induced by the action of h on Z as in Definition 8.

(v) Translation vectors $c_h \in Q$ for $h \in H$, such that the maps

$$\alpha_h(u) := L_h(u) + c_h \text{ for } u \in Q.$$

define a group action $\alpha : H \times Q \to Q$.

(vi) Two points $x, y \in Q$.

The task is to compute the transporter

$$\mathrm{Trans}_{H,\alpha}(x, y) = \{h \in H : \alpha_h(x) = y\}.$$

The output is emptiness, or a witness g_0 and generators of $H_x = \{h \in H : \alpha_h(x) = x\}$, so that $\mathrm{Trans}_{H,\alpha}(x, y) = g_0 H_x$.

Note that the problem in Definition 13 is a special case of the point affine transporter problem in Definition 12, with additional structure on the space Q and its linear action L_h. The task is still to compute the transporter between the two given points $x, y \in Q$ under an affine group action.

This motivates us to construct an invariant flag that satisfies Proposition 4.5, so we can directly apply it to get the claimed algorithm solving this special point affine transporter problem. The following lemma constructs such a flag.

Lemma 4.8 (An invariant flag for degree-1 quotient tensor modules)**.** *Consider an instance of Definition 13 with original dimension $d \geq 1$ and input length* InSize*. Let t be the number of summands in the given direct-sum decomposition of Z. Number these summands from 1 to t in any fixed order, define*

$$Z_0 := 0, \quad Z_j := \bigoplus_{a=1}^{j} \bigotimes_{b=1}^{k_a} W_{a,b} \quad \text{for } j \in [t].$$

For $0 \leq j \leq t$, define

$$Q_j := ((N \cap Z_j) + P)/P.$$

Then

$$0 = Q_0 \leq Q_1 \leq \cdots \leq Q_t = Q,$$

and

$$L_h(Q_j) = Q_j \text{ for every } h \in H, \quad \dim_{\mathbb{F}_q}(Q_j/Q_{j-1}) \leq d \text{ for } j \in [t].$$

The subspaces Q_j and their bases can be computed deterministically in time $\mathrm{poly}(\mathrm{InSize}, \log q)$*. In particular, upon deleting repeated terms, this chain satisfies the invariant-flag conditions of Proposition 4.5.*

Proof. By Definition 8, the action of H on Z preserves each specified summand, since its label is fixed by H. Thus every Z_j and $N \cap Z_j$ is H-invariant. The image of $N \cap Z_j$ under the quotient map $N \to N/P$ is exactly Q_j, so $L_h(Q_j) = Q_j$ for every $h \in H$. The inclusions $Z_{j-1} \leq Z_j$ give $Q_{j-1} \leq Q_j$, while $Z_0 = 0$ and $Z_t = Z$ give $Q_0 = 0$ and $Q_t = Q$.

For each $j \in [t]$, inclusion into Z_j induces an injective linear map from $(N \cap Z_j)/(N \cap Z_{j-1})$ to Z_j/Z_{j-1}. The quotient map $N \to Q$ also induces a surjective linear map from $(N \cap Z_j)/(N \cap Z_{j-1})$ onto Q_j/Q_{j-1}. Consequently,

$$\dim_{\mathbb{F}_q}(Q_j/Q_{j-1}) \leq \dim_{\mathbb{F}_q}\big((N \cap Z_j)/(N \cap Z_{j-1})\big) \leq \dim_{\mathbb{F}_q}(Z_j/Z_{j-1}).$$

The last space is isomorphic to the j-th summand of Z. If $k_j = 0$, its dimension is $1 \leq d$. If $k_j = 1$, it is a given V_i, its dual, or a Frobenius twist of either, and hence has dimension $\dim_{\mathbb{F}_q} V_i \leq d$ by the original-dimension convention of Section 3.5. This proves the required bound.

Finally, the given direct-sum coordinates provide bases of the Z_j. Gaussian elimination computes $N \cap Z_j$; applying the supplied quotient-coordinate map $N \to Q$ and extracting a basis gives a basis of Q_j. There are t such computations, and all matrices have size polynomial in the explicit input length, so the total cost is $\mathrm{poly}(\mathrm{InSize}, \log q)$. Removing repeated terms preserves invariance and the bound on successive quotient dimensions, giving the claimed flag. When $Q = 0$, the resulting flag consists only of 0. □

Now we are ready to prove the following algorithm that solves Defnition 13.

Theorem 4.9 (Degree-1 quotient tensor module transporter)**.** *Given an instance of Definition 13 with original dimension d and input length* InSize*, its complete transporter can be computed deterministically in time* $q^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log q)$*.*

Proof. Suppose first that $d \geq 1$. Lemma 4.8 constructs, in time $\mathrm{poly}(\mathrm{InSize}, \log q)$, a flag in Q preserved by every L_h, with successive quotient dimensions at most d. Using Definition 8 and the supplied quotient coordinates, compute the matrices of L_h for the given generators of H in polynomial time. Together with the given translations, these matrices specify the affine action on Q. By the encoding convention of Section 3.5, the matrix coordinates of H lie in $\mathrm{GL}(c, q)$ with $c \leq d$. Thus all hypotheses of Proposition 4.5 hold over $\mathbb{F}_q$.

Apply that proposition to $x, y \in Q$. It returns either emptiness or a witness g_0 and generators of the full source stabilizer H_x, in the original group encoding, such that $\mathrm{Trans}_{H,\alpha}(x, y) = g_0 H_x$. This is precisely the required output, without any faithfulness assumption on the representations defining Z. The constructed affine data and flag bases have total bit length $\mathrm{poly}(\mathrm{InSize}, \log q)$, so the running time is $q^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log q)$.

If $d = 0$, every L_h is the identity on Q, so the group-action identity makes $h \mapsto c_h$ a homomorphism from H to $(Q, +)$. View Q as an $\mathbb{F}_p$-vector space of dimension $f \dim_{\mathbb{F}_q} Q$. In these coordinates, the additive-image and kernel computations in Section 3.5 compute the image as an $\mathbb{F}_p$-subspace, lifts of an $\mathbb{F}_p$-basis, and generators of the kernel in time $\mathrm{poly}(\mathrm{InSize}, \log q)$, since there are no matrix coordinates. Test whether $y - x$ lies in this image by Gaussian elimination over $\mathbb{F}_p$. If it does not, return emptiness; otherwise, the computed lifts give a witness g_0 with $c_{g_0} = y - x$, and the transporter is $g_0 \ker(h \mapsto c_h)$. This also satisfies the claimed bound. □

5 Computing matrix group structures

Let $H \leq \mathrm{GL}(d_q, q)$ be given by generators, where $q = p^f$ and p is prime. Our goal is to compute its structure in time $q^{O(d_q)} \cdot \mathrm{poly}(\mathrm{InSize})$. Restriction of scalars embeds $\mathrm{GL}(d_q, q)$ into $\mathrm{GL}(f d_q, p)$. Henceforth set $d := f d_q$ and work over $\mathbb{F}_p$, so d always denotes the prime-field dimension in this section and $p^{O(d)} = q^{O(d_q)}$. All group computations retain words in the original generators, action maps, and their full kernels. An action with a kernel is processed on its faithful image, and the resulting subgroups are pulled back to the original group.

5.1 Step 1: invariant subspaces and imprimitive systems

Let $H \leq \mathrm{GL}(d, p)$. We say that H is irreducible if there are no non-trivial invariant subspaces of H. An invariant composition flag of H is a sequence of invariant subspaces

$$0 = V_0 < V_1 < \cdots < V_\ell = \mathbb{F}_p^d$$

such that for any $i \in [\ell]$, V_i/V_{i-1} is irreducible under the induced action of H. The associated graded representation is $\oplus_{i \in [\ell]} V_i/V_{i-1}$, with the natural induced action.

A linear imprimitivity system of $H \leq \mathrm{GL}(d, p)$ is a direct sum decomposition $\mathbb{F}_p^d = U_1 \oplus \cdots \oplus U_k$, with $k \geq 2$ and $U_i \neq 0$ for all i, such that for any $h \in H$ and $i \in [k]$, $h(U_i) = U_j$ for some $j \in [k]$. An irreducible action is linearly primitive if no such decomposition exists. We use "primitive" in this linear sense for matrix actions; primitivity on a finite index set has its usual permutation-group meaning.

Let Alt_m be the alternating group on $[m]$ with the natural action on $\mathbb{F}_p^m$. Let $P = \{(x_1, \ldots, x_m) \in \mathbb{F}_p^m : \sum_i x_i = 0\}$ and $T = \mathbb{F}_p(1, \ldots, 1)$. The *fully deleted permutation module* of Alt_m over $\mathbb{F}_p$ is $U_m = P/(P \cap T)$ with the natural induced action (see e.g. [KL90b, pp. 185–187]). Its dimension is $m - 1$ if $p \nmid m$ and $m - 2$ otherwise.

We use *standard module* to mean a natural module over the defining matrix field K of a classical group in characteristic p, viewed over $\mathbb{F}_p$ by restriction of scalars, or the fully deleted permutation module for Alt_m over $\mathbb{F}_p$.

Fix sufficiently large absolute constants c_0 and C_*, and put

$$T_{\mathrm{cl}} = p^{C_* d}. \tag{5.1}$$

The initial dimension budget d and the constant C_* remain fixed under subgroup restriction and passage to invariant subquotients.

The proof of the following theorem is given in Appendix B.2.

Theorem 5.1 (Invariant composition flags and linear primitive systems)**.** *Let $H \le \mathrm{GL}(d,p)$ be given by generators. The following can be computed in time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$.*

1. *An invariant composition flag $0 = V_0 < V_1 < \cdots < V_\ell = \mathbb{F}_p^d$, such that the action of H on the associated graded space has kernel $\mathrm{O}_p(H)$.*
2. *For each V_i/V_{i-1}, a direct-sum decomposition into blocks permuted transitively by $G = H/\mathrm{O}_p(H)$, with each block stabiliser acting irreducibly and linearly primitively on its block.*

5.2 Step 2: Large local components and diagonal links

In Theorem 5.1, if the action on V_i/V_{i-1} is already linearly primitive, the decomposition consists of a single block. For each j, write $V_j/V_{j-1} = \bigoplus_{a=1}^{t_j} W_{j,a}$. Relabel all the spaces $W_{j,a}$ as $W_1, \dots, W_t$, where $t = \sum_j t_j$.

After Theorem 5.1, it remains to describe how the actions on different primitive blocks are linked. For each block W_i, let $P_i \le \mathrm{GL}(W_i)$ be the image of its stabiliser in $G = H/\mathrm{O}_p(H)$, and put $d_i = \dim_{\mathbb{F}_p} W_i$. Let $N_{\mathrm{blk}} \trianglelefteq G$ be the kernel of the permutation action on $\{W_1, \dots, W_t\}$. Restriction to the blocks gives homomorphisms $\rho_i : N_{\mathrm{blk}} \to P_i$, whose joint map embeds N_{blk} into $\prod_i P_i$.

We call P_i *bounded* if $|P_i| \le p^{c_0 d_i}$, and *unbounded* otherwise. For $A \le N_{\mathrm{blk}}$, its *block support* is $\mathrm{supp}(A) = \{i : \rho_i(A) \ne 1\}$. A quasisimple subgroup of N_{blk} is called *classical* if all its nontrivial block images are classical groups in characteristic p.

Whenever $C \le N_{\mathrm{blk}}$ is a central product of quasisimple subgroups with pairwise disjoint block supports, let $\mathcal{I}$ denote the family of these supports. We write $C = \prod_{I \in \mathcal{I}} Q_I$, where $\mathrm{supp}(Q_I) = I$. For such a product, we wish to single out those with large enough central quotients. Therefore, define

$$D(C) = \prod_{\substack{I \in \mathcal{I}:\ Q_I \text{ classical} \\ |Q_I/\mathrm{Z}(Q_I)| > T_{\mathrm{cl}}}} Q_I \le C.$$

The following theorem constructs such a product C and its subgroup $D = D(C)$.

The *stable range* is that of Section 6.1: natural dimension at least 64, and Witt index at least 32 for symplectic, unitary, and orthogonal modules. Orthogonal natural modules in characteristic 2 are even-dimensional. We choose C_* sufficiently large that every classical component selected in D lies in this range.

The proof of the following theorem relies on structural results on primitive linear groups due to Jaikin-Zapirain and Pyber [JZP11, Proposition 5.7], in the form given by Halasi et al. [HMPQ19, Theorems 3.1–3.2]. The full proof is given in Appendix B.3.

Theorem 5.2 (Large local components and diagonal links)**.** *Given generators for $H \le \mathrm{GL}(d,p)$ and the decompositions from Theorem 5.1, the following can be computed in time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$.*

1. *For every unbounded P_i, a normal quasisimple subgroup $S_i \trianglelefteq P_i$ satisfying $[P_i : S_i] \le p^{5d_i}$, and an explicit S_i-module isomorphism $W_i \cong X_i^{\oplus m_i}$, where X_i is a standard module for S_i.*
2. *A central product $C = \prod_{I \in \mathcal{I}} Q_I$ (Q_I quasisimple) of the form described above, satisfying*

$$C \le N_{\mathrm{blk}}, \qquad C \trianglelefteq G, \qquad [N_{\mathrm{blk}} : C] \le p^{(c_0+1)d}.$$

The output includes $\mathcal{I}$ and generators for each Q_I and its centre. Every $I \in \mathcal{I}$ contains only indices of unbounded blocks, and $\rho_i(Q_I) = S_i$ for every $i \in I$. Explicit isomorphisms identify the simple central quotients $S_i/\mathrm{Z}(S_i)$ diagonally within each I.

3. *Generators for the subgroup $D = D(C)$. This subgroup satisfies*

$$D \le C \le N_{\mathrm{blk}} \le G, \qquad D \trianglelefteq G.$$

Every local classical component S_i satisfying $|S_i/\mathrm{Z}(S_i)| > T_{\mathrm{cl}}$ is the image under ρ_i of a factor selected in D.

4. *For every block W_i on which D acts nontrivially, the defining field K_i and explicit natural K_i-coordinates for the standard module X_i from Item (1), in the prescribed stable range. The output includes the natural action matrices, coordinate identifications realising the diagonal links in Item (2), and projection maps from the corresponding full preimages in H.*

5.3 Step 3: Alternating-dominant actions

A *simple section* of a group is a simple quotient of one of its subgroups. An action is *alternating-dominant* if every nonalternating simple section of its image has order at most T_{cl}. We use the same term for a module with such an action. In particular, every action with image order at most T_{cl} is alternating-dominant. The definition requires neither an alternating section nor irreducibility; a decomposition into permuted direct summands is separate data.

After Theorem 5.2, the selected classical components are handled by the classical degree-2 solver. The remaining actions are alternating-dominant. We construct their representations and the additional permutation structure used by the recursion. Call an unbounded block W_i *alternating* if $X_i = U_{a_i}$ is the fully deleted permutation module for Alt_{a_i}. Choose c_0 sufficiently large that every such $a_i \ge 8$. For these blocks, the earlier S_i-module decomposition is extended to a presentation of the full block-stabiliser action on $\mathbb{F}_p^{a_i} \otimes \mathbb{F}_p^{m_i}$.

For the alternating blocks, we use the standard description of automorphisms of alternating groups [Wil09b, Theorem 2.3]. The bounds on nonalternating simple sections use the composition-factor bound of Guralnick, Maróti, and Pyber [GMP17, Proposition 6.1]. The proof of the following theorem is given in Appendix B.4.

Theorem 5.3 (Alternating-dominant actions)**.** *Given generators for $H \le \mathrm{GL}(d,p)$ and the data from Theorem 5.2, the following* alternating-dominant actions *can be computed in time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$.*

1. *The actions given by the bounded primitive images P_i; the classical primitive images P_i with $|S_i/\mathrm{Z}(S_i)| \le T_{\mathrm{cl}}$; the quotient N_{blk}/C; and the permutation actions of G on the blocks W_i and factors Q_I, each of degree $O(d)$.*

2. *For each alternating block W_i, an auxiliary representation of the block stabiliser on $A_i = \mathbb{F}_p^{a_i} \otimes_{\mathbb{F}_p} M_i$, where $M_i = \mathbb{F}_p^{m_i}$, given by*

$$h \longmapsto \mathrm{perm}(\pi_h) \otimes \mu_h, \qquad h \in \mathrm{Stab}_G(W_i), \quad \pi_h \in \mathrm{Sym}_{a_i}, \quad \mu_h \in \mathrm{GL}(M_i).$$

The action permutes the summands in $A_i = \bigoplus_{j=1}^{a_i}(\mathbb{F}_p e_j \otimes M_i)$. The multiplicity action $\mu : h \mapsto \mu_h$ factors through P_i/S_i, so $|\operatorname{im}\mu| \le [P_i : S_i] \le p^{5d_i}$. This representation induces the given block-stabiliser action on

$$W_i \cong \frac{P_{\mathrm{aug},i} \otimes M_i}{(P_{\mathrm{aug},i} \cap \mathbb{F}_p\mathbf{1}) \otimes M_i},$$

where $P_{\mathrm{aug},i}$ is the sum-zero subspace of $\mathbb{F}_p^{a_i}$. The equivariant inclusion and quotient maps are supplied, and these presentations assemble over each block orbit into a G-equivariant presentation.

Every nonalternating simple section L of an image in Items (1)–(2) satisfies $|L| \le T_{\mathrm{cl}}$. This bound also holds for all subgroup images and quotients of these images. The auxiliary spaces satisfy $\sum_{i \text{ alternating}} \dim_{\mathbb{F}_p} A_i \le 4d/3$. All actions retain the original generator matrices, maps from the relevant full preimages in H, projection kernels, and equivariant inclusion and quotient maps.

5.4 Using the selected classical components

Let $V_{\mathrm{gr}} = \bigoplus_{i=1}^{\ell} V_i/V_{i-1}$ and let $D \trianglelefteq G$ be the full selected classical product from Theorem 5.2. Put $V_{\mathrm{fix}} = V_{\mathrm{gr}}^{D}$ and $V_D = \sum_{g \in S_D} (g-1)V_{\mathrm{gr}}$, where S_D generates D. In time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$, one can compute explicit decompositions

$$V_{\mathrm{gr}} = V_{\mathrm{fix}} \oplus V_D, \qquad V_D \cong \bigoplus_{j \in J} X_j \otimes_{K_j} M_j,$$

where the X_j are pairwise non-isomorphic nontrivial simple D-modules, $K_j = \mathrm{End}_{\mathbb{F}_p[D]}(X_j)$, and $M_j = \mathrm{Hom}_{\mathbb{F}_p[D]}(X_j, V_D)$. Equivalent copies across different blocks are collected in the same M_j, with the original inclusion and quotient maps retained. The space V_{fix} retains its alternating-dominant presentations from Theorem 5.3. We identify D with its faithful image on V_D. The same notation applies to a fixed number of copies and duals, collecting equivalent constituents again and keeping the original d and T_{cl} fixed. The image of G on V_D is an example of the normaliser subgroup A below. A space fixed by one selected component but moved by another belongs to V_D; the fixed part is taken with respect to the full product D.

The proof of the following lemma is given in Appendix B.5.

Lemma 5.4 (Actions on tensor factors)**.** *Given generators for $A \le N_{\mathrm{GL}_{\mathbb{F}_p}(V_D)}(D)$ and the above decomposition, the following can be computed in time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$: a group $\widehat{A}$ and a surjection $\pi_{\mathrm{fac}} : \widehat{A} \twoheadrightarrow A$ recording the maps on the natural and multiplicity factors, field automorphisms, and permutations of the isotypic summands. Its kernel is*

$$\ker \pi_{\mathrm{fac}} = \prod_{j \in J} \{(cI_{X_j}, c^{-1} I_{M_j}) : c \in K_j^{\times}\}.$$

The group $\widehat{A}$ has a faithful $\mathbb{F}_p$-representation on $\bigoplus_j (X_j \oplus M_j)$ of dimension at most $2 \dim_{\mathbb{F}_p} V_D = O(d)$; all action maps and projection kernels are supplied. For a given surjection $\rho : H \twoheadrightarrow A$, the construction also retains the group of pairs $(h, \widehat{a})$ satisfying $\rho(h) = \pi_{\mathrm{fac}}(\widehat{a})$ and the full kernel of ρ.

The proof of the following lemma is given in Appendix B.6.

Lemma 5.5 (Multiplicity and normaliser bounds)**.** *Fix a constant $\kappa \ge 1$ and suppose $\dim_{\mathbb{F}_p} V_D \le \kappa d$. Put $f_j = [K_j : \mathbb{F}_p]$, $a_j = \dim_{K_j} X_j$, and $b_j = \dim_{K_j} M_j$. Then*

$$|\operatorname{Cent}_{\mathrm{GL}_{\mathbb{F}_p}(V_D)}(D)| \le p^{\sum_j f_j b_j^2} \le p^{\kappa^2 d / C_*}.$$

Given generators for a subgroup $D \le A_0 \le N_{\mathrm{GL}_{\mathbb{F}_p}(V_D)}(D)$, one has

$$[A_0 : D] \le p^{O(d)}, \qquad A_{\mathrm{in}} = D \operatorname{Cent}_{A_0}(D),$$

where A_{in} consists of the elements inducing inner automorphisms of D. Let $\pi_{\mathrm{fac}} : \widehat{A}_0 \twoheadrightarrow A_0$ be the cover from Lemma 5.4. Its joint image under the action on $\bigoplus_j M_j$ and the map $\widehat{a} \mapsto \pi_{\mathrm{fac}}(\widehat{a})D$ can be enumerated in time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$, retaining original lifts and projection kernels.

The proof of the following lemma is given in Appendix B.7.

Lemma 5.6 (Separation in a simultaneous image)**.** *Let Γ be a finite group given by generators, let $\alpha : \Gamma \twoheadrightarrow A$ and $\beta : \Gamma \twoheadrightarrow B$ be explicit homomorphisms, and let $E \leq A$ be a given central product of quasisimple groups. If no quotient of a subgroup of B is isomorphic to the simple central quotient of a factor of E, then*

$$E \times 1 \leq (\alpha, \beta)(\Gamma) \leq A \times B.$$

Generators of $E \times 1$ and its full preimage $\alpha^{-1}(E) \cap \ker \beta$ can be computed using polynomially many constructive kernel, membership and lifting operations. If these operations take time $p^{O(d)} \cdot$ poly(InSize)*, the construction has the same bound.*

For simultaneous classical and alternating-dominant actions of the same original group, take E to be the selected classical image and B the joint alternating-dominant image. The simple-section exclusion in Theorem 5.3 supplies the hypothesis of Lemma 5.6.

6 Classical group representations and cohomologies

6.1 Main results

Classical groups over finite fields. In this section we prove several results about classical groups in their natural representations. We follow the notation and form conventions of [Tay92].

Let $\mathbb{F}$ be a finite field. For the unitary groups, let $\mathbb{K}/\mathbb{F}$ be a quadratic extension, with nontrivial automorphism σ. We consider the following groups:

- The special linear group $\mathrm{SL}(n, \mathbb{F})$.
- The symplectic group $\mathrm{Sp}(2m, \mathbb{F})$, preserving a nondegenerate alternating form on $\mathbb{F}^{2m}$.
- The unitary group $\mathrm{U}(n, \mathbb{K})$, preserving a nondegenerate σ-Hermitian form on $\mathbb{K}^n$, and its determinant-one subgroup $\mathrm{SU}(n, \mathbb{K})$.
- The orthogonal group $\mathrm{O}(Q)$, preserving a specified nondegenerate quadratic form Q on $\mathbb{F}^n$, and the subgroup $\Omega(Q)$. In the stable range considered below, $\Omega(Q) = [\mathrm{O}(Q), \mathrm{O}(Q)]$.

Recall that a quadratic form satisfies $Q(av) = a^2 Q(v)$ and has a bilinear *polar form* $b_Q(v, w) = Q(v + w) - Q(v) - Q(w)$

Remark 1 (Conventions of classical groups)**.** Recall that the *Witt index* of a bilinear form is the largest dimension of a subspace on which the form vanishes identically, using totally singular subspaces for a quadratic form. Orthogonal groups in characteristic 2 have even natural dimension. We define the *stable range* of a classical group to mean natural dimensions at least 64, and Witt indices at least 32 for symplectic, unitary, and orthogonal factors.

For all running-time bounds in this section, InSize denotes the total explicit input length, and $d = \sum_i [\mathbb{K}_i : \mathbb{F}_p] n_i$ counts each natural representation in the input product cover once. Repeated tensor summands and their fixed multiplicities contribute to InSize.

The first result in this section is concerned about quotients of degree-2 tensor modules. The goal is to find a nice matrix "model" of such modules. The proof of the following theorem is given in Subsubsection C.5.1 of Appendix C.

Theorem 6.1 (Constructive reduction of degree-two subquotients)**.** *Let $D = \prod_{i=1}^{t} \mathrm{Cl}(n_i, \mathbb{K}_i)$, where the $\mathbb{K}_i$ are finite fields of characteristic p,* Cl *denotes a derived classical group type, and each factor is in the stable range above. Let $d = \sum_{i=1}^{t} [\mathbb{K}_i : \mathbb{F}_p] n_i$. Let M be a degree-2 tensor module over their natural modules, and let $P \leq N \leq M$ be a chain of invariant subspaces. Set $Q = N/P$. There is an algorithm that, in time $p^{O(d)}\mathrm{poly}(\mathrm{InSize})$, constructs the following data, where* InSize *is the total explicit input length.*

(i) *Two D-modules I, E of dimension polynomial in* InSize*, and D-equivariant linear maps*

$$Q \xhookrightarrow{\iota} I \xrightarrow{\pi} E,$$

where ι is injective and $H^1(D, E) = 0$. Both modules have explicit matrix and tensor-square quotient presentations: each quotient A/B is specified by bases for $B \leq A$ in the ambient matrix or tensor space, together with its coordinate maps. All field maps and linear relations among scalar coordinates are included in the output presentations.

(ii) *If p is odd, $E = I$ and $\pi = \mathrm{id}_I$, where I has the matrix quotient presentation of Definition 14.*

(iii) *If $p = 2$, the target E has the coordinate models of Definition 15, with the specified multiplicities, field maps, and scalar relations. These presentations include explicit actions of the containing full classical groups.*

(iv) *An explicit realization of $\ker \pi$ as a degree-1 tensor module, given by an equivariant isomorphism*

$$\ker \pi \cong \bigoplus_{\lambda} U_\lambda \otimes_{\mathbb{F}_p} B_\lambda.$$

Each U_λ is one of the input natural modules, its dual, a Frobenius twist of either, or a trivial line; each B_λ has trivial D-action. The distinct underlying natural spaces have total $\mathbb{F}_p$-dimension $O(d)$. Their multiplicities contribute only to the polynomial dependence on InSize*.*

All modules are viewed over $\mathbb{F}_p$. Their actions are specified on group generators, and all constructed linear maps are given in the input and output bases. For a central-product image given with its product cover, pull the action back to that cover. The cover map is part of the output data so that later transporter computations can project to the original group.

Definition 14 (Coordinate models for the odd-characteristic target)**.** Assume p is odd, and let D be as in Theorem 6.1. Let

$$B = \bigoplus_{j=1}^{s} B_j, \qquad B_j = \mathrm{Hom}_{\mathbb{K}_j}(X_j, Y_j) \cong \mathrm{Mat}(a_j \times b_j, \mathbb{K}_j),$$

where each $\mathbb{K}_j$ is a finite field of characteristic p, and X_j, Y_j are explicitly represented natural, dual, Frobenius-twisted, or trivial modules, with restriction of scalars where required, and $a_j = \dim_{\mathbb{K}_j} Y_j$, $b_j = \dim_{\mathbb{K}_j} X_j$. Writing ρ_{X_j}, ρ_{Y_j} for their actions, the action on B_j is

$$g \cdot A = \rho_{Y_j}(g) A \rho_{X_j}(g)^{-1}.$$

Define the fixed scalar subspaces

$$\Sigma_j = \begin{cases} (\mathbb{K}_j \,\mathrm{Id}_{a_j}) \cap B_j^D, & a_j = b_j, \\ 0, & a_j \neq b_j, \end{cases} \qquad \Sigma = \bigoplus_{j=1}^{s} \Sigma_j.$$

An *odd-characteristic target model* is

$$B/C, \qquad C \leq_{\mathbb{F}_p} \Sigma.$$

All spaces are viewed over $\mathbb{F}_p$. An explicit presentation includes the field and action data, quotient coordinate maps, and one $\mathbb{F}_p$-basis for C in the coordinates of B; relations in C may involve different summands.

Definition 15 (Coordinate models for the characteristic-two target)**.** Assume $p = 2$, and let D be as in Theorem 6.1. For each square block, let $V = \mathbb{K}^n$ be the natural module of its classical factor, and put

$$T = V \otimes_{\mathbb{K}} V, \qquad \tau(v \otimes w) = w \otimes v, \qquad A = \mathrm{im}(1 + \tau).$$

In the symplectic and orthogonal cases, $n = 2m$. Let β be the preserved alternating form in the symplectic case, and the polar form of the preserved quadratic form in the orthogonal case. For a symplectic basis e_i, f_i of β, put

$$\omega = \sum_{i=1}^{m} (e_i \otimes f_i + f_i \otimes e_i).$$

For the symplectic case, define the linear map $\ell : A \to \mathbb{K}$ by $\ell(v \otimes w + w \otimes v) = \beta(v, w)$, and set

$$I_0 = \begin{cases} T/\ker \ell, & m \text{ odd}, \\ (T \oplus T)/\{(a, \ell(a)\omega) : a \in A\}, & m \text{ even}. \end{cases}$$

Here $\ker \ell$ is a subspace of $A \leq T$. The square models are

Classical factor	Target square modules
Special linear or special unitary	$T,\ T/A$
Symplectic	$T/\mathbb{K}\omega,\ T/A,\ I_0$
Derived orthogonal	$T/\mathbb{K}\omega$

The other models are full matrix or Hermitian matrix spaces modulo fixed scalar subspaces, simple two-factor tensor modules in their tensor bases, and trivial modules. The same formulas define the models for the dual and Frobenius-twisted natural actions. Fixed multiplicity spaces have trivial D-action.

All defining-field spaces are viewed over $\mathbb{F}_2$ by restriction of scalars. An explicit presentation includes bases for every subspace being quotiented out, all field maps, and all $\mathbb{F}_2$-linear relations among scalar coordinates, including relations between different summands. It also includes matrices on generators for the actions of the containing full classical groups under the corresponding natural, dual, and twisted tensor actions.

For an affine action $\alpha_g(v) = L_g(v) + c_g$ on Q, write L_g^E for the induced linear action on E. Theorem 6.1 computes a change of origin in E by solving $\pi\iota(c_g) = (L_g^E - \mathrm{id}_E)b$ on generators. Thus the projected equation is $L_g^E(\pi\iota x + b) = \pi\iota y + b$. Once its *complete* linear transporter has been returned, Lemma 6.6 gives one residual degree-one affine equation under the full returned stabilizer. The theorem constructs the representations and this reduction; the later algorithms solve the projected matrix equation. The cohomology assertion is for the full input product D, not for every subgroup that may occur as a returned stabilizer. When the original action is on a central-product image, its pullback cocycle vanishes on the kernel of the input product cover, and the final witness and stabilizer project through the given quotient map.

Definition 16 (Pairing maps and pairing submodules)**.** Let D be a product of classical groups with common defining field $\mathbb{K}$, and let $T = T_1 \otimes_{\mathbb{K}} T_2 \otimes_{\mathbb{K}} T_3$, where each T_i is a natural, dual, or Frobenius-twisted module of one factor of D. The field, bases, and actions on the T_i are part of the input. For $i \in \{1,2,3\}$, let $j < \ell$ be the other two indices and put $B_i = (T_j \otimes_{\mathbb{K}} T_\ell)^D$. For $\eta \in B_i$, let $\kappa_{i,\eta} : T_i \to T$ be the linear map that inserts its argument in position i and η in the other two positions. Define the *pairing parameter space* and its insertion map by

$$\mathcal{P}(T) = \bigoplus_{i=1}^{3} T_i \otimes_{\mathbb{K}} B_i, \qquad \mathcal{J}_T : \mathcal{P}(T) \longrightarrow T, \qquad \mathcal{J}_T|_{T_i \otimes_{\mathbb{K}} B_i}(v \otimes \eta) = \kappa_{i,\eta}(v).$$

The *pairing submodule* is $J(T) = \operatorname{im} \mathcal{J}_T$. Each B_i has trivial D-action, and the insertion maps are D-equivariant. Tensor products, Hom spaces, and submodules in the defining-field cubic statements are over $\mathbb{K}$; after restriction to $\mathbb{F}_p$, the defining-field linearity relations remain part of their coordinate presentations.

A simple module *occurs in degree at most two* if it is a composition factor of a tensor product of at most two natural, dual, or twisted modules from the given representation data for D; degree zero includes the trivial module. The *socle* of a module is the sum of its simple submodules.

The following cubic results identify the pairing submodule and remove the affine obstruction after quotienting by it.

The proof of the following theorem is given in Subsubsection C.5.2 of Appendix C.

Theorem 6.2 (Cubic pairing submodules and their maps)**.** *Let D be a product of classical derived groups in the stable range above, and let T and $\mathcal{J}_T$ be as in Definition 16. The following assertions hold in every characteristic.*

(i) *The map $\mathcal{J}_T$ is injective. Thus $J(T) \cong \bigoplus_i T_i \otimes_{\mathbb{K}} B_i$, with each parameter occurrence indexed by a basis of its actual invariant-pairing space.*

(ii) *If every composition factor of a D-submodule $P \le T$ occurs in degree at most two, then $P \le J(T)$. The socle of $T/J(T)$ has no simple constituent occurring in degree at most two.*

(iii) *For a natural, dual, or twisted-natural module U given by its matrices on the generators of D,*

$$\operatorname{Hom}_D(U,T) = \sum_{i=1}^{3} \operatorname{Span}_{\mathbb{K}}\{\kappa_{i,\eta} \circ f : \eta \in B_i,\ f \in \operatorname{Hom}_D(U,T_i)\}.$$

Bases of these spaces, the pairing maps, their images and kernels, and the indicated coordinate identifications are computable by linear equations on the input matrices in time $p^{O(d)}\operatorname{poly}(\mathrm{InSize}, \log p)$.

These assertions apply componentwise with fixed multiplicities. For a prime-field tensor module, apply them to its defining-field components after scalar extension and descend the resulting maps, with the intertwiner and endomorphism fields, twist exponents, and central-product quotient map included in the coordinate descriptions. The direct-sum parameters in (i) count invariant pairings, rather than choices of intertwiners into an already counted natural occurrence.

The proof of the following theorem is given in Subsubsection C.5.3 of Appendix C.

Theorem 6.3 (Cohomology after quotienting by invariant pairings)**.** *Let D and T satisfy the hypotheses of Theorem 6.2. Then*

$$(T/J(T))^D = 0, \qquad H^1(D, T/J(T)) = 0.$$

The assertions hold after prime-field descent, with fixed multiplicities, and for a central-product image given with its quotient map. Consequently a cocycle $c : D \to T/J(T)$ specified by its values on the generators has a unique $b \in T/J(T)$ with $c(g) = (g-1)b$, computable by solving these linear equations on generators.

6.2 Notions and facts supporting classical group results

Injective modules. Let A be a unital algebra over a field $\mathbb{F}$. Modules are finite-dimensional left modules over A. An A-module I is *injective* if, for every inclusion of A-modules $U \hookrightarrow W$ and every A-linear map $f : U \to I$, there is an A-linear map $\bar{f} : W \to I$ restricting to f. An inclusion $M \hookrightarrow I$ is *essential* if every nonzero submodule of I meets M nontrivially. An *injective envelope* $E_A(M)$ is an essential inclusion into an injective module; it is unique up to an isomorphism fixing M.

Cogenerators. An A-module C is a *cogenerator* if, for every A-module M and every $0 \neq v \in M$, some $f \in \mathrm{Hom}_A(M, C)$ satisfies $f(v) \neq 0$. Equivalently, every M embeds in a finite direct sum of copies of C.

We collect some standard finite-dimensional duality facts [Ben91, Section 1.5]. For a finite-dimensional algebra A, the dual $D(A) = \mathrm{Hom}(A, \mathbb{F})$, with $(a \cdot f)(b) = f(ba)$, is an injective cogenerator. The modules $E_A(S)$, one for each simple type S, are exactly the indecomposable injective types, and $\bigoplus_S E_A(S)$ is an injective cogenerator. In particular the types are indexed by their simple socles. If e is a primitive idempotent, the dual right ideal $D(eA) = \mathrm{Hom}_{\mathbb{F}}(eA, \mathbb{F})$ gives the corresponding injective left module.

Joint enveloping algebras. Let D be a finite group and M a finite-dimensional D-module over a field $\mathbb{F}$. Suppose $M = \bigoplus_{i=1}^{s} M_i$ is a direct sum of D-submodules, with actions $\rho_i : D \to \mathrm{GL}(M_i)$, and let $A_i = \mathrm{span}_{\mathbb{F}} \rho_i(D)$ be their enveloping algebras. Let $P \leq N \leq M$ be a chain of D-invariant subspaces, and set $Q = N/P$. The *joint enveloping algebra* is

$$A = \mathrm{span}_{\mathbb{F}} \left\{ \bigoplus_{i=1}^{s} \rho_i(g) : g \in D \right\} \leq \bigoplus_{i=1}^{s} A_i.$$

This inclusion can be strict. For two identical nonzero representations, $A = \{(a, a) : a \in A_1\}$, so the projections onto individual copies do not belong to A. Invariant subspaces may therefore satisfy relations between different copies, including trivial summands.

Modules are *composition-disjoint* if they have no isomorphic simple composition factors over $\mathbb{F}$. The proof of the following proposition is given in Subsubsection C.1.1 of Appendix C.

Proposition 6.4 (Splitting composition-disjoint summands)**.** *Suppose $M = U \oplus W$ is a direct sum of composition-disjoint D-submodules. Then the two coordinate projections belong to A, and every D-invariant subspace $X \leq M$ satisfies*

$$X = (X \cap U) \oplus (X \cap W).$$

In particular, P and N split across these summands.

All composition factors are compared over the chosen coefficient field; after restriction of scalars, that field is $\mathbb{F}_p$.

Embedding an arbitrary subquotient. The proof of the following lemma is given in Subsubsection C.1.2 of Appendix C.

Lemma 6.5 (Embedding into injective modules)**.** *Let $A \le \mathrm{End}_{\mathbb{F}_p}(M)$ be a finite-dimensional $\mathbb{F}_p$-algebra. Let $k \ge 0$ and let $P \le N \le M^{\oplus k}$ be arbitrary explicitly given A-submodules. Set $Q = N/P$. Neither P nor N is assumed to be a direct sum of copies of M. Suppose C contains one copy of every indecomposable injective left A-module. If $f_1, \dots, f_s$ is a basis of $\mathrm{Hom}_A(Q, C)$, then*

$$\iota : Q \longrightarrow C^s, \qquad v \longmapsto (f_1(v), \dots, f_s(v))$$

is injective. It and the algebra A can be computed by linear algebra. The target has $\mathbb{F}_p$-dimension at most $\dim_{\mathbb{F}_p} Q(\dim_{\mathbb{F}_p} C)^2$.

Passing to an affine fibre. The next statement applies the quotient–fiber reduction of Lemma 3.5 after a change of origin in the target. It turns the projected equation into one residual equation under its full source stabilizer.

The proof of the following lemma is given in Subsubsection C.1.3 of Appendix C.

Lemma 6.6 (Removing a translation after an equivariant map)**.** *Let Q, I, E be $\mathbb{F}_p[G]$-modules, let $\iota : Q \hookrightarrow I$ and $\pi : I \to E$ be equivariant $\mathbb{F}_p$-linear maps, and let $c : G \to Q$ be a cocycle. Write $c_g = c(g)$ and let L_g, L_g^I, and L_g^E denote the linear actions on Q, I, and E, respectively. Suppose $H^1(G, E) = 0$. Given $x, y \in Q$, the complete affine transporter for $\alpha_g(v) = L_g(v) + c_g$, $g \in G$, is obtained by one complete linear transporter in E and one complete affine transporter in $\ker \pi$, under the full source stabilizer returned by the first transporter. The residual cocycle is computed from the original affine action.*

Scalar ambiguities. Matrix comparisons modulo fixed scalar subspaces produce the following simultaneous equations. The bound controls the number of complete tuples of shifts in subsequent matrix comparisons.

The proof of the following lemma is given in Subsubsection C.1.4 of Appendix C.

Lemma 6.7 (Solving scalar ambiguities simultaneously)**.** *Let $\mathbb{K}/\mathbb{F}_p$ be a finite extension and let $A_j, B_j \in \mathrm{Mat}(n, \mathbb{K})$ for $1 \le j \le s$. Every solution with $0 \ne X \in \mathrm{Mat}(n, \mathbb{K})$ and $(\lambda_1, \dots, \lambda_s) \in \mathbb{K}^s$ of*

$$XA_j - B_jX = \lambda_j X \qquad (1 \le j \le s)$$

belongs to one of at most n^2 nonzero common scalar eigenspaces. All tuples $(\lambda_1, \dots, \lambda_s)$ of scalar shifts that occur can be found in $|\mathbb{K}|\mathrm{poly}(n, s, \log|\mathbb{K}|)$ operations. For independent matrix factors with $\sum_i [\mathbb{K}_i : \mathbb{F}_p] n_i \le d$, all joint choices number at most 4^d, and their computation costs $p^{O(d)}\mathrm{poly}(\mathrm{InSize})$.

A prescribed correlated additive subspace of allowable scalar tuples is imposed only after computing the complete tuples in Lemma 6.7. All remaining matrix, bilinear, and quadratic equations are imposed simultaneously, so one returned transformation satisfies every tensor coordinate.

Full and derived classical groups. The proof of the following lemma is given in Subsubsection C.1.5 of Appendix C.

Lemma 6.8 (Classical-group characters)**.** *In the stable range, the specified derived classical groups are kernels of the following characters of their containing full groups:*

$$\mathrm{SL}_n(\mathbb{K}) = \ker(\det : \mathrm{GL}(n,\mathbb{K}) \to \mathbb{K}^\times), \qquad \mathrm{SU}_n(\mathbb{K}/\mathbb{K}_0) = \ker(\det : \mathrm{U}_n(\mathbb{K}/\mathbb{K}_0) \to \mathbb{K}^\times),$$

$$\Omega(V,Q) = \begin{cases} \ker(\det, \mathrm{sp}), & p \neq 2, \\ \ker \Delta, & p = 2, \end{cases} \qquad \Delta(g) = \operatorname{rank}(I-g) \bmod 2.$$

In the unitary formula, $\mathbb{K}_0 = \mathbb{K}^\sigma$ is the fixed field, denoted by $\mathbb{F}$ at the start of Section 6. Here sp *is the spinor norm with values in $\mathbb{K}^\times/(\mathbb{K}^\times)^2$, and the stable symplectic group requires no additional character condition. These characters are computable in polynomial time from the matrices. For a product of the input natural spaces, their joint image has size $p^{O(d)}$.*

Classical-group notation used in the proofs. The following calculations use the standard block decomposition of a classical group [Tay92, Chapters 8–11]. The stabilizer of a totally isotropic subspace (totally singular for a quadratic form) is a *parabolic subgroup*. Its block-diagonal subgroup is a *Levi subgroup*, and its block-unitriangular kernel is the *unipotent radical*. The parabolic associated with a maximal totally isotropic subspace is called a *Siegel parabolic*. We first solve the equation on the diagonal blocks and then determine the blocks joining them.

The simple-module names $L(\lambda)$ in these proofs are the standard highest-weight names: ω_i denotes the ith fundamental weight, and $L(\lambda)$ the corresponding absolutely simple algebraic-group module. We use the cited classification only to distinguish simple types and their Frobenius twists. The algorithm obtains the actual maps from matrix equations. An algebraic classical group here allows the same defining matrices over an algebraic closure of $\mathbb{K}$. Comparisons with that group concern linear spans of its representation matrices; they do not assume that the finite group itself equals it.

Cohomology and field conventions. We use the module and extension conventions of Subsection 3.7. For group cohomology, write $H^i(G,M) = \mathrm{Ext}^i_{\mathbb{F}G}(\mathbb{F},M)$, where $\mathbb{F}$ has trivial action. Thus $H^0(G,M) = M^G$, and the degree-one group agrees with the cocycle quotient defined there. The proofs use degrees $0,1,2$. Long exact sequences and inflation–restriction are used in their usual module-theoretic form [Bro82, Chapters III and VII].

For an exact sequence $0 \to J \to T \to Q \to 0$, the portion needed below is

$$T^D \longrightarrow Q^D \longrightarrow H^1(D,J) \longrightarrow H^1(D,T) \longrightarrow H^1(D,Q) \longrightarrow H^2(D,J) \longrightarrow H^2(D,T).$$

Exactness means that the image of each arrow is the kernel of the next. For example, surjectivity of $H^1(D,J) \to H^1(D,T)$ and injectivity of $H^2(D,J) \to H^2(D,T)$ force $H^1(D,Q) = 0$. Thus several proofs below compare the cohomology of J and T, rather than calculating cocycles in the quotient directly.

Tensor products are formed over their stated defining fields. After restriction to the prime field, their coordinate presentations include the equations expressing defining-field linearity. In the following calculations $\mathbb{K}$ is the common coefficient field and its prime field is $\mathbb{F}_p$.

We distinguish a twisted representation from a coordinate map. For a based representation $\rho : D \to \mathrm{GL}(V)$ and an integer $a \geq 0$, $V^{[a]}$ carries the matrices $\rho^{[a]}(g) = \rho(g)^{p^a}$ for $g \in D$, with the power applied entrywise. The notation $x^{[a]}$ for a vector means the coordinate Frobenius map $\mathrm{Fr}_a : V \to V^{[a]}$, $\sum_i x_i e_i \mapsto \sum_i x_i^{p^a} e_i^{[a]}$. This map is additive, p^a-semilinear over $\mathbb{K}$, and $\mathbb{F}_p$-linear. For a $\mathbb{K}$-linear map $f : V \to W$ between based spaces, $f^{[a]} : V^{[a]} \to W^{[a]}$ is the $\mathbb{K}$-linear map obtained by taking entrywise p^a powers of its matrix. Duals and tensor products in a defining-field

calculation are over $\mathbb{K}$, or over an algebraic closure $\overline{\mathbb{K}}$ when scalar extension is explicitly made. Such a closure contains $\mathbb{K}$ and roots of every polynomial over it. Extending scalars replaces V by $\overline{\mathbb{K}} \otimes_{\mathbb{K}} V$ and keeps the same representing matrices. In the prime-field algorithm they are over $\mathbb{F}_p$, with defining-field linearity expressed by additional linear equations.

7 Degree-2 quotient tensor module transporter

We solve the degree-2 quotient tensor module transporter problem. The matrix-group results of Section 5 separate the selected large classical actions from alternating-dominant actions. The algorithm divides the permuted summands of the latter and completes the remaining comparisons with Proposition 7.2.

The constructions and proofs for this section are given in Appendix D, including the controlled extension of Subsection 7.6.

7.1 Problem and main results

We follow the group encoding and input/output conventions of Section 3.5.

Definition 17 (Degree-2 quotient tensor module transporter)**.** An instance over $\mathbb{F}_p$, with p prime, consists of the following data.

(i) Vector spaces $V_1, \dots, V_s$ over $\mathbb{F}_p$.

(ii) A finite group H with representations $\rho_i : H \to \mathrm{GL}(V_i)$ for $i \in [s]$.

(iii) A tensor module of degree at most 2,

$$Z = \bigoplus_{a \in I} \bigotimes_{j=1}^{k_a} W_{a,j}, \quad k_a \in \{0, 1, 2\},$$

where I is a finite set of summand labels fixed by H, and each $W_{a,j}$ is a specified V_i, its dual, or a Frobenius twist of either. Repetitions are allowed, and an empty tensor product means the trivial module $\mathbb{F}_p$. All tensor products are over $\mathbb{F}_p$.

(iv) Two H-invariant subspaces $P \leq N \leq Z$, with quotient coordinates for $Q := N/P$. For each $h \in H$, let $L_h : Q \to Q$ be the linear map induced by the action of h on Z as in Definition 8.

(v) Translation vectors $c_h \in Q$ for $h \in H$, such that the maps

$$\alpha_h(u) := L_h(u) + c_h \text{ for } u \in Q.$$

define a group action $\alpha : H \times Q \to Q$.

(vi) Two points $x, y \in Q$.

The task is to compute the transporter

$$\mathrm{Trans}_{H,\alpha}(x, y) = \{h \in H : \alpha_h(x) = y\}.$$

The output is emptiness, or a witness g_0 and generators of $H_x = \{h \in H : \alpha_h(x) = x\}$, so that $\mathrm{Trans}_{H,\alpha}(x, y) = g_0 H_x$.

Each original representation is counted once in d. Repeated summands and fixed multiplicities contribute to the explicit input length InSize. A multiplicity space with nontrivial action is instead an original representation and counts as a moving tensor factor. The input length includes all matrices, field and twist data, subspace bases, quotient maps, points, and generator translations.

The promise in (v) means that the translations extend to a map $c : H \to Q$ with $c_{gh} = c_g + L_g c_h$ and $\alpha_h(z) = L_h z + c_h$. This identity is required in Q; no cocycle lift to N is assumed. Field-linearity, symmetry, and specified linear identifications are encoded in the invariant subspaces and their defining maps.

Theorem 7.1 (Degree-2 quotient tensor module transporter)**.** *Given an instance of Definition 17 with original dimension d and input length* InSize*, its complete transporter can be computed deterministically in time $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$.*

Theorem 7.1 is based on a recursive algorithm. A key terminal case of the recursion is when the group H is a central product of classical groups (see Definition 19). The following proposition has a proof outline in Section 7.5 and its full proof in Section D.5.

Proposition 7.2 (Degree-2 quotient tensor module transporter under classical group actions)**.** *There is a deterministic algorithm in time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$ for the degree-2 quotient tensor module transporter problem of Definition 17 when H is an explicit central product of classical groups in the prescribed stable range, with natural, dual, or Frobenius-twisted factor actions.*

Matrix-group and coset intersection. Given two matrix groups and a coset of each, we ask whether the cosets have a common element and, if they do, for their complete intersection. Even when the individual groups are given by short generating sets, listing their elements need not fit a bound exponential only in the matrix dimension. Theorem 7.1 gives that bound and computes a witness and the entire intersection subgroup.

Theorem 7.3 (Intersection of matrix groups and cosets)**.** *Let p be a prime. The input consists of the following data:*

(i) a based finite-dimensional $\mathbb{F}_p$-space W;

(ii) two groups $G_1, G_2 \leq \mathrm{GL}(W)$, given by generators, together with any original comparison representations included in the input under the original-group encoding convention above;

(iii) two matrices $u, v \in \mathrm{GL}(W)$.

Let d be the total prime-field dimension of the original representations, including the two displayed copies of W, and let InSize *be the total explicit input length. There is a deterministic $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$-time algorithm that computes $uG_1 \cap vG_2$. Its output is either emptiness, or a matrix $w \in uG_1 \cap vG_2$ and generators of $G_1 \cap G_2$, representing*

$$uG_1 \cap vG_2 = w(G_1 \cap G_2).$$

When the displayed groups are images of originally encoded groups, the algorithm also computes the full kernels and lifts of the comparison actions. In particular, for groups given by their matrices in $\mathrm{GL}(n, \mathbb{F}_q)$, the bound is $q^{O(n)}\mathrm{poly}(\mathrm{InSize}, \log q)$.

Proof of Theorem 7.3 from Theorem 7.1. We construct an instance of Definition 17. Let W_1, W_2 be copies of W with their fixed coordinate identifications, and put

$$H = G_1 \times G_2, \qquad Z = \mathrm{Hom}_{\mathbb{F}_p}(W_2, W_1) \cong W_1 \otimes_{\mathbb{F}_p} W_2^*,$$
$$N = Z, \qquad P = 0, \qquad Q = N/P = Z, \qquad x = I, \qquad y = D := u^{-1}v, \qquad c_h = 0 \quad (h \in H).$$

The element I is the identity matrix, viewed as the identity map $W_2 \to W_1$ under these identifications. The action is

$$L_{(g_1,g_2)}(X) = g_1 X g_2^{-1} \qquad (g_i \in G_i,\ X \in Z).$$

If $\mathcal{S}_i$ generates G_i, generators of H are (s, I) for $s \in \mathcal{S}_1$ and (I, t) for $t \in \mathcal{S}_2$. Thus all the transporter data are explicit.

The constructed equation is

$$(g_1, g_2) \cdot I = D \quad \Longleftrightarrow \quad g_1 g_2^{-1} = u^{-1} v \quad \Longleftrightarrow \quad u g_1 = v g_2.$$

Consequently its solutions correspond bijectively to the elements of $uG_1 \cap vG_2$, by $(g_1, g_2) \mapsto ug_1 = vg_2$; the inverse sends w to $(u^{-1}w, v^{-1}w)$. The source stabilizer is

$$H_I = \{(s, s) : s \in G_1 \cap G_2\}.$$

If the complete transporter is $(a_1, a_2)H_I$, set $w = ua_1 = va_2$ and project its stabilizer generators to their first coordinates. These generate $G_1 \cap G_2$ and give exactly $uG_1 \cap vG_2 = w(G_1 \cap G_2)$. An empty transporter gives an empty intersection. In particular, taking $u = v = I$ computes the intersection subgroup.

For $\dim_{\mathbb{F}_p} W = n$, the original representation is on $W_1 \oplus W_2$ of dimension $2n$, although the tensor module Z has dimension n^2. The latter enters only the explicit input length. For matrices over $\mathbb{F}_q$, where $q = p^f$, use the underlying prime-field spaces, set $Z = \mathrm{Hom}_{\mathbb{F}_p}(W_2, W_1)$ and $N = \mathrm{Hom}_{\mathbb{F}_q}(W_2, W_1) \leq Z$, and set $P = 0$. Commuting with scalar multiplication by a field basis gives linear conditions defining N, which is invariant under H. The original dimension is $2fn$, giving $p^{O(fn)} = q^{O(n)}$. If original groups map onto the displayed G_i, carry out the construction in their original product encoding and project the answer only at the end. The full preimage of H_I, including all elements invisible in the displayed actions, is computed as the full source stabilizer in the original product group; the dimension parameter includes the original matrix representations. □

7.2 Group structure and recursive data

The algorithm follows the matrix-group structure in Theorems 5.1 to 5.3. Item (1) of Theorem 5.1 gives invariant composition flags. These reduce a tensor factor to irreducible sections, using a recursive transporter call on a quotient followed by a recursive call on its fiber. Item (2) gives systems of imprimitivity on the irreducible sections and their primitive block actions. Theorem 5.2 identifies the local module types, diagonal links, and selected classical product D. Theorem 5.3 gives the remaining actions and their auxiliary presentations.

The remaining actions are alternating-dominant in the sense of Section 5.3, with the fixed threshold T_{cl} from (5.1).

The remaining modules have direct-sum presentations whose summands are permuted by the group. Restricting to the stabilizer of half of these summands gives two smaller recursive transporter problems. The index of this stabilizer is at most exponential in the number of summands. Once that number is bounded, the image of the remaining actions has order $p^{O(d)}$, and the classical solver completes the transporter. The recurrence in Subsection 7.4 accounts for these subgroup restrictions and recursive calls.

Auxiliary module presentations. For an H-module B, an auxiliary presentation consists of an H-module R, invariant subspaces $K_B \leq Y_B \leq R$, and an explicit H-module isomorphism $Y_B/K_B \cong B$. In this section, R also has a specified decomposition into nonzero direct summands permuted by H. The presentations of Theorem 5.3(2) include the full permutation modules from

which fully deleted alternating modules are obtained. For bounded primitive modules and unselected classical modules, the auxiliary module can be the original module itself. The substitution into the current tensor space is explained in Appendix D.1.

Write Γ for the original input group. A recursive call has a current subgroup $H \le \Gamma$, an element $t \in \Gamma$, and data

$$P \le N \le Z, \qquad Q = N/P, \qquad \alpha_h(z) = L_h z + c_h, \qquad x, y \in Q.$$

It computes $t\,\mathrm{Trans}_{H,\alpha}(x, y)$. The element t is the accumulated left multiplier from earlier coset choices and witnesses. If the transporter in H is nonempty, write it as aH_x, where $a \in H$ sends x to y and H_x is the full affine stabilizer of x. The call returns ta and generators of H_x. The current spaces and actions may change, while the original dimension d and input length InSize remain fixed.

Factors and selected summands in a call. To implement these two treatments, we distinguish the active factors, the auxiliary modules, and their summands still used by the current transporter problem. Let $\mathcal{A}$ and $\mathcal{B}$ be finite index sets for this call. The tensor positions are $i \in [2]$; they are positions within a tensor product, not indices of the original spaces $V_1, \dots, V_s$. All modules below have specified bases and generator matrices.

$\mathbf{U} = (U_a)_{a\in\mathcal{A}}$:
The distinct *active factors*. Initially these are the original representation spaces and the duals or twists used in the input. Step R1 replaces them by invariant sections. Step R2 represents the remaining types using auxiliary modules; the factors left in $\mathbf{U}$ support the selected large classical components. Their multiplicity coordinates and diagonal identifications come from Theorem 5.2(1),(2),(4). After a subgroup restriction, the algorithm prepares these factors again if necessary. Each factor comes with the maps identifying it with its preceding representation or invariant section.

$\mathbf{R} = (R_\nu)_{\nu\in\mathcal{B}}$:
The *alternating-dominant auxiliary modules*. Each has a finite summand index set I_ν and a decomposition

$$R_\nu = \bigoplus_{\alpha\in I_\nu} R_{\nu,\alpha}, \qquad R_{\nu,\alpha} \neq 0, \qquad hR_{\nu,\alpha} = R_{\nu,h\alpha} \quad (h \in H).$$

The data include the summand bases, the permutations of I_ν, the linear maps between summands, and the maps expressing the replaced factor as a subquotient of R_ν. An individual $R_{\nu,\alpha}$ is a module for its stabilizer. The whole R_ν is an H-module.

$\mathbf{S} = (S_{i,\nu})_{i\in[2],\,\nu\in\mathcal{B}}$:
The *selected summand indices*, with $S_{i,\nu} \subseteq I_\nu$ and

$$R_\nu[S_{i,\nu}] = \bigoplus_{\alpha\in S_{i,\nu}} R_{\nu,\alpha}.$$

Each set $S_{i,\nu}$ is invariant under the current group and is empty when R_ν is unused in position i. The same auxiliary module can occur in both positions with different selections. Its action is stored once, while the selections depend on the position. A child can restrict the group and reduce a selection; it cannot restore indices excluded by an earlier recursive call whose fiber remains to be solved.

Initially $\mathcal{B} = \varnothing$, so $\mathbf{R}$ and $\mathbf{S}$ are empty. The lists describe the factors available to a call; the tensor module below specifies how they occur.

Scalar and semilinear spaces. The auxiliary data also include a finite family $(\mathbb{K}_\ell, L_\ell, \tau_\ell)_{\ell\in\mathcal{L}}$, where $L_\ell = \mathbb{K}_\ell = \mathbb{F}_{p^{e_\ell}}$ and $\tau_\ell(h)(a) = \lambda_\ell(h)a^{p^{j_\ell(h)}}$ for $a \in \mathbb{K}_\ell$. These are the semilinear lines of Definition 20. These spaces express the scalar and field-automorphism actions in the structural coordinates. Their underlying $\mathbb{F}_p$-dimensions are e_ℓ and count towards the representation dimension budget. The family may be empty. It is stored with the auxiliary data $\mathbf{R}$ in the notation for a call, and all its actions are restricted when H is replaced by a subgroup. Initially $\mathcal{L} = \varnothing$ for the input of Definition 17.

The tensor module of a call. Let Σ be a finite set of summand labels fixed by H. For each $\sigma \in \Sigma$, let $a_\sigma, b_\sigma \geq 0$ with $a_\sigma + b_\sigma \leq 2$, and write

$$Z = \bigoplus_{\sigma\in\Sigma} \left(\bigotimes_{j=1}^{a_\sigma} E_{\sigma,j} \right) \otimes \left(\bigotimes_{k=1}^{b_\sigma} W_{\sigma,k} \right). \tag{7.1}$$

Each $W_{\sigma,k}$ is an entry of $\mathbf{U}$. Each $E_{\sigma,j}$ is a selected module $R_\nu[S_{i,\nu}]$ or its dual. Every displayed factor is assigned a distinct position $i \in [2]$ within its summand. The assignment is part of the data, and is unchanged by rearranging tensor factors or restricting the group. Empty tensor products mean $\mathbb{F}_p$ with trivial action. Write $\rho_{E_{\sigma,j}}$ and $\rho_{W_{\sigma,k}}$ for the actions on the respective factors. The induced action on Z is

$$\rho_Z(h) = \bigoplus_{\sigma\in\Sigma} \left(\bigotimes_{j} \rho_{E_{\sigma,j}}(h) \right) \otimes \left(\bigotimes_{k} \rho_{W_{\sigma,k}}(h) \right).$$

The same $h \in H$ acts in all occurrences. Repeated summands are fixed multiplicities and contribute to InSize, not to the original dimension d. The group permutes the direct summands inside auxiliary factors and fixes the labels in Σ. The subspaces $P \leq N \leq Z$ can impose relations between any of these summands. The position assignments are recorded for each labelled tensor summand.

For an isotypic factor $X \otimes_{\mathbb{K}} M$ from Theorem 5.2(1),(4), $\mathbb{K}$ is its defining field, X is its natural classical module, and M is its multiplicity space. The whole factor still occupies one original position. Only when the local classical subgroup fixes M can a basis of M replace this factor by fixed copies of X. This convention prevents the original tensor-position count from increasing during structural preparation.

Definition 18 (Data and answer of a recursive call)**.** For the original group Γ and dimension d, a recursive call is

$$\mathtt{D}(t, H; \mathbf{U}, \mathbf{R}, \mathbf{S}; N, P, c; x, y), \tag{7.2}$$

where $t \in \Gamma$, $H \leq \Gamma$, and the lists, scalar and semilinear spaces, and tensor module are as defined above. The subspaces $P \leq N \leq Z$ are H-invariant, $Q = N/P$, and $x, y \in Q$. The generator values of $c : H \to Q$ define the affine action

$$\alpha_h(z) = L_h z + c_h, \qquad c_{hk} = c_h + L_h c_k, \tag{7.3}$$

where L_h is induced by $\rho_Z(h)$. The answer is

$$\mathtt{D}(t, H; \mathbf{U}, \mathbf{R}, \mathbf{S}; N, P, c; x, y) = t \operatorname{Trans}_{H,\alpha}(x, y). \tag{7.4}$$

It is represented by emptiness, or a witness ta and generators of the full source stabilizer H_x, giving the coset taH_x. All elements, including kernel generators, use the original encoding of Γ and have words in its original generators.

Affine data in quotient coordinates. Each call stores the matrices L_s and vectors $c_s \in Q = N/P$ on its current generators s of H. When passing to a subgroup, evaluate words for its generators in the parent generators using (7.3), and then transfer the resulting data to the child quotient or fiber. Retain separate words in the original generators of Γ for the output. The current quotient need only be invariant under H, so its affine action is evaluated on words in H, not on arbitrary elements of Γ. Coordinate changes act directly on these quotient vectors. Every recursive group remains a subgroup of Γ.

At the root, $t = 1$, $H = \Gamma$, and N, P, c, x, y are the input data. Every call must return the complete coset in (7.4), including its full stabilizer. A recursive call on a quotient returns all elements satisfying the projected equation. Its parent keeps the original data for the subsequent fiber call. Appendix D.1 proves the quotient and fiber identities used by the algorithm.

Structural guarantees for the recursive data. The next two lemmas concern the auxiliary presentations constructed from Theorems 5.2 and 5.3 and their subsequent restrictions. The basic quadratic problem has two tensor positions. We state the next two lemmas for any fixed $r \geq 2$, where r bounds the total number of positions in each tensor summand, including evaluation arguments. The selections are $S_{i,\nu}$ for $i \in [r]$ and $\nu \in \mathcal{B}$. Subsection 7.6 permits additional evaluation arguments while keeping at most two active factors. The basic cubic problem has three positions. Proposition 7.8 gives the corresponding conditional recursion bound.

Fix a structural dimension constant κ_r so that the active factors, distinct auxiliary modules, and scalar and semilinear spaces have total prime-field dimension at most $\kappa_r d$. In particular, $\sum_{\ell \in \mathcal{L}} e_\ell \leq \kappa_r d$. Choose $C_* = C_*(r)$ sufficiently large for this bound, and fix $T_{\mathrm{cl}} = p^{C_* d}$ throughout the recursion.

Let $\rho_{\mathrm{aux}} : H \to B$ be the joint action on the selected modules $R_\nu[S_{i,\nu}]$, their summand indices, and the spaces $(L_\ell)_{\ell \in \mathcal{L}}$, with $B = \rho_{\mathrm{aux}}(H)$. Of each auxiliary module R_ν, only its selected summands enter this image.

Lemma 7.4 (Simple sections of the alternating-dominant action)**.** *For every $B_0 \leq B$ and $K_0 \trianglelefteq B_0$,*

$$B_0/K_0 \text{ nonalternating nonabelian simple} \quad \Longrightarrow \quad |B_0/K_0| \leq T_{\mathrm{cl}}.$$

The assertion persists after restricting H to a subgroup and replacing these representations by invariant subquotients.

After structural preparation, let $\rho_{\mathbf{U}} : H \to A \leq \mathrm{GL}(\bigoplus_{a \in \mathcal{A}} U_a)$ be the joint action on the remaining active factors, with $A = \rho_{\mathbf{U}}(H)$. These factors consist of the nontrivial part for the full selected classical product $D \trianglelefteq A$ from Theorem 5.2(3),(4), with equivalent constituents collected as in Section 5.4. Its quasisimple factors have nonalternating simple central quotients of order greater than T_{cl}. Write $J = (\rho_{\mathbf{U}}, \rho_{\mathrm{aux}})(H) \leq A \times B$ for the simultaneous image of the same group. The matrix action of A is faithful, whereas both maps from H may have kernels.

The following uses Lemmas 5.5 and 5.6, together with the auxiliary-image bound above.

Lemma 7.5 (An actual subgroup and a bounded quotient)**.** *One has $[A : D] \leq p^{O_r(d)}$; this bound uses only the hypotheses on A and D. Moreover, $D \times 1 \leq J$. Thus the selected classical action can be realized while fixing all selected auxiliary modules and the spaces L_ℓ pointwise. It also acts trivially on the multiplicity spaces in the natural coordinates of Theorem 5.2(1),(4). Identify D with the normal subgroup $D \times 1 \trianglelefteq J$. If $s = \sum_{i \in [r]} \sum_{\nu \in \mathcal{B}} |S_{i,\nu}| = O_r(1)$, then*

$$|B| \leq p^{O_r(d)}, \qquad [J : D] \leq [A : D]|B| \leq p^{O_r(d)}. \tag{7.5}$$

Generators of $D \times 1$ and its full preimage in H can be computed in $p^{O_r(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$ time. Under the same bounded-summand hypothesis, a complete list of coset representatives for J/D, with lifts to H, can be computed in this time. All lifts and kernel generators use the original group encoding.

The proofs are given with the structural preparation and the final classical subgroup construction in Appendix D.2.

7.3 The recursive algorithm

We state the algorithm on the data of Definition 18. The quotient maps used in its recursive calls are defined below. Their correctness and the coordinate substitutions are proved after the running-time analysis, beginning in Appendix D.1.

The induced flag. Composition flags of the active factors give a filtration of each tensor summand by filtering one factor at a time. For flags $0 = U_0 < \cdots < U_a = U$ and $0 = V_0 < \cdots < V_b = V$ in two active factors, use $U_{i-1} \otimes V + U_i \otimes V_j$ for $1 \le i \le a$, $0 \le j \le b$, ordered first by i and then by j, deleting repetitions. With one active factor U and an auxiliary factor R, use $U_i \otimes R$; take $R = \mathbb{F}_p$ when the other factor is absent. With no active factor use the one-step flag of the whole summand. Concatenate these filtrations by adding each term to the preceding whole summands, and write the result as $0 = Z_0 < \cdots < Z_{\ell_Z} = Z$. Appendix D.2.1 derives these terms and their quotient presentations.

The corresponding subspaces of the current quotient $Q = N/P$ are

$$((N \cap Z_j) + P)/P \le Q.$$

Delete repeated terms and call the resulting flag $0 = Q_0 < \cdots < Q_\ell = Q$, recording the ambient grade for each nonzero quotient. These subspaces are invariant under the linear action L_h; the affine translations need not fix them. Step R1 makes a recursive transporter call on $Q/Q_{\ell-1}$ and then one on the fiber with direction space $Q_{\ell-1}$. Lemma D.2 gives exact quotient coordinates for both recursive calls, including relations imposed by P.

Selected coordinates and the splitting parameter. For each position $i \in [2]$, let I_i be the disjoint union of the sets $\{\nu\} \times S_{i,\nu}$, for $\nu \in \mathcal{B}$. If some tensor summand has an active factor or no factor in position i, add a single fixed symbol $\perp_i$. The pair (ν, α) identifies a summand of a particular auxiliary module; $\perp_i$ groups the alternatives that have no auxiliary factor in that position. Expand each auxiliary factor into its selected direct summands; for a dual factor use the dual summands $R^*_{\nu,\alpha}$. The resulting coordinate blocks have one index in I_i. For $A \subseteq I_i$, write Z_A for the direct sum of the coordinate blocks with index in A, and

$$\mathrm{pr}_A : Z \longrightarrow Z_A$$

for their coordinate projection. It commutes with the action of the set stabilizer $H_A = \{h \in H : hA = A\}$. The induced quotient map is

$$q_A : Q \twoheadrightarrow Q_A = \mathrm{pr}_A(N)/\mathrm{pr}_A(P), \qquad q_A(n + P) = \mathrm{pr}_A(n) + \mathrm{pr}_A(P).$$

Its kernel is isomorphic to $(N \cap \ker \mathrm{pr}_A)/(P \cap \ker \mathrm{pr}_A)$, so the remaining fiber uses only the other coordinate blocks. Indeed, a representative whose projection lies in $\mathrm{pr}_A(P)$ can be corrected by an element of P to have zero projection.

Fix the structural threshold at the root, and choose a constant $b_0 \geq 6$ sufficiently large for Lemma D.4 and the contraction estimate in Proposition 7.8. At a call, define

$$b_i = |I_i| \ (i \in [2]), \qquad b = b_1 + b_2, \qquad v = \sum_{a \in \mathcal{A}} \dim_{\mathbb{F}_p} U_a, \qquad m = \dim_{\mathbb{F}_p} Q.$$

Thus b counts selected summand indices, v counts distinct active factor dimensions, and m is the dimension of the current equation. An auxiliary module used in both positions contributes once to the stored dimension and at most twice to the index count. These are current parameters; d always denotes the original dimension budget.

Algorithm 7.6 (Recursive quadratic transporter)**. Input:** the data $\mathtt{D}(t, H; \mathbf{U}, \mathbf{R}, \mathbf{S}; N, P, c; x, y)$ of Definition 18. **Output:** the complete coset $t \operatorname{Trans}_{H,\alpha}(x, y)$, or emptiness.

B1. **If** $Q = N/P = 0$, **return** tH.

B2. **If** $d = 0$:

(a) Compute $c(H)$ and $\ker c$ by Lemma 3.1.

(b) **If** $y - x \notin c(H)$, **return** emptiness.

(c) **Otherwise**, find $a \in H$ with $c_a = y - x$ and **return** $ta \ker c$.

For $d > 0$ and $Q \neq 0$, execute R1–R4 below.

R1. Invariant subspaces (Theorem 5.1(1)).

(a) Compute composition flags of all active factors, including the inclusion and quotient maps. Form the induced flags of Z and Q defined above, with $0 = Q_0 < \cdots < Q_\ell = Q$.

(b) **If** $\ell = 1$, express Q in its single nonzero tensor grade. Replace the active factors by the corresponding composition factors and transfer c, x, y through the quotient isomorphism. Delete unused factors and **continue to R2**.

(c) For $\ell \geq 2$, **call this algorithm recursively** on $Q/Q_{\ell-1}$ with source $x + Q_{\ell-1}$, target $y + Q_{\ell-1}$, the induced affine action of H, and multiplier 1. Use the factor sections in the recorded ambient grade.

(d) **If** that call returns emptiness, **return** emptiness. **Otherwise**, write its answer as aK, with K the full stabilizer of $x + Q_{\ell-1}$.

(e) Set

$$F = Q_{\ell-1}, \qquad N_F = \{n \in N : n + P \in F\}, \qquad w = \alpha_{a^{-1}}(y) - x,$$
$$\beta_k(z) = L_k z + \alpha_k(x) - x \qquad (k \in K, \ z \in F).$$

Return the recursive call on $F = N_F/P$ with group K, action β, source 0, target w, and multiplier ta. Its factor lists are the parent's lists restricted to K.

R2. Systems of imprimitivity and primitive module types.

(a) Write ρ_a for the action on U_a. Compute the joint image

$$\rho_{\mathbf{U}} : H \longrightarrow \mathrm{GL}\left(\bigoplus_{a \in \mathcal{A}} U_a\right), \qquad \rho_{\mathbf{U}}(h) = \bigoplus_{a \in \mathcal{A}} \rho_a(h), \qquad A = \rho_{\mathbf{U}}(H), \tag{7.6}$$

with its kernel and generator lifts in H. The displayed direct sum is semisimple and faithful for A after R1. Thus $O_p(A) = 1$, by Theorem 5.1(1), and we use this direct sum as the associated graded representation.

(b) **Systems of imprimitivity:** apply Theorem 5.1(2) to compute, for each active factor,

$$U_a = \bigoplus_{j \in J_a} B_{a,j}, \qquad hB_{a,j} = B_{a,hj},$$

with primitive block-stabilizer actions. Here J_a is the block index set; $|J_a| = 1$ is allowed. Record the maps between blocks and keep each whole orbit of blocks as an H-module.

(c) **Module types and links:** use Theorem 5.2(1) to distinguish bounded primitive images and unbounded classical or alternating types. Use Items (2)–(3) to compute their diagonal links and the selected normal classical product $D \trianglelefteq A$.

(d) **Natural and auxiliary presentations:** keep in **U** the modules on which D acts nontrivially, with the blockwise natural coordinates of Theorem 5.2(1),(4). For each remaining module, compute its auxiliary presentation from Theorem 5.3 and Appendix D.2.2. Add its auxiliary module to **R** and initialize its selection $S_{i,\nu}$ to I_ν in each position where the factor occurs and to $\varnothing$ in unused positions. Leave every previous selection unchanged.

(e) Substitute these presentations into the whole tensor module, using the map of Appendix D.1. Pull back N, P to $\widehat{N}, \widehat{P}$, and let $u : \widehat{N}/\widehat{P} \xrightarrow{\sim} N/P$ be the induced equivariant isomorphism. Replace c_h, x, y by $u^{-1}(c_h), u^{-1}(x), u^{-1}(y)$ in the new quotient coordinates. Keep t, H unchanged, update b_i, b, v, and **continue to R3**. This step makes no recursive call.

R3. Permuted summands of the alternating-dominant modules.

(a) **If** $b \le b_0$, **continue to R4**.

(b) Choose i with maximum b_i and $A \subseteq I_i$ with $|A| = \lfloor b_i/2 \rfloor$. Compute the set stabilizer H_A and a left transversal $H = \bigsqcup_{j=1}^{k} r_j H_A$, with $k \le 2^{b_i}$.

(c) Prepare the two child presentations as follows. In the quotient child, restrict the selection in position i to A; in the fiber child, restrict it to $I_i \setminus A$. Restrict both children's actions to their groups and delete the coordinate blocks on the other side. A side excluding $\perp_i$ also excludes every labelled summand with an active factor or no factor in position i. Selections in other positions and all prior exclusions remain in force.

(d) **For each** $j \in [k]$, perform the following operations.

(1) Put $y_j = \alpha_{r_j^{-1}}(y)$. **Call this algorithm recursively** on $Q_A = \mathrm{pr}_A(N)/\mathrm{pr}_A(P)$, with source $q_A(x)$, target $q_A(y_j)$, the induced affine action of H_A, and multiplier 1.

(2) **If** the call returns emptiness, **skip this coset**. **Otherwise**, write its answer as $a_j K_j$, with K_j the full stabilizer of $q_A(x)$, and put

$$F_A = \ker q_A, \qquad w_j = \alpha_{a_j^{-1}}(y_j) - x,$$
$$\beta_k(z) = L_k z + \alpha_k(x) - x \qquad (k \in K_j,\ z \in F_A).$$

(3) **Call this algorithm recursively** on F_A, with group K_j, action β, source 0, target w_j, and multiplier $tr_j a_j$. Use the presentation

$$F_A \cong (N \cap \ker \mathrm{pr}_A)/(P \cap \ker \mathrm{pr}_A)$$

on the remaining coordinate blocks, as in Lemma D.3. Store this call's answer for coset $r_j H_A$.

(e) **If** all coset answers are empty, **return** emptiness. **Otherwise**, combine the nonempty cosets by the identity (3.1) and **return** their complete coset.

R4. The remaining classical action.

(a) Using the linked components and natural coordinates of Theorem 5.2(2)–(4), compute the joint image $\theta : H \twoheadrightarrow J$ on the active factors, selected auxiliary modules, and the scalar and semilinear spaces L_ℓ. Lemma 7.5 gives a normal classical subgroup $D \trianglelefteq J$ with $[J : D] = p^{O(d)}$, where D fixes the selected auxiliary modules, multiplicity spaces, and the spaces L_ℓ pointwise.

(b) Center the source: replace c_h by $L_h x + c_h - x$, y by $y - x$, and x by 0. Compute $\Delta = c(\ker \theta)$ and the induced affine action of J on $Q_{\mathrm{tr}} = Q/\Delta$.

(c) **For each coset of D in J**, normalize the target by the inverse affine action of its representative. Expand the spaces fixed by D in their bases, and invoke Proposition 7.2 on Q_{tr}. This is a call to the completed classical solver.

(d) Combine the coset answers and restore the witness, stabilizer lifts, and translation kernel by Lemma 3.6, as specified in Appendix D.3. **Return** emptiness or the resulting coset, with its witness multiplied by t.

Every recursive call first applies the base cases and then starts at R1 if neither applies. Reuse completed flags and decompositions while their subgroup and factor actions are unchanged. After a subgroup restriction, compute new flags as needed. A parent keeps its current data until its quotient call and ensuing fiber call have returned.

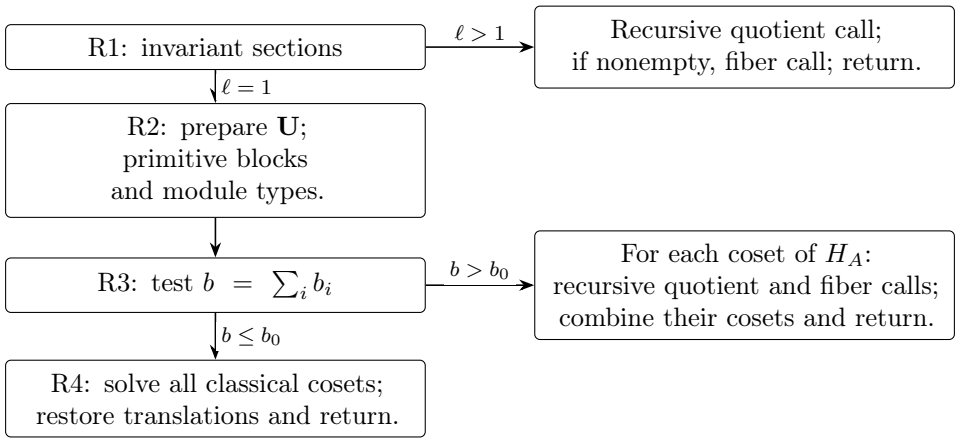


Figure 1: Control flow after the base cases. Every child checks the base cases before starting at R1. The right-hand nodes make recursive transporter calls. R2 changes the presentation without making a recursive call.

7.4 Running-time analysis

All bounds in this subsection use the dimension budget d and complete explicit input length InSize of the initial call; neither is reset during recursion. Fix the number r of original tensor positions, with $r = 2$ for the quadratic problem. At a current call, let $H \le \Gamma$ be the acting subgroup and $Q = N/P$ the quotient on which its affine equation is posed. For each position $i \in [r]$, let I_i be the tagged set of surviving block indices, together with any fixed symbol for an active factor or an

unused position. The changing parameters are

$$b = \sum_{i=1}^{r} |I_i|, \qquad v = \sum_{a} \dim_{\mathbb{F}_p} U_a, \qquad m = \dim_{\mathbb{F}_p} Q.$$

Distinct active factors are counted once in v. An index used in two tensor positions is counted twice in b; the fixed symbols contribute at most r. When $Q = 0$, the transporter is the entire current group, so the path bounds below concern $m > 0$. Lemmas D.6 and 7.7, proved after the constructions in the appendix, bound unchanged-parameter preparation and the cost of each operation. The following analysis uses those bounds and the recursive formulas; Appendix D verifies their hypotheses.

Bounds used by the recursion. Use the constants κ_2, C_* and threshold $T_{\mathrm{cl}} = p^{C_* d}$ fixed in Subsection 7.2. They are not reset in a child. The construction establishes the following properties.

(i) The distinct active spaces, their required dual copies, and the auxiliary modules have total prime-field dimension at most $\kappa_2 d$. Repeated tensor coordinates and stored maps contribute to the explicit input length.

(ii) On each selected summand of an auxiliary module, the image of its stabilizer has order $p^{O(d)}$. The joint image of the alternating-dominant actions has no nonalternating nonabelian simple section of order greater than T_{cl}. These bounds also hold after subgroup restriction.

(iii) The spaces L_ℓ of Subsection 7.2 satisfy $\sum_{\ell \in \mathcal{L}} e_\ell \le \kappa_2 d$. Their actions are included in the auxiliary image B, so the simple-section bound in (ii) includes them.

The dimension bound is proved in (D.11); the summand-stabilizer bounds follow from Appendix D.2.2, and the simple-section bound from Lemma 7.4. The quadratic input has no additional auxiliary modules. Later applications with a fixed number r of tensor positions use $i \in [r]$ in the definitions above and must establish these bounds at their root, with a constant κ_r. Subsection 7.6 specifies those additional inputs and proves the required local comparison theorem.

The following bound applies to every local operation, including the symbolic affine-group operations used in later sections. Its proof is given in Appendix D.4.

Lemma 7.7 (Local operations and additive kernels)**.** *Fix the number r of tensor positions. Under the dimension and encoding bounds above, all presentations and local operations in Algorithm 7.6 have a common bound*

$$p^{O_r(d)} \mathrm{poly}(\mathrm{InSize}, \log p), \tag{7.7}$$

measured from the initial call, provided every previously established local algorithm has this bound on original representation spaces of total dimension $O_r(d)$. The recursive calls retain subgroups of the original input group; their affine data are stored in the current quotient coordinates.

In particular, let G satisfy these original dimension and encoding bounds, let M be an explicit $\mathbb{F}_p G$-module, and let $E \le M^+ \rtimes G$ be given by generators as pairs, where M^+ is the additive group of M. Write $\pi : E \to G$ for projection, and include the additive coordinates and action matrices in InSize*. Let $S \le E$ be given by generators. Include in* InSize *the generator descriptions of E, S and any subgroup J below, and any linear correction equations. The following can be computed within* (7.7)*:*

(i) generators of $\pi(S)$ with lifts in S, and a basis of $S \cap M$ with words in the generators of S;

(ii) membership tests and generator compression in S, and generators of $S \cap \pi^{-1}(J)$ for a given subgroup $J \le \pi(S)$ given by generators;

(iii) solutions of linear correction equations in the computed additive kernels, with the corresponding group elements.

The translation space M is handled by linear algebra; its dimension enters the polynomial input factor.

7.4.1 The recursion-tree bound

A quotient and its fiber have dimensions adding to m. A summand division repeats such a pair over at most 2^b cosets, but decreases the summand count in both children. A conversion may increase b; it pays for that increase by decreasing v. For the fixed-position variant, converting total active dimension $c > 0$ in R2 gives $v' = v - c$ and $b' \le b + a_r c$ for a constant a_r, by (D.10). A split in R3 gives $b_{\text{child}} \le \gamma_r b$, where $\gamma_r = 1 - 1/(3r)$, by Lemma D.4. Choose an integer $\lambda > a_r$ and put $e = b + \lambda v$ and $\eta = 1 - \gamma_r$. Let $T_0 = p^{O_r(d)}\text{poly}(\text{InSize}, \log p)$ bound the work between calls that change these parameters, including the preparation work of Lemma D.6. Write $T(e, m)$ for the maximum remaining work over all valid recursive calls with the original parameters p, d, InSize, with $b + \lambda v \le e$ and current quotient dimension at most m. Zero-dimensional children are charged to T_0, so set $T(e, 0) = 0$. Taking the maximum over the admissible operations and parameters gives the recurrence bound

$$T(e,m) \le T_0 + \max \left\{ \begin{array}{ll} T(e,m_1) + T(e,m_2), & \text{R1}, \\ T(e-(\lambda - a_r)c, m), & \text{R2}, \\ 2^b \max\limits_{\substack{m_1,m_2 \ge 0 \\ m_1+m_2=m}} \big(T(e-\eta b, m_1) + T(e-\eta b, m_2)\big), & \text{R3}, \\ 0, & \text{R4}. \end{array} \right\}$$

In R1, $m_1, m_2 > 0$ and $m_1 + m_2 = m$; in R2, $c > 0$. Each line bounds the remaining work for the indicated operation; the arguments of T are upper bounds on the new parameters. The conversion line describes continuation after R2, which makes no recursive call. The split line applies above the fixed cutoff for b. For a local solver that finishes only a quotient of dimension $0 < h \le m$, replace the last line by $T(e, m - h)$. In a split, the two quotient dimensions can differ between cosets; the maximum bounds every case. The proof below uses the weight $p^{ke}m$: quotient and fiber weights add, while $2^b p^{-k\eta b} < 1$ controls a split for sufficiently large k. It also counts failed branches and the complete stabilizer outputs.

Step	Change to the recursive data	Progress or cost
R1	Recursively solve the top quotient transporter, then the exact fiber transporter.	$\dim(Q/Q_{\ell-1}) + \dim Q_{\ell-1} = m$; no coset branching.
R2	Compute primitive blocks and module types; replace remaining factors by auxiliary presentations.	$v' \le v$, $b' \le b + a_2(v - v')$; same equation.
R3	Restrict the group to a subset stabilizer; restrict $\mathbf{S}$ and the tensor layout to each side.	At most 2^b coset cases, each with two successive calls; $b' \le \gamma_2 b$ in each child.
R4	Quotient out kernel translations, solve the classical cosets, and restore all lifts.	Complete answer in $p^{O(d)}\text{poly}(\text{InSize}, \log p)$ time.

Table 1: Data changes and progress in the quadratic recursion. Here $m = \dim(N/P)$, v is the active-factor dimension, and b counts surviving summand indices.

The conditional interface for later sections. The ordinary quadratic algorithm has two positions and its local step finishes the equation. The same count applies with a fixed number r of positions if the initial extra modules satisfy the bounds above and a local algorithm meeting the complete-output and cost guarantees below is specified. Such a local algorithm may instead finish a positive-dimensional quotient, reunite all its coset answers, and leave one fiber. These are explicit hypotheses below; no unrestricted degree-r transporter is being asserted.

Proposition 7.8 (The complete recursion-tree bound)**.** *Fix $r \geq 2$. Consider Algorithm 7.6 with at most r fixed tensor positions in each summand and the exact working data above. The active original and alternating-dominant representations have total dimension $O_r(d)$. Any auxiliary alternating-dominant modules included in the initial input have the permuted-summand data and bounds specified above; the quadratic input has none and uses $r = 2$. Suppose the local algorithm in Step R4 either solves its affine equation completely, or solves a positive-dimensional quotient equation with its full stabilizer, combines all quotient coset cases, and returns exactly one remaining fiber in the original presentation. Suppose its work apart from that fiber is $p^{O_r(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$, using only previously established complete algorithms. Then Algorithm 7.6 computes the complete original transporter in $p^{O_r(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$ time, including failed cases, stabilizers, and lifts. Here* InSize *is the complete explicit input length.*

Proof of Proposition 7.8. Correctness of a return. Induct on the completed calls, each interpreted by (7.4). A quotient child recursively computes the complete transporter on its quotient space. Once that coset has returned, Lemma 3.5 identifies the full parent equation with the single remaining affine fiber under its full stabilizer. The auxiliary quotient presentations of Fact 3.4, the invariant flag construction of Lemma D.2, and the selected-coordinate presentation of Lemma D.3 give the exact coordinate maps for these recursive calls and their returns. The child translations are quotient vectors obtained by these maps or by the fiber formula; their cocycle identities follow from the parent affine action. Thus each child has the data specified in Definition 18, with its group contained in the original group.

Every coset case searches its specified original candidate coset and is completed before reunion by the identity (3.1). The hypothesis on the local algorithm gives the same complete quotient-then-fiber return when it does not finish outright. Thus the completed return, including every stabilizer generator, gives exactly the parent's transporter.

A weight for recursive progress. Choose fixed integers $\lambda > a_r$ and then k_0 sufficiently large. For each call with a positive-dimensional quotient, put

$$\Phi = p^{k_0(b+\lambda v)}m, \qquad b, v \in \mathbb{Z}_{\geq 0}, \quad m = \dim_{\mathbb{F}_p} Q > 0. \tag{7.8}$$

The exponent pays for the selected summands and for future conversions of the current active factors. The factor m pays for successive recursive calls on quotients and their exact fibers. A conversion of dimension $c > 0$ satisfies

$$(b' + \lambda v') - (b + \lambda v) \leq -(\lambda - a_r)c < 0$$

by (D.10). It therefore decreases the weight even though it may add summand indices.

For an invariant filtration or an image–fiber decomposition, the positive quotient dimensions add to the original m, while the inherited exponents do not increase. The sum of their weights is therefore at most Φ. A local solver that computes a complete transporter on a quotient of dimension $h > 0$ leaves one fiber of dimension $m - h$; when positive, that fiber has strictly smaller weight.

For a summand division, Lemma D.4 gives at most 2^b coset cases and at most $\gamma_r b$ indices in each child, before any new conversion. In one coset case the two positive quotient dimensions add

to at most m; if the quotient call returns emptiness, there is no fiber call. Consequently the sum of all child weights over all cosets is at most

$$2^b p^{-k_0(1-\gamma_r)b}\Phi \le \tfrac{1}{2}\Phi$$

above a fixed cutoff. Choose $k_0(1-\gamma_r) > 1$ and enlarge the cutoff if necessary, using $2^b \le p^b$. Any later conversion decreases its own child's weight. Evaluating on all vectors of the same selected summands leaves their indices unchanged, so it cannot undo the decrease. The fixed symbols account for labelled tensor summands with an active factor or no factor in the divided position.

Counting the entire tree. Contract maximal unary chains on which the weight is unchanged. Along every path, both $b+\lambda v$ and m are nonincreasing; conversions and divisions decrease the former, and a proper quotient fiber decreases the latter. Thus unchanged-weight chains have unchanged b, v, m, and Lemma D.6 bounds their lengths by P_0. The contracted tree has positive integer weights, the sum of child weights at most the parent weight, and a strict decrease on every unary edge.

Assign Φ_0 distinct units to the root, and partition them among children in accordance with the child weights. Charge each strict unary vertex to a unit not passed to its child; assign a unit to every leaf. These charges use distinct units, so

$$\#\{\text{leaves}\} + \#\{\text{strict unary vertices}\} \le \Phi_0.$$

After deleting unary vertices, there are at most one fewer branching vertices than leaves. Restoring the contracted chains gives $O((P_0+1)\Phi_0)$ vertices in the complete recursion tree. Initially $b+v = O_r(d)$, and m is bounded by the explicit input size. The total is therefore $p^{O_r(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$.

Failed positive children are leaves and are counted by their assigned weights. Zero-dimensional children are direct checks; their construction and the complete lists of at most 2^b coset cases belong to the local bound. The estimate therefore includes all-failing divisions. Reunion includes all branch stabilizers and witness ratios, compressed in the original representation. Translation restoration corrects both the witness and every projected stabilizer generator. These are among the operations bounded in Lemma 7.7.

Total cost and depth. Multiply the node count by the common local bound once. A previously completed algorithm contributes its own fixed single-exponential bound at a node; the dependency order used below has fixed length. The resulting total remains $p^{O_r(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$.

Each strict-weight edge decreases the nonnegative integer $b+\lambda v$ or the positive integer m, and each intervening unchanged run has length at most P_0. Hence the depth is polynomial in the initial explicit input length. Keeping one symbolic affine presentation for each open call uses polynomial space per path, apart from the already counted explicit evaluation data. □

7.5 A subroutine: classical group degree-2 quotient tensor module transporter

Definition 19 (Explicit central product of classical groups)**.** Fix a prime p. Let $G_1, \dots, G_t$ be groups of type SL, SU, Sp, or Ω, with each G_i in its natural matrix representation on a based space $U_i = \mathbb{K}_i^{n_i}$, using the conventions of Section 6.1. An *explicit central product of classical groups* in characteristic p is the group

$$H := \widetilde{H}/C, \qquad \widetilde{H} := \prod_{i=1}^{t} G_i, \qquad C \le \prod_{i=1}^{t} Z(G_i),$$

where the input specifies matrices $c_{\ell i} \in \mathrm{Z}(G_i)$ for $1 \le \ell \le r$ and $1 \le i \le t$, with $C = \langle (c_{\ell 1}, \dots, c_{\ell t}) : 1 \le \ell \le r \rangle$. The input includes coordinate representations of the finite fields $\mathbb{K}_i$ of characteristic p,

matrix generators of the G_i, the defining forms, and the involutions in the unitary cases. Elements of H are represented by tuples of factor elements modulo C, and the images of the factor generators generate H. Write $\pi_{\mathrm{cov}} : \widetilde{H} \twoheadrightarrow H$ for the canonical quotient map.

Our convention also allows central quotients of individual factors. The factors themselves embed in H when $C \cap G_i = 1$ for every i, identifying G_i with its copy in $\widetilde{H}$.

Natural, dual, and twisted factor actions refer to the covering factors G_i, with all changes of coordinates and twist data supplied. An input action of H is pulled back along π_{cov}, so C acts trivially on it. The individual natural modules, and auxiliary modules constructed on the cover, need not descend to H.

Proposition 7.2 (restated). *There is a deterministic algorithm in time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$ for the degree-2 quotient tensor module transporter problem of Definition 17 when H is an explicit central product of classical groups in the prescribed stable range, with natural, dual, or Frobenius-twisted factor actions.*

For a central-product input $H = \widetilde{H}/C$, first pull the action and its cocycle back along the canonical quotient map $\pi_{\mathrm{cov}} : \widetilde{H} \twoheadrightarrow H$. Write $D := \widetilde{H}$ for the working product cover, retaining $H = D/C$ for the input group. In the argument below, α is the pulled-back affine action of D. All auxiliary modules are constructed on the cover and need not descend to H. The covering kernel acts trivially on the input quotient and has zero pulled-back cocycle, so the final witness and stabilizer generators project through π_{cov} to the complete transporter in H, as justified in Section D.5.5.

The algorithm for Proposition 7.2 is *not* of a recursive nature. The main technical part of this algorithm is to identify a nice model of the quotient space Q.

Classical representation reduction. Write $\alpha_h(v) = L_h(v) + c_h$ for the affine action of D on $Q = N/P$. Two difficulties arise: the quotient need not admit an equivariant choice of representatives in N, and the translation cocycle need not be removable by a change of origin in Q.

Example 3. Let $D = \mathrm{GL}_2(\mathbb{F}_2) \cong S_3$ act by conjugation on $N = \mathrm{End}(\mathbb{F}_2^2)$, and set $P = \ker(\mathrm{tr})$ and $Q = N/P \cong \mathbb{F}_2$. The induced action on Q is trivial. There is no equivariant section $Q \to N$: a representative of $1 \in Q$ fixed by conjugation would have to be scalar, whereas every scalar 2×2 matrix over $\mathbb{F}_2$ has trace zero.

Let $\epsilon : D \to (\mathbb{F}_2, +)$ be the parity homomorphism, and give Q the affine action $\alpha_g(q) = q + \epsilon(g)$. Since the linear action on Q is trivial, every coboundary is zero. Thus this nonzero translation cocycle cannot be removed by a change of origin in Q.

We construct equivariant linear maps

$$Q \overset{\iota}{\hookrightarrow} I \overset{\pi}{\to} E.$$

The embedding ι realizes the arbitrary subquotient $Q = N/P$ inside the structured classical-group module models provided by Theorem 6.1, proved in Section C.5.1. Its prescribed quotient presentations admit the subsequent matrix/form comparisons. This construction does not require an equivariant choice of representatives in N. The map π is chosen so that the projected cocycle $h \mapsto \pi\iota(c_h)$ can be removed by a change of origin in E, while $\ker \pi$ has a degree-1 presentation. Both $\iota(c_h)$ and $\pi\iota(c_h)$ are obtained by applying equivariant linear maps to the cocycle in Q, so they satisfy the 1-cocycle identity in their respective modules. This reduces the original problem to a linear transporter in E, followed by a degree-1 quotient tensor module transporter in the remaining fiber under the returned stabiliser.

Example 4 (Example 3 continued)**.** Take $I = E = \mathrm{End}(\mathbb{F}_2^2)$ with the conjugation action, $\pi = \mathrm{id}$, and

$$\iota(q) = q\mathrm{Id}_2, \qquad b = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

The map ι is injective and equivariant, although it is not a section of $N \to Q$. Even permutations centralize b, while odd permutations conjugate it to $b^{-1} = b + \mathrm{Id}_2$. Hence

$$\rho_E(g)b - b = gbg^{-1} - b = \epsilon(g)\mathrm{Id}_2 = \pi\iota(c_g).$$

Therefore $\eta(q) = \pi\iota(q) + b = q\mathrm{Id}_2 + b$ satisfies $\eta(\alpha_g(q)) = g\eta(q)g^{-1}$. The original affine comparison is consequently equivalent to the linear matrix comparison

$$\alpha_g(x) = y \quad \Longleftrightarrow \quad g(x\mathrm{Id}_2 + b)g^{-1} = y\mathrm{Id}_2 + b.$$

Thus the embedding avoids the need for equivariant representatives, and the translation becomes removable in the larger module. Here $\ker \pi = 0$, so no residual degree-1 comparison is needed.

An algorithm outline. The algorithm has three main ingredients: an explicit matrix model, a simultaneous matrix tuple algorithm, and a degree-1 affine transporter. The following steps describe the data passed between them.

(1) Construct an explicit matrix model. From $Q = N/P$ and its affine action $\alpha_h(v) = L_h(v) + c_h$, construct

$$Q \overset{\iota}{\hookrightarrow} I \overset{\pi}{\to} E, \qquad U = \ker \pi, \qquad H^1(D, E) = 0,$$

where E has explicit matrix and form coordinates and U has a degree-1 presentation. Thus the model separates a matrix comparison from a possible degree-1 remainder (see Theorem 6.1 and Section C.5.1). Let ρ_I, ρ_E denote the induced linear actions. To remove the translations in E, solve $\pi\iota(c_g) = (\rho_E(g) - 1)b$ for $b \in E$ on the generators $g \in \mathcal{S}$; the displayed cohomology vanishing guarantees a solution. Set $\eta(v) = \pi\iota(v) + b$ and put $X = \eta(x)$, $Y = \eta(y)$. Then $\eta(\alpha_h(v)) = \rho_E(h)\eta(v)$, so every desired transporter must carry X to Y (see Section D.5.2).

(2) Solve the projected problem by matrix tuple algorithms. Compute the full linear transporter $T_E = \{h \in D : \rho_E(h)X = Y\}$ in the explicit coordinates of E. For a simple instance, let $V = \mathbb{K}^n$ and $D = \mathrm{Sp}(V, J) = \{g \in \mathrm{GL}(V) : gJg^{\mathrm{t}} = J\}$, where J is a nonsingular alternating matrix. Suppose

$$E = \mathrm{Mat}_n(\mathbb{K})^m / C_\Lambda, \qquad C_\Lambda = \{(\lambda_1 J, \ldots, \lambda_m J) : \lambda \in \Lambda\}, \qquad \Lambda \leq_{\mathbb{F}_p} \mathbb{K}^m,$$

with D acting by simultaneous congruence. Writing $X = [(A_1, \ldots, A_m)]$ and $Y = [(B_1, \ldots, B_m)]$, the projected comparison asks for $\lambda \in \Lambda$ and $g \in \mathrm{GL}(V)$ such that

$$gA_jg^{\mathrm{t}} = B_j + \lambda_j J \quad (1 \leq j \leq m), \qquad gJg^{\mathrm{t}} = J.$$

For each candidate λ, this is exactly simultaneous congruence of the augmented tuples $(A_1, \ldots, A_m, J)$ and $(B_1 + \lambda_1 J, \ldots, B_m + \lambda_m J, J)$. The form equation converts the other equations into $g(A_jJ^{-1}) - (B_jJ^{-1})g = \lambda_j g$. The scalar-shift routine therefore enumerates the possible joint shifts; we retain those in Λ before solving the corresponding matrix tuple problems (see Lemma 6.7).

The general target also involves mixed matrix and form equations, field constraints, and twists; these are handled simultaneously by the matrix-algebra algorithms in Section D.5.3. In characteristic 2, matching polar forms is followed by a degree-1 correction of the zero-polar differences, together with the remaining additive scalar and trace conditions. We then impose the determinant, spinor-norm, or Dickson-invariant conditions defining D and reunite all surviving branches (see Lemma 6.8 and Section D.5.4). If T_E is empty, the original transporter is empty. Otherwise this step returns a witness a and the full projected source stabilizer D_0, giving

$$T_E = aH_0, \qquad D_0 = \{h \in D : \rho_E(h)X = X\}.$$

(3) Solve the remaining degree-1 transporter. Only the elements of aH_0 can solve the original equation. Undoing a makes the source and target agree in E, so the remaining difference, embedded by ι, lies in $U = \ker \pi$. Translate the source to zero and define

$$w = \iota(\alpha_{a^{-1}}(y) - x), \qquad \delta_h = \iota(\alpha_h(x) - x) \quad (h \in D_0).$$

Both vectors lie in U, and $\beta_h(u) = \rho_I(h)u + \delta_h$ defines an affine action of D_0 on U. Thus $\alpha_{ah}(x) = y$ if and only if $\beta_h(0) = w$ (see Lemma 6.6). The cocycle is evaluated on the returned elements only after the matrix step has restricted them to D.

The degree-1 presentation from the first step is

$$U \cong N_1/P_1, \qquad P_1 \le N_1 \le \bigoplus_\lambda U_\lambda \otimes B_\lambda,$$

where each U_λ is a natural module, its dual, a specified Frobenius twist, or a trivial module, and D acts trivially on each B_λ. The distinct underlying natural spaces have total prime-field dimension $O(d)$. Choosing bases in the fixed multiplicity spaces induces an invariant flag with factors of dimension $O(d)$. We solve the affine transporter from 0 to w successively along this flag, using the original faithful representation of D_0; this degree-1 algorithm accepts arbitrary subgroups D_0 (see Proposition 4.5 and Section D.5.2). If the remaining transporter is empty, so is the original one. If it returns $h_0 S_x$, restore the witness a to obtain

$$\mathrm{Trans}_{D,\alpha}(x, y) = a\,\mathrm{Trans}_{D_0,\beta}(0, w) = ah_0 S_x,$$

where S_x is the full stabilizer of x under α. This final fiber imposes the original affine equation; the earlier degree-1 correction in the matrix step imposes the quadratic-form equations.

In odd characteristic, $\pi = \mathrm{id}_I$, so $U = 0$ and the matrix step already gives the complete answer. For a central-product input, project the final witness and stabilizer generators through the specified cover map, as described above.

7.6 A generalisation: controlled degree-2 transport

The degree-2 quotient tensor module transporter allows at most two moving factors in each summand. We now allow additional factors whose actions are alternating-dominant in the sense of Subsection 5.3, with bounded actions on their individual summands. These factors are viewed as evaluation arguments. The original algorithm already constructs such auxiliary modules during structural reduction; here they may be part of the input. The extra work is to verify that the same recursion handles these modules and a fixed number of additional tensor positions. The terminal solver is unchanged. The detailed constructions and proofs are in Appendix D.6.

Use $p, H, \mathcal{S}, (V_i), d$ and the original-group encoding from Definition 17. Throughout this subsection, $r \geq 2$ and $\kappa_r, \beta_r > 0$ are fixed constants, independent of d and the input size. Any fixed r is allowed; constants in $O(\cdot)$ and polynomial bounds may depend on these parameters. The quadratic calls arising in Algorithm 8.6 from (8.1) fit the fixed budget $r = 8$. Lemma E.2 makes the value space M a trivial multiplicity space throughout these calls. The presentation in Lemma E.3 uses at most three positions, and the pairing-parameter presentation in Lemma E.7 uses one active factor and one semilinear evaluation line. Its subquotients retain that budget in Lemmas E.8 and E.9. The remaining constructions in Subsubsection E.1.7 and Subsection E.2 use bilinear observations, fixed-field-value comparisons, and the subspace comparisons of Corollary 7.10. The latter have the two active positions V, E^* and trivial multiplicities in these applications; see Lemmas E.15, E.20, E.29, and E.31. Fixed marks and value-basis coordinates index direct summands rather than additional tensor positions. Choose the structural threshold $T_{\mathrm{cl}} = p^{C_* d}$ as in Section 5, with $C_* = C_*(r) \geq \kappa_r$ large enough for the enlarged $O(d)$ representation budget, and keep it fixed throughout the recursion. All modules and tensor products below are over $\mathbb{F}_p$.

Definition 20 (Semilinear line)**.** Let $\mathbb{K} = \mathbb{F}_{p^e}$. A *semilinear line* for H over $\mathbb{K}$ is the one-dimensional $\mathbb{K}$-space $L = \mathbb{K}$ with an H-action of the form

$$\tau(h)(a) = \lambda(h) a^{p^{j(h)}} \qquad (a \in \mathbb{K}), \qquad \lambda(h) \in \mathbb{K}^\times, \quad j(h) \in \mathbb{Z}/e\mathbb{Z}.$$

The action is $\mathbb{F}_p$-linear, so L is an $\mathbb{F}_p H$-module of dimension e. It is specified by the field $\mathbb{K}$ and the pairs $(\lambda(h), j(h))$ for $h \in \mathcal{S}$.

Definition 21 (Controlled catalogue)**.** A *controlled catalogue* consists of the following data and conditions.

- A finite family of H-modules $\mathcal{R} = (R_t)_{t \in \mathcal{T}}$, with actions η_t and decompositions $R_t = \bigoplus_{\alpha \in I_t} R_{t,\alpha}$ into nonzero summands permuted by H. The generator actions and summand permutations are given. Write $B_{t,\alpha} = \eta_t(\mathrm{Stab}_H(\alpha))|_{R_{t,\alpha}}$.
- A finite, possibly empty, family $(L_\ell)_{\ell \in \mathcal{L}}$ of semilinear lines $L_\ell = \mathbb{K}_\ell = \mathbb{F}_{p^{e_\ell}}$, with actions τ_ℓ as in Definition 20.
- The dimension and summand-stabilizer bounds

$$\sum_{t \in \mathcal{T}} \dim R_t + \sum_{\ell \in \mathcal{L}} e_\ell \leq \kappa_r d, \qquad |B_{t,\alpha}| \leq p^{\beta_r d}.$$

- The *joint auxiliary image* $B = \beta(H)$ is *alternating-dominant*: every nonalternating simple section of B has order at most T_{cl}, where

$$\beta = \bigoplus_t \eta_t \oplus \bigoplus_\ell \tau_\ell : H \longrightarrow \mathrm{GL}\left(\bigoplus_t R_t \oplus \bigoplus_\ell L_\ell \right).$$

  Here $\beta(h)$ records the action of the same h on every auxiliary module, including the induced permutations of the nonzero summands.

Thus control consists of alternating-dominance together with a dimension budget and bounds on the summand-stabilizer images. It does not require $|B| \leq p^{O(d)}$. For example, the coordinate permutation action on $\mathbb{F}_p^t$, with $t = O(d)$, has trivial coordinate-stabilizer images, but its full image can have order $t!$. Its nonalternating simple sections have order at most 3^t by the argument in Appendix B.4, so it is controlled after the root threshold is chosen sufficiently large.

Definition 22 (Controlled subquotient presentation)**.** Let $\mathcal{R} = (R_t)_{t\in\mathcal{T}}$ be a controlled catalogue as in Definition 21, with its specified semilinear lines $(L_\ell)_{\ell\in\mathcal{L}}$. A *controlled subquotient presentation* of a finite-dimensional $\mathbb{F}_pH$-module M consists of a given H-equivariant isomorphism

$$\theta_0 : U_0/K_0 \xrightarrow{\sim} M, \qquad K_0 \le U_0 \le C_0 = \bigoplus_{\tau\in\Omega} \bigotimes_{j=1}^{a_\tau} D_{\tau,j}.$$

Here K_0, U_0 are H-invariant, Ω is a finite set fixed by H, and the a_τ are nonnegative integers. Each $D_{\tau,j}$ is a specified R_t, R_t^*, L_ℓ, or L_ℓ^* from this catalogue. All tensor products and duals are over $\mathbb{F}_p$, and an empty tensor product is the trivial module $\mathbb{F}_p$. The presentation includes the subspace bases, the generator actions, and the map θ_0 in the given coordinates.

Definition 23 (Degree-2 quotient tensor module transporter with evaluation arguments)**.** Take the data and task of Definition 17, together with a controlled catalogue, and replace its ambient module in item (iii) by

$$Z = \bigoplus_{\sigma\in\Sigma} \mathrm{Hom}_{\mathbb{F}_p}\left(\bigotimes_{j=1}^{a_\sigma} E_{\sigma,j}, \bigotimes_{k=1}^{b_\sigma} W_{\sigma,k}\right), \qquad a_\sigma + b_\sigma \le r, \quad b_\sigma \le 2.$$

Here Σ is fixed by H, each $E_{\sigma,j}$ is a module R_t or a semilinear line L_ℓ from the catalogue, or its dual, and each $W_{\sigma,k}$ is an original module V_i, its dual, or a Frobenius twist of either, as before. The nonnegative integers a_σ, b_σ count argument and active factors, respectively. Factor occurrences in each summand have distinct position labels in $[r]$ that stay fixed throughout the recursion. The action on Z is the induced Hom action.

The quotient $Q = N/P$, affine action α, points x, y, and complete original-group output are unchanged. The input length InSize includes all additional explicit data. Each catalogue module and semilinear line is charged once to the dimension budget; repeated occurrences and fixed multiplicities contribute only to InSize.

With $a_\sigma = 0$ and an empty catalogue this is the original degree-2 problem. An argument contributes the tensor factor $E_{\sigma,j}^*$, so the displayed module can have degree greater than two. The restriction on its action, rather than the Hom notation alone, permits the following extension.

Theorem 7.9 (Degree-2 transport with controlled evaluation arguments)**.** *The complete transporter in Definition 23 can be computed deterministically in time* $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$.

A weaker alternative to the alternating-dominance hypothesis is given in Remark 3 with the local proof.

Algorithm overview. First extend the flag and summand-index recursion of Section 7 from two positions to r positions, including the input catalogue modules. Since r is fixed and the shared catalogue has total dimension $O(d)$, the same recursion count applies (Proposition 7.11).

At a local call only $O(1)$ auxiliary summands remain selected, so their joint image has order $p^{O(d)}$. Alternating-dominance ensures that each transformation in the selected classical product D can be realised while fixing these auxiliary spaces pointwise (Lemma 5.6). After a search through $p^{O(d)}$ cosets (Lemma 7.5), the remaining comparison is under D. For example, if E is a tensor product of evaluation arguments, then

$$\mathrm{Hom}_{\mathbb{F}_p}(E, V\otimes W)\big|_D \cong (V\otimes W)^{\oplus \dim E},$$

by evaluation on a basis of E. These are fixed copies of a degree-2 module, so the existing classical solver (Proposition 7.2) applies with the exact numerator, denominator, and affine action. Combining the complete coset answers recovers the original transporter.

Subspace transport. Definition 23 already gives the expanded ambient module and quotient needed by the algorithm. The following application starts with another H-module M; its given realisation over a controlled catalogue converts the application into that form.

Corollary 7.10 (Transporting subspaces using injective linear maps)**.** *Let V be an original H-module and let M be an H-module with a given controlled subquotient presentation as in Definition 22, satisfying $a_\tau \le r-2$ for every $\tau \in \Omega$. Given injective linear maps $f, g : E \hookrightarrow V \otimes M$, with $\dim E = O(d)$, the full H-transporter from $\operatorname{im} f$ to $\operatorname{im} g$ is computable in time $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$, in the original-group encoding.*

Trivial modules M are included by taking $a_\tau = 0$. More generally, the catalogue factors realising M become evaluation arguments, via $T \otimes D \cong \mathrm{Hom}_{\mathbb{F}_p}(D^*, T)$, and the relations defining M enter the numerator and denominator. The two active positions are V and E^*, leaving at most $r-2$ catalogue factors (six when $r = 8$). The exact quotient changes are given in Appendix D.6.

The later higher-degree algorithms also use the recursion independently of the degree-2 local solver. We record that interface separately.

Proposition 7.11 (The recursive template with a fixed number of positions)**.** *Run Algorithm 7.6 on the data of Definition 23, allowing any $b_\sigma \ge 0$ with $a_\sigma + b_\sigma \le r$, using its exact recursive data with position set $[r]$. Suppose the local step either computes the complete affine transporter, or computes a positive-dimensional quotient transporter with its full stabilizer, combines every quotient coset answer, and returns exactly one remaining affine fiber in the original presentation. If a previously established local algorithm does this in $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$ time apart from that fiber, the full recursion computes the complete original transporter within the same bound.*

This is conditional on the stated local algorithm. The preceding theorem gives one when $b_\sigma \le 2$; the proposition alone does not assert an unrestricted degree-r transporter algorithm.

8 3-tensor isomorphism

8.1 Problem and main results

Definition 1 (restated). *Let q be a prime power, let $U_i = \mathbb{F}_q^{n_i}$ for $i \in \{1,2,3\}$, and let $\mathtt{T}, \mathtt{T}' \in U_1 \otimes_{\mathbb{F}_q} U_2 \otimes_{\mathbb{F}_q} U_3$. The tensors $\mathtt{T}$ and $\mathtt{T}'$ are isomorphic if there exist $g_i \in \mathrm{GL}(U_i)$ such that*

$$(g_1 \otimes g_2 \otimes g_3)\mathtt{T} = \mathtt{T}'.$$

We prove the following strengthening of Theorem 1.2.

Theorem 8.1 (Complete 3-tensor isomorphism)**.** *Given $\mathtt{T}, \mathtt{T}' \in U_1 \otimes_{\mathbb{F}_q} U_2 \otimes_{\mathbb{F}_q} U_3$ and generators of $\Gamma_{\mathrm{in}} \le \prod_{i=1}^3 \mathrm{GL}(U_i)$, there is a deterministic algorithm computing*

$$\mathrm{Trans}_{\Gamma_{\mathrm{in}}}(\mathtt{T}, \mathtt{T}') := \{(g_1, g_2, g_3) \in \Gamma_{\mathrm{in}} : (g_1 \otimes g_2 \otimes g_3)\mathtt{T} = \mathtt{T}'\}$$

in time $q^{O(n_1+n_2+n_3)}\mathrm{poly}(\mathrm{InSize})$, where InSize *is the total input bit length. The output is emptiness, or a representative u and generators of $\mathrm{Stab}_{\Gamma_{\mathrm{in}}}(\mathtt{T}) := \mathrm{Trans}_{\Gamma_{\mathrm{in}}}(\mathtt{T}, \mathtt{T})$, representing the set $u\,\mathrm{Stab}_{\Gamma_{\mathrm{in}}}(\mathtt{T})$.*

8.2 The recursive 3-tensor isomorphism algorithm and its analysis

The recursive input and output. We follow the group encoding and input/output conventions of Section 3.5. Let $q = p^e$, where p is prime and $e \geq 1$, and let $U_i = \mathbb{F}_q^{n_i}$ for $i \in \{1, 2, 3\}$. Fix the ambient group $\Gamma := \prod_{i=1}^{3} \mathrm{GL}(U_i)$, where each $\mathrm{GL}(U_i)$ consists of the $\mathbb{F}_q$-linear automorphisms of U_i. We regard $U_i^* = \mathrm{Hom}_{\mathbb{F}_q}(U_i, \mathbb{F}_q)$ as an $\mathbb{F}_p$-space with the contragredient action of Γ as in Definition 6. Unless specified otherwise, subsequent tensor products and duals are over $\mathbb{F}_p$.

Definition 24 (Inputs to the recursive tensor procedure)**.** The input consists of the following data.

(i) A subgroup $H \leq \Gamma$ and an element $g \in \Gamma$.

(ii) For each $i \in \{1, 2, 3\}$, an H-invariant flag in U_i^* containing $P_i \leq N_i$, and the quotient map

$$\pi_i : N_i \twoheadrightarrow A_i = N_i/P_i.$$

Write $\mathbf{A} = ((N_i, P_i))_{i=1}^3$ for these quotient presentations, together with their flags and maps.

(iii) A $\mathbb{F}_p H$-module

$$Z = Z_{\mathrm{quad}} \oplus Z_{\mathrm{tri}},$$
$$Z_{\mathrm{tri}} = A_1^* \otimes A_2^* \otimes A_3^* \otimes M \cong \mathrm{Hom}(A_1 \otimes A_2 \otimes A_3, M).$$

Here Z_{tri} is the space of M-valued trilinear maps on $A_1 \times A_2 \times A_3$, and Z_{quad} is a possibly zero H-module with a presentation as in Definition 23. Write ρ for the induced H-action on Z as in Definition 8.

M is a finite-dimensional $\mathbb{F}_p H$-module with a given controlled subquotient presentation as in Definition 22, using the same controlled catalogue as Z_{quad}.

(iv) Two H-invariant subspaces $P \leq N \leq Z$ and an element $z \in Z$, defining

$$\mathcal{A} := (z + P) + N/P \subseteq Z/P.$$

We also use ρ for the induced action on Z/P.

(v) A family of translation vectors $c = (c_h)_{h \in H}$, with $c_h \in Z/P$, such that the maps

$$\alpha_h(v) := \rho_h(v) + c_h \text{ for } v \in \mathcal{A}$$

define a group action $\alpha : H \times \mathcal{A} \to \mathcal{A}$.

(vi) Two points $x, y \in \mathcal{A}$.

The task is to compute

$$\mathtt{C}(gH; \mathbf{A}; Z, \rho; P, N, z; c; x, y) := g\,\mathrm{Trans}_{H,\alpha}(x, y) = g\{h \in H : \alpha_h(x) = y\}.$$

The output is emptiness, or a representative gu and generators of $K = \mathrm{Stab}_{H,\alpha}(x)$, representing the set guK.

The initial comparison. We first express the original 3-tensor comparison as an input to Definition 24. The tensors are over $\mathbb{F}_q$, whereas the recursive input uses $\mathbb{F}_p$-spaces. For each input tensor $\mathtt{T}$, define

$$\widehat{\mathtt{T}}(a_1, a_2, a_3) := \mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p}\big((a_1 \otimes_{\mathbb{F}_q} a_2 \otimes_{\mathbb{F}_q} a_3)(\mathtt{T})\big), \quad a_i \in U_i^*,$$

where $\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(t) = \sum_{j=0}^{e-1} t^{p^j}$ is the field trace. Thus $\widehat{\mathtt{T}}$ is an $\mathbb{F}_p$-valued trilinear map on the three $\mathbb{F}_p$-spaces U_i^*. The values $\widehat{\mathtt{T}}(\lambda a_1, a_2, a_3)$ for $\lambda \in \mathbb{F}_q$ recover the corresponding $\mathbb{F}_q$-valued evaluation by the nondegenerate trace pairing. Hence $\mathtt{T} \mapsto \widehat{\mathtt{T}}$ is injective. It also commutes with the action of $\prod_{i=1}^3 \mathrm{GL}(U_i)$, so comparing $\widehat{\mathtt{T}}$ and $\widehat{\mathtt{T}'}$ preserves exactly the transformations between $\mathtt{T}$ and $\mathtt{T}'$.

Take $N_i = U_i^*$ and $P_i = 0$, with trivial flags and identity quotient maps, so $A_i = U_i^*$. Take $M = \mathbb{F}_p$ with trivial action and $Z_{\mathrm{quad}} = 0$. Then $Z = Z_{\mathrm{tri}}$ with its induced action ρ, and the initial call is

$$\mathtt{C}\big(H; (U_i^*, 0)_{i=1}^3; Z_{\mathrm{tri}}, \rho; 0, Z_{\mathrm{tri}}, 0; 0; \widehat{\mathtt{T}}, \widehat{\mathtt{T}'}\big). \tag{8.1}$$

Here $g = 1$, $H = \Gamma$, $P = 0$, $N = Z$, $z = 0$, and $c_h = 0$, so $\mathcal{A} = Z$ and the action is linear. For the prescribed subgroup Γ_{in} of Theorem 8.1, first compute the complete solution coset in Γ, then intersect it with Γ_{in} using Theorem 7.3.

Splitting along invariant subspaces. We next explain how matching part of a tensor changes the remaining comparison. Consider $\rho_h(\mathtt{T}) = \mathtt{T}'$ with $\mathtt{T}, \mathtt{T}' \in Z_{\mathrm{tri}}$ and $h \in H$. Suppose that $0 < W < A_1$ is H-invariant, and let

$$r : Z_{\mathrm{tri}} \longrightarrow \mathrm{Hom}(W \otimes A_2 \otimes A_3, M)$$

be restriction to $W \times A_2 \times A_3$. Explicitly, for $\mathtt{F} \in Z_{\mathrm{tri}}$, define

$$[r(\mathtt{F})](a_1, a_2, a_3) := \mathtt{F}(a_1, a_2, a_3) \quad (a_1 \in W,\ a_2 \in A_2,\ a_3 \in A_3).$$

Since W is invariant, restriction commutes with the H-action. We first compare $r(\mathtt{T})$ and $r(\mathtt{T}')$. If their transporter is empty, the original comparison is empty. Otherwise, write the returned transporter as uK, where $K = \mathrm{Stab}_H(r(\mathtt{T}))$. Every transformation that can match the full tensors must therefore have the form uk with $k \in K$.

Applying u^{-1} to the target makes its restriction equal to that of $\mathtt{T}$. Consequently,

$$w := \rho_{u^{-1}}(\mathtt{T}') - \mathtt{T}, \quad c_k := \rho_k(\mathtt{T}) - \mathtt{T} \text{ for } k \in K$$

both vanish on $W \times A_2 \times A_3$. They belong to

$$\ker r \cong \mathrm{Hom}((A_1/W) \otimes A_2 \otimes A_3, M),$$

and the candidate uk matches the full tensors exactly when $c_k = w$. On this space, the action on differences from $\mathtt{T}$ is

$$\beta_k(v) := \rho_k(\mathtt{T} + v) - \mathtt{T} = \rho_k(v) + c_k \text{ for } k \in K,\ v \in \ker r.$$

These maps form an affine action of K on $\ker r$. Thus the remaining comparison has source 0, target w, and action β, and

$$\mathrm{Trans}_{H,\rho}(\mathtt{T}, \mathtt{T}') = u\,\mathrm{Trans}_{K,\beta}(0, w).$$

The first comparison uses W as its first argument, while the remaining comparison uses A_1/W. This step explains both the change of subgroup and the translation term in the recursive input.

The preceding example leaves an affine action β on the remaining differences. To split the remaining comparison again, we therefore need a reduction that applies when the current action is already affine, as in Definition 24. Given a subspace V preserved by the linear part, we first compare $x+V$ and $y+V$. If this returns uK, we must express the remaining conditions as an affine comparison on V under K, and recover all original solutions from its answer. The following lemma constructs the new target and action and gives the identities for both the transporter and the source stabilizer. It is Lemma 3.5 written in our notation; in the preceding example, $V = \ker r$.

Lemma 8.2 (Splitting an affine comparison)**.** *Consider the comparison*

$$\mathtt{C}(gH; \mathbf{A}; Z, \rho; P, N, z; c; x, y) = g \operatorname{Trans}_{H,\alpha}(x, y)$$

with input as in Definition 24. Let $V \le N/P$ satisfy $\rho_h(V) = V$ for every $h \in H$, and equip $\mathcal{A}/V$ with the induced action

$$h \cdot (v + V) = \alpha_h(v) + V \text{ for } h \in H,\ v \in \mathcal{A}.$$

Then the following statements hold.

(i) If $\operatorname{Trans}_H(x+V, y+V) = \varnothing$, then $g \operatorname{Trans}_{H,\alpha}(x, y) = \varnothing$.

(ii) Suppose $\operatorname{Trans}_H(x+V, y+V) = uK$, where $u \in H$ and $K = \operatorname{Stab}_H(x+V)$ for the quotient action. Define

$$w := \alpha_{u^{-1}}(y) - x,$$
$$\beta_k(v) := \rho_k(v) + \alpha_k(x) - x \text{ for } k \in K,\ v \in V.$$

Then $w \in V$, the maps β_k define an affine action of K on V, and

$$\begin{aligned} g \operatorname{Trans}_{H,\alpha}(x, y) &= gu \operatorname{Trans}_{K,\beta}(0, w), \\ \operatorname{Stab}_{H,\alpha}(x) &= \operatorname{Stab}_{K,\beta}(0). \end{aligned} \tag{8.2}$$

Proof. Every element taking x to y also takes $x+V$ to $y+V$, which proves (i). In (ii), the equality $\alpha_u(x) + V = y + V$ gives $w \in V$. Since K preserves $x+V$, taking x as the origin gives the affine action $\beta_k(v) = \alpha_k(x+v) - x$ on V. Every transformation taking x to y lies in uK, and for $k \in K$,

$$\alpha_{uk}(x) = y \quad\Longleftrightarrow\quad \alpha_k(x) = \alpha_{u^{-1}}(y) \quad\Longleftrightarrow\quad \beta_k(0) = w.$$

Every element fixing x belongs to K, and $\beta_k(0) = 0$ exactly when $\alpha_k(x) = x$. This proves the stabilizer equality. □

We now use the input flags to repeat the split above. For example, suppose $0 < F_1 < F_2 < A_1$ is H-invariant. In the tensor comparison above, we first compare the restrictions to $F_1 \times A_2 \times A_3$. A nonempty answer leaves an affine comparison on trilinear maps whose first argument is A_1/F_1. Since F_2/F_1 is invariant under the returned subgroup, we can apply Lemma 8.2 to this remaining comparison: first compare the restrictions to $(F_2/F_1) \times A_2 \times A_3$ and, if the answer is nonempty, continue with first argument

$$(A_1/F_1)/(F_2/F_1) \cong A_1/F_2.$$

Thus we successively compare the parts with first arguments F_1, F_2/F_1, and A_1/F_2, updating the subgroup and affine action after each comparison.

For a general induced flag $0 = F_0 < \cdots < F_s = A_1$, we want to carry out the same process along the successive argument spaces F_a/F_{a-1}. To apply Lemma 8.2, we need the corresponding invariant subspaces of the tensor module. For $0 \le a \le s$, let

$$r_a : Z_{\mathrm{tri}} \longrightarrow \operatorname{Hom}(F_a \otimes A_2 \otimes A_3, M)$$

be restriction to $F_a \times A_2 \times A_3$. Two tensors agree on this domain exactly when their difference lies in $\ker r_a$. As we match the restrictions on larger argument spaces, the remaining differences therefore lie in successively smaller subspaces:

$$Z_{\mathrm{tri}} = \ker r_0 \geq \ker r_1 \geq \cdots \geq \ker r_s = 0.$$

All these kernels are H-invariant. Once the restriction on F_{a-1} has been matched, the next comparison takes place on $\ker r_{a-1}/\ker r_a$, and a nonempty answer leaves a comparison on $\ker r_a$. Indeed, a trilinear map vanishing on $F_{a-1} \times A_2 \times A_3$ restricts to a map on $F_a \times A_2 \times A_3$ that depends only on the first argument's class in F_a/F_{a-1}. Restriction gives the H-equivariant identification

$$\frac{\ker r_{a-1}}{\ker r_a} \cong \mathrm{Hom}\big((F_a/F_{a-1}) \otimes A_2 \otimes A_3, M\big) \text{ for } 1 \leq a \leq s.$$

In the example above, the first comparison is on $Z_{\mathrm{tri}}/\ker r_1$, corresponding to restriction to F_1; the second is on $\ker r_1/\ker r_2$, corresponding to the next argument space F_2/F_1.

Within each of these quotients, use the flags on A_2 and A_3 in the same way to compare their successive argument spaces. Taking inverse images of the resulting subspaces gives an H-invariant refined chain in Z_{tri}. To include Z_{quad}, add it to every subspace of this refined chain and adjoin 0. Writing the resulting chain in increasing order and removing repeated subspaces gives

$$0 = \mathcal{Z}_0 < \cdots < \mathcal{Z}_t = Z.$$

Each quotient $\mathcal{Z}_j/\mathcal{Z}_{j-1}$ is either Z_{quad} or a space of M-valued trilinear maps whose three arguments are successive quotients of the flags on A_1, A_2, A_3.

For the full recursive input, differences between points of $\mathcal{A}$ lie in N/P. We therefore intersect each $\mathcal{Z}_j$ with N and take its image modulo P. These images form a chain in the difference space preserved by ρ. After each nonempty quotient comparison, Lemma 8.2 gives the remaining affine comparison under the returned stabilizer. Its linear part is a restriction of ρ, so the remaining subspaces of the chain are still invariant under that linear part. The following lemma records the sequence of quotient comparisons and shows how their complete answers recover the original solution set.

Lemma 8.3 (Comparison along argument flags)**.** *Consider the comparison*

$$\mathtt{C}(gH; \mathbf{A}; Z, \rho; P, N, z; c; x, y) = g\,\mathrm{Trans}_{H,\alpha}(x, y)$$

with input as in Definition 24. Let $0 = \mathcal{Z}_0 < \cdots < \mathcal{Z}_t = Z$ be the chain constructed above, and put

$$Q_j := \big((N \cap \mathcal{Z}_j) + P\big)/P \text{ for } 0 \leq j \leq t.$$

Then

$$0 = Q_0 \leq Q_1 \leq \cdots \leq Q_t = N/P,$$

and computing the original solution set reduces to at most t affine transporter problems on the nonzero quotients Q_j/Q_{j-1}, in decreasing order of j. Each problem is under a subgroup of H; its input can be constructed from the original data and the complete answers to the preceding problems.

If any of these transporters is empty, then $g\,\mathrm{Trans}_{H,\alpha}(x, y) = \varnothing$. Otherwise, their complete answers give an element $u \in H$ and generators of a subgroup $K \leq H$ satisfying

$$g\,\mathrm{Trans}_{H,\alpha}(x, y) = guK, \quad K = \mathrm{Stab}_{H,\alpha}(x).$$

Proof. Each Q_j is preserved by every ρ_h, since N, P, and $\mathcal{Z}_j$ are H-invariant. If $Q_t = 0$, the affine space has one point and the solution set is gH, so no comparisons are needed. Suppose $Q_t \neq 0$. Taking x as the origin identifies $\mathcal{A}$ with Q_t, with source 0, target $y - x$, and action

$$v \longmapsto \rho_h(v) + \alpha_h(x) - x \text{ for } h \in H,\ v \in Q_t.$$

Apply Lemma 8.2 with $V = Q_{t-1}$. The quotient comparison is on Q_t/Q_{t-1}, and a nonempty answer leaves an affine comparison on Q_{t-1} under the returned source stabilizer. Its linear action is the restriction of ρ, so it preserves the remaining chain. Repeating this step gives the comparisons on Q_j/Q_{j-1} in decreasing order; zero quotients require no transporter computation. The transporter and stabilizer identities in Lemma 8.2 give the claimed complete output. □

The group reductions. Lemma 8.3 recovers the original solution set from complete answers to the comparisons on the flag layers. We still need to solve those comparisons and to make progress when the flags give no useful split. For example, in the tensor comparison $\rho_h(\mathtt{T}) = \mathtt{T}'$ considered above, suppose each A_i is an irreducible H-module. Then its only invariant flag is $0 < A_i$, and the argument-flag construction leaves a single layer equal to the original tensor space. In this case the comparison has not become smaller.

A subgroup of H can preserve a subspace that H itself does not preserve. This can make another application of the flag lemma possible. For example, suppose A_1 has a direct-sum decomposition

$$A_1 = D_1 \oplus \cdots \oplus D_m, \qquad m > 1, \quad D_\nu \neq 0,$$

and H permutes the subspaces $D_1, \ldots, D_m$ transitively. Put

$$K := \{h \in H : hD_1 = D_1\}.$$

Then $0 < D_1 < A_1$ is a K-invariant flag. To cover every possible solution when passing from H to K, choose $u_\nu \in H$ with $u_\nu D_1 = D_\nu$. We have

$$H = \bigsqcup_{\nu=1}^{m} u_\nu K.$$

For $k \in K$, the action identity gives

$$\alpha_{u_\nu k}(x) = y \quad \Longleftrightarrow \quad \alpha_k(x) = \alpha_{u_\nu^{-1}}(y).$$

Consequently, the current comparison decomposes as

$$g \operatorname{Trans}_{H,\alpha}(x, y) = \bigcup_{\nu=1}^{m} g u_\nu \operatorname{Trans}_{K,\alpha|_K}\big(x, \alpha_{u_\nu^{-1}}(y)\big).$$

Each term now uses the subgroup K, so Lemmas 8.2 and 8.3 can use D_1 to split the first argument into D_1 and A_1/D_1. The target in that term is $\alpha_{u_\nu^{-1}}(y)$, and the remaining affine action is constructed by Lemma 8.2. This identity explains how information about the group action can provide a new split even when the original H-invariant flags are trivial.

We obtain invariant flags and decompositions into permuted subspaces by applying Theorem 5.1 to the joint image of H on $A_1 \oplus A_2 \oplus A_3$, pulling the computed subgroups back to H. Whenever the current group is restricted, we can examine its invariant subspaces again and apply Lemma 8.3 to the resulting flags. For the comparisons that remain, we next specify when the existing degree-2 algorithm applies and how we measure progress when the argument spaces become smaller.

To formulate the degree-2 case, we first specify which classical components are considered large. Throughout the recursion, keep fixed

$$d := \sum_{i=1}^{3} \bigl(\dim_{\mathbb{F}_p} U_i + \dim_{\mathbb{F}_p} U_i^*\bigr) = 2e(n_1 + n_2 + n_3),$$

and let InSize be the total bit length of the initial input. Theorems 5.2 and 5.3 describe the actions on the resulting module factors; we use these data to check the hypotheses of Definition 23. As in Section 5, use the threshold $T_{\mathrm{cl}} = p^{C_* d}$ with C_* fixed sufficiently large. A *large classical component* is a classical quasisimple component S obtained from Theorem 5.2 satisfying

$$|S/\mathrm{Z}(S)| > T_{\mathrm{cl}}.$$

A *large natural occurrence* is a tensor position carrying a natural, dual, or Frobenius-twisted natural module for a large classical component. Repeated appearances of the same module are counted separately.

If each summand has at most two large natural occurrences, and the remaining factors give a controlled presentation satisfying Definition 23, then Theorem 7.9 computes the complete transporter. The presentation must include the given relations and affine action, and satisfy the bound on the total number of tensor positions in that definition.

For a comparison that is not solved in this way, we measure the decrease in its argument dimensions against reference dimensions a_i, with $\dim_{\mathbb{F}_p} A_i \le a_i$ for $i = 1, 2, 3$. In the initial call, set $a_i = \dim_{\mathbb{F}_p} A_i$. Recursive calls on resulting comparisons use the same reference dimensions unless new values are specified. For a resulting comparison with argument spaces A_i', put $a_i' = \dim_{\mathbb{F}_p} A_i'$. We reset the reference dimensions for the child call to a_i' only when

$$a_i' \le a_i \quad (i = 1, 2, 3), \quad a_j' \le \frac{127}{128} a_j \quad \text{for some } j \text{ with } a_j > 0. \tag{8.3}$$

Recursive calls that do not reset the reference dimensions use the same values a_i as their parent call. Resetting the values for one child call does not change those used by its parent or its other children.

Call a comparison *ordinary* if $Z = Z_{\mathrm{tri}}$, $P = 0$, and $c_h = 0$ for every $h \in H$. Its equation is $\rho_h(\mathtt{T}) = \mathtt{T}'$ for tensors $\mathtt{T}, \mathtt{T}' \in Z_{\mathrm{tri}}$. Before entering the tensor reductions, we further reduce a nonordinary comparison. These reductions either complete the comparison, produce recursive comparisons, or leave an ordinary comparison.

The following proposition combines these reductions, states the form of the comparisons still to be solved, and specifies how their complete answers recover the original solution set.

Proposition 8.4 (Reduction by the current group action)**.** *Consider a comparison*

$$\mathtt{C}(gH; \mathbf{A}; Z, \rho; P, N, z; c; x, y)$$

with input as in Definition 24 arising in the recursion, and current reference dimensions a_i. Steps G1–G7 return emptiness or reduce the comparison to the following three types of output. Resulting comparison data are denoted by $(H'; \mathbf{A}'; Z', \rho'; P', N', z'; c'; x', y')$.

(i) Degree-2 completion. *The complete answer is obtained from comparisons with presentations as in Definition 23, using Theorem 7.9.*

(ii) Recursive comparisons. *Each resulting comparison either satisfies* (8.3) *or has $H' < H$. In the former case, Step G3 resets the reference dimensions to $\dim_{\mathbb{F}_p} A_i'$ for the child call. Otherwise, make a recursive call starting at Step G1, with the same reference dimensions a_i.*

(iii) Ordinary input to the tensor reductions. *Each A_i' is a single natural, dual, or Frobenius-twisted natural module for a large classical component, in the computed coordinates. The comparison satisfies*

$$Z' = Z'_{\mathrm{tri}}, \qquad P' = 0, \qquad c'_h = 0 \quad (h \in H').$$

It is passed to the tensor reductions as an ordinary comparison.

The complete answers recover the original solution set as emptiness or guK, where $u \in H$ and $K = \mathrm{Stab}_{H,\alpha}(x)$ is given by generators.

Every output satisfies

$$H' \le H, \quad \dim_{\mathbb{F}_p} A_i' \le \dim_{\mathbb{F}_p} A_i \quad (i = 1, 2, 3).$$

The reduction and recovery take $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, d, \log p)$ time, where $d = 2e(n_1 + n_2 + n_3)$ and InSize *is the initial input bit length. Each resulting input has length* $\mathrm{poly}(\mathrm{InSize}, d, \log p)$*, uniformly throughout the recursion.*

Steps G1–G7 and Proposition 8.4 are proved in Appendix E.1.

The tensor reductions. In case (iii) of Proposition 8.4, the remaining comparison is ordinary. We now use the tensor coefficients to make further progress. The following proposition states the possible outcomes and how their complete answers recover the original solution set.

Proposition 8.5 (Tensor reductions)**.** *Consider an ordinary comparison*

$$g\,\mathrm{Trans}_{H,\rho}(\mathtt{T}, \mathtt{T}'), \qquad \mathtt{T}, \mathtt{T}' \in Z_{\mathrm{tri}},$$

arising in case (iii) of Proposition 8.4, with the current reference dimensions a_i. The tensor reductions return emptiness or give one of the following three outcomes. Each resulting comparison has the input format of Definition 24, with data denoted by $(H'; \mathbf{A}'; Z', \rho'; P', N', z'; c'; x', y')$.

(i) Degree-2 completion. *The complete answer is obtained from controlled degree-2 comparisons using Theorem 7.9.*

(ii) Restriction to proper subgroups. *The comparison reduces to a family of comparisons with*

$$A_i' = A_i \quad (i = 1, 2, 3), \qquad H' < H, \qquad |H'| \le |H|/2.$$

Every transformation in a resulting candidate coset satisfies all conditions matched in earlier steps. For each comparison, make a recursive call starting at Step G1, with the same reference dimensions a_i.

(iii) A sequence leading to smaller comparisons. *The comparison reduces to at most three successive tensor comparisons with*

$$\dim_{\mathbb{F}_p} A_i' \le \dim_{\mathbb{F}_p} A_i \quad (i = 1, 2, 3).$$

Each later input is constructed from the complete answers to the preceding comparisons. Applying Proposition 8.4 to each input yields only degree-2 comparisons or comparisons satisfying (8.3)*. For the latter, Step G3 makes recursive calls with their argument dimensions as the new reference dimensions.*

The complete answers recover the original solution set as emptiness or guK, with $u \in H$ and generators of $K = \mathrm{Stab}_{H,\rho}(\mathtt{T})$.

The local reduction and recovery take $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, d, \log p)$ time, where $d = 2e(n_1 + n_2 + n_3)$ and InSize *is the initial input bit length. Each resulting input has length* $\mathrm{poly}(\mathrm{InSize}, d, \log p)$*, uniformly throughout the recursion.*

We prove Proposition 8.5 in Subsubsection E.2.5. Algorithm 8.6 combines the group and tensor reductions and specifies their input updates.

Algorithm 8.6 (The recursive tensor comparison)**.** The comparison data and required output are those of Definition 24. The algorithm also carries reference dimensions a_1, a_2, a_3, with $\dim_{\mathbb{F}_p} A_i \le a_i$. In the initial call, set $a_i = \dim_{\mathbb{F}_p} A_i$ for $i = 1, 2, 3$. Recursive calls on resulting comparisons start at Step G1 and use the same reference dimensions unless Step G3 specifies new values for that child call. Updates of the current comparison also use the same reference dimensions. Start at Step G1. If $N = P$, return gH.

Whenever a step centers the current comparison at its source x, define

$$\delta_h := \alpha_h(x) - x \quad (h \in H), \quad w := y - x. \tag{8.4}$$

Taking x as the origin gives the affine space N/P, source 0, target w, and action $v \mapsto \rho_h(v) + \delta_h$. These quantities are defined from the current input each time this convention is used.

Whenever a step restricts the candidates to a coset $uK \subseteq H$, the corresponding input is

$$\mathtt{C}\big(guK; \mathbf{A}; Z, \rho|_K; P, N, z; c|_K; x, \alpha_{u^{-1}}(y)\big). \tag{8.5}$$

For a subspace $V \le N/P$ preserved by every ρ_h, first compute $\mathrm{Trans}_H(x + V, y + V)$. If it is empty, return emptiness. Otherwise, let its answer be uK, and let w, β be as in Lemma 8.2. Then

$$g\,\mathrm{Trans}_{H,\alpha}(x, y) = gu\,\mathrm{Trans}_{K,\beta}(0, w).$$

Compute the remaining transporter on the right. If its answer is vS, return $guvS$; if it is empty, return emptiness.

If a step produces several candidate cosets covering all possible solutions, compute every branch. If every branch is empty, return emptiness. Otherwise write the nonempty completed answers as $gu_\nu K_\nu$, with $u_\nu \in H$ and $K_\nu \le H$. Then

$$g\,\mathrm{Trans}_{H,\alpha}(x, y) = \bigcup_\nu gu_\nu K_\nu.$$

Choose one index ν_0 and return

$$gu_{\nu_0}\left\langle \bigcup_\nu K_\nu,\ u_{\nu_0}^{-1}u_\nu\ (\nu) \right\rangle. \tag{8.6}$$

Group reductions.

G1. Compare along the argument flags. Apply Theorems 5.1, 5.2, and 5.3 to the current group action, refine the argument flags, and form $0 = \mathcal{Z}_0 < \cdots < \mathcal{Z}_t = Z$ as in Lemma 8.3. For each successive argument quotient, also record the direct-sum decomposition into subspaces permuted by H supplied by Theorem 5.1. If there is a single nonzero layer and its argument spaces are unchanged, go to Step G2. Otherwise put

$$Q_j := \big((N \cap \mathcal{Z}_j) + P\big)/P \quad (0 \le j \le t),$$

and initialize $(g_t, H_t, w_t) = (g, H, w)$, with δ, w as in (8.4). At index j, the remaining comparison is on Q_j, with source 0, target w_j, and action

$$v \longmapsto \rho_h(v) + \delta_h \qquad (h \in H_j).$$

For $j = t, t-1, \ldots, 1$, skip $Q_j = Q_{j-1}$ using unchanged data. On each nonzero quotient Q_j/Q_{j-1}, compute the transporter

$$g_j\{h \in H_j : \delta_h - w_j \in Q_{j-1}\}.$$

Use the subquotient presentation of Q_j/Q_{j-1} induced by $P \leq N$ in $\mathcal{Z}_j/\mathcal{Z}_{j-1}$. For the layer coming from Z_{quad}, use Step G2. Otherwise write

$$\mathcal{Z}_j/\mathcal{Z}_{j-1} \cong \mathrm{Hom}\big(A_1' \otimes A_2' \otimes A_3', M\big)\,,$$

where each A_i' is the successive quotient of the current flag on A_i used in this layer ($i = 1, 2, 3$). Solve the comparison on Q_j/Q_{j-1} by a recursive call from Step G1, with argument spaces A_i', the induced input data of Lemma 8.3, and the same reference dimensions a_i.

If the answer is empty, return emptiness. Otherwise write it as $g_j u_j H_{j-1}$. By Lemma 8.2, update

$$\begin{aligned} g_{j-1} &:= g_j u_j, \\ w_{j-1} &:= \rho_{u_j^{-1}}(w_j) + \delta_{u_j^{-1}}. \end{aligned} \tag{8.7}$$

After this quotient has been matched, the remaining comparison is on Q_{j-1}, with target w_{j-1} and the original argument presentations $\mathbf{A}$. Its input is

$$\begin{aligned} &\mathtt{C}\big(g_{j-1}H_{j-1}; \mathbf{A}; Z, \rho|_{H_{j-1}}; \\ &\quad P, (N \cap \mathcal{Z}_{j-1}) + P, 0; \delta|_{H_{j-1}}; 0, w_{j-1}\big). \end{aligned}$$

Continue with index $j-1$; after the last quotient, return $g_0 H_0$.

G2. Solve a degree-2 comparison. With δ, w as in (8.4), the current task is to compare source 0 with target w in N/P under the action $v \mapsto \rho_h(v) + \delta_h$, and return $g\{h \in H : \delta_h = w\}$.

Using the module data of the current comparison, construct, when available, a presentation satisfying Definition 23, with

$$P' \leq N' \leq Z', \qquad \phi : N/P \xrightarrow{\sim} N'/P'.$$

Here ϕ is an explicitly computed $\mathbb{F}_p$-linear bijection, and $\phi\rho_h\phi^{-1}$ is the action on N'/P' induced from Z'. If no such presentation is obtained, go to Step G3.

Otherwise apply Theorem 7.9 on N'/P' with

$$\begin{aligned} &\text{source}: \ 0, \\ &\text{target}: \ \phi(w), \\ &\text{action}: \ v \longmapsto \phi\rho_h\phi^{-1}(v) + \phi(\delta_h) \quad (h \in H). \end{aligned}$$

Since $\phi(\delta_h) = \phi(w)$ if and only if $\delta_h = w$, this computes the same transporter in H. Return emptiness if the answer is empty; otherwise return guK for the returned transporter uK.

G3. Restart with smaller reference dimensions. For the current unsolved comparison, put $a_i' := \dim_{\mathbb{F}_p} A_i$ for $i = 1, 2, 3$, and test the following condition against the inherited reference dimensions a_i:

$$a_i' \le a_i \quad (i = 1, 2, 3), \qquad a_j' \le \frac{127}{128} a_j \quad \text{for some } j \text{ with } a_j > 0.$$

If this condition fails, go to Step G4. Otherwise, make a recursive call from Step G1 on the current comparison

$$\mathtt{C}(gH; \mathbf{A}; Z, \rho; P, N, z; c; x, y),$$

using (a_1', a_2', a_3') as this child call's reference dimensions, and return its complete answer. The calling computation keeps its reference dimensions a_i. At the start of the child call, the argument dimensions equal the new reference dimensions; another reset therefore requires a further decrease in the argument dimensions.

G4. Compare smaller summands within candidate cosets. When H permutes the summands of a nontrivial direct sum, split the candidates into cosets according to the image of a summand and compare on that smaller space under its stabilizer. The summand is invariant under its stabilizer, although it need not be invariant under H.

Step G1 uses the invariant argument flags from Theorem 5.1 and makes recursive comparisons with their successive quotients as the argument spaces. The same theorem also computes a direct-sum decomposition of each such quotient into subspaces permuted by H. When a quotient becomes a current argument space A_k, this decomposition is inside A_k; for a trivial flag $0 < A_k$, the quotient is A_k itself.

If the computed decomposition of some current argument A_k, with $k \in \{1, 2, 3\}$, has more than one summand, write

$$A_k = D_1 \oplus \cdots \oplus D_s, \qquad s > 1, \quad D_\nu \neq 0.$$

Otherwise go to Step G5. Each subspace in the flag used by Step G1 is H-invariant, whereas here we only require

$$\{h(D_1), \ldots, h(D_s)\} = \{D_1, \ldots, D_s\} \qquad (h \in H).$$

An individual D_ν need not be H-invariant, so this decomposition can give smaller arguments even when the invariant flag is $0 < A_k$. Each h still acts linearly on A_k, including the linear maps between the corresponding summands.

For example, take $k = 1$, $s = 2$, and ordinary tensors $x = \mathtt{T}$, $y = \mathtt{T}'$ with $P = 0$, $c = 0$, and $Z = Z_{\mathrm{tri}}$. For $i, j \in \{1, 2\}$, put

$$C_{ij} := \left\{ h \in H : \begin{array}{l} h(D_i) = D_j, \\ (\rho_h \mathtt{T})|_{D_j \times A_2 \times A_3} = \mathtt{T}'|_{D_j \times A_2 \times A_3} \end{array} \right\}.$$

Then

$$\mathrm{Trans}_H(\mathtt{T}, \mathtt{T}') = (C_{11} \cap C_{22}) \cup (C_{12} \cap C_{21}).$$

The two intersections correspond to fixing or exchanging the summands; each requires the same h to match both restrictions. Thus we first solve the comparisons on smaller argument spaces, then combine their answers to match all summands simultaneously.

For the general input, use the chosen decomposition and compare each pair $1 \le \nu, \omega \le s$ as follows. Find $r \in H$ with $r(D_\nu) = D_\omega$ and put $K := \mathrm{Stab}_H(D_\nu)$. If no such r exists, record emptiness and skip this pair. Otherwise all candidates for this pair lie in rK: pull the target back by $\alpha_{r^{-1}}$ and compare on D_ν under K.

Set $A'_k = D_\nu$ and $A'_i = A_i$ for $i \neq k$, and let $\mathbf{A}'$ be the induced quotient presentations. Put $Z' := \mathrm{Hom}(A'_1 \otimes A'_2 \otimes A'_3, M)$. Restriction to these arguments, extended by zero on Z_{quad}, sends P to a subspace $P' \le Z'$ and induces $q : Z/P \to Z'/P'$. Let ρ' be the induced K-action on Z', and set

$$c'_h := q(c_h)\ (h \in K), \qquad x' := q(x), \qquad y' := q(\alpha_{r^{-1}}(y)).$$

Make a recursive call from Step G1, with the same reference dimensions a_i, on

$$\mathtt{C}(rK; \mathbf{A}'; Z', \rho'; P', Z', 0; c'; x', y'), \tag{8.8}$$

and record its complete answer for this pair.

Combine the pair answers by requiring one common $h \in H$ and one common representative $v \in Z$ of c_h, with $v + P = c_h$, to match all restrictions simultaneously. This computes

$$\mathrm{Trans}_H(x + V, y + V), \qquad V := \big((N \cap Z_{\mathrm{quad}}) + P\big)/P.$$

If it is empty, return emptiness. Otherwise write it as uK. The remaining comparison is on V, a subquotient of Z_{quad}: use Lemma 8.2 with candidate coset guK, solve the resulting comparison by Step G2, and return its complete answer.

G5. Look for repeated copies to reduce an argument. If the structural computation expresses a current argument as at least two copies of one natural module, record the tensor restrictions to those copies as separate value coordinates to obtain equivalent comparisons with a smaller argument.

Suppose Theorem 5.2, applied in Step G1, expresses A_k, as a module for a large classical component, as $m \ge 2$ copies of a natural module B (X_i in the theorem; see Definition 5). Over the defining field $\mathbb{K}$ of B, write

$$A_k = B \otimes_{\mathbb{K}} \mathbb{K}^m, \qquad k \in \{1, 2, 3\}, \quad m \ge 2,$$

with $\mathbb{K}/\mathbb{F}_p$ finite.

If none of the three current argument spaces A_1, A_2, A_3 has a presentation of the above form with $m \ge 2$, go to Step G6.

The factor $\mathbb{K}^m$ records m copies of B. The group H may mix these copies linearly, so they need not form the permuted summands required by Step G4. We will replace the argument A_k by B and record the m restrictions together in the value space $M^{\oplus m}$.

For example, take $k = 1$ and $m = 2$, and choose a basis (e_1, e_2) of $\mathbb{K}^2$. Rewrite a tensor $\mathtt{T}$ as

$$(\Phi\mathtt{T})(b, a_2, a_3) = \big(\mathtt{T}(b \otimes e_1, a_2, a_3), \mathtt{T}(b \otimes e_2, a_2, a_3)\big) \in M \oplus M.$$

The new first argument is B. The two coordinates record both restrictions, which determine $\mathtt{T}$ by linearity on $A_1 = (B \otimes e_1) \oplus (B \otimes e_2)$.

For general m, choose an ordered basis $(e_1, \ldots, e_m)$ of $\mathbb{K}^m$. Using the actions on B and $\mathbb{K}^m$ supplied by Lemma 5.4 and the enumeration in Lemma 5.5, compute cosets $uK \subseteq H$ covering

H for the possible images of this basis. Here K is obtained by projecting the stabilizer of the ordered basis to H. In each branch, pull the target back by $\alpha_{u^{-1}}$. With the basis fixed, the remaining transformations act on every copy through the same action on B.

Set $A'_k = B$, $A'_i = A_i$ for $i \neq k$, and $M' := M^{\oplus m}$, with K acting on each coordinate as on M. Put

$$Z' := Z_{\mathrm{quad}} \oplus \mathrm{Hom}(A'_1 \otimes A'_2 \otimes A'_3, M').$$

Let $\Phi : Z \to Z'$ be the linear isomorphism that is the identity on Z_{quad} and is given on Z_{tri} by

$$(\Phi \mathtt{T})(\ldots, b, \ldots) = \big(\mathtt{T}(\ldots, b \otimes e_j, \ldots)\big)_{j=1}^{m}.$$

Let $\overline{\Phi} : Z/P \to Z'/\Phi(P)$ be induced by Φ, put $\rho'_h = \Phi \rho_h \Phi^{-1}$ for $h \in K$, and let $\mathbf{A}'$ replace (N_k, P_k) by $(\pi_k^{-1}(B \otimes e_1), P_k)$, using $b \otimes e_1 \mapsto b$. Pass the transformed comparison

$$\begin{aligned} &\mathtt{C}\big(guK; \mathbf{A}'; Z', \rho'; \Phi(P), \Phi(N), \Phi(z); \\ &\quad \overline{\Phi}(c|_K); \overline{\Phi}(x), \overline{\Phi}(\alpha_{u^{-1}}(y))\big) \end{aligned} \tag{8.9}$$

to Step G3. Indeed,

$$\dim_{\mathbb{F}_p} A'_k = \frac{\dim_{\mathbb{F}_p} A_k}{m} \leq \frac{a_k}{2},$$

and the other argument dimensions are unchanged. Thus Step G3 resets the reference dimensions and makes a recursive call from Step G1. Return the combined answers to all branches for the images of the basis using (8.6).

G6. Make the quotient action linear. Using the pairing maps of Definition 16, construct an H-equivariant map

$$\mathcal{J} : \mathcal{S} \longrightarrow Z_{\mathrm{tri}}, \qquad J(Z) := Z_{\mathrm{quad}} \oplus \mathrm{im}\, \mathcal{J},$$

where $\mathcal{S}$ has a controlled degree-2 presentation as in Definition 23. The construction gives $P \leq J(Z)$. We change the origin so that all translation terms lie in $J(Z)/P$, making the induced action on $Z/J(Z)$ linear.

For the current action $\alpha_h(v) = \rho_h(v) + c_h$, seek $b \in Z$ such that the coordinate change $v \mapsto v + \overline{b}$, with $\overline{b} = b + P$, gives new translations

$$c_h^b := c_h + \overline{b} - \rho_h(\overline{b}) \in J(Z)/P \quad (h \in H).$$

Let $\pi_J : Z/P \to Z/J(Z)$ be the quotient map and S_H the given generating set of H. To obtain the required b, solve the linear equations

$$(\rho_s - I_Z)b + J(Z) = \pi_J(c_s) \quad (s \in S_H).$$

Choose $b = 0$ when $c = 0$.

Continue at Step G7 with the equivalent comparison

$$\mathtt{C}\big(gH; \mathbf{A}; Z, \rho; P, N, z + b; c^b; x + \overline{b}, y + \overline{b}\big). \tag{8.10}$$

G7. Handle the remaining affine comparison. The input is the comparison produced by Step G6, with its argument modules, pairing map $\mathcal{J}$, reference dimensions a_i, and

$$P \leq J(Z), \qquad c_h \in J(Z)/P \quad (h \in H).$$

If it is ordinary, enter the tensor reductions below. Otherwise use one of the following three exits.

(a) **Complete the comparison.** Compute the complete answer using the degree-2 algorithm and return it, including emptiness when there is no solution.

(b) **Make recursive comparisons.** Construct resulting comparisons whose argument dimensions do not increase and which either satisfy (8.3) or use a proper subgroup of H. Pass a comparison satisfying (8.3) to Step G3; solve every other resulting comparison from Step G1 with the same reference dimensions a_i. For a restriction to a candidate coset, use (8.5). Recover the complete answer from the returned answers; for a family of candidate cosets, use (8.6).

(c) **Obtain an ordinary comparison.** Keeping H and the argument spaces unchanged, construct an equivalent remaining comparison

$$\mathtt{C}\big(g'H; \mathbf{A}; Z_{\mathrm{tri}}, \rho_{\mathrm{tri}}; 0, Z_{\mathrm{tri}}, 0; 0; \mathtt{T}, \mathtt{T}'\big), \tag{8.11}$$

where $g' \in gH$, $\mathtt{T}, \mathtt{T}' \in Z_{\mathrm{tri}}$, and ρ_{tri} is the induced action on Z_{tri}. Enter the tensor reductions with this input. If obtaining the ordinary comparison also restricts the group, use exit (b) instead.

Tensor reductions. The current input is ordinary by Step G7 and satisfies case (iii) of Proposition 8.4. Write $x = \mathtt{T}$ and $y = \mathtt{T}'$. If either tensor is zero, return gH when both are zero and emptiness otherwise. If both are nonzero, start at Step T0. Unresolved output comparisons start at Step G1 with the same reference dimensions a_i.

T0. Compare the action types. Permute the argument positions, their presentations, and the reference dimensions a_i together so that $\dim_{\mathbb{F}_p} A_1 = \min_i \dim_{\mathbb{F}_p} A_i$.

Let D be the subgroup supplied by Theorem 5.2 in Step G1 for the action on $A_1 \oplus A_2 \oplus A_3$, and let $\sigma_i : D \to \mathrm{GL}(A_i)$ denote its induced actions. The actions on A_2 and A_3^* have the *same type* if there is an $\mathbb{F}_p$-linear bijection $\iota : A_2 \to A_3^*$ satisfying

$$\iota\, \sigma_2(d) = \sigma_3(d)^{-T} \iota \qquad \text{for every } d \in D.$$

Here $\sigma_3(d)^{-T}$ is the action on A_3^* given by $\lambda \mapsto \lambda \circ \sigma_3(d)^{-1}$. If no such ι exists, the actions have *different types*.

If the types differ, go to Step T1. If they agree, compute and record such an ι and go to Step T2.

T1. Match a high-rank contraction. The types of A_2 and A_3^* differ by Step T0. For $\mathtt{F} \in Z_{\mathrm{tri}}$ and $v \in A_1$, define the contraction

$$\mathcal{C}_v(\mathtt{F}) : A_2 \longrightarrow A_3^* \otimes M, \qquad b \longmapsto \mathtt{F}(v, b, -).$$

Its rank is the $\mathbb{F}_p$-dimension of its image. For each $\mathtt{F} \in \{\mathtt{T}, \mathtt{T}'\}$, test whether

$$\operatorname{rank} \mathcal{C}_v(\mathtt{F}) \geq \frac{\dim_{\mathbb{F}_p} A_1}{64} \quad \text{for some } v \in A_1.$$

If this holds for exactly one tensor, return emptiness. If it holds for neither, go to Step T4.

Otherwise choose $v \in A_1$ satisfying the rank bound for $\mathtt{T}$, and put $H_v := \mathrm{Stab}_H(v)$. Enumerate the possible images $v' \in H \cdot v$; there are at most p^{a_1}. For each v', find $r \in H$ with $rv = v'$, so that all candidates taking v to v' form the coset rH_v.

Contraction at v is H_v-equivariant. Use Theorem 7.9 to compare the bilinear maps $\mathcal{C}_v(\mathtt{T})$ and $\mathcal{C}_v(\rho_{r^{-1}}(\mathtt{T}'))$ under the induced action of H_v, computing

$$\{k \in H_v : \mathcal{C}_v(\rho_k(\mathtt{T})) = \mathcal{C}_v(\rho_{r^{-1}}(\mathtt{T}'))\}. \tag{8.12}$$

Discard an empty answer. For a nonempty answer uK, the remaining candidates lie in ruK, with $K \leq H_v < H$. Apply (8.5) to this coset and make a recursive call from Step G1, with the same reference dimensions a_i. This is case (ii) of Proposition 8.5. Combine the completed branches by (8.6); if every branch is empty, return emptiness.

T2. Match several contractions of the same type. The types of A_2 and A_3^* agree, with the identification ι recorded in Step T0. For each $\mathtt{F} \in \{\mathtt{T}, \mathtt{T}'\}$, test tuples $\mathbf{v} = (v_1, \ldots, v_t) \in A_1^t$ with $1 \leq t \leq 3$. Put $H_{\mathbf{v}} := \mathrm{Stab}_H(\mathbf{v})$, fixing each entry, and use the degree-2 algorithm to compute

$$K_{\mathbf{v}}(\mathtt{F}) := \{h \in H_{\mathbf{v}} : \mathcal{C}_{v_j}(\rho_h(\mathtt{F})) = \mathcal{C}_{v_j}(\mathtt{F}) \text{ for } 1 \leq j \leq t\}.$$

Using the structural computation from Step G1, test whether there is an index $k \in \{1, 2, 3\}$ such that every natural module factor B arising from A_k under $K_{\mathbf{v}}(\mathtt{F})$ satisfies

$$\dim_{\mathbb{F}_p} B \leq \frac{63}{64} \dim_{\mathbb{F}_p} A_k.$$

If such a tuple exists for exactly one tensor, return emptiness. If neither tensor has such a tuple, go to Step T3.

Otherwise fix one such tuple for $\mathtt{T}$ and write $K = K_{\mathbf{v}}(\mathtt{T})$. Enumerate its images $\mathbf{v}' \in H \cdot \mathbf{v}$, at most p^{3a_1} in number. For each image, choose $r \in H$ with $r\mathbf{v} = \mathbf{v}'$. As in Step T1, use the degree-2 algorithm to match all contractions with the same $h \in H_{\mathbf{v}}$, computing

$$\{h \in H_{\mathbf{v}} : \mathcal{C}_{v_j}(\rho_h(\mathtt{T})) = \mathcal{C}_{v_j}(\rho_{r^{-1}}(\mathtt{T}')) \text{ for } 1 \leq j \leq t\}.$$

Discard an empty answer. Each nonempty answer has the form uK, with $K < H$. Apply (8.5) with candidate coset ruK and make a recursive call from Step G1, with the same reference dimensions a_i. The group reductions yield only degree-2 comparisons or comparisons satisfying (8.3), since

$$\frac{63}{64} \dim_{\mathbb{F}_p} A_k \leq \frac{63}{64} a_k < \frac{127}{128} a_k.$$

Combine the completed branches by (8.6); if every branch is empty, return emptiness.

T3. Use the kernel of a linear map. Both tensors have failed the tuple test in Step T2. Put $\mathbb{F} := \mathrm{End}_{\mathbb{F}_p[D]}(A_2)$, the endomorphism field of the current D-action. From their contractions, construct finite extensions $\mathbb{K}_{\mathtt{T}}, \mathbb{K}_{\mathtt{T}'}/\mathbb{F}$, each of degree at most $(\dim_{\mathbb{F}_p} A_2)^2$ over $\mathbb{F}_p$, and $\mathbb{F}_p$-linear maps

$$\lambda_{\mathtt{T}} : A_1 \longrightarrow \mathbb{K}_{\mathtt{T}} \otimes M, \qquad \lambda_{\mathtt{T}'} : A_1 \longrightarrow \mathbb{K}_{\mathtt{T}'} \otimes M.$$

Every tensor solution induces an $\mathbb{F}_p$-isomorphism between these fields. If their degrees differ, return emptiness; otherwise identify them over $\mathbb{F}$ with a common field $\mathbb{K}$. For $\kappa \in \mathbb{F}^\times$ and $0 \leq j < [\mathbb{K} : \mathbb{F}_p]$, put $\psi_{\kappa,j}(t) := \kappa t^{p^j}$. The maps are constructed so that every tensor solution $h \in H$ satisfies

$$\lambda_{\mathtt{T}'} \circ h_{A_1} = (\psi_{\kappa,j} \otimes h_M) \circ \lambda_{\mathtt{T}}$$

for some pair (κ, j). Here h_{A_1} and h_M denote the induced actions; κ allows for the scalar action on the pairing, and j for the field automorphism.

Put $W := \ker \lambda_{\mathtt{T}}$ and $W' := \ker \lambda_{\mathtt{T}'}$. If their dimensions differ, return emptiness. Otherwise use the following three cases.

(a) If $0 < W < A_1$, compute $\mathrm{Trans}_H(W, W')$. Return emptiness if it is empty. Otherwise write it as uH_W, where $H_W = \mathrm{Stab}_H(W) < H$. Apply (8.5) with candidate coset uH_W and make a recursive call from Step G1, with the same reference dimensions a_i. Return its complete answer.

(b) If $W = 0$, both maps are injective. For each pair (κ, j) above, use the degree-2 algorithm to compute

$$\{h \in H : \lambda_{\mathtt{T}'} \circ h_{A_1} = (\psi_{\kappa,j} \otimes h_M) \circ \lambda_{\mathtt{T}}\}.$$

Discard an empty answer. For a nonempty answer uK, apply (8.5) to this coset. On K, injectivity determines the action on A_1 from the action on M; the remaining tensor comparison has a controlled degree-2 presentation. Solve it by Step G2, and combine the completed answers by (8.6). If every branch is empty, return emptiness.

(c) If $W = A_1$, both maps vanish. The construction then gives the uniform rank bound in Step T4; continue at that step.

T4. Construct a common zero block and compare the remaining parts. This step is reached from Step T1 or T3 with

$$\operatorname{rank} \mathcal{C}_v(\mathtt{F}) \le \frac{1}{32} \min\bigl(\dim_{\mathbb{F}_p} A_2, \dim_{\mathbb{F}_p} A_3\bigr) \quad (v \in A_1,\ \mathtt{F} \in \{\mathtt{T}, \mathtt{T}'\}).$$

(a) **Construct the two zero blocks.** Construct pairs (X, E) and (X', E'), with $X, X' \le A_2$ and $E, E' \le A_3^* \otimes M$, such that

$$\mathtt{T}(a, b, -) \in E \quad (a \in A_1,\ b \in X),$$
$$\mathtt{T}'(a, b', -) \in E' \quad (a \in A_1,\ b' \in X'),$$

and both $\mathrm{codim}_{A_2} X + \dim_{\mathbb{F}_p} E$ and $\mathrm{codim}_{A_2} X' + \dim_{\mathbb{F}_p} E'$ are at most $\frac{1}{8}\min(\dim_{\mathbb{F}_p} A_2, \dim_{\mathbb{F}_p} A_3)$. The pairs are chosen so that every tensor solution satisfies $hX = X'$ and $hE = E'$. Put

$$Y := \bigcap_{e \in E} \ker e \le A_3, \qquad Y' := \bigcap_{e \in E'} \ker e \le A_3.$$

Then $\mathtt{T}$ vanishes on $A_1 \times X \times Y$, and $\mathtt{T}'$ vanishes on $A_1 \times X' \times Y'$. We next match the pairs to put these zero blocks on the same argument subspaces.

(b) **Match the pairs and obtain a common zero block.** Compute

$$\mathrm{Trans}_H\bigl((X, E), (X', E')\bigr) = \{h \in H : hX = X',\ hE = E'\}.$$

Return emptiness if it is empty. Otherwise write its complete answer as uK, where $K = \mathrm{Stab}_H(X, E)$. Every tensor solution lies in uK, so

$$g \,\mathrm{Trans}_{H,\rho}(\mathtt{T}, \mathtt{T}') = gu \,\mathrm{Trans}_{K,\rho|_K}\bigl(\mathtt{T}, \rho_{u^{-1}}(\mathtt{T}')\bigr).$$

Thus the remaining candidates form the coset guK. As in (8.5), make the simultaneous input update

$$(g, H, \rho, \mathtt{T}') \longleftarrow \bigl(gu, K, \rho|_K, \rho_{u^{-1}}(\mathtt{T}')\bigr),$$

with the argument and value actions restricted to K and the same argument presentations and reference dimensions a_i. The source $\mathtt{T}$ is unchanged. Since $u^{-1}(X', E') = (X, E)$ and $u^{-1}Y' = Y$, both tensors in the updated input vanish on the common block $A_1 \times X \times Y$, and the current group H preserves X, E, and Y.

(c) **Compare the remaining parts successively.** First compare the restrictions with third argument in Y; their second argument descends to A_2/X. Within the returned candidate coset, compare the restrictions with second argument in X; their third argument descends to A_3/Y. After these two comparisons, the remaining differences descend to both quotients, giving a final comparison on $A_1 \times (A_2/X) \times (A_3/Y)$.

For tensors $\mathtt{F}$ vanishing on $A_1 \times X \times Y$, define

$$f_1(\mathtt{F})(a, b + X, y) := \mathtt{F}(a, b, y),$$
$$f_2(\mathtt{F})(a, x, c + Y) := \mathtt{F}(a, x, c),$$

where $a \in A_1$, $b \in A_2$, $x \in X$, $y \in Y$, and $c \in A_3$. Their target spaces, and the space for the remaining differences, are

$$Z_1 := \mathrm{Hom}(A_1 \otimes (A_2/X) \otimes Y, M),$$
$$Z_2 := \mathrm{Hom}(A_1 \otimes X \otimes (A_3/Y), M),$$
$$Z_3 := \mathrm{Hom}(A_1 \otimes (A_2/X) \otimes (A_3/Y), M).$$

Let $\mathbf{A}^{(1)}, \mathbf{A}^{(2)}, \mathbf{A}^{(3)}$ be the induced quotient presentations of the respective argument triples, and write $\rho^{(j)}$ for the induced action on Z_j. Let $\jmath : Z_3 \hookrightarrow Z_{\mathrm{tri}}$ be pullback along the quotient maps; its image is $\ker f_1 \cap \ker f_2$.

Make the following two recursive calls successively from Step G1, with the same reference dimensions a_i. After the first nonempty answer, pull back the target by u_1^{-1} and make the second comparison under H_1:

$$\begin{aligned} &\mathtt{C}(gH; \mathbf{A}^{(1)}; Z_1, \rho^{(1)}; 0, Z_1, 0; 0; f_1(\mathtt{T}), f_1(\mathtt{T}')) = gu_1H_1, \\ &\mathtt{C}\big(gu_1H_1; \mathbf{A}^{(2)}; Z_2, \rho^{(2)}|_{H_1}; 0, Z_2, 0; 0; \\ &\qquad\qquad f_2(\mathtt{T}), \rho^{(2)}_{u_1^{-1}} f_2(\mathtt{T}')\big) = gu_1u_2H_2. \end{aligned} \tag{8.13}$$

If either answer is empty, return emptiness. Otherwise put $u = u_1u_2$. Every candidate in uH_2 matches both images, and H_2 fixes both source images. Hence the differences below lie in $\mathrm{im}\,\jmath$, and we can define

$$w := \jmath^{-1}\big(\rho_{u^{-1}}(\mathtt{T}') - \mathtt{T}\big), \qquad \mu_h := \jmath^{-1}\big(\rho_h(\mathtt{T}) - \mathtt{T}\big) \quad (h \in H_2).$$

By Lemma 8.2, the remaining task is the affine comparison

$$\mathtt{C}(guH_2; \mathbf{A}^{(3)}; Z_3, \rho^{(3)}|_{H_2}; 0, Z_3, 0; \mu; 0, w). \tag{8.14}$$

Make a recursive call on this input from Step G1, with the same reference dimensions a_i, and return its complete answer. This is case (iii) of Proposition 8.5: on each of the three inputs, the group reductions yield only degree-2 comparisons or comparisons satisfying (8.3). In the latter case, Step G3 resets the reference dimensions for the recursive solution.

The cost of the recursion. For one reduction step, let T be the total time needed to complete the current comparison. Let w be the work outside the resulting tensor comparisons, including input construction, all auxiliary comparisons completed locally, and recovery after the resulting comparisons return. For m resulting tensor comparisons, write their costs as $T_1, \ldots, T_m$; for recursive

candidate cosets indexed by Ξ, write them as T_ξ; and for a single updated comparison, write its cost as T'. The three forms of reduction give

$$\begin{aligned} T &\le w + \sum_{j=1}^{m} T_j \quad \text{(successive comparisons)}, \\ T &\le w + \sum_{\xi\in\Xi} T_\xi \quad \text{(candidate cosets)}, \\ T &\le w + T' \qquad \text{(one updated comparison)}. \end{aligned} \tag{8.15}$$

The first bound covers the flag comparisons in Step G1, the pairs of summands in Step G4, and the at most three successive comparisons in Step T4. Each pair in Step G4 is computed once and reused. The candidate lists in the second bound come from Step G5, the marked contraction cases of Step G7, and Steps T1–T2. A restriction to one coset or the ordinary comparison produced by Step G7(c) has the third form. A local completion satisfies $T \le w$. The contraction and tuple searches, the field and scalar cases completed by the degree-2 algorithm, and the subset computation in Step G4 are included in w.

We group the computation by its reference dimensions. Only Step G3 assigns new reference dimensions to a child call. Other returns to Step G1 and the ordinary conversion in Step G7 keep the current values. For counting only, fixing candidate choices means retaining one child at each recursive candidate list and all comparisons in the first bound. The retained inputs and complete answers are those of the actual computation. The following lemma counts all work with the same reference dimensions, including the work after a call made at Step G3 returns.

Lemma 8.7 (Cost bounds with fixed reference dimensions)**.** *Start with the initial call to Algorithm 8.6, or with a call made at Step G3, whose reference dimensions are $a_i = \dim_{\mathbb{F}_p} A_i$ for $i = 1, 2, 3$. Follow all input updates and recursive calls with these reference dimensions. At a further call made at Step G3, count that call once and exclude its internal work. Include all work performed by its caller after it returns, including later comparisons and answer recovery. The simultaneous permutation in Step T0 only reindexes the reference dimensions.*

Let $d = 2e(n_1 + n_2 + n_3)$ and let InSize *be the initial input bit length. For an absolute constant C, the three operations in* (8.15)*, counted over the part of the computation just specified, satisfy the following bounds.*

(i) Successive comparisons. *For any fixed candidate choices, the total number of resulting tensor comparisons in this part of the computation is at most* $\mathrm{poly}(d, \log p)$*. This includes all successive comparisons and counts each unexpanded call made at Step G3 once.*

(ii) Candidate cosets. *For a path starting at the chosen call's input and following one resulting comparison at each reduction, let $\Xi_1, \dots, \Xi_t$ be the lists from which candidates are chosen. Stop at a local computation or at the next call made at Step G3; in the latter case, write a_i' for that call's new reference dimensions. Then*

$$\prod_{\nu=1}^{t} |\Xi_\nu| \le \begin{cases} p^{C\sum_{i=1}^{3}(a_i - a_i')}, & \text{call made at Step G3}, \\ p^{C\sum_{i=1}^{3} a_i}, & \text{local computation}. \end{cases}$$

(iii) Input updates. *For any fixed candidate choices, the cumulative number of input updates in this part of the computation is at most* $\mathrm{poly}(d, \log p)$*, including updates within nested recursive calls and after candidate choices.*

Every call made at Step G3 satisfies (8.3) *relative to the reference dimensions inherited by its caller. Assuming that these calls return their complete answers, the chosen call and all its nested calls using the same reference dimensions terminate. Each local cost satisfies*

$$w \le p^{Cd}\mathrm{poly}(\mathrm{InSize}, d, \log p).$$

Every resulting recursive tensor input of Definition 24 has bit length $\mathrm{poly}(\mathrm{InSize}, d, \log p)$*, with fixed polynomial bounds throughout the recursion.*

The proof is given in Subsubsection E.2.6.

Theorem 8.8 (Correctness of the recursive tensor algorithm)**.** *On every input of Definition 24 arising from* (8.1)*, Algorithm 8.6 terminates and computes* $g\,\mathrm{Trans}_{H,\alpha}(x, y)$*. Its output is emptiness, or a representative* gu *and generators of* $K = \mathrm{Stab}_{H,\alpha}(x)$ *such that*

$$g\,\mathrm{Trans}_{H,\alpha}(x, y) = guK.$$

In particular, the initial call computes $\mathrm{Trans}_{\prod_{i=1}^{3}\mathrm{GL}(U_i)}(\mathtt{T}, \mathtt{T}')$.

Proof. We prove together that each call terminates and returns its complete answer, by strong induction on $a_1 + a_2 + a_3$ for the initial call and the calls made at Step G3. Fix one such call and stop each recursive path at its first further call made at Step G3. Every such call has smaller reference-dimension sum by (8.3). By induction, these calls terminate and return their complete answers. Lemma 8.7 then makes the intervening computation finite, including all nested calls with unchanged reference dimensions and all work after a reset call returns. If $a_1 + a_2 + a_3 = 0$, no further reset is possible, and the same lemma gives finiteness directly.

If $N = P$, the affine space has one point and the answer is gH. Otherwise, Propositions 8.4 and 8.5 recover each comparison from the complete answers to its resulting comparisons. Within the finite computation just described, locally completed comparisons have their complete answers, and reset calls have theirs by induction. Lemmas 8.2 and 8.3 give the transporter and stabilizer identities for successive comparisons. Applying these identities from the completed comparisons back to their callers recovers the current call's answer. This includes the three successive comparisons in Step T4, whose final input may be affine.

For a family of candidates, write the nonempty completed answers as $gu_\nu K_\nu$ and choose an index ν_0. Their union is the original solution set. A nonempty transporter is a coset of the source stabilizer, so, with $K = \mathrm{Stab}_{H,\alpha}(x)$,

$$K = \bigcup_\nu u_{\nu_0}^{-1}u_\nu K_\nu = \left\langle \bigcup_\nu K_\nu, \quad u_{\nu_0}^{-1}u_\nu\ (\nu) \right\rangle.$$

Thus (8.6) returns a representative and generators of the full source stabilizer. If every branch is empty, the original transporter is empty. This proves the complete output for the current call and closes the induction. Every recursive call belongs to the computation starting at the initial call or at its most recent reset, so the assertion holds for all inputs reached by the algorithm.

The trace map in (8.1) is injective and equivariant under Γ. Hence the initial call preserves exactly the transformations between $\mathtt{T}$ and $\mathtt{T}'$, giving $\mathrm{Trans}_\Gamma(\mathtt{T}, \mathtt{T}')$. □

Theorem 8.9 (Running time of the recursive tensor algorithm)**.** *The total running time of Algorithm 8.6 on* (8.1) *is*

$$p^{O(d)}\mathrm{poly}(\mathrm{InSize}, d, \log p),$$

where $d = 2e(n_1 + n_2 + n_3)$ *and* InSize *is the initial input bit length. Thus the initial* 3*-tensor comparison takes* $q^{O(n_1+n_2+n_3)}\mathrm{poly}(\mathrm{InSize})$ *time.*

Proof. The case $d = 0$ is immediate, so assume $d \geq 1$. Along a path in the computation, the reference dimensions change only when Step G3 passes new values to a child call, apart from simultaneous reindexing in Step T0. At each such reset, (8.3) decreases $\prod_{i=1}^{3}(a_i + 1)$ by a fixed factor below 1. Hence the maximum number of resets along a path is $R = O(\log(d + 2))$. This bounds the number of resets, rather than the depth of all recursive calls.

By Theorem 8.8, the complete computation is finite. Organize it as a tree using the three recurrences in (8.15). All successive comparisons are children of their caller, with their actual inputs fixed by the completed computation; answer recovery is charged to the caller. For counting, retain one child at each candidate list and retain all successive comparisons. This selection changes no input or returned answer.

For each initial or reset call, Lemma 8.7(i),(iii) bounds the local computations and the next reset calls in the retained tree by $\mathrm{poly}(d, \log p)$. It includes calls with unchanged reference dimensions and all later work by their callers. Across at most R resets, the total number of retained local computations is therefore at most

$$\big(\mathrm{poly}(d, \log p)\big)^{R+1} = p^{O(d)}.$$

The last equality follows from $(\log(d + 2))^2 = O(d)$ and $\log(d + 2)\log(\log p + 2) = O(d \log p)$.

To count all candidate branches, retain a child independently and uniformly at each node with a nonempty candidate list. Index the local costs w, including answer recovery, by v. The probability that v is retained is

$$\omega_v := \prod_{\Xi \text{ at strict ancestors of } v} |\Xi|^{-1},$$

with an empty product equal to 1. The path to v follows caller–child edges in the fixed tree. Recovery and later comparisons remain attached to their caller, so choices inside an earlier completed child are not included on their paths. Taking expectations of the preceding bound gives

$$\sum_v \omega_v \leq \big(\mathrm{poly}(d, \log p)\big)^{R+1} = p^{O(d)}.$$

Let s_0 be the initial reference-dimension sum, and let $s_1, \ldots, s_r$ be its values immediately after the successive resets at Step G3 along the path to v. Apply Lemma 8.7(ii) to each portion of this path between resets, including the initial and final portions. If $r = 0$, use the lemma's local-computation bound directly from the initial call to v. In both cases, with an empty sum interpreted as zero, we obtain

$$\omega_v^{-1} \leq p^{C\left(\sum_{\ell=1}^{r}(s_{\ell-1} - s_\ell) + s_r\right)} = p^{Cs_0} \leq p^{Cd}.$$

Therefore the total number of local computations is at most

$$\sum_v 1 \leq p^{Cd} \sum_v \omega_v = p^{O(d)}.$$

Each costs at most $p^{Cd}\mathrm{poly}(\mathrm{InSize}, d, \log p)$ by Lemma 8.7. Its uniform input-size bound keeps the polynomial dependence on the original InSize fixed throughout the recursion. Summing these costs proves the asserted running time. Finally, $q = p^e$ and $d = 2e(n_1 + n_2 + n_3)$ give the stated bound over $\mathbb{F}_q$. □

Proof of Theorem 8.1. Theorems 8.8 and 8.9 compute either emptiness or the complete ambient transporter $aK = \mathrm{Trans}_\Gamma(\mathtt{T}, \mathtt{T}')$, where $K = \mathrm{Stab}_\Gamma(\mathtt{T})$. In the nonempty case, apply Theorem 7.3 to

aK and Γ_{in}. An empty intersection gives an empty prescribed transporter; otherwise the returned witness b and generators of $K \cap \Gamma_{\text{in}}$ describe exactly

$$\text{Trans}_{\Gamma_{\text{in}}}(\mathtt{T}, \mathtt{T}') = b(K \cap \Gamma_{\text{in}}), \qquad K \cap \Gamma_{\text{in}} = \text{Stab}_{\Gamma_{\text{in}}}(\mathtt{T}).$$

Represent both groups block-diagonally on $W = U_1 \oplus U_2 \oplus U_3$ and restrict scalars to $\mathbb{F}_p$. The two copies of W used by the intersection algorithm have total prime-field dimension $2e(n_1+n_2+n_3) = d$. Thus the intersection costs $p^{O(d)}\text{poly}(\text{InSize}, \log p)$ time, and the complete computation retains the claimed bound $q^{O(n_1+n_2+n_3)}\text{poly}(\text{InSize})$. □

8.3 A generalisation: degree-3 quotient tensor module transporter

We follow the group encoding and input/output conventions of Section 3.5. The input below is the degree-3 version of Definition 17. The representation-list and Frobenius-twist conventions of that definition also apply here.

Definition 25 (Degree-3 quotient tensor module transporter)**.** An instance over $\mathbb{F}_p$, with p prime, consists of the following data.

(i) Vector spaces $V_1, \ldots, V_s$ over $\mathbb{F}_p$.

(ii) A finite group H with representations $\rho_i : H \to \text{GL}(V_i)$ for $i \in [s]$.

(iii) A tensor module of degree at most 3,

$$Z = \bigoplus_{a \in I} \bigotimes_{j=1}^{k_a} W_{a,j}, \qquad k_a \in \{0, 1, 2, 3\},$$

where I is a finite set of summand labels fixed by H, and each $W_{a,j}$ is a specified V_i, its dual, or a Frobenius twist of either. Repetitions are allowed, and an empty tensor product means the trivial module $\mathbb{F}_p$. All tensor products are over $\mathbb{F}_p$.

(iv) Two H-invariant subspaces $P \leq N \leq Z$, with quotient coordinates for $Q := N/P$. For each $h \in H$, let $L_h : Q \to Q$ be the linear map induced by the action of h on Z as in Definition 8.

(v) Translation vectors $c_h \in Q$ for $h \in H$, specified on the given generators, such that the maps

$$\alpha_h(u) := L_h(u) + c_h \qquad (u \in Q)$$

define a group action $\alpha : H \times Q \to Q$.

(vi) Two points $x, y \in Q$.

The task is to compute

$$\text{Trans}_{H,\alpha}(x, y) = \{h \in H : \alpha_h(x) = y\}.$$

The output is emptiness, or a witness g_0 and generators of $H_x = \{h \in H : \alpha_h(x) = x\}$, so that $\text{Trans}_{H,\alpha}(x, y) = g_0 H_x$. All output elements are given in the original group encoding, with words in the original generators.

Set $d = \sum_i \dim_{\mathbb{F}_p} V_i$, including every representation needed for the original group encoding, and let InSize be the total input length, as in Definition 17. Each original representation is counted once in d; repeated summands and fixed multiplicities contribute to InSize. A multiplicity space with nontrivial action is an original representation and counts as a tensor factor. The relations in P may connect different summands. The cocycle identity

$$c_{gh} = c_g + L_g c_h$$

is required only in Q; no lift to an N-valued cocycle is assumed.

Theorem 8.10 (Degree-3 quotient tensor module transporter)**.** *Given an instance of Definition 25 with original dimension d and input length* InSize*, its complete transporter can be computed deterministically in time $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$.*

The proof of Theorem 8.10 and its supporting results are given in Subsection E.3.

9 Group isomorphism

9.1 Problem and main results

We prove the group-isomorphism result of Theorem 1.1, including computation of the isomorphism coset for groups given by their multiplication tables.

Theorem 9.1 (Group isomorphism in time polynomial in group order)**.** *Let G and H be two finite groups of order N. There is an algorithm that computes the coset* $\mathrm{Iso}(G, H)$ *in time* $\mathrm{poly}(N)$.

We prove this theorem in §9.6 from the three transport results below. We use the group-extension and cohomology conventions of Subsection 3.6. An extension with elementary abelian kernel A and quotient Q determines a conjugation action $\theta : Q \to \mathrm{GL}(A)$ and a class in $H^2(Q, A)$. The extension-isomorphism criterion (see [GQ17, Lemma 2.3]) first asks for compatible actions on A and Q, and then for a match between the two extension classes under those actions. As recalled in Subsection 3.7, first cohomology enters when comparing module extensions: for fixed Q-modules U, T, the classes in $H^1(Q, \mathrm{Hom}_{\mathbb{F}_p}(T, U))$ describe the ways of joining U and T into a module extension. Thus first-cohomology transport supports action compatibility, which in turn determines the allowed actions for second-cohomology transport.

Using the tensor-module conventions of Definition 17, the recursive comparisons take place on invariant subquotients and may be affine. We therefore use the following common problem for both cohomological degrees.

Definition 26 (Cohomology transport)**.** Fix a prime p and $j \in \{1, 2\}$. An instance consists of:

(i) a finite table group Q and an $\mathbb{F}_p Q$-module B;

(ii) a group $\Gamma = \langle \mathcal{S} \rangle \leq \mathrm{Sym}(\Omega)$ with actions $\beta : \Gamma \to \mathrm{Aut}(Q)$ and $\alpha : \Gamma \to \mathrm{GL}(B)$, given on generators and satisfying $\alpha_g(qb) = \beta_g(q)\alpha_g(b)$;

(iii) for $j = 1$, a degree-at-most-two tensor module Z_B on spaces $V_1, \ldots, V_s$ and their duals, with invariant subspaces $P_B \leq N_B \leq Z_B$ presenting $B = N_B/P_B$ and the compatible actions specified in §9.3.1; for $j = 2$, the single argument space $V_1 = B$;

(iv) a Γ-invariant subquotient $W = N/P$, specified by $P \leq N \leq H^j(Q, B)$ and quotient coordinate maps;

(v) generator values of a map $c : \Gamma \to W$ satisfying $c_{gh} = c_g + L_g c_h$, where L_g is the induced linear action on W, and two points $x, y \in W$.

The task is to compute the full transporter $\mathrm{Trans}_{\Gamma,\psi}(x, y)$ for the affine action $\psi_g(w) = L_g w + c_g$, in the original group Γ.

In this section, α denotes the compatible linear action on the coefficient module, so we use ψ for the affine action on the current quotient. Thus $\psi_h(w) = L_h(w) + c_h$ is the action denoted by α_h in the general affine-transporter conventions; the quotient linear part remains L_h.

The action on cohomology is induced by $(gf)(q_1, \ldots, q_j) = \alpha_g f(\beta_g^{-1} q_1, \ldots, \beta_g^{-1} q_j)$. Compatibility makes this action commute with the cochain differential. The translation identity is required in W; it does not require a cocycle lift to N. Write $n = |Q|$, and use InSize for the total input size as in Section 3. This includes the tensor presentations and the actions of Q in §9.3.1. For $j = 1$, put $d = \sum_{k=1}^{s} \dim_{\mathbb{F}_p} V_k$; for $j = 2$, put $d = \dim_{\mathbb{F}_p} B$. Each argument space is counted once in d, including when it or its dual occurs repeatedly. Tensor dimensions, fixed multiplicities, and quotient coordinates contribute to InSize.

Theorem 9.2 (First-cohomology transport)**.** *For the data of Definition 26 with $j = 1$, the complete affine transporter can be computed deterministically in time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$. Its terminal cubic transporter calls have total argument dimension at most $C(d + \log_p n)$, for an absolute constant C. The algorithm may use any fixed subgroup series constructed by successively choosing nontrivial characteristic elementary abelian or nonabelian simple-product subgroups in the remaining quotients, as in §9.2.1; the same bound applies when starting at a quotient corresponding to a term of this series.*

Action compatibility compares two module structures while allowing an automorphism of the group. Unlike cohomology transport, its input need not already have compatible actions on the two modules.

Definition 27 (Module action compatibility)**.** Let p be prime, let Q be a finite table group, and let M, M' be a-dimensional $\mathbb{F}_p$-spaces with representations $\theta : Q \to \mathrm{GL}(M)$ and $\theta' : Q \to \mathrm{GL}(M')$ given on a specified generating set of Q. Also given are generators of $\Gamma \leq \mathrm{Sym}(\Omega)$ and their images under $\beta : \Gamma \to \mathrm{Aut}(Q)$. Write $\beta_g = \beta(g)$. The call

$$\mathtt{M}(Q, \theta, \theta'; \Gamma, \beta) \tag{9.1}$$

asks for all pairs $(\alpha, g) \in \mathrm{Iso}_{\mathbb{F}_p}(M, M') \times \Gamma$ satisfying

$$\alpha\theta(q) = \theta'(\beta_g q)\alpha \qquad (q \in Q). \tag{9.2}$$

The answer is represented as a witness times the full source stabilizer in $\mathrm{GL}(M) \times \Gamma$, or by emptiness. The homomorphism β need not be injective, and output elements keep their matrix components and original elements of Γ.

Theorem 9.3 (Module action compatibility)**.** *Given the data of Definition 27, with $n = |Q|$ and total explicit input length* InSize*, the complete coset of compatible pairs can be computed deterministically in time $p^{O(a)} \cdot \mathrm{poly}(\mathrm{InSize})$.*

For group extensions, take $M = M' = A$ and $\Gamma = \mathrm{Aut}(Q)$. After one compatible pair identifies the two actions, its source stabilizer acts on the common space $H^2(Q, A)$. The remaining task is the second-cohomology transporter.

Theorem 9.4 (Second-cohomology transport)**.** *For the data of Definition 26 with $j = 2$, the complete affine transporter can be computed deterministically in time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$. The algorithm makes polynomially many cubic transporter calls, each with total argument dimension $O(d + \log_p n)$.*

When $A \cong \mathbb{F}_p^a$ is an elementary abelian kernel, the factor $p^{O(a)}$ in the last two theorems is polynomial in $|A|$. This gives polynomial work at each group-extension step. The group recursion stops at the quotient by the solvable radical, where the algorithm of Babai, Codenotti, and Qiao applies [BCQ12]; its full-coset output is recorded in [GQ15a, Introduction and proof of Corollary 2].

The cubic transport input. We use the complete cubic affine transport theorem (Theorem 8.10). For an affine action on an invariant subquotient of a tensor module of degree at most three, it computes the complete transporter in deterministic time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$, where d is the sum of the original argument dimensions. It allows nonfaithful argument actions and returns the answer in the original group. We refer to this result below as the *cubic transport theorem*; its proof is given in Subsubsection E.3.1, with the supporting results in Subsection E.3.

Organisation of the section. Subsection 9.2 gives the characteristic subgroup series, extension reconstruction, augmentation ideals, and inflation, and relates the shared quotient and kernel operations to these tasks. Subsections 9.3, 9.4, and 9.5 then give the first-cohomology, action-compatibility, and second-cohomology algorithms, in that order, with their recursive data and running-time analyses. Subsection 9.6 combines them to prove Theorem 9.1. Appendix F contains the technical results and proofs in the corresponding order.

9.2 Preliminaries

We use the notation and computing conventions of Section 3, in particular the cohomology conventions of Subsection 3.6 (Group extensions and cohomology), the module and affine-transport constructions of Sections 3.7 and 3.8, and the group algorithms of Section 3.5. All dimensions below are over $\mathbb{F}_p$ unless another field is specified. For an S-module B, write $\mathrm{Fix}_S(B) = \{b \in B : sb = b \text{ for all } s \in S\}$.

9.2.1 A characteristic subgroup series

For an initial table group Q, construct a series

$$1 = N_0 < \cdots < N_j = \mathrm{srad}(Q) < \cdots < N_\ell = Q \tag{9.3}$$

of subgroups characteristic in Q, where $0 \le j \le \ell$; either segment may be empty. The construction is as follows. For a nontrivial group K with $\mathrm{srad}(K) \neq 1$, let D be the last nonidentity term of the derived series of $\mathrm{srad}(K)$, let r be the least prime dividing $|D|$, and let D_r be the unique Sylow r-subgroup of D. As in the overview, set $\mathrm{AC}(K) = \{x \in D_r : x^r = 1\}$ and $S(K) = \mathrm{AC}(K)$. If $\mathrm{srad}(K) = 1$, set $S(K) = \mathrm{Soc}(K)$, a direct product of nonabelian simple groups. Define N_{i+1} as the preimage of $S(Q/N_i)$ in Q. Both choices make $S(K)$ characteristic in K: in the first case, the derived-series term D, its unique Sylow r-subgroup, and the subgroup of elements of order dividing r are all characteristic; in the second case, the socle is characteristic. Now $N_0 = 1$ is characteristic in Q. If N_i is characteristic in Q, every automorphism of Q induces an automorphism of Q/N_i preserving $S(Q/N_i)$, and hence preserves its preimage N_{i+1}. Thus every N_i is characteristic in the original group Q. The selection is preserved by isomorphisms, every inclusion is strict, and $\ell \le \lfloor \log_2 |Q| \rfloor$.

The series and its subgroup and quotient tables and maps are computable in polynomial table time by Section 3.5.

To see why the series reaches $\operatorname{srad}(Q)$, suppose $N_i < \operatorname{srad}(Q)$. Then $\operatorname{srad}(Q/N_i) = \operatorname{srad}(Q)/N_i \neq 1$, so the construction chooses a nontrivial elementary abelian subgroup $S(Q/N_i) \leq \operatorname{srad}(Q)/N_i$. Its preimage therefore satisfies $N_i < N_{i+1} \leq \operatorname{srad}(Q)$. These strict inclusions must reach $\operatorname{srad}(Q)$, giving the index j in (9.3). Thus all factors below N_j are elementary abelian; $j = 0$ when $\operatorname{srad}(Q) = 1$, and $j = \ell$ when Q is solvable. If $j < \ell$, the next factor is $N_{j+1}/N_j = \operatorname{Soc}(Q/\operatorname{srad}(Q))$, a product of nonabelian simple groups. Later quotients can again have nontrivial solvable radicals. The series depends only on Q, not on the module or its field characteristic.

The group-isomorphism algorithm uses this construction for G only through $\operatorname{srad}(G)$, and applies Babai–Codenotti–Qiao to $G/\operatorname{srad}(G)$ [BCQ12], using its full isomorphism-coset output. The cohomology algorithms use all of (9.3), stopping at $N_\ell = Q$, or earlier when the current cohomology subquotient is zero. In an extension step with $Q = G/\operatorname{AC}(G)$, its series is obtained by taking the images of the terms above $\operatorname{AC}(G)$ in the series for G.

At position i, put $R_i = Q/N_i$ and write $\nu_i : Q \twoheadrightarrow R_i$ for the quotient map. For $i < \ell$, put $S_i = N_{i+1}/N_i = S(R_i) \leq R_i$ and write $\pi_i : R_i \twoheadrightarrow R_{i+1}$ for the quotient by S_i. Thus $\nu_{i+1} = \pi_i \nu_i$. Every automorphism of Q induces automorphisms of these quotients preserving the selected factors. The series and maps are fixed background data throughout an execution: advancing to $i+1$ changes the domain to R_{i+1}, whereas replacing the module by a subquotient keeps i fixed. A call starting at position i uses the induced series $1 = N_i/N_i < \cdots < N_\ell/N_i = R_i$.

9.2.2 Extension coordinates and reconstruction

For an extension of $A \cong \mathbb{F}_p^a$ by Q, use a section $s : Q \to G$ with $s(1) = 1$, as in Subsection 3.6. Writing (u, q) for $us(q)$ gives

$$(u, q)(v, r) = (u + \theta(q)v + f(q, r), qr), \tag{9.4}$$

where θ and f are its action and normalised 2-cocycle. Within the cochain spaces $C^j(R, B)$ of Subsection 3.6, we compute using the subcomplex of normalised cochains, which vanish whenever an argument is the identity; in degree zero, put $C^0(R, B) = B$. This subcomplex computes the same cohomology [Bro82, Chapters I–III]. Degree-three cochains are needed only for the 2-cocycle equations. The resulting linear systems have at most $a|Q|^3$ coordinates; their matrices and quotient maps contribute to InSize.

After identifying characteristic kernels with A and quotients with Q, the extension-isomorphism criterion is [GQ17, Lemma 2.3]. In the coordinates above, $F(u, q) = (\alpha u + b(q), \beta q)$ is an isomorphism precisely when

$$\alpha\theta_G(q) = \theta_H(\beta q)\alpha, \tag{9.5}$$

$$b(q) + \theta_H(\beta q)b(r) - b(qr) = \alpha f_G(q, r) - f_H(\beta q, \beta r). \tag{9.6}$$

Here $(\alpha, \beta) \in \mathrm{GL}(A) \times \mathrm{Aut}(Q)$ and $b \in C^1(Q, A)$. A pair satisfying (9.5) is *compatible*. Its induced map on cochains is $(T_{\alpha,\beta} f)(q_1, \ldots, q_j) = \alpha f(\beta^{-1} q_1, \ldots, \beta^{-1} q_j)$. It commutes with the differentials, and (9.6) is solvable exactly when $[T_{\alpha,\beta} f_G] = [f_H]$. Thus the algorithm matches the actions, matches the extension classes, and solves this linear equation to reconstruct the isomorphisms.

9.2.3 Augmentation ideals

Definition 28 (Augmentation ideal)**.** For a finite group S and a prime p, the *augmentation map* $\epsilon : \mathbb{F}_p[S] \to \mathbb{F}_p$ sends $\sum_{s \in S} a_s s$ to $\sum_{s \in S} a_s$. Its kernel $I_S = \operatorname{Span}_{\mathbb{F}_p}\{s - 1 : s \in S\}$ is the *augmentation*

ideal. For an $\mathbb{F}_pS$-module B, $I_SB = \mathrm{Span}_{\mathbb{F}_p}\{sb - b : s \in S,\ b \in B\}$, and its powers give $B = I_S^0B \geq I_SB \geq I_S^2B \geq \cdots$. In particular, $I_SB = 0$ exactly when S acts trivially on B.

For the elementary abelian p-groups used below, Lemma 9.8 shows that this filtration reaches zero and gives its algorithmic properties.

9.2.4 Restriction and inflation

Fix $i < \ell$ and an $\mathbb{F}_pR_i$-module B. Restriction of cochains to S_i induces $\mathrm{res} : H^j(R_i, B) \to H^j(S_i, B)$ for $j = 1, 2$. The space $\mathrm{Fix}_{S_i}(B)$ is an R_{i+1}-module. The *inflation* of a cocycle $\bar{f} : R_{i+1}^j \to \mathrm{Fix}_{S_i}(B)$ is its pullback along π_i, followed by the inclusion into B: $(\inf \bar{f})(q_1, \ldots, q_j) = \bar{f}(\pi_i(q_1), \ldots, \pi_i(q_j))$. For $j = 1$ this is $q \mapsto \bar{f}(qS_i)$, and for $j = 2$ it is $(q, r) \mapsto \bar{f}(qS_i, rS_i)$ [Wei94, Section 6.7, especially 6.7.3 and 6.7.6].

Both maps commute with the differential. In degree one, inflation identifies $H^1(R_{i+1}, \mathrm{Fix}_{S_i}(B))$ with the restriction kernel, as used in Step R3. In degree two, it need not be injective or exhaust that kernel; Lemma F.1 specifies the remaining obstruction and image quotients. All these maps, kernels, and images are computed by linear algebra in cochain coordinates.

9.2.5 Applying the quotient–fibre operation

Every quotient match is followed by the affine equation in its kernel from Lemma 3.5, using the full returned coset. Induced filtrations are those of Lemma D.2 and (D.9); changes of quotient presentation use Fact 3.4. If the linear action has a kernel, Lemmas 3.6 and 3.1 compute its translation image and restore all original group elements. On tensor arguments of total dimension d, these kernel computations cost $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$ by Section 3.5. After a partial quotient condition, use the full inverse image of its transporter; complete the kernel correction after imposing the whole current condition.

9.3 First-cohomology transport

9.3.1 Recursive data and algorithm

The algorithm first compares a restriction to a characteristic subgroup, then solves the remaining affine equation over the next quotient. If an elementary abelian p-subgroup acts nontrivially on B, first filter B into smaller modules on which that subgroup acts trivially. We specify the calls here; their exact quotient formulas follow the algorithm. The technical proofs appear in §F.1.

Background data fixed throughout the recursion. Fix a prime p, the initial table group Q, and the original group $\Gamma \leq \mathrm{Sym}(\Omega)$ with its action $\beta : \Gamma \to \mathrm{Aut}(Q)$. Construct the characteristic subgroup series $1 = N_0 < \cdots < N_\ell = Q$ of (9.3) once. At position i, the current group is $R_i = Q/N_i$ and B is an $\mathbb{F}_pR_i$-module; for $i < \ell$, the next factor is $S_i = N_{i+1}/N_i \leq R_i$, with $R_i/S_i = R_{i+1}$. Store the subgroups, quotient tables, and natural maps with the series. The series, Q, Γ, and β stay fixed throughout the execution. Write β_i for the induced action of Γ on R_i, so $\nu_i\beta_g = \beta_i(g)\nu_i$. Advancing from i to $i + 1$ uses the next term N_{i+1}; replacing B by a filtration grade D_j keeps i fixed. The tensor arguments are indexed separately as V_k, $1 \leq k \leq s$.

Actions on the tensor arguments and on B. For the tensor presentation in Definition 26, the summand labels are fixed and repeated occurrences of a V_k use the same action, with the induced action on its dual. The argument spaces carry actions ρ_k of the fixed initial group Q. Their action

matrices $\alpha_k(g)$ for $g \in \Gamma$ satisfy $\alpha_k(g)\rho_k(q) = \rho_k(\beta_g(q))\alpha_k(g)$ for $q \in Q$. The numerator N_B and denominator P_B are invariant under these Q- and Γ-actions. The induced action on $B = N_B/P_B$ is the given R_i-action pulled back along ν_i, and its induced Γ-action is α. Thus $N_i = \ker \nu_i$ acts trivially on B, although it may act nontrivially on the individual V_k. As i increases, the V_k keep their Q-actions; only the quotient R_i changes.

At a recursive call, let $H \leq \Gamma$ be the current subgroup and let $t \in \Gamma$ represent the current candidate coset tH. This coset consists of the transformations still allowed by the previous comparisons. Write

$$\mathcal{P}_B = (\nu_i; (V_k, \rho_k, \alpha_k)_{k=1}^{s}; Z_B, N_B, P_B)$$

for the explicit tensor presentation of B in Definition 26: ν_i maps Q to R_i, ρ_k is the action of Q on V_k, α_k is the action of H on V_k, and $B = N_B/P_B$. All inclusions, quotient maps, bases, and generator matrices belong to this tuple. These data accompany every call. If $\pi_B : N_B \to B$ is the quotient projection and $D = B'/B''$ is an invariant subquotient of B, its inherited presentation has numerator $\pi_B^{-1}(B')$, denominator $\pi_B^{-1}(B'')$, and the same argument spaces and actions.

Definition 29 (Data and answer of a first-cohomology call)**.** With this fixed background, a recursive call is

$$\mathtt{H}_1(t, H; i, B, \mathcal{P}_B; W = N/P, c; x, y), \tag{9.7}$$

where $0 \leq i \leq \ell$ and R_i is the table group at position i, $W = N/P$ includes its H-invariant numerator and denominator $P \leq N \leq H^1(R_i, B)$, with their bases and quotient projection and lifting coordinate maps. The entry c specifies the affine action $\psi_h(z) = L_h z + c_h$ on W. Its answer is $t \operatorname{Trans}_{H,\psi}(x, y)$. It is represented by emptiness or a witness ta and generators of the full source stabilizer H_x, giving the coset taH_x in the original group Γ. At the root, $t = 1$ and $H = \Gamma$.

Outline of the algorithm. At a recursive call, the task is to find the elements $h \in H$ for which $\psi_h(x) = y$ in $W = N/P$. Algorithm 9.5 uses the next factor $S_i = N_{i+1}/N_i \leq R_i$ to split this task into a restriction comparison and a comparison over R_{i+1}. Step R1 first prepares the module when necessary; Steps R2 and R3 then perform these two comparisons. Restriction and inflation are the maps of §9.2.4.

R1. Prepare the module. If S_i is elementary abelian of characteristic p and acts nontrivially on B, use the augmentation filtration of B. Its grades D have trivial S_i-action and induce a filtration of W. Compare the induced quotients, from the highest grade downwards, by recursive calls on subquotients of $H^1(R_i, D)$, using each returned coset to pass to the next grade. These calls keep i fixed and use smaller modules. If this filtration is unnecessary, proceed directly to R2.

R2. Compare the restrictions. The restriction map $\operatorname{res} : H^1(R_i, B) \to H^1(S_i, B)$ sends a cocycle class $[z]$ to $[z|_{S_i}]$. Its images $\operatorname{res}(N)$ and $\operatorname{res}(P)$ define the quotient map $q : W \to \operatorname{res}(N)/\operatorname{res}(P)$, given by $q(u + P) = \operatorname{res}(u) + \operatorname{res}(P)$ for $u \in N$. First solve $q(\psi_h(x)) = q(y)$ for $h \in H$. For an elementary abelian p-factor, R1 ensures that S_i acts trivially on B, so the ambient space is $H^1(S_i, B) = B \otimes S_i^*$, and the comparison is a terminal tensor call. For an elementary factor of characteristic different from p, the restriction space is zero. For a product of nonabelian simple groups, use Proposition 9.10.

R3. Continue over the quotient. If the restrictions match, use the full returned coset to form the remaining affine equation on a subquotient of $\ker(\operatorname{res})$. Inflation identifies this kernel with $H^1(R_{i+1}, \operatorname{Fix}_{S_i}(B))$. If the remaining subquotient is zero, return the coset immediately.

Otherwise transfer the equation through this identification and make one recursive call at position $i+1$. The module becomes $\mathrm{Fix}_{S_i}(B)$, with its inherited tensor presentation on the same argument spaces.

All cubic-solver calls in this algorithm occur in R2, including those made through Proposition 9.10. R1 and R3 reach the cubic solver only through recursive calls. An empty answer at any stage ends the current call. The exact quotients, affine updates, and supporting statements are given in §9.3.2; their longer proofs are in §F.1.

Progress and termination. The lexicographic measure $(\ell - i, \dim B)$ strictly decreases at every recursive call: R1 keeps i fixed and decreases $\dim B$, whereas R3 increases i and hence decreases $\ell - i$. R2 is a terminal comparison. The base case is $W = 0$, which always holds when $i = \ell$ and $R_i = 1$. When R_i has trivial solvable radical and $W \neq 0$, R2 compares the restriction to its socle; any remaining comparison continues to the next quotient in R3.

Relation to quotient tensor-module transport. The degree-two quotient tensor-module transporter of §7 uses the same quotient–fibre procedure: first solve a projected equation, then use its full returned coset to solve the remaining affine equation in the kernel (Lemma 3.5). Both algorithms also use invariant module filtrations. The additional structure here is the domain of a cocycle: its arguments lie in $R_i = Q/N_i$, and the action changes both those arguments and the values in B. The fixed characteristic series decomposes this domain group Q, while Γ records the allowed transformations. The required quotient maps now come from module maps on cohomology, restriction, and inflation; their images and kernels must be computed in cocycle and coboundary coordinates.

This domain decomposition leads to two kinds of restriction problem. For an elementary abelian p-factor acting trivially on a grade D, the identity $H^1(S_i, D) = D \otimes S_i^*$ adds one tensor argument to the presentation of D. Thus a presentation of degree at most two leads to a cubic transporter call. Products of nonabelian simple groups require a different construction: Proposition 9.10 accounts for inner automorphisms and permutations of the simple factors to obtain the required tensor presentation. These simple groups occur in the cocycle's domain, a different role from simple groups acting on tensor arguments in the tensor-module algorithm. In both cases the extra arguments are used only for the restriction comparison; the inflation continuation uses the original argument spaces.

Notation reminder. For $i < \ell$, recall that $\mathrm{Fix}_{S_i}(B) = \{b \in B : sb = b \text{ for all } s \in S_i\}$ and that I_{S_i} is the augmentation ideal of $\mathbb{F}_p[S_i]$ (Definition 28). The map res is restriction from R_i to S_i. When S_i is elementary abelian of characteristic p, write $S_i^* = \mathrm{Hom}_{\mathbb{F}_p}(S_i, \mathbb{F}_p)$. For a quotient cocycle $\bar{z} : R_i/S_i \to \mathrm{Fix}_{S_i}(B)$, its inflation, as in §9.2.4, is $q \mapsto \bar{z}(qS_i)$ on R_i. The coset representative t records earlier matches; it need not act on B.

Algorithm 9.5 (First-cohomology transport along a fixed subgroup series)**. Fixed background:** the prime p; the characteristic subgroup series $1 = N_0 < \cdots < N_\ell = Q$, with subgroup and quotient tables and natural maps; and the original group $\Gamma \leq \mathrm{Sym}(\Omega)$, with its action $\beta : \Gamma \to \mathrm{Aut}(Q)$ given on generators. Write $\nu_i : Q \twoheadrightarrow R_i$ for the natural quotient map.
Recursive input: a call

$$\mathtt{H}_1(t, H; i, B, \mathcal{P}_B; W = N/P, c; x, y),$$

with the following data.

- A position $0 \leq i \leq \ell$, with current group $R_i = Q/N_i$. For $i < \ell$, the selected subgroup is $S_i = N_{i+1}/N_i$, and the next group is $R_{i+1} = Q/N_{i+1} = R_i/S_i$.

- A current subgroup $H \leq \Gamma$, given by generators, and a current coset representative $t \in \Gamma$.
- An $\mathbb{F}_p R_i$-module B with compatible H-actions on R_i and B specified on generators. Its explicit tensor presentation $\mathcal{P}_B$ specifies $B = N_B/P_B$ in a tensor module of degree at most two on Q-modules V_k and their duals, the factor actions, and the quotient map $\nu_i : Q \twoheadrightarrow R_i$, with the bases, coordinate maps, and compatibility data of §9.3.1.
- A quotient $W = N/P$, specified by H-invariant subspaces $P \leq N \leq H^1(R_i, B)$ and quotient coordinates, and points $x, y \in W$. Write L_h for the induced linear action of $h \in H$ on W. Generator values of a cocycle $c : H \to W$, satisfying $c_{hk} = c_h + L_h c_k$, specify the affine action $\psi_h(z) = L_h z + c_h$.

Output: the set $t\,\mathrm{Trans}_{H,\psi}(x, y) = \{th : h \in H,\ \psi_h(x) = y\}$, represented by emptiness or a witness ta and generators of the source stabilizer $K = \{h \in H : \psi_h(x) = x\}$, giving the coset taK in the original group Γ.

B1. **If** $W = 0$ or $i = \ell$, **return** tH.

Otherwise execute R1–R3 below. Compute the maps induced by module homomorphisms, together with restriction and inflation, in normalised cochain coordinates.

R1. Filter B when an elementary abelian p-subgroup acts nontrivially.

(a) **If** S_i is not elementary abelian of characteristic p, or acts trivially on B, **continue to R2**. Otherwise compute the augmentation filtration $B_0 = B$, $B_{k+1} = I_{S_i} B_k$, ending at zero. Order its nonzero grades $D_1, \ldots, D_e$ from bottom to top, as in Lemma 9.8.

(b) Compute the induced filtration $0 = W_0 \leq \cdots \leq W_e = W$ and the subquotient presentations of its grades by (9.10). For each j, record the quotient map $q_j : W_j \to W_j/W_{j-1}$, its realization in $H^1(R_i, D_j)$, and the inherited tensor presentation of D_j. If $W_j/W_{j-1} = 0$, B1 handles the recursive call.

(c) Initialize the current tuple on W_e with coset representative t, group H, source x, target y, and cocycle c. For $j = e, e-1, \ldots, 1$, do the following:

(i) **Call** $\mathtt{H}_1$ recursively with coset representative 1, the current group, position i, module D_j, and quotient W_j/W_{j-1} with its presentation in $H^1(R_i, D_j)$. The affine data are the images of the current data under q_j.

(ii) **If** the answer is empty, **return** emptiness. Otherwise write it as aK. If $j = 1$, **return** the current coset representative times aK.

(iii) Replace the current tuple by its fibre tuple on W_{j-1} using (9.8): multiply its coset representative on the right by a, replace the group by K, replace the source by 0 and the target by $\psi_{a^{-1}}(y) - x$, and replace the cocycle by $k \mapsto \psi_k(x) - x$. Here x, y, ψ are those before this update.

R2. Restriction to S_i.

(a) Set

$$F = \ker\big(H^1(R_i, B) \longrightarrow H^1(S_i, B)\big), \qquad W_F = ((N \cap F) + P)/P \leq W.$$

Compute $q : W \to W/W_F$ and identify its target with $\mathrm{res}(N)/\mathrm{res}(P)$ in (9.12).

(b) Compare $q(x), q(y)$ with group H, coset representative 1, and cocycle $q \circ c$. If S_i is elementary abelian of characteristic p, use $H^1(S_i, B) = B \otimes S_i^*$ and the cubic solver of the cubic transport theorem. If S_i is a product of nonabelian simple groups, use Proposition 9.10. In the coprime elementary case, set the answer to $1H$.

(c) **If** the answer is empty, **return** emptiness. Otherwise write it as aK. **If** $W_F = 0$, **return** taK. Otherwise **continue to R3** with this aK.

R3. Inflation from the next quotient.

(a) Set $w = \psi_{a^{-1}}(y) - x$ and $\beta_k(z) = L_k z + \psi_k(x) - x$ for $k \in K$ and $z \in W_F$. Pull this affine comparison back through the inflation isomorphism $H^1(R_{i+1}, \mathrm{Fix}_{S_i}(B)) \longrightarrow F$ of (9.11). Use the numerator and denominator in (9.13) and denote the resulting affine data by $(W' = N'/P', c'; 0, w')$.

(b) **Return the recursive call**

$$\mathtt{H}_1(ta, K; i+1, \mathrm{Fix}_{S_i}(B), \mathcal{P}_{\mathrm{Fix}_{S_i}(B)}; W' = N'/P', c'; 0, w').$$

The tensor presentation of $\mathrm{Fix}_{S_i}(B)$ uses the old argument spaces and Q-actions, with numerator the preimage of $\mathrm{Fix}_{S_i}(B)$ and denominator P_B. Replace ν_i by $\nu_{i+1} = \pi_i \nu_i : Q \to R_{i+1}$.

Every recursive call starts at B1. The parent keeps its current affine data until the quotient answer and subsequent fibre answer have both returned.

In Step R2, quotient images and kernels are computed before forming the affine comparison. Thus the denominator includes the image of P, and the full restriction space is not substituted for the actual restriction image. In Step R3, N_{i+1} acts trivially on $\mathrm{Fix}_{S_i}(B)$, although it may still act nontrivially on the individual tensor arguments. Those original actions remain part of the tensor presentation, as required in Definition 26.

9.3.2 Operations used by the algorithm

We now specify the quotient spaces and affine equations used in Algorithm 9.5. Invariant subspaces below are invariant under the linear part L_h of the H-action. The affine translations are projected or changed by the quotient–fibre formula; they need not preserve the individual subspaces of a filtration.

The common quotient–fibre update. Let $q : W \to W'$ be an equivariant surjection, put $F = \ker q$, and compare $q(x), q(y)$ under the induced affine action of H. If that comparison returns aK, then for $k \in K$ set

$$w = \psi_{a^{-1}}(y) - x, \qquad \beta_k(z) = L_k z + \psi_k(x) - x \quad (z \in F). \tag{9.8}$$

The vector w and the cocycle values $\psi_k(x) - x$ lie in F. The remaining call has group K, coset representative ta, space F, source 0, target w, and affine action β. If it returns $(ta)bK_0$, that is the parent's answer. An empty quotient or fibre answer makes the parent answer empty. For $F = 0$, return taK. When F is represented by a different cohomology subquotient, transfer all these data through its computed coordinate isomorphism. This update is exactly Lemma 3.5; it applies to both cohomological degrees.

Module grades in Step R1. Reindex the descending filtration of B from Step R1 in ascending order as $0 = B_0 \leq \cdots \leq B_e = B$, with $D_k = B_k/B_{k-1}$. At the fixed position i, let $j \in \{1, 2\}$,

since the same formulas will also be used in §9.5. Applying cohomology to a short exact sequence of modules gives a sequence exact at its middle term:

$$H^j(R_i, B') \longrightarrow H^j(R_i, B) \longrightarrow H^j(R_i, B''). \tag{9.9}$$

Here $0 \to B' \to B \to B'' \to 0$ is any short exact sequence of $\mathbb{F}_p R_i$-modules. For the chosen filtration define

$$\phi_k : H^j(R_i, B_k) \longrightarrow H^j(R_i, B), \qquad \lambda_k : H^j(R_i, B_k) \longrightarrow H^j(R_i, D_k), \qquad F_k = \operatorname{im} \phi_k.$$

For the actual input $W = N/P$, the recursive quotients are

$$\begin{aligned} W_k &= ((N \cap F_k) + P)/P, \\ W_k/W_{k-1} &\cong N^{(k)}/P^{(k)}, \\ N^{(k)} &= \lambda_k(\phi_k^{-1}(N)), \qquad P^{(k)} = \lambda_k(\phi_k^{-1}(P)). \end{aligned} \tag{9.10}$$

In particular, $P^{(k)}$ includes $\lambda_k(\ker \phi_k)$. For $W = H^j(R_i, B)$ this reduces to $N^{(k)} = \operatorname{im} \lambda_k$ and $P^{(k)} = \lambda_k(\ker \phi_k)$. The recursive call uses $j = 1$. It compares the highest remaining grade, applies (9.8), and proceeds to the next grade with the returned stabilizer. Proposition 9.6 justifies these formulas without assuming that the filtration of B splits.

Restriction and inflation in Steps R2–R3. Let res and inf denote restriction and inflation. The exact sequence

$$0 \longrightarrow H^1(R_{i+1}, \operatorname{Fix}_{S_i}(B)) \xrightarrow{\inf} H^1(R_i, B) \xrightarrow{\operatorname{res}} H^1(S_i, B) \tag{9.11}$$

identifies $F = \ker \operatorname{res}$ with $H^1(R_{i+1}, \operatorname{Fix}_{S_i}(B))$. For the current $W = N/P$, set $W_F = ((N \cap F) + P)/P$. Then

$$W/W_F \cong \operatorname{res}(N)/\operatorname{res}(P), \qquad W_F \cong (N \cap F)/(P \cap F). \tag{9.12}$$

After the restriction match returns aK, the quotient–fibre formula gives the remaining target and affine action on W_F. The next call uses the fixed submodule $\operatorname{Fix}_{S_i}(B)$ and

$$N' = \inf^{-1}(N \cap F), \qquad P' = \inf^{-1}(P \cap F). \tag{9.13}$$

Transfer the target and cocycle through the induced isomorphism $W' = N'/P' \cong W_F$. Thus there is only one continuing cohomology call after each restriction comparison. Its quotient map is $\nu_{i+1} = \pi_i \nu_i$; the tensor presentation uses the original argument actions and the preimage of $\operatorname{Fix}_{S_i}(B)$ in its numerator. Proposition 9.6 justifies the restriction and inflation identifications.

Exactness and coprime inflation. The following claims specify the cohomological identifications used in Steps R1–R3. The elementary and simple-product restriction constructions follow below.

Proposition 9.6 (Exact subquotients for the recursive calls)**.** *The formulas for module grades* (9.10) *hold in degrees $j = 1, 2$. The restriction and inflation formulas* (9.12)–(9.13) *hold in degree one. All these identifications commute with compatible transformations of R_i and the modules involved, and are computable by linear algebra on normalised cochains.*

The remaining elementary case in Step R2 has $|S_i|$ prime to p. Its restriction cohomology vanishes, and the following shared fact justifies passing directly to inflation.

Lemma 9.7 (Coprime inflation)**.** *Let $S \trianglelefteq R$ and let B be an $\mathbb{F}_p R$-module. If $p \nmid |S|$, inflation gives an equivariant isomorphism $H^j(R/S, \mathrm{Fix}_S(B)) \cong H^j(R, B)$ for $j = 1, 2$. The isomorphism and its inverse are computable by linear algebra on normalised cochains.*

Proof. Averaging gives $H^k(S, B) = 0$ for every $k > 0$. Inflation–restriction, or the corresponding collapse of the Lyndon–Hochschild–Serre spectral sequence, then gives the stated isomorphism [Bro82, Chapter III]. Apply the inflation map to cocycles and invert its induced matrix by elimination. □

The maps induced by module homomorphisms and the restriction maps above commute with the compatible actions on R and its modules. Applying them to the input numerator and denominator therefore gives invariant subquotients. The affine cocycle is projected by the same maps; after each quotient match its fibre cocycle is computed by Lemma 3.5. No equivariant choice of cocycle representatives is required.

Elementary restrictions and the augmentation filtration. Step R1 of Algorithm 9.5 filters B into smaller modules on which the elementary subgroup acts trivially. Step R2 then compares the restriction image on each grade using one additional tensor argument. The following two lemmas justify these constructions.

Let R be a finite group and $C \trianglelefteq R$ an elementary abelian normal subgroup, with $C \cong \mathbb{F}_p^r$, and let B be an $\mathbb{F}_p R$-module. The Γ-action on R preserves C and is compatible with its action on B. The next two lemmas apply when C is an elementary abelian p-group. The coprime case follows from Lemma 9.7.

Lemma 9.8 (Module filtration for Step R1)**.** *Let I_C be the augmentation ideal of $\mathbb{F}_p[C]$ from Definition 28. Set $B_k = I_C^k B$ for $k \geq 0$ and $D_k = B_k/B_{k+1}$. Then $B_0 = B$, and the B_k form a descending filtration by R-submodules preserved by Γ. Every nonzero B_k properly contains B_{k+1}, and C acts trivially on every grade D_k. There are at most $\dim_{\mathbb{F}_p} B$ nonzero grades, and their dimensions sum to $\dim_{\mathbb{F}_p} B$. The filtration and its induced actions are computable by linear algebra on the given action matrices.*

When C acts nontrivially on B, every nonzero D_k has smaller dimension than B. These are the smaller modules used by Step R1. The step applies the maps in (9.9) to obtain image filtrations in cohomology; it does not replace B by a direct sum of its grades. In particular, the numerator, denominator, and quotient maps of each induced cohomology subquotient are part of the computation. This use of the filtration of B applies to both H^1 and H^2.

Lemma 9.9 (Elementary restriction for Step R2)**.** *Let D be a filtration grade of B on which C acts trivially. There is an equivariant identification $H^1(C, D) = \mathrm{Hom}_{\mathbb{F}_p}(C, D) = D \otimes_{\mathbb{F}_p} C^*$, where $C^* = \mathrm{Hom}_{\mathbb{F}_p}(C, \mathbb{F}_p)$. Under this identification, $g \in \Gamma$ acts by $f \mapsto \alpha_g \circ f \circ \beta_g^{-1}$. Consequently the restriction comparison adds one tensor occurrence, with new moving argument of dimension $r = \log_p |C|$. For $P \leq N \leq H^1(R, D)$, the quotient compared in Step R2 is $\mathrm{res}(N)/\mathrm{res}(P)$, as a subquotient of $H^1(C, D)$. The remaining fibre is a subquotient of $H^1(R/C, D)$ and is the only continuing cohomology task for this grade.*

The elementary restriction is therefore a terminal tensor comparison. Its new argument contributes $p^{O(r)} = |C|^{O(1)}$ to the exponential running time. The dimension of D is handled by its inherited tensor presentation; it is not charged as a new argument dimension. The corresponding degree-two restriction formulas are given in §9.5.

Restriction to products of simple groups. This is the remaining construction used in Step R2. It gives the terminal comparison bound used in §9.3.3. We give the common interface for both cohomological degrees so that §9.5 can use the same construction.

Proposition 9.10 (Restriction to a product of simple groups)**.** *Let $S \trianglelefteq R$ be a direct product of nonabelian finite simple groups, and let B be an $\mathbb{F}_p R$-module. Let Γ act compatibly on R and B, preserving S. Suppose B has an explicit tensor subquotient presentation of degree at most e on argument spaces of total dimension d, with compatible actions of an initial group Q and a quotient map $Q \twoheadrightarrow R$ as in §9.3.1. Let $j \in \{1, 2\}$ and $e + j \leq 3$. Every affine transport problem on an invariant subquotient of $H^j(S, B)$ reduces to at most $|S|$ cubic affine transporter calls. Each has moving argument dimension at most $C(d + \log_p |S|)$ and explicit input length polynomial in the original data, apart from any faithful vector-set realization of these moving arguments.*

The complete answers are combined and returned in the original group Γ before any subsequent quotient step. Temporary covering groups and corrected actions are not passed to that step. The only additional moving argument is the permutation module $Y = \mathbb{F}_p^{\tau_p}$ on the simple factors whose orders are divisible by p. It satisfies

$$\dim_{\mathbb{F}_p} Y = \tau_p, \qquad p^{\tau_p} \leq |S|. \tag{9.14}$$

A degree-j restriction uses at most j occurrences of Y; its dimension is counted once, and fixed element labels contribute only to the explicit coordinate size.

The longer proofs, including the construction of the simple-product cochain presentation and its comparison with ordinary cochains, are given in §F.1.

9.3.3 Correctness and running time

Proposition 9.11 (Correctness and termination of the recursion)**.** *Assuming the local constructions in §9.3.2, Algorithm 9.5 returns the complete coset specified in Definition 29, and every call terminates.*

Together with Proposition 9.11, the following bounds prove Theorem 9.2. We use the construction guarantees of Lemmas 9.8 and 9.9 and Proposition 9.10, whose proofs appear in §F.1. Put $b = \dim B$ for the initial module, $n = |Q|$, and use ℓ for the length of the fixed subgroup series (9.3). If $b = 0$ or $n = 1$, the algorithm returns immediately. Otherwise b is bounded by the explicit input length even when it is much larger than d.

The two running-time recurrences. Write $T_i(B)$ for the cost of a call at position i, suppressing its affine data and current subgroup H. Each term below uses the data returned by the preceding comparison. The recurrences apply when $i < \ell$; at the final position B1 returns immediately. Let $A_i(B)$ bound the local linear algebra, quotient–fibre updates, and encoding operations; let $C_i(B)$ bound the terminal restriction comparison. In Step R1, the nonzero module grades satisfy

$$T_i(B) \leq A_i(B) + \sum_{k=1}^{e} T_i(D_k), \qquad \sum_{k=1}^{e} \dim D_k = \dim B, \qquad 0 < \dim D_k < \dim B. \tag{9.15}$$

Each D_k has trivial S_i-action, so its call proceeds directly to restriction. Whenever Step R1 is skipped, Steps R2–R3 give

$$T_i(B) \leq A_i(B) + C_i(B) + T_{i+1}(\mathrm{Fix}_{S_i}(B)), \qquad \dim \mathrm{Fix}_{S_i}(B) \leq \dim B. \tag{9.16}$$

The last term is omitted if the fibre is zero or the restriction match is empty. A simple-product restriction uses at most $|S_i|$ cubic calls, whose answers are combined before the single call at position $i+1$. Thus this factor multiplies $C_i(B)$, not $T_{i+1}(\mathrm{Fix}_{S_i}(B))$. The dimension and encoding bounds below give a common bound for all local costs in terms of the initial input.

Closure and moving dimensions. Every continuing module is a subquotient of the initial module B. Its quadratic tensor presentation uses the original arguments. Taking a grade or the fixed space $\mathrm{Fix}_{S_i}(B)$ changes only the numerator and denominator, without introducing a new argument. A terminal elementary restriction adds one occurrence of S_i^*, of dimension $\log_p |S_i|$. A terminal nonabelian restriction adds one occurrence of the factor-index space Y, with $\dim Y \le \log_p |S_i|$ by (9.14). Thus its tensor degree is at most three, and, after the cubic solver's realization change, the total argument dimension d_{call} in a terminal call satisfies

$$d_{\mathrm{call}} \le C(d + \log_p n), \qquad p^{O(d_{\mathrm{call}})} = p^{O(d)} n^{O(1)}. \tag{9.17}$$

The constant C allows a fixed-factor realization change in the cubic transport theorem. No previous restriction argument is passed to the next quotient call. Large cochain coordinate spaces and fixed multiplicity labels therefore contribute to the polynomial input-size factor, not to d_{call}.

Total number of tasks. At position i, expand each module filtration into its nonzero grades before counting the tasks to which Steps R2–R3 apply. Let μ_i be the sum of the dimensions of the modules in these tasks. The grades of a filtration have the same total dimension as the module being filtered, and each processed grade has only one continuing module, $\mathrm{Fix}_{S_i}(B)$, of no larger dimension. Consequently

$$\mu_0 \le b, \qquad \mu_{i+1} \le \mu_i, \qquad \sum_{i=0}^{\ell} \mu_i \le (\ell+1)b. \tag{9.18}$$

Tasks omitted after an empty answer only reduce these bounds. Each nonzero module has dimension at least one, so this also bounds the number of processed grades. A call that constructs a nontrivial filtration coordinates the calls on its grades; there is at most one such parent per module arriving at this position. Including these coordinating calls changes the count by at most a constant factor. There are at most n outer-action cosets per nonabelian restriction. All their answers are combined before its single inflation continuation, so this factor does not multiply across positions in the series. The total number of cubic calls is therefore $O(n(\ell+1)b)$.

Coordinates, original elements, and encodings. Every continuing module has dimension at most b. Ordinary cochains, maps induced by module homomorphisms, and resolution comparisons through degree two have polynomially many coordinates in bn; for example, $O(bn^4)$ is a sufficient common bound for the vector-space dimensions of these calculations. All coordinates and quotient maps are computed by elimination. The algorithm never enumerates the vectors of B, a tensor-product module, or a cochain space.

The input already gives a faithful permutation encoding of Γ. A temporary covering group in Proposition 9.10 adds only the domain Q, whose table belongs to the input, and is projected away before continuation. If a faithful realization of moving matrices is needed, use the vector sets of their argument spaces; their total degree is $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$ by (9.17). Handle kernels with nonzero affine displacement using Lemma 3.6.

After each return, compute a succinct generating set and store their expressions in earlier generators by shared straight-line programs. There are polynomially many calls by (9.18), and each appends a fixed polynomial number of operations in its explicit local input size. Thus the aggregate encoding length, cached evaluation cost, and all lift computations are $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$, with exponents independent of the chain depth. Applying the cubic transport bound to each call and adding the linear algebra and permutation computations gives the same bound for the whole algorithm. Together with Proposition 9.11, this establishes the bound in Theorem 9.2.

Proof of Theorem 9.2. Use Algorithm 9.5 along the fixed subgroup series, with $t = 1$ and $H = \Gamma$. Proposition 9.11 proves termination and the complete coset output. The recurrences (9.15) and (9.16) and the bound on the total module dimension (9.18) give at most $O(n(\ell + 1)b)$ terminal cubic calls, where $b = \dim B$ and ℓ is the length of this series. The construction lemmas in §9.3.2 give their explicit tensor presentations. Formula (9.17) bounds their moving argument dimensions by $C(d + \log_p n)$. The coordinate and group-encoding analysis in §9.3.3 therefore gives total time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$. These arguments use only the stated factor types of the subgroup series, so they also apply to the induced series in any current quotient $R_i = Q/N_i$. □

Application to module extensions.

Corollary 9.12 (Comparing module extension classes)**.** *Let Q be a finite table group, and let U, V be explicitly given $\mathbb{F}_p Q$-modules with $\dim U + \dim V \le a$. Given a group Γ acting compatibly on Q, U, V, and affine subquotient data in $H^1(Q, \mathrm{Hom}_{\mathbb{F}_p}(V, U))$ as in Definition 26, the complete affine transporter is computable in time $p^{O(a)} \cdot \mathrm{poly}(\mathrm{InSize})$.*

Proof. Use the quadratic tensor presentation $\mathrm{Hom}_{\mathbb{F}_p}(V, U) = U \otimes_{\mathbb{F}_p} V^*$ in Theorem 9.2. □

For example, take $Q = \mathbb{F}_p^r$ in Corollary 9.12, and suppose that the induced Q-action on $B = \mathrm{Hom}_{\mathbb{F}_p}(V, U)$ is trivial. Then $H^1(Q, B) = \mathrm{Hom}_{\mathbb{F}_p}(Q, B) = U \otimes_{\mathbb{F}_p} V^* \otimes_{\mathbb{F}_p} Q^*$. A compatible triple (g_U, g_V, β) sends f to $x \mapsto g_U f(\beta^{-1}x) g_V^{-1}$. The third argument has dimension $r = \log_p |Q|$, so its exponential contribution is $p^{O(r)} = |Q|^{O(1)}$. For a general Q, the construction uses small sections of Q to achieve this logarithmic bound. Treating the full cochain-coordinate space as one moving argument would not give the theorem's running time.

9.4 Module-action compatibility

We prove Theorem 9.3 by keeping Q fixed and recursing on module dimension. The algorithm matches the radical and the semisimple quotient, then uses first-cohomology transport to compare the off-diagonal blocks joining them. Here semisimplicity refers to a module; no assumption on the solvable radical of Q is made. The first-cohomology subroutine follows the subgroup series of Q.

In the group-extension application, take $M = M' = \mathbb{F}_p^a$ and $\Gamma = \mathrm{Aut}(Q)$ acting on Q, with β the inclusion. The result then gives the pairs required by (9.5). Inside the recursion, the input group Γ also records maps chosen on earlier layers; its homomorphism β need not be injective. Every subsequent comparison must carry these records and its full projection kernel.

The same procedure handles a coset $g_0 K \subseteq \Gamma$ of allowed transformations. Replace the target action by $q \mapsto \theta'(\beta_{g_0} q)$, solve using K and $\beta|_K$, and replace each returned $h \in K$ by $g_0 h$. When a coset also records maps on previously matched layers, compose with its recorded witness maps at the same time.

9.4.1 Recursive data and algorithm

We use the module and semisimplicity terminology from the preliminaries. The *module radical* $\operatorname{rad}(M) \leq M$ is the intersection of the maximal proper submodules, and $M/\operatorname{rad}(M)$ is semisimple. For a nonzero finite-dimensional M, choose a proper submodule N of largest dimension. It is maximal, so $\operatorname{rad}(M) \leq N < M$. Thus the radical is always proper, but it can be zero: $\operatorname{rad}(M) = 0$ exactly when M is semisimple. The algorithm uses this as a terminal case and recurses only when $0 < \dim \operatorname{rad}(M) < \dim M$. This module radical is distinct from the solvable radical of a group.

If $\mathcal{A} = \operatorname{span}_{\mathbb{F}_p}\{\theta(q) : q \in Q\}$ is the enveloping algebra, then $\operatorname{rad}(M) = J(\mathcal{A})M$, where $J(\mathcal{A})$ is its Jacobson radical. The matrix ideal $J(\mathcal{A})$ and the module subspace $J(\mathcal{A})M$ are different objects; Proposition 9.15 computes the latter. Matching the simple summands of $M/\operatorname{rad}(M)$ compares the diagonal blocks of the action matrices. The first-cohomology calculation then matches the off-diagonal blocks joining this quotient to $\operatorname{rad}(M)$. Both calculations use the same quotient automorphism.

For semisimple M, M', write $\mathtt{S}(Q, \theta, \theta'; \Gamma, \beta)$ for Algorithm 9.14. It uses the same answer and output convention as `M`.

Algorithm 9.13 (Module action compatibility by radical layers)**. Input:** a call $\mathtt{M}(Q, \theta, \theta'; \Gamma, \beta)$, with the following data and notation.

- A prime p and a finite group Q, given by its multiplication table and a generating set.
- $\mathbb{F}_p$-spaces M, M' of the same dimension a, with representations $\theta : Q \to \mathrm{GL}(M)$ and $\theta' : Q \to \mathrm{GL}(M')$ given by matrices on these generators of Q.
- A permutation group $\Gamma \leq \mathrm{Sym}(\Omega)$ on a finite set Ω, given by generators, and a homomorphism $\beta : \Gamma \to \mathrm{Aut}(Q)$ specified on those generators. Write $\beta_g = \beta(g)$.

The notation $\operatorname{rad}(M)$ denotes the intersection of the maximal proper Q-submodules of M, and similarly for M'. The routine `S` denotes the semisimple comparison procedure of Algorithm 9.14, with the same input and output convention as `M`.
Output: all pairs (α, g) with an invertible linear map $\alpha : M \to M'$ and $g \in \Gamma$ satisfying $\alpha\theta(q) = \theta'(\beta_g q)\alpha$ for every $q \in Q$. Return emptiness, or one such pair (α_0, g_0) and generators of the full source stabilizer $K_\theta = \{(u, g) \in \mathrm{GL}(M) \times \Gamma : u\theta(q) = \theta(\beta_g q)u \text{ for all } q \in Q\}$, giving the coset $(\alpha_0, g_0)K_\theta$ in the original bases and permutation encoding.

B1. **If** $a = 0$, **return** $\{\mathrm{id}_0\} \times \Gamma$.

B2. Compute $U = \operatorname{rad}(M)$ and $U' = \operatorname{rad}(M')$ by Proposition 9.15. **If** $\dim U \neq \dim U'$, **return** emptiness. **If** $U = U' = 0$, **return** $\mathtt{S}(Q, \theta, \theta'; \Gamma, \beta)$.

For $0 < \dim U = \dim U' < a$, execute R1–R4 below. Use bases of U, U' and quotient bases of $V = M/U, V' = M'/U'$; write $\theta_U, \theta'_U, \theta_V, \theta'_V$ for the induced actions.

R1. Recursive radical comparison.

(a) **Call this algorithm recursively** on

$$\mathtt{M}(Q, \theta_U, \theta'_U; \Gamma, \beta).$$

Only the two represented spaces and their actions change; Q, Γ, β are unchanged.

(b) **If** its answer is empty, **return** emptiness. **Otherwise**, write the answer as $(u_{\mathrm{rad}}, g_{\mathrm{rad}})K_U$, with K_U the full source stabilizer of the radical action. Each element $k = (u, g) \in K_U$ records both a matrix on U and an original permutation $g \in \Gamma$.

R2. Semisimple quotient comparison under the radical stabilizer.

(a) Put $\beta_U(u, g) = \beta_g$ for $(u, g) \in K_U$. Normalize the target by the radical witness:

$$\theta'^0_U(q) = u_{\mathrm{rad}}^{-1}\theta'_U(\beta_{g_{\mathrm{rad}}}q)u_{\mathrm{rad}} = \theta_U(q), \qquad \theta'^0_V(q) = \theta'_V(\beta_{g_{\mathrm{rad}}}q).$$

The same quotient automorphism is applied on both target layers; θ'^0_V is still an action on V'. Represent K_U faithfully with its recorded matrices and original permutations as in Section 9.4.2.

(b) **Call** the semisimple procedure

$$\mathtt{S}(Q, \theta_V, \theta'^0_V; K_U, \beta_U).$$

If its answer is empty, **return** emptiness. **Otherwise**, denote its full source stabilizer by Δ. Write its elements as triples (u, v, g), recording the radical map, the quotient map, and the original element of Γ, respectively. For a returned witness (u_1, v_0, g_1), set $h_0 = (u_0, v_0, g_0) = (u_{\mathrm{rad}}u_1, v_0, g_{\mathrm{rad}}g_1)$. Here $u_0 : U \to U'$ and $v_0 : V \to V'$ are invertible. The coset $h_0\Delta$ consists of all compatible diagonal triples for the original two modules, by Proposition 9.18; use it as the input to R3.

R3. First-cohomology comparison.

(a) Recall that $V = M/U$ and $V' = M'/U'$ are the semisimple quotients compared in R2. Choose linear sections of $M \to V$ and $M' \to V'$, giving vector-space decompositions $M \cong U \oplus V$ and $M' \cong U' \oplus V'$. The upper-right action blocks are maps $c(q) : V \to U$ and $c'(q) : V' \to U'$. Form the cocycles

$$z(q) = c(q)\theta_V(q)^{-1} \in X = \mathrm{Hom}_{\mathbb{F}_p}(V, U),$$
$$z'(q) = c'(q)\theta'_V(q)^{-1} \in X' = \mathrm{Hom}_{\mathbb{F}_p}(V', U'),$$

as in (9.19).

(b) Use the diagonal witness $h_0 = (u_0, v_0, g_0)$ from R2 to transport the target cocycle into $X = \mathrm{Hom}_{\mathbb{F}_p}(V, U)$ by setting $z'_0(q) = u_0^{-1}z'(\beta_{g_0}q)v_0 \in X$. In this composition, v_0 maps V to V', the target cocycle maps V' to U', and u_0^{-1} maps U' to U. Give X the action $q \cdot b = \theta_U(q)b\theta_V(q)^{-1}$. Both z, z'_0 are now cocycles for this same source action.

(c) Use the homomorphism $\Delta \to \mathrm{Aut}(Q)$, $(u, v, g) \mapsto \beta_g$, and the action on $H^1(Q, X)$ induced by $((u, v, g)z)(q) = u\, z(\beta_g^{-1}q)\, v^{-1}$ for $(u, v, g) \in \Delta$. **Call** Algorithm 9.5, as in Corollary 9.12, with group Δ, source $[z]$, target $[z'_0]$, and zero affine cocycle; take numerator $H^1(Q, X)$ and denominator 0. **If** its answer is empty, **return** emptiness. **Otherwise**, write the answer as h_1K, with $K = \mathrm{Stab}_\Delta([z])$. By Lemma 9.19, the surviving diagonal coset is h_0h_1K.

R4. Lift and return the complete compatible-pair coset.

(a) Write $h_0h_1 = (u_*, v_*, g_*)$. Solve (9.21) for the upper-right block $t_* \in \mathrm{Hom}_{\mathbb{F}_p}(V, U')$, and put $\alpha_* = \left(\begin{smallmatrix} u_* & t_* \\ 0 & v_* \end{smallmatrix}\right)$.

(b) For each generator $k = (u_k, v_k, g_k)$ of K, solve the source-to-source version of (9.21) for $t_k \in \mathrm{Hom}_{\mathbb{F}_p}(V, U)$, replacing θ' by θ. Obtain a source lift $\alpha_k = \left(\begin{smallmatrix} u_k & t_k \\ 0 & v_k \end{smallmatrix}\right)$. These systems have solutions by the class comparison in R3.

(c) Compute a basis $\mathcal{B}$ of $\mathrm{Hom}_Q(V, U) = \{b \in \mathrm{Hom}_{\mathbb{F}_p}(V, U) : b\theta_V(q) = \theta_U(q)b \text{ for all } q \in Q\}$ by linear equations. Transfer all matrices from the chosen layer bases to the original module bases, and **return**

$$(\alpha_*, g_*) \left\langle (\alpha_k, g_k) \ (k \text{ a generator of } K), \quad \left(\begin{pmatrix} I & b \\ 0 & I \end{pmatrix}, 1_\Gamma \right) \ (b \in \mathcal{B}) \right\rangle.$$

The shear matrices in this display are likewise transferred to the original source basis. Their inclusion recovers the full projection kernel, as proved in Proposition 9.20.

Every recursive call starts with B1–B2. A parent stores its original module actions until the radical call returns; R2–R4 use the entire returned coset, including all kernel generators.

The progress measure is $\dim M$. At a nonsplit node, $0 < \dim U < \dim M$, and the only recursive module call is on U, U'. The remaining calls are the semisimple comparison and the already established H^1 algorithm. In particular, Q is not replaced by a quotient in this recursion. The semisimple quotient and the extension class are compared under the full group left by the preceding step, so a single element of Γ, and hence a single automorphism of Q, governs all layers.

The semisimple case uses the method of Babai–Qiao [BQ12, Sections 5.1.2–5.2]: compare multiplicities of simple types under quotient automorphisms, then construct the corresponding module isomorphisms. The closure of the occurring types and the construction of the full compatibility coset are also explicit in Grochow–Qiao [GQ15a, Section 5.1, Proposition 9 and Theorem 10]. Algorithm 9.14 specializes this method to semisimple inputs, for which the closure has at most $|Q|$ simple types in any characteristic. Here the allowed automorphisms are specified by $\beta : \Gamma \to \mathrm{Aut}(Q)$. In recursive calls, Γ can also record maps on previously matched layers, so the answer must be lifted back to this group. The additional extension-class test for nonsplit modules is step R3 of Algorithm 9.13.

Algorithm 9.14 (Semisimple module comparison)**. Input:** a prime p; a finite group Q of order n, given by its multiplication table and generators; semisimple $\mathbb{F}_pQ$-modules M, M' of dimension a, with representations θ, θ' given on those generators; a permutation group $\Gamma \leq \mathrm{Sym}(\Omega)$, given by generators on an explicit finite set Ω; and a homomorphism $\beta : \Gamma \to \mathrm{Aut}(Q)$ given on generators. Write $\beta_g = \beta(g)$ and let InSize be the total explicit input length.
Output: emptiness, or a compatible pair (α_0, g_0) and generators of $K_\theta = \{(u, g) \in \mathrm{GL}(M) \times \Gamma : u\theta(q) = \theta(\beta_g q)u \text{ for all } q \in Q\}$, whose coset $(\alpha_0, g_0)K_\theta$ is the full answer.

S1. If $a = 0$, return $\{\mathrm{id}_0\} \times \Gamma$. Otherwise compute decompositions of M, M' into simple summands by Proposition 9.15. Store each simple type by representative generator matrices and its multiplicity.

S2. Close the occurring simple types under the generators of $\beta(\Gamma)$. Use the simple-module Hom test to identify equal types. Write $\mathcal{S}$ for the resulting set and compute the permutation action $\lambda_g[S] = [\rho_S \circ \beta_g^{-1}]$, where ρ_S is the chosen representation of type $[S]$. Store the original generator of Γ with each image generator.

S3. For $e = 1, \ldots, a$, let $\mathcal{S}_e, \mathcal{S}'_e$ be the types occurring with multiplicity e in M, M'. If their cardinalities differ for any e, return emptiness. Start with the allowed coset Γ and compare these subsets successively by Lemma 9.16 and (3.2), normalizing each comparison by the preceding coset witness. If a comparison is empty, return emptiness. Lift the final witness and stabilizer generators through λ, including generators of $\ker \lambda$, to obtain the complete multiplicity transporter $g_0K \subseteq \Gamma$.

S4. Match simple summands to construct an intertwiner α_0 above g_0 and a source intertwiner u_h above every generator h of K. Compute generators u of $\mathrm{Aut}_Q(M)$ by Proposition 9.15, and return

$$(\alpha_0, g_0)\big\langle (u_h, h)\ (h \text{ a generator of } K), \quad (u, 1_\Gamma)\ (u \text{ a generator of } \mathrm{Aut}_Q(M))\big\rangle.$$

The group on $\mathcal{S}$ is stored by permutations with generator preimages; all module maps are stored by matrices. There are at most n types and at most a positive-multiplicity types in either module. The subset comparisons cost $2^{O(a)} \cdot \mathrm{poly}(\mathrm{InSize})$ in total, and the module computations and lifts cost $p^{O(a)} \cdot \mathrm{poly}(\mathrm{InSize})$. The procedure makes no recursive module call.

9.4.2 Data structures and running time

Represent each group of compatible pairs or triples faithfully on the disjoint union of the original permutation domain Ω and the vector sets of its recorded module layers. An element acts on Ω by its original permutation and on each vector set by its recorded matrix. Fixing every point means that every recorded component is the identity, so this action is faithful. The projection to Γ, and hence the homomorphism to $\mathrm{Aut}(Q)$, remains explicitly available. Store the generator matrices and changes of module bases with this representation.

There are at most a nonsplit levels, and all these layer spaces have dimension at most the original a. Thus the total permutation degree needed for their records is at most $|\Omega| + O(ap^a)$. The vector spaces $\mathrm{Hom}(V, U)$ and their cochains are instead stored in linear coordinates. In particular, we never enumerate the $p^{(\dim U)(\dim V)}$ elements of a Hom space. The cohomology algorithm uses the original arguments U, V, of total dimension at most a, as quantified in Corollary 9.12.

Let a denote the original dimension and put $n = |Q|$. The only recursive module call at a nonsplit node is on its proper radical, so there are at most a such nodes and at most one semisimple comparison and one cohomology comparison per node. The module computations use $p^{O(a)} \cdot \mathrm{poly}(\mathrm{InSize})$ time. The closure of the simple types has size at most n; there are at most a multiplicity classes to compare and each small-set transporter has cost $2^{O(a)}$ times a polynomial factor. Intertwiner and shear calculations are linear systems of size polynomial in n, a and the generator records.

Every cohomology root has argument parameter at most a, so Corollary 9.12 bounds its cost by $p^{O(a)} \cdot \mathrm{poly}(\mathrm{InSize}_{\mathrm{loc}})$, where $\mathrm{InSize}_{\mathrm{loc}}$ is its explicit input length. The faithful permutation domains above, layer matrices, and cochain coordinates give a common bound $p^{O(a)} \cdot \mathrm{poly}(\mathrm{InSize})$ on all local input sizes. Compute a succinct generating set after each return and represent their expressions in earlier generators by shared straight-line programs. Each module level adds a polynomial number of records within these bounds; there are at most a levels. Thus the sizes of lifted matrices, permutations, and expressions have the same bound, with exponents independent of the recursion depth. Summing the costs proves $p^{O(a)} \cdot \mathrm{poly}(\mathrm{InSize})$ for the complete algorithm.

The correctness proof of Theorem 9.3 is in Section F.2; the technical statements below isolate the module computations and the three reconstruction steps used by that proof.

9.4.3 Module computations and the semisimple terminal case

Proposition 9.15 (Module computations for the base cases)**.** *Let Q be a table group of order n, and let M be an $\mathbb{F}_p Q$-module of dimension at most a, given by generator matrices, with total explicit input length* InSize*. In deterministic time $p^{O(a)} \cdot \mathrm{poly}(\mathrm{InSize})$ one can compute:*

- *a composition series of M, with the induced actions on its factors;*
- *the radical $\mathrm{rad}(M)$ and the induced actions on $\mathrm{rad}(M)$ and $M/\mathrm{rad}(M)$;*
- *if M is semisimple, a decomposition into simple summands, with the change-of-basis matrix;*
- *if M is semisimple, generators of $\mathrm{Aut}_Q(M)$ as matrices on the original basis;*
- *for two simple modules S, T of dimensions at most a, a basis of $\mathrm{Hom}_Q(S, T)$ and, when they are isomorphic, an invertible intertwiner $S \to T$, with* InSize *including both inputs.*

For $M \neq 0$, the radical is a proper, possibly zero, submodule, and $M/\mathrm{rad}(M)$ is nonzero and semisimple. The radical is zero exactly when M is semisimple, and is preserved by module isomorphisms and twists by automorphisms of Q.

These are standard constructive module computations. The semisimple computations in the coprime setting appear in Babai–Qiao [BQ12, Sections 2.4 and 5.2]. For arbitrary characteristic, apply Lemma 3.2 to the algebra A generated by the action of Q: the identity $\mathrm{rad}(M) = \mathrm{rad}(A)M$ gives the module radical [CIK97, Section 2.2], and decomposing successive semisimple radical layers gives a composition series [CIK97, Theorem 5]. The same decomposition algorithm gives simple summands when M is semisimple. Applying Lemma 3.2 to $\mathrm{End}_Q(M)$ gives generators of $\mathrm{Aut}_Q(M) = \mathrm{End}_Q(M)^\times$. Hom spaces and induced actions are computed by linear algebra; see also [BQ12, Proposition 2]. The deterministic bounds are polynomial in the explicit input size and p, and hence satisfy the stated bound.

The bound is sufficient for group extensions because p^a divides the current table-group order. It is not a lower bound: for a fixed quotient automorphism, constructive module isomorphism has a deterministic polynomial-time algorithm [CIK97, Corollary 3]. For a fixed quotient automorphism,

Cannon–Holt give the twisted-module test and its module-automorphism kernel [CH03, Sections 5.2–5.3]. Here the additional search is over the allowed quotient automorphisms and the extension classes that they must preserve.

Lemma 9.16 (Small-set transport, [BQ12, Proposition 4])**.** *Let Ω be finite, let $P \leq \mathrm{Sym}(\Omega)$ be given by a succinct generating set, and let $S, T \subseteq \Omega$ have the same size $b \in \mathbb{Z}_{\geq 0}$. The complete set transporter $\{g \in P : gS = T\}$ is computable in $2^{O(b)} \cdot \mathrm{poly}(|\Omega|)$ time.*

This imported result is applied to the image of Γ on simple types. Generator expressions and their preimages, together with generators of the image kernel, recover the complete coset in Γ. The permutation-group preimage operations are charged to the explicit input and output records.

Proposition 9.17 (The semisimple terminal comparison)**.** *On semisimple inputs, Algorithm 9.14 returns the complete answer of Definition 27. It runs deterministically in time $p^{O(a)} \cdot \mathrm{poly}(\mathrm{InSize})$, where $n = |Q|$ and* InSize *is the total explicit input length. It closes the occurring simple types under $\beta(\Gamma)$ to a set $\mathcal{S}$ of size at most n, compares their multiplicities, and lifts the complete resulting coset to compatible pairs. The output includes the full kernel of projection to the allowed quotient automorphisms.*

9.4.4 Technical statements for nonsplit layers

Example 5 (A module extension has off-diagonal data)**.** Let $Q = \langle q \rangle \cong C_p$, let $M = \mathbb{F}_p^2$, and define

$$\theta_\lambda(q) = \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix} \qquad (\lambda \in \mathbb{F}_p).$$

Each matrix has pth power one. The line $U = \langle e_1 \rangle$ and quotient M/U both have trivial Q-action, for every λ. Nevertheless $\lambda = 0$ gives a direct sum of trivial modules, whereas $\lambda \neq 0$ gives a nonsplit extension. Comparing the two diagonal actions alone therefore loses information. The first-cohomology step R3 records exactly this off-diagonal block.

The next three results correspond to R1–R2, R3, and R4 of Algorithm 9.13. Throughout their statements, the radical dimensions agree and satisfy $0 < \dim U < \dim M$, where $U = \mathrm{rad}(M)$, $U' = \mathrm{rad}(M')$, $V = M/U$, and $V' = M'/U'$. Write θ_U, θ_V for the induced source actions and use primes for the target actions.

Proposition 9.18 (The diagonal comparison in steps R1–R2)**.** *Assume that the recursive call in step R1 returns the complete coset of compatible radical pairs. Then step R2 returns exactly all triples (u, v, g) with $u : U \to U'$ and $v : V \to V'$ invertible and $g \in \Gamma$ satisfying*

$$\begin{aligned} u\theta_U(q) &= \theta'_U(\beta_g q)u, \\ v\theta_V(q) &= \theta'_V(\beta_g q)v \qquad (q \in Q). \end{aligned}$$

If this set is nonempty, it is the coset $h_0\Delta$ used in step R3, where $h_0 = (u_0, v_0, g_0)$ is one such triple and Δ is its full source stabilizer. Thus $\Delta \leq \mathrm{GL}(U) \times \mathrm{GL}(V) \times \Gamma$ satisfies the same identities with the primed actions replaced by the unprimed actions. Triples compose componentwise.

Matching the two layers does not yet ensure that a triple lifts to M, M'. To express the missing condition, make the choices in step R3(a): choose linear sections of $M \to V$ and $M' \to V'$. They give vector-space identifications $M = U \oplus V$ and $M' = U' \oplus V'$, without asserting that either is a direct sum of Q-modules. In these coordinates write

$$\theta(q) = \begin{pmatrix} \theta_U(q) & c(q) \\ 0 & \theta_V(q) \end{pmatrix}, \qquad z(q) = c(q)\theta_V(q)^{-1}. \tag{9.19}$$

and define $c'(q), z'(q)$ analogously for the target. Their types are $c(q), z(q) \in X := \mathrm{Hom}_{\mathbb{F}_p}(V, U)$ and $c'(q), z'(q) \in X' := \mathrm{Hom}_{\mathbb{F}_p}(V', U')$. The off-diagonal blocks measure the failure of the chosen sections to have invariant images. Give X the action $q \cdot b = \theta_U(q) b \theta_V(q)^{-1}$, and give X' the analogous primed action. The relevant source cohomology space is

$$H^1(Q, X) = H^1(Q, \mathrm{Hom}_{\mathbb{F}_p}(V, U)). \tag{9.20}$$

Lemma 9.19 (The extension-class test in step R3)**.** *The maps z, z' are cocycles, and their cohomology classes do not depend on the chosen sections. For a diagonal witness $h_0 = (u_0, v_0, g_0)$, the formula $z'_0(q) = u_0^{-1} z'(\beta_{g_0} q) v_0$ defines a cocycle in the source module X. The source diagonal group Δ acts on $H^1(Q, X)$ by the cochain formula $((u, v, g)z)(q) = u\, z(\beta_g^{-1} q)\, v^{-1}$. For a diagonal triple $h = (u, v, g) \in h_0\Delta$, a map lifting it has the form $\alpha = \left(\begin{smallmatrix} u & t \\ 0 & v \end{smallmatrix}\right)$ with $t \in \mathrm{Hom}_{\mathbb{F}_p}(V, U')$. It is invertible because u, v are invertible. Its remaining compatibility condition is the linear system*

$$uc(q) + t\theta_V(q) = \theta'_U(\beta_g q) t + c'(\beta_g q) v \qquad (q \in Q). \tag{9.21}$$

Equivalently, with $b = u^{-1} t \in \mathrm{Hom}_{\mathbb{F}_p}(V, U)$, this condition is

$$z(q) - u^{-1} z'(\beta_g q) v = q \cdot b - b \qquad (q \in Q). \tag{9.22}$$

The triples in $h_0\Delta$ that lift to compatible maps $M \to M'$ are exactly $h_0 \,\mathrm{Trans}_\Delta([z], [z'_0])$.

This lemma gives precisely the input and output of step R3(c), after the normalization in (b). Apply Corollary 9.12 to the action of Δ on $H^1(Q, X)$, taking this whole space as numerator, zero denominator, and zero affine cocycle. The module X has the tensor presentation $X = U \otimes_{\mathbb{F}_p} V^*$, with $\dim U + \dim V = a$. The dimension $\dim X = (\dim U)(\dim V)$ enters through explicit coordinates and linear algebra; the tensor-argument parameter is $\dim U + \dim V = a$. Corollary 9.12 therefore gives a cost $p^{O(a)} \cdot \mathrm{poly}(\mathrm{InSize}_{\mathrm{loc}})$, where $\mathrm{InSize}_{\mathrm{loc}}$ includes the recorded diagonal group. In particular, the cochain coordinates indexed by Q do not cause a factor $p^{O(|Q|)}$.

Proposition 9.20 (Complete reconstruction in step R4)**.** *If step R3 returns the nonempty transporter $h_1 K$, with $K = \mathrm{Stab}_\Delta([z])$, restriction to U and passage to V give the exact sequence*

$$1 \longrightarrow \mathrm{Hom}_Q(V, U)_{\mathrm{add}} \longrightarrow K_\theta \longrightarrow K \longrightarrow 1.$$

Step R4 therefore returns the complete coset of compatible pairs for M, M'.

In the group-isomorphism application, the module witness identifies the two actions on the elementary abelian kernel A. The full source stabilizer acts on $H^2(Q, A)$; Corollary 9.22 computes the transporter of the two extension classes under that action. The proof of Theorem 9.1 combines these costs in terms of the extension order $p^a|Q|$.

9.5 Second-cohomology transport

After the action of the quotient Q on the kernel A has been matched, its remaining data are a class in $H^2(Q, A)$. The group acting in this transport problem acts on the kernel A and the quotient Q simultaneously. This subsection computes the complete transporter of two such classes, including the affine comparisons on invariant subquotients that arise during the computation.

The additional issue beyond first cohomology is a middle obstruction. Restriction to a normal subgroup $S \trianglelefteq R$ can vanish without the extension coming from R/S: a splitting over S may fail to be preserved by conjugation from R. The failure is a class in $H^1(R/S, H^1(S, B))$. We use the

first-cohomology theorem of §9.3 to compare this class. Thus the order of the three comparisons is restriction, this first-cohomology obstruction, and inflation from the quotient. All three are performed on their exact images and quotients.

Each second-cohomology task represents its module by one vector-space argument. The cochain spaces indexed by elements of Q may be much larger. Their dimensions enter the polynomial factor of the cost; the additional moving arguments have total dimension $O(\log_p |Q|)$. The analysis below gives both bounds. The technical constructions and correctness proof are collected in §F.3, independently of the group-isomorphism theorem.

9.5.1 Recursive data and algorithm

Background data fixed throughout the recursion. Fix a prime p, the initial table group Q, and the original group $\Gamma_0 \le \mathrm{Sym}(\Omega)$. Use the characteristic subgroup series $1 = N_0 < \cdots < N_\ell = Q$ from (9.3), with $R_i = Q/N_i$ and $S_i = N_{i+1}/N_i \le R_i$ for $i < \ell$. Its subgroup and quotient tables and natural maps are fixed throughout the execution. A recursive call at position i uses the induced series $1 = N_i/N_i < \cdots < N_\ell/N_i = R_i$; the first-cohomology calls use the corresponding induced series as well.

Definition 30 (Data and answer of a second-cohomology call)**.** With this fixed background, a call at position i has current group R_i and notation

$$\mathtt{H}_2(t, H; i, B; W = N/P, c; x, y),$$

where $t \in \Gamma_0$, $H \le \Gamma_0$, $0 \le i \le \ell$, B is an $\mathbb{F}_p R_i$-module, and H acts compatibly on R_i and B. The entry $W = N/P$ includes its H-invariant numerator and denominator $P \le N \le H^2(R_i, B)$, with their bases and quotient projection and lifting coordinate maps. The remaining entries are a cocycle $c : H \to W$ given on generators and points $x, y \in W$. Write $\psi_h(z) = L_h z + c_h$ for the resulting affine action. The answer is $t \,\mathrm{Trans}_{H,\psi}(x, y)$, represented by emptiness or a witness ta and generators of the full source stabilizer H_x, so that the answer is taH_x. Every group element is represented in Γ_0. At the root, $t = 1$, $H = \Gamma_0$, $i = 0$, and all other entries are the input data.

The coset representative t records previously matched conditions. It need not act on the current module B. The parent stores its own space and action while a quotient call is running. When that call returns aK, the common quotient–fibre update restores the parent and uses the target $w = \psi_{a^{-1}}(y) - x$ and action $\beta_k(z) = L_k z + \psi_k(x) - x$ for $k \in K$ on the kernel of the quotient map, with new coset representative ta.

Data representation. The module B is stored by a basis and action matrices. Normalized cochains are arrays indexed by group tuples with entries in this basis; cycles, boundaries, invariant subspaces, quotient bases, and maps between them are stored as matrices. Degree-three cochains are used only to impose the degree-two cocycle equations. There is no transporter problem in third cohomology.

Each recursive record stores the presentation of $W = N/P$, its affine cocycle on generators, and maps back to its parent. Tensor presentations describe their argument spaces and invariant subquotients explicitly; large cochain and tensor spaces are handled by coordinates, without listing all their vectors. The cost of these records is bounded below.

Notation reminder. For $i < \ell$, recall that $\mathrm{Fix}_{S_i}(B) = \{b \in B : sb = b \text{ for all } s \in S_i\}$ and that I_{S_i} is the augmentation ideal of $\mathbb{F}_p[S_i]$ (Definition 28). For a quotient cocycle $\bar{f} : R_{i+1}^2 \to \mathrm{Fix}_{S_i}(B)$, its inflation from §9.2.4 is $(q, r) \mapsto \bar{f}(qS_i, rS_i)$ on R_i. The coset representative t records earlier matches; it need not act on B.

Algorithm 9.21 (Recursive second-cohomology transporter)**. Fixed background:** the prime p; the characteristic subgroup series $1 = N_0 < \cdots < N_\ell = Q$, with subgroup and quotient tables and natural maps; and the original group $\Gamma_0 \le \mathrm{Sym}(\Omega)$, given by generators.
Recursive input: a call

$$\mathtt{H}_2(t, H; i, B; W = N/P, c; x, y),$$

with the following data.

- A position $0 \le i \le \ell$, with current group $R_i = Q/N_i$. For $i < \ell$, the selected subgroup is $S_i = N_{i+1}/N_i$, and the next group is $R_{i+1} = Q/N_{i+1} = R_i/S_i$.
- A current subgroup $H \le \Gamma_0$, given by generators, and a current coset representative $t \in \Gamma_0$.
- An $\mathbb{F}_p R_i$-module B, with compatible H-actions on R_i and B specified on generators.
- A quotient $W = N/P$, specified by H-invariant subspaces $P \le N \le H^2(R_i, B)$ and quotient coordinates, and points $x, y \in W$. Write L_h for the induced linear action of $h \in H$ on W. Generator values of a cocycle $c : H \to W$, satisfying $c_{hk} = c_h + L_h c_k$, specify the affine action $\psi_h(z) = L_h z + c_h$.

Output: the set $t\,\mathrm{Trans}_{H,\psi}(x, y) = \{th : h \in H,\ \psi_h(x) = y\}$, represented by emptiness or a witness ta and generators of the source stabilizer $K = \{h \in H : \psi_h(x) = x\}$, giving the coset taK in the original group Γ_0.

B1. **If** $W = 0$, **return** tH.

B2. **If** $i = \ell$ or $B = 0$, then $W = 0$ by the input conditions; **return** tH.

Otherwise execute R1–R3 below.

R1. Coprime restriction.

(a) **If** $p \mid |S_i|$, **continue to R2**.

(b) Compute the inverse inflation isomorphism $\sigma : H^2(R_i, B) \to H^2(R_{i+1}, \mathrm{Fix}_{S_i}(B))$ of Lemma 9.7, and its induced quotient map $\bar{\sigma} : W \to W'$, where $W' = \sigma(N)/\sigma(P)$.

(c) **Return the recursive call**

$$\mathtt{H}_2(t, H; i+1, \mathrm{Fix}_{S_i}(B); W' = \sigma(N)/\sigma(P), \bar{\sigma}c; \bar{\sigma}x, \bar{\sigma}y).$$

R2. Filter B **until** S_i **acts trivially on each grade.**

(a) **If** S_i is nonabelian, or S_i acts trivially on B, **continue to R3**. Otherwise S_i is elementary abelian of characteristic p. Compute the augmentation filtration $(I_{S_i}^k B)_{k \ge 0}$ and reindex it as $0 = B_0 < B_1 < \cdots < B_e = B$, with $e \ge 2$. Put $D_k = B_k/B_{k-1}$.

(b) Compute $F_k = \mathrm{im}\big(H^2(R_i, B_k) \to H^2(R_i, B)\big)$ and $W_k = ((N \cap F_k) + P)/P$. Thus $0 = W_0 \le \cdots \le W_e = W$. Using (9.9), compute, for each k, a quotient map

$$q_k : W_k \longrightarrow N^{(k)}/P^{(k)}, \quad \ker q_k = W_{k-1}, \quad P^{(k)} \le N^{(k)} \le H^2(R_i, D_k).$$

(c) Initialize $t_e = t$, $H_e = H$, $\psi^{(e)} = \psi$, $x_e = x$, and $y_e = y$. For $k = e, e-1, \dots, 1$, let $c_h^{(k)} = \psi_h^{(k)}(0)$ and perform the following operations.

(1) **Call this algorithm recursively** with coset representative 1:

$$\mathtt{H}_2(1, H_k; i, D_k; W^{(k)} = N^{(k)}/P^{(k)}, q_k c^{(k)}; q_k x_k, q_k y_k).$$

(2) **If** the answer is empty, **return** emptiness. Otherwise write it as $a_k K_k$, where K_k is the full stabilizer of $q_k x_k$.

(3) Set

$$t_{k-1} = t_k a_k, \qquad H_{k-1} = K_k, \qquad x_{k-1} = 0,$$
$$y_{k-1} = \psi^{(k)}_{a_k^{-1}}(y_k) - x_k \in W_{k-1},$$
$$\psi_h^{(k-1)}(z) = L_h^{(k)} z + \psi_h^{(k)}(x_k) - x_k \quad (h \in K_k,\ z \in W_{k-1}).$$

Here $L_h^{(k)}$ is the linear part of $\psi_h^{(k)}$.

(d) **Return** $t_0 H_0$.

R3. Restriction, the middle obstruction, and inflation.

(a) Compute the restriction kernel and inflation image

$$F_1 = \ker\big(H^2(R_i, B) \longrightarrow H^2(S_i, B)\big), \qquad F_2 = \mathrm{im}\big(H^2(R_{i+1}, \mathrm{Fix}_{S_i}(B)) \longrightarrow H^2(R_i, B)\big)$$

of Lemma F.1, with $F_2 \le F_1$, and put $U_1 = ((N \cap F_1) + P)/P$ and $U_2 = ((N \cap F_2) + P)/P$. Compute the corresponding quotient maps and presentations, with $E = H^1(S_i, B)$:

$$\begin{aligned} q_{\mathrm{res}} : W \twoheadrightarrow N_{\mathrm{res}}/P_{\mathrm{res}}, \quad & \ker q_{\mathrm{res}} = U_1, \\ & N_{\mathrm{res}} \le H^2(S_i, B), \\ q_{\mathrm{mid}} : U_1 \twoheadrightarrow N_{\mathrm{mid}}/P_{\mathrm{mid}}, \quad & \ker q_{\mathrm{mid}} = U_2, \\ & N_{\mathrm{mid}} \le H^1(R_{i+1}, E), \\ q_{\mathrm{inf}} : U_2 \xrightarrow{\sim} N_{\mathrm{inf}}/P_{\mathrm{inf}}, \quad & N_{\mathrm{inf}} \le H^2(R_{i+1}, \mathrm{Fix}_{S_i}(B)). \end{aligned}$$

All three denominators are subspaces of their numerators. The inflation kernel and the kernel F_2 of the obstruction map $\omega_{S_i} : F_1 \to H^1(R_{i+1}, E)$ are included in these presentations.

(b) **Restriction:** compute

$$\mathrm{Trans}_{H,\bar\psi}(q_{\mathrm{res}} x, q_{\mathrm{res}} y), \qquad \bar\psi_h(q_{\mathrm{res}} z) = q_{\mathrm{res}} \psi_h(z),$$

using the cubic transport theorem with the cubic presentation of §F.3.2 if S_i is elementary abelian, or using Proposition 9.10 otherwise. **If** the answer is empty, **return** emptiness. Otherwise write it as aK, set $w = \psi_{a^{-1}}(y) - x \in U_1$, and define $\beta_k(z) = L_k z + \psi_k(x) - x$ for $k \in K$ and $z \in U_1$.

(c) **Middle obstruction:** construct the tensor presentation $\mathcal{P}_E$ of E from §F.3.2 and set $c_k^\beta = \beta_k(0)$. Call the first-cohomology algorithm with coset representative 1:

$$\mathtt{H}_1(1, K; i+1, E, \mathcal{P}_E; W_{\mathrm{mid}} = N_{\mathrm{mid}}/P_{\mathrm{mid}}, q_{\mathrm{mid}} c^\beta; 0, q_{\mathrm{mid}} w).$$

If the answer is empty, **return** emptiness. Otherwise write it as bJ, set $v = \beta_{b^{-1}}(w) \in U_2$, and define $\gamma_j(z) = L_j z + \beta_j(0)$ for $j \in J$ and $z \in U_2$.

(d) **Inflation:** set $c_j^\gamma = \gamma_j(0)$ and **return the recursive call**

$$\mathtt{H}_2(tab, J; i+1, \mathrm{Fix}_{S_i}(B); W_{\mathrm{inf}} = N_{\mathrm{inf}}/P_{\mathrm{inf}}, q_{\mathrm{inf}}c^\gamma; 0, q_{\mathrm{inf}}v).$$

Each call records the maps to its parent presentation. All returned elements and stabilizer generators use the original group encoding.

The progress measure for second-cohomology calls is the lexicographically ordered pair $(\ell - i, \dim B)$. The calls in R1 and R3 increase i. Each call in R2 keeps i and uses the module D_k of strictly smaller dimension; its S_i-action is trivial, so that child does not repeat R2 at the same position. The middle call in R3 also increases i and changes to cohomological degree one; the first-cohomology algorithm never raises that degree again. Restriction comparisons are terminal tensor calls. The mass calculation below bounds the number of all these calls, beyond proving termination.

9.5.2 Recursive formulas and running time

The filtration step R2 uses the exact subquotient formulas of Proposition 9.6, valid also in degree two. In R3 the maps identify the following three subquotients of W

$$W/U_1 \;\cong\; N_{\mathrm{res}}/P_{\mathrm{res}}, \qquad U_1/U_2 \;\cong\; N_{\mathrm{mid}}/P_{\mathrm{mid}}, \qquad U_2 \;\cong\; N_{\mathrm{inf}}/P_{\mathrm{inf}}.$$

Lemma F.1 identifies these with the restriction, middle-obstruction, and inflation factors. Lemma 3.5 determines the affine action on each successive fiber. In particular, a successful quotient comparison never discards its full source stabilizer. The elementary and nonabelian presentations used in these formulas are given with their proofs in §F.3.2.

Fix the initial input group Q, write $n = |Q|$, and put $b = \dim_{\mathbb{F}_p} B = d$. Cases with $Q = 1$ or $b = 0$ have already been treated. The following bounds concern one complete invocation of Algorithm 9.21, including the first-cohomology calls it creates.

Dimensions of the moving arguments. Let d_{call} be the total argument dimension used by one terminal cubic call, including its fixed-factor realization change. Repeated occurrences of the same action, including its dual, are counted once. A second-cohomology restriction represents its current module by one argument of dimension at most d, and adds a section argument of dimension at most $\log_p |S_i|$.

An elementary middle term starts a first-cohomology call with arguments B, S_i of total dimension $\dim B + \log_p |S_i|$, where B is the current module after any filtration step. A later restriction in that call adds at most one further section argument. The two sections occur at different positions of the fixed subgroup series, so their orders have product at most n. Their logarithmic contributions sum to at most $\log_p n$. A nonabelian middle term instead starts with one argument of dimension at most $2d$, and its later restriction adds at most $\log_p n$. No branch makes a second transition from cohomological degree two to one.

Thus, including the fixed-factor realization change permitted by the cubic transport theorem, there is an absolute constant $C \geq 1$ such that every terminal tensor call satisfies

$$d_{\mathrm{call}} \leq C(2d + \log_p n), \qquad p^{d_{\mathrm{call}}} \leq (p^{2d} n)^C. \tag{9.23}$$

The table-element labels within a simple group, fixed multiplicity labels, and ordinary cochain coordinates do not contribute further moving arguments. They contribute to the explicit input length, which is bounded separately below. This is the reason the quotient part costs a polynomial in $|Q|$, rather than $p^{O(|Q|)}$.

Number of continuing tasks. For the fixed subgroup series (9.3), put

$$\eta_i = \begin{cases} \dim_{\mathbb{F}_p} S_i, & S_i \text{ is elementary abelian of characteristic } p, \\ 2, & S_i \text{ is a product of nonabelian simple groups}, \\ 0, & S_i \text{ is elementary abelian of characteristic different from } p. \end{cases}$$

Let v be the sum of the elementary dimensions in the first case, and let s count the nonabelian positions. Then $\ell \le \lfloor \log_2 n \rfloor$, $v \le \log_p n$, $s \le \ell$, and $\sum_{i=0}^{\ell-1} \eta_i = v + 2s$. After splitting the module filtrations into grades, let $\mu_i^{(2)}$ and $\mu_i^{(1)}$ be the sums of the dimensions of the modules in the degree-two and degree-one tasks at position i. The initial values satisfy $\mu_0^{(2)} \le b$ and $\mu_0^{(1)} = 0$, allowing tasks with zero transport space to be omitted. The same estimates allow multiple initial tasks, with their dimensions added in these two values.

Passing from a module to its filtration grades preserves the sum of dimensions. A degree-one task has just its inflation continuation, whose module dimension cannot increase. A degree-two task on D continues with $\mathrm{Fix}_{S_i}(D)$ for inflation and $H^1(S_i, D)$ for the middle term. These have dimensions at most $\dim D$ and $\eta_i \dim D$, respectively. The latter bound follows from $H^1(S_i, D) = D \otimes S_i^*$ in the elementary case, Lemma F.3 in the nonabelian case, and coprime vanishing. Therefore

$$\mu_{i+1}^{(2)} \le \mu_i^{(2)}, \qquad \mu_{i+1}^{(1)} \le \mu_i^{(1)} + \eta_i \mu_i^{(2)}. \tag{9.24}$$

It follows that

$$\sum_{i=0}^{\ell} (\mu_i^{(2)} + \mu_i^{(1)}) \le (\ell + 1)\big[\mu_0^{(1)} + \mu_0^{(2)}(1 + v + 2s)\big]. \tag{9.25}$$

For example, $\Phi_i = \mu_i^{(1)} + \mu_i^{(2)}(1 + \sum_{h=i}^{\ell-1} \eta_h)$ is nonincreasing by (9.24), and bounds each level's total module dimension. This proves (9.25).

A nonzero module has positive integer dimension, and the number of its nonzero filtration grades is at most that dimension. Consequently (9.25) bounds the number of continuing tasks and grade calculations by a polynomial in n, InSize. Keeping a parent record as well as its grades changes this count by only a constant factor. At a nonabelian restriction there are at most $|S_i| \le n$ outer-action cases. Their complete answers are combined before continuing, so they multiply the number of terminal tensor calls by at most n, and do not multiply continuation branches at successive chain positions.

Explicit coordinates and group encodings. The module in a continuing degree-two task has dimension at most b. A degree-one task created by an elementary middle term uses a module of dimension at most bv; in the nonabelian case its dimension is at most $2b$. Later degree-one continuations take only subquotients. Put $b_* = \max(1, b, bv, 2b)$. Ordinary cochains, the nested systems computing Lemma F.1, and the resolution comparisons through degree three have polynomially many coordinates in $b_* n$. For example, $O(b_* n^4)$ bounds their vector-space dimensions. All maps are computed by elimination in specified bases and composed as matrices. The vectors of a cochain space or of a tensor product are never enumerated.

The group Γ_0 uses its given faithful permutation encoding. A first-cohomology middle call uses the same initial group Q and the quotient map $\nu_{i+1} : Q \to R_{i+1}$. A local covering group from Proposition 9.10 adds only Q, of order n, as an extra permutation domain; it is projected away before continuation. If a faithful permutation realization of a moving matrix action is required, use the vector sets of its underlying arguments. By (9.23), their total degree is at most $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$. In particular, the nonabelian middle-term argument has at most p^{2d} vectors. A large module

with a quadratic presentation stays in that presentation. Original action kernels and their affine displacements are handled by Lemma 3.6.

Compute a succinct generating set after each return and store expressions in earlier generators as shared straight-line programs. There are polynomially many calls by (9.25); all coordinates and concrete permutations have the bounds just given. Each call adds polynomially many records and instructions within these bounds. Thus total encoding length, cached evaluation time, and the data needed to lift returned transformations are $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$, with exponents independent of the chain depth.

Total running time. Apply the cubic transport theorem to each terminal tensor call. The dimension bound (9.23) and the local input length above give cost $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$ for that call. Equations (9.24)–(9.25) give polynomially many such calls and continuing records. The remaining linear algebra, permutation-group operations, and lifts have the same total bound. Thus the complete invocation, including all first-cohomology subcalls, runs deterministically in $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$, with a fixed exponent independent of the chain depth.

Proof of Theorem 9.4. Correctness and completeness follow from Proposition F.5. Each terminal cubic call has cost $p^{O(d)} n^{O(1)}$ times a fixed polynomial in its explicit input length, by (9.23) and the cubic transport theorem. The mass and encoding bounds in the analysis of §9.5.2 show that there are polynomially many calls and that every local input has length $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$. Applying the fixed polynomial input-size bound of the cubic theorem, and adding the linear algebra and permutation-group calculations, gives the claimed total time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$. □

Application to group extensions.

Corollary 9.22 (Comparing group-extension classes)**.** *Let p be prime, let Q be a finite table group of order n, and let A be an $\mathbb{F}_p Q$-module of dimension at most a. Given a group with compatible actions on Q and A, and affine subquotient data as in Definition 26, complete transport in $H^2(Q, A)$ can be computed in time $p^{O(a)} \cdot \mathrm{poly}(\mathrm{InSize})$.*

Proof. Apply Theorem 9.4 with $B = A$ and $d \le a$. □

For an extension with kernel $A \cong \mathbb{F}_p^a$, this exponential factor is polynomial in the kernel order. If the kernel is written as $\mathbb{F}_q^a$, with $q = p^f$, use its $\mathbb{F}_p$-dimension fa; the bound becomes $q^{O(a)} \cdot \mathrm{poly}(\mathrm{InSize})$. General automorphisms of the additive kernel need only be $\mathbb{F}_p$-linear.

9.6 Proof of the group-isomorphism theorem

The transport results now complete the characteristic-extension strategy of Cannon and Holt [CH03, Sections 2.1–2.2 and 5], in the form recorded by Grochow and Qiao [GQ15a, Introduction, Problem 1]. The only terminal group-isomorphism algorithm required is that of Babai, Codenotti, and Qiao [BCQ12] for groups with trivial solvable radical. Its polynomial-time computation of the full isomorphism coset is explicitly recorded in [GQ15a, Introduction and proof of Corollary 2].

Proof of Theorem 9.1. Proceed by induction on the common group order m; unequal orders and the identity case are immediate. Compute the solvable radicals. If exactly one is trivial, the groups are not isomorphic. If both are trivial, apply the Babai–Codenotti–Qiao algorithm. Otherwise choose the characteristic elementary abelian subgroups $\mathrm{AC}(G), \mathrm{AC}(H)$ of §9.2.1. Their definitions are preserved by every isomorphism. Reject different primes or dimensions; otherwise identify both

kernels with $A = \mathbb{F}_p^a$. Recursively compute the full isomorphism coset of the quotients. An empty answer rules out $G \cong H$; a quotient witness identifies both quotients with $Q = G/\mathrm{AC}(G)$, while the accompanying generators give $\mathrm{Aut}(Q)$. All remaining quotient identifications are represented by this group. Thus the group recursion follows the elementary abelian part of the characteristic series and stops at the quotient by the solvable radical.

Choose normalised representatives and compute the two actions θ_G, θ_H and extension cocycles f_G, f_H as in §9.2.2. Apply Theorem 9.3 with $\Gamma = \mathrm{Aut}(Q)$ to obtain the coset of compatible pairs (α, β) on A, Q. If it is empty, reject. Otherwise write it as $k_0 K_\theta$, put $f_* = T_{k_0}^{-1} f_H$, and apply Corollary 9.22 with allowed group K_θ to match $[f_G]$ and $[f_*]$ in the source space $H^2(Q, A)$. By Grochow and Qiao [GQ17, Lemma 2.3], multiplying the returned coset by k_0 gives exactly the pairs that lift to isomorphisms $G \to H$. Theorem 9.2 is used within these transport algorithms; no further first-cohomology transport call is needed here.

For a nonempty answer, solve (9.6) to lift its witness to F_0 and its source stabilizer generators to automorphisms of G, using the source extension on both sides for the latter systems. The kernel on A, Q consists of the maps $(u, q) \mapsto (u + z(q), q)$ for $z \in Z^1(Q, A)$, so a basis of this space gives the remaining generators. These lifts and kernel generators generate $\mathrm{Aut}(G)$ by the elementary group-homomorphism principle recorded in [GQ17, Observation 5.2]. Here the characteristic choice of $\mathrm{AC}(G)$ ensures that every automorphism preserves the kernel. Linear algebra constructs the lifts and the Z^1 basis; standard permutation-group algorithms produce a succinct generating set [Ser03].

For the running time, put $n = |Q|$, so $m = p^a n$. All tables, cochain coordinates, and lift systems have size polynomial in m. Compatible pairs act faithfully on the disjoint union of the vector set of A and the elements of Q, of size $p^a + n$; succinct generating sets therefore keep every transport input length InSize polynomial in m. The action and second-cohomology bounds give $p^{O(a)} \cdot \mathrm{poly}(\mathrm{InSize}) = m^{O(1)}$ for all nonrecursive transport work, including their first-cohomology subroutines. The other constructions and the BCQ terminal case also take polynomial time. There is just one recursive group-isomorphism call, at order $n \le m/2$. Thus, for the worst-case time $T_{\le}(m)$ on orders at most m, we have $T_{\le}(m) \le T_{\le}(\lfloor m/2 \rfloor) + O(m^c)$ for an absolute constant c, after enlarging c to include the terminal case. This proves the stated polynomial bound, including the full coset output. □

Remark 2. The algorithm uses [BCQ12] for the terminal case of groups with trivial solvable radical. This avoids a separate treatment of outer automorphism groups of nonabelian simple groups.

A Proofs for preliminaries

We use the notation of Subsections 3.7 and 3.8. Let G be a finite group, let $V \le U$ be finite-dimensional $\mathbb{F}_p G$-modules, and put $W = U/V$. Write $\pi : U \twoheadrightarrow W$ for the quotient map, ρ for the action on U, and $\hat{\rho}$ for the induced action on W. Let $c : G \to W$ be a 1-cocycle, let $\alpha_h(w) = \hat{\rho}_h(w) + c_h$, and fix $x, y \in W$.

A.1 Removing translations of a linear kernel

For both results below, take $x = 0$.

Let J be a finite group and $\lambda : G \to J$ a surjective homomorphism. Set $K_{\mathrm{lin}} = \ker \lambda$, and assume $\hat{\rho}_u = 1_W$ for every $u \in K_{\mathrm{lin}}$, where 1_W is the identity map on W. Define $\tau : K_{\mathrm{lin}} \to W^+$ by $\tau(u) = c_u$, and put $E = \tau(K_{\mathrm{lin}})$. The computational input consists of the actions and λ on generators, generators of K_{lin}, and lifts in G of generators of J.

Lemma 3.6 (restated). *The map τ is a homomorphism, and E is a $\hat{\rho}(G)$-invariant $\mathbb{F}_p$-subspace of W. The affine action on W/E factors through J. Given the complete transporter from $0+E$ to $y+E$ in J, linear algebra and the group operations of Lemma 3.1 compute $\mathrm{Trans}_{G,\alpha}(0,y)$ with its full source stabilizer.*

Proof. The cocycle identity gives

$$c_{uu'} = c_u + c_{u'}, \qquad c_{huh^{-1}} = \hat{\rho}_h(c_u) \quad (u, u' \in K_{\mathrm{lin}},\ h \in G).$$

Thus τ is additive, E is invariant, and

$$\overline{\alpha}_{\lambda(h)}(z+E) = \hat{\rho}_h(z) + c_h + E$$

is well defined. An empty quotient transporter implies an empty transporter in G.

Otherwise, write the quotient transporter as $a\overline{S}$, where $\overline{S} = \mathrm{Stab}_{J,\overline{\alpha}}(0+E)$. Choose a lift h_0 of a and a lift h_s of each generator $\bar{s}$ of $\overline{S}$. Since $y - c_{h_0}, c_{h_s} \in E$, choose $u_0, u_s \in K_{\mathrm{lin}}$ such that

$$\tau(u_0) = y - c_{h_0}, \qquad \tau(u_s) = -c_{h_s}.$$

Then $g_0 = u_0 h_0$ maps 0 to y, and each $u_s h_s$ fixes 0. The same correction applies to every element of $\overline{S}$. Consequently $S = \mathrm{Stab}_{G,\alpha}(0)$ maps onto $\overline{S}$ with kernel $\ker\tau$, and

$$S = \langle \ker\tau,\ u_s h_s : \bar{s} \text{ ranges over generators of } \overline{S} \rangle, \qquad \mathrm{Trans}_{G,\alpha}(0,y) = g_0 S.$$

Lemma 3.1 computes a basis of E, word lifts of its basis vectors, and generators of $\ker\tau$. Expressing the correction vectors in this basis gives u_0 and the u_s. The construction also applies when $E = W$. □

A.2 Translations from a flag kernel

Suppose $G \le \prod_{a=1}^{s} \mathrm{GL}(Y_a)$, where the Y_a are finite-dimensional $\mathbb{F}_p$-spaces with specified G-invariant flags. Put

$$0 = Y_{a,0} < Y_{a,1} < \cdots < Y_{a,\ell_a} = Y_a, \qquad X_{a,i} = Y_{a,i}/Y_{a,i-1}, \qquad d = \sum_{a=1}^{s} \dim_{\mathbb{F}_p} Y_a.$$

Write $h_{a,i}$ for the induced action of h on $X_{a,i}$, and define

$$\lambda_{\mathrm{gr}} : G \longrightarrow \prod_{a=1}^{s} \prod_{i=1}^{\ell_a} \mathrm{GL}(X_{a,i}), \qquad \lambda_{\mathrm{gr}}(h) = (h_{a,i})_{a,i}, \qquad R = \ker\lambda_{\mathrm{gr}}.$$

Take $\lambda = \lambda_{\mathrm{gr}}$ with codomain $J = \mathrm{im}\,\lambda_{\mathrm{gr}} \cong G/R$, so $K_{\mathrm{lin}} = R$, and assume $\hat{\rho}_r = 1_W$ for $r \in R$. Kernel generators and lifts are computed by the group operations of Subsection 3.5. No relation between W and the spaces $Y_a, X_{a,i}$ is assumed. The action of G/R on $\mathrm{Hom}_{\mathbb{F}_p}(X_{a,j}, X_{a,i})$ is $f \mapsto h_{a,i} f h_{a,j}^{-1}$.

Lemma 3.7 (restated). *The space $E = \tau(R)$ is an $\mathbb{F}_p[G/R]$-module. Every composition factor of E occurs among those of*

$$\bigoplus_{a=1}^{s} \bigoplus_{1\le i<j\le \ell_a} \mathrm{Hom}_{\mathbb{F}_p}(X_{a,j}, X_{a,i}).$$

A basis of E, word lifts in R of its basis vectors, and generators of $\ker\tau$ can be computed in $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$ time, where InSize *is the total explicit input length.*

Proof. By Lemma 3.6, τ is additive and $\tau(hrh^{-1}) = \hat{\rho}_h(\tau(r))$. Since $\hat{\rho}_r = 1_W$ for $r \in R$, the action on E factors through G/R.

Let

$$I = \bigoplus_{a=1}^{s} \{f \in \mathrm{End}_{\mathbb{F}_p}(Y_a) : f(Y_{a,i}) \subseteq Y_{a,i-1} \text{ for } 1 \le i \le \ell_a\}.$$

This is a nilpotent algebra normalized by G, with $R \le 1 + I$. For $k \ge 1$, put $R_k = R \cap (1 + I^k)$. The map $r \mapsto r - 1$ induces an injection of $\mathbb{F}_p[G/R]$-modules

$$R_k/R_{k+1} \hookrightarrow I^k/I^{k+1} \cong \bigoplus_{a=1}^{s} \bigoplus_{i=1}^{\ell_a - k} \mathrm{Hom}_{\mathbb{F}_p}(X_{a,i+k}, X_{a,i}),$$

with empty sums when $\ell_a - k < 1$. Indeed, products of two terms in I^k lie in I^{k+1}, and $[1+I, 1+I^k] \le 1 + I^{k+1}$. The induced action on each Hom summand is $f \mapsto h_{a,i} f h_{a,i+k}^{-1}$.

Each $\tau(R_k)/\tau(R_{k+1})$ is a quotient of R_k/R_{k+1}. Since $R_1 = R$ and the filtration terminates at 1, the composition factors of E occur in the displayed Hom spaces. No splitting of the successive extensions is needed.

Lemma 3.1 computes the image, word lifts, and kernel of $\tau : R \to W$. Applied to successive leading-block maps, it also computes the groups R_k. The faithful action on $\bigsqcup_a Y_a$ has degree $\sum_a p^{\dim Y_a}$; hence the group operations cost $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$. □

B Technical results for matrix group computation

We prove the results of Section 5, keeping its notation and the fixed dimension budget d. All constructions are over $\mathbb{F}_p$; restriction of scalars gives the stated bounds over $\mathbb{F}_q$. The constants c_0 and C_* are chosen sufficiently large once and remain fixed throughout the constructions.

B.1 Computational operations and original lifts

We use the deterministic permutation-group and quotient-group algorithms of Kantor–Luks [KL90a, Sections 3–4 and 9] and Seress [Ser03]. These compute orders, constructive membership, point stabilisers, normal closures, presentations, composition series, socles, solvable radicals, prime cores, and centralisers of normal subgroups in time polynomial in the permutation degree and input length. Constructive membership returns a word in the given generators, stored as a straight-line program of products and inverses. A matrix group on an r-dimensional prime-field space acts faithfully on its p^r vectors. Consequently these operations take time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$ whenever $r = O(d)$. Additional actions of degree $p^{O(d)}$ are represented on a disjoint union, retaining the original faithful action. A kernel is the pointwise stabiliser of the corresponding additional domain.

When an image is represented separately, we recover its full original kernel by the presentation construction of Luks [Luk92, Lemma 4.2]. Lift the image generators, evaluate its defining relators on

those lifts, and include each original generator corrected by a lifted word for its image. Their normal closure is exactly the kernel: after factoring it out, the image presentation gives an inverse to the induced surjection. Thus all subgroups and quotients below retain original words, projection kernels, and full preimages; no splitting of an extension is assumed. Generator lists are compressed using constructive membership, adding an element only when it increases the subgroup. Each addition at least doubles the order, so at most $\log_2 |H|$ generators are needed for a subgroup of the original group H.

All intertwiner spaces and spaces of invariant forms are computed by linear equations on the generator matrices. An endomorphism field represented on an r-dimensional simple module has at most p^r elements; enumerating that field, when necessary, also fits the same bound. In particular, the arguments below do not require enumerating a large classical group.

B.2 Invariant composition flags and primitive blocks

Theorem 5.1 (restated). *Let $H \leq \mathrm{GL}(d,p)$ be given by generators. The following can be computed in time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$.*

1. *An invariant composition flag $0 = V_0 < V_1 < \cdots < V_\ell = \mathbb{F}_p^d$, such that the action of H on the associated graded space has kernel $\mathrm{O}_p(H)$.*
2. *For each V_i/V_{i-1}, a direct-sum decomposition into blocks permuted transitively by $G = H/\mathrm{O}_p(H)$, with each block stabiliser acting irreducibly and linearly primitively on its block.*

Proof. We use the permutation-group operations described in Section B.1, applied to the action on vectors. For every nonzero v in the current space U, close $\mathbb{F}_p v$ under its generator matrices by repeated linear-span operations. This finds $\mathrm{Span}_{\mathbb{F}_p}\{hv : h \in H\}$ with polynomially many linear-algebra operations. If U has a proper nonzero invariant subspace, a vector in that subspace produces a proper invariant span. Thus exhausting the at most $p^{\dim U}$ vectors either finds such a subspace or certifies irreducibility. Recurse on a subspace found in this way and on its quotient. The recursion has at most $2d-1$ nodes; the dimensions add at every split. Record bases, inclusions and quotient maps at each node. This constructs the required flag within the stated time, and the same procedure refines a given invariant flag.

Let R be the kernel on its associated graded space. In a basis adapted to the flag, R is a group of block unitriangular matrices, so it is a normal p-subgroup of H. Conversely, every finite p-group acting in characteristic p has a nonzero fixed vector: its orbits on a nonzero finite vector space have p-power sizes, so the number of fixed vectors is divisible by p and cannot consist of the zero vector alone. On each irreducible flag factor, the fixed space of $\mathrm{O}_p(H)$ is also H-invariant, by normality. It is therefore the whole factor. Consequently $R = \mathrm{O}_p(H)$. Its generators are obtained as the kernel of the graded vector action, retaining the original vector action and its lift words. Both vector domains have at most p^d points.

It remains to decompose each irreducible graded factor U under its image A. For every $u \in U \backslash \{0\}$ compute the orbit $\Omega = u^A$, which spans U, and choose a basis $B \subseteq \Omega$. For $x \in \Omega \backslash B$, its fundamental circuit consists of x and the basis vectors with nonzero coefficients in the expansion of x in B. Form a graph on Ω by joining the members of each fundamental circuit. Different connected components use disjoint subsets of B, so their spans form a direct sum. Every minimal linear dependency lies in one of those summands: projection to the summands would otherwise produce a proper dependency. Conversely, the fundamental circuits are minimal dependencies. Thus the connected components

are determined by all minimal dependencies, independently of B, and are permuted by A. This permutation action is transitive, since Ω is one orbit.

Suppose $U = \bigoplus_{a=1}^{k} U_a$ is any nontrivial direct-sum system permuted by A. For a seed $u \in U_1 \setminus \{0\}$, its orbit lies in $\bigcup_a U_a$, and every minimal dependency lies in a single U_a. The graph constructed from this seed therefore has more than one component. Exhausting all seeds consequently either finds a nontrivial system or certifies linear primitivity. Each orbit and graph has at most $p^{\dim U}$ vertices, so the complete search costs $p^{O(\dim U)} \cdot \mathrm{poly}(\mathrm{InSize})$.

For a system found in this way, compute a block stabiliser and transversal words from the permutation action on the at most $\dim U$ blocks. The stabiliser acts irreducibly on its block: translating a proper nonzero invariant subspace to every other block would give a proper nonzero A-invariant subspace of U. Recurse inside one block stabiliser and transport its decomposition to the other blocks using the transversal words. There are linearly many nodes in the resulting refinement tree. Generator compression as in Section B.1 keeps the encodings polynomial in the vector-action degree and the original input. The final blocks are permuted transitively, their stabilisers act irreducibly and linearly primitively, and the sum of all their dimensions is d. □

B.3 Large local components and diagonal links

Theorem 5.2 (restated). *Given generators for $H \leq \mathrm{GL}(d,p)$ and the decompositions from Theorem 5.1, the following can be computed in time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$.*

1. *For every unbounded P_i, a normal quasisimple subgroup $S_i \trianglelefteq P_i$ satisfying $[P_i : S_i] \leq p^{5d_i}$, and an explicit S_i-module isomorphism $W_i \cong X_i^{\oplus m_i}$, where X_i is a standard module for S_i.*

2. *A central product $C = \prod_{I \in \mathcal{I}} Q_I$ (Q_I quasisimple) of the form described above, satisfying*

$$C \leq N_{\mathrm{blk}}, \qquad C \trianglelefteq G, \qquad [N_{\mathrm{blk}} : C] \leq p^{(c_0+1)d}.$$

   The output includes $\mathcal{I}$ and generators for each Q_I and its centre. Every $I \in \mathcal{I}$ contains only indices of unbounded blocks, and $\rho_i(Q_I) = S_i$ for every $i \in I$. Explicit isomorphisms identify the simple central quotients $S_i/\mathrm{Z}(S_i)$ diagonally within each I.

3. *Generators for the subgroup $D = D(C)$. This subgroup satisfies*

$$D \leq C \leq N_{\mathrm{blk}} \leq G, \qquad D \trianglelefteq G.$$

   Every local classical component S_i satisfying $|S_i/\mathrm{Z}(S_i)| > T_{\mathrm{cl}}$ is the image under ρ_i of a factor selected in D.

4. *For every block W_i on which D acts nontrivially, the defining field K_i and explicit natural K_i-coordinates for the standard module X_i from Item (1), in the prescribed stable range. The output includes the natural action matrices, coordinate identifications realising the diagonal links in Item (2), and projection maps from the corresponding full preimages in H.*

We first record the constructive ingredients. All group computations below use Section B.1, and retain original generator words and the kernels of the displayed actions.

Lemma B.1 (Extracting quasisimple components)**.** *Given generators of $A \leq \mathrm{GL}(r,p)$, its quasisimple components, their centres, and their simple central quotients can be constructed in time $p^{O(r)} \cdot \mathrm{poly}(\mathrm{InSize})$.*

Proof. Compute the Fitting subgroup F of A, namely its largest nilpotent normal subgroup, as the product of the prime cores $\mathrm{O}_\ell(A)$ over primes ℓ dividing $|A|$. Every such prime divides $p\prod_{j=1}^{r}(p^j-1)$; trial division of these integers costs $p^{O(r)}$. Put $B = C_A(F)$ and $Z = \mathrm{Z}(F)$. Since $B \trianglelefteq A$, its Fitting subgroup is normal in A, and hence $F(B) \leq F \cap B = Z$. Conversely Z is central in B, giving $F(B) = Z$. A nilpotent normal subgroup of B/Z lifts to a nilpotent normal subgroup of B, because its kernel is central. Therefore $F(B/Z) = 1$. The solvable radical of B/Z is also trivial, since a nontrivial finite solvable normal subgroup has a nontrivial characteristic Fitting subgroup.

Compute the socle of B/Z. It is a direct product of nonabelian simple groups. Separate its simple direct factors by finding a minimal normal subgroup inside this direct product and then its centraliser there, repeatedly. For a factor T_j, let A_j be its full preimage in B, and put $Q_j = A_j'$. Since T_j is perfect, $A_j = A_j'Z$, and consequently

$$A_j' = [A_j'Z, A_j'Z] = [A_j', A_j'].$$

Thus Q_j is perfect with simple central quotient T_j and is quasisimple. Distinct Q_j commute: their commutators lie in Z, and a central commutator map from a perfect group is trivial. These groups are components of A, because the simple socle factors are subnormal in B/Z, and $B \trianglelefteq A$. Conversely, a component of a finite group centralises its Fitting subgroup; its image in the radical-free group B/Z is a simple socle factor. Hence this procedure finds every component. Its centre and central quotient are then obtained by the same normal-subgroup operations. The faithful vector action has degree p^r, so all these computations have the required bound. □

Lemma B.2 (Homogeneous coordinates)**.** *Let $P \leq \mathrm{GL}(W)$ be irreducible and linearly primitive, and let $S \trianglelefteq P$. Then $W{\downarrow_S}$ is semisimple and homogeneous. If S is quasisimple, one can construct a faithful simple constituent X, its actual endomorphism field $K = \mathrm{End}_{\mathbb{F}_p[S]}(X)$, and the isomorphism*

$$X \otimes_K \mathrm{Hom}_{\mathbb{F}_p[S]}(X, W) \longrightarrow W, \qquad x \otimes f \longmapsto f(x),$$

in time $p^{O(\dim_{\mathbb{F}_p} W)} \cdot \mathrm{poly}(\mathrm{InSize})$. Moreover, $|\mathrm{Z}(S)| < p^{\dim_{\mathbb{F}_p} W}$.

Proof. Clifford's theorem makes the restriction semisimple and makes its isotypic components a system permuted by P [Miy99, Theorem 4.3.1(i)–(ii)]. Linear primitivity forces one isotypic component. A simple constituent is faithful, since all copies have the same kernel and the given action of S on W is faithful. Find $X \leq W$ by the invariant-span search in Section B.2. Intertwiner equations compute K and $M = \mathrm{Hom}_{\mathbb{F}_p[S]}(X, W)$. Schur's lemma and the finiteness of K make it a field. Its elements are actual operators on X, so its arithmetic and compatible bases can be computed in these coordinates. The evaluation map displayed above is an isomorphism by semisimplicity and homogeneity; verify its rank. A K-basis of M also gives an explicit direct sum $W \cong X^{\oplus \dim_K M}$ over $\mathbb{F}_p$. The searches use at most $p^{O(\dim W)}$ vectors or field elements, and all other operations are linear algebra. Finally a central element of S acts on faithful X as an element of $K^\times$, so $|\mathrm{Z}(S)| \leq |K| - 1 < p^{\dim W}$. □

The local structural input and its construction. The structural theorem of Jaikin-Zapirain and Pyber, in the form of Halasi et al., gives the following for a sufficiently large absolute c_0: if $P \leq \mathrm{GL}(r, p)$ is irreducible and linearly primitive with $|P| > p^{c_0 r}$, it has a normal quasisimple subgroup S such that $[P : S] \leq p^{5r}$; its moving tensor factor is a natural classical module in characteristic p or a fully deleted alternating module [JZP11, Proposition 5.7] [HMPQ19, Theorems 3.1–3.2]. Extension scalars in that tensor description can be absorbed into the multiplicity space. In the notation of

Lemma B.2, its simple $\mathbb{F}_p$-constituent X is therefore a standard module in the sense of the main text.

Choose $c_0 > 12$. Since $|\mathrm{Z}(S)| < p^r$, the promised component satisfies

$$|S/\mathrm{Z}(S)| > p^{(c_0-6)r} > p^{5r}.$$

Every other component commutes with S, and its simple central quotient occurs in P/S, whose order is at most p^{5r}. Thus the component above the bound p^{5r} is unique and is found by Lemma B.1; denote it by S. This uniqueness makes the choice invariant under conjugation. Apply Lemma B.2 to construct the constituent and its multiplicity coordinates. Try the classical and alternating families below, accepting only the verified group and module identifications. For an alternating component the explicit recognition of X as a fully deleted module is supplied by Lemma B.4. That construction requires only this local component and its homogeneous coordinates.

For a classical component, the endomorphism field K of X is its defining matrix field. Test its representation against $\mathrm{SL}_a(K)$, $\mathrm{Sp}_a(K, b)$, $\mathrm{SU}_a(K/K_0, h)$ and $\Omega_a(K, Q)$, where $a = \dim_K X$ and, in the unitary case, $[K : K_0] = 2$. Invariant bilinear or Hermitian forms are obtained from equations on the actual generator matrices. For Hermitian forms these are linear equations over K_0; the field involution is known. Irreducibility makes a nonzero invariant form nondegenerate. In characteristic 2, invariant quadratic forms are found by solving $Q(gx) = Q(x)$ for their coefficients. A zero-polar quadratic form is the square of a linear functional; on a nontrivial irreducible module it cannot be nonzero and invariant. Thus polarization is injective on the invariant quadratic forms here. In particular an odd-dimensional orthogonal natural module in characteristic 2, with its invariant polar radical, cannot be an irreducible constituent of this kind.

These candidate groups can be generated within the required time. Use elementary generators for $\mathrm{SL}_a(K)$. For an isometry group, choose a nondegenerate one- or two-dimensional subspace and enumerate the images of its ordered basis, at most $|K|^{2a}$ possibilities. Test its restricted form and extend each admissible frame to an isometry by formed-space elimination. The transversal isometries, together with a recursively constructed pointwise stabiliser on the orthogonal complement, generate the full isometry group. Reuse the same stabiliser for every frame. Witt extension and the standard formed-space elimination justify these operations [Tay92, Chapters 7–11]; field norm equations can be solved by enumeration in K. Taking the derived group of the full unitary or orthogonal group gives SU or Ω in the stable dimensions. The constant c_0 may be enlarged so that the classical constituents arising here have those dimensions: from $|S| > p^{(c_0-5)r}$ and $|S| \le p^{[K:\mathbb{F}_p]a^2}$ one gets $a > c_0 - 5$. Test candidate-generator membership in the actual component and compare exact orders. Classification guarantees a successful candidate, and these checks certify equality. Finally, formed-space elimination puts the successful form into standard coordinates. The frame enumeration has cost $|K|^{O(a)} = p^{O(\dim_{\mathbb{F}_p} X)}$.

Lemma B.3 (Central product and diagonal identifications)**.** *For the local components constructed above, there is a normal central product $C = \prod_{I\in\mathcal{I}} Q_I \le N_{\mathrm{blk}}$ with the support and diagonal properties in Theorem 5.2, and $[N_{\mathrm{blk}} : C] \le p^{(c_0+1)d}$. Its factors, centres and diagonal identifications can be constructed in time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$.*

Proof. For bounded blocks put $S_i = 1$ and $B_i = P_i$; otherwise put $B_i = P_i/S_i$. Then

$$|B_i| \le p^{c_0 d_i}, \qquad \prod_i |B_i| \le p^{c_0 d}, \qquad \sum_i d_i = d.$$

Enumerate these quotients by coset breadth-first search and membership in S_i. Their regular actions

have total degree at most $dp^{c_0 d}$. Let

$$M_{\mathrm{blk}} = \ker\bigl(N_{\mathrm{blk}} \longrightarrow \prod_i B_i\bigr).$$

Compute this kernel on the disjoint union of those quotient domains. The conjugation-invariant choice of the local components makes $M_{\mathrm{blk}} \trianglelefteq G$. In particular $M_i = \rho_i(M_{\mathrm{blk}}) \leq S_i$ is normal in P_i. A normal subgroup of a quasisimple group is central or is the whole group: if it surjects onto the simple central quotient, then $S_i = M_i Z(S_i)$, and perfectness gives $S_i \leq M_i$. Thus every M_i is central or equals S_i.

Set $Z_0 = M_{\mathrm{blk}} \cap \prod_i Z(S_i)$. It is central in M_{blk}. Omitting central coordinates, M_{blk}/Z_0 is a subgroup of a product of nonabelian simple groups surjecting onto every factor. Such a subgroup is a direct product of full diagonal subgroups on disjoint classes of coordinates. One proves this by adjoining the simple coordinates successively: the kernel of the projection to the preceding coordinates is either the whole new simple factor, giving an independent factor, or trivial; in the latter case the new factor is a simple quotient of the preceding product and hence is identified with exactly one of its simple direct factors. This is also the subdirect-product argument used in [HMPQ19, proof of Theorem 3.3].

It follows that M_{blk}/Z_0 is perfect, so $M_{\mathrm{blk}} = M'_{\mathrm{blk}} Z_0$. Put $C = M'_{\mathrm{blk}}$; then $C = C'$ and $C \trianglelefteq G$. Moreover

$$[M_{\mathrm{blk}} : C] \leq |Z_0| \leq \prod_{i : S_i \neq 1} |\mathrm{Z}(S_i)| \leq p^d, \qquad [N_{\mathrm{blk}} : C] \leq p^{(c_0+1)d}.$$

Perfectness also implies that a central block image of C is trivial, whereas a noncentral image is all of S_i.

For every nontrivial local image compute $J_i = \ker(C \longrightarrow S_i/\mathrm{Z}(S_i))$. For a classical factor use its action on the natural projective space, with at most p^{d_i} points; for an alternating factor use the letter action of Lemma B.4. Two indices lie in the same diagonal class exactly when $J_i = J_j$, which is tested by mutual generator membership. For such a class I put

$$R_I = \bigcap_{j \notin I} J_j, \qquad Q_I = R'_I,$$

where the intersection ranges over the nontrivial local images. The image of R_I in $C/\mathrm{Z}(C)$ is precisely the simple direct factor associated with I; its kernel is central. The same commutator argument as above makes Q_I perfect with that simple central quotient. Its images off I are central and hence trivial, and its image at every $i \in I$ equals S_i. Therefore $\mathrm{supp}(Q_I) = I$. Distinct Q_I commute, since their commutators are central and they are perfect. Their product maps onto $C/\mathrm{Z}(C)$; the residual quotient is abelian and perfect, hence trivial. This proves $C = \prod_I Q_I$ with disjoint supports.

The equality $J_i = J_j$ also constructs the diagonal isomorphism: lift a generator of $S_i/\mathrm{Z}(S_i)$ to C, and read its image in $S_j/\mathrm{Z}(S_j)$. Equality of the kernels makes this independent of the lift, and the reverse construction is its inverse. Centres are computed by the operations in Section B.1. Every kernel used is a kernel of a displayed action of degree $p^{O(d)}$, and the retained words give all the original lifts. □

Proof of Theorem 5.2. The local construction and Lemma B.2 give Item (1), with $m_i = \dim_{K_i} \mathrm{Hom}_{\mathbb{F}_p[S_i]}(X_i, W_i)$. Lemma B.3 gives Item (2). Select the classical factors Q_I whose simple central quotients have order greater than $T_{\mathrm{cl}} = p^{C_* d}$, and generate their product D. Conjugation by G permutes the factors, preserving the classical type and their orders. Thus $D \trianglelefteq G$.

We verify that every local classical component above this threshold survives in D. The permutation-group bound of Guralnick, Maróti and Pyber implies that every nonalternating simple section of a subgroup of $\mathrm{Sym}(t)$ has order less than 3^t [GMP17, Proposition 6.1]. Indeed, the product of its nonalternating composition-factor orders is at most $24^{(t-1)/3} < 3^t$, and the bound can be applied to the subgroup in which the section occurs. Let $N_i = \rho_i(N_{\mathrm{blk}}) \trianglelefteq P_i$. If N_i does not contain S_i, then $N_i \cap S_i$ is central, so $S_i/\mathrm{Z}(S_i)$ is a section of P_i/N_i. This last group is a quotient of a subgroup of $G/N_{\mathrm{blk}} \leq \mathrm{Sym}(t)$. For $|S_i/\mathrm{Z}(S_i)| > T_{\mathrm{cl}}$ the simple classical group is in stable range, and in particular is nonalternating; taking $C_* > \log_2 3$ contradicts the permutation-group bound, since $t \leq d$. Thus N_i contains S_i.

If the image of C on this block were central, then the same simple quotient would be a section of N_{blk}/C, contradicting its order bound $p^{(c_0+1)d}$ when $C_* > c_0 + 1$. Hence a unique diagonal factor maps onto S_i. Its simple central quotient is isomorphic to $S_i/\mathrm{Z}(S_i)$. All its nontrivial local images are classical: an alternating image would identify this large simple classical quotient with an alternating group, which is excluded in the stable range. The factor is therefore selected in D. This proves Item (3).

For a selected local component, write $a_i = \dim_{K_i} X_i$ and $f_i = [K_i : \mathbb{F}_p]$. The elementary classical order estimate gives

$$p^{C_* d} < |S_i/\mathrm{Z}(S_i)| < p^{f_i a_i^2}, \qquad f_i a_i \leq d_i \leq d.$$

Thus $a_i > C_*$; choosing the fixed constant C_* sufficiently large enforces natural dimension at least 64 and Witt index at least 32 where prescribed. The local construction already gives its actual defining field and natural coordinates. Characteristic 2 orthogonal constituents are even-dimensional, as proved above.

To realise the diagonal identifications in these coordinates, enumerate isomorphisms of the actual defining fields, try the natural and dual alternatives, and solve the resulting intertwiner equations on the linked generator images. The stable-rank classical automorphism theorem states that inner, diagonal, field and graph automorphisms exhaust the possibilities [Ste60, Sections 3.2–3.6]. In this range graph automorphisms give the natural/dual alternative or a formed-space isometry; the triality case has been excluded. Inner and diagonal changes are included among the intertwiner solutions. When two lifted representations have the same projective action, their discrepancy is a scalar homomorphism from a perfect component and is therefore trivial. Check every accepted intertwiner on the actual generators and for full rank. This supplies the coordinate identifications and natural action matrices in Item (4).

Throughout the construction, every subgroup of the graded image is pulled back to H with the full kernel of the graded action; no splitting of $H \longrightarrow G$ is used. Inclusions, quotients, projection kernels and generator words from the flag and subsequent actions supply the required projection maps from these full preimages. All vector, field, frame and quotient domains have size $p^{O(d)}$, all refinement trees have polynomially many nodes, and the remaining work is linear algebra or the group operations of Section B.1. This proves the stated overall running time and completes Item (4). □

B.4 Alternating-dominant actions

Theorem 5.3 (restated). *Given generators for $H \leq \mathrm{GL}(d, p)$ and the data from Theorem 5.2, the following* alternating-dominant actions *can be computed in time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$.*

1. *The actions given by the bounded primitive images P_i; the classical primitive images P_i with $|S_i/\mathrm{Z}(S_i)| \leq T_{\mathrm{cl}}$; the quotient N_{blk}/C; and the permutation actions of G on the blocks W_i and factors Q_I, each of degree $O(d)$.*

2. *For each alternating block W_i, an auxiliary representation of the block stabiliser on $A_i = \mathbb{F}_p^{a_i} \otimes_{\mathbb{F}_p} M_i$, where $M_i = \mathbb{F}_p^{m_i}$, given by*

$$h \longmapsto \mathrm{perm}(\pi_h) \otimes \mu_h, \qquad h \in \mathrm{Stab}_G(W_i), \quad \pi_h \in \mathrm{Sym}_{a_i}, \quad \mu_h \in \mathrm{GL}(M_i).$$

The action permutes the summands in $A_i = \bigoplus_{j=1}^{a_i} (\mathbb{F}_p e_j \otimes M_i)$. The multiplicity action $\mu : h \mapsto \mu_h$ factors through P_i/S_i, so $|\operatorname{im}\mu| \le [P_i : S_i] \le p^{5d_i}$. This representation induces the given block-stabiliser action on

$$W_i \cong \frac{P_{\mathrm{aug},i} \otimes M_i}{(P_{\mathrm{aug},i} \cap \mathbb{F}_p\mathbf{1}) \otimes M_i},$$

where $P_{\mathrm{aug},i}$ is the sum-zero subspace of $\mathbb{F}_p^{a_i}$. The equivariant inclusion and quotient maps are supplied, and these presentations assemble over each block orbit into a G-equivariant presentation.

Every nonalternating simple section L of an image in Items (1)–(2) satisfies $|L| \le T_{\mathrm{cl}}$. This bound also holds for all subgroup images and quotients of these images. The auxiliary spaces satisfy $\sum_{i \text{ alternating}} \dim_{\mathbb{F}_p} A_i \le 4d/3$. All actions retain the original generator matrices, maps from the relevant full preimages in H, projection kernels, and equivariant inclusion and quotient maps.

The alternating construction below is independent of the diagonal-link construction: it needs only the normal local component and its action on one primitive block. Thus it also supplies the letter actions used to compute the simple-quotient kernels in the preceding subsection.

Lemma B.4 (Realising the full alternating block action)**.** *Let $P \le \mathrm{GL}(W)$ have a specified normal quasisimple subgroup S whose faithful action on W is a direct sum of copies of the fully deleted permutation module of Alt_a, with $a \ge 8$. Suppose $[P : S] \le p^{5\dim_{\mathbb{F}_p} W}$. In time $p^{O(\dim_{\mathbb{F}_p} W)} \cdot \mathrm{poly}(\mathrm{InSize})$ one can construct the letter action of S, coordinates $W \cong U_a \otimes_{\mathbb{F}_p} M$, and an action of P on $A = \mathbb{F}_p^a \otimes_{\mathbb{F}_p} M$ of the form*

$$h \longmapsto \mathrm{perm}(\pi_h) \otimes \mu_h, \qquad \pi_h \in \mathrm{Sym}_a, \quad \mu_h \in \mathrm{GL}(M).$$

The multiplicity homomorphism is trivial on S. The action on A presents W as the quotient of $P_{\mathrm{aug}} \otimes M$ by $(P_{\mathrm{aug}} \cap \mathbb{F}_p\mathbf{1}) \otimes M$, with explicit equivariant maps, and $\dim_{\mathbb{F}_p} A \le 4 \dim_{\mathbb{F}_p} W/3$.

Proof. For $a \ge 8$ the fully deleted permutation module U_a is faithful and absolutely irreducible. Its invariant dot product is nondegenerate: the radical of the dot product on P_{aug} is precisely $P_{\mathrm{aug}} \cap \mathbb{F}_p\mathbf{1}$. These are the standard deleted-module properties in [KL90b, pp. 185–187]. Find a simple S-submodule $X \le W$ by the invariant-span search: for each nonzero vector, repeatedly close its span under the generators, and recurse whenever the resulting nonzero invariant subspace is proper. The promised semisimplicity and homogeneity give $X \cong U_a$. Solve the intertwiner equations to obtain $M = \mathrm{Hom}_{\mathbb{F}_p[S]}(X, W)$. Since X is absolutely irreducible, $\mathrm{End}_{\mathbb{F}_p[S]}(X) = \mathbb{F}_p$, and the evaluation map

$$X \otimes_{\mathbb{F}_p} M \longrightarrow W, \qquad x \otimes f \longmapsto f(x),$$

is an isomorphism. Compute its matrix and verify its rank. These operations cost $p^{O(\dim_{\mathbb{F}_p} W)} \cdot \mathrm{poly}(\mathrm{InSize})$.

We next recover the letters in the actual coordinates of X. Try $a = \dim_{\mathbb{F}_p} X + 1$ and $a = \dim_{\mathbb{F}_p} X + 2$, discarding candidates inconsistent with the characteristic-dependent dimension formula for U_a or with $|S| = a!/2$. Compute a nonzero invariant bilinear form on X by linear equations. It

is a scalar multiple of the transported dot product, since X is absolutely irreducible. Enumerate the orbits of S on the projective points of X. The desired orbit has $\binom{a}{2}$ points and corresponds to the lines

$$\mathbb{F}_p(e_j - e_k) \quad (1 \le j < k \le a)$$

in U_a. These lines remain distinct in the quotient: a purported equality modulo $\mathbb{F}_p\mathbf{1}$ would equate a vector supported on at most four coordinates to a constant vector, forcing that constant to vanish because $a \ge 8$. Distinct lines have nonzero pairing exactly when their two-element supports intersect. Thus their pairing graph is the line graph of the complete graph on a letters.

Recover the letters as stars in this graph. For two adjacent vertices $jk, j\ell$, their common neighbours consist of a clique of size $a-3$ (the other edges through j) and the isolated vertex $k\ell$. The clique together with $jk, j\ell$ is the star of j. Collect these stars and check that there are exactly a, that each has size $a-1$, that each graph vertex belongs to exactly two stars, and that adjacency agrees with intersection of the resulting letter pairs. Check that the generators permute the stars, that their permutations are even, and that their permutation image has order $a!/2$. This produces a faithful letter action of S. Construct the associated fully deleted permutation matrices and solve their intertwiner equations with the matrices on X. Accept only when an invertible intertwiner exists, and retain it. The genuine coordinate-difference orbit passes all these tests. All orbit and graph searches have size $p^{O(\dim_{\mathbb{F}_p} X)}$; permutation-group operations use Subsection B.1.

For a generator h of P, normality of S gives a known automorphism $s \mapsto hsh^{-1}$ of its faithful letter action. Every automorphism of Alt_a, for $a \ge 8$, is conjugation by a unique $\pi_h \in \mathrm{Sym}_a$ [Wil09b, Theorem 2.3]. This permutation can be found without enumerating Sym_a. For each proposed image of one letter, take its pair orbit under the two letter actions on $[a] \times [a]$. An orbit is the graph of an intertwining bijection exactly when it has a points and both coordinate projections are bijective. Test each resulting permutation against the component generators. Uniqueness gives $\pi_{hh'} = \pi_h \pi_{h'}$.

In the coordinates $W \cong U_a \otimes M$, remove $\mathrm{del}(\pi_h) \otimes I_M$ from h, where del denotes the induced fully deleted permutation representation. The remaining matrix centralises S. Its unique form is $I_{U_a} \otimes \mu_h$, because the commuting algebra is $I_{U_a} \otimes \mathrm{End}_{\mathbb{F}_p}(M)$. Read μ_h by linear equations. Consequently

$$h = \mathrm{del}(\pi_h) \otimes \mu_h, \qquad \mu_{hh'} = \mu_h \mu_{h'}.$$

For $s \in S$, its letter permutation already supplies its action on U_a, so $\mu_s = I_M$. Thus μ factors through P/S and $|\operatorname{im} \mu| \le [P : S]$.

The matrices $\mathrm{perm}(\pi_h) \otimes \mu_h$ now give the required action on A. They preserve both $P_{\mathrm{aug}} \otimes M$ and $(P_{\mathrm{aug}} \cap \mathbb{F}_p\mathbf{1}) \otimes M$. Their quotient action is exactly the original action on W, through the computed intertwiner and evaluation map. Finally,

$$\dim_{\mathbb{F}_p} A \le \frac{a}{a-2} \dim_{\mathbb{F}_p} W \le \frac{4}{3} \dim_{\mathbb{F}_p} W.$$

All group computations retain words in the original generators, so the construction also retains the corresponding maps from full preimages and their kernels. □

Proof of Theorem 5.3. The block stabilisers, their images P_i, and their maps from H are obtained from the block action. The groups C and Q_I and the local components S_i are supplied by Theorem 5.2. Conjugating the computed block subspaces and the factors Q_I gives the two permutation actions in Item (1). There are at most d blocks and at most d disjoint nonempty factor supports, so these actions have degree at most d.

Bounded primitive images have order at most $p^{c_0 d_i}$. For an unselected classical block, the homogeneous standard-module coordinates embed $\mathrm{Z}(S_i)$ into the multiplicative group of its endomorphism field. Hence $|\mathrm{Z}(S_i)| < p^{d_i}$ and

$$|P_i| = [P_i : S_i]\,|\mathrm{Z}(S_i)|\,|S_i/\mathrm{Z}(S_i)| < p^{6d_i} T_{\mathrm{cl}} \le p^{(C_*+6)d}.$$

These images can therefore be enumerated within the claimed bound. Likewise $[N_{\mathrm{blk}} : C] \le p^{(c_0+1)d}$ allows us to enumerate N_{blk}/C by coset breadth-first search, using membership in C, and to give its regular action. Use the original vector actions alongside these actions when computing kernels, as in Subsection B.1. This constructs Item (1) with its required maps and words.

Apply Lemma B.4 to the full stabiliser image of each representative alternating block. It gives all the data in Item (2) for that stabiliser, including the multiplicity bound and the equivariant inclusion and quotient maps. To assemble an orbit of blocks, choose original group words t_i carrying a fixed representative block to its ith block. If $gW_i = W_j$, use the constructed representative-stabiliser action of $t_j^{-1} g t_i$ as the map between the ith and jth auxiliary copies. In a product of two group elements the intermediate transversals cancel, proving the group law. Transporting the original block coordinates by the same t_i makes the inclusion and quotient maps equivariant. The local dimension bounds add to give

$$\sum_{i \text{ alternating}} \dim_{\mathbb{F}_p} A_i \le \frac{4}{3} \sum_{i \text{ alternating}} d_i \le \frac{4}{3} d.$$

It remains to prove the assertion about simple sections. Here a *simple section* of a finite group means a simple quotient of one of its subgroups. We first record why the bounds below persist through the group constructions being used. If $B \trianglelefteq A$ and $J \le A$ has a simple quotient J/K, the image of $J \cap B$ in J/K is either trivial or all of J/K. In the first case J/K is a section of $J/(J \cap B) \le A/B$; in the second it is a section of $J \cap B \le B$. Thus every simple section of an extension is a simple section of its normal subgroup or its quotient. Repeated coordinate projections give the same conclusion for subgroups of finite direct products. Moreover, a section of a subgroup or quotient is already a section of the original group, by taking the appropriate preimage of its defining subgroup.

For $J \le \mathrm{Sym}(t)$, Guralnick, Maróti, and Pyber [GMP17, Proposition 6.1] bound the product of the orders of its composition factors other than alternating groups of degree greater than 4 by $24^{(t-1)/3}$. Apply this to the subgroup in which a specified simple section occurs. It follows that every nonalternating simple section of $\mathrm{Sym}(t)$ has order less than 3^t. This also bounds the nonalternating simple sections of the block and factor permutation images.

For a bounded primitive image the group order itself is at most $p^{c_0 d_i}$. For an unselected classical primitive image, use

$$\mathrm{Z}(S_i) \trianglelefteq S_i \trianglelefteq P_i.$$

The first group has order less than p^{d_i}, the middle quotient has order at most T_{cl}, and the last quotient has order at most p^{5d_i}. The extension observation therefore bounds every simple section of P_i by T_{cl}, once C_* dominates c_0 and 5. The quotient N_{blk}/C has order at most $p^{(c_0+1)d}$ and satisfies the same bound.

For the assembled alternating auxiliary actions, let the axes mean the summands $\mathbb{F}_p e_j \otimes M_i$. Their total number is at most $\sum_i a_i \le \sum_i \dim A_i \le 4d/3$. Their permutation image therefore has nonalternating simple sections of order less than $3^{4d/3}$. The kernel of this axis action embeds in the product of the actual multiplicity images on the individual blocks: fixing all axes forces the letter permutations to be trivial, leaving only the matrices μ_h. Each such image has order at most p^{5d_i},

so their product has order at most p^{5d}. The extension observation applies to the actual joint image, including any correlations between the permutation and multiplicity actions. Taking C_* also larger than $c_0 + 1$ and $\frac{4}{3}\log_2 3$ proves the required bound, including when $p = 2$. Its persistence under subgroup images and quotients follows from the same section observation.

Every vector space introduced has total dimension $O(d)$, and all explicitly enumerated groups have order $p^{O(d)}$. There are at most d local constructions. The operations in Subsection B.1, carried out simultaneously with the original generator matrices, retain all original lift words, full projection kernels, and the displayed equivariant maps. The complete running time is $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$. □

B.5 Actions on tensor factors

We first justify the isotypic coordinates used in Lemmas 5.4 and 5.5. On each irreducible summand of V_{gr}, restriction to the normal subgroup D is semisimple by Clifford's theorem [Miy99, Theorem 4.3.1(i)]. Consequently

$$V_{\mathrm{fix}} = V_{\mathrm{gr}}^{D}, \qquad V_D = [D, V_{\mathrm{gr}}] = \sum_{g \in S_D} (g-1)V_{\mathrm{gr}}, \qquad V_{\mathrm{gr}} = V_{\mathrm{fix}} \oplus V_D.$$

Indeed, the commutator space is zero on every trivial simple summand and is the whole summand on every nontrivial simple summand. Fixed spaces, commutator spaces, and the resulting projections are computed by linear algebra. Both summands are invariant under every transformation normalising D.

The product D here is the full selected classical product. A space fixed by one selected component but moved by another belongs to V_D. The fixed part V_{fix} retains the actual alternating-dominant presentations from Theorem 5.3.

Write $\rho_D : D \to \mathrm{GL}(V_D)$ for the action on the nontrivial part. Its enveloping algebra $A_D = \mathrm{span}_{\mathbb{F}_p} \rho_D(D) \le \mathrm{End}_{\mathbb{F}_p}(V_D)$ is semisimple. Find a simple submodule X, using the invariant-subspace computation of Theorem 5.1, and compute $\mathrm{Hom}_{\mathbb{F}_p[D]}(X, V_D)$ by solving the intertwining equations for the generators of D. The sum of the images of these homomorphisms is the entire isotypic component of X. A D-equivariant projection onto that component is found by linear equations: require the projection to restrict to the identity on the component and to commute with the generators of D. Semisimplicity guarantees a solution. Repeat on its kernel. Schur's lemma and the fact that a finite division ring is a field give

$$K_j = \mathrm{End}_{\mathbb{F}_p[D]}(X_j), \qquad M_j = \mathrm{Hom}_{\mathbb{F}_p[D]}(X_j, V_D), \qquad V_D \cong \bigoplus_{j \in J} X_j \otimes_{K_j} M_j.$$

The last map is evaluation, $x \otimes \varphi \mapsto \varphi(x)$. All equivalent copies, including copies in different original blocks, are therefore collected in the same multiplicity space. The inclusions, projections, and original block labels are retained. There are at most $O(d)$ simple summands to process, so this construction takes $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$ time. A fixed number of copies and duals has the same bound.

Lemma 5.4 (restated). *Given generators for $A \le N_{\mathrm{GL}_{\mathbb{F}_p}(V_D)}(D)$ and the above decomposition, the following can be computed in time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$: a group $\widehat{A}$ and a surjection $\pi_{\mathrm{fac}} : \widehat{A} \twoheadrightarrow A$ recording the maps on the natural and multiplicity factors, field automorphisms, and permutations of the isotypic summands. Its kernel is*

$$\ker \pi_{\mathrm{fac}} = \prod_{j \in J} \{(cI_{X_j}, c^{-1} I_{M_j}) : c \in K_j^{\times}\}.$$

The group $\widehat{A}$ has a faithful $\mathbb{F}_p$-representation on $\bigoplus_j (X_j \oplus M_j)$ of dimension at most $2\dim_{\mathbb{F}_p} V_D = O(d)$; all action maps and projection kernels are supplied. For a given surjection $\rho : H \twoheadrightarrow A$, the construction also retains the group of pairs $(h, \widehat{a})$ satisfying $\rho(h) = \pi_{\mathrm{fac}}(\widehat{a})$ and the full kernel of ρ.

Proof. Write $a_j = \dim_{K_j} X_j$ and $b_j = \dim_{K_j} M_j$. The image of the enveloping algebra on the jth isotypic component is

$$\mathrm{End}_{K_j}(X_j) \otimes_{K_j} I_{M_j}.$$

Every $h \in A$ permutes these simple algebra factors and their isotypic spaces. If it sends the jth component to the kth, conjugation induces an algebra isomorphism between the displayed factors. On their centres this gives a field isomorphism $\sigma : K_j \to K_k$, fixing $\mathbb{F}_p$. In particular $a_j = a_k$ and $b_j = b_k$.

After removing the entrywise action of σ, the remaining isomorphism α of full matrix algebras is inner. Its implementing matrix u is computed from the linear equations

$$uT = \alpha(T)u \qquad (T \text{ in a basis of } \mathrm{Mat}(a_j, K_j)).$$

There is a nonzero solution, and every nonzero solution is invertible: its kernel and image are submodules of the simple natural modules. Removing this natural-factor map and the common field map from h leaves a map commuting with the full matrix algebra. It is therefore $I_{X_j} \otimes v$, where v is an invertible multiplicity-factor map, again determined by linear equations. Thus the map on the homogeneous component has the form

$$x \otimes m \longmapsto u_h(x) \otimes v_h(m),$$

where $u_h : X_j \to X_k$ and $v_h : M_j \to M_k$ are semilinear with respect to the same field isomorphism σ.

Two such factorizations of the same map have the same field map, and their natural-factor maps differ by a scalar, since the centraliser of a full matrix algebra on its natural module is its scalar field. The multiplicity-factor maps then differ by the reciprocal scalar. This proves the asserted description of $\ker \pi_{\mathrm{fac}}$.

Let $\widehat{A}$ be the group of all these factorizations of elements of A, including their type permutations. Composition uses the common field maps on both tensor factors, so it respects the tensor product and makes π_{fac} a homomorphism. Choose one factorization of each given generator of A and adjoin generators of each scalar kernel $K_j^{\times}$. These generate all of $\widehat{A}$: any factorization differs from the corresponding word in the chosen lifts by an element of the scalar kernel. The field maps are read from the centres of the algebra factors; all field arithmetic and scalar-generator computations take $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$ time.

The factor maps act faithfully on $\bigoplus_j (X_j \oplus M_j)$. They are $\mathbb{F}_p$-linear, even when semilinear over K_j, and the summands are permuted by the recorded type permutation. Since $a_j, b_j \geq 1$, we have $a_j + b_j \leq 2a_jb_j$, whence

$$\sum_j [K_j : \mathbb{F}_p](a_j + b_j) \leq 2 \sum_j [K_j : \mathbb{F}_p] a_j b_j = 2\dim_{\mathbb{F}_p} V_D.$$

The action and projection kernels can now be computed on this faithful representation using the group operations of Section B.1.

Finally, for $\rho : H \twoheadrightarrow A$ take the fibre product

$$\{(h, \widehat{a}) \in H \times \widehat{A} : \rho(h) = \pi_{\mathrm{fac}}(\widehat{a})\}.$$

Pair each chosen lifted image generator with a preimage in H, and adjoin the generators $(1, z)$ of the scalar kernel and $(k, 1)$ for generators k of $\ker \rho$. These generate the fibre product, because any two original lifts with the same image differ by $\ker \rho$. The full original kernel and its original encoding are retained; the claimed small faithful representation is that of $\widehat{A}$. This also proves the running-time claim. □

B.6 Multiplicity and normaliser bounds

Lemma 5.5 (restated). *Fix a constant $\kappa \geq 1$ and suppose $\dim_{\mathbb{F}_p} V_D \leq \kappa d$. Put $f_j = [K_j : \mathbb{F}_p]$, $a_j = \dim_{K_j} X_j$, and $b_j = \dim_{K_j} M_j$. Then*

$$|\operatorname{Cent}_{\mathrm{GL}_{\mathbb{F}_p}(V_D)}(D)| \leq p^{\sum_j f_j b_j^2} \leq p^{\kappa^2 d / C_*}.$$

Given generators for a subgroup $D \leq A_0 \leq N_{\mathrm{GL}_{\mathbb{F}_p}(V_D)}(D)$, one has

$$[A_0 : D] \leq p^{O(d)}, \qquad A_{\mathrm{in}} = D \operatorname{Cent}_{A_0}(D),$$

where A_{in} consists of the elements inducing inner automorphisms of D. Let $\pi_{\mathrm{fac}} : \widehat{A}_0 \twoheadrightarrow A_0$ be the cover from Lemma 5.4. Its joint image under the action on $\bigoplus_j M_j$ and the map $\widehat{a} \mapsto \pi_{\mathrm{fac}}(\widehat{a})D$ can be enumerated in time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$, retaining original lifts and projection kernels.

Proof. Each X_j is a natural constituent, or one of its specified duals, for a selected classical factor. That factor has a simple central quotient of order greater than $p^{C_* d}$, while its image on X_j lies in $\mathrm{GL}(a_j, K_j)$. Hence

$$C_* d < f_j a_j^2, \qquad \sum_j f_j a_j b_j \leq \kappa d.$$

For each j, these inequalities imply

$$\frac{b_j}{a_j} = \frac{f_j a_j b_j}{f_j a_j^2} < \frac{\kappa}{C_*}.$$

Schur's lemma identifies the full commuting algebra with $\prod_j \mathrm{End}_{K_j}(M_j)$, including maps mixing equivalent copies from different original blocks. Therefore

$$|\operatorname{Cent}_{\mathrm{GL}_{\mathbb{F}_p}(V_D)}(D)| = \prod_j |\mathrm{GL}(b_j, K_j)| \leq p^{\sum_j f_j b_j^2},$$

$$\sum_j f_j b_j^2 = \sum_j \frac{b_j}{a_j} f_j a_j b_j \leq \frac{\kappa^2 d}{C_*}.$$

We next bound the outer automorphisms that can occur in the normaliser. The dimension of every distinct nontrivial type is at least $\sqrt{C_* d}$, because $f_j a_j \geq \sqrt{f_j C_* d} \geq \sqrt{C_* d}$. Thus there are $O(\sqrt{d})$ such types. The same bound holds for the number of selected classical factors: choose one constituent from each factor, using their disjoint block supports. Permutations of these factors have order at most $(O(\sqrt{d}))! = p^{O(d)}$.

The quotient $D/\mathrm{Z}(D)$ is the direct product of the selected nonabelian simple classical groups. Its outer automorphisms consist of permutations of isomorphic factors and the diagonal, field, and graph automorphisms of the individual factors [Ste60, Sections 3.2–3.6]. For a classical factor of natural dimension a over a field of prime-field degree f, the number of diagonal and field automorphisms

is at most $p^{O(fa)}$, and the graph-automorphism group has bounded order. The field and natural dimensions are charged to distinct selected block supports, whose dimensions sum to at most d. Consequently

$$|\operatorname{Out}(D/\mathrm{Z}(D))| = p^{O(d)}.$$

An automorphism of the perfect group D acting trivially on $D/\mathrm{Z}(D)$ is the identity: writing its value on g as gz_g, with $z_g \in \mathrm{Z}(D)$, makes $g \mapsto z_g$ a homomorphism from D to an abelian group, and such a homomorphism is trivial. It follows that the kernel of $A_0 \to \operatorname{Out}(D/\mathrm{Z}(D))$ is precisely $D \operatorname{Cent}_{A_0}(D)$. In particular

$$[A_0 : D] \le |\operatorname{Cent}_{\mathrm{GL}_{\mathbb{F}_p}(V_D)}(D)|\,|\operatorname{Out}(D/\mathrm{Z}(D))| = p^{O(d)}.$$

If $a \in A_{\mathrm{in}}$, choose $u \in D$ inducing the same conjugation on D. Then $u^{-1}a \in \operatorname{Cent}_{A_0}(D)$. Conversely, every element of $D \operatorname{Cent}_{A_0}(D)$ induces an inner automorphism. This proves the asserted equality for A_{in}.

For the final assertion, put $t = |J|$. Every factorization in $\widehat{A}_0$ acts on the multiplicity spaces by a type permutation, a compatible field isomorphism on each type, and an invertible linear map over that field. After choosing reference isomorphisms between equal fields, the total number of possible multiplicity actions is at most

$$t! \prod_j f_j |\mathrm{GL}(b_j, K_j)|.$$

Here $t = O(\sqrt{d})$, $\sum_j f_j \le \kappa d$, and the product of the general-linear orders has already been bounded. Since $f_j \le p^{f_j}$, the displayed expression is $p^{O(d)}$. The simultaneous image with the quotient A_0/D therefore has order $p^{O(d)}$ as well.

Enumerate this actual joint image breadth first using the original generators and retain a lift word for each discovered element. Equality of two records is tested by equality on the multiplicity spaces and membership of the quotient of their projected matrices in D. The operations of Section B.1 give the claimed running time. Schreier generators obtained during the same enumeration give the full projection kernel. If the action comes from an original group, carry its lifts throughout and adjoin its full kernel as in Lemma 5.4. Thus the enumeration preserves simultaneous actions and all original kernels. □

B.7 Separation in a simultaneous image

Lemma 5.6 (restated). *Let Γ be a finite group given by generators, let $\alpha : \Gamma \twoheadrightarrow A$ and $\beta : \Gamma \twoheadrightarrow B$ be explicit homomorphisms, and let $E \le A$ be a given central product of quasisimple groups. If no quotient of a subgroup of B is isomorphic to the simple central quotient of a factor of E, then*

$$E \times 1 \le (\alpha, \beta)(\Gamma) \le A \times B.$$

Generators of $E \times 1$ and its full preimage $\alpha^{-1}(E) \cap \ker \beta$ can be computed using polynomially many constructive kernel, membership and lifting operations. If these operations take time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$, the construction has the same bound.

Proof. Put $R = \ker \alpha$ and $\Gamma_E = \alpha^{-1}(E)$. Since $R \trianglelefteq \Gamma_E$, the group

$$Q = \beta(\Gamma_E)/\beta(R)$$

is defined and is both a quotient of $\Gamma_E/R \cong E$ and a quotient of a subgroup of B. Every nontrivial quotient of E has a simple section isomorphic to the simple central quotient of one of its quasisimple factors. Indeed, if the quotient is nontrivial, some factor has nontrivial image. The kernel on that factor is a proper normal subgroup and hence is central, so its image has the same simple central quotient. Taking the preimage of this subgroup under $\beta(\Gamma_E) \twoheadrightarrow Q$ would exhibit the forbidden simple quotient of a subgroup of B. The hypothesis therefore forces $Q = 1$.

For every $h \in \Gamma_E$ there is consequently $r \in R$ such that $\beta(r) = \beta(h)$. The element hr^{-1} belongs to $\Gamma_E \cap \ker \beta$ and has the same α-image as h. Thus

$$\alpha(\Gamma_E \cap \ker \beta) = E,$$

which proves the required containment.

Constructively, compute R, lift generators of E through α, and adjoin these lifts to generators of R to obtain Γ_E. Compute the kernel of β restricted to this group. The preceding argument shows that its image under α is all of E, so constructive membership and lifting give, for each chosen generator e of E, an original element mapping to $(e, 1)$. These simultaneous images generate $E \times 1$. Together with generators of $\ker(\alpha, \beta)$, their lifts generate its full preimage $\alpha^{-1}(E) \cap \ker \beta$. This uses only polynomially many of the stated operations and proves the algorithmic assertions. □

C Technical results for classical group representations and cohomologies

We prove the results of Section 6, recalling each statement before its proof. The stable range in those statements is the range specified in Subsection 6.1: natural dimension at least 64, and Witt index at least 32 for symplectic, unitary, and orthogonal factors. The supporting lemmas state their individual rank hypotheses explicitly.

We retain the input fields, natural actions, forms, multiplicities, and central-product quotient maps from the main statements. Throughout, InSize is the total explicit input length, and $d = \sum_i [\mathbb{K}_i : \mathbb{F}_p] n_i$ counts each natural representation in the input product cover once. Repeated tensor summands contribute to the input length rather than adding new natural spaces to d. The proofs use the joint enveloping algebra and the cohomology and field conventions in Subsection 6.2.

C.1 Preliminaries and basic properties

C.1.1 Splitting composition-disjoint summands

We retain the joint enveloping algebra A and the decomposition $M = U \oplus W$ from Subsection 6.2.

Proposition 6.4 (restated). *Suppose $M = U \oplus W$ is a direct sum of composition-disjoint D-submodules. Then the two coordinate projections belong to A, and every D-invariant subspace $X \leq M$ satisfies*

$$X = (X \cap U) \oplus (X \cap W).$$

In particular, P and N split across these summands.

Proof. Let $J = \{a \in A : aU = 0\}$ and $J' = \{a \in A : aW = 0\}$ be the annihilator ideals. Since A acts faithfully on M, $J \cap J' = 0$. We claim that $J + J' = A$. Otherwise, a simple module of $A/(J + J')$ would occur as a composition factor of both A/J and A/J'. Evaluation on a basis

embeds A/J into a finite direct sum of copies of U, and similarly for A/J' and W. This contradicts composition-disjointness. The two-sided Chinese remainder theorem now gives

$$A \cong A/J \times A/J'.$$

The elements corresponding to $(1,0)$ and $(0,1)$ act as the two coordinate projections. Every D-invariant subspace is A-invariant, so these projections give the asserted splitting. □

C.1.2 Embedding into injective modules

Lemma 6.5 (restated). *Let $A \le \mathrm{End}_{\mathbb{F}_p}(M)$ be a finite-dimensional $\mathbb{F}_p$-algebra. Let $k \ge 0$ and let $P \le N \le M^{\oplus k}$ be arbitrary explicitly given A-submodules. Set $Q = N/P$. Neither P nor N is assumed to be a direct sum of copies of M. Suppose C contains one copy of every indecomposable injective left A-module. If $f_1, \dots, f_s$ is a basis of $\mathrm{Hom}_A(Q, C)$, then*

$$\iota : Q \longrightarrow C^s, \qquad v \longmapsto (f_1(v), \dots, f_s(v))$$

is injective. It and the algebra A can be computed by linear algebra. The target has $\mathbb{F}_p$-dimension at most $\dim_{\mathbb{F}_p} Q (\dim_{\mathbb{F}_p} C)^2$.

Proof. By projective–injective duality [Ben91, Section 1.5], C is a cogenerator, so a basis of $\mathrm{Hom}_A(Q, C)$ separates points. Compute this basis by intertwining equations and A by closing the input generator matrices under linear combinations and products. The bounds $\dim \mathrm{Hom}_A(Q, C) \le \dim Q \dim C$ and $\dim A \le (\dim M)^2$ give the claimed size and linear-algebra cost. □

C.1.3 Removing translations after equivariant maps

Lemma 6.6 (restated). *Let Q, I, E be $\mathbb{F}_p[G]$-modules, let $\iota : Q \hookrightarrow I$ and $\pi : I \to E$ be equivariant $\mathbb{F}_p$-linear maps, and let $c : G \to Q$ be a cocycle. Write $c_g = c(g)$ and let L_g, L_g^I, and L_g^E denote the linear actions on Q, I, and E, respectively. Suppose $H^1(G, E) = 0$. Given $x, y \in Q$, the complete affine transporter for $\alpha_g(v) = L_g(v) + c_g$, $g \in G$, is obtained by one complete linear transporter in E and one complete affine transporter in $\ker \pi$, under the full source stabilizer returned by the first transporter. The residual cocycle is computed from the original affine action.*

Proof of Lemma 6.6. Since $H^1(G, E) = 0$, solve $\pi\iota(c_g) = (L_g^E - \mathrm{id}_E)b$ on generators for $b \in E$. The cocycle identity extends these equations to G, and

$$\psi(v) = \pi\iota(v) + b, \qquad \psi(\alpha_g(v)) = L_g^E(\psi(v)).$$

Put $W = \iota(Q)$ and $T = W \cap \ker \pi$, and give W the affine action with translation $\iota(c_g)$. The map $w + T \mapsto \pi(w) + b$ identifies its quotient action on W/T with the linear action on the invariant affine subset $b + \pi(W) \subseteq E$. If the linear transporter of $\psi(x)$ to $\psi(y)$ is empty, so is the required transporter. Otherwise write it as aS, with S the full source stabilizer. Apply Lemma 3.5 to $T \le W$. It returns the residual translation and target

$$c_s^{\mathrm{res}} = \iota(\alpha_s(x) - x), \qquad w = \iota(\alpha_{a^{-1}}(y) - x), \qquad c_s^{\mathrm{res}}, w \in T.$$

The same cocycle defines $\beta_s(u) = L_s^I(u) + c_s^{\mathrm{res}}$ on all of $\ker \pi$. Its transporter from 0 to w and stabilizer of 0 are unchanged by this enlargement from T. Lemma 3.5 therefore gives

$$\mathrm{Trans}_{G,\alpha}(x, y) = a\mathrm{Trans}_{S,\beta}(0, w),$$

and the residual source stabilizer is exactly the original affine stabilizer of x, as required. □

C.1.4 Simultaneous scalar ambiguities

Lemma 6.7 (restated). *Let $\mathbb{K}/\mathbb{F}_p$ be a finite extension and let $A_j, B_j \in \mathrm{Mat}(n, \mathbb{K})$ for $1 \le j \le s$. Every solution with $0 \neq X \in \mathrm{Mat}(n, \mathbb{K})$ and $(\lambda_1, \dots, \lambda_s) \in \mathbb{K}^s$ of*

$$XA_j - B_jX = \lambda_j X \qquad (1 \le j \le s)$$

belongs to one of at most n^2 nonzero common scalar eigenspaces. All tuples $(\lambda_1, \dots, \lambda_s)$ of scalar shifts that occur can be found in $|\mathbb{K}|\mathrm{poly}(n, s, \log|\mathbb{K}|)$ operations. For independent matrix factors with $\sum_i [\mathbb{K}_i : \mathbb{F}_p] n_i \le d$, all joint choices number at most 4^d, and their computation costs $p^{O(d)}\mathrm{poly}(\mathrm{InSize})$.

Proof. For $L_j(X) = XA_j - B_jX$, successively intersect each current subspace with $\ker(L_j - \lambda I)$ for every $\lambda \in \mathbb{K}$, discarding zero intersections. Distinct eigenspaces have independent sum, so the total dimension never increases and at most n^2 subspaces survive at each stage; Gaussian elimination gives the stated computation time. The product bounds follow from $\prod_i n_i^2 \le 4^{\sum_i n_i} \le 4^d$ and $|\mathbb{K}_i| \le p^d$. □

C.1.5 Characters of classical groups

Lemma 6.8 (restated). *In the stable range, the specified derived classical groups are kernels of the following characters of their containing full groups:*

$$\mathrm{SL}_n(\mathbb{K}) = \ker(\det : \mathrm{GL}(n, \mathbb{K}) \to \mathbb{K}^\times), \qquad \mathrm{SU}_n(\mathbb{K}/\mathbb{K}_0) = \ker(\det : \mathrm{U}_n(\mathbb{K}/\mathbb{K}_0) \to \mathbb{K}^\times),$$

$$\Omega(V, Q) = \begin{cases} \ker(\det, \mathrm{sp}), & p \neq 2, \\ \ker \Delta, & p = 2, \end{cases} \qquad \Delta(g) = \mathrm{rank}(I - g) \bmod 2.$$

In the unitary formula, $\mathbb{K}_0 = \mathbb{K}^\sigma$ is the fixed field, denoted by $\mathbb{F}$ at the start of Section 6. Here sp *is the spinor norm with values in $\mathbb{K}^\times/(\mathbb{K}^\times)^2$, and the stable symplectic group requires no additional character condition. These characters are computable in polynomial time from the matrices. For a product of the input natural spaces, their joint image has size $p^{O(d)}$.*

Proof. The character formulas are standard [Tay92, Chapters 8–11, especially Theorems 11.43 and 11.50–11.51]. Compute the spinor norm as the determinant modulo squares of the Wall form on $\mathrm{im}(I - g)$ [Tay92, Theorem 11.32 and the definition preceding Theorem 11.50]; this uses Gaussian elimination and a square test. Determinants and Dickson rank parity are also polynomial-time computable. Each linear or unitary factor contributes at most $|\mathbb{K}|$ values and each orthogonal factor at most four, giving $p^{O(d)}$ jointly. □

C.2 Finite-field comparison and descent

The following tools identify finite-group cohomology and joint enveloping algebras over the input prime field. Published comparison theorems are cited directly; the specializations to the input tensor modules and their explicit field descent are recorded here.

Derived and full groups. The following elementary support comparison concerns the entire joint enveloping algebra, including its nonsplit extensions and shared scalar coordinates.

Lemma C.1 (Derived and full enveloping algebras)**.** *Let $\mathbb{K}$ be a finite extension of $\mathbb{F}_p$; in the unitary case let $\mathbb{K}_0 \leq \mathbb{K}$ be the fixed field of the defining involution. On a joint sum of tensor products of degree at most r, the prime-field enveloping algebras of $\mathrm{SL}_n(\mathbb{K})$ and $\mathrm{GL}(n,\mathbb{K})$, and of $\mathrm{SU}_n(\mathbb{K}/\mathbb{K}_0)$ and $\mathrm{U}_n(\mathbb{K}/\mathbb{K}_0)$, coincide if $n > 2r$. In odd characteristic the enveloping algebras of $\Omega_n(\mathbb{K})$ and $\mathrm{O}_n(\mathbb{K})$ coincide if $n \geq 4r+3$. All occurrences, including trivial coordinates, must be included jointly.*

The classical torus and determinant/spinor-norm descriptions are standard [Tay92, Chapters 8–11]. The weight-projector argument below is needed because the conclusion concerns the joint prime-field enveloping algebra on whole modules, including nonsplit extensions and shared trivial coordinates.

Proof of Lemma C.1. Let $\rho : T \to \mathrm{GL}(M)$ be the restriction of the joint representation under consideration to a prime-to-p abelian group T, and let $S \leq T$. After scalar extension to an algebraic closure $\overline{\mathbb{F}}_p$, suppose restriction to S is injective on the characters of T occurring in $\overline{\mathbb{F}}_p \otimes_{\mathbb{F}_p} M$. For each such character $\chi : T \to \overline{\mathbb{F}}_p^{\times}$, its weight projector is

$$|S|^{-1} \sum_{s \in S} \chi(s)^{-1} \rho(s) \ \in \mathrm{span}_{\overline{\mathbb{F}}_p} \rho(S) \leq \mathrm{End}_{\overline{\mathbb{F}}_p}(\overline{\mathbb{F}}_p \otimes_{\mathbb{F}_p} M).$$

Thus every operator $\rho(t)$, $t \in T$, belongs to the extended enveloping algebra of $S \leq T$. For general linear and unitary groups use the diagonal and diagonal norm-one tori, respectively. A character trivial on the determinant-one torus has constant exponent vector; a difference of two length-r weights has support at most $2r$, so cannot be a nonzero such vector. The full groups are their special subgroups times these tori.

In odd characteristic diagonalize a quadratic form with entries in the two square classes. Both classes can be made to occur more than $2r$ times: every nonzero element is a sum of two squares, so replacing two equal diagonal entries by entries in the other square class is possible; the interval $[2r+1, n-2r-1]$ contains two consecutive integers. In the diagonal sign group the determinant and spinor norm are respectively the total parity and the parity on the nonsquare positions. In the character group of this diagonal sign group, the *annihilator* of its intersection with $\Omega_n(\mathbb{K})$ has three nonzero supports: all positions and the two square-class sets. Each has size exceeding $2r$. The same interpolation applies. These assertions concern the operators on the entire modules, including nonsplit extensions. Linear descent proves prime-field equality. Restricting these operators to an invariant N and passing to N/P preserves that equality for invariant $P \leq N$. In characteristic 2, the Dickson quotient between Ω and the full quadratic orthogonal group is treated separately. □

C.2.1 General-linear cohomology and intertwining maps

Two representation facts are needed: a projected cocycle can be removed by a change of origin, and only specified maps can occur between the small tensor representations. We obtain their stable values from functor cohomology [FFSS99], and use twisted homological stability [Put23] for a uniform finite-dimensional range. We verify their hypotheses for the coefficient systems used here, and then compute the required degree-1 and intertwining spaces.

For a finite field $\mathbb{K}$ of characteristic p, let $\mathcal{F}(\mathbb{K})$ denote the category of *ordinary functors* from finite-dimensional $\mathbb{K}$-vector spaces and $\mathbb{K}$-linear maps to $\mathbb{K}$-vector spaces; its morphisms are natural

transformations. The adjective "ordinary" distinguishes this category from the category of strict-polynomial functors. The *finite functors* used in the comparison below have finite ordinary difference degree and finite-dimensional values, in the sense of [FFSS99, Appendix]. For an integer $a \geq 0$, write $I_a \in \mathcal{F}(\mathbb{K})$ for base change along the field automorphism $t \mapsto t^{p^a}$ of $\mathbb{K}$. The functor I_a is additive and has ordinary degree 1, regardless of a. We use ordinary difference degree in this comparison.

Here a functor is a rule assigning a vector space to every $\mathbb{K}$-space and a compatible linear map to every $\mathbb{K}$-linear map. A natural transformation is a family of maps commuting with all those assignments. *Additive* means that the rule preserves direct sums. Ordinary difference degree measures how many new summands can interact when a space is enlarged: the natural functor has degree 1 and its tensor square has degree 2. Frobenius twists remain additive. These terms specify the coefficient systems to which we apply the comparison theorems; the algorithm itself computes the resulting Hom spaces by linear equations on the finite input matrices.

Lemma C.2 (Restriction to a determinant subgroup)**.** *Let $\mathbb{K}$ be a finite field of characteristic p, let $\mathbb{F}$ be a field containing $\mathbb{K}$, and let W_n be a tensor product over $\mathbb{F}$ of $e \geq 1$ natural modules $\mathbb{F} \otimes_{\mathbb{K}} \mathbb{K}^n$, their duals, and specified Frobenius twists. For $i \geq 0$ and $n \geq 2i + 2e + 2$, the determinant quotient $\mathrm{GL}(n, \mathbb{K})/\mathrm{SL}_n(\mathbb{K})$ acts trivially on $H^i(\mathrm{SL}_n(\mathbb{K}), W_n)$. Consequently, for every group G' with $\mathrm{SL}_n(\mathbb{K}) \leq G' \leq \mathrm{GL}(n, \mathbb{K})$, restriction gives isomorphisms*

$$H^i(\mathrm{GL}(n, \mathbb{K}), W_n) \xrightarrow{\cong} H^i(G', W_n) \xrightarrow{\cong} H^i(\mathrm{SL}_n(\mathbb{K}), W_n).$$

The same assertion holds for the symmetric square of the natural module, its dual, and their field twists, with $e = 2$, and for finite direct sums of the listed modules.

Proof. Put $M_n = W_n^*$. Split injections act on natural factors by inclusion and on dual factors by extension by zero on the specified complement. Thus M_n is a polynomial VIC-module of degree at most e, starting at zero; twists are additive, and the symmetric-square decomposition $\mathrm{Sym}^2(\mathbb{F}^n \oplus \mathbb{F}) = \mathrm{Sym}^2(\mathbb{F}^n) \oplus \mathbb{F}^n \oplus \mathbb{F}$ gives the same assertion with $e = 2$ for symmetric squares and their duals. By [Put23, Theorem 12.1 and Example 9.2], the stabilization $j_* : H_i(\mathrm{SL}_n(\mathbb{K}), M_n) \to H_i(\mathrm{SL}_{n+1}(\mathbb{K}), M_{n+1})$ is injective for $n \geq 2i + 2e + 2$; fields have stable-rank parameter 2.

For $g \in \mathrm{GL}(n, \mathbb{K})$, put $h = \mathrm{diag}(g, \det(g)^{-1}) \in \mathrm{SL}_{n+1}(\mathbb{K})$. The coefficient inclusion satisfies $hj = jg$, so naturality and the triviality of simultaneous inner conjugation on homology give $j_*(gz) = hj_*(z) = j_*(z)$ [Bro82, Proposition III.8.1]. Injectivity of j_* and finite-dimensional duality prove that the quotient acts trivially on $H^i(\mathrm{SL}_n(\mathbb{K}), W_n)$. Since $G'/\mathrm{SL}_n(\mathbb{K})$ has order prime to p, the extension spectral sequence identifies restriction with invariants under this trivial action. This proves both restriction isomorphisms, also for finite sums. □

Lemma C.3 (Finite general-linear computations)**.** *Let $\mathbb{K}$ be any finite field and $V = \mathbb{K}^n$. Tensor products, symmetric squares, and duals in this lemma are over $\mathbb{K}$. The length of a tensor product is its number of factors.*

(i) If $n \geq 8$, then $H^1(\mathrm{GL}(n, \mathbb{K}), M) = 0$ for every tensor product M of length at most two and for $\mathrm{Sym}^2 V$, its dual, and their common field twists.

(ii) If $n \geq 10$, Hom spaces between tensor products without dual factors whose total length is at most four are the finite evaluations of the ordinary-functor Hom spaces. In particular tensor products of different lengths have no nonzero maps; between two products of two additive functors the maps are the permutation matchings of equal field twists.

(iii) These assertions, and their Hom analogues with dual factors, hold for every intermediate group $\mathrm{SL}_n(\mathbb{K}) \leq G \leq \mathrm{GL}(n, \mathbb{K})$ in the respective stable ranges $n \geq 8$ and $n \geq 10$ of (i) and (ii). The same vanishing holds for the trivial coefficient in degree 1.

The comparison with ordinary functor cohomology is [FFSS99, Appendix, Theorem A.1], and the finite-dimensional range follows from [Put23, Theorem 12.1]. The following specialization verifies the coefficient systems and computes the required low-degree Ext and Hom spaces, including the smallest fields.

Proof. Use the split-injection coefficient systems verified in Lemma C.2: a tensor product with e factors has polynomial degree at most e, including dual and twisted factors; symmetric and *divided squares* have degree 2. Putman's general-linear stability theorem [Put23, Theorem B] gives the same range $n \geq 2i+2e+2$. Apply it to the dual system and use finite *chain/cochain duality*. For covariant functors $P, Q \in \mathcal{F}(\mathbb{K})$, let $f : V \hookrightarrow V'$ be a split injection of finite-dimensional $\mathbb{K}$-spaces, with its specified retraction $r : V' \to V$. The coefficient map on $Q(V) \otimes_{\mathbb{K}} P(V)^*$ is $Q(f) \otimes_{\mathbb{K}} P(r)^*$, with target $Q(V') \otimes_{\mathbb{K}} P(V')^*$. Its dual sends $h \in \mathrm{Hom}_{\mathbb{K}}(Q(V'), P(V'))$ to $P(r)hQ(f) \in \mathrm{Hom}_{\mathbb{K}}(Q(V), P(V))$. This is the restriction/evaluation transition for $\mathrm{Ext}_{\mathrm{GL}(n,\mathbb{K})}(Q(\mathbb{K}^n), P(\mathbb{K}^n))$. Thus the resulting stable values agree with the comparison of ordinary finite functors in [FFSS99, Appendix, Theorem A.1], over every finite field $\mathbb{K}$. The stated bounds are $2+4+2 = 8$ for first cohomology and $0+8+2 = 10$ for spaces of intertwining maps involving four tensor factors.

The constant functor $\mathbb{K}$ is both projective and injective: it is the linearized functor represented by the zero object, and also its dual corepresentable functor. Thus it has zero positive Ext to or from any *reduced functor*, meaning a functor taking the zero space to zero. For a reduced functor R, its *second cross-effect* is

$$\mathrm{cr}_2\, R(U, W) = \ker\big(R(U \oplus W) \xrightarrow{(R(\pi_U), R(\pi_W))} R(U) \oplus R(W)\big),$$

where π_U, π_W are the coordinate projections. It is the image of a natural split idempotent, so this construction is exact in R. For an extension of two additive functors, the middle functor is reduced and has zero second cross-effect, hence is additive. Additive functors are modules over $\mathbb{K} \otimes_{\mathbb{F}_p} \mathbb{K}$, a product of fields, so the extension splits. This proves all first-cohomology claims for $\mathrm{GL}(n, \mathbb{K})$.

To spell out the coefficients in this last step, a natural tensor times a dual natural tensor is read as a Hom space: its first cohomology is an Ext group between the corresponding functors. Two covariant factors, or a symmetric square, give Ext from the constant functor to a reduced functor; two dual factors give the reverse Ext. One natural and one dual factor give Ext between two additive functors. These are precisely the cases just shown to vanish.

We compute the Hom spaces using the sum/diagonal adjunction. For products of r and s additive functors, use the sum/diagonal adjunction between the ordinary functor category $\mathcal{F}(\mathbb{K})$ and the corresponding t-variable ordinary functor category, with $t = \max(r, s)$, and expand the smaller product. If $r \neq s$, every term is constant in a variable where the other side is reduced, so its Hom is zero. For $r = s = 2$, the two surviving terms are the two permutations. Finally $\mathrm{Hom}_{\mathcal{F}(\mathbb{K})}(I_a, I_b)$ is $\mathbb{K}$ for equal Frobenius exponents modulo $[\mathbb{K} : \mathbb{F}_p]$ and zero otherwise, as follows by evaluating at $\mathbb{K}$ and commuting with its scalar multiplications. Dual variances are moved across Hom by tensor duality.

Lemma C.2 transfers the preceding calculations to every group G with $\mathrm{SL}_n(\mathbb{K}) \leq G \leq \mathrm{GL}(n, \mathbb{K})$ in the stated stable ranks. Apply it to the coefficient module in (i), with $i = 1$ and $e \leq 2$, and to the tensor realization of a space of linear maps in (ii), with $i = 0$ and $e \leq 4$. This proves the assertion for natural, dual, and field-twisted factors, including the symmetric square and its dual. For a trivial coefficient, $\mathrm{SL}_n(\mathbb{K})$ is perfect in the stated rank and the determinant quotient has order prime to the characteristic, so its first cohomology is zero. □

Lemma C.4 (Maps from symmetric and alternating matrices)**.** *Let $\mathbb{K} = \mathbb{F}_{p^f}$, let $\mathbb{F}$ be an algebraic closure of $\mathbb{K}$, and let $E = \mathbb{F} \otimes_{\mathbb{K}} \mathbb{K}^m$ with $m \geq 10$. For $0 \leq s, a, b < f$, put $P_s = \Gamma^2(E^{[s]})$ or*

$P_s = \bigwedge^2(E^{[s]})$, where $\Gamma^2 E = (E \otimes_{\mathbb{F}} E)^{\langle \mathrm{swap} \rangle}$ is the divided square; the swap sends $u \otimes_{\mathbb{F}} v$ to $v \otimes_{\mathbb{F}} u$ for $u, v \in E$. All tensor products and duals in this lemma are over $\mathbb{F}$. Then

$$\mathrm{Hom}_{\mathrm{GL}(m,\mathbb{K})}(P_s, E^{[a]} \otimes_{\mathbb{F}} E^{[b]}) = \begin{cases} \mathbb{F}\iota, & a = b = s, \\ 0, & \text{otherwise}, \end{cases}$$

where $\iota : P_s \to E^{[s]} \otimes_{\mathbb{F}} E^{[s]}$ is the symmetric or alternating matrix inclusion. Replacing one or both target factors by their duals gives zero. In characteristic 2 one also has

$$\mathrm{Hom}_{\mathrm{GL}(m,\mathbb{K})}(\Gamma^2(E^{[s]}), \mathrm{Sym}^2(E^{[a]})) = \begin{cases} \mathbb{F}\delta, & a = s, \\ 0, & a \neq s, \end{cases}$$

where $e_1, \ldots, e_m$ is the coordinate basis of E (and of its indicated twists), and $\delta : \Gamma^2(E^{[s]}) \to \mathrm{Sym}^2(E^{[s]})$ satisfies $\delta(e_i \otimes e_i) = e_i^2$ and $\delta(e_i \otimes e_j + e_j \otimes e_i) = 0$ for $i \neq j$; the corresponding Hom into the dual symmetric square is zero. Finally $\mathrm{Hom}_{\mathrm{GL}(m,\mathbb{K})}(\bigwedge^2(E^{[s]}), E^{[a]}) = 0$. In odd characteristic, $\mathrm{Hom}_{\mathrm{GL}(m,\mathbb{K})}(\Gamma^2(E^{[s]}), E^{[a]}) = 0$ as well. The same formulas hold with $\mathrm{GL}(m,\mathbb{K})$ replaced by any intermediate group G' satisfying $\mathrm{SL}_m(\mathbb{K}) \leq G' \leq \mathrm{GL}(m,\mathbb{K})$. These statements also determine all additive prime-field maps from matrix parameter spaces, by taking the direct sum over s.

These formulas are obtained by the ordinary-functor comparison in Lemma C.3, whose inputs are [FFSS99, Appendix, Theorem A.1] and [Put23, Theorem 12.1]. The square-functor calculation below specifies the allowable maps, their Frobenius twists, and the characteristic-two diagonal quotient.

Proof. Symmetric or alternating matrices form a submodule of the tensor square. To compute maps from that submodule, we first calculate its ordinary-functor Hom spaces and then justify evaluation on $\mathbb{K}^m$.

Write I_s for the additive Frobenius functor defined above and P for either $\Gamma^2 I_s$ or $\bigwedge^2 I_s$. For finite-dimensional $\mathbb{K}$-spaces V, W, the coordinate inclusions and projections give the decomposition

$$P(V \oplus W) = P(V) \oplus (I_s(V) \otimes_{\mathbb{K}} I_s(W)) \oplus P(W).$$

Its inclusions and projections commute with linear maps on V and W. The sum/diagonal adjunction used in Lemma C.3 identifies maps from P to $I_a \otimes_{\mathbb{K}} I_b$ with maps from this decomposition to $I_a(V) \otimes_{\mathbb{K}} I_b(W)$. The first and last summands have zero maps because the target is reduced in both variables. The middle summand contributes $\mathbb{K}$ precisely when $a = b = s$, and its associated map is ι. These are calculations in $\mathcal{F}(\mathbb{K})$; extension of scalars to $\mathbb{F}$ is performed after finite evaluation below. For a mixed dual target, tensor–Hom adjunction instead asks for maps $P \otimes_{\mathbb{K}} I_b \to I_a$. The same sum/diagonal adjunction makes these maps zero: the source is reduced in both variables, whereas $I_a(V \oplus W) = I_a(V) \oplus I_a(W)$. Two dual factors ask for a map from the reduced functor $P \otimes_{\mathbb{K}} I_a \otimes_{\mathbb{K}} I_b$ to the constant functor, which is zero. A map $\bigwedge^2 I_s \to I_a$ is zero as well: each coordinate projection sends $e_i \wedge e_j$ to zero, and coordinate projections detect every vector in the additive target.

For the characteristic-2 symmetric-square target, naturality with the maps $\mathbb{K} \to V$ forces the value on a diagonal tensor $v \otimes v$ to be a scalar multiple of the corresponding square. The scalar maps on $\mathbb{K}$ force $a = s$ modulo f; for $\mathbb{K} = \mathbb{F}_2$ there is only this one twist. When $a = s$, diagonal tensors span $\Gamma^2 V$, and evaluating on $v + w$ forces every mixed symmetric tensor to map to zero. This gives exactly δ. For its dual target, tensor–Hom adjunction asks for a map from $\Gamma^2 I_s \otimes_{\mathbb{K}} \Gamma^2 I_a$ to the constant functor, again zero.

For the additional natural target in odd characteristic, the central element $-I \in \mathrm{GL}(m,\mathbb{K})$ acts as $+1$ on $\Gamma^2(E^{[s]})$ and as -1 on $E^{[a]}$, which proves the claimed vanishing. In characteristic 2 the Frobenius quotient of $\Gamma^2(E^{[s]})$ instead gives a nonzero map to $E^{[s+1]}$.

The comparison argument in the proof of Lemma C.3 applies to these ordinary finite functors, which have finite difference degree and finite-dimensional values. For maps $Q \to P$, use the coefficient system $Q(V) \otimes_{\mathbb{K}} P(V)^*$. Each square has the displayed direct-sum decomposition, its dual has the dual decomposition, and each natural factor has degree 1. Thus these split-injection coefficient systems have degree at most four, starting at zero. In homological degree zero, finite-dimensional duality and the restriction/evaluation compatibility from that proof identify the computed functor Hom spaces with the $\mathrm{GL}(m, \mathbb{K})$-Hom spaces for $m \geq 2 \cdot 4 + 2 = 10$, without a field-size restriction. Extending scalars from $\mathbb{K}$ to $\mathbb{F}$ commutes with these Hom spaces. Finally, restriction of scalars from $\mathbb{K}$ to $\mathbb{F}_p$, followed by extension to $\mathbb{F}$, decomposes into the f Frobenius conjugates indexed by s; this proves the assertion for additive maps.

For an intermediate group G', use the proof of Lemma C.2 for the coefficient of each Hom space. Its argument applies here because the displayed square decompositions and their duals define split-injection coefficient systems of degree at most four, and the determinant-compensating last coordinate acts trivially on their included subspaces. The same bound $m \geq 2 \cdot 4 + 2 = 10$ therefore proves the asserted restriction isomorphisms. □

The next fact is used when a fixed scalar subspace is divided out. For an exact sequence $0 \to T \to B \to B/T \to 0$ of G-modules, the cohomology sequence contains

$$H^1(G, B) \longrightarrow H^1(G, B/T) \longrightarrow H^2(G, T).$$

Thus $H^1(G, B) = 0$ and $H^2(G, T) = 0$ imply that translations can also be removed in B/T. Here T is a trivial module, so every element of G fixes every vector of T. This is the specific second-cohomology consequence we need; the general exact sequence is standard [Ben91, Section 2.5].

Lemma C.5 (Trivial coefficients in degrees one and two)**.** *For the stable natural groups* SL*,* SU*,* Sp*, and* Ω *in defining characteristic* p*, and for their products,* $H^1(G, T) = H^2(G, T) = 0$ *for every finite-dimensional trivial characteristic-*p *module* T*. This also holds for a prime-to-*p *extension of such a product.*

Proof. The stable simply connected covers are perfect with trivial Schur multiplier [Ste81, Theorem 1.1 and the twisted-group statement on p. 528]; all exceptional ranks are excluded here. Universal coefficients give $H^1 = H^2 = 0$ for trivial coefficients. The kernels on the natural modules have order prime to p, so inflation transfers this vanishing to the natural images. The Künneth formula gives the assertion for products, and averaging in the extension spectral sequence gives it for prime-to-p extensions. □

C.2.2 Descent of the whole enveloping algebra

The input subspaces are over $\mathbb{F}_p$, even when the classical action is defined over $\mathbb{K}$. The next lemma shows that the prime-field enveloping algebra contains the entire defining-field augmentation ideal, including its radical.

Lemma C.6 (Augmentation and field descent)**.** *Let* $\mathbb{K} = \mathbb{F}_{p^e}$*, let* G *be perfect, and let* $\rho : G \to \mathrm{GL}(W)$ *define the* $\mathbb{K}[G]$*-module* W*, with a nonzero fixed vector. Here* $\mathbb{K}[G]$ *denotes the abstract group algebra. Suppose distinct matrix-Frobenius twists* $W^{[i]}$ *and* $W^{[j]}$*,* $0 \leq i < j < e$*, have no common nontrivial absolutely simple composition factor. Write* $A_{\mathbb{K}} = \mathrm{span}_{\mathbb{K}} \rho(G) \leq \mathrm{End}_{\mathbb{K}}(W)$ *and let* $\alpha : A_{\mathbb{K}} \to \mathbb{K}$ *be the character determined by* $av = \alpha(a)v$ *for* $a \in A_{\mathbb{K}}$ *and a chosen nonzero fixed vector* $v \in W$*. Then*

$$\mathrm{span}_{\mathbb{F}_p} \rho(G) = \mathbb{F}_p I \oplus \ker \alpha, \qquad \ker \alpha = \mathrm{span}_{\mathbb{F}_p} \{\rho(g) - I : g \in G\}.$$

In particular the prime-field enveloping algebra contains the entire $\mathbb{K}$-linear image of the augmentation ideal, including the radical of $A_{\mathbb{K}}$.

The argument uses augmentation ideals and separation by annihilator ideals in the group algebra. The descent calculation below is needed because the required prime-field equality concerns the full nonsimple enveloping algebra, including its radical, rather than only its simple constituents.

Proof. The case $e = 1$ is immediate, so assume $e > 1$. Put $R = \mathbb{K}[G]$, the abstract group algebra of formal sums, and define its *augmentation map* by

$$\epsilon : R \to \mathbb{K}, \qquad \epsilon\Big(\sum_{g\in G} a_g g\Big) = \sum_{g\in G} a_g \qquad (a_g \in \mathbb{K}).$$

For $0 \le i < e$, extend the twisted representation $\mathbb{K}$-linearly to

$$\rho_i : R \to \mathrm{End}_{\mathbb{K}}(W^{[i]}), \qquad \rho_i\Big(\sum_{g\in G} a_g g\Big) = \sum_{g\in G} a_g \rho(g)^{(p^i)}.$$

The matrices in this formula use the same chosen basis and its twisted copies. In particular ρ_0 extends ρ. Let $\Delta = \ker \epsilon \lhd R$ be the *augmentation ideal*, and let $J_i = \ker \rho_i = \mathrm{ann}_R(W^{[i]}) \lhd R$ be the annihilator ideal for $0 \le i < e$. The map $G \to \Delta/\Delta^2$, $g \mapsto g - 1 + \Delta^2$, is a homomorphism to an additive abelian group. Perfectness gives $\Delta = \Delta^2$, hence $\Delta^t = \Delta$ for all $t \ge 1$. Fixed vectors give $J_i \subseteq \Delta$. For a $\mathbb{K}$-basis $w_1, \dots, w_s$ of $W^{[i]}$, the map $R/J_i \to (W^{[i]})^s$ sending $a + J_i \mapsto (aw_1, \dots, aw_s)$ for $a \in R$ is injective. Thus the composition factors of R/J_i occur in $W^{[i]}$. The quotient $R/(J_i + J_j)$ has only trivial composition factors after scalar extension. Some power of Δ annihilates it, and therefore Δ itself annihilates it. Hence $J_i + J_j = \Delta$. Let $J = \bigcap_{i=1}^{e-1} J_i$. Expanding the ordered product $\prod_{i=1}^{e-1}(J_0 + J_i) = \Delta^{e-1} = \Delta$ shows $\Delta = J_0 + J$: every term containing J_0 lies in J_0 and the remaining product lies in every J_i, because each J_i is a two-sided ideal. Given $a \in \ker \alpha$, choose $w \in J$ with $\rho_0(w) = a$. Define coefficient Frobenius $\sigma : R \to R$ by $\sigma(\sum_g a_g g) = \sum_g a_g^p g$ for $a_g \in \mathbb{K}$, and put $u = \sum_{s=0}^{e-1} \sigma^s(w) \in R$. Then u belongs to $\mathbb{F}_p[G]$, has augmentation zero, and

$$\rho_0(\sigma^s(w)) = \rho_{e-s}(w)^{(p^s)} = 0 \quad (w \in J;\ 0 < s < e).$$

Thus $\rho_0(u) = a$. This proves the augmentation assertion, and adjoining $I \in \mathrm{End}_{\mathbb{K}}(W)$ proves the first. The abstract algebra $R = \mathbb{K}[G]$ is used only in the proof; the algorithm computes the concrete enveloping algebras by matrix-span closure. □

C.3 Odd characteristic

Throughout this subsection p is odd. We compute the joint matrix algebra, realize its injectives as matrix quotients, and prove Theorem C.9. For a Hermitian matrix, $X = \bar{X}^{\mathrm{t}}$; a form-self-adjoint endomorphism satisfies $b(Xu, v) = b(u, Xv)$. The identity matrix spans the scalar submodule.

C.3.1 Matrix enveloping algebras and injective quotients

The scalar and simple constituents of the classical square and adjoint modules are described in [GT05, Section 3.2]. The following calculation also determines the nonsplit extensions and the prime-field enveloping algebra, including its shared scalar coordinate.

Proposition C.7 (The enveloping algebra on the nonsimple matrix modules)**.** *Assume the stable degree-two rank bounds of Subsection 6.1, and let p be odd. The only nonsemisimple same-factor components of a joint sum of tensor products of degree at most two are the following modules M, when p divides $n = \dim_{\mathbb{K}} V$, the dimension of the natural module over the displayed matrix group's field $\mathbb{K}$:*

Group	M	*Defining field of* M
$\mathrm{SL}_n(\mathbb{K})$	$\mathrm{End}_{\mathbb{K}}(V)$	$\mathbb{K}$
$\mathrm{SU}_n(\mathbb{K}/\mathbb{K}_0)$	*Hermitian matrices*	$\mathbb{K}_0$
$\mathrm{O}_n(\mathbb{K})$	*form-self-adjoint matrices*	$\mathbb{K}$
$\mathrm{Sp}_n(\mathbb{K})$	*form-self-adjoint matrices*	$\mathbb{K}$

Index these components by i, with defining field $\mathbb{K}_i$ as in the last column. They have the nonsplit chain $\mathbb{K}_i I < M_{i,0} < M_i$, where $M_{i,0} = \ker(\mathrm{tr} : M_i \to \mathbb{K}_i)$ and $H_i = M_{i,0}/\mathbb{K}_i I$ is simple. Its dimension is $n^2 - 2$, $n^2 - 2$, $n(n+1)/2 - 2$, or $n(n-1)/2 - 2$, respectively. If $p \nmid n$ the scalar line splits and the component is semisimple. After identifying all isomorphic occurrences, the prime-field joint algebra on the nonsemisimple components is

$$\mathbb{F}_p I + \bigoplus_i R_i, \qquad R_i = \left\{ \begin{pmatrix} 0 & b & c \\ 0 & D & e \\ 0 & 0 & 0 \end{pmatrix} : b, D, e, c \text{ arbitrary over } \mathbb{K}_i \right\}. \tag{C.1}$$

Here $h_i = \dim_{\mathbb{K}_i} H_i$, and the three matrix blocks correspond to the scalar submodule, the quotient H_i, and the scalar quotient. Thus $D \in \mathrm{Mat}(h_i, \mathbb{K}_i)$, $b \in \mathrm{Mat}(1 \times h_i, \mathbb{K}_i)$, $e \in \mathrm{Mat}(h_i \times 1, \mathbb{K}_i)$, and $c \in \mathbb{K}_i$. The remaining nontrivial simple module types give separate semisimple algebra factors. Repeated occurrences and distinct classical factors share the displayed scalar coordinate.

Proof. The degree-two constituents can be checked from the defining representations: natural squares have highest weights $2\omega_1$ and ω_2, adjoints have $\omega_1 + \omega_{n-1}$, and orthogonal and symplectic form-self-adjoint spaces have their trivial contraction and the corresponding simple quotient of the trace-zero module. The algebraic-group constituents are described in [GT05, Section 3.2]. For the finite groups, use the tensor-product and restricted-weight restriction statements of [Ste67, Theorems 41 and 43]: products with distinct Frobenius-twist exponents are simple, and the equal-twist weights are restricted even at $p = 3$. Unitary duality identifies the $\mathbb{K}$-dual $V^* = \mathrm{Hom}_{\mathbb{K}}(V, \mathbb{K})$ with $V^{[f]}$ when $|\mathbb{K}_0| = p^f$. The general finite-restriction argument in [GT05, proofs of Theorems 2.9 and 2.11] applies for $|\mathbb{K}| \geq 4$. We verify the nonsplitting, coefficient-field, and joint enveloping-algebra assertions over all the fields in the statement.

The pairing $M_i \times M_i \to \mathbb{K}_i$ given by $(a, b) \mapsto \mathrm{tr}(ab)$ is nondegenerate and invariant, and its fixed space is $\mathbb{K}_i I$. If the nontrivial simple quotient split off as a submodule, its dimension greater than half of $\dim_{\mathbb{K}_i} M_i$ would make the restricted pairing nondegenerate: its radical is an invariant submodule of a simple module. Its two-dimensional orthogonal complement would have only trivial composition factors. On a perfect derived group its upper-unitriangular image is trivial; on the full group the remaining quotient has order prime to p, so the image is again trivial. This contradicts the one-dimensional fixed space. Thus both adjacent extensions are nonsplit, the second by duality. When $p \nmid n$, trace splits the scalar line instead. For the coefficient fields, use the diagonal matrix with entries $s, r, 1, \ldots, 1$, where $r = \pm 1$. The difference of adjoint traces is $2(s + s^{-1}) + 4(n-2)$. For a unitary adjoint take norm-one s; the same expression is in $\mathbb{K}_0$. In the orthogonal and symplectic cases use two hyperbolic pairs with eigenvalues s, s^{-1}, r, r^{-1}; the difference on the relevant square component is $4(s + s^{-1} + n - 4)$. Removing trivial factors does not change these differences. The set of $s + s^{-1}$ has at least $(|\mathbb{K}| - 1)/2$ values, or at least $(|\mathbb{K}_0| + 1)/2$ in the norm-one case. These sets

exceed every proper subfield when a proper subfield is possible. Thus ordinary traces generate the required defining field. The enveloping algebra of an absolutely irreducible finite-field representation is a full matrix algebra over its coefficient subfield; these trace calculations force that subfield to be exactly $\mathbb{K}_i$. Choose a separable semisimple complement in the enveloping algebra. The nonsplit bottom extension forces a nonzero radical block $H_i \to \mathbb{K}_i I$. The full matrix algebra on H_i makes this block all of $\mathrm{Hom}_{\mathbb{K}_i}(H_i, \mathbb{K}_i) \cong \mathbb{K}_i^{1\times h_i}$. The top extension similarly forces the entire block $\mathbb{K}_i \to H_i$, and products give every scalar top-to-bottom entry. The two endpoint diagonal entries are the same element of $\mathbb{F}_p$, since group elements fix both pointwise. This proves the single-chain algebra. For separate chains their only common absolutely simple type is trivial, after the stated identifications. The annihilator-product argument of Lemma C.6 therefore separates their augmentation ideals, even for two chains of the same acting group. Applying it jointly to Frobenius conjugates and taking the Frobenius sum gives the whole defining-field ideals over $\mathbb{F}_p$. They have zero cross-products on the joint sum, with just one common identity operator. Composition-disjoint nontrivial simple types separate by the two-sided Chinese remainder theorem: the sum of their annihilator ideals is the whole algebra, since any nonzero common quotient would have a simple factor of both types. Field-conjugate, dual, and repeated occurrences are identified before this step; they repeat the same algebra action. This proves joint equality, including the radical. □

Lemma C.8 (Injective modules for the matrix representations)**.** *For the algebra of Proposition C.7, the injective modules associated with the nontrivial simple socles are $M_i/\mathbb{K}_i I$; its single trivial-type injective is*

$$\left(\bigoplus_i M_i\right)/P_*, \qquad P_* = \{(c_i I)_i : \sum_i \mathrm{Tr}_{\mathbb{K}_i/\mathbb{F}_p}(c_i) = 0\}.$$

The formula for the trivial-type injective assumes that at least one nonsimple chain is present. If none is present but the joint module under consideration has a trivial constituent, take $\mathbb{F}_p$ as its trivial-type injective; if there is no trivial constituent, omit that injective. Each embeds in a full matrix module modulo a fixed scalar subspace. Consequently every quotient N/P of invariant subspaces of the stated tensor sums of degree 2 embeds in such a module, with polynomial auxiliary dimension.

Finite-dimensional projective–injective duality identifies the injective types as duals of primitive right ideals [Ben91, Section 1.5]. The field-trace pairing below computes their concrete quotient models, with the common prime-field scalar relation included in the denominator.

Proof of Lemma C.8. Let $\mathcal{A} = \mathbb{F}_p I + \bigoplus_i R_i$ be the chain algebra of Proposition C.7. If the chain index set is empty, the algebra is semisimple, and its trivial factor, when present, has injective module $\mathbb{F}_p$. Otherwise let $e_0 \in \mathcal{A}$ be the primitive idempotent that is the identity on both scalar positions of every chain and zero on every H_i. The right ideal $e_0\mathcal{A}$ consists of its common $\mathbb{F}_p$-scalar, all blocks $H_i \to \mathbb{K}_i I$, and all top-to-bottom coordinates. For $a \in e_0\mathcal{A}$ and $(v_i)_i \in \bigoplus_i M_i$, pair them by summing $\mathrm{Tr}_{\mathbb{K}_i/\mathbb{F}_p}$ of the first coordinate of av_i. The full row and scalar entries detect all nontrivial simple and top coordinates; the common scalar detects precisely the one remaining scalar quotient. Its annihilator is P_*. This identifies its dual with the first displayed quotient. For a primitive idempotent $e_i \in \mathcal{A}$ in the matrix component acting on H_i, the same pairing identifies the dual of $e_i\mathcal{A}$ with $M_i/\mathbb{K}_i I$, by nondegeneracy of the field trace. Each M_i is a submodule of a full matrix module on a natural representation with its scalar line fixed. Embed the remaining semisimple factors in their original tensor summands and apply Lemma 6.5. These embeddings are constructed on N/P using both input subspaces $P \le N$; they do not require an invariant complement to P. □

C.3.2 Vanishing for full matrix spaces and their quotients

The target models require quotients by pointwise fixed subspaces, including correlated scalar subspaces. We first prove vanishing on the full matrix spaces, using the functor-cohomology comparison [FFSS99, Appendix, Theorem A.1] in the finite-rank range of [Put23, Theorem 12.1] for the general-linear groups and the calculations below for the form groups. Trivial-coefficient H^2 then gives vanishing on the required quotients.

Theorem C.9 (First cohomology of matrix modules in odd characteristic)**.** *Let G be a product of the stable natural derived classical groups in odd characteristic p. Let B be a direct sum of full spaces of linear maps between natural, dual, Frobenius-twisted natural, and trivial modules, viewed over $\mathbb{F}_p$ by restriction of scalars. For every $\mathbb{F}_p$-subspace $C \leq B^G$,*

$$H^1(G, B) = H^1(G, B/C) = 0.$$

Proof of Theorem C.9. General-linear tensor products follow from Lemma C.3. For the symplectic group, use the parabolic with Levi $\mathrm{GL}(m, \mathbb{K})$ and additive radical consisting of symmetric matrices in $\mathrm{Mat}(m, \mathbb{K})$. For the split even orthogonal derived group, the Levi is $\{g \in \mathrm{GL}(m, \mathbb{K}) : \det(g) \in (\mathbb{K}^\times)^2\}$ and the radical consists of alternating matrices in $\mathrm{Mat}(m, \mathbb{K})$. The determinant condition follows from the spinor norm of the block-diagonal isometry [Tay92, Chapter 11]. Both Levi groups contain $\mathrm{SL}_m(\mathbb{K})$, so Lemmas C.3 and C.4 apply. In either case the parabolic index is prime to p. Normalize a cocycle on the Levi. Its first radical grade is an additive equivariant map from the parameter module; Lemma C.4 leaves only the top map $Z \mapsto aZ^{[i]}$ when the two twists match, and none otherwise. This value is the coboundary of the first-half projector. No first grade remains, so restriction to the parabolic is zero. For even unitary groups the parameter is antihermitian and its prime-field components are $E^{[s]} \otimes_{\mathbb{K}} E^{[s+f]}$; the identical matching calculation gives the same conclusion. For passage from the full to the special unitary group, let T be the diagonal norm-one torus and $S = T \cap \mathrm{SU}_n(\mathbb{K}/\mathbb{K}_0)$. Coordinate SU_3 subgroups generate the special group by Witt extension [Tay92, Theorem 7.4]: for a norm-one vector, three nonzero coordinate norms cannot have all pair sums zero in odd characteristic, so successive two-coordinate elimination applies. A cocycle normalized on S is determined by its values on these subgroups. Its T-weights, and hence those on first cohomology, have support at most $2 + 3$ for the present degree-two coefficients. Since S acts trivially on cohomology, these are constant-coordinate characters; $n \geq 64$ excludes every nontrivial such character of support at most five. Inflation–restriction across the prime-to-p quotient $\mathrm{U}_n/\mathrm{SU}_n$ gives the same vanishing for the even-dimensional special unitary group.

It remains to treat odd-dimensional unitary groups, odd-dimensional orthogonal groups, and nonsplit even-dimensional orthogonal groups. All symplectic dimensions were treated by the Siegel-parabolic argument above. In the remaining cases choose the largest even integer m not exceeding the Witt index and write $V = E \oplus E^\vee \oplus Z$, where $\dim_{\mathbb{K}} E = m$ and $\dim_{\mathbb{K}} Z \leq 4$. Here $E^\vee$ is $E^* = \mathrm{Hom}_{\mathbb{K}}(E, \mathbb{K})$, or $E^{\bar{*}}$ in the unitary case, paired with E by the preserved form. In the following block equations $A^\dagger = A^{\mathrm{t}}$ in the orthogonal case and $A^\dagger = \bar{A}^{\mathrm{t}}$ in the unitary case. Choose the hyperbolic pairing blocks to be identity matrices, and let $H_Z = H_Z^\dagger$ be the matrix of the restricted form on Z. Thus the preserved form has block matrix $\left(\begin{smallmatrix} 0 & I & 0 \\ I & 0 & 0 \\ 0 & 0 & H_Z \end{smallmatrix}\right)$ in the order $E, E^\vee, Z$. The split even derived group on the first two spaces contains both its central $-I$ and an isometry interchanging E and $E^\vee$: even m gives determinant one and, in the orthogonal case, square spinor norm. Its full matrix blocks have the already proved vanishing, blocks joining Z with $E \oplus E^\vee$ have vanishing by the central involution, and blocks in $\mathrm{End}_{\mathbb{K}}(Z)$ have vanishing by perfectness. Normalize there. For

$X \in \mathrm{Hom}_{\mathbb{K}}(Z, E)$ and $B \in \mathrm{Hom}_{\mathbb{K}}(E^\vee, E)$, the upper radical has matrices

$$u(X,B) = \begin{pmatrix} I & B - \frac{1}{2} X H_Z^{-1} X^\dagger & X \\ 0 & I & 0 \\ 0 & -H_Z^{-1} X^\dagger & I \end{pmatrix}, \qquad B^\dagger = -B.$$

The center $u(0, B)$ belongs to the normalized even group. The cocycle commuting identities with this center force blocks $(2,1), (2,3), (3,1)$ to zero. The remaining diagonal blocks are additive in $X \in \mathrm{Hom}_{\mathbb{K}}(Z, E)$. The central element acting by $-I$ on $E \oplus E^\vee$ and by I on Z sends X to $-X$, while its induced action on every diagonal endomorphism block is the identity. Equivariance therefore forces these additive blocks to vanish, since p is odd. Natural intertwiner matching leaves, for source and target twists i, j, only $X^{[j]} D_0$ and $D_1 (X^{[i]})^\dagger$ in positions $(1,3), (3,2)$, where $D_0, D_1 \in \mathrm{Hom}_{\mathbb{K}}(Z^{[i]}, Z^{[j]})$. Here $X^{[i]} : Z^{[i]} \to E^{[i]}$ is the entrywise Frobenius twist, and similarly for j. For $X, Y \in \mathrm{Hom}_{\mathbb{K}}(Z, E)$, comparing the two multiplication orders gives

$$X^{[j]} \big(D_1 - D_0 (H_Z^{[i]})^{-1}\big) (Y^{[i]})^\dagger = Y^{[j]} \big(D_1 - D_0 (H_Z^{[i]})^{-1}\big) (X^{[i]})^\dagger.$$

Choosing distinct rows indexed by a basis of E and arbitrary coordinates in Z forces $D_1 = D_0 (H_Z^{[i]})^{-1}$. The remaining top block is the corresponding quadratic block $X^{[j]} D_0 (H_Z^{[i]})^{-1} (X^{[i]})^\dagger$; after subtraction its additive remainder is again excluded by the central sign. Direct multiplication shows that these are exactly the coboundary of the constant matrix D_0 supported on Z. It is fixed by the derived group on $E \oplus E^\vee$, so subtraction preserves normalization. The isometry interchanging those summands handles the opposite radical, and these radicals and the even group generate the full derived group. For a natural coefficient the same equations leave $X^{[i]} v_0$, the coboundary of a vector $v_0 \in Z^{[i]}$. Duals follow by the form identifications. This proves every full matrix-module case.

For independent factors $G_1, G_2 \leq G$, let X and Y be their respective nontrivial simple coefficient modules. The first-cohomology product formula gives zero on the *external tensor product* $X \boxtimes_{\mathbb{F}_p} Y$, in which each group acts on its own factor, since $X^{G_1} = 0$ and $Y^{G_2} = 0$. Trivial coefficients are handled by Lemma C.5. For the quotient assertion, the exact sequence $0 \to C \to B \to B/C \to 0$ gives

$$H^1(G, B) \longrightarrow H^1(G, B/C) \longrightarrow H^2(G, C).$$

The first term vanishes by the preceding calculation. Since $C \leq B^G$ is a trivial G-module, the last term vanishes by Lemma C.5. Thus $H^1(G, B/C) = 0$. The subspace C may impose arbitrary $\mathbb{F}_p$-linear relations between fixed coordinates in different summands. □

C.3.3 Cohomology of odd tensor degree

The cubic pairing quotient has no fixed vectors and no first cohomology in odd characteristic. The following stronger statement proves both facts directly on invariant subquotients.

Lemma C.10 (Odd-degree sign argument)**.** *Let $\mathbb{K}$ be a finite field of odd characteristic, let D be a stable natural derived classical group, and let M be a sum of tensor products with an odd number of the specified natural, dual, or twisted factors, at most three. Let $P \leq U \leq M$ be D-invariant subspaces, so D acts on U/P by $u + P \mapsto gu + P$. Then $(U/P)^D = 0$ and $H^1(D, U/P) = 0$. The same holds for each nonzero parity summand for a product of such groups.*

Proof. Put $W = U/P$, a $\mathbb{K}[D]$-module, and let $c : D \to W$ be a cocycle. For $D = \mathrm{SL}_n(\mathbb{K})$ use the determinant-one diagonal sign group $S \leq D$. Average to make c vanish on S. The elementary

matrix $u_{ij}(t)$ commutes with $S_{ij} = \{s \in S : s_i = s_j\}$, so $c(u_{ij}(t))$ is fixed by S_{ij}. An occurring sign character has odd support at most three. The annihilator of S_{ij} has representatives of supports $0, 2, n, n-2$, none of the required odd size when $n \geq 6$. Thus this fixed space is zero, also on every invariant quotient U/P, because $|S_{ij}|$ is prime to the characteristic. Elementary matrices generate SL, proving the assertion. The same sign argument already gives the absence of fixed vectors.

For a symplectic factor D, its central element $z = -I \in D$ acts as -1 on W. Comparing $c(zg)$ and $c(gz)$ for $g \in D$ gives $2c(g) = c(z) - gc(z)$ in W, an explicit coboundary. For a unitary factor D, choose an orthonormal basis of its natural $\mathbb{K}$-module, and write $a \mapsto \bar{a}$ for the defining involution of $\mathbb{K}$. The coordinate SU_2 groups generate: if $a, b, d \in \mathbb{K}$, $N = \bar{a}a + \bar{b}b \neq 0$, and $\bar{d}d = N$, then

$$\begin{pmatrix} \bar{a}/\bar{d} & \bar{b}/\bar{d} \\ -b/d & a/d \end{pmatrix}$$

has determinant one, is unitary, and sends (a, b) to $(d, 0)$. For a vector with at least three nonzero coordinates, some pair of coordinate norms has nonzero sum, since the characteristic is odd. These transformations reduce a unit vector to one coordinate; induction on its stabilizer proves generation. Each coordinate group commutes with the corresponding S_{ij}, so the special-linear sign proof applies unchanged.

For an orthogonal factor $D = \Omega(V, Q)$, where $Q : V \to \mathbb{K}$ is its defining quadratic form, diagonalize Q with coordinate norms 1 and a fixed nonsquare ξ. Arrange at least nine coordinates of each norm class, indexed by $I_0, I_1 \subseteq \{1, \ldots, n\}$, where $n = \dim_{\mathbb{K}} V$. This is possible in dimension at least twenty: every $\xi \neq 0$ is a sum of two squares, and the change of basis with rows $(x, y), (-y, x)$ changes two square coordinates to two nonsquare coordinates. The parity prescribed by the discriminant is preserved. The orthogonal group is generated by reflections supported on at most three coordinates. Indeed an anisotropic vector with coordinates from both norm classes has either a cross-class pair of nonzero norm or a three-coordinate block with nonzero norm. A vector supported on one norm class has a pair of nonzero total norm, unless its support is one; include a zero coordinate of the other class. In each case the block has an axis in the norm class of that total and its support can be reduced. Vectors $v, w \in V$ of equal nonzero norm are carried to each other by reflection in $v - w$ when that vector is anisotropic; otherwise reflection in $v + w$ followed by reflection in w suffices, since $Q(v + w) = 4Q(v)$. Induction on an orthogonal complement proves the generation claim. The last complement can be ordered so that both norm classes occur until dimension two.

The determinant and spinor-norm kernel is $\Omega(Q)$ [Tay92, Theorem 11.51]. Choose coordinate reflections r_0, r_1 of the two norm classes. The four elements $1, r_0, r_1, r_0 r_1$ are a transversal of this kernel. Their Schreier generators relative to the three-coordinate reflections are supported on at most five coordinates. Take $S \leq \Omega$ to be the signs with an even number of minus signs in each I_j. For a generator supported on B, $|B| \leq 5$, let S_B be its signs constant on $B \subseteq \{1, \ldots, n\}$. In the sign-character space $\mathbb{F}_2^n$, the annihilator is

$$\langle \mathbf{1}_{I_0}, \mathbf{1}_{I_1} \rangle + \{v \in \mathbb{F}_2^n : \mathrm{supp}(v) \subseteq B,\ |\mathrm{supp}(v)| \text{ is even}\}.$$

A nonzero class-indicator coefficient leaves at least four coordinates outside B; without one, the weight is even. Thus no odd weight of size at most three is trivial on S_B. Normalize on S, and the commuting cocycle equation kills every generator. This proves both orthogonal signs and odd-dimensional orthogonal groups.

For a product, the parity projectors preserve all sections by joint image comparison, Lemma C.1. Choose a factor with odd occurrence number. Normalize the cocycle to zero there by the proved one-factor result. All other factors commute with it, so their cocycle values lie in its zero fixed space. A central product is treated by pullback to its product cover; inflation is injective in degree 1. □

C.4 Characteristic two

Throughout this subsection $p = 2$. Symmetric and alternating squares need not split, so the full enveloping algebra and its injective quotient models are required. We first treat adjoints and nonorthogonal squares, then the invariant-tensor input and orthogonal quotients.

C.4.1 Adjoint modules

The adjoint constituents are the scalar and simple trace-zero types of [GT05, Section 3.2], with finite restrictions as in [Ste67, Theorems 41 and 43]. As in Proposition C.7, we additionally compute their coefficient fields and simultaneous prime-field enveloping algebra.

Lemma C.11 (Adjoint modules in characteristic two)**.** *Assume the stable degree-two rank bounds of Subsection 6.1, with $p = 2$. For $\mathrm{SL}_n(\mathbb{K})$ take $M = \mathrm{End}_{\mathbb{K}}(\mathbb{K}^n)$ over $\mathbb{K}$; for $\mathrm{SU}_n(\mathbb{K}/\mathbb{K}_0)$ take the Hermitian matrix module M over $\mathbb{K}_0$. If n is even, each has a nonsplit chain $\mathbb{K}_i I < M_0 < M$, where $M_0 = \ker \mathrm{tr}$, $\mathbb{K}_i = \mathbb{K}$ or $\mathbb{K}_0$, respectively, and $H = M_0/\mathbb{K}_i I$ is simple of $\mathbb{K}_i$-dimension $n^2 - 2$. If n is odd, $M = \mathbb{K}_i I \oplus M_0$ is semisimple. On a joint sum of the nonsimple components, after identifying isomorphic occurrences, the prime-field enveloping algebra is $\mathbb{F}_2 I + \bigoplus_i R_i$, with the blocks R_i of* (C.1)*. The scalar coordinate is shared; composition-disjoint nontrivial simple types give separate semisimple factors. The indecomposable injective modules are the quotient models of Lemma C.8, with $p = 2$.*

Proof. The matrix commutator description of the trace-zero subspace gives the scalar and adjoint-quotient constituents. When n is even, the invariant nondegenerate trace pairing has fixed space $\mathbb{K}_i I$; the nonsplitting argument in Proposition C.7 applies without a change of characteristic. When n is odd, trace splits the scalar line.

For the general-linear coefficient field put

$$C(a,b) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & b \\ 0 & 1 & a \end{pmatrix}, \qquad C(a,b)^{-1} = \begin{pmatrix} b & 1 & 0 \\ a & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

Both the determinant and inverse are checked by multiplication. For $\epsilon = (n-3) \bmod 2$, set $g(a) = \mathrm{diag}(C(a, 1+\epsilon), I_{n-3})$. Its determinant is one, its traces on the natural module and its dual are $a+\epsilon$ and 1, and its adjoint-quotient trace is $a+1$. These run over all of $\mathbb{K}$. For the unitary adjoint use $\mathrm{diag}(t, t^{-1}, 1, \ldots, 1)$ with norm-one t: the trace on the nontrivial simple quotient, up to a prime-field constant, is $(n + t + t^{-1})^2$. The set of $t + t^{-1}$ has $|\mathbb{K}_0|/2 + 1$ values, more than any proper subfield; the case $\mathbb{K}_0 = \mathbb{F}_2$ is immediate. Thus the coefficient fields are exactly $\mathbb{K}$ and $\mathbb{K}_0$, respectively.

The full matrix algebra on the simple quotient, the two nonsplit adjacent extensions, and multiplication of their radical blocks now give each chain algebra exactly as in Proposition C.7. Its two endpoint diagonal coordinates are the same prime-field scalar. The annihilator-product and Frobenius-descent argument of Lemma C.6 separates the defining-field augmentation ideals of distinct chain types while their scalar coordinates remain equal. Isomorphic occurrences have the same operators; composition-disjoint simple types separate by the two-sided Chinese remainder theorem. This proves the joint assertion. Dualizing the same primitive right ideals then gives the injective quotients of Lemma C.8; its field-trace pairing argument is independent of the characteristic. □

C.4.2 General-linear, symplectic, and unitary tensor squares

For the following characteristic-2 calculations, $\mathbb{K}$ is a finite field and V is a finite-dimensional $\mathbb{K}$-space. On $T = V \otimes_{\mathbb{K}} V$, define

$$\tau \in \mathrm{End}_{\mathbb{K}}(T), \quad \tau(v \otimes_{\mathbb{K}} w) = w \otimes_{\mathbb{K}} v \quad (v, w \in V), \qquad x = I_T + \tau \in \mathrm{End}_{\mathbb{K}}(T).$$

Put $A = \operatorname{im} x \leq T$ and $\Gamma = \ker x \leq T$. Thus $A \cong \bigwedge^2_{\mathbb{K}} V$, $\Gamma = \Gamma^2_{\mathbb{K}} V$, and $T/A = \mathrm{Sym}^2_{\mathbb{K}} V$, the symmetric quotient of T. The algebra $\mathbb{K}[x] \leq \mathrm{End}_{\mathbb{K}}(T)$ is generated by x. Duals are $\mathbb{K}$-linear: $U^* = \mathrm{Hom}_{\mathbb{K}}(U, \mathbb{K})$.

For a coordinate pair e_i, e_j with $i \neq j$, x sends $e_i \otimes e_j$ to $e_i \otimes e_j + e_j \otimes e_i$ and kills that sum. It also kills $e_i \otimes e_i$. Thus each unordered off-diagonal pair gives a length-two Jordan block, and each diagonal entry gives a length-one block. The quotient Γ/A records precisely those diagonal coordinates, with the Frobenius-twisted natural action. This explicit filtration replaces the symmetric/alternating direct sum available in odd characteristic.

Lemma C.12 (The general linear group on the tensor square)**.** *For $n = \dim_{\mathbb{K}} V \geq 3$, the defining-field enveloping algebra $\mathrm{span}_{\mathbb{K}}\{g \otimes_{\mathbb{K}} g : g \in \mathrm{GL}(V)\} \leq \mathrm{End}_{\mathbb{K}}(T)$, where $T = V \otimes_{\mathbb{K}} V$, is $\mathrm{End}_{\mathbb{K}[x]}(T)$. Its indecomposable injective types are T and $\mathrm{Sym}^2_{\mathbb{K}} V$, with their respective simple socles $\bigwedge^2_{\mathbb{K}} V$ and $V^{[1]}$.*

The enveloping-algebra identity is degree-two Schur–Weyl duality: [BD09, Corollary 4.4] proves it for $|\mathbb{K}| > 2$. We give the remaining binary case and the explicit injective models.

Proof. Assume first that $\mathbb{K} = \mathbb{F}_2$. A homogeneous quadratic polynomial in the n^2 matrix entries which vanishes on every invertible matrix can be made multilinear by replacing squares by their variables. For two distinct positions $(i, j), (k, l)$ in an $n \times n$ matrix, choose a permutation matrix $P \in \mathrm{GL}(n, \mathbb{F}_2)$ such that $P^{-1}E_{ij}, P^{-1}E_{kl}$ are off-diagonal units with acyclic directed edges. Here $E_{ij}, E_{kl} \in \mathrm{Mat}(n, \mathbb{F}_2)$ are matrix units. If $i = k$, choose the common new row outside $\{j, l\}$; if $j = l$, choose distinct new rows outside $\{j\}$; otherwise choose $c \notin \{j, l\}$ and new rows c, j. In the last case the edges form the path $c \to j \to l$. Complete these choices to a permutation. All four matrices $P + sE_{ij} + tE_{kl}$, with $s, t \in \mathbb{F}_2$, are invertible. Summing their polynomial values isolates the indicated mixed coefficient. Hence all mixed coefficients vanish. An invertible pair $P, P + E_{ij}$ similarly kills each remaining linear coefficient. The polynomial was zero.

The quadratic coordinate functions of $g \otimes_{\mathbb{F}_2} g$ therefore have rank $n^2(n^2+1)/2$. The operator x has Jordan type $2^h 1^n$, where $h = n(n-1)/2$, and

$$\dim_{\mathbb{K}} \mathrm{End}_{\mathbb{K}[x]}(T) = (h+n)^2 + h^2 = n^2(n^2+1)/2.$$

The enveloping algebra commutes with x, proving equality over $\mathbb{F}_2$ and, together with the cited result, over every finite field $\mathbb{K}$ of characteristic 2.

Over any such $\mathbb{K}$, choose coordinates adapted to $\operatorname{im} x \subseteq \ker x$. The algebra $\mathcal{A} = \mathrm{End}_{\mathbb{K}[x]}(T)$ consists of

$$\begin{pmatrix} B & C & D \\ 0 & E & F \\ 0 & 0 & B \end{pmatrix},$$

where $B, D \in \mathrm{Mat}(h, \mathbb{K})$, $E \in \mathrm{Mat}(n, \mathbb{K})$, $C \in \mathrm{Mat}(h \times n, \mathbb{K})$, and $F \in \mathrm{Mat}(n \times h, \mathbb{K})$ are arbitrary. For a primitive idempotent e in either diagonal matrix component, the dual right ideal $(e\mathcal{A})^*$ is an indecomposable injective left $\mathcal{A}$-module. These duals are T and T/A, respectively: the displayed blocks give their essential simple socles $\bigwedge^2_{\mathbb{K}} V$ and $V^{[1]}$, and the corresponding right-ideal dimensions are $2h + n$ and $h + n$. Thus these are exactly the two indecomposable injective types. □

For a general-linear square with no fixed constituent, its prime-field enveloping algebra equals its defining-field enveloping algebra: scalar matrices act by a^2, and the squares of $\mathbb{K}$ span $\mathbb{K}$. Passage from GL to SL uses Lemma C.1 on the entire module, including any natural summands already appearing as its Frobenius-twisted natural constituent.

Lemma C.13 (The symplectic group on the tensor square)**.** *Let V be a $2m$-dimensional $\mathbb{K}$-space with a nonsingular alternating form $b : V \times V \to \mathbb{K}$, where $m \geq 5$. Choose a symplectic basis $e_1, \dots, e_m, f_1, \dots, f_m$ of V. On $T = V \otimes_{\mathbb{K}} V$, with $x = I_T + \tau$ and $A = \operatorname{im} x$, define*

$$\omega = \sum_{i=1}^{m} (e_i \otimes_{\mathbb{K}} f_i + f_i \otimes_{\mathbb{K}} e_i) \in A,$$

$$\ell : A \to \mathbb{K}, \qquad \ell(v \otimes_{\mathbb{K}} w + w \otimes_{\mathbb{K}} v) = b(v, w) \quad (v, w \in V).$$

$$y \in \operatorname{End}_{\mathbb{K}}(T), \qquad y(t) = \ell(xt)\omega \quad (t \in T).$$

Then $x^2 = y^2 = xy = yx = 0$, and the defining-field enveloping algebra on T is $\operatorname{End}_C(T)$, where $C = \mathbb{K}[x, y]/(x^2, y^2, xy)$ acts through the displayed operators. Its prime-field enveloping algebra is $\mathbb{F}_2 I_T + \ker \alpha$, where $\alpha : \operatorname{End}_C(T) \to \mathbb{K}$ is determined by $a\omega = \alpha(a)\omega$ for $a \in \operatorname{End}_C(T)$.

Over an algebraic closure, the enveloping-algebra identity is the degree-two case of [DDH08, Proposition 1.3]. The proof below establishes the finite-group span comparison and prime-field descent needed here.

Proof of Lemma C.13. Let $G = \operatorname{Sp}(V, b)$ and write $\rho(g) = g \otimes_{\mathbb{K}} g$ for its action on T. Use the Siegel Levi $\operatorname{GL}(E)$ on $V = E \oplus E^*$, where $\dim_{\mathbb{K}} E = m$ and $E^* = \operatorname{Hom}_{\mathbb{K}}(E, \mathbb{K})$. The three components of T are $E \otimes_{\mathbb{K}} E$, $(E \otimes_{\mathbb{K}} E^*) \oplus (E^* \otimes_{\mathbb{K}} E)$, and $E^* \otimes_{\mathbb{K}} E^*$. They have pairwise disjoint simple types for $m \geq 5$. The first has the exterior-square type $L(\omega_2)$ and the Frobenius-twisted natural type $L(2\omega_1)$; the last has their dual types, and the middle has only trivial and adjoint types. These highest weights refer to the algebraic special linear group of the Levi. For the finite Levi, first reduce each Frobenius exponent modulo $e = [\mathbb{K} : \mathbb{F}_2]$ and then apply [Ste67, Theorems 41 and 43] to the resulting restricted weights. In particular, over $\mathbb{F}_2$ the type $L(2\omega_1)$ restricts to the natural module, and its dual to the dual natural module; the three lists are still disjoint. The joint enveloping-algebra observation in Subsection 6.2 therefore gives the three tensor-grade projectors in the finite Levi algebra.

For a symmetric matrix $B \in \operatorname{Mat}(m, \mathbb{K})$, let $N_B \in \operatorname{End}_{\mathbb{K}}(V)$ be the block map $E^* \to E$ given by B, zero on E. The upper Siegel element is $u(B) = I_V + N_B$. Its action $\rho(u(B)) = u(B) \otimes_{\mathbb{K}} u(B) \in \operatorname{End}_{\mathbb{K}}(T)$ is

$$\rho(u(B)) = I_T + L_B + Q_B,$$

$$L_B = N_B \otimes_{\mathbb{K}} I_V + I_V \otimes_{\mathbb{K}} N_B, \qquad Q_B = N_B \otimes_{\mathbb{K}} N_B.$$

Here $L_B, Q_B \in \operatorname{End}_{\mathbb{K}}(T)$ raise the tensor-block grade by one and two, respectively. The projections isolate both coefficients. After scalar extension the algebra therefore contains $I + tL_B + t^2 Q_B = \rho(u(tB))$ for every scalar t in the extension field. Symmetric matrices with entries in $\mathbb{F}_2$ span the symmetric matrices over $\mathbb{K}$, and $u(B + B') = u(B)u(B')$. The same applies to the opposite radical. These radicals generate the algebraic symplectic group (the mixed root subgroups follow by elementary commutators). Over the algebraic closure, apply [DDH08, Proposition 1.3], which identifies the enveloping algebra of the symplectic similitude group over any infinite field with the commutant of the Brauer-algebra action on tensor space. In tensor degree 2 that action is generated by the signed swap and contraction. In characteristic 2 these are $\tau = I_T + x$ and y, so its commutant is $\operatorname{End}_C(T)$. Every similitude over the algebraic closure is a scalar times a symplectic matrix, so the

two group spans coincide there (also [DDH08, Equation (1.1)]). The cited infinite-field theorem is used only after the preceding finite-to-algebraic span comparison. Centralizers commute with scalar extension, proving the asserted finite-field identity.

For the prime-field assertion apply Lemma C.6 to descend the projectors constructed over $\mathbb{K}$. The group is perfect in this range and fixes $\omega \neq 0$. The nontrivial algebraic simple types in its square are $L(\omega_2)$ and $L(2\omega_1)$: the former is the nontrivial factor of the contraction kernel in A, and the latter is $\Gamma/A \cong V^{[1]}$. This description also follows from the block centralizer just computed. For $0 \leq i < e = [\mathbb{K} : \mathbb{F}_2]$, their finite Frobenius conjugates have types $L(2^i\omega_2)$ and $V^{[i+1]}$, with natural twist indices reduced modulo e. These lists for distinct i are disjoint by [Ste67, Theorems 41 and 43]; the fundamental-weight positions distinguish the two families. The descent lemma gives $\operatorname{span}_{\mathbb{F}_2} \rho(G) = \mathbb{F}_2 I_T + \ker \alpha$, including its radical. □

Lemma C.14 (Symplectic injective modules)**.** *The algebra* $\operatorname{End}_C(T)$ *of Lemma C.13, acting on* $T = V \otimes_{\mathbb{K}} V$*, has three indecomposable injective types. With* $A = \operatorname{im} x$*,* $\omega \in A$*, and* $\ell : A \to \mathbb{K}$ *as in that lemma, they are*

$$I_H = T/\mathbb{K}\omega, \qquad I_W = T/A = \operatorname{Sym}^2_{\mathbb{K}} V,$$

and

$$I_0 = \begin{cases} T/\ker \ell, & m \text{ odd}, \\ (T \oplus T)/\{(a, \ell(a)\omega) : a \in A\}, & m \text{ even}. \end{cases}$$

The injective construction is projective–injective duality [Ben91, Section 1.5]. The calculation below identifies its primitive right-ideal duals with the displayed parity-dependent quotients.

Proof of Lemma C.14. Put $h = \dim_{\mathbb{K}} A$. Choose a $\mathbb{K}$-linear complement of A in $\Gamma = \ker x$. Since both x and y vanish on Γ, this complement consists of $2m$ trivial C-modules. On the remaining space, x identifies a top copy of A with its bottom copy, while y induces the rank-one map $a \mapsto \ell(a)\omega$ on A. Its square is m times itself, so the C-module is

$$T \cong \begin{cases} (C/(y))^{h-1} \oplus C/(x+y) \oplus \mathbb{K}^{2m}, & m \text{ odd}, \\ (C/(y))^{h-2} \oplus R \oplus \mathbb{K}^{2m}, & m \text{ even}, \end{cases}$$

where R has generators u, v and relations $yu = 0$, $yv = xu$. Let $e_i \in \mathcal{A} = \operatorname{End}_C(T)$ be the projection onto an indecomposable summand $C_i \leq T$. The primitive right ideal $e_i\mathcal{A}$ is $\operatorname{Hom}_C(T, C_i)$, with its values included in T. Its $\mathbb{K}$-dual is $T \otimes_C C_i^*$, where $C_i^* = \operatorname{Hom}_{\mathbb{K}}(C_i, \mathbb{K})$, by *tensor–Hom duality*. All three C_i are self-dual. For $C/(y)$ this quotient is $T/\mathbb{K}\omega$; for $\mathbb{K}$ it is T/A; for $C/(x+y)$ it is $T/\operatorname{im}(x+y) = T/\ker \ell$. In the even case the presentation gives relations $(yt, 0)$ and (xt, yt) in $T \oplus T$, for $t \in T$. Since $\ell(\omega) = 0$, $(\omega, 0)$ already occurs in the second family, and the denominator is precisely the displayed graph. This derives all three injectives, including their nonsplit structure, from primitive right ideals. □

Lemma C.15 (The unitary group on the tensor square)**.** *Let* $\mathbb{K}/\mathbb{K}_0$ *be a quadratic extension of finite fields of characteristic two, and let* $V = \mathbb{K}^n$ *be the natural unitary module, with* n *in the stable degree-two range of Subsection 6.1. Its ordinary tensor square* $T = V \otimes_{\mathbb{K}} V$ *has defining-field enveloping algebra* $\operatorname{End}_{\mathbb{K}[x]}(T)$ *and the same two injective types as Lemma C.12. Its prime-field enveloping algebra contains the full scalar field. The special unitary group has the same joint enveloping algebra on tensor products of bounded length in the range of Lemma C.1.*

The Levi and root-subgroup descriptions are standard [Tay92, Chapter 10], and the finite constituent types use [Ste67, Theorems 41 and 43]. The proof includes the finite-span and coefficient-field calculations before applying the general-linear quotient models.

Proof of Lemma C.15. First work with the full group $\mathrm{U}_n(\mathbb{K}/\mathbb{K}_0)$; restriction to $\mathrm{SU}_n(\mathbb{K}/\mathbb{K}_0)$ is made at the end. Write $\bar{a} = a^{|\mathbb{K}_0|}$ for $a \in \mathbb{K}$, and put $\rho(g) = g \otimes_{\mathbb{K}} g$ on T. For the natural $\mathbb{K}$-space E of the Levi $\mathrm{GL}(m, \mathbb{K})$, let $E^{\bar{*}}$ denote the *conjugate-dual* $\mathbb{K}$-module: $g \in \mathrm{GL}(m, \mathbb{K})$ acts on it by $\bar{g}^{-\mathrm{t}}$. In even dimension $2m$, use $V = E \oplus E^{\bar{*}}$. The tensor-square components are $E \otimes_{\mathbb{K}} E$, $(E \otimes_{\mathbb{K}} E^{\bar{*}}) \oplus (E^{\bar{*}} \otimes_{\mathbb{K}} E)$, and $E^{\bar{*}} \otimes_{\mathbb{K}} E^{\bar{*}}$. They have disjoint composition types: the mixed grade is the simple tensor product with distinct Frobenius exponents, and the two squares have the exterior and Frobenius-twisted natural types with their indicated variances. This remains true for $\mathbb{K}_0 = \mathbb{F}_2$ and $m \geq 5$, by the restricted weights in [Ste67, Theorems 41 and 43]. Thus their projections onto these components belong to the finite Levi algebra. For a Hermitian matrix $B \in \mathrm{Mat}(m, \mathbb{K})$, define the block map $N_B : E^{\bar{*}} \to E$, extended by zero on E. On $T = V \otimes_{\mathbb{K}} V$, its radical element again acts as $I_T + L_B + Q_B$, where $L_B = N_B \otimes_{\mathbb{K}} I_V + I_V \otimes_{\mathbb{K}} N_B$ and $Q_B = N_B \otimes_{\mathbb{K}} N_B$. The component projections separate these terms. The $\mathbb{K}$-linear span of the Hermitian matrices is $\mathrm{Mat}(m, \mathbb{K})$: use the diagonal units and, for $i < j$, $E_{ij} + E_{ji}$ and $\theta E_{ij} + \bar{\theta} E_{ji}$ with $\theta \in \mathbb{K}$ and $\theta \neq \bar{\theta}$. Over an algebraic closure $\overline{\mathbb{K}}$, the scalar-extended algebra contains the tensor-square actions of all upper and lower general-linear block unipotents. These unipotents generate $\mathrm{SL}_{2m}(\overline{\mathbb{K}})$, whose enveloping algebra on the tensor square is the swap centralizer; the latter also follows by scalar extension of Lemma C.12.

In odd dimension write $V = E \oplus F \oplus Z$, where $F = E^{\bar{*}}$ is the conjugate-dual $\mathbb{K}$-module and $\dim_{\mathbb{K}} Z = 1$. The six Levi summands $E \otimes_{\mathbb{K}} E$, $(E \otimes_{\mathbb{K}} F) \oplus (F \otimes_{\mathbb{K}} E)$, $F \otimes_{\mathbb{K}} F$, $(E \otimes_{\mathbb{K}} Z) \oplus (Z \otimes_{\mathbb{K}} E)$, $(F \otimes_{\mathbb{K}} Z) \oplus (Z \otimes_{\mathbb{K}} F)$, and $Z \otimes_{\mathbb{K}} Z$ have disjoint sets of composition factors and hence projections in the finite Levi enveloping algebra. Choose $\theta \in \mathbb{K}$ with $\theta + \bar{\theta} = 1$ and a basis vector $z \in Z$. Use the finite unitary root element sending $z \mapsto z + e_i$ and $f_i \mapsto f_i + z + \theta e_i$. Put $a = E_{i,z}, b = E_{z,f_i} \in \mathrm{End}_{\mathbb{K}}(V)$, so a maps z to e_i and b maps f_i to z, with both zero on the other basis vectors. Write $P_{EE}, P_{EF}, P_{FF}, P_{EZ}, P_{FZ}, P_{ZZ}$ for the six Levi projections, with EF denoting $(E \otimes_{\mathbb{K}} F) \oplus (F \otimes_{\mathbb{K}} E)$ and similarly for EZ, FZ. For the chosen root element u, direct block multiplication gives

$$\begin{aligned} a \otimes I_V + I_V \otimes a &= P_{EE}\rho(u)P_{EZ} + P_{EF}\rho(u)P_{FZ} + P_{EZ}\rho(u)P_{ZZ}, \\ a \otimes a &= P_{EE}\rho(u)P_{ZZ}, \\ b \otimes I_V + I_V \otimes b &= P_{FZ}\rho(u)P_{FF} + P_{ZZ}\rho(u)P_{FZ} + P_{EZ}\rho(u)P_{EF}, \\ b \otimes b &= P_{ZZ}\rho(u)P_{FF}. \end{aligned}$$

All tensor products in this display are over $\mathbb{K}$. The other root terms have different source or target grades. These identities place the tensor-square actions of the algebraic elementary matrices $I_V + ta$, $I_V + tb$ in the scalar extension of the finite unitary enveloping algebra; together with $\mathrm{SL}_{2m}(\overline{\mathbb{K}})$ on the scalar extension of $E \oplus F$, they generate $\mathrm{SL}_{2m+1}(\overline{\mathbb{K}})$. Linear descent proves the equality. A central norm-one scalar of order $|\mathbb{K}_0| + 1$ acts on the square by its square, and generates $\mathbb{K}$ over $\mathbb{F}_2$: every proper subfield has at most $|\mathbb{K}_0|$ elements. Thus the full scalar field lies in the prime-field enveloping algebra. Finally use the joint derived transfer lemma. □

C.4.3 First cohomology in characteristic two

The following calculations establish the change of origin on the nonorthogonal injectives and on the orthogonal target $T/\mathbb{K}\omega$.

Lemma C.16 (Special-unitary transfer in degree 1)**.** *Let $U = \mathrm{U}_n(\mathbb{K}/\mathbb{K}_0)$, $D = \mathrm{SU}_n(\mathbb{K}/\mathbb{K}_0)$, in characteristic 2, and let M be the coefficient module. If every diagonal norm-one-torus weight of M is supported on at most r coordinates and $n > r + 3$, restriction identifies $H^1(U, M)$ with $H^1(D, M)$.*

The extension step is the standard inflation–restriction argument, and the coordinate isometries use Witt extension [Tay92, Theorem 7.4]. The torus-support calculation below verifies that the

determinant quotient acts trivially on the stated degree-one cohomology, including over the smallest field.

Proof. Coordinate SU_3 subgroups generate D. Here are the small-field details. A norm-one vector with two coordinates of unequal nonzero $\mathbb{K}_0$-norm can have those two coordinates replaced by one using a two-dimensional unitary isometry. If all its nonzero coordinate norms coincide, their number is odd, and any three of them have nonzero total norm; a three-dimensional isometry replaces that triple by one coordinate. Witt extension in the corresponding coordinate subspace justifies each operation [Tay92, Theorem 7.4]. Successive column elimination shows that U is generated by these coordinate unitary groups and its diagonal torus T. Each local unitary group is its special subgroup times a diagonal element. The determinant-one torus $S = T \cap D$ is generated by two-coordinate $\mathrm{diag}(t, t^{-1})$ and is inside the generated special subgroup. Thus the latter is exactly D. Normalize a D-cocycle to vanish on S. Evaluation on the generating coordinate SU_3 subgroups is injective and shows that its T-weights have support at most $r + 3$. Since $|T|$ is odd, this bound passes to the cohomology quotient. Inner D-action is trivial there, so all its characters are constant-coordinate characters trivial on S. The inequality on n makes them zero. The quotient U/D has odd order; inflation–restriction finishes the proof. □

Lemma C.17 (First cohomology of the injective modules)**.** *Assume the stable degree-two rank bounds of Subsection 6.1. Every general-linear and special-unitary injective in Lemmas C.12 and C.15, including the adjoint modulo scalars modules of Lemma C.11, has zero first cohomology. All three symplectic injectives in Lemma C.14 have zero first cohomology. The tensor products with unequal Frobenius exponents also have zero first cohomology. These assertions are for the full stated groups.*

The general-linear case follows from Lemma C.3, based on [FFSS99, Appendix, Theorem A.1] and [Put23, Theorem 12.1]. For the form groups the explicit Levi and radical calculation below is needed: it verifies the finite-group vanishing for the displayed quotient models and specified twists, rather than an algebraic-group cohomology assertion.

Proof of Lemma C.17. For general linear groups this is Lemma C.3; quotients of adjoint modules by fixed scalar subspaces additionally use Lemma C.5 and the long exact sequence.

For $U = \mathrm{U}_n(\mathbb{K}/\mathbb{K}_0)$ and $D = \mathrm{SU}_n(\mathbb{K}/\mathbb{K}_0)$, put $|\mathbb{K}_0| = q_0 = 2^f$. On the coefficient module $V^{[a]} \otimes_{\mathbb{K}} V^{[b]}$, a central norm-one scalar $\lambda \in \mathbb{K}^\times$ acts as multiplication by $\lambda^{2^a+2^b}$. The exponent $2^a + 2^b$ is zero modulo $q_0 + 1$ exactly when $b - a = f$ modulo $2f$. Otherwise its prime-to-two cyclic subgroup has no fixed vector, so all cohomology of U is zero by averaging; Lemma C.16 passes degree 1 to D. It remains to treat the adjoint case.

In even dimension $2m$, the Siegel parabolic of D has odd index, Levi $L = \{g \in \mathrm{GL}(m, \mathbb{K}) : \det g \in \mathbb{K}_0^\times\}$, and additive Hermitian radical $u(Z) = \left(\begin{smallmatrix} I & Z \\ 0 & I \end{smallmatrix}\right)$, where $Z \in \mathrm{Mat}(m, \mathbb{K})$ satisfies $Z = \bar{Z}^{\mathrm{t}}$. Every Levi block of $\mathrm{End}_{\mathbb{K}}(V)$ is a tensor product of length at most two, so normalize a cocycle to zero on L. The radical raises the usual three endomorphism grades. At its first nonzero grade the cocycle is an additive L-map from the Hermitian parameter space. Over a splitting field that parameter has types $E^{[s]} \otimes_{\mathbb{K}} E^{[s+f]}$, $0 \le s < f$. The four-factor matching in Lemma C.3 excludes the opposite and diagonal blocks and leaves only the top block $Z \mapsto aZ$. This is the coboundary of a times the projector onto the first half. After subtracting it no first grade remains. Restriction to an odd-index subgroup is injective in characteristic 2, proving the claim.

In odd dimension write $V = E \oplus \mathbb{K}t \oplus E^{\bar{*}}$, with $\dim_{\mathbb{K}} E = m$ and $E^{\bar{*}}$ the conjugate dual. Here $A^\dagger = \bar{A}^{\mathrm{t}}$ for a matrix A over $\mathbb{K}$. The Levi is $\mathrm{GL}(m, \mathbb{K})$, acting by $\mathrm{diag}(g, \chi(g), g^{-\dagger})$ for $g \in \mathrm{GL}(m, \mathbb{K})$,

where $\chi : \mathrm{GL}(m, \mathbb{K}) \to \mathbb{K}^\times$ is $\chi(g) = (\det g)^{q_0 - 1}$. For $X \in \mathrm{Hom}_{\mathbb{K}}(\mathbb{K}t, E)$ and $Z \in \mathrm{Hom}_{\mathbb{K}}(E^{\bar{*}}, E)$, its radical consists of

$$u(X, Z) = \begin{pmatrix} I & X & Z \\ 0 & 1 & X^\dagger \\ 0 & 0 & I \end{pmatrix}, \qquad Z + Z^\dagger = XX^\dagger.$$

Its center is $X = 0$ and its quotient is the additive space E with action $g \cdot v = \chi(g)^{-1} gv$. Every adjoint Levi block has zero first cohomology. For the blocks joining $\mathbb{K}t$ with E or $E^{\bar{*}}$, normalize on the prime-to-two torus: a nonzero determinant twist has support m, whereas the possible natural or cocycle-root weight has support at most three, so the same proof as in Lemma C.3 excludes it. Kill the center value aZ by the first-half projector, as above. A first remaining grade then factors through the radical quotient and has exactly the two possibilities

$$aX \in \mathrm{Hom}_{\mathbb{K}}(\mathbb{K}t, E), \qquad bX^\dagger \in \mathrm{Hom}_{\mathbb{K}}(E^{\bar{*}}, \mathbb{K}t), \qquad a, b \in \mathbb{K}.$$

They are the two field components, since $\chi^{[f]} = \chi^{-1}$. Commutators lie in the center, where the cocycle is now zero. The cocycle identity for two radical elements gives in the top block $(a+b)(XY^\dagger + YX^\dagger) = 0$ in $\mathrm{Hom}_{\mathbb{K}}(E^{\bar{*}}, E)$, for $X, Y \in \mathrm{Hom}_{\mathbb{K}}(\mathbb{K}t, E)$. Distinct coordinate vectors force $a = b$. The projector onto $\mathbb{K}t$ removes these values. No later first grade is allowed by the same Hom matching. Again odd-index restriction proves $H^1(D, \mathrm{End}_{\mathbb{K}}(V)) = 0$.

The $\mathbb{K}_0$-space H of Hermitian endomorphisms of V satisfies $\mathrm{End}_{\mathbb{K}}(V) = H \oplus \theta H$ for $\theta \in \mathbb{K} \setminus \mathbb{K}_0$. Thus $H^1(D, H) = 0$, and every quotient by a fixed scalar subspace has vanishing first cohomology by Lemma C.5.

For symplectic groups use the odd-index Siegel parabolic with Levi $\mathrm{GL}(E)$, where $\dim_{\mathbb{K}} E = m$, and additive radical $\Gamma^2_{\mathbb{K}} E$, identified with the symmetric matrices in the chosen basis. On $T = V \otimes_{\mathbb{K}} V$, where $V = E \oplus E^*$ and $E^* = \mathrm{Hom}_{\mathbb{K}}(E, \mathbb{K})$, its 3-tensor-block grades are $E \otimes_{\mathbb{K}} E$, $(E \otimes_{\mathbb{K}} E^*) \oplus (E^* \otimes_{\mathbb{K}} E)$, and $E^* \otimes_{\mathbb{K}} E^*$. After Levi normalization the lower and middle radical values are zero by Lemma C.4. The top maps $\Gamma^2_{\mathbb{K}} E \to E \otimes_{\mathbb{K}} E$ are scalar multiples of its inclusion: this follows directly from the block algebra in Lemma C.12. Over a proper extension field scalar squares force prime-field intertwiners to be $\mathbb{K}$-linear. For $r = \sum_i e_i \otimes_{\mathbb{K}} f_i \in T$ and $B \in \Gamma^2_{\mathbb{K}} E$, the identity $u(B)r - r = B$ holds in T under this inclusion. Thus this last value is a coboundary. Hence $H^1(\mathrm{Sp}_{2m}(\mathbb{K}), T) = 0$. For $\mathrm{Sym}^2_{\mathbb{K}} V$ the same argument leaves only $B \mapsto \sum_i B_{ii} e_i^2$; it is the coboundary of $\sum_i e_i f_i \in \mathrm{Sym}^2_{\mathbb{K}} V$, where products denote classes in the symmetric quotient. Tensor products with distinct relative Frobenius-twist exponents have a simple tensor product in the top component and admit no map from $\Gamma^2_{\mathbb{K}} E$ into that component, so their first cohomology is also zero. The quotient $T/\mathbb{K}\omega$ uses Lemma C.5.

Finally, for odd m there is the exact sequence $0 \to \mathbb{K} \to I_0 \to \mathrm{Sym}^2_{\mathbb{K}} V \to 0$; for even m there is $0 \to T \to I_0 \to \mathrm{Sym}^2_{\mathbb{K}} V \to 0$, with first map $t \mapsto [(0, t)]$ and last map $[(s, t)] \mapsto s + A$. The displayed denominators in Lemma C.14 verify exactness. In either case both exterior first-cohomology terms are zero, so $H^1(\mathrm{Sp}_{2m}(\mathbb{K}), I_0) = 0$. □

Lemma C.18 (Orthogonal square and natural cohomology)**.** *Let $G = \Omega(V, Q)$ in characteristic 2, with either quadratic sign and stable Witt index as in Subsection 6.1. Set $T = V \otimes_{\mathbb{K}} V$, and let $\mathbb{K}\omega \leq T$ be the invariant line determined by the polar form of $Q : V \to \mathbb{K}$. For Frobenius-twist exponents a, b,*

$$H^1(G, V^{[a]}) = 0, \qquad H^1(G, V^{[a]} \otimes_{\mathbb{K}} V^{[b]}) = 0, \qquad H^1(G, T/\mathbb{K}\omega) = 0.$$

Restriction to a parabolic of odd index and the associated cohomology sequences are standard group-cohomology tools [Bro82, Chapter III]; the classical parabolic coordinates are those of [Tay92,

Chapter 11]. The radical values are calculated below for both quadratic signs, since the conclusion includes arbitrary specified Frobenius twists and the scalar quotient.

Proof of Lemma C.18. In split type the odd-index parabolic has Levi $\mathrm{GL}(m,\mathbb{K})$ and additive alternating radical $\bigwedge^2_{\mathbb{K}} E$. Normalize on the Levi. Lemma C.4 permits no radical map to a natural coefficient; for a tensor coefficient it permits only $Z \mapsto aZ$ in the top block, with equal Frobenius twists. That map is the coboundary of the first-half projector. This proves the assertions before scalar quotient in split type.

For nonsplit type write $V = E \oplus Z \oplus E^*$, with Z an anisotropic $\mathbb{K}$-plane and $E^* = \mathrm{Hom}_{\mathbb{K}}(E,\mathbb{K})$. The Levi is $\mathrm{GL}(m,\mathbb{K}) \times T_0$, where the rotation torus T_0 has order $q+1$, $q = |\mathbb{K}|$. The radical has center $\bigwedge^2_{\mathbb{K}} E$ and quotient $\mathrm{Hom}_{\mathbb{K}}(Z,E)$. Let $Q_Z = Q|_Z$ and let J_Z be the matrix of its polar form in the chosen basis of Z. For $X \in \mathrm{Hom}_{\mathbb{K}}(Z,E)$ and $B \in \mathrm{Hom}_{\mathbb{K}}(E^*,E)$, write its elements as

$$\begin{pmatrix} I & X & B \\ 0 & I & X^\dagger \\ 0 & 0 & I \end{pmatrix}, \qquad X^\dagger = J_Z^{-1} X^{\mathrm{t}}, \quad t^{\mathrm{t}} B t = Q_Z(X^\dagger t) \quad (t \in E^*).$$

Every X occurs. The parabolic has odd index because it contains a *Sylow two-subgroup*. Levi cohomology vanishes by Lemma C.3 and the odd-order torus. Normalize there, and kill the only possible center map aB in the matched case by the first-half projector. No center map exists otherwise. Over a splitting field the two T_0-weights on $Z^{[a]}$ are $\chi^{\pm 2^a}$. A first quotient-radical grade in a two-slot tensor requires $2^a = \pm 2^b$ modulo $q+1$. For $0 \le a, b < \log_2 q$ this means $a = b$: a nonzero difference has absolute value below $q+1$, and the sum is at most q. Hence unequal twists have no value. For a natural coefficient no matching T_0-character is available.

For the matched square identify $T = V \otimes_{\mathbb{K}} V$ with $\mathrm{End}_{\mathbb{K}}(V)$ by the polar form. The only possible first grade is $XD_0 \in \mathrm{Hom}_{\mathbb{K}}(Z,E)$ and $D_1 X^\dagger \in \mathrm{Hom}_{\mathbb{K}}(E^*,Z)$, with $D_0, D_1 \in \mathrm{End}_{\mathbb{K}[T_0]}(Z)$. Commutators lie in the center, and the cocycle identity gives $X(D_1+D_0)Y^\dagger + Y(D_1+D_0)X^\dagger = 0$. Choose distinct coordinates of E for the single nonzero rows of $X, Y \in \mathrm{Hom}_{\mathbb{K}}(Z,E)$, and arbitrary row functionals on Z. Their matrix positions in $\mathrm{Hom}_{\mathbb{K}}(E^*,E)$ differ, so $D_1 + D_0 = 0$. The endomorphism supported by D_0 on Z removes the remaining values by a coboundary. There is no other first grade, proving vanishing on the parabolic and hence on G. Finally apply the long exact sequence and Lemma C.5 to $\mathbb{K}\omega$. □

C.4.4 Invariant tensors needed for the orthogonal calculation

We prove Lemma C.20 and Corollary C.21 under their separate rank bounds stated in the following lemma. The orthogonal square calculation needs their Hom vanishing. Each factor has the dual or Frobenius-twisted action specified in the input. All tensor products in this calculation are over the common coefficient field $\mathbb{K}$.

The coordinate twist $v \mapsto v^{[a]}$ is additive and p^a-semilinear, whereas the pairing maps $\kappa_{i,\eta}$ are $\mathbb{K}$-linear maps between the specified based modules. The following elementary lemma identifies a trilinear form supported on pairs. A hyperbolic plane has basis e, f with $b(e,f) = 1$; distinct planes in a hyperbolic basis are orthogonal.

Lemma C.19 (Trilinear forms vanishing on orthogonal triples)**.** *Let V be a vector space over a field $\mathbb{K}$ with $\dim_{\mathbb{K}} V \ge 6$, and let $b : V \times V \to \mathbb{K}$ be a nondegenerate alternating bilinear form. If a trilinear form $F : V^3 \to \mathbb{K}$ vanishes on every pairwise b-orthogonal triple $(x,y,z) \in V^3$, then there exist $\ell_1, \ell_2, \ell_3 \in V^*$ such that, for $(x,y,z) \in V^3$,*

$$F(x,y,z) = \ell_1(x)b(y,z) + \ell_2(y)b(x,z) + \ell_3(z)b(x,y).$$

In characteristic 2 the same assertion holds for the standard dot product $b : \mathbb{K}^n \times \mathbb{K}^n \to \mathbb{K}$ *with* $n \geq 6$.

This is an elementary calculation in hyperbolic coordinates, using the standard form conventions of [Tay92, Chapter 7]. We include the coordinate proof to cover the characteristic-two dot product as well as alternating forms, without an infinite-field hypothesis.

Proof. Choose a hyperbolic basis $e_i, f_i \in V$, with $1 \leq i \leq m$ and $\dim_{\mathbb{K}} V = 2m$. A coefficient on three basis vectors containing no paired e_i, f_i is zero. For a free basis vector w in plane j, compare (e_i+e_k, f_i+f_k, w) in characteristic 2, and (e_i+e_k, f_i-f_k, w) otherwise, for $i, k \neq j$. This shows that the coefficient with its paired positions in plane i is independent of i. The triple $(e_i + f_i, e_i + f_i, w)$ identifies the two orders of the pair, with the sign prescribed by b. Repeat in each free position and subtract the three resulting pairing terms. The remainder has support only on the six mixed triples inside one hyperbolic plane. For distinct i, j, k, the triple

$$e_i + e_j, \quad e_i + e_k, \quad f_i + f_j + f_k$$

is pairwise orthogonal in characteristic 2 and isolates the coefficient (e_i, e_i, f_i). In other characteristics change the two spare coefficients in the last vector to their negatives. Permuting positions and interchanging e_i, f_i kills the other five coefficients.

For the dot product in characteristic 2, three distinct basis indices give zero coefficients. The orthogonal triple $(e_i + e_k, e_i + e_k, e_j)$ shows that the coefficient of (e_i, e_i, e_j) is independent of $i \neq j$. Subtract the three pairing terms. Only diagonal coefficients d_i remain. Evaluating on three copies of $e_i + e_j$ gives $d_i = d_j$. Finally the following three vectors of $\mathbb{K}^n$, with zero coordinates appended when $n > 6$,

$$(1, 1, 1, 1, 0, 0), \quad (1, 1, 0, 0, 1, 1), \quad (1, 0, 1, 0, 1, 0)$$

are pairwise orthogonal and have common support of size one. Their value is the common d_i, which is therefore zero. □

A finite subgroup of order prime to the characteristic acts semisimply: averaging preserves invariant subspaces and quotients. For an abelian such subgroup, after extending the field its action is diagonal, and the simultaneous eigenspaces are its *weight spaces*. A *character* records the scalar by which each element acts on one of these spaces. In the proof, a product of order-three groups acts on separate coordinate blocks. A tensor with three different blocks has a character involving all three, whereas degree-2 modules involve at most two. This remains valid for a submodule with nonsplit extensions, because taking these weight spaces is exact.

The proof first uses this support test for symplectic groups. It then uses the corresponding form-preserving blocks in orthogonal and unitary groups, and restricts special linear groups to a symplectic subgroup. The last step tests scalings from the full defining field, so only pairings compatible with the specified Frobenius twists remain. For a quadratic form, a vector is *isotropic* or *singular* when its quadratic value is zero; a subspace is totally isotropic when the form vanishes on it. The *Witt index* is the maximum dimension of such a subspace. Witt extension says that an isometry between subspaces extends to an isometry of the whole nondegenerate space.

Witt extension gives the required changes of coordinates [Tay92, Theorem 7.4]; averaging makes the prime-to-two weight-space restrictions exact. The support and full-field scaling arguments below are needed because the statement concerns submodules with nonsplit extensions, specified Frobenius twists, and both finite orthogonal signs.

Lemma C.20 (Cubic submodules in characteristic 2)**.** *Let $\mathbb{K}$ be a finite field of characteristic 2, and let D be a product of the derived classical groups defined above, with natural dimensions at least 36 and, for unitary and orthogonal factors, Witt indices at least 12. Let $T = T_1 \otimes_{\mathbb{K}} T_2 \otimes_{\mathbb{K}} T_3$ be a cubic tensor module of natural, dual, or Frobenius-twisted D-modules. If every composition factor of a D-submodule $P \leq T$ occurs in degree at most two, then $P \leq J(T)$. This holds for special linear, special unitary, symplectic, and both even orthogonal derived groups, with arbitrary twists and fixed multiplicities.*

Proof of Lemma C.20. First consider $D = \mathrm{Sp}_{2m}(\mathbb{K})$ on $V = \mathbb{K}^{2m}$, with polar form $b : V \times V \to \mathbb{K}$. On each hyperbolic plane take the order-three subgroup of $\mathrm{SL}_2(\mathbb{F}_2)$. Their product is an odd-order torus. After extending the coefficient field, a natural module has characters supported on one plane, whereas a tensor product of at most two natural modules, duals, or twists has characters supported on at most two planes. Taking invariants and weight spaces for this torus is exact. Thus P has no component whose three positions lie in three different planes. This remains true after every group conjugation. Witt extension [Tay92, Theorem 7.4] therefore implies that each vector $u \in P$, viewed as a trilinear form through the polar identification, vanishes on independent totally isotropic triples in V^3. The independent ordered triples span the third tensor power of a three-dimensional space, even over $\mathbb{F}_2$: with two independent arguments fixed, the vectors outside their span span the space; apply this successively in all three positions. Extending an isotropic subspace to dimension three removes the independence condition. Lemma C.19 now gives the three expressions formed using invariant pairings.

Every Frobenius twist has the same matrices on the subgroup over $\mathbb{F}_2$, so the preceding argument applies to each fixed choice of twists. Isolate a free coordinate in one plane and a pairing in another. Scaling e_i by $t \in \mathbb{K}^\times$ and f_i by t^{-1} makes the two orders of that pair acquire factors $t^{2^a-2^b}$ and $t^{2^b-2^a}$. Since the displayed pairing identity holds for every $t \in \mathbb{K}^\times$, their equality forces $2^a = 2^b$ modulo $|\mathbb{K}| - 1$ or the coefficient is zero. The oddness of $|\mathbb{K}| - 1$ justifies cancelling the factor two. This is exactly the eligible-twist condition.

For an even quadratic orthogonal group, use hyperbolic four-spaces $A_i \oplus A_i^*$ with $\dim_{\mathbb{K}} A_i = 2$ and the subgroup $g \oplus g^{-\mathrm{t}}$ for $g \in \mathrm{SL}_2(\mathbb{F}_2)$. Its order-three subgroup has no fixed natural vector and lies in Ω, since the rank of $g \oplus g^{-\mathrm{t}} - I$ is even. The preceding character-support argument excludes three different four-blocks. Any independent triple with zero mutual polar products can be moved into these three blocks: each block represents each prescribed quadratic value, and Witt extension applies to the isometry of their spans. A symmetry in an unused hyperbolic plane corrects the Dickson invariant without changing the three images. For twisted positions, let $u \in P$ and let $\widehat{u}$ denote its trilinear form on the three twisted natural spaces, using their polar identifications. Apply the argument to the evaluation $F_u(x, y, z) = \widehat{u}(x^{[a]}, y^{[b]}, z^{[c]})$ for $x, y, z \in V$. Here $x^{[a]} \in V^{[a]}$, $y^{[b]} \in V^{[b]}$, and $z^{[c]} \in V^{[c]}$ are coordinate Frobenius images; the evaluation is semilinear in the original coordinates, with the indicated exponents. Additive replacements with spare orthogonal vectors remove dependent triples. Choose a symplectic basis for the polar form, also in the nonsplit case. The coordinate proof of Lemma C.19 uses only addition and coefficients zero and one, so the Frobenius twists do not alter it. Hyperbolic scalings prune the same ineligible pairings; their plane can be chosen disjoint from a free coordinate in the anisotropic two-space. This proves both signs.

For a unitary group on $V = \mathbb{K}^n$, with $|\mathbb{K}| = 2^{2f}$ and $\sigma : \mathbb{K} \to \mathbb{K}$, $t \mapsto t^{2^f}$, use the same hyperbolic four-block subgroups. Each represents every Hermitian norm, since $t + t^\sigma$ maps the defining field onto its fixed field. Hermitian Witt extension moves independent mutually orthogonal triples into the blocks; a spare anisotropic line corrects the determinant. The determinant and Dickson-invariant corrections place the resulting isometries in the required derived groups. If a repetition occurs, its repeated vector must be isotropic. For example choose $w \in x^\perp \cap z^\perp$ outside $\langle x, z \rangle$ and subtract the

values on $(x, x+w, z)$ and (x, w, z). Three repeated arguments are handled in a three-dimensional totally isotropic subspace. In an orthonormal basis the zero-one coordinate calculation is the dot-product part of Lemma C.19. For twist indices $a, b \in \mathbb{Z}_{\geq 0}$, a norm-one diagonal element in a paired coordinate, with determinant compensation on a spare coordinate, forces

$$2^a + 2^b \equiv 0 \pmod{2^f + 1}, \qquad |\mathbb{K}| = 2^{2f}.$$

The order of 2 modulo $2^f + 1$ is $2f$: a smaller period would divide $2f$ and be at most f, which is impossible. The unique power equal to -1 is therefore the fth. The displayed condition is exactly the Hermitian dual-twist condition.

For $\mathrm{SL}_n(\mathbb{K})$ in even dimension restrict first to the symplectic subgroup over $\mathbb{F}_2$, identifying duals by its form. An elementary determinant-one transvection outside that symplectic subgroup takes a natural–natural pairing coefficient, with its free coordinate in a third plane, to a three-distinct-plane coefficient. Its free-position origin is unique, so it cannot cancel with another diagram. Consequently only natural–dual evaluation pairings survive. Diagonal elements with determinant compensation on spare coordinates impose matching twists. In odd dimension apply this argument in every even-dimensional coordinate subspace. For a mixed tensor product represented by a bilinear map $B : V \times V \to V$, it gives $B(x, y) \in \langle x, y \rangle$ for the zero-one vectors after every zero-one change of basis. Write $B(e_i, e_j) = a_{ij} e_i + b_{ij} e_j$. Evaluating on $(e_i, e_j + e_k)$ and $(e_i + e_k, e_j)$ makes the respective coefficients depend only on the free index. Evaluating on $(e_i + e_j, e_i + e_k)$ determines the diagonal coefficient. Thus $B(x, y) = \alpha(x)y + \beta(y)x$, exactly the two evaluation pairing terms. For a tensor product of three natural or three dual factors, the submodule P is zero, since every three coordinates lie in an even coordinate subspace. Full-field diagonal elements again impose the twists.

When two positions belong to one group factor and the third to an independent factor, let V be the repeated natural space, with invariant pairing $b : V \times V \to \mathbb{K}$, and apply the product of the same tori. The repeated-factor bilinear form must vanish on all orthogonal pairs, hence be a multiple of the pairing: for fixed $x \in V \setminus \{0\}$ its second functional is proportional to $b(x, -)$, and comparison at independent $x \in V$ makes the scalar constant. The free factor has no fixed natural vector; conjugating its torus tests every free coordinate. Three independent factors give zero immediately by support three. The arguments are coefficientwise in passive multiplicities. They place P inside the sum of the pairing images; the translation subspace is still P.

Finally extend scalars from the prime field, decompose its natural modules into summands with specified Frobenius twists, and apply the proved containment to each projection of P. All maps obtained by tensoring with invariant pairings are defined over the prime field using trace pairings. Containment descends because it is a linear-system rank statement. The denominator remains the original prime-field subspace. □

This is a consequence of Lemma C.20. The short check below shows that the pairing maps have independent natural parameters; polar self-duality then gives the stated Hom-space vanishing.

Corollary C.21 (Invariant tensors of degree 3)**.** *For the finite group D and cubic $\mathbb{K}$-module $T = T_1 \otimes_{\mathbb{K}} T_2 \otimes_{\mathbb{K}} T_3$ in Lemma C.20, $T^D = 0$. In particular, for every stable even orthogonal derived group D, its natural module V, and Frobenius exponents $a, b, c \in \mathbb{Z}_{\geq 0}$,*

$$\mathrm{Hom}_D(V^{[a]}, V^{[b]} \otimes_{\mathbb{K}} V^{[c]}) = 0.$$

Proof of Corollary C.21. A fixed vector $u \in T^D$ generates a trivial D-submodule of T, which occurs in degree zero and lies in $J(T) \leq T$. The eligible maps obtained by tensoring with invariant pairings form a direct sum of natural modules. To check injectivity, put the free basis coordinate in one block and a nonzero pairing coefficient in another. This isolates each pairing position, so a zero

sum has every parameter zero. The natural parameters have no fixed vectors. The Hom assertion follows by the actual polar self-duality of the twisted natural module. □

C.4.5 Orthogonal enveloping algebras and quotient maps

Return to the degree-two range, with orthogonal Witt index at least 32. It satisfies the bounds used for the preceding cubic vanishing. We use the notation T, x, A, Γ from the tensor-square calculations.

Lemma C.22 (Finite orthogonal density)**.** *The defining-field enveloping algebra of finite $\Omega^{\pm}_{2m}(\mathbb{K})$ on its tensor square, in the stated stable range, becomes the enveloping algebra of the connected algebraic orthogonal group on that square after scalar extension.*

The root-subgroup and parabolic descriptions are standard [Tay92, Chapter 11]. The finite-span argument below is needed because algebraic-group density alone does not establish equality of tensor-square enveloping algebras over a small finite field.

Proof. Let $T = V \otimes_{\mathbb{K}} V$, with tensor-square action ρ. Let $h(t)$ be a one-parameter group action, called a *cocharacter*, with $T = \bigoplus_i T_i$ and $h(t)$ acting as t^i on T_i. If the enveloping algebra contains the projections $P_i : T \to T_i \leq T$, then, after scalar extension, it contains

$$\rho(h(t)uh(t)^{-1}) = \sum_{i,j} t^{i-j} P_i \rho(u) P_j$$

for every finite-group element u and every nonzero scalar t in the extension field. In split type, $V = E \oplus E^*$, where $E^* = \mathrm{Hom}_{\mathbb{K}}(E, \mathbb{K})$. The Levi $\mathrm{GL}(E)$ has the three composition-disjoint tensor components $E \otimes_{\mathbb{K}} E$, $(E \otimes_{\mathbb{K}} E^*) \oplus (E^* \otimes_{\mathbb{K}} E)$, and $E^* \otimes_{\mathbb{K}} E^*$ from Lemma C.13; their projections lie in its enveloping algebra. The cocharacter $\mathrm{diag}(tI, t^{-1}I)$ scales an upper alternating radical parameter by t^2. Thus scaling each finite alternating matrix unit gives its whole algebraic root subgroup. The opposite radical and commutators generate all connected type-D root subgroups. In nonsplit type use $E \oplus Z \oplus E^*$ and Levi $\mathrm{GL}(m, \mathbb{K}) \times T_0$, where $q = |\mathbb{K}|$, $|T_0| = q + 1$, and the cocharacter weights on E, Z, E^* are $2, 1, 0$, respectively. The five square grades have disjoint finite Levi types. Grades one and three are dual-natural and natural $\mathrm{GL}(m, \mathbb{K})$-modules with nontrivial T_0-characters; this separates them from the extreme square factors even when $\mathbb{K} = \mathbb{F}_2$. The middle grade consists of $E \otimes_{\mathbb{K}} E^*$, $E^* \otimes_{\mathbb{K}} E$, and $Z \otimes_{\mathbb{K}} Z$. Thus all five projectors exist. The cocharacter takes $u(X, B)$ to $u(tX, t^2B)$ in the radical of Lemma C.18. These curves contain every scalar multiple of every finite central alternating matrix, and modulo the center every scalar multiple of every finite X. The latter span $\mathrm{Hom}_{\mathbb{K}}(Z, E)$. Their generated subgroup is the whole algebraic radical; similarly it contains the opposite radical. Together they generate the connected group. The reverse containment of enveloping algebras is immediate. □

Let $Q_V : V \to \mathbb{K}$ be the specified nondegenerate quadratic form, and put $n = \dim_{\mathbb{K}} V = 2m$ and $h = \dim_{\mathbb{K}} A = n(n-1)/2$. Let b be its polar form. Use T, x, A, Γ from Subsection C.4. The polar form determines ω, ℓ, and the symplectic square algebra $\mathrm{End}_C(T)$ of Lemma C.13. In particular, for a symplectic basis e_i, f_i,

$$\omega = \sum_{i=1}^{m} (e_i \otimes f_i + f_i \otimes e_i), \qquad \ell(v \otimes w + w \otimes v) = b(v, w).$$

Use b to identify the quadratic class of Q_V with an element of T/A, and choose a representative $X \in T$, so that $xX = \omega$. The tensor-product polar pairing is specified on pure tensors by

$$\langle u_1 \otimes u_2, v_1 \otimes v_2 \rangle = b(u_1, v_1) b(u_2, v_2).$$

For an operator a on T, write $a^\dagger$ for its adjoint under this pairing. Define

$$f_{Q_V} : \Gamma \to \mathbb{K}, \qquad f_{Q_V}(s) = \langle X, s \rangle.$$

Then $f_{Q_V}(v \otimes v) = Q_V(v)$ and $f_{Q_V}|_A = \ell$. The following statements write the quadratic form as Q and this functional as f_Q. The scalar $\alpha(a)$ is determined by $a\omega = \alpha(a)\omega$ on the invariant line. The Dickson invariant is the characteristic-2 orthogonal character $\mathrm{O}(Q) \to \mathbb{F}_2$ whose kernel is $\Omega(Q)$; it is computed by the parity of $\mathrm{rank}(I - g)$ [Tay92, Theorem 11.51].

Lemma C.23 (The Dickson invariant as a matrix coefficient)**.** *For $g \in \mathrm{O}(Q)$, $gX - X \in A$ and $\ell(gX - X)$ is its Dickson invariant in $\mathbb{F}_2 \subseteq \mathbb{K}$.*

Proof. Under the standard identification $\Phi : T \to \mathrm{End}_{\mathbb{K}}(V)$, $\Phi(v \otimes w)(z) = vb(w, z)$, put $L = \Phi(X)$ and let σ be the adjoint involution. Then $\Phi(A) = \mathrm{Alt}(\mathrm{End}_{\mathbb{K}}(V), \sigma)$, and the quadratic-pair functional $f(s) = \mathrm{tr}(Ls)$ satisfies $f(\Phi(a)) = \ell(a)$ for $a \in A$ [KMRT98, §5.A]. Preservation of the quadratic class gives $gX - X \in A$. The published Dickson formula $\Delta(u) = f(u^{-1}Lu - L)$ [KMRT98, Example (12.31), pp. 166–167], applied to $u = g^{-1}$, therefore gives

$$\ell(gX - X) = f(gLg^{-1} - L) = \Delta(g^{-1}) = \Delta(g).$$

□

Proposition C.24 (The orthogonal enveloping algebra and its injective modules)**.** *Assume the stable degree-two rank bounds of Subsection 6.1. Let $\mathcal{B} \leq \mathrm{End}_C(T)$ be the subalgebra of the symplectic enveloping algebra on the tensor square consisting of operators a such that*

$$aX - \alpha(a)X \in \ker \ell, \qquad a^\dagger X - \alpha(a)X \in \ker \ell.$$

Here $T = V \otimes_{\mathbb{K}} V$, $X \in T$ is the representative of Q fixed above, $\alpha(a) \in \mathbb{K}$ is the scalar by which a acts on $\mathbb{K}\omega$, and $a^\dagger \in \mathrm{End}_{\mathbb{K}}(T)$ is its adjoint for the tensor-product polar pairing. The defining-field enveloping algebra of $\Omega(V, Q)$ is exactly $\mathcal{B}$. Its injectives are

$$I_H = T/\mathbb{K}\omega, \qquad I_W = T/(A + \mathbb{K}X).$$

$$I_0 = \begin{cases} T/(\ker f_Q + \mathbb{K}X), & m \text{ odd}, \\ (T \oplus T)/\{(s + cX, f_Q(s)\omega) : s \in \Gamma, c \in \mathbb{K}\}, & m \text{ even}. \end{cases}$$

Its prime-field enveloping algebra is $\mathbb{F}_2 I + \ker \alpha$.

The containment uses the Dickson matrix-coefficient identity of Lemma C.23; finite irreducibility uses [Ste67, Theorem 43], and injective duality is standard [Ben91, Section 1.5]. The block calculation below establishes the full radical, the two parity-dependent injective models, and the prime-field enveloping algebra.

Proof of Proposition C.24. Put $G = \Omega(V, Q)$. Containment follows from Lemma C.23 and self-duality. Suppose first that m is odd, and put $H = \ker \ell$, of dimension $h - 1$. The nondegenerate C-summand $\mathbb{K}X + \mathbb{K}\omega \leq T$ has type $C/(x + y)$. Choose coordinates $H_t, W, \mathbb{K}X, H_b, \mathbb{K}\omega$ on T, where H_t, H_b are the top and bottom copies of H and $W \cong \Gamma/A \cong V^{[1]}$ has $\mathbb{K}$-dimension n. In these coordinates, $\mathcal{B}$ has the same arbitrary H-matrix on H_t, H_b, an arbitrary W-matrix, and a common scalar on X, ω. Its remaining blocks are precisely

$$H_t \to H_b, \quad H_t \to W, \quad W \to H_b, \quad H_t \to \mathbb{K}\omega, \quad \mathbb{K}X \to H_b.$$

This follows by commuting with x, y and imposing the two conditions. The radical R has $R^2 = \mathrm{Hom}_{\mathbb{K}}(H_t, H_b)$ and $R^3 = 0$. The module H is absolutely simple: the nonzero weight vectors of $\bigwedge^2_{\mathbb{K}} V$ are the root vectors, one of which generates all root spaces by Weyl transitivity and all diagonal differences by root operations. A zero-weight vector killed by all roots has equal diagonal coefficients and lies in $\mathbb{K}\omega$, whereas $\ell(\omega) = 1$. Lemma C.22 gives the same assertion for the finite group. The other distinct simple module types are $W \cong V^{[1]}$ and $\mathbb{K}$, so the enveloping algebra has the full semisimple diagonal. Conjugate a separable semisimple complement by $1 + R$ to make it the displayed diagonal. A missing block $W \to H_b$ or $H_t \to W$ would make W a submodule or quotient of T, contrary to Corollary C.21. That invariant calculation is independent of both algorithms. A missing block $\mathbb{K}X \to H_b$ or its dual would give a second invariant of T or $T^* = \mathrm{Hom}_{\mathbb{K}}(T, \mathbb{K})$, whereas $T \cong \mathrm{End}_{\mathbb{K}}(V)$ has just one by absolute irreducibility of V. Thus all four block bimodules occur in full. Their products through W span R^2, proving equality of the enveloping algebra and $\mathcal{B}$.

For even m, put $H = (\ker \ell)/\mathbb{K}\omega$, of dimension $h - 2$. Choose $a \in A$ with $\ell(a) = 1$ and $Y \in T$ with $xY = a$. The nondegenerate C-summand $\langle X, Y, \omega, a\rangle$ is the four-dimensional module of Lemma C.14. Use coordinates $H_t, W, \mathbb{K}X, \mathbb{K}Y, H_b, \mathbb{K}\omega, \mathbb{K}a$ on T, with H_t, H_b the two copies of H and $W \cong \Gamma/A$. In the block lists, a vector name denotes its $\mathbb{K}$-span. The diagonal algebra is $\mathrm{Mat}(h-2, \mathbb{K}) \times \mathrm{Mat}(n, \mathbb{K}) \times \mathbb{K}$, with common scalar on the four scalar positions. The radical blocks are

$$\begin{gathered} n_H : H_t \to H_b, \quad \beta : (Y \to X,\ a \to \omega), \quad \pi_1 : X \to \omega, \quad \pi_2 : Y \to \omega, \quad t : Y \to a, \\ a_1 : (H_t \to X,\ H_b \to \omega), \quad a_2 : H_t \to \omega, \quad a_3 : H_t \to a, \\ b_1 : (Y \to H_t,\ a \to H_b), \quad b_2 : X \to H_b, \quad b_3 : Y \to H_b, \\ H_t \to W, \quad W \to H_b, \quad W \to \omega, \quad Y \to W. \end{gathered}$$

The two maps in each parenthesized block have the same coefficient; all rectangular coefficients are arbitrary. To verify the list, commuting with x, y on the four-dimensional summand gives the same top/bottom block $\left(\begin{smallmatrix} \alpha & \beta \\ 0 & \alpha \end{smallmatrix}\right)$ and an arbitrary top-to-bottom block. The additional conditions remove $X \to a$, $X \to W$, and $W \to a$, exactly $2n + 1$ dimensions. For the radical $R \le \mathcal{B}$, the quotient R/R^2 has the $H \leftrightarrows W$ blocks and two copies of each $H \leftrightarrows \mathbb{K}$ block, represented by a_1, a_3, b_1, b_2. They generate R, since

$$a_1 b_1 = \beta, \quad b_2 a_1 = n_H, \quad a_3 b_1 = t, \quad a_1 b_2 = \pi_1, \quad a_1 n_H = a_2, \quad b_2 \beta = b_3, \quad \pi_1 \beta = \pi_2,$$

and the remaining blocks are $a_1(W \to H_b)$ and $(H_t \to W)b_1$. Matrix units realize every allowed coefficient. Levels $Y = 4, H_t = 3, W = X = a = 2, H_b = 1, \omega = 0$ show $R^5 = 0$.

The root argument proves simplicity of H in this parity too. Perfectness and $T^G = \mathbb{K}\omega$ imply $(T/\mathbb{K}\omega)^G = 0$, using $H^1(G, \mathbb{K}) = 0$. The invariant module $(A + \mathbb{K}X)/\mathbb{K}\omega$ is an extension of $\mathbb{K}^2$ by H, annihilated by R^2. Fewer than two independent $\mathbb{K} \to H$ blocks would leave a nonzero combination of the two top coordinates killed by the radical, hence a fixed vector. Thus both occur, as do both reverse blocks by duality. The extension $(\ker f_Q)/\mathbb{K}\omega$ of W by H cannot split: otherwise W maps to $T/\mathbb{K}\omega$, but $\mathrm{Hom}_G(W, T) = 0$ by Corollary C.21 and $\mathrm{Ext}^1_G(W, \mathbb{K}) = H^1(G, W^*) = 0$ by Lemma C.18. This gives both blocks joining H and W. The displayed products and nilpotence now give the full radical.

For odd m the primitive-right-ideal dual dimensions are $2(h-1)+n+1, h-1+n, h$. The stated quotients have these dimensions and simple socles $H, W, \mathbb{K}$, read directly from the five-coordinate list. For even m the dimensions are $2(h-2) + n + 3, h - 2 + n + 1, 3(h-2) + n + 5$. The radical annihilator in $T/\mathbb{K}\omega$ is H_b; in $T/(A + \mathbb{K}X)$ its surviving coordinates H_t, W, Y have annihilator W. In the last quotient use first-copy coordinates H_t, Y and the whole second copy; first-copy a equals second-copy ω. The block maps a_3, b_1 eliminate the first-copy coordinates in a socle vector, and

the remaining block maps eliminate all second-copy coordinates except ω. Its socle is therefore $\mathbb{K}$. Every simple socle is essential; mapping into its *injective envelope* and comparing these dimensions proves the claimed identifications.

Finally the square has only the nontrivial algebraic factors $L(\omega_2)$ and $L(2\omega_1)$. Its Frobenius conjugates have factors $L(2^i\omega_2)$ and $L(2^{i+1}\omega_1)$ with indices modulo $\log_2|\mathbb{K}|$, pairwise distinct by [Ste67, Theorem 43]. In nonsplit type the graph automorphism fixes both displayed fundamental weights; the same period applies. Lemma C.6 now proves the prime-field equality including the radical. □

Lemma C.25 (Equivariant maps for the orthogonal quotient modules)**.** *For the orthogonal algebra of Proposition C.24, use $T = V \otimes_{\mathbb{K}} V$, $x = I_T + \tau$, $A = \operatorname{im} x$, and $\Gamma = \ker x$. Each of its injectives I_H, I_W, I_0 has an equivariant map into one or two copies of $I_H = T/\mathbb{K}\omega$ with a Frobenius-twisted natural or trivial kernel. For representatives $Y, Z \in T$, the maps are given by*

$$I_H \xrightarrow{1} I_H, \qquad I_W \longrightarrow I_H,\ [Y] \longmapsto xY + \mathbb{K}\omega,$$

$$I_0 \longrightarrow I_H,\ [Y] \longmapsto xY + \mathbb{K}\omega \quad (m \text{ odd}),$$

and

$$I_0 \longrightarrow I_H^2,\ [(Y,Z)] \longmapsto (xY + \mathbb{K}\omega, Z + \mathbb{K}\omega) \quad (m \text{ even}).$$

The second kernel is $\Gamma/A \cong V^{[1]}$; the last two kernels are a scalar line. These maps kill the full scalar socle whenever one is present.

The maps are explicit consequences of the quotient models in Proposition C.24. We include the denominator and kernel checks because these specify the actual degree-one remainder used by the affine reduction.

Proof of Lemma C.25. Each formula sends the denominator defining its source in Proposition C.24 to zero, so it is independent of the chosen representatives. The operator x and the quotient maps commute with the group action, proving equivariance. If $Y \in T$ satisfies $xY = c\omega$ for $c \in \mathbb{K}$, then $Y - cX \in \Gamma$, since the fixed representative $X \in T$ has $xX = \omega$. For I_W the remaining denominator is exactly A, giving Γ/A. For odd I_0 it is $\ker f_Q$, giving a trivial one-dimensional quotient. For even I_0, if also $Z = b\omega$ with $b \in \mathbb{K}$, subtract $(cX, 0)$ and then the relation $(Y - cX, f_Q(Y - cX)\omega)$ in $T \oplus T$; only a scalar multiple of $(0, \omega)$ remains. This class is nonzero because a defining relation with zero first coordinate has $s = cX \in \Gamma$, forcing $c = s = 0$. □

C.4.6 The combined characteristic-two action

The local algebras are combined with their shared simple and scalar coordinates intact. This gives the injective cogenerator for an arbitrary joint tensor sum.

Proposition C.26 (The enveloping algebra and its injective modules in characteristic 2)**.** *For a product of stable characteristic-2 classical groups, the enveloping algebra on any joint sum of tensor products of degree at most two is obtained from Lemmas C.12, C.13, C.15, Proposition C.24, and the adjoint cases of Lemma C.11, by identifying isomorphic occurrences, using the induced quotient action on every natural simple module N/P already occurring in a square with invariant $P \le N$, adding only composition-disjoint nontrivial semisimple factors, and identifying the single trivial character. Its injective cogenerator has polynomial dimension in d, with the explicit quotient realizations in the cited results.*

The simple tensor-product constituents use [Ste67, Theorems 41 and 43], and the injective construction uses projective–injective duality [Ben91, Section 1.5]. The assembly argument below identifies repeated natural types inside squares and preserves the single correlated prime-field scalar denominator.

Proof of Proposition C.26. Over a splitting field, a tensor product with two factors is a same-factor square, a same-factor adjoint, a simple tensor with distinct Frobenius-twist exponents, or a simple external tensor product for two independent factors. The distinct-twist and external tensor product cases follow from [Ste67, Theorems 41 and 43]; scalar restriction preserves semisimplicity since the field extension is separable. In unitary type, write $|\mathbb{K}_0| = 2^f$ and $|\mathbb{K}| = 2^{2f}$. The relative twist exponents are taken modulo $2f$: difference f gives an adjoint and difference zero gives an ordinary square. At $f = 1$ the Frobenius-twisted natural constituent of the square is the dual natural module, so those occurrences are the same algebra module. All common twists are identified as modules over the prime field, not as independent factors. In particular, adjoining a natural occurrence whose simple type already occurs in a square adds no new algebra factor: its matrices are the quotient matrices already determined by the square action. Indeed the annihilator of the square annihilates every such simple module, so projection from the joint enveloping algebra onto the enveloping algebra on the square is injective as well as surjective. For symplectic and orthogonal groups polar duality makes the corresponding dual identifications. Independent factors are formed only after any diagonal identifications of the acting simple groups have been made.

Disjoint composition types separate by the two-sided Chinese remainder argument. For blocks sharing only the trivial type, apply the annihilator-product construction of Lemma C.6: the group is perfect, their annihilator sums are the augmentation ideal, and distinct Frobenius conjugates share no nontrivial simple type. That construction separates their full defining-field augmentation ideals J_i, even for blocks of the same group. Thus their joint algebra is exactly $\mathcal{A} = \mathbb{F}_2 I + \bigoplus_i J_i$, with zero cross-products and one common scalar character. For a primitive idempotent $e \in \mathcal{A}$ associated with a nontrivial simple module, the right ideal $e\mathcal{A}$ is unchanged by prime-field restriction. Its $\mathbb{F}_2$-dual is identified with the same defining-field injective by the field-trace pairing. Let $e_0 \in \mathcal{A}$ be the primitive idempotent associated with the common trivial constituent. Dualizing $e_0\mathcal{A}$ gives

$$I_{\mathrm{triv}} = \left(\bigoplus_i I_{0,i} \right) / \{ (c_i s_i)_i : \sum_i \mathrm{Tr}_{\mathbb{K}_i/\mathbb{F}_2}(c_i) = 0 \},$$

where $\mathbb{K}_i$ is the defining field of the summand $I_{0,i}$, $s_i \in I_{0,i}$ spans its scalar socle over $\mathbb{K}_i$, and $c_i \in \mathbb{K}_i$. Two tuples in the scalar socles have the same quotient class exactly when their difference has total trace zero. Thus the quotient has one common $\mathbb{F}_2$-scalar coordinate. If there is a trivial constituent but no block $I_{0,i}$ in this sum, take $I_{\mathrm{triv}} = \mathbb{F}_2$; if there is no trivial constituent, no trivial-type injective is needed. In the former case the trivial type is composition-disjoint from every remaining block, so the two-sided Chinese remainder argument splits off a direct algebra factor $\mathbb{F}_2$. There are no radical maps joining that factor to the other blocks. For each remaining semisimple component, take a basis of the space of equivariant maps into the tensor summands in its representation and combine those maps into a single direct-sum map, as in Lemma 6.5. This handles smaller endomorphism fields and field-conjugate identifications without constructing or enumerating a compositum. The listed modules have dimension $O(d^2)$ up to a fixed multiplicity for each distinct module type, and the Hom construction of Lemma 6.5 is polynomial. □

C.4.7 Cohomology of cubic pairing quotients

The remaining characteristic-2 proofs need low-degree calculations for special linear groups. We state precisely which calculations are used, and which ranks are required. Their external input is a comparison between stable finite-group cohomology and cohomology of functors. In this setting a *functor* assigns a vector space to every finite-dimensional $\mathbb{K}$-space and a linear map to every linear map, preserving composition and identities. A natural transformation is a family of linear maps commuting with all those maps. The ordinary functor category $\mathcal{F}(\mathbb{K})$ allows these functors without a specified strict-polynomial presentation. The Frobenius functor $U \mapsto U^{[a]}$ is additive: it takes a direct sum to the direct sum of its values. Hom and Ext in this category refer to natural transformations and their extensions; the comparison theorem, rather than an identification by notation, passes from them to the displayed finite groups.

Lemma C.27 (Finite general-linear coefficient calculations)**.** *Let $\mathbb{K}$ be a finite field of characteristic 2, let $V = \mathbb{K}^n$, and let $G = \mathrm{GL}(n, \mathbb{K})$. The tensor products in this lemma are over $\mathbb{K}$, with specified natural modules, duals, or Frobenius twists as factors; their degree is the number of factors. In stable rank $n \geq 12$: (i) every such tensor product of odd length at most five has no invariants; (ii) every tensor product of three natural modules, duals, or twists has zero H^1; (iii) every natural factor has zero H^1 and H^2. The same assertions for these modules hold for every intermediate group $\mathrm{SL}_n(\mathbb{K}) \leq G' \leq G$. For $G = \mathrm{GL}(n, \mathbb{K})$, a nontrivial determinant character times a tensor product of length r has zero H^0 for $n > r$ and zero H^1 for $n > r + 2$. Assertions for quotient modules will be deduced separately from the cohomology exact sequence.*

Proof. Use the ordinary functor category $\mathcal{F}(\mathbb{K})$ just described and used before Lemma C.2. There are two parts to the proof: compute the stable functor groups, then check that the finite-group comparison and determinant restriction are valid at the stated ranks. The Hom and Ext groups in the next paragraph are taken there; the later $H^i(G, -)$ are finite-group cohomology groups with the displayed $\mathbb{K}[G]$-modules as coefficients. The natural Frobenius functors are additive in this category, regardless of their strict-polynomial degree. The exponential-functor identities of [FFSS99, Theorem 1.7 and Corollary 1.8] imply that tensor products with unequal numbers of additive factors have zero Ext in either direction. The constant and reduced parts are separate exact factors, so Ext from the constant functor to a reduced functor also vanishes. Consequently a pure cubic has zero Ext from the constant functor, and a mixed cubic gives Ext between one and two additive factors, which is zero. Natural coefficients satisfy the same vanishing in degrees one and two. We now pass from these functor calculations to the specific finite groups, keeping track of the required ranks. The tensor products of additive functors used above are ordinary finite functors, so apply the comparison of [FFSS99, Appendix, Theorem A.1] with the split-injection systems of Lemma C.2 and the restriction/evaluation transition maps verified in the proof of Lemma C.3. For tensor length $r \geq 1$, those systems have degree at most r, starting at zero, including dual and Frobenius-twisted factors. The general-linear stability range used there is $n \geq 2i + 2r + 2$ in cohomological degree $i \geq 0$. Thus $(i, r) = (1, 3), (2, 1), (0, 5)$ give rank ten for cubic H^1, rank eight for natural H^2, and rank twelve for degree-five H^0, respectively, without a field-size restriction.

To check the determinant assertion, let W be the indicated $\mathbb{K}[G]$-module: a tensor product of length r tensored with a nontrivial determinant character. Normalize a cocycle $c : G \to W$ on the full odd-order diagonal torus of $G = \mathrm{GL}(n, \mathbb{K})$. Evaluation on elementary root groups embeds the normalized cocycle space in the direct sum of their function spaces. Each weight is a coefficient weight plus a character supported on the two root coordinates. It therefore has support at most $r + 2$, whereas a nontrivial determinant character is a nonzero constant exponent vector. These cannot cancel when $n > r + 2$. The invariant-vector version omits the two root coordinates. These

weight arguments also apply to a specified invariant quotient, because taking weight spaces for the odd-order torus is exact.

For restriction from $\mathrm{GL}(n,\mathbb{K})$ to a group G' with $\mathrm{SL}_n(\mathbb{K}) \le G' \le \mathrm{GL}(n,\mathbb{K})$, apply Lemma C.2 to the whole coefficient modules. It applies to degree-five invariants at rank twelve, cubic first cohomology at rank ten, and natural second cohomology at rank eight. This gives both the first- and second-cohomology restriction isomorphisms needed here. For an invariant quotient N/P, we use the cohomology exact sequence for its specified numerator and denominator in each application.

The Schur-multiplier argument in the proof of Lemma C.5 applies to $\mathrm{SL}_n(\mathbb{K})$ already for $n \ge 5$, since every special-linear exception in its cited list has $n \le 4$. It gives $H^1(\mathrm{SL}_n(\mathbb{K}),\mathbb{K}) = H^2(\mathrm{SL}_n(\mathbb{K}),\mathbb{K}) = 0$ in the present range $n \ge 12$. A determinant character $\theta : G \to \mathbb{K}^\times$ is trivial on $\mathrm{SL}_n(\mathbb{K})$, and $G/\mathrm{SL}_n(\mathbb{K})$ has odd order. The Lyndon–Hochschild–Serre spectral sequence therefore gives $H^1(G,\mathbb{K}_\theta) = H^2(G,\mathbb{K}_\theta) = 0$, where $\mathbb{K}_\theta$ is the one-dimensional $\mathbb{K}[G]$-module with action $g \cdot x = \theta(g)x$. □

For symplectic groups, the cubic module itself can have nonzero first cohomology. The point is that its pairing submodule accounts for all of it. The proof computes the possible first-cohomology classes according to which two positions pair, and then separates the pairing positions by restricting to an orthogonal complement of one hyperbolic plane. This gives the surjectivity in degree 1 and injectivity in degree 2 needed by the cohomology exact sequence in Subsection 6.2.

Two functor constructions used in that calculation have concrete meanings. $\mathbb{K}[X]$ is the vector space with the elements of a finite set X as a basis. Its augmentation sends every such basis vector to 1; the kernel is spanned by differences $[x] - [0]$ when $0 \in X$. The notation $P \boxtimes R$ denotes the functor in separate variables with value $P(U) \otimes R(W)$. Sum–diagonal adjunction compares a functor at $U_1 \oplus \cdots \oplus U_s$ with a tensor product of functors in the separate variables. It allows the proof to keep track of which variables occur in each reduced, or zero-at-zero, summand.

Lemma C.28 (Symplectic cubic quotients in characteristic 2)**.** *Let $\mathbb{K}$ be a finite field of characteristic 2, let $V = \mathbb{K}^{2m}$ with $m \ge 18$, and let $G = \mathrm{Sp}_{2m}(\mathbb{K})$. For any tensor product of three field-twisted natural modules $T = T_1 \otimes_{\mathbb{K}} T_2 \otimes_{\mathbb{K}} T_3$ on the natural module V, with $J(T) \le T$ the sum of pairing-map images,*

$$(T/J(T))^G = 0, \qquad H^1(G, T/J(T)) = 0.$$

Proof. In the ordinary functor category $\mathcal{F}(\mathbb{K})$, let $I_a : U \mapsto U^{[a]}$ be the additive Frobenius functor and let $A(U) = \mathbb{K}[\bigwedge^2_{\mathbb{K}} U]$ for each finite-dimensional $\mathbb{K}$-space U. Here $\mathbb{K}[X]$ is the free $\mathbb{K}$-space on the finite set X; its augmentation sends every basis element to $1 \in \mathbb{K}$. Write $E_a = \mathrm{Ext}^1_{\mathcal{F}(\mathbb{K})}(A, I_a)$ for $a \in \mathbb{Z}_{\ge 0}$. Each functor Hom or Ext group below is taken in the ordinary functor category on all variables occurring in its two arguments; for one variable this is $\mathcal{F}(\mathbb{K})$. The symplectic comparison [DV10, Corollaries 3.32–3.33] applies over every finite field to ordinary polynomial degree r, in rank $m \ge 2i + r + 6$. The comparison reduces finite symplectic cohomology to the following calculation of the relevant Ext group. For twist indices $(a_1, a_2, a_3) = (a, b, c) \in \mathbb{Z}^3_{\ge 0}$, compared modulo $[\mathbb{K} : \mathbb{F}_2]$, we compute

$$\mathrm{Ext}^1(A, I_a \otimes I_b \otimes I_c) = \bigoplus_{i:\text{the other two twists agree}} E_{a_i}. \tag{C.2}$$

Here is the full degree-1 argument. Sum–diagonal adjunction in three variables U_1, U_2, U_3, each a finite-dimensional $\mathbb{K}$-space, replaces the source A by

$$\bigotimes_{i=1}^{3} \mathbb{K}[\bigwedge^2_{\mathbb{K}} U_i] \otimes_{\mathbb{K}} \bigotimes_{1 \le i < j \le 3} \mathbb{K}[U_i \otimes_{\mathbb{K}} U_j].$$

Split each factor into its constant and augmentation summands. A summand missing a variable has zero Ext to the reduced target in that variable. A variable in two reduced factors also gives zero Ext: duplicate that variable, leave all other shared parameters unchanged, apply exact sum–diagonal adjunction, and use additivity of the target. Thus the only survivors are a singleton augmentation in one variable and a pair augmentation in the other two, or the three singleton augmentations.

Put $\bar{A} = \ker(A \to \mathbb{K})$, using the augmentation to the constant functor $\mathbb{K}$. Then $\mathrm{Hom}(\bar{A}, I_a) = 0$, since $\bar{A}(\mathbb{K}) = 0$ and all $\mathbb{K}$-linear functionals $U \to \mathbb{K}$ separate vectors in a finite-dimensional $\mathbb{K}$-space U. For pointwise finite-dimensional parameter functors R, Q in the remaining variables, this gives

$$\mathrm{Ext}^1(P \boxtimes R, I_a \boxtimes Q) = \mathrm{Hom}(R, \mathrm{Ext}^1(P, I_a) \otimes Q), \qquad P = \bar{A}.$$

To prove this identity, restrict an extension to each parameter object. A morphism between its source and target extensions exists precisely when the associated Ext classes agree under pushout and pullback. It is unique because its possible difference belongs to $\mathrm{Hom}(P, I_a)$ tensored with finite-dimensional multiplicities, which is zero. These unique morphisms satisfy composition, so the parameterwise classes assemble uniquely to the required functor extension.

For finite-dimensional $\mathbb{K}$-spaces U, W, put $B(U, W) = \mathbb{K}[U \otimes_{\mathbb{K}} W]$ and let $\bar{B} = \ker(B \to \mathbb{K})$ be its augmentation subfunctor. For a passive finite-dimensional $\mathbb{K}$-space N, its Hom space is

$$\mathrm{Hom}(\bar{B}, N \otimes I_b \boxtimes I_c) = \begin{cases} N & b = c \pmod{[\mathbb{K} : \mathbb{F}_2]}, \\ 0 & b \neq c. \end{cases}$$

Indeed, for a natural transformation $f : \bar{B} \to N \otimes I_b \boxtimes I_c$, identify $\mathbb{K} \otimes_{\mathbb{K}} \mathbb{K}$ with $\mathbb{K}$ and the target at $(\mathbb{K}, \mathbb{K})$ with N using the standard bases. The vectors $[t] - [0]$, for $t \in \mathbb{K}$, span $\bar{B}(\mathbb{K}, \mathbb{K})$. Scalar multiplication by $s \in \mathbb{K}$ in either variable gives

$$(s^{2^b} - s^{2^c}) f_{\mathbb{K},\mathbb{K}}([t] - [0]) = 0.$$

Matching twists force $f_{\mathbb{K},\mathbb{K}}([t]-[0]) = t^{2^b} f_{\mathbb{K},\mathbb{K}}([1]-[0])$. Coordinate functionals determine every $f_{U,W}$. Conversely, for matching twists each $n \in N$ gives a natural transformation by $[z] - [0] \mapsto n \otimes z^{[b]}$, using the canonical identification $(U \otimes_{\mathbb{K}} W)^{[b]} \cong U^{[b]} \otimes_{\mathbb{K}} W^{[b]}$. The triple-singleton term vanishes by the same one-dimensional test in one of its remaining variables. This proves (C.2).

The pairing maps embed a direct sum of natural twists, as checked in Corollary C.21. Their first cohomology therefore has the same dimension as $H^1(G, T)$ by the comparison and (C.2). To prove injectivity also in degree 2, write $V = W \perp U$ with U a hyperbolic plane and restrict to $\mathrm{Sp}(W)$. For free tensor position i, project that position to W and the other two to U, and apply any linear functional taking the invariant tensor of U to one. The resulting $\mathrm{Sp}(W)$-equivariant map $D_i : T \to W^{[a_i]}$ satisfies $D_i \kappa_j = \delta_{ij} \mathrm{pr}_W$, where $\kappa_j : V^{[a_j]} \to T$ is the eligible pairing insertion in position j. For $i = j$ the right side is the orthogonal-sum projection $\mathrm{pr}_W : V^{[a_i]} \to W^{[a_i]}$; for $i \neq j$ it is the zero map $V^{[a_j]} \to W^{[a_i]}$. The chosen linear functional on the hyperbolic plane U gives an $\mathrm{Sp}(W)$-equivariant map because that group fixes U. The induced composite in degrees one and two is diagonal with the natural-coefficient restriction/projection stabilization maps. The smaller rank $m - 1 \geq 11$ is in the comparison range, so these maps are isomorphisms. Thus $H^i(G, J(T)) \to H^i(G, T)$ is injective for $i = 1, 2$, and is surjective for $i = 1$ by the dimension equality. The cohomology exact sequence proves $H^1(G, T/J(T)) = 0$. Its earlier terms, using $T^G = 0$ and this injectivity in degree 1, prove $(T/J(T))^G = 0$. □

For the unitary and orthogonal calculations, we work in a subgroup stabilizing a maximal totally isotropic subspace. Such a subgroup is a *parabolic*. Its diagonal blocks form a *Levi subgroup*; its

upper unitriangular blocks form its normal unipotent radical. The displayed matrices below specify these groups, so no choice of an abstract splitting is needed. Their index in the full group is odd. Restriction on characteristic-2 cohomology is injective in this case: restriction followed by transfer multiplies by that odd index, an invertible scalar.

The strategy is to remove a cocycle on the Levi subgroup first. A weight on the argument blocks is then chosen so that every nonidentity radical action raises weight. The lowest nonzero component of the remaining cocycle would have to be an additive Levi-equivariant map from the radical to a specified coefficient layer. Low-degree Hom and Ext vanishing exclude most such maps. The exceptional odd-dimensional case has one possible form; the cocycle commutation identity forces it to be a coboundary.

Lemma C.29 (Unitary cubic quotients in characteristic 2)**.** *Let $f \geq 1$, let $\mathbb{K} = \mathbb{F}_{2^{2f}}$, and let $V = \mathbb{K}^n$ with $n \geq 36$. For $G = \mathrm{SU}_n(2^f)$ on V and every tensor product over $\mathbb{K}$ of three natural modules, duals, or twists $T = T_1 \otimes_{\mathbb{K}} T_2 \otimes_{\mathbb{K}} T_3$, with $J(T) \leq T$ the sum of pairing-map images,*

$$(T/J(T))^G = 0, \qquad H^1(G, T/J(T)) = 0.$$

Proof. Put $\sigma : \mathbb{K} \to \mathbb{K}$, $x \mapsto x^{2^f}$, and $m = \lfloor n/2 \rfloor$. The parabolic $P \leq G$ stabilizing a maximal isotropic subspace has odd index, so restriction on cohomology is injective. We give its radical calculation explicitly; all coefficient scalars may be extended for this calculation.

Even dimension. For $n = 2m$ put $A = \mathbb{K}^m$ and write $V = A \oplus A^{\sigma,*}$, where $A^{\sigma,*} = (A^{[f]})^*$ is the $\mathbb{K}$-linear dual of the σ-twisted natural module. For a matrix $Z \in \mathrm{Mat}(m, \mathbb{K})$, write $Z^\dagger = (Z^\sigma)^{\mathrm{t}}$. The Levi subgroup of P is $L' = \{g \in \mathrm{GL}(m, \mathbb{K}) : \det(g) \in \mathbb{F}_{2^f}^*\}$, and its abelian radical $U \trianglelefteq P$ consists of

$$u(Z) = \begin{pmatrix} I & Z \\ 0 & I \end{pmatrix}, \qquad Z + Z^\dagger = 0, \qquad Z \in \mathrm{Mat}(m, \mathbb{K}).$$

Its additive module, after prime-field scalar extension, is a direct sum of whole products $A^{[a]} \otimes A^{[a+f]}$. The Levi restriction of T is a sum of general-linear tensor products with three factors; that of $J(T)$ is a sum of natural or dual modules with their specified Frobenius twists. Lemma C.27 and the exact sequence give $H^1(L', T/J(T)) = 0$. Normalize a cocycle $c : P \to T/J(T)$ on $L' \leq P$. Give the lower summand weight zero and A weight one. The radical strictly raises weight and $J(T)$ is homogeneous. Write $T_d \leq T$ for the weight-d subspace, $J_d = J(T) \cap T_d$, and $(T/J(T))_d = T_d/J_d$ for this grading; all these spaces are L'-modules. A first nonzero weight of $c|_U : U \to T/J(T)$ is an additive L'-equivariant map from one of the whole quadratic products $S = A^{[a]} \otimes A^{[a+f]}$ to $(T/J(T))_d$. But

$$\mathrm{Hom}_{L'}(S, T_d) = 0, \qquad \mathrm{Ext}^1_{L'}(S, J_d) = 0$$

by odd-length-five invariants and cubic H^1, respectively. The Hom exact sequence forbids such a map. Thus the radical cocycle is zero.

Odd dimension. For $n = 2m + 1$, put $A = \mathbb{K}^m$, choose a nonzero vector t on the middle anisotropic line, order $V = A \oplus \mathbb{K}t \oplus A^{\sigma,*}$, and put $\chi(g) = (\det g)^{2^f - 1}$. Let $L \leq P$ be the Levi subgroup isomorphic to $\mathrm{GL}(m, \mathbb{K})$, embedded as $\mathrm{diag}(g, \chi(g), g^{-\dagger})$. The radical $U \trianglelefteq P$ is

$$u(X, Z) = \begin{pmatrix} I & X & Z \\ 0 & 1 & X^\dagger \\ 0 & 0 & I \end{pmatrix}, \qquad Z + Z^\dagger = XX^\dagger,$$

with product $u(X, Z)u(Y, W) = u(X + Y, Z + W + XY^\dagger)$. Here $X \in A$ and $Z \in \mathrm{Mat}(m, \mathbb{K})$. Its center $C \leq U$ is the Hermitian $X = 0$ subgroup, and $U/C \cong A \otimes_{\mathbb{K}} \mathbb{K}_{\chi^{-1}}$ as an L-module; the

notation $\mathbb{K}_{\chi^{-1}}$ denotes its one-dimensional character module. Give the summands $A, \mathbb{K}t, A^{\sigma,*}$ of V weights two, one, zero. For this grading, set $T_d \le T$, $J_d = J(T) \cap T_d$, and $(T/J(T))_d = T_d/J_d$. For twist exponent $r_i \in \mathbb{Z}_{\ge 0}$ their Levi constituents are $A^{[r_i]}, \mathbb{K}_{\chi^{2^{r_i}}}, (A^{[r_i+f]})^*$.

Levi H^1 of T vanishes on each summand. A summand with three factors that are twists of A or its dual uses cubic vanishing. Two such factors and one $\mathbb{K}t$ factor have a nontrivial determinant twist. With one factor that is a twist of A or its dual, the coefficient is either a nontrivial determinant twist or a natural or dual module with its specified Frobenius twist; three $\mathbb{K}t$ factors give a determinant character. The same lemma gives $H^2(L, J(T)) = 0$, including the one-dimensional characters. Thus normalize the cocycle $c : P \to T/J(T)$ on $L = \mathrm{GL}(m, \mathbb{K})$. A first central-radical weight would be a map from a Hermitian quadratic product S to $(T/J(T))_d$. With three factors that are twists of A or its dual, the coefficient module $\mathrm{Hom}_{\mathbb{K}}(S, T_d)$ is a tensor product of five natural or dual factors with their specified Frobenius twists; with two such factors it has a nontrivial determinant character; with one it is either such a character or a tensor product of three natural or dual factors with their specified Frobenius twists; with none it is a determinant twist of a tensor product of two dual natural modules with their specified Frobenius twists. Each has zero L-invariants. Also $\mathrm{Ext}^1_L(S, J_d) = 0$: a constituent that is a twist of A or its dual gives cubic H^1, while a $\mathbb{K}t$ constituent gives a nontrivial determinant twist of a tensor product of two natural or dual factors with their specified Frobenius twists. Therefore $c(C) = 0$.

A first nonzero weight of $c(U)$ now factors through U/C. For its scalar component $W_a = A^{[a]} \otimes_{\mathbb{K}} \mathbb{K}_{\chi^{-2^a}}$ one has $\mathrm{Hom}_L(W_a, J_d) = \mathrm{Ext}^1_L(W_a, J_d) = 0$. A target that is a twist of A or its dual has a nontrivial determinant twist, and a $\mathbb{K}t$ target reduces to natural or dual coefficients with or without such a twist. Consequently every map to $(T/J(T))_d$ lifts uniquely to T_d. The possible maps into T_d have exactly one factor that is a twist of A or its dual, which must be the matching $A^{[a]}$; their other two factors are twists of $\mathbb{K}t$. Such maps occur only if

$$\chi^{2^{r_1}+2^{r_2}+2^{r_3}} = 1,$$

and only in weight four. In the following two displays the ambient space is $\overline{\mathbb{K}} \otimes_{\mathbb{K}} T$ and tensor products are over $\overline{\mathbb{K}}$. Each $X^{[r_i]} \in A^{[r_i]}$ is viewed there via scalar extension; the symbol t in position i denotes the corresponding basis vector of $(\mathbb{K}t)^{[r_i]}$. Their value has the form

$$\gamma_1 X^{[r_1]} \otimes t \otimes t + \gamma_2 t \otimes X^{[r_2]} \otimes t + \gamma_3 t \otimes t \otimes X^{[r_3]}.$$

Here $\gamma_i \in \overline{\mathbb{K}}$ and $X, Y \in A$. For $X \in A$, write $u(X)$ for any lift of its class in U/C, that is, $u(X, Z)$ with any admissible Z. For $z \in C$ and $u \in U$, centrality and $c(C) = 0$ give $c(uz) = c(u) = zc(u)$. Thus $c(U) \subseteq (T/J(T))^C$, its values depend only on the quotient class, and the action on this fixed subspace is independent of the lift. The induced cocycle on the abelian group U/C therefore satisfies $(u(X) - I)c(u(Y)) = (u(Y) - I)c(u(X))$. In weight five, project to the summand with A factors in positions i, j. Its value is

$$(\gamma_i - \gamma_j)(Y^{[r_i]} \otimes X^{[r_j]} + X^{[r_i]} \otimes Y^{[r_j]}) = 0.$$

Two different coordinate vectors make the summands independent, hence all three parameters agree. The equality holds in this weight because $J(T)$ has weights two, three, and four. Subtract the coboundary of the corresponding tensor in the product of the three $\mathbb{K}t$ factors, which is Levi-fixed by the character condition. It removes the only possible first weight and preserves $c(L) = c(C) = 0$. Thus $c(U) = 0$. Odd-index restriction proves the claimed H^1 in both parities.

The same argument on natural coefficients proves $H^1(G, V^{[a]}) = 0$. In even dimension a quadratic radical has no natural target. In odd dimension first kill the center similarly; the quotient

radical $A \otimes_{\mathbb{K}} \mathbb{K}_{\chi^{-1}}$ has no map to a constituent that is a twist of A, its dual, or $\mathbb{K}t$, by the determinant-character test. Since $J(T)$ is a direct sum of natural twists and $T^G = 0$ by Corollary C.21, the exact sequence gives $(T/J(T))^G = 0$. □

The orthogonal proof follows the same route. In split type there is no anisotropic middle block, and the radical is the additive space of alternating matrices. In nonsplit type the middle block is a quadratic plane. The radical then has an alternating-matrix center and a quotient of maps from that plane to the isotropic block. We kill the center first, classify the remaining lowest-weight maps, and use commutation to show that their three position parameters are one common tensor. Subtracting its coboundary completes the calculation. Here $\bigwedge^2 A$ is the exterior square, generated by symbols $x \wedge y$ with bilinearity and $x \wedge x = 0$; in a basis it is the space parametrizing alternating matrices. The notation $\Omega(V)$ denotes the even orthogonal derived group on the stated quadratic space.

Lemma C.30 (Orthogonal cubic quotients in characteristic 2)**.** *Let $\mathbb{K}$ be a finite field of characteristic 2, let $V = \mathbb{K}^{2n}$, with $2n \geq 36$, carry a nondegenerate quadratic form, and let $G = \Omega(V)$ be either even orthogonal derived group, with Witt index at least twelve. For every tensor product $T = T_1 \otimes_{\mathbb{K}} T_2 \otimes_{\mathbb{K}} T_3$ of three natural modules on V, duals, or twists, the quotient $T/J(T)$ has zero invariants and zero first cohomology.*

Proof. Use the polar form to identify dual occurrences. Write $V = A \oplus Z \oplus A^*$, where $A = \mathbb{K}^m$ and m is the Witt index of V. In split type $m = n$ and $Z = 0$; otherwise $m = n - 1$ and Z is an anisotropic quadratic plane. The maximal-singular-space parabolic $P \leq G$ has odd index. Choose a Levi subgroup $L \leq P$. In split type, $L \cong \mathrm{GL}(m, \mathbb{K})$; in nonsplit type, $L \cong \mathrm{GL}(m, \mathbb{K}) \times S$, where $S = \mathrm{SO}(Z)$ is cyclic of order $|\mathbb{K}| + 1$. In split type the radical $U \trianglelefteq P$ is the additive $\bigwedge^2 A$, with its alternating matrix action. In nonsplit type it is

$$u(X, B) = \begin{pmatrix} I & X & B \\ 0 & I & J_Z^{-1} X^{\mathrm{t}} \\ 0 & 0 & I \end{pmatrix}, \quad b^{\mathrm{t}} B b = Q_Z(J_Z^{-1} X^{\mathrm{t}} b) \quad (b \in A^*),$$

where $X \in \mathrm{Hom}_{\mathbb{K}}(Z, A)$, $B \in \mathrm{Mat}(m, \mathbb{K})$, $Q_Z : Z \to \mathbb{K}$ is the restricted quadratic form, and $J_Z \in \mathrm{Mat}(2, \mathbb{K})$ is its polar matrix. In this matrix formula, $b \in A^*$ is represented by its column of dual-basis coordinates. Every X occurs; the remaining freedom in B is an alternating matrix. Multiplication adds $X J_Z^{-1} Y^{\mathrm{t}}$ to the upper-right block. Thus the central subgroup $C \cong \bigwedge^2_{\mathbb{K}} A$ has quotient $U/C \cong \mathrm{Hom}_{\mathbb{K}}(Z, A)$ as an L-module. These displayed elements lie in Ω: if $r = \operatorname{rank} X$, block elimination gives $\operatorname{rank}(u - I) = 2r + \operatorname{rank} \beta$, where the induced form β on $\ker X^{\mathrm{t}}$ is alternating. Its rank is even. The radical orders, respectively $|\mathbb{K}|^{m(m-1)/2}$ and $|\mathbb{K}|^{m(m-1)/2+2m}$, multiplied by the two-part of the Levi order, are $|\mathbb{K}|^{m(m-1)}$ and $|\mathbb{K}|^{m(m+1)}$, exactly the two-parts of the corresponding orthogonal derived-group orders [Tay92, Chapter 11]. This also verifies odd index.

Let $Q = T/J(T)$ be the quotient module, with the action of the Levi subgroup $L \leq P$. Lemma C.27 and the exact sequence of L-modules $0 \to J(T) \to T \to Q \to 0$ give $H^1(L, Q) = 0$ in split type. In nonsplit type diagonalize the odd cyclic S after extending scalars. Each twisted Z factor has two nontrivial characters. A summand with three A or A^* factors uses cubic H^1. With two such factors and one Z factor, its nontrivial S-character kills cohomology. With one A or A^* factor, use natural H^1 when the character of the two Z factors is trivial. With three Z factors, use the absence of homomorphisms from $\mathrm{GL}(m, \mathbb{K})$ to an additive characteristic-2 group. Also $H^2(L, J(T)) = 0$: the A and A^* constituents of $J(T)$ are natural or dual modules with their specified Frobenius twists, and its Z constituents have nontrivial S-character. Normalize a cocycle $c : P \to Q$ so that $c(L) = 0$.

In split type give A weight one and A^* weight zero. Let $T_d \le T$ be the weight-d subspace, $J_d = J(T) \cap T_d$, and $Q_d = T_d/J_d \le Q$. A first nonzero radical value of c is an L-equivariant map $\bigwedge^2 A^{[a]} \to Q_d$. Precompose with $A^{[a]} \otimes A^{[a]}$. Its Hom into T_d is an odd invariant problem of length five, and its Ext into J_d is cubic H^1. Both vanish. Therefore $c(U) = 0$.

In nonsplit type give the summands A, Z, A^* of V weights two, one, zero, and use T_d, $J_d = J(T) \cap T_d$, and $Q_d = T_d/J_d$ for the components of this grading. First kill $c(C)$ by the same precomposition. In the Hom calculation, three A or A^* factors give length-five odd invariants. Two such factors give a nontrivial character from the remaining Z factor; one gives length-three odd invariants when the character of the two Z factors is trivial; three Z factors give a tensor product of two dual natural modules with their specified Frobenius twists. For Ext into $J(T)$, an A or A^* constituent gives cubic H^1, and a Z constituent has nontrivial S-character. All vanish.

Now $c(U)$ is constant on C-cosets and is C-fixed. A first nonzero weight is an L-equivariant map $W_a \to Q_d$, where W_a is a scalar-extension component of $\mathrm{Hom}_{\mathbb{K}}(Z, A)$ with its specified Frobenius twist. The groups $\mathrm{Hom}_L(W_a, J_d)$ and $\mathrm{Ext}^1_L(W_a, J_d)$ vanish for the weight-d part $J_d \le J(T)$: an A or A^* target leaves the nontrivial S-character of W_a; a Z target leaves a natural dual coefficient. Hence it lifts uniquely to T_d. The only possible maps into T_d have one matching A factor and two Z factors, in weight four. Write them

$$F(X) = X_1(b_1) + X_2(b_2) + X_3(b_3), \qquad b_i \in (Z^{[a_1]} \otimes Z^{[a_2]} \otimes Z^{[a_3]})^S.$$

For $X \in \mathrm{Hom}_{\mathbb{K}}(Z, A)$ and $i \in \{1, 2, 3\}$, the map X_i applies $X^{[a_i]} : Z^{[a_i]} \to A^{[a_i]}$ in position i and the identity in the other positions. Thus its domain is $Z^{[a_1]} \otimes_{\mathbb{K}} Z^{[a_2]} \otimes_{\mathbb{K}} Z^{[a_3]}$ and its codomain is the weight-four tensor summand with $A^{[a_i]}$ in position i. If scalars have been extended to diagonalize S, all these domains and codomains are extended to the same field. This description includes every character on the Z factors; an incompatible position has $b_i = 0$. Uniqueness follows by varying every coordinate functional $Z \to \mathbb{K}$ in the corresponding output coordinate of A. The commutator identity, projected to weight five with A factors in positions i, j, gives

$$(Y_i X_j + X_i Y_j)(b_i + b_j) = 0.$$

Choose $X, Y \in \mathrm{Hom}_{\mathbb{K}}(Z, A)$ with images on different coordinate axes of A. One output coordinate isolates $Y_i X_j$. Varying the two maps $Z \to \mathbb{K}$ separates both positions occupied by Z factors and forces $b_i = b_j$. Thus all are one S-fixed tensor $b \in Z^{[a_1]} \otimes Z^{[a_2]} \otimes Z^{[a_3]}$. Subtract its coboundary. It is Levi- and C-fixed, has weight three, and removes exactly F. Every pairing term has weight at most four, because a pairing has weight two and the free occurrence has weight at most two. The preceding classification forbids any later first weight. This kills $c(U)$ and proves $H^1(G, Q) = 0$ by odd-index restriction.

On natural coefficients the identical argument is shorter: neither the exterior-square center nor $A \otimes Z$ has an equivariant map to an A, A^*, or Z constituent, by the vanishing of invariants in tensor products of an odd number (at most five) of natural or dual factors with their specified Frobenius twists and the nontrivial characters of the Z constituents. Thus $H^1(G, V^{[a]}) = 0$. The exact sequence, the direct sum of the parameter spaces of the pairing maps, and Corollary C.21 give $Q^G = 0$. Inflation–restriction also proves these conclusions for the full orthogonal group, since its normal Ω has zero coefficient fixed space. □

C.5 Proofs of the main results

C.5.1 Constructive reduction of degree-two subquotients

Theorem 6.1 (restated). *Let $D = \prod_{i=1}^{t} \mathrm{Cl}(n_i, \mathbb{K}_i)$, where the $\mathbb{K}_i$ are finite fields of characteristic p,* Cl *denotes a derived classical group type, and each factor is in the stable range above. Let $d = \sum_{i=1}^{t} [\mathbb{K}_i : \mathbb{F}_p] n_i$. Let M be a degree-2 tensor module over their natural modules, and let $P \leq N \leq M$ be a chain of invariant subspaces. Set $Q = N/P$. There is an algorithm that, in time $p^{O(d)}\mathrm{poly}(\mathrm{InSize})$, constructs the following data, where* InSize *is the total explicit input length.*

(i) *Two D-modules I, E of dimension polynomial in* InSize*, and D-equivariant linear maps*

$$Q \xhookrightarrow{\iota} I \xrightarrow{\pi} E,$$

where ι is injective and $H^1(D, E) = 0$. Both modules have explicit matrix and tensor-square quotient presentations: each quotient A/B is specified by bases for $B \leq A$ in the ambient matrix or tensor space, together with its coordinate maps. All field maps and linear relations among scalar coordinates are included in the output presentations.

(ii) *If p is odd, $E = I$ and $\pi = \mathrm{id}_I$, where I has the matrix quotient presentation of Definition 14.*

(iii) *If $p = 2$, the target E has the coordinate models of Definition 15, with the specified multiplicities, field maps, and scalar relations. These presentations include explicit actions of the containing full classical groups.*

(iv) *An explicit realization of $\ker \pi$ as a degree-1 tensor module, given by an equivariant isomorphism*

$$\ker \pi \cong \bigoplus_{\lambda} U_\lambda \otimes_{\mathbb{F}_p} B_\lambda.$$

Each U_λ is one of the input natural modules, its dual, a Frobenius twist of either, or a trivial line; each B_λ has trivial D-action. The distinct underlying natural spaces have total $\mathbb{F}_p$-dimension $O(d)$. Their multiplicities contribute only to the polynomial dependence on InSize*.*

All modules are viewed over $\mathbb{F}_p$. Their actions are specified on group generators, and all constructed linear maps are given in the input and output bases. For a central-product image given with its product cover, pull the action back to that cover. The cover map is part of the output data so that later transporter computations can project to the original group.

We now construct the maps and the degree-one kernel of Theorem 6.1, using the coordinate models of Definitions 14 and 15. The inputs are the representation matrices and the invariant subquotient; the construction uses the preceding enveloping algebras, cohomology, and linear algebra.

Proof of Theorem 6.1. We work with the input product D, with all diagonal identifications already made. A central-product action is first pulled back to this product. Every action used below is therefore an action of the specified full product; no assertion about cohomology of its arbitrary subgroups is needed. Put

$$A_\rho = \mathrm{span}_{\mathbb{F}_p} \rho(D) \leq \mathrm{End}_{\mathbb{F}_p}(M),$$

where ρ is the input action on the ambient tensor sum. Closing the generator matrices under linear combinations and multiplication computes A_ρ. Keep this ambient algebra fixed: the invariant subspaces P, N and the quotient $Q = N/P$ are its modules even when the image on Q is smaller. All intertwining maps are computed on these quotient coordinates.

Odd characteristic. Proposition C.7 and Lemma C.8 give the injective types and their embeddings in full matrix spaces modulo fixed scalar subspaces. The semisimple types are already summands of the input matrix spaces. Applying Lemma 6.5 and these embeddings yields $\iota : Q \hookrightarrow I$,

where I has the matrix quotient presentation of Definition 14. Its scalar denominators impose all correlations between coordinates, including the trace relation joining trivial-type occurrences. In particular, this does not choose a section of $N \to N/P$. Writing this presentation as $I = B/C$, its denominator satisfies $C \leq B^D$. Theorem C.9 therefore gives $H^1(D, I) = 0$, so set $E = I$ and $\pi = \mathrm{id}_I$. The kernel is zero and has the empty degree-one presentation.

Characteristic two: the embedding. Proposition C.26 constructs the joint enveloping algebra and its injective cogenerator, with the explicit quotient models from Lemmas C.12, C.14, C.15, and Proposition C.24, together with the adjoint models of Lemma C.11. Lemma 6.5 then gives an A_ρ-linear injection $\iota : Q \hookrightarrow I$. The grouping is by isomorphism as prime-field representations, including all dual and Frobenius-twisted coincidences. In particular, a natural type already occurring in a square has the action induced by that square; it is not treated as an independent algebra factor.

Separate only the semisimple direct factors whose simple modules occur in degree at most one and are composition-disjoint from the remaining ambient blocks. Their central idempotent $e_{\mathrm{lin}} \in A_\rho$ is computed by linear equations in the concrete enveloping algebra: it is the identity on these factors and zero on the complementary blocks. Thus

$$I = I_{\mathrm{lin}} \oplus I_{\mathrm{rem}}, \qquad I_{\mathrm{lin}} = e_{\mathrm{lin}} I, \quad I_{\mathrm{rem}} = (1 - e_{\mathrm{lin}})I.$$

Take $\pi_{\mathrm{lin}} : I_{\mathrm{lin}} \to 0$ to be zero. A natural constituent inside a nonsimple square remains in that square's full injective in I_{rem}.

Characteristic two: the projection. On a nonorthogonal nontrivial injective in I_{rem} use its identity map. Its first cohomology vanishes by Lemma C.17; for special linear groups the intermediate-group assertion of Lemma C.3 gives the same vanishing. For adjoint modules modulo scalar subspaces, also use Lemma C.5 and the long exact cohomology sequence, as in the proof of Lemma C.17. The simple same-factor modules with unequal twists are included there. For the orthogonal simple tensor modules with unequal twists, use the identity and Lemma C.18. On the orthogonal tensor-square injectives use the maps of Lemma C.25 into one or two copies of $(V \otimes_{\mathbb{K}} V)/\mathbb{K}\omega$, whose first cohomology vanishes by Lemma C.18. The kernels of these maps are the explicit twisted-natural or trivial modules in that lemma.

For completeness, the common trivial-type injective has the form

$$I_{\mathrm{triv}} = \left(\bigoplus_i I_{0,i} \right) / R_0, \qquad R_0 = \left\{ (c_i s_i)_i : \sum_i \mathrm{Tr}_{\mathbb{K}_i/\mathbb{F}_2}(c_i) = 0 \right\},$$

where s_i spans the defining-field scalar socle in $I_{0,i}$. This is the single trace denominator in the proof of Proposition C.26; it includes all the shared scalar coordinates. On a nonorthogonal $I_{0,i}$, first quotient by its entire scalar socle $\mathbb{K}_i s_i$, and then embed the quotient in a sum of that family's injectives, using Lemma 6.5. Their first cohomology vanishes by Lemma C.17. The composite has kernel exactly $\mathbb{K}_i s_i$. On an orthogonal $I_{0,i}$ use Lemma C.25, whose displayed maps also kill this scalar socle. The resulting maps vanish on R_0 and hence define a map on I_{triv}, independent of the chosen representatives. Its kernel is the quotient of the sum of the displayed natural or scalar kernels by R_0, so the scalar coordinates satisfy this common relation. An isolated trivial algebra factor has already been put in I_{lin}.

These maps assemble to $\pi_{\mathrm{rem}} : I_{\mathrm{rem}} \to E$, and we set $\pi = 0 \oplus \pi_{\mathrm{rem}}$. The target is a direct sum of the listed matrix and tensor-square quotient modules with vanishing first cohomology. If a module involves two independent factors, the degree-one product formula gives

$$\begin{aligned} H^1(D_1 \times D_2, X \boxtimes_{\mathbb{F}_2} Y) \cong\ & H^1(D_1, X) \otimes_{\mathbb{F}_2} Y^{D_2} \\ & \oplus X^{D_1} \otimes_{\mathbb{F}_2} H^1(D_2, Y) = 0 \end{aligned}$$

for nontrivial simple X, Y. Restriction of scalars and the input field identifications preserve this calculation. A factor acting trivially on a summand contributes no first cohomology because the stable derived groups in D are perfect. Thus $H^1(D, E) = 0$.

Target coordinates and ambient actions. We verify that the target has the presentation required by Definition 15. The nonorthogonal square models are exactly $T, T/A$ in the linear and unitary cases, and $T/\mathbb{K}\omega, T/A, I_0$ in the symplectic case. The adjoint constructions use matrix or Hermitian matrix spaces modulo the fixed scalar subspaces computed in those constructions. For every orthogonal square, the preceding projection has target copies of $T/\mathbb{K}\omega$; its other injective models occur in I, rather than in E. The maps on the common trivial-type component annihilate its trace relation. Their target coordinates are the same nonorthogonal models and orthogonal copies just listed.

The fixed denominators and the displayed square denominators are preserved by the corresponding full classical groups. For the linear, unitary and odd orthogonal actions this also follows from Lemma C.1; for characteristic-two orthogonal squares it follows directly from invariance of $\mathbb{K}\omega$ under the full polar symplectic group. Thus the formulas define the claimed ambient actions on E. Equivariance of ι, π is asserted only for D. Equality tests in these quotients are membership tests in their displayed denominators; for the symplectic I_0 models these are the subspaces displayed in Definition 15.

The kernel presentation and bounds. The preceding explicit maps identify

$$\ker \pi = I_{\mathrm{lin}} \oplus \ker \pi_{\mathrm{rem}}$$

as a subquotient of a sum of natural, dual, Frobenius-twisted natural, and trivial modules. Compute this subquotient and its action matrices from the kernel formulas, quotient maps, and common scalar relation R_0. The ambient degree-one module is semisimple, so $\ker \pi$ is a direct sum of the same simple types. For each such type U, solve intertwining equations for a basis of $\mathrm{Hom}_D(U, \ker \pi)$, and successively choose maps whose images are not contained in the sum already constructed. Simplicity makes each new image disjoint from that sum. Gaussian elimination therefore computes the direct-sum isomorphism in the statement in polynomial time. Prime-field Frobenius twists of a natural action are identified by their coordinate Frobenius isomorphisms. Repeated actions remain in the trivial multiplicity spaces. The underlying representation spaces therefore consist of a fixed number of copies of the original natural actions and their duals, together with at most one trivial line, of total dimension $O(d)$. No bound on the multiplicity dimensions by d is used.

The individual injective models and their joint cogenerator have polynomial dimension in d. Lemma 6.5 gives polynomial dimension in InSize for I and for every auxiliary embedding used to construct E. All additional operations are explicit field operations, matrix spans, quotient maps, kernels, and intertwining linear equations in these dimensions. They fit the asserted $p^{O(d)}\mathrm{poly}(\mathrm{InSize})$ bound. All constructions use the input fields and carry the input product-cover map as part of their output data. If an affine action was originally given on a central-product image, its pullback translation is zero on the cover kernel: an element there acts as the identity affine map. Thus that kernel fixes every original point, and any later full affine transporter on the cover projects through the given quotient map. This construction computes only representations and maps. This proves all outputs without using a recursive affine or tensor algorithm. □

C.5.2 Cubic pairing submodules and their maps

Theorem 6.2 (restated). *Let D be a product of classical derived groups in the stable range above, and let T and $\mathcal{J}_T$ be as in Definition 16. The following assertions hold in every characteristic.*

(i) *The map $\mathcal{J}_T$ is injective. Thus $J(T) \cong \bigoplus_i T_i \otimes_{\mathbb{K}} B_i$, with each parameter occurrence indexed by a basis of its actual invariant-pairing space.*

(ii) *If every composition factor of a D-submodule $P \leq T$ occurs in degree at most two, then $P \leq J(T)$. The socle of $T/J(T)$ has no simple constituent occurring in degree at most two.*

(iii) *For a natural, dual, or twisted-natural module U given by its matrices on the generators of D,*

$$\mathrm{Hom}_D(U,T) = \sum_{i=1}^{3} \mathrm{Span}_{\mathbb{K}}\{\kappa_{i,\eta} \circ f : \eta \in B_i,\ f \in \mathrm{Hom}_D(U,T_i)\}.$$

Bases of these spaces, the pairing maps, their images and kernels, and the indicated coordinate identifications are computable by linear equations on the input matrices in time $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$.

These assertions apply componentwise with fixed multiplicities. For a prime-field tensor module, apply them to its defining-field components after scalar extension and descend the resulting maps, with the intertwiner and endomorphism fields, twist exponents, and central-product quotient map included in the coordinate descriptions. The direct-sum parameters in (i) count invariant pairings, rather than choices of intertwiners into an already counted natural occurrence.

We prove the containment, the direct-sum parameter description, and the explicit map assertions of Theorem 6.2. Invariant pairings also underlie the classical centralizer description of tensor powers; see the symplectic centralizer theorem over arbitrary infinite fields of Dipper, Doty, and Hu [DDH08, Theorem 1.4]. Here the acting groups are finite and the fields may be small. We obtain the required containment by restricting to finite tori and comparing their characters.

The characteristic-2 containment is proved in Lemma C.20, in Subsubsection C.4.4. That calculation compares weights on disjoint coordinate blocks and then identifies the permitted pairing terms by Lemma C.19. It also gives $T^D = 0$ in Corollary C.21. The calculation is included among these shared representation tools because its orthogonal special case is also needed by the orthogonal square calculation above. These are representation-theoretic arguments independent of the tensor algorithms.

In odd characteristic, signs distinguish odd tensor degree from even tensor degree. The only possible lower-degree simple factors in a cubic module are therefore natural ones. The earlier Hom-space cohomology calculation makes their extensions split. We can then classify maps from a natural module into the cubic module by moving one tensor factor across the Hom symbol. This operation, called *currying*, changes it into a degree-2 intertwiner problem already solved in the paper.

Proof of Theorem 6.2. The containment in characteristic two is Lemma C.20. In odd characteristic use the joint derived-to-full enveloping-algebra comparison in Lemma C.1 on the sum of the tensor products being compared. The central signs in the full classical groups separate the eigenspaces of those signs. A constituent occurring in degree at most two in a tensor product of three natural modules, duals, or twists must therefore be a degree-1 natural, dual, or twisted-natural constituent. For two such constituents S, S', viewed as $\mathbb{K}[D]$-modules, the vanishing on the full space of linear maps in Theorem C.9 gives $\mathrm{Ext}^1_D(S,S') = H^1(D, \mathrm{Hom}_{\mathbb{K}}(S,S')) = 0$. Hence P is semisimple. Every D-equivariant $\mathbb{K}$-linear map $S \to T_1 \otimes_{\mathbb{K}} T_2 \otimes_{\mathbb{K}} T_3$ is, by currying, a map

$$T_3^* \otimes_{\mathbb{K}} S \longrightarrow T_1 \otimes_{\mathbb{K}} T_2.$$

The degree-two intertwiner calculations of Lemma C.3 and Proposition C.7 give three pairing diagrams: the straight identification, its switch, and contraction followed by insertion of an invariant pairing tensor. These are the three Brauer pairings of the four tensor positions, interpreted through the specified natural-module intertwiners. For a nonsplit bottom–simple–top component, commutation with its simple-block idempotent and full simple matrices forces a scalar on the simple block. Subtract the corresponding whole component map. Commutation with all simple-to-bottom and top-to-simple matrices then forces the remainder to factor from trivial top to trivial bottom. Thus there is no additional isolated simple-block map. On squares, let τ swap the two positions. The projections are $(1\pm\tau)/2$, and different twists allow only the actual natural intertwiners. Uncurrying gives precisely the three free tensor positions. This argument divides by two only, so it includes characteristic three.

Since $J(T)$ itself has composition factors occurring in degree at most two, the inverse image of a simple submodule occurring in degree at most two of $T/J(T)$ would be a larger submodule of composition factors occurring in degree at most two, contradicting the proved containment. For each invariant-pairing space B_i, choose a basis. In the atomic natural factors such a space is zero or one-dimensional by the natural commutant, with the dualities and twists specified in the input. The resulting insertion images have independent natural parameters: put a free basis coordinate in one coordinate block and a nonzero pairing coefficient in a disjoint block. This isolates the chosen free position. This is the check used in Corollary C.21; the same coordinate test works in odd characteristic. For fixed multiplicities apply it to each labeled coefficient. It proves injectivity of $\mathcal{J}_T$ and the asserted direct-sum description.

The image of a map from a natural simple module U has its composition factors in degree one, hence lies in $J(T)$ by the containment already proved. Using the direct-sum identification, its components are exactly the maps $U \to T_i$ followed by the basis insertions. This proves the Hom formula. Intertwining equations compute each $\mathrm{Hom}_D(U, T_i)$ and each B_i; tensoring these matrices and Gaussian elimination compute the image, kernel and all displayed maps in polynomial time in their explicit sizes. Multiplicity coordinates remain passive. Extending scalars and then descending these linear systems gives maps over the original endomorphism fields satisfying the prime-field relations, as in the last paragraph of the characteristic-two containment proof. For a central quotient given as part of the input, the representations factor through its map, so all these Hom spaces and submodules are unchanged on pullback. The dimensions are polynomial in the explicit input length and the natural dimensions, giving the stated bound. □

C.5.3 Cohomology after quotienting by invariant pairings

Theorem 6.3 (restated). *Let D and T satisfy the hypotheses of Theorem 6.2. Then*

$$(T/J(T))^D = 0, \qquad H^1(D, T/J(T)) = 0.$$

The assertions hold after prime-field descent, with fixed multiplicities, and for a central-product image given with its quotient map. Consequently a cocycle $c : D \to T/J(T)$ specified by its values on the generators has a unique $b \in T/J(T)$ with $c(g) = (g-1)b$, computable by solving these linear equations on generators.

We now assemble the one-group calculations. The three positions can belong to one recognized factor, to a repeated factor and an independent factor, or to three independent factors. In the last two arrangements, fixed-space factors make the degree-1 product formula particularly simple. Field descent then returns the original prime-field module. Finally, pulling back a cocycle along the product cover handles central identifications without requiring a complement in the original group.

Proof of Theorem 6.3. Odd characteristic follows from Lemma C.10. For one factor in characteristic 2 use Lemmas C.28, C.29, and C.30. For the special-linear case Lemma C.27 gives $H^1(D,T)=0$ and $H^2(D,J(T))=0$, since $J(T)\le T$ is a direct sum of natural twists; the cohomology exact sequence for $0\to J(T)\to T\to T/J(T)\to 0$ therefore gives $H^1(D,T/J(T))=0$. Vanishing of $H^1(D,J(T))$ and $T^D=0$ also give the fixed-space assertion. When the three positions belong to three distinct group factors, the degree-1 product formula has a fixed-space factor from another nontrivial natural module in every summand and is zero. When two positions belong to one factor D_1 and the third to an independent factor D_2, let W be the quadratic $\mathbb{K}[D_1]$-module, let $I\le W$ be its invariant pairing subspace, and let V_2 be the natural or twisted $\mathbb{K}[D_2]$-module. The quotient module is $(W/I)\otimes_{\mathbb{K}} V_2$. The natural factor V_2 has no invariants. Also $(W/I)^{D_1}=0$: for a matching pair identify W with an endomorphism module and I with its scalar field. A fixed coset $X+I$ gives an additive homomorphism $g\mapsto gXg^{-1}-X$ from the perfect classical subgroup to the scalar field. It vanishes, and the actual natural commutant is the scalar field. For a nonmatching pair its ordinary invariant space is zero. The product formula proves $H^1=0$. Apply these product arguments to each recognized factor identification in the current group. Scalar extension is exact and cohomology is the kernel modulo image of finite-dimensional linear systems, so all assertions descend. Inflation is injective in degree 1 for the product-to-central-product map, and fixed vectors are unchanged. For a cocycle given by its values on generators, $H^1=0$ and $H^0=0$ give existence and uniqueness of b. Solve $(g-1)b=c(g)$ on generators in the quotient coordinates computed above. The cocycle identity extends these equations to all of D. □

D Technical results for degree-2 quotient module transporter

We give the constructions and proofs for the degree-2 quotient tensor module transporter of Section 7, using its notation and retaining the original group encoding throughout.

D.1 Quotient maps and changes of presentation

The recursive calls in R1 and R3 use the current affine equation $\alpha_h(x)=y$ on $Q=N/P$. We now justify their quotient and fiber inputs, and the coordinate changes made in R2.

A quotient call followed by its fiber call. Let $q:Q\twoheadrightarrow Q'$ be a surjective linear map commuting with the linear H-action, and put $F=\ker q$. The induced affine action on Q' is $\alpha'_h(q(z))=q(\alpha_h(z))$. Recursively compute the complete transporter of $q(x)$ to $q(y)$. If it is empty, return emptiness. Otherwise write its answer as

$$\mathrm{Trans}_{H,\alpha'}(q(x),q(y))=aK, \tag{D.1}$$

where K is the full stabilizer of $q(x)$. The remaining candidates are ak, $k\in K$. Define

$$w=\alpha_{a^{-1}}(y)-x\in F,\qquad \delta_k=\alpha_k(x)-x\in F,\qquad \beta_k(z)=L_kz+\delta_k\quad(k\in K).$$

For $k,h\in K$, the affine-action identity gives $\delta_{kh}=\delta_k+L_k\delta_h$. Thus β is an affine action of K on F, and

$$\alpha_{ak}(x)=y\quad\Longleftrightarrow\quad\beta_k(0)=w, \tag{D.2}$$

and hence

$$\mathrm{Trans}_{H,\alpha}(x,y)=a\,\mathrm{Trans}_{K,\beta}(0,w). \tag{D.3}$$

Recursively compute the transporter on the right. For parent multiplier t, this fiber call has multiplier ta, group K, source 0, and target w. Its answer is the complete answer to the parent call. Both recursive spaces have quotient presentations:

$$N_F = \{n \in N : n + P \in F\}, \qquad Q' \cong N/N_F, \qquad F \cong N_F/P. \tag{D.4}$$

The later construction lemmas express these quotients in their new tensor coordinates. The translations δ_k are defined in F; a cocycle lift to N_F is unnecessary.

Subgroup restriction. Let $H' \leq H$. Compute a left transversal $H = \bigsqcup_{j=1}^{k} r_j H'$. In coset $r_j H'$, write the unknown as $r_j h'$. The equation becomes $\alpha_{h'}(x) = \alpha_{r_j^{-1}}(y)$, giving

$$t \operatorname{Trans}_{H,\alpha}(x, y) = \bigcup_{j=1}^{k} t r_j \operatorname{Trans}_{H',\alpha|_{H'}}(x, \alpha_{r_j^{-1}}(y)). \tag{D.5}$$

Solve every coset case with its full stabilizer. Identity (3.1) combines these answers into one witness and generators of the full source stabilizer. This is the operation used after restricting to H_A in R3.

Coordinate changes. An equivariant linear isomorphism $u : N/P \to N'/P'$ changes the recursive data by

$$(\mathbf{U}, \mathbf{R}, \mathbf{S}; N, P, c; x, y) \longmapsto (\mathbf{U}', \mathbf{R}', \mathbf{S}'; N', P', uc; ux, uy), \qquad (uc)_h = u(c_h). \tag{D.6}$$

The group H and multiplier t remain unchanged. Compute u and its inverse; this operation does not make a recursive call.

Substituting an auxiliary presentation. Suppose an H-module factor B has a presentation $K_B \leq Y_B \leq R$ and $Y_B/K_B \cong B$, as in Subsection 7.2. Let $\eta_B : Y_B \twoheadrightarrow B$ be the corresponding surjection. On each labelled tensor summand, tensor this map with the maps for its other replaced factors and the identity maps on unchanged factors. Take the direct sum over all labelled summands. This gives a surjection

$$\eta : \widehat{Z}_0 \twoheadrightarrow Z, \qquad \widehat{Z}_0 \leq \widehat{Z},$$

where $\widehat{Z}$ uses the auxiliary factors R, and $\widehat{Z}_0$ uses their invariant numerator subspaces Y_B. Pull back the entire current numerator and denominator:

$$\widehat{N} = \eta^{-1}(N), \qquad \widehat{P} = \eta^{-1}(P), \qquad \widehat{N}/\widehat{P} \cong N/P.$$

Thus all relations between labelled summands remain in the quotient, and $\ker \eta \leq \widehat{P}$. Transfer the points and affine action through this isomorphism. For dual occurrences use the dual subquotient presentation, as specified in Subsubsection D.2.2. Applying Fact 3.4 to $\eta|_{\widehat{N}} : \widehat{N} \twoheadrightarrow N$ shows that the transporter is unchanged.

D.2 Exact presentations for the recursive calls

The operations R1–R4 must produce new instances of the same affine transporter problem in the original group. We verify this first for flag quotients, then for subquotients of auxiliary modules and coordinate projections, and finally construct the classical subgroup used by the local solver. Throughout, $P \leq N \leq Z$ is the current tensor presentation, $Q = N/P$, and all changes preserve (7.4).

The quadratic case and the fixed-position extension. For Theorem 7.1, each summand has at most two original tensor positions, and the alternating-dominant list is initially empty. Thus $r = 2$ in the arguments below. The same presentation lemmas apply to any fixed number r of positions. This extension is used in Subsection 7.6, with its additional input bounds and local solver verified in Appendix D.6. Proposition 7.8 is conditional on such a solver; the presentation lemmas alone do not solve arbitrary degree-r tensor transport. In the extension, the permuted-summand, dimension, summand-stabilizer, and simple-section bounds remain hypotheses on all additional modules.

D.2.1 R1: quotient flags

From factor flags to a filtration of Z and Q. Consider a tensor summand $U \otimes V$ with two active factors and composition flags $0 = U_0 < \cdots < U_{\ell_U} = U$ and $0 = V_0 < \cdots < V_{\ell_V} = V$. First use the invariant flag

$$0 = U_0 \otimes V < \cdots < U_{\ell_U} \otimes V = U \otimes V.$$

Its ith quotient is $(U_i/U_{i-1}) \otimes V$. The flag of V refines this quotient by the subspaces $(U_i/U_{i-1}) \otimes V_j$. Taking their preimages in $U_i \otimes V$ gives

$$F_{i,j} = U_{i-1} \otimes V + U_i \otimes V_j \qquad (1 \le i \le \ell_U,\ 0 \le j \le \ell_V).$$

Here $F_{i,0} = U_{i-1} \otimes V$ and $F_{i,\ell_V} = U_i \otimes V = F_{i+1,0}$ when $i < \ell_U$. Ordering first by i and then by j, and deleting repetitions, therefore gives one invariant flag of $U \otimes V$, with grades

$$F_{i,j}/F_{i,j-1} \cong (U_i/U_{i-1}) \otimes (V_j/V_{j-1}) \qquad (1 \le j \le \ell_V).$$

These tensor products need not be irreducible; only their individual active factors are composition factors.

For a summand $U \otimes R$ with only U active, use $U_i \otimes R$ and leave the auxiliary factor R unchanged; take $R = \mathbb{F}_p$ if there is no other factor. A summand with no active factor has the one-step flag from zero to the whole summand. Finally, order the labelled direct summands of Z. Add each term of a summand's flag to all preceding whole summands. Relabel this combined flag as $0 = Z_0 < \cdots < Z_{\ell_Z} = Z$.

R1 applies invariant flags of the active factors to the current quotient N/P. A flag in the ambient tensor module does not by itself specify the quotient presentation of each grade. The two lemmas below give its numerator and denominator, including the case where N/P is represented by the image of an evaluation map.

For finite-dimensional $\mathbb{F}_pH$-modules U, V, write ρ_U and ρ_{V^*} for the actions on U and V^*. The canonical map is

$$U \otimes V \cong \mathrm{Hom}_{\mathbb{F}_p}(V^*, U), \qquad z = \sum_j u_j \otimes v_j \longmapsto f_z, \quad f_z(\lambda) = \sum_j \lambda(v_j) u_j \tag{D.7}$$

where $\lambda \in V^*$. It commutes with the H-actions: $f_{hz} = \rho_U(h) f_z \rho_{V^*}(h)^{-1}$. An individual f_z need not commute with H. In a tensor summand, let T be the tensor product of its active factors and let W be the tensor product of the duals of its actual alternating-dominant factors. Reordering factors only for this notation identifies the summand with $T \otimes W^* \cong \mathrm{Hom}_{\mathbb{F}_p}(W, T)$; their assigned original positions are unchanged. Empty products are $\mathbb{F}_p$.

Evaluation labels and block projections. An evaluation of $f : W \to T$ is a value $f(w)$ at a specified vector $w \in W$. We use a finite H-set Λ of labels and specified vectors $w_\lambda \in W$ satisfying $hw_\lambda = w_{h\lambda}$. Its evaluation table is $(f(w_\lambda))_{\lambda\in\Lambda}$. This notation also permits repeated vectors with different labels. The labels record the summand and tensor position as well as the vector.

Each factor of W has the direct-sum decomposition induced by the selected auxiliary module or its dual. Take all vectors in each summand of each such factor. An evaluation label records the labelled tensor summand, the original positions, the auxiliary-module and summand indices, and a tuple of vectors; w_λ is their tensor product. These vectors span W, so full evaluation is injective. At a position with an active factor or no factor, use its fixed index $\perp_i$. Every evaluation label then has exactly one index in each position. If b_i is the number of selected indices in position i, including $\perp_i$ when present, put $b = \sum_{i=1}^{r} b_i$.

A block split $A \subseteq I_i$ partitions these labels according to their ith block index. On the full vector domains, evaluation on the labels in A is precisely full evaluation after the tensor projection pr_A. Thus the ordinary quadratic algorithm can use the coordinate projections of Subsection 7.3. The more general evaluation notation is useful for the map-valued applications in Subsection 7.6 and for evaluation on restricted families of vectors. Its kernel specifies the directions that must be included in the denominator.

An isotypic factor $X \otimes_{\mathbb{K}} M$ still occupies one original position while both factor actions are nontrivial. Suppose a factor B has the explicit invariant subquotient presentation $B \cong Y/K$, where $K \le Y \le R$ and R is an auxiliary module. R2 replaces each occurrence using $Y \twoheadrightarrow B$ and pulls back N and P through the resulting map on the whole ambient tensor module; its kernel lies in the new denominator. The tensor factor is then represented inside R, so its evaluation argument is R^*. For several labelled tensor summands, use the direct sum of their evaluation maps, with their distinct labels and the common N and P. All relations between these summands remain in one quotient.

Lemma D.1 (Adding the kernel of the evaluation map)**.** *Let H act linearly on finite-dimensional $\mathbb{F}_p$-spaces W, T. Let Λ be a finite H-set with specified vectors $w_\lambda \in W$ such that $hw_\lambda = w_{h\lambda}$. Set $Z = \mathrm{Hom}_{\mathbb{F}_p}(W, T)$, with action $(h \cdot f)(w) = hf(h^{-1}w)$. Let $\mathrm{Fun}(\Lambda, T)$ be the vector space of tables indexed by Λ, with action $(h \cdot F)_\lambda = hF_{h^{-1}\lambda}$. Define*

$$\mathrm{ev}_\Lambda : Z \longrightarrow \mathrm{Fun}(\Lambda, T), \qquad \mathrm{ev}_\Lambda(f) = (f(w_\lambda))_{\lambda\in\Lambda},$$
$$K_\Lambda = \ker \mathrm{ev}_\Lambda \le Z.$$

The evaluation map is H-equivariant. Let $P \le N \le Z$ be H-invariant and suppose $N \cap K_\Lambda \le P$, equivalently that the induced map $N/P \to \mathrm{ev}_\Lambda(N)/\mathrm{ev}_\Lambda(P)$ is injective. Then

$$N/P \cong (N + K_\Lambda)/(P + K_\Lambda)$$

equivariantly. For $\Lambda_0 \subseteq \Lambda$, let ev_{Λ_0} denote evaluation on those labels. The change leaves both $\mathrm{ev}_{\Lambda_0}(N)$ and $\mathrm{ev}_{\Lambda_0}(P)$ unchanged, whether or not Λ_0 is invariant. For every subspace $T' \le T$,

$$\mathrm{ev}_\Lambda^{-1}(\mathrm{Fun}(\Lambda, T')) = \mathrm{Hom}_{\mathbb{F}_p}(W, T') + K_\Lambda. \tag{D.8}$$

Proof. The map $n + P \mapsto n + (P + K_\Lambda)$ is surjective and has kernel $(P + (N \cap K_\Lambda))/P = 0$. Since $K_\Lambda \le \ker \mathrm{ev}_{\Lambda_0}$ for every $\Lambda_0 \subseteq \Lambda$, adding K_Λ changes neither $\mathrm{ev}_{\Lambda_0}(N)$ nor $\mathrm{ev}_{\Lambda_0}(P)$.

Put $S = \mathrm{span}_{\mathbb{F}_p}\{w_\lambda : \lambda \in \Lambda\} \le W$. If $f(w_\lambda) \in T'$ for every λ, then $f(S) \subseteq T'$. Extend $f|_S : S \to T'$ to a linear map $f' : W \to T'$ by extending a basis of S. The difference $f - f'$ belongs to K_Λ, which proves the nontrivial inclusion in (D.8); the other inclusion is immediate. This extension need not commute with H: it proves an equality of subspaces. If T' is invariant, both sides are H-invariant. □

For full evaluation, $K_\Lambda = 0$. For evaluation on a restricted family, under the injectivity hypothesis of Lemma D.1, adding K_Λ to both numerator and denominator preserves Q. Consider an invariant flag (T_i) of T induced by the active-factor flags. The spaces $Z_i = \mathrm{Hom}_{\mathbb{F}_p}(W, T_i) \cong T_i \otimes W^*$ identify with the ambient tensor flag used in R1, written in map notation. For full evaluation its induced quotient flag is $Q_i = ((N \cap Z_i) + P)/P$, as in the algorithm. The additional K_Λ terms only accommodate restricted evaluation families.

Lemma D.2 (A flag on the current quotient)**.** *Use the data and hypotheses of Lemma D.1. Put $N' = N + K_\Lambda$ and $P' = P + K_\Lambda$. For an H-invariant flag $0 = T_0 \le \cdots \le T_{\ell_T} = T$, set $Z_i = \mathrm{Hom}_{\mathbb{F}_p}(W, T_i)$ and $U_i = N' \cap Z_i$. Under the preceding identification $Q = N/P \cong N'/P'$,*

$$Q_i = (P' + U_i)/P' \qquad (0 \le i \le \ell_T)$$

is a flag preserved by the linear part of the current action. For $1 \le i \le \ell_T$, let $q_i : Z_i \to \mathrm{Hom}_{\mathbb{F}_p}(W, T_i/T_{i-1})$ be postcomposition with $T_i \twoheadrightarrow T_i/T_{i-1}$. The grades have the exact presentations

$$Q_i/Q_{i-1} \cong \frac{U_i}{(P' \cap Z_i) + U_{i-1}} \cong \frac{q_i(U_i)}{q_i(P' \cap Z_i)}, \tag{D.9}$$

Evaluation on Λ, modulo the evaluated denominator, is injective on each displayed quotient. The positive grade dimensions sum to $\dim_{\mathbb{F}_p} Q$.

Proof. The kernel of q_i is Z_{i-1}, and its intersection with U_i is U_{i-1}. Moreover,

$$U_i \cap (P' + U_{i-1}) = (P' \cap Z_i) + U_{i-1}.$$

These identities prove both isomorphisms in (D.9).

Suppose $u \in U_i$ evaluates to zero modulo T_{i-1}. Equation (D.8) gives $u = w + z_{\mathrm{ker}}$ with $w \in Z_{i-1}$ and $z_{\mathrm{ker}} \in K_\Lambda$. Since $u, z_{\mathrm{ker}} \in N'$, we have $w \in U_{i-1}$; since $u, w \in Z_i$, we have $z_{\mathrm{ker}} \in P' \cap Z_i$. Thus $q_i(u)$ represents zero. If the evaluation of $q_i(u)$ belongs to the evaluated denominator, first subtract a representative from $q_i(P' \cap Z_i)$ and apply the same argument. This proves the claimed injectivity. Dimension additivity follows from $0 = Q_0 \le \cdots \le Q_{\ell_T} = Q$. □

For several active factors, form the tensor-product filtration of their invariant flags, in a fixed order with successive tensor products of the actual quotients. The alternating-dominant factors are unchanged. Apply (D.9) to the whole N/P, including relations between summands.

Only the linear part preserves these flag subspaces; the affine action may translate them. R1 recursively computes the full affine transporter on the top quotient. If it is nonempty, R1 then recursively computes the full transporter on its exact affine fiber, under the source stabilizer returned by the first call, as in Lemma 3.5 and (3.3). If a fiber class represented by $n \in N'$ belongs to $(P' + U_i)/P'$, solve for $a \in P'$ with $n - a \in U_i$. Two choices differ by $P' \cap Z_i$, which specifies the representative correction at that step.

The graded presentations do not split the original module. Let K be the source stabilizer returned by the quotient call, and let $F \le Q$ be the direction space of the remaining affine fiber. Its numerator is $N_F = \{n \in N : n + P \in F\}$. The fiber call initially uses $\mathbf{U}|_K, \mathbf{R}|_K, \mathbf{S}$, N_F, and the parent's denominator P. Any later change of its tensor layout requires an explicit equivariant isomorphism of that fiber quotient.

D.2.2 R2: large classical and alternating-dominant modules

R2 starts with the invariant sections computed by R1. It identifies their large classical components and moves the other factors to the alternating-dominant auxiliary-module list **R**. The active list **U** then consists of the nontrivial modules for the selected large classical components. For each converted factor, the construction below gives an invariant subquotient presentation inside an auxiliary module with permuted direct summands, adds the summand indices to **S**, and computes an isomorphism from the new quotient to the old N/P. This isomorphism transports both points and the affine action.

Separate the actions on the current factors. Apply Theorems 5.1, 5.2, and 5.3 to the joint matrix image of H on the active factors **U**, as in (7.6), with the labeled factors and flags. The primitive blocks are the summands of systems of imprimitivity on the irreducible sections, with primitive actions of their stabilizers. Record the permutations and actual maps between blocks. The threshold is

$$T_{\mathrm{cl}} = p^{C_* d},$$

with the root constant $C_* = C_*(r) \geq 128$ of Section 5, chosen large enough for all structural bounds. Both d and C_* remain fixed when the current dimensions decrease. Use the stable ranges of Subsection 6.1: natural dimension at least 64 for special linear and unitary factors, and Witt index at least 32 for symplectic and even orthogonal factors. Here Witt index means the largest dimension of a totally isotropic subspace for a bilinear or Hermitian form, or a totally singular subspace for a quadratic form. In particular, the nonsplit even orthogonal condition is on the Witt index. Each fixed-rank exception has order $p^{O(d)}$ with an absolute constant and is left in the alternating-dominant part by the choice of C_*.

These direct sums lie in the successive quotients produced by R1; no invariant complement in an earlier flag space is required. Record all diagonal, dual, and Frobenius identifications between selected component actions. Each isotypic summand remains one tensor factor.

Present each alternating-dominant factor inside an auxiliary module. For each factor B being converted, construct an auxiliary H-module R, invariant subspaces $K \leq Y \leq R$, and an equivariant surjection $q : Y \twoheadrightarrow B$ with kernel K. Construct these data over whole block orbits using the maps between blocks; an individual block is a module only for its stabilizer. The action on R is specified by generator matrices. There are the following cases.

(i) For a bounded primitive factor, set $R = Y = B$ and $K = 0$.

(ii) For an unselected classical block, also use its actual representation, giving $R = Y = B$ and $K = 0$ over the whole block orbit. Its block-stabilizer image has the global bound $p^{(C_*+6)d}$ proved in Appendix B.4, measured using the original d; it gives one alternating-dominant summand with that bound.

(iii) For a fully deleted permutation representation, use the auxiliary permutation module of Theorem 5.3(2). Record its multiplicity action, its maps between blocks, and the inclusion and quotient maps identifying B with Y/K.

In every case $\dim R$ is at most a fixed multiple of the dimension being replaced. For a stable fully deleted permutation module, this multiple is at most $4/3$. The summands of R are the actual bounded primitive or unselected classical summands, or the coordinate summands of the auxiliary permutation module. Record both their summand permutations and their linear maps between summands for each group generator.

Transport the quotient and its affine equation. Add R, or R^* for a dual occurrence, as a new module in $\mathbf{R}$, in the converted factor's original position. For a dual occurrence, use the surjection $K^\perp \twoheadrightarrow B^*$ with kernel $Y^\perp$, where $K^\perp, Y^\perp \leq R^*$ are the annihilators of K, Y. In each labelled tensor summand, tensor the quotient maps for its replaced factors with the identity maps on the other factors. Take their direct sum, using the identity on every unchanged tensor summand. This defines the surjection $\eta : \widehat{Z}_0 \twoheadrightarrow Z$, where $\widehat{Z}_0$ is the invariant subspace of the new auxiliary ambient module $\widehat{Z}$ obtained by using the corresponding numerators Y or $K^\perp$ in each replaced position. Pull back the whole N and P under η, so the new denominator contains $\ker \eta$. Fact 3.4 shows that the resulting quotient, points, and affine action give the same original transporter. The matrices and summand indices of existing alternating-dominant modules are unchanged; only active factors undergo this conversion.

Charge the conversion to the dimension it removes. Let $v = \sum_{a \in \mathcal{A}} \dim_{\mathbb{F}_p} U_a$, with each distinct active factor counted once. Converting total dimension $c > 0$ gives, for a constant a_r depending only on r,

$$v' = v - c, \qquad b' \leq b + a_r c. \tag{D.10}$$

The fixed index $\perp_i$ remains exactly when some surviving labelled tensor summand has an active factor or no factor in position i. The pullback uses only labelled tensor summands and coordinate blocks present in the current call. Every added index is therefore charged to the converted dimension c, and no coordinate block excluded by an unfinished ancestor call is restored.

Along a path, these conversions consume disjoint portions of the remaining representation dimension. Passing to an invariant subquotient decreases dimension, and taking a direct-sum decomposition preserves its sum. Summing the bounded increase from the auxiliary modules therefore gives

$$v + \sum_{R \in \mathbf{R}} \dim_{\mathbb{F}_p} R \leq O_r(d). \tag{D.11}$$

The number of original positions is unchanged. After a subgroup restriction, new flags may replace the old decompositions of the current active factors. The previously selected alternating-dominant summands use the restricted actions and the same summand indices, including every ancestor's exclusions.

Recall the structural auxiliary data of Subsection 7.2: $B = \rho_{\mathrm{aux}}(H)$ is the joint image on the selected modules, their summand indices, and the spaces L_ℓ; the threshold is $T_{\mathrm{cl}} = p^{C_* d}$, fixed at the root.

Lemma 7.4 (restated). *For every $B_0 \leq B$ and $K_0 \trianglelefteq B_0$,*

$$B_0/K_0 \text{ nonalternating nonabelian simple} \quad \Longrightarrow \quad |B_0/K_0| \leq T_{\mathrm{cl}}.$$

The assertion persists after restricting H to a subgroup and replacing these representations by invariant subquotients.

Proof of Lemma 7.4. The individual assertions for alternating-dominant actions and alternating auxiliary actions follow from Theorem 5.3 and its concluding simple-section bound. The subspace fixed pointwise by the selected classical product has these alternating-dominant presentations, as specified before Lemma 5.4. For the alternating auxiliary modules, the kernel of the permutation action on their coordinate indices has order $p^{O_r(d)}$: its local multiplicity images have exponents bounded by their original disjoint dimensions. Its permutation degree is $O_r(d)$ by (D.11), so the

hereditary simple-section bound of Theorem 5.3 applies. Each image on L_ℓ lies in the solvable group $\mathbb{K}_\ell^\times \rtimes \mathrm{Gal}(\mathbb{K}_\ell/\mathbb{F}_p)$.

The subgroup-and-extension argument in the proof of Theorem 5.3 in Appendix B.4 now combines these bounds for the actual joint image, including nonsplit extensions. Because that argument already quantifies over all subgroups, it also proves persistence after restriction. The argument applies to the joint image of the current group, including all correlations between the stored actions. After restriction and passage to invariant subquotients, the new joint image is a quotient of a subgroup of B, so it has no new simple sections. □

D.2.3 R3: selected summands and exact fibers

R3 chooses block indices $A \subseteq I_i$ in one original position. After restricting to H_A, both tensor projections pr_A and $\mathrm{pr}_{I_i \setminus A}$ are equivariant. The first recursive call computes the full affine transporter of the projected points in $\mathrm{pr}_A(N)/\mathrm{pr}_A(P)$ under H_A. If its answer is nonempty, the second recursive call computes the full transporter on the remaining affine fiber under the source stabilizer returned by the first call. The next lemma gives the fiber presentation on its remaining coordinate blocks, including when the incoming quotient is represented by evaluation tables.

Lemma D.3 (Restriction to selected summands)**.** *Let $P \le N \le Z$ be invariant under the current subgroup $H \le \Gamma$, where Z is the current tensor module. Equip $Q = N/P$ with an affine H-action having the induced linear part, and let $x, y \in Q$. Let $\Lambda = \Lambda_0 \sqcup \Lambda_1$ be an H-invariant partition of its evaluation labels, and suppose the induced map $N/P \to \mathrm{ev}_\Lambda(N)/\mathrm{ev}_\Lambda(P)$ is injective. Suppose an H-invariant coordinate summand $Z_1 \le Z$ and an H-equivariant coordinate projection $\mathrm{pr}_1 : Z \to Z_1$ are given, with $(\mathrm{ev}_{\Lambda_1}|_{Z_1})\,\mathrm{pr}_1 = \mathrm{ev}_{\Lambda_1}$. Recursively compute the full affine transporter of the images of x, y under the induced quotient map to $\mathrm{ev}_{\Lambda_0}(N)/\mathrm{ev}_{\Lambda_0}(P)$. If its answer is nonempty, the remaining affine fiber, under the returned source stabilizer, has direction space N_0/P_0, where*

$$N_0 = N \cap \ker \mathrm{ev}_{\Lambda_0}, \qquad P_0 = P \cap \ker \mathrm{ev}_{\Lambda_0} \le N_0.$$

Put

$$\begin{aligned} U_1 &= \mathrm{pr}_1(N_0) \le Z_1, \\ D_1 &= \mathrm{pr}_1(P_0) + (U_1 \cap \ker(\mathrm{ev}_{\Lambda_1}|_{Z_1})) \le U_1. \end{aligned} \tag{D.12}$$

Then $N_0/P_0 \cong U_1/D_1$ equivariantly under the returned source stabilizer. Transport the induced fiber action and its source and target along this isomorphism. Subsequent full evaluation on these same coordinate blocks preserves the entire original equation.

Proof. The map $N_0/P_0 \to U_1/D_1$ given by $n + P_0 \mapsto \mathrm{pr}_1(n) + D_1$ is well defined and surjective. If its value is zero, choose $a \in P_0$ with $\mathrm{ev}_{\Lambda_1}(n - a) = 0$. Both n and a vanish under ev_{Λ_0}, so $n - a$ vanishes on all of Λ. Injectivity of the induced evaluation map on N/P gives $n - a \in P$, and hence $n \in P_0$. Thus the map is injective.

Full multilinear evaluation is injective on a tensor module: values on tuples of basis vectors determine every tensor entry. Distinct tensor-summand labels give injectivity on their direct sum as well. It follows that full evaluation is injective on U_1/D_1 modulo its evaluated denominator. The displayed isomorphism transports the action, the points, and their transporter. The projection pr_1 first removes all excluded coordinate blocks. Excluding $\perp_i$ also removes labelled tensor summands with an active factor or no factor in position i. Enlarging the families of evaluation vectors in surviving blocks therefore cannot restore an excluded coordinate block or labelled tensor summand. □

For the block split $A \subseteq I_i$, let Λ_0 and Λ_1 be the labels with ith index in A and in $I_i \setminus A$, respectively. Then $\mathrm{pr}_1 = \mathrm{pr}_{I_i \setminus A}$. On full vector domains, each restricted evaluation has the same kernel as its corresponding tensor projection. In particular the first child is represented directly by

$$\mathrm{pr}_A(N)/\mathrm{pr}_A(P),$$

and the second by

$$\frac{\mathrm{pr}_{I_i \setminus A}(N \cap \ker \mathrm{pr}_A)}{\mathrm{pr}_{I_i \setminus A}(P \cap \ker \mathrm{pr}_A)}.$$

These are the coordinate presentations used in R3.

For a restricted family of evaluations, suppose likewise that $Z_0 \leq Z$ is an H-invariant coordinate summand with equivariant projection $\mathrm{pr}_0 : Z \to Z_0$ satisfying $(\mathrm{ev}_{\Lambda_0}|_{Z_0})\,\mathrm{pr}_0 = \mathrm{ev}_{\Lambda_0}$. The first recursive call then uses

$$\frac{\mathrm{pr}_0(N)}{\mathrm{pr}_0(P) + (\mathrm{pr}_0(N) \cap \ker(\mathrm{ev}_{\Lambda_0}|_{Z_0}))}.$$

Thus both recursive calls use only their selected coordinate blocks. Their shared auxiliary modules remain in $\mathbf{R}$; update $S_{i,\nu}$ only in the divided position. Selections in other positions are unchanged, even when they use the same auxiliary module. The quotient isomorphisms also transport both points and the induced image or fiber action.

A balanced split. The alternating-dominant image can be large because it permutes many summands. R3 reduces their number by choosing a balanced subset and enumerating its orbit. In each coset, recursively compute the full transporter on the projected quotient and then on its exact affine fiber. The following lemma bounds both the number of cosets and the size of each child.

Lemma D.4 (Splitting the selected summands)**.** *For a recursive call of Definition 18 with at most r tensor positions and $b = \sum_{i=1}^{r} |I_i| \geq 3r$, the candidates can be partitioned into at most 2^b coset cases. Each case is solved by a quotient transporter call and, if its answer is nonempty, a subsequent fiber transporter call. The quotient and fiber direction dimensions sum to $\dim_{\mathbb{F}_p} Q$; zero-dimensional spaces are resolved directly. Each child uses at most $\gamma_r b$ selected indices, where $\gamma_r = 1 - 1/(3r) < 1$. The construction and coset representatives are explicit.*

Proof. Choose a position i with $b_i \geq b/r$ and a subset $A \subseteq I_i$ of size $\lfloor b_i/2 \rfloor$. The orbit–stabilizer algorithm computes H_A, the orbit of A, and a left transversal

$$H = \bigsqcup_j a_j H_A, \qquad |H : H_A| \leq 2^{b_i} \leq 2^b,$$

with words in the original generators. In the case $a_j H_A$, write the unknown as $a_j h$, with $h \in H_A$, and replace the target by $\alpha_{a_j^{-1}}(y)$.

The block indices in A select a direct summand of the current tensor layout; the other indices select the remaining summand. The fixed index $\perp_i$, when present, lies on exactly one side. The equivariant projection pr_A induces the quotient map $q_A : Q \twoheadrightarrow \mathrm{pr}_A(N)/\mathrm{pr}_A(P)$. Recursively compute the full affine transporter of the projected points under H_A. If nonempty, recursively compute the full affine transporter on its fiber under the returned source stabilizer. Fact 3.3, applied to $\mathrm{pr}_A|_N$, and the full-domain case of Lemma D.3 give the displayed tensor presentations. When the incoming quotient is represented by evaluation tables, use the corresponding partition $\Lambda = \Lambda_0 \sqcup \Lambda_1$ and the general form of the same lemma. The direction spaces are the image and

kernel of the induced quotient map, so their dimensions sum to $\dim_{\mathbb{F}_p} Q$. An empty first answer makes that coset case empty without a second call.

Both children use at most $\lceil b_i/2 \rceil$ indices in position i, and at most the old number in every other position. Let b_{child} be the total number of indices in either child. Then

$$b_{\text{child}} \le b - \lfloor b_i/2 \rfloor \le b - \frac{b}{2r} + \frac{1}{2} \le \left(1 - \frac{1}{3r}\right) b \qquad (b \ge 3r).$$

Resolve a zero-dimensional direction space directly; this also covers an empty set of coordinate blocks. By Lemma D.3, full evaluation on surviving coordinate blocks cannot restore an excluded index. Compute the full transporter in every coset case and apply (3.1) to their union. The cosets exhaust H, so their union is the complete transporter. □

D.2.4 R4: the joint classical subgroup

Recall $A = \rho_{\mathbf{U}}(H)$, $B = \rho_{\text{aux}}(H)$, and $J = (\rho_{\mathbf{U}}, \rho_{\text{aux}})(H)$ from Subsection 7.2. The active representation is the nontrivial part for the full selected classical product $D \trianglelefteq A$; all representation dimensions use the fixed bound $\kappa_r d$.

Lemma 7.5 (restated). *One has $[A : D] \le p^{O_r(d)}$; this bound uses only the hypotheses on A and D. Moreover, $D \times 1 \le J$. Thus the selected classical action can be realized while fixing all selected auxiliary modules and the spaces L_ℓ pointwise. It also acts trivially on the multiplicity spaces in the natural coordinates of Theorem 5.2(1),(4). Identify D with the normal subgroup $D \times 1 \trianglelefteq J$. If $s = \sum_{i\in[r]} \sum_{\nu\in\mathcal{B}} |S_{i,\nu}| = O_r(1)$, then*

$$|B| \le p^{O_r(d)}, \qquad [J : D] \le [A : D]|B| \le p^{O_r(d)}.$$

Generators of $D \times 1$ and its full preimage in H can be computed in $p^{O_r(d)}\text{poly}(\text{InSize}, \log p)$ time. Under the same bounded-summand hypothesis, a complete list of coset representatives for J/D, with lifts to H, can be computed in this time. All lifts and kernel generators use the original group encoding.

Proof of Lemma 7.5. The subgroup fixes the selected alternating-dominant modules. Lemma 7.4 excludes the selected nonabelian simple quotients from every subgroup of B. Applying Lemma 5.6 to $J \le A \times B$ and $D \trianglelefteq A$ therefore gives $D \times 1 \le J$. Its constructive conclusion gives generators and original preimages, using homomorphism kernels, membership, and lifts in the original group. Adjoining the kernel of $(\rho_{\mathbf{U}}, \rho_{\text{aux}})$ to lifts of the generators gives the full preimage of $D \times 1$ in H. In the natural coordinates of Theorem 5.2(1),(4), D acts trivially on the multiplicity spaces.

The remaining active action has bounded index. The subspaces fixed pointwise by D were converted to alternating-dominant modules using the invariant subquotient presentations constructed in Subsubsection D.2.2. Every remaining active factor is therefore in the nontrivial part of that product. Apply Lemma 5.5, aggregating equivalent isotypic types across the labeled original spaces and the fixed number of required dual copies. Let $X_j \otimes_{\mathbb{K}_j} M_j$ be their nontrivial isotypic summands, where X_j represents a simple $\mathbb{F}_p D$-module, $\mathbb{K}_j = \text{End}_{\mathbb{F}_p[D]}(X_j)$ is its endomorphism field, and M_j is its multiplicity space. Put $f_j = [\mathbb{K}_j : \mathbb{F}_p]$ and $b_j = \dim_{\mathbb{K}_j} M_j$. With the fixed dimension-budget constant κ_r, the lemma gives

$$\sum_j f_j b_j^2 \le \frac{\kappa_r^2}{C_*} d. \tag{D.13}$$

Consequently the centralizer of D on this direct sum has order $p^{O_r(d)}$. The index bound in the same lemma applies to $D \trianglelefteq A$ with $D \le A$, and gives $[A : D] \le p^{O_r(d)}$. This bound is independent of the number of selected auxiliary summands.

The alternating-dominant image has bounded order. Use $s = \sum_{i\in[r]} \sum_{\nu\in\mathcal{B}} |S_{i,\nu}|$, counting each original position separately. For each such occurrence a, let B_a be the image of its stabilizer in H on that summand, and let B_{scal} be the joint image on the spaces L_ℓ. The subgroup of B fixing all summand indices embeds in the product of the B_a and B_{scal}. Each B_a has order $p^{O_r(d)}$ by the structural construction; for a coordinate summand of an auxiliary permutation module, this is its bounded multiplicity image. The maps between summands are among the explicit maps in the auxiliary-module data. Thus

$$|B| \le s! \prod_{a \text{ selected}} |B_a|\,|B_{\mathrm{scal}}|.$$

Since $[\mathbb{K}_\ell : \mathbb{F}_p] = e_\ell$ and $\sum_{\ell\in\mathcal{L}} e_\ell \le \kappa_r d$, their image satisfies

$$|B_{\mathrm{scal}}| \le \prod_{\ell\in\mathcal{L}} e_\ell(p^{e_\ell} - 1) \le p^{2\sum_{\ell\in\mathcal{L}} e_\ell} = p^{O_r(d)}.$$

The scalar and field-automorphism tuples extracted from the matrix actions on the current factors have the same bound by the constructions in Section 5. Correlations can only reduce the image. Since $s = O_r(1)$, these inequalities give $|B| = p^{O_r(d)}$.

Both bounds concern the projections of the same joint image. Since $D \times 1 \le J$,

$$[J : D] \le [A : D]|B| \le p^{O_r(d)}.$$

Breadth-first coset enumeration in J/D, using constructive membership in D, lists all coset representatives in the stated time. Their permutation, scalar, and field actions are the actions of actual common elements of H, expressed as words in the original generators. □

D.3 The final classical step

Consider a call at Step R4, with current subgroup $H \le \Gamma$, quotient $Q = N/P$, and affine action $\alpha_h(z) = L_h z + c_h$. Its equation is $\alpha_h(x) = y$. Only boundedly many alternating-dominant summands remain. Lemma 7.5 computes the joint action on the active factors, selected modules $R_\nu[S_{i,\nu}]$, and spaces L_ℓ:

$$\theta : H \twoheadrightarrow J, \qquad K = \ker\theta, \qquad D \trianglelefteq J, \qquad [J : D] = p^{O(d)},$$

where D is the selected classical subgroup. The linear action on Q factors through θ, but the affine action may not: elements of K can translate Q.

First center the source at 0 by replacing the cocycle and target with

$$c_h^\circ = L_h x + c_h - x, \qquad y^\circ = y - x.$$

Write c, y for these centered data below. Since K acts trivially through the linear action, $c|_K$ is unchanged and is a homomorphism into the additive group of Q. Quotienting by its image makes the affine action descend to J. We then compute the complete transporter in each of the $p^{O(d)}$ cosets of D and correct its lifts to recover the complete answer in H. The only problem-specific call is the classical affine transporter of Subsection 7.5; it completes its equation without leaving a fiber for the common recursion.

(L1) **Remove the actual translation image.** Compute $\Delta = c(K) \le Q$ and set $Q_{\mathrm{tr}} = Q/\Delta$, using Lemma 3.6. Write $y_{\mathrm{tr}} = y + \Delta$ and α^{tr} for the induced affine action of J on Q_{tr}. If $Q_{\mathrm{tr}} = 0$, the same lemma finishes the call by translation correction.

(L2) **Complete every classical coset case.** Enumerate all cosets uD in J using Lemma 7.5. For the case uD, solve $\alpha_g^{\mathrm{tr}}(0) = (\alpha_u^{\mathrm{tr}})^{-1}(y_{\mathrm{tr}})$ for $g \in D$. Expand the multiplicity spaces and selected modules $R_\nu[S_{i,\nu}]$ fixed by D in their specified bases. Apply Proposition 7.2 to the resulting quotient of matrix spaces under the actual classical product D, with its numerator, denominator, and quotient-valued cocycle.

(L3) **Return the full original transporter.** Reunite all successful coset answers in J by (3.1). Lift the witness and every stabilizer generator to H and correct their translations using K, as in Lemma 3.6. Include generators of $\ker(c|_K)$, the kernel of the projection from the recovered stabilizer to its image in J.

Proposition 7.2 computes the complete affine transporter on Q_{tr} in each coset, including its degree-1 fiber in characteristic 2. Step L3 then restores the kernel translations.

Proposition D.5 (The local algorithm for quadratic affine transport)**.** *Given a call of Definition 18 at Step R4 of Algorithm 7.6, after the structural preparation in Step R2 and with $b \le b_0$, Steps L1–L3 compute its complete affine transporter in the original group in $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$ time.*

Proof. The quotient passed to the classical solver. Let $q : N \twoheadrightarrow Q = N/P$ be the quotient map. Replacing P by $q^{-1}(\Delta)$ gives the exact presentation

$$Q_{\mathrm{tr}} = Q/\Delta \cong N/q^{-1}(\Delta)$$

in the original tensor space, including all relations between its summands. The induced action and cocycle factor through J by Lemma 3.6.

The subgroup D fixes every selected module $R_\nu[S_{i,\nu}]$ and every space L_ℓ. On an isotypic factor $X \otimes_{\mathbb{K}} M$, it acts on X and fixes M. Expanding bases of those fixed spaces leaves at most two natural, dual, or Frobenius-twisted occurrences in each summand. They give full matrix spaces, with a common trivial one-dimensional space when there are one or zero occurrences. Symmetry conditions, field balancing, and the original numerator and denominator remain invariant linear constraints in their joint direct sum.

Each coset call therefore satisfies Proposition 7.2: the acting group is the actual stable central-product image, the module is the explicit quotient $N/q^{-1}(\Delta)$, and the cocycle is defined on that quotient. The solver uses the parent call's diagonal identifications. If a surjection $\lambda : \widetilde{D} \twoheadrightarrow D$ is used, pull the input affine action α_D on Q_{tr} back to $\widetilde{\alpha} = \alpha_D \circ \lambda$. For any $x', y' \in Q_{\mathrm{tr}}$,

$$\mathrm{Trans}_{\widetilde{D},\widetilde{\alpha}}(x', y') = \lambda^{-1}\big(\mathrm{Trans}_{D,\alpha_D}(x', y')\big)\,.$$

Hence projecting a witness and generators of the full source stabilizer on $\widetilde{D}$ gives the complete transporter in D. The projection is made after the full input affine equation is solved; the auxiliary modules used inside the classical solver need not individually descend to D. If $D = 1$, test equality of the source and target.

Completeness and return to the original group. The proposition completes each coset case with its full stabilizer. The hypotheses of the reunion and translation-restoration lemmas therefore hold, and Step L3 returns exactly the original transporter. No incomplete coset case or unresolved fiber is passed back to the common recursion.

Cost. There are $p^{O(d)}$ cosets. In each classical call, the total prime-field dimension of the natural modules is $O(d)$; fixed multiplicities contribute only to InSize. The classical calls, reunion, and translation restoration consequently have total cost $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$. □

D.4 Cost of the constructions

We prove the two operation bounds used in Subsection 7.4.

D.4.1 Preparation without a decrease in the parameters

Computing a flag or changing coordinates need not decrease b, v, or m. The computation records completed preparation steps, so an unchanged presentation cannot restart one. The following bound makes that rule quantitative, using only flags and subgroup orders.

Lemma D.6 (Bounded equal-parameter sequences)**.** *For Algorithm 7.6, the number of successive operations on a path that leave b, v, and the positive dimension m of the current quotient unchanged is bounded by a fixed polynomial in the initial explicit input length and* $\log p$.

Proof. First fix the current subgroup H. For each current active factor V, Step R1 computes a composition flag

$$0 = V_0 < V_1 < \cdots < V_\ell = V$$

and records its quotient maps and the list of completed successive quotients. There are at most $\dim_{\mathbb{F}_p} V$ quotients in this flag. With at most r tensor positions, the number of tuples of such quotients is polynomial in the original input size. Coordinate changes carry these flags and their completed slots to the new presentation. A composition quotient is simple, and a direct sum of such quotients is semisimple, so further invariant subquotients require no new composition flags while H is fixed.

On the path under consideration, a conversion cannot occur because it lowers v, a summand division cannot occur because it lowers b, and a positive quotient cannot leave a nontrivial continuing fiber of the same dimension m. The algorithm can therefore only advance through the previously computed flag quotients and preparation steps, finish a local calculation, or restrict H properly. Its explicit reuse rule bounds the number of the first kind of operations by a fixed polynomial.

For the subgroup changes, let n_0 be the number of points in the original external permutation action, and let t_0 be the dimension over $\mathbb{F}_p$ of the original elementary abelian kernel represented by additive coordinates. Set the corresponding parameter to zero if that part of the encoding is absent. The original encoding gives

$$\log_2 |\Gamma| \le O\big(d^2 \log p + n_0 \log(n_0 + 1) + t_0 \log p\big). \tag{D.14}$$

Every proper subgroup restriction has index at least two. There are consequently at most $\log_2 |\Gamma|$ such restrictions along a path. They may require new flags, but only on the current invariant subquotients, with explicit maps to the preceding presentations. Their dimensions and original equation are unchanged by these maps.

Multiplying the fixed-subgroup bound by the subgroup-chain bound proves the result. For example, a bound of the form

$$P_0 = (1 + \log_2 |\Gamma|)(d + 1)^{r+2} \mathrm{poly}(\mathrm{InSize}, \log p)$$

suffices. This counts all unchanged-parameter preparations without charging them as new recursive branches. □

D.4.2 The cost of one operation

A uniform local bound must control both tensor coordinates and group encodings. The dimension budget bounds the former. Each recursive group is retained as a subgroup of the original input group, with its affine action stored in the current quotient coordinates. Any additive kernel already present in the input is handled by linear algebra. The following proof accounts for these operations using the initial parameters d, InSize.

Use the original dimension and encoding bounds of Subsection 7.4, with a fixed number r of tensor positions.

Lemma 7.7 (restated). *Fix the number r of tensor positions. Under the dimension and encoding bounds above, all presentations and local operations in Algorithm 7.6 have a common bound*

$$p^{O_r(d)}\mathrm{poly}(\mathrm{InSize}, \log p),$$

measured from the initial call, provided every previously established local algorithm has this bound on original representation spaces of total dimension $O_r(d)$. The recursive calls retain subgroups of the original input group; their affine data are stored in the current quotient coordinates.

In particular, let G satisfy these original dimension and encoding bounds, let M be an explicit $\mathbb{F}_pG$-module, and let $E \le M^+ \rtimes G$ be given by generators as pairs, where M^+ is the additive group of M. Write $\pi : E \to G$ for projection, and include the additive coordinates and action matrices in InSize*. Let $S \le E$ be given by generators. Include in* InSize *the generator descriptions of E, S and any subgroup J below, and any linear correction equations. The following can be computed within* (7.7)*:*

(i) generators of $\pi(S)$ with lifts in S, and a basis of $S \cap M$ with words in the generators of S;

(ii) membership tests and generator compression in S, and generators of $S \cap \pi^{-1}(J)$ for a given subgroup $J \le \pi(S)$ given by generators;

(iii) solutions of linear correction equations in the computed additive kernels, with the corresponding group elements.

The translation space M is handled by linear algebra; its dimension enters the polynomial input factor.

Proof of Lemma 7.7. Tensor spaces and evaluation tables. By (D.11), the active factors and auxiliary modules have total dimension $O_r(d)$. Since the number of positions is fixed, their tensor-space dimension is polynomial in d times the initial passive multiplicity size. Numerators, denominators, quotient maps, and evaluations are computed by images, kernels, intersections, and inverse images in these spaces. When a factor is replaced by a subquotient of an auxiliary module, the entire kernel of the presenting surjection is included in the denominator. A completed classical or lower-degree call uses its own stated polynomial coordinate bound and returns transformations in the caller's original encoding.

The full set of evaluation tuples has size $p^{O_r(d)}$: a product of vector sets has size p raised to the sum of their dimensions, and there are only polynomially many tagged summand choices. Enumerating these tuples and performing exact linear algebra on their tables has cost (7.7). This is local work; the recursive division is on summand indices.

Affine data in quotient coordinates. Every recursive group is a subgroup H of the original input group Γ. The data attached to a generator h of H are its linear map L_h on the current quotient $Q = N/P$ and its translation $c_h \in Q$. Evaluate words using

$$(L_g, c_g)(L_h, c_h) = (L_gL_h, c_g + L_gc_h), \qquad (L_g, c_g)^{-1} = (L_g^{-1}, -L_g^{-1}c_g).$$

When a call restricts to a subgroup, evaluate its new generators as straight-line programs in the parent subgroup's generators, where the parent affine action is defined. Store the resulting affine data and retain separate words in the original generators for the output. A current quotient need only be invariant under its current subgroup, so its affine data are not evaluated on arbitrary generators of Γ. Shared word computations are stored once.

Coordinate changes transport the translations through the specified quotient isomorphisms. In a fiber call, the translations are $\alpha_k(x)-x \in \ker q$, as in Lemma 3.5. These operations use polynomially many field operations per word node in spaces of polynomial dimension in the input. All enumerated vector domains belong to the structural representations of total dimension $O_r(d)$; the vectors of Q itself are not enumerated. Thus affine word evaluation has cost (7.7), using the word-size bounds of the group operations and the stored coordinate maps.

R1 and R3 restrict subgroups and pass to quotient and fiber coordinates. R2 replaces a presentation through the equivariant quotient isomorphism of Fact 3.4. R4 computes in an image and restores the original subgroup elements by Lemma 3.6. These updates preserve the original group encoding. Temporary covers used in structural calculations or in a completed local solver are projected back after the full input comparison, before the calling recursion resumes. Their stated bounds use $O_r(d)$-dimensional structural representations. Consequently the recursive updates introduce no additive group extension.

An additive kernel in the input group. Let $B \trianglelefteq \Gamma$ be the elementary abelian kernel recorded in additive coordinates in the initial encoding; Γ/B is represented by the visible matrix and permutation actions. For $H \le \Gamma$, compute its image $\overline{H}$ in these actions. Their total degree is $p^{O_r(d)} + O(\mathrm{InSize})$. Compute a presentation of $\overline{H}$ and retain generator lifts in H. Evaluate its relators on those lifts and form the correction words for the given generators of H. By the kernel identity in Subsection 3.5, their closure under conjugation by H generates $H \cap B$. Since B is elementary abelian, compute this closure by enlarging a linear span inside B; at most $\dim_{\mathbb{F}_p} B$ strict enlargements occur. Keep word lifts for a basis. Membership in H is then tested by membership in $\overline{H}$, lifting an image word, and checking the remaining additive coordinate in $H \cap B$.

The extension construction of Luks [Luk92, Lemma 4.3] also gives a presentation of H: record the power and commutator relations for the additive basis, its conjugation by the chosen image lifts, and the actual additive values of the image relators. These corrected relations present H even when this input group extension is nonsplit. All operations have cost (7.7); the dimension of B contributes to the polynomial dependence on InSize. This is the fixed additive kernel allowed in the initial encoding.

The stated operations for a symbolic additive group. For the additional interface in the lemma, let $E_0 \le M^+ \rtimes G$ be given by generator pairs and put $G_0 = \pi(E_0)$. Compute a presentation of G_0, using the preceding construction if its encoding includes an additive kernel, and retain generator preimages in E_0. Evaluate the presentation relators and the generator-correction words on these preimages. Close their translation span under conjugation by the generators of E_0. The same kernel identity identifies this span with $E_0 \cap M$; at most $\dim_{\mathbb{F}_p} M$ strict enlargements occur. Store word lifts for its basis. Membership in E_0 is tested by lifting a word for the image in G_0 and testing its remaining translation by linear algebra. A lift of each compressed image generator, together with the kernel basis, generates E_0.

Apply this construction to $E_0 = S$ to obtain Item (i). For $J \le \pi(S)$, lifts in S of its generators together with the basis of $S \cap M$ generate $S \cap \pi^{-1}(J)$, giving Item (ii). Linear equations in the computed additive kernels give Item (iii), including word lifts of the required corrections. The translation space M is never enumerated, so its dimension enters only the polynomial input factor. This proves the additional interface independently of the recursive affine data.

Finally, original words share directed acyclic graphs. To compute a succinct generating set,

add an element only if it enlarges the represented subgroup, which occurs at most $\log_2 |\Gamma|$ times by (D.14). For a symbolic group $E \leq M^+ \rtimes G$, use instead $\log_2 |E| \leq \log_2 |G| + (\dim_{\mathbb{F}_p} M) \log_2 p$, which is also polynomial in the original input length. Charge each new graph node to the local operation that creates it. This includes witnesses and kernel lifts and proves the uniform bound. □

Theorem 7.1 (restated). *Given an instance of Definition 17 with original dimension d and input length* InSize*, its complete transporter can be computed deterministically in time $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$.*

Proof of Theorem 7.1. Initialize Algorithm 7.6 from Definition 17, with the original group, original active factors, and empty $\mathbf{R}, \mathbf{S}$. The exact updates preserve its solution set and its presentation class. The structural and conversion lemmas establish the required dimension and simple-section bounds. When the selected-summand count is bounded, Proposition D.5 computes the complete transporter at the local step. Proposition 7.8, with $r = 2$, now gives the full original transporter and the stated global running time.

This proves the basic quadratic algorithm using only the established classical solver and the structural results. Appendix D.6 verifies the additional initial-data conditions for the controlled extension stated in Subsection 7.6. □

D.5 Classical degree-2 quotient tensor module transporter

We give the detailed constructions and proof for the algorithm outlined in Section 7.5. Throughout, InSize is the total explicit input length and d counts the original natural prime-field dimensions.

Proposition 7.2 (restated). *There is a deterministic algorithm in time $p^{O(d)} \cdot \mathrm{poly}(\mathrm{InSize})$ for the degree-2 quotient tensor module transporter problem of Definition 17 when H is an explicit central product of classical groups in the prescribed stable range, with natural, dual, or Frobenius-twisted factor actions.*

For an input $H = D/C$, where D is the product of the specified classical factors, pull the input module and affine action back along $\pi_{\mathrm{cov}} : D \twoheadrightarrow H$. Write $\alpha_g(v) = L_g(v) + c_g$ for this action on $Q = N/P$. Write $c_g = c(g)$, and denote the induced linear actions on I and E by L_g^I and L_g^E, respectively. Store the cover map for the final projection of the transporter. The auxiliary target modules are constructed on D and need not descend to H. The cocycle c is zero on C, and all output witnesses and stabilizer generators are projected only after the original affine equation has been solved.

D.5.1 Plan of the classical-group solver

The classical-group procedure. The working group D is the full product of derived classical groups specified by the input, using the product cover when necessary. We apply Theorem 6.1 to this group and the invariant quotient $Q = N/P$. Its cohomology conclusion is used for D; we do not assume the same conclusion for an arbitrary subgroup returned later. The final degree-one solver, by contrast, accepts that returned subgroup with its original faithful action.

The two reductions before solving matrices. As explained in Section 7.5, an invariant quotient need not admit equivariant representatives, and its cocycle need not be a coboundary. The maps $Q \hookrightarrow I \xrightarrow{\pi} E$ address these two difficulties and give the following procedure.

(C1) Apply Theorem 6.1 to construct $\iota : Q \hookrightarrow I$ and $\pi : I \to E$, with their explicit matrix and quotient coordinates. The spaces have polynomial dimension in InSize, the target satisfies $H^1(D, E) = 0$, and $K = \ker \pi$ has a degree-one presentation using total natural prime-field dimension $O(d)$. For odd p, $E = I$ and $\pi = \mathrm{id}_I$.

(C2) Remove the projected translations by solving

$$\pi\iota(c_g) = (L_g^E - \mathrm{id}_E)b \quad (g \in \mathcal{S}_D,\ b \in E),$$

where $\mathcal{S}_D$ generates D. The projected problem is now $L_g^E(\pi\iota x + b) = \pi\iota y + b$.

(C3) Solve these conditions simultaneously as matrix and form equations, including correlated scalar conditions. Use an *algebra with involution*: an algebra with an adjoint operation reversing products and squaring to the identity.

(C4) Impose the determinant, spinor-norm, or Dickson-invariant conditions defining D inside the ambient full classical groups. Do this before evaluating a cocycle defined only on D.

(C5) Under the returned stabilizer, solve the residual affine equation in $\ker \pi \leq I$ by Proposition 4.5; restore the earlier witness and project through the central cover in the input.

Subsubsections D.5.2, D.5.3, and D.5.4 apply this shared reduction, solve the matrix equations, and restrict to D, respectively. The matrix step includes a degree-1 correction making polar-form isometries preserve the quadratic forms in characteristic 2. This differs from the final degree-1 correction in $\ker \pi$, which imposes the original affine equation by Lemma 6.6 and Proposition 4.5. That correction is made after restriction to D, the domain of the cocycle.

The simultaneous matrix step uses the algebra-with-involution framework of [BW12, IQ19], as developed in Subsection 4.2. The unitary generators and norm-equation solutions are constructed in Subsection 4.5, using Lemma 4.3 and the subsequent radical lifting. Corollary 4.7 records the complete transporter output for simultaneous matrix and form equations in characteristic 2; the odd-characteristic matrix step is the algorithm of [IQ19, Sections 3.3–3.5]. The concrete equations used here are encoded on prime-field spaces of total dimension $O(d)$ in Subsubsection D.5.3, so these computations take $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$ time. The scalar-shift choices, subgroup restrictions, and quotient affine corrections are carried out explicitly below.

D.5.2 Applying the shared representation theorem

The input classical actions and the based subspaces $P \leq N \leq Z$ satisfy the hypotheses of Theorem 6.1. Apply that theorem to obtain the equivariant maps

$$Q = N/P \xhookrightarrow{\iota} I \xrightarrow{\pi} E, \qquad K = \ker \pi.$$

The output includes the explicit matrix and tensor-square quotient presentations of E, with their scalar identifications and field actions, as well as a degree-one presentation of K. These are the data used by the matrix comparison and the final fiber comparison, respectively. The construction of the maps and the verification of $H^1(D, E) = 0$ are contained in the shared theorem; they need not be repeated for each local call.

The degree-one calls below use Proposition 4.5. To see that it applies, choose a basis in each trivial multiplicity space and flag the resulting degree-one ambient module Z_1 by its individual

natural, dual, twisted, or trivial copies. Each flag factor has prime-field dimension $O(d)$. For an invariant subquotient N_1/P_1 the induced flag is

$$Q_j = ((N_1 \cap Z_{1,j}) + P_1)/P_1.$$

Its nonzero factors are subquotients of the ambient flag factors, so they satisfy the same dimension bound. The construction uses only the triviality of the multiplicity actions, and remains valid for subgroups of the containing full classical groups as well as for subgroups of D. Use their faithful action on the original natural spaces, of total prime-field dimension $O(d)$. The proposition therefore gives the complete affine transporter in $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$ time, including its witness and full stabilizer.

Push the input cocycle forward to $g \mapsto \pi\iota(c_g) : D \to E$. By $H^1(D, E) = 0$, the generator equations in Step C2 have a solution $b \in E$, and the projected equation is

$$L_g^E(\pi\iota x + b) = \pi\iota y + b.$$

In odd characteristic this already imposes the original equation, since $\pi = \mathrm{id}_I$. In characteristic 2, solve it first to obtain its complete transporter aS inside D. Lemma 6.6 then constructs the remaining affine equation in K under S. The degree-one presentation from Theorem 6.1 uses total natural prime-field dimension $O(d)$ and fixed multiplicities of polynomial explicit size, so Proposition 4.5 solves this equation within the claimed bound. No vanishing of $H^1(S, E)$ is needed.

D.5.3 Solving the projected classical equation

The input consists of the two points $\pi\iota x+b$ and $\pi\iota y+b$ in the $\mathbb{F}_p[D]$-module E of Subsubsection D.5.2, after the change of origin in Step C2. The output consists of the complete ambient solution cosets for all scalar-shift tuples satisfying the specified correlated constraints. Subsubsection D.5.4 restricts these answers to the specified product and reunites them.

Use the actions of the containing full classical groups on the target coordinates constructed by Theorem 6.1. Pair natural representations with their duals where necessary. The product of the containing full bilinear and Hermitian isometry groups is the unitary group of one represented semisimple $\mathbb{F}_p$-algebra $\mathcal{R}$ with involution. For a characteristic-2 quadratic orthogonal factor $\mathrm{O}(V, Q_V)$, with $Q_V : V \to \mathbb{K}$, write $b_{Q_V}(u, v) = Q_V(u + v) - Q_V(u) - Q_V(v)$ for $u, v \in V$. Use its polar symplectic group $\mathrm{Sp}(V, b_{Q_V})$ at this stage, and impose the additional equation $gQ_V = Q_V$ on the quadratic form itself. Include the specified field-linearity constraints and all forms that define the classical groups in the simultaneous system. Whole bilinear tensors, external tensor products, and fixed Frobenius twists give simultaneous matrix equations on these representation spaces. For a tensor summand with twist exponents i, j, use the $\mathbb{F}_p$-algebra representations $a \mapsto a^{[i]}$ and $a \mapsto a^{[j]}$ of $a \in \mathcal{R}$ for its left and right actions. Here the bracketed powers apply the stated Frobenius automorphisms to matrix entries in the chosen bases. The block is thus a bimodule for the same algebra $\mathcal{R}$ on the input representation space, shared by the twisted actions. Choose matrix representatives of the source and target classes. For a tensor square modulo its invariant pairing line, or an adjoint module modulo scalars, equality after applying g is a matrix equation modulo a scalar. For example, conjugation on $\mathrm{End}_{\mathbb{K}}(V)/\mathbb{K}I$ gives

$$gAg^{-1} - B = \lambda I \quad \Longleftrightarrow \quad gA - Bg = \lambda g, \qquad A, B \in \mathrm{End}_{\mathbb{K}}(V), \quad \lambda \in \mathbb{K}.$$

Collect all scalar shifts at once using Lemma 6.7, and select exactly the complete tuples belonging to the specified correlated scalar subspace. For each fixed scalar-shift tuple, encode the simultaneous matrix and bilinear equations as follows. View the original natural spaces over $\mathbb{F}_p$ and put

$W = \bigoplus_i (V_i \oplus V_i^*)$, with all duals in this display taken over $\mathbb{F}_p$. Then $\dim_{\mathbb{F}_p} W = O(d)$. Let J be the standard hyperbolic pairing matrix on W. Include J and CJ among the fixed matrices for every block projector and field-multiplication map C: together, $gJg^{\mathrm{t}} = J$ and $g(CJ)g^{\mathrm{t}} = CJ$ enforce $gC = Cg$. These constraints preserve each natural block and force the corresponding dual block to carry the inverse-transpose action. Encode matrix tensors on the corresponding blocks of W. For a defining-field-valued bilinear or Hermitian form, use its $\mathbb{F}_p$-coordinates obtained from a trace-dual basis; preserving these coordinates is equivalent to preserving the original form. The fixed coordinate Frobenius maps identify twisted occurrences with the original prime-field representations. Thus additional coordinates and twists enlarge only the matrix tuple, not the dimension of W. In characteristic 2, apply Corollary 4.7 with an empty quadratic tuple to the augmented matrix and polar-form equations. In odd characteristic, apply the augmented-equivalence reduction of Subsection 4.2, whose paired-algebra identities are characteristic-independent, and then the represented-algebra norm and unitary-generator construction of Subsection 4.5; quadratic forms are determined by their polars. To make the complete output explicit, write the current source tuple as $\mathbf{B}$ and the target as $\mathbf{A}$. In the notation of Subsection 4.2, a norm solution $x_0^* x_0 = c^* c$ gives the witness $a = \pi_1(c x_0^{-1})$, where π_1 is first-component projection, and generators of $\mathrm{U}(F)$ give generators of $S = \pi_1(\mathrm{U}(F))$. A unitary pair is $(g, g^{-\mathrm{t}})$, so the equations defining F identify S with the full stabilizer of $\mathbf{B}$. Thus the matrix transporter is exactly aS; failure of either the equivalence or norm step returns emptiness. The same description applies to the characteristic-2 output above. The characteristic-2 quotient zero-polar correction is performed next. Quadratic equations are included through their polars, followed by one simultaneous degree-1 correction for forms with zero polar form. The ordered quadratic tuple includes each pair (Q_V, Q_V) from the characteristic-2 orthogonal factors, as well as all other source and target quadratic forms in the current fixed scalar-shift tuple. The coordinates for (Q_V, Q_V) must match exactly: if P_{quad} is the allowed correlated denominator in the original zero-polar coordinates, the enlarged tuple has denominator $P_{\mathrm{quad}} \oplus 0$. After the simultaneous matrix and polar equations have been solved, let aS be the complete returned coset, and let $\mathbf{f}, \mathbf{f}'$ be the source and target quadratic tuples for the current scalar branch. Write $\mathcal{Z}$ for the space of tuples of zero-polar quadratic forms and $P' = P_{\mathrm{quad}} \oplus 0$ for its allowed denominator, which is invariant under the ambient group. For $s \in S$, the vectors

$$e_s = s\mathbf{f} - \mathbf{f} + P', \qquad w = a^{-1}\mathbf{f}' - \mathbf{f} + P'$$

belong to $\mathcal{Z}/P'$. The maps $\beta_s(u) = su + e_s$ form an affine action of S, and

$$as\mathbf{f} - \mathbf{f}' \in P' \quad \Longleftrightarrow \quad \beta_s(0) = w.$$

Over a finite field of characteristic 2, a zero-polar quadratic form is a sum of scalar multiples of coordinate squares. Its coefficient space is a Frobenius-twisted dual natural module. Thus $\mathcal{Z}/P'$ has the degree-one flag described above, and Proposition 4.5 computes this correction under the full returned subgroup, with all representation kernels. Restore the polar witness a. The resulting complete coset preserves every Q_V and lies in the intended product of full quadratic orthogonal groups on those factors. Perform this correction before imposing the derived-group characters. These added form equations use the original natural spaces; their zero-polar differences are twisted-dual degree-1 coordinates with fixed multiplicities, so the total faithful realization dimension remains $O(d)$.

For an orthogonal factor, the target coordinates constructed by Theorem 6.1 are copies of $(V \otimes_{\mathbb{K}} V)/\mathbb{K}\omega$, as specified in Definition 15. Their comparison is already among the matrix equations modulo the invariant pairing line. The natural and scalar coordinates removed by the projection belong to $\ker \pi$ and are handled in the final affine fibre.

Additive residual equations in a fixed scalar branch. The remaining residual formulas concern symplectic factors in characteristic 2. Let $\dim_{\mathbb{K}} V = 2m$, let β be its symplectic form, and put $T = V \otimes_{\mathbb{K}} V$ and $A = \mathrm{im}(1+\tau)$, where $\tau(v \otimes w) = w \otimes v$. Write $\omega \in A$ for the invariant pairing tensor and $\ell : A \to \mathbb{K}$ for the invariant linear map $\ell(v \otimes w + w \otimes v) = \beta(v, w)$. The target coordinate I_0 of Definition 15 is

$$I_0 = \begin{cases} T/\ker \ell, & m \text{ odd}, \\ (T \oplus T)/\{(a, \ell(a)\omega) : a \in A\}, & m \text{ even}. \end{cases}$$

For representatives $Y, Y', Z, Z' \in T$, the odd case requires $gY - Y' \in A$ and $\ell(gY - Y') = 0$. The even case requires

$$gY - Y' \in A, \qquad gZ - Z' = b\omega, \qquad b = \ell(gY - Y'), \qquad b \in \mathbb{K}.$$

The class in T/A is a quadratic-form coordinate. First match this class by the preceding form procedure and, in the even case, fix the scalar shift b of Z. Keep the entire scalar-shift tuple fixed.

For a resulting transporter coset aS, every $s \in S$ satisfies $sY - Y \in A$. Since ℓ is invariant under the symplectic group, $\delta(s) = \ell(sY - Y)$ is additive on S, and

$$\ell(asY - Y') = \ell(aY - Y') + \delta(s).$$

Thus impose $\delta(s) = -\ell(aY - Y')$ in the odd case and $\delta(s) = b - \ell(aY - Y')$ in the even case. Impose all scalar and trace conditions jointly by Lemma 3.1 before reuniting the branches.

D.5.4 Restricting to the specified derived group

Return from the ambient full classical groups to the specified derived factors D. For SL, impose determinant one; for SU, impose unitary determinant one. The stable symplectic group needs no additional character condition. For odd-characteristic Ω, impose determinant one and spinor norm one; for characteristic-2 Ω, impose Dickson invariant zero. These defining characters and their kernels are given in Lemma 6.8.

Let $\mathcal{G}$ be the product of ambient classical groups on the input factor representations, and let χ be the product of these characters on $\mathcal{G}$. Its image has size $p^{O(d)}$: the determinant groups have order at most the product of the original field sizes, and each orthogonal factor adds at most four character values. For an ambient answer aS, where $a \in \mathcal{G}$ and $S \leq \mathcal{G}$, the permitted elements satisfy $\chi(s) = \chi(a)^{-1}$ for $s \in S$. Orbit and kernel computation on the $p^{O(d)}$-element character image (Subsection 3.5) gives either emptiness or $as_0 \ker(\chi|_S)$.

This filtering places every remaining witness and generator in D, where the input cocycle is defined. Constructive membership in D expresses these elements as words in its input generators before the cocycle is evaluated. The preceding ambient equations use the two points after the change of origin and the equations for preserving the quadratic forms. For a central-product image, these computations take place on its input product cover. Store the quotient map and generators of the full central kernel for projection after the final affine comparison. The shift cases exhaust the projected equation, and each returns its full stabilizer. Identity (3.1) therefore gives their complete reunion before the final affine fiber is formed.

D.5.5 Correctness and running time

Proof of Proposition 7.2. For a central-product input, pull the module and cocycle back to the input product cover, and write D for that working product until the final projection. The pulled-back

cocycle is zero on the covering kernel. All auxiliary representations and cohomology calculations in the construction are on this product; their descent to the input image is unnecessary.

For Steps C1–C2, apply Theorem 6.1 as described in Subsubsection D.5.2. It constructs $\iota : Q \hookrightarrow I$ and $\pi : I \to E$, with $H^1(D, E) = 0$ and a degree-one presentation of $K = \ker \pi$ on total natural prime-field dimension $O(d)$. Solve the generator equations for the change of origin $b \in E$. Lemma 6.6 then reduces the original affine comparison to its projected linear comparison and the remaining affine equation in K.

In Step C3, Subsubsection D.5.3 solves the resulting linear transporter in the containing full groups. Its simultaneous constraints enforce field linearity, the correlated scalar relations, and exact preservation of the forms defining the classical groups. Every scalar-shift branch is kept fixed through its additive residual conditions. Step C4 then uses Subsubsection D.5.4 to impose the specified character conditions and reunite all projected answers inside this call. If the filtered answer is empty, so is the original affine transporter. Otherwise write the filtered answer as aS, with $a \in D$ and $S \leq D$ the full stabilizer of the projected source in the $\mathbb{F}_p[D]$-module E.

For Step C5, apply Lemma 6.6 to this complete coset aS, with $a \in D$ and $S \leq D$. Only now evaluate the original cocycle $c : D \to Q = N/P$ on the required witnesses and subgroup generators. The lemma computes the residual affine equation in $\iota(Q) \cap \ker \pi \leq I$ and proves that testing it in the projection kernel $\ker \pi \leq I$ gives exactly the same solutions. The degree-1 presentation constructed by Theorem 6.1 therefore permits Proposition 4.5 under the returned subgroup $S \leq D$. Restore the projected witness $a \in D$ to the residual answer. Lemma 6.6 gives either emptiness or one original witness and generators of the full affine source stabilizer. This final fiber is distinct from the earlier zero-polar correction: the latter imposed the quadratic-form equations before character filtering, whereas this fiber imposes the original affine equation after filtering.

Running time and full output. All module dimensions remain polynomial; the exponent uses only a fixed number of copies of the original natural representation spaces. There are at most $4^{O(d)}$ shift tuples and $p^{O(d)}$ finite quotient cases, all completed and reunited inside this call. The caller therefore receives a completed answer. Matrix-span and Hom calculations use linear algebra. For the original central-product input $H = D/C$, let $\pi_{\mathrm{cov}} : D \twoheadrightarrow H$ be the specified cover, α_H the input affine action, and $\widetilde{\alpha} = \alpha_H \circ \pi_{\mathrm{cov}}$ its pullback. Then, directly from this definition,

$$\mathrm{Trans}_{D,\widetilde{\alpha}}(x, y) = \pi_{\mathrm{cov}}^{-1}\big(\mathrm{Trans}_{H,\alpha_H}(x, y)\big), \qquad D_x = \pi_{\mathrm{cov}}^{-1}(H_x).$$

Consequently the covering kernel is contained in the final stabilizer, and projecting the returned witness and stabilizer generators gives exactly the complete transporter in H. This argument uses only the pulled-back action on the input quotient; no descent of the auxiliary modules or lift of the cocycle to a numerator is required. □

D.6 Controlled degree-2 quotient tensor module transporter

We prove the results of Subsection 7.6, using its fixed constants and notation.

Actions and exact quotient changes. Put $T_\sigma = \bigotimes_k W_{\sigma,k}$ and let $\eta_{\sigma,j}$ denote the action on $E_{\sigma,j}$. The action used above is

$$(\rho_Z(h)f)_\sigma = \rho_{T_\sigma}(h) \circ f_\sigma \circ \left(\bigotimes_j \eta_{\sigma,j}(h)^{-1} \right).$$

For the realisation in Corollary 7.10, let $q_0 : U_0 \twoheadrightarrow M$ be the map induced by θ_0, with kernel K_0. Tensor q_0 with the other factors and pull back the full numerator and denominator through

this surjection. The pulled-back denominator contains its kernel, and the resulting quotient is equivariantly isomorphic to the original one. This is precisely the construction of Fact 3.4; it preserves relations between different summands as well as relations within a summand.

For an argument $A = U/F$, where $F \le U \le R$ are invariant and R is a catalogue module or its dual, use

$$\mathcal{D} = \{g \in \mathrm{Hom}_{\mathbb{F}_p}(R, T) : g(F) = 0\}, \qquad q : \mathcal{D} \twoheadrightarrow \mathrm{Hom}_{\mathbb{F}_p}(A, T), \quad q(g)(u + F) = g(u).$$

Restriction and descent make q equivariant. It is surjective by linear extension from U to R, and its kernel consists of the maps vanishing on U. Thus N_A/P_A becomes exactly $q^{-1}(N_A)/q^{-1}(P_A)$. With several arguments, restrict to the product of the U-spaces and require the restriction to vanish whenever one argument lies in its F-space. The resulting multilinear restriction-and-descent map is again an equivariant surjection. Apply the direct sum of these maps to the entire quotient presentation. Transport the generator translations through the quotient isomorphism; the cocycle identity is required only in that quotient, not in its numerator. All these changes use linear algebra in polynomial-size spaces for fixed r.

Alternating-dominance and inherited control. The joint auxiliary image has a faithful representation of dimension $n = \sum_t \dim R_t + \sum_\ell e_\ell \le \kappa_r d$; the permutations of nonzero summands are determined by this representation. Every cyclic simple section has prime order dividing $|\mathrm{GL}_n(p)|$, hence at most $p^n \le T_{\mathrm{cl}}$. Consequently, in the controlled definition it is equivalent to bound only the nonalternating *nonabelian* simple sections. This identifies the condition with the alternating-dominant terminology of Section 5.

Restriction to a subgroup preserves the summand-stabilizer order bounds and all simple-section exclusions. The new alternating-dominant modules produced by the structural step satisfy the same types of bounds by Lemma 7.4. A simple section of a subgroup of a direct product is a section of one of its coordinate groups: intersect with the kernel of one projection and use simplicity, then induct. Thus adjoining these modules preserves the joint exclusion. The semilinear line images have order at most $e_\ell(p^{e_\ell} - 1)$; their total contribution is $p^{O(d)}$.

Proposition 7.11 (restated). *With at most r positions, the controlled input and a local algorithm with the stated complete-output and single-fiber guarantees give a complete transporter algorithm in time $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$.*

Proof. Apply the fixed-position recursion bound of Proposition 7.8. In the factor presentation of Subsection 7.2, replace [2] by $[r]$: the selections are $S_{i,\nu} \subseteq I_\nu$ for $i \in [r]$ and $\nu \in \mathcal{B}$, and coordinate projections select whole tensor blocks. An argument in the Hom presentation contributes its dual factor. Adjoin each catalogue module and the required duals to the formal representation list, at most once per position rather than once per occurrence. The total representation dimension, initial number of summand indices, and remaining active dimension are $O(d)$. The root choice of C_* absorbs this enlarged budget.

The preceding hereditary argument verifies the additional group-action hypotheses. For fixed r, flag combinations and all numerator and denominator matrices have size polynomial in the explicit input. Evaluation enumerates products of vector domains of total dimension $O(d)$, not arbitrary coefficient arrays. The local size bound of Lemma 7.7 therefore applies. These are exactly the additional input and size hypotheses of Proposition 7.8; its assumed local-solver guarantee is the one in the statement. That proposition gives the running time and preserves the full original transporter through its exact quotient, fiber, and coset operations. The dimension and summand-index count use only the dimension and summand-stabilizer bounds. The simple-section condition

is needed for classical separation at the local step. Hence the same count applies under the weaker exclusion of Remark 3 whenever the local algorithm has the stated guarantees. □

Theorem 7.9 (restated). *The complete degree-2 transporter with controlled evaluation arguments is computable deterministically in time* $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$.

Proof. Apply Proposition 7.11 with at most two active factors. The argument modules enter with their specified decompositions; they are not reclassified as unrestricted active factors. Their maps, including any quotient changes above, remain in the exact numerator and denominator. We verify the local-solver hypothesis.

Let $G \le H$ be the current subgroup. Write $\rho : G \to A$ for its action on the selected classical modules and $\beta_G : G \to B$ for its action on the selected auxiliary summands and semilinear lines; both codomains are the actual images. Thus here B is the joint auxiliary image for this call. Put $J = (\rho, \beta_G)(G) \le A \times B$. Let $D \trianglelefteq A$ be the selected classical product. Each simple central quotient of a factor of D is nonalternating and exceeds T_{cl}. The auxiliary exclusion and Lemma 5.6 give $D \times 1 \le J$. This is a statement about the simultaneous image: it requires no complement in the original group.

There are $s = O(1)$ selected summands. The permutation action on them has order at most $s!$, and its kernel acts through their summand-stabilizer images B_a. Hence

$$|B| \le s! \prod_{a \text{ selected}} |B_a| \prod_{\ell \in \mathcal{L}} e_\ell(p^{e_\ell} - 1) = p^{O(d)}.$$

Lemma 7.5 gives $[A : D] \le p^{O(d)}$, so $[J : D \times 1] \le [A : D]|B| = p^{O(d)}$. This bound uses the preceding index recursion; the full auxiliary image before that recursion can be much larger.

First quotient out translations of the kernel of the map to J, recording its original preimages, by Lemma 3.6. Enumerate the actual cosets of $D \times 1$ in J and normalize the target in each. On $D \times 1$, all argument factors are fixed pointwise. Evaluating on their bases therefore identifies the module with fixed copies of tensor modules having at most two moving active factors. Keep the exact images of the numerator, denominator, and quotient-valued affine action under this identification. Proposition 7.2 computes the complete classical transporter in each coset; the number of fixed copies contributes only to the explicit input length. Reunite all coset answers and restore the kernel translations and original preimages. This gives a complete local answer within the required bound, so Proposition 7.11 finishes the proof. □

Remark 3 (A weaker separation hypothesis)**.** Use the local notation G, A, B, J, D from the preceding proof, and write D as a central product of quasisimple factors C_i. Theorem 7.9 still holds if alternating-dominance is replaced by the requirement that, at every local call,

$$B_0/K \ncong C_i/\mathrm{Z}(C_i) \quad \text{for all } i,\ B_0 \le B,\ K \trianglelefteq B_0.$$

In other words, no simple section of the action on the selected auxiliary spaces may be one of the selected classical simple groups. The dimension and summand-stabilizer bounds are still required. The condition concerns the factors selected in each call, including those arising after subgroup restriction.

This is exactly the hypothesis of Lemma 5.6: it gives $D \times 1 \le J$, so every $d \in D$ has an original preimage fixing all selected auxiliary spaces. Under alternating-dominance the condition is automatic, since each $C_i/\mathrm{Z}(C_i)$ is nonalternating and has order greater than T_{cl}. The recursion count uses only the dimension and summand-stabilizer bounds; hence Proposition 7.11 also holds under this weaker condition whenever its local algorithm has the stated guarantees.

Corollary 7.10 (restated). *Let V be an original H-module and let M have a given equivariant realisation as a subquotient of a direct sum of tensor products of at most $r-2$ controlled catalogue modules or their duals. For injections $f, g : E \hookrightarrow V \otimes M$ with $\dim E = O(d)$, the full H-transporter from $\operatorname{im} f$ to $\operatorname{im} g$ is computable in time $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$, in the original-group encoding.*

Proof. Let $H \times \mathrm{GL}(E)$ act on the injections by $(h, a)f = hfa^{-1}$. In $\mathrm{Hom}_{\mathbb{F}_p}(E, V \otimes M) \cong V \otimes M \otimes E^*$, the two active factors are V and E^*. Expand the catalogue realisation of M and move its catalogue factors to arguments, pulling back the entire quotient as above. There are at most r positions in each summand. If M is trivial, its basis simply indexes fixed copies. The additional original representation on E increases the dimension budget by $O(d)$, and $\mathrm{GL}(E)$ acts trivially on the catalogue. The hypotheses of Theorem 7.9 therefore hold.

For each $h \in H$ carrying $\operatorname{im} f$ to $\operatorname{im} g$, there is a unique $a \in \mathrm{GL}(E)$ with $hf = ga$, because f and g are injective. Projecting the complete transporter in $H \times \mathrm{GL}(E)$ onto H thus gives exactly the desired subspace transporter, including its full stabilizer and all original action kernels. □

E Technical results for 3-tensor isomorphism and degree-3 quotient tensor module transporter

We give the constructions and proofs for the group and tensor reductions of Section 8, followed by the proof and supporting results for the degree-3 quotient tensor module transporter of Subsection 8.3.

E.1 Group reductions

We justify Steps G1–G7 in their current order, then combine their correctness and cost bounds to prove Proposition 8.4. The current task is $g\,\mathrm{Trans}_{H,\alpha}(x, y)$, where $\alpha_h(v) = \rho_h(v) + c_h$ and the input data are those of Definition 24. We consider all inputs encountered during Algorithm 8.6, starting from the initial comparison (8.1), including those passed from Steps T1–T4 back to Step G1.

Unless a field is specified, dimensions, tensor products, and linear maps are over $\mathbb{F}_p$. The parameter $d = 2e(n_1 + n_2 + n_3)$, the initial input bit length InSize, and the threshold $T_{\mathrm{cl}} = p^{C_* d}$ remain fixed. The current reference dimensions are $a_i \geq \dim A_i$; only Step G3 assigns new reference dimensions to a child call.

For the local bounds in Steps G1–G7, let $\mathrm{InSize}_{\mathrm{cur}}$ be the bit length of the current explicit input, including the module presentations and generator actions supplied to the step. This length may change between calls. We first bound each step in terms of $\mathrm{InSize}_{\mathrm{cur}}$. In Lemma E.23(iv), we then prove a uniform bound on $\mathrm{InSize}_{\mathrm{cur}}$ in terms of the initial length InSize and use it to express the costs in terms of InSize.

E.1.1 Step G1: comparisons along the argument flags

Lemma 8.3 recovers the original transporter from comparisons on successive quotients of the induced flag in N/P. To use this reduction in Step G1, we must give each quotient an input of the form in Definition 24: argument spaces, invariant subspaces $P' \leq N' \leq Z'$, and an affine action on N'/P'.

To prepare the flags, apply Theorem 5.1 to the successive quotients of the recorded flags in the full original spaces U_i^*. This refines them to composition flags. On a current argument $A_i = N_i/P_i$, write the induced flag as

$$0 = F_{i,0} < F_{i,1} < \cdots < F_{i,r_i} = A_i.$$

The same theorem supplies a decomposition of each successive quotient into blocks permuted by H. Apply Theorems 5.2 and 5.3 to the joint action on all the original flag quotients, including those outside the current arguments. Their total dimension is at most d. These computations take $p^{O(d)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$ time. The homomorphisms from H, their kernels, and preimages in H of computed image-group elements are stored as in Section 3.5.

The following lemma constructs these subquotient presentations and affine actions. It specifies how the answer for one quotient determines the comparison on the next, and when the new argument dimensions satisfy (8.3). It applies to all H-invariant subspaces $P \le N \le Z$ in Definition 24.

Lemma E.1 (Recursive inputs for successive flag quotients)**.** *Consider a current input with H-invariant subspaces $P \le N \le Z$ and affine action α. Form the chain $0 = \mathcal{Z}_0 < \cdots < \mathcal{Z}_t = Z$ from the argument flags as in Lemma 8.3, and set*

$$Q_j := ((N \cap \mathcal{Z}_j) + P)/P \quad (0 \le j \le t).$$

For $1 \le j \le t$, let $\eta_j : \mathcal{Z}_j \twoheadrightarrow E_j := \mathcal{Z}_j/\mathcal{Z}_{j-1}$ and put

$$\overline{N}_j := \eta_j(N \cap \mathcal{Z}_j), \qquad \overline{P}_j := \eta_j(P \cap \mathcal{Z}_j).$$

Then there is an explicitly computable H-equivariant isomorphism

$$\psi_j : Q_j/Q_{j-1} \xrightarrow{\sim} \overline{N}_j/\overline{P}_j, \qquad (u + P) + Q_{j-1} \longmapsto \eta_j(u) + \overline{P}_j \quad (u \in N \cap \mathcal{Z}_j). \tag{E.1}$$

In particular, $P = 0$ implies $\overline{P}_j = 0$. The subspace maps and the induced affine actions are computable by linear algebra. Complete answers to the comparisons on the nonzero quotients Q_j/Q_{j-1}, processed in decreasing order of j, recover the original transporter and its full source stabilizer. For fixed reference dimensions $a_i \ge \dim_{\mathbb{F}_p} A_i$, at most one quotient E_j consisting of trilinear maps has argument dimensions that fail (8.3)*.*

Proof. The map ψ_j realizes Q_j/Q_{j-1} as the subquotient $\overline{N}_j/\overline{P}_j$ of E_j. The map from $N \cap \mathcal{Z}_j$ to $\overline{N}_j/\overline{P}_j$ has kernel

$$(N \cap \mathcal{Z}_{j-1}) + (P \cap \mathcal{Z}_j),$$

which gives exactly Q_j/Q_{j-1}. All maps commute with H. To compute $\psi_j((v + P) + Q_{j-1})$ for $v + P \in Q_j$, solve for $v_P \in P$ such that $v - v_P \in N \cap \mathcal{Z}_j$, and apply η_j to $v - v_P$. The quotient makes the answer independent of that choice.

When E_j is a space of trilinear maps, choose $\ell_i \in \{1, \dots, r_i\}$ to specify the successive quotient of the flag on each A_i. Then

$$E_j \cong \mathrm{Hom}(A_1' \otimes A_2' \otimes A_3', M), \qquad A_i' = F_{i,\ell_i}/F_{i,\ell_i-1}.$$

Using the input map $\pi_i : N_i \twoheadrightarrow A_i$, their presentations in U_i^* are

$$N_i' := \pi_i^{-1}(F_{i,\ell_i}), \qquad P_i' := \pi_i^{-1}(F_{i,\ell_i-1}), \qquad N_i'/P_i' \cong A_i'.$$

The subspaces $\overline{P}_j \le \overline{N}_j \le E_j$ give the required presentation of the quotient $\overline{N}_j/\overline{P}_j$. If $E_j = Z_{\mathrm{quad}}$, use its given controlled degree-2 presentation instead.

We now specify the affine data and their updates. Center the current comparison at its source x, writing

$$\delta_h := \alpha_h(x) - x \quad (h \in H), \qquad w := y - x.$$

Then $\delta_h \in N/P$ and $\delta_{hk} = \delta_h + \rho_h(\delta_k)$. Initialize $(g_t, H_t, w_t) = (g, H, w)$. At index j, the remaining comparison on Q_j has source 0, target w_j, and action $v \mapsto \rho_h(v) + \delta_h$ for $h \in H_j$. On the quotient,

use (E.1); its translation and target are respectively $\psi_j(\delta_h + Q_{j-1})$ and $\psi_j(w_j + Q_{j-1})$. If its answer is empty, the original answer is empty. Otherwise write the complete answer, including its left multiplier, as $g_j u_j H_{j-1}$. The source stabilizer and the next target satisfy

$$\begin{aligned} H_{j-1} &= \{h \in H_j : \delta_h \in Q_{j-1}\}, \\ g_{j-1} &= g_j u_j, \\ w_{j-1} &= \rho_{u_j^{-1}}(w_j) + \delta_{u_j^{-1}} \in Q_{j-1}. \end{aligned}$$

The remaining action is the restriction of the same centered action to Q_{j-1}. These are precisely the transporter and stabilizer identities of Lemma 8.2. Zero quotients are skipped with unchanged data, and the final answer is $g_0 H_0$. This proves that these successive comparisons recover the full transporter.

Finally, the dimensions of the successive quotients of the flag on A_i sum to $\dim A_i \le a_i$. At most one such quotient can have dimension greater than $127a_i/128$. Hence at most one triple of argument quotients can fail (8.3). □

The construction above shows that $P = 0$ gives $\overline{P}_j = \eta_j(P \cap \mathcal{Z}_j) = 0$ in every quotient comparison. The trilinear maps still take values in M. We now check the corresponding properties for all steps of the algorithm, including Steps T1–T4. The resulting global invariant is used in the degree-2 presentation of Step G2, the cost of combining cosets in Step G4, and the change of origin in Step G6.

Lemma E.2 (Properties preserved by the recursion)**.** *Every tensor comparison encountered during Algorithm 8.6, starting from the initial comparison* (8.1)*, satisfies*

$$P = 0, \qquad Z_{\mathrm{quad}} = 0, \qquad H \text{ acts trivially on } M. \tag{E.2}$$

These assertions concern the inputs to the tensor procedure. They do not require $P_i = 0$ in $A_i = N_i/P_i$, or $P' = 0$ in a quotient N'/P' used by an auxiliary degree-2 comparison.

Proof. The assertions hold initially with $M = \mathbb{F}_p$. For Step G1, Lemma E.1 shows that $P = 0$ gives $\overline{P}_j = 0$ for every j. Its construction keeps M, and every E_j is a space of trilinear maps when $Z_{\mathrm{quad}} = 0$. Thus Step G1 preserves all three properties. Subgroup restriction and changes of origin also preserve them. Step G4 sends P to its restriction image, and Step G5 sends it to $\Phi(P)$. The latter step replaces M by $M^{\oplus m}$ with the coordinatewise action specified there, which is again trivial. The ordinary comparison (8.11) used in Step G7 has $P = Z_{\mathrm{quad}} = 0$ and keeps M. The tensor reductions either keep these data, complete a degree-2 comparison, or give the inputs (8.13) and (8.14), which also have $P = 0$ and keep M. Each step that produces another tensor comparison also keeps $Z_{\mathrm{quad}} = 0$ or uses only Z_{tri}. Thus every step preserves the three properties. All recursive outputs of the construction for Step G7 below use exactly these formulas: they restrict the original input to a candidate coset, or replace it by an ordinary comparison with the same argument and value spaces. This observation concerns the input formulas, independently of the proof that the construction covers all solutions. Thus each input update preserves the three properties, and induction from the initial input proves the assertion. □

E.1.2 Step G2: completion by the degree-2 algorithm

After taking x as the origin, Step G2 solves the current comparison when N/P, with its induced affine action, has a presentation satisfying Definition 23. The following lemma gives both the correctness of that call and the structural case in which the group reductions construct such a presentation.

Lemma E.3 (Reduction to the degree-2 transporter problem)**.** *Suppose the current input has an explicitly given equivariant isomorphism*

$$\phi : N/P \xrightarrow{\sim} N'/P', \qquad P' \le N' \le Z',$$

where the right-hand side satisfies Definition 23, with original representation dimension $O(d)$. Then Step G2 computes the complete answer, with its full source stabilizer, in time $p^{O(d)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$.

For a tensor input generated from (8.1)*, such a presentation can be constructed if the current arguments are irreducible and linearly primitive, meaning that no nontrivial direct-sum decomposition has its summands permuted by H, and at most two arguments have a large classical component. Its trilinear part needs at most three tensor positions and at most two active factors. Consequently, if Steps G1 and G4 leave irreducible primitive arguments, Step G2 does not obtain such a presentation, and Step G5 finds no repeated copies, then all three arguments are single natural modules for large classical components, with their specified duals or Frobenius twists.*

Proof. We first verify that changing the quotient presentation preserves the comparison. Recall locally that

$$\delta_h := \alpha_h(x) - x, \qquad w := y - x.$$

Centering at x gives source 0, target w, and action $v \mapsto \rho_h(v) + \delta_h$ on N/P. Under ϕ, these become source 0, target $\phi(w)$, and action

$$v \longmapsto \phi\rho_h\phi^{-1}(v) + \phi(\delta_h).$$

Its transporter is exactly $\{h \in H : \delta_h = w\}$. Theorem 7.9 computes this set and the full source stabilizer in H, represented as in Section 3.5. Restoring the left multiplier g proves the first assertion and its time bound.

For the structural assertion, we will present each argument without a large classical component as a subquotient of an H-module R_i, and verify that these modules form a controlled catalogue. They will serve as evaluation arguments in Definition 23; the remaining A_i^* give at most two active factors. For example, when A_2^*, A_3^* are the active factors, the proposed ambient module is

$$Z' = \mathrm{Hom}_{\mathbb{F}_p}(R_1, A_2^* \otimes A_3^*)^{\oplus \dim M}.$$

Here Lemma E.2 gives $Z_{\mathrm{quad}} = 0$ and a trivial action on M, so a basis of M indexes the $\dim M$ direct summands, each preserved by H. Here R_1 is an evaluation argument, so the three tensor positions contain only two active factors. The same construction uses several evaluation arguments when fewer than two factors are active. The cases below construct the R_i and verify their dimension bounds, summand-stabilizer bounds, and alternating-dominance.

Let $G_i \le \mathrm{GL}(A_i)$ be the image of the current H-action on A_i. These images are primitive by hypothesis. Here "bounded" means $|G_i| \le p^{c_0 \dim A_i}$, with c_0 the fixed constant of Section 5. For each unbounded image, let $S_i \trianglelefteq G_i$ be the quasisimple subgroup supplied by Theorem 5.2. An argument without a large classical component has the following auxiliary presentation. If G_i is bounded, take $R_i = A_i$ as a catalogue module with its one-summand decomposition and the identity quotient map. If S_i is classical and $|S_i/\mathrm{Z}(S_i)| \le T_{\mathrm{cl}}$, then

$$[G_i : S_i] \le p^{5 \dim A_i}, \qquad |S_i/\mathrm{Z}(S_i)| \le T_{\mathrm{cl}},$$

so $|G_i| \le p^{O(d)}$: the center acts by scalars on its defining-field natural module. Again take $R_i = A_i$ with its one-summand decomposition.

If S_i is alternating, Theorem 5.3(2) supplies integers $s_i \geq 8$ and $m_i \geq 1$, and an equivariant subquotient presentation of A_i in

$$R_i = \mathbb{F}_p^{s_i} \otimes \mathbb{F}_p^{m_i} = \bigoplus_{\nu=1}^{s_i} (\mathbb{F}_p e_\nu \otimes \mathbb{F}_p^{m_i}),$$

where $e_1, \ldots, e_{s_i}$ is the standard basis of $\mathbb{F}_p^{s_i}$. The first factor is the permutation module covering the fully deleted permutation module; the second records its multiplicity. The entire R_i is one catalogue module. Its dimension is at most $4 \dim A_i/3$. The setwise stabilizer in H of $\mathbb{F}_p e_\nu \otimes \mathbb{F}_p^{m_i}$ has image of order at most $p^{5 \dim A_i}$ on that subspace. The same theorem makes the action on R_i alternating-dominant. Lemma 7.4 gives the corresponding condition on the joint auxiliary image and on subsequent subgroup restrictions. Thus the at most three auxiliary presentations have total dimension $O(d)$ and the required summand-stabilizer bounds.

Dualizing these presentations of the A_i and using the R_i as evaluation arguments gives an ambient module Z' with at most three tensor positions and at most two active factors in each summand. The representations on the active factors, together with the original group representation needed for lifts, have total dimension $O(d)$.

To represent the original quotient N/P, these substitutions give an invariant subspace $\widetilde{U} \leq Z'$ and an equivariant surjection $q : \widetilde{U} \twoheadrightarrow Z$. Put

$$N' := q^{-1}(N), \qquad P' := q^{-1}(P).$$

The induced isomorphism $N'/P' \xrightarrow{\sim} N/P$ gives ϕ in the reverse direction. If $P = 0$, then $P' = \ker q$, which need not be zero. These are the exact quotient changes from Appendix D.6. Every tensor factor uses the action of the same element $h \in H$, and $q(h \cdot u) = h \cdot q(u)$ for $u \in \widetilde{U}$. The solver also receives the kernels of the action homomorphisms from H and lifts to H of image-group elements. We have verified the catalogue conditions and the bounds on the number of tensor positions and active factors in Definition 23; counting large natural occurrences alone would not verify these conditions. The expanded presentation has bit length $\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$: there are at most three tensor positions, the auxiliary spaces have total dimension $O(d)$, and the number of direct summands is $\dim M$. The structural computations and the linear algebra on these data take $p^{O(d)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$ time.

If Step G2 does not obtain such a presentation, all three primitive arguments have a large classical component. Theorem 5.2 expresses each as copies of its specified natural module. A multiplicity greater than one is the case of Step G5; if that step does not apply, every multiplicity is one. This proves the last assertion. □

E.1.3 Step G3: resetting the reference dimensions

Step G3 replaces the reference dimensions a_i by $a_i' = \dim A_i$ once (8.3) holds. The comparison data are unchanged. The lemma below bounds how many times this replacement can occur along a recursive path.

Lemma E.4 (Correctness and progress of a reference reset)**.** *Let a_i be the inherited reference dimensions and set $a_i' := \dim_{\mathbb{F}_p} A_i$ for the current arguments. Suppose*

$$a_i' \leq a_i \quad (i = 1, 2, 3), \qquad a_j' \leq 127 a_j/128 \quad \textit{for some } j \textit{ with } a_j > 0.$$

Passing the current comparison to a child call with reference dimensions a_i' preserves its complete answer and satisfies

$$\sum_i a_i' < \sum_i a_i, \qquad \prod_{i=1}^{3} (a_i' + 1) \leq \frac{255}{256} \prod_{i=1}^{3} (a_i + 1).$$

There are at most $O(\log(d+2))$ such resets along any path starting from the initial input.

Proof. The child receives exactly the current tuple

$$\mathtt{C}(gH; \mathbf{A}; Z, \rho; P, N, z; c; x, y).$$

Only its reference dimensions change, so its complete answer is the answer required by its parent. The parent and its other children keep their own reference dimensions. At the start of the child, argument and reference dimensions agree; another reset requires another decrease of an argument dimension.

For the index j in the hypothesis, $a_j \geq 1$ and $a'_j < a_j$. The other dimensions do not increase, proving the strict decrease of the sum. Moreover,

$$\frac{a'_j + 1}{a_j + 1} \leq \frac{127a_j/128 + 1}{a_j + 1} = 1 - \frac{a_j}{128(a_j + 1)} \leq \frac{255}{256}.$$

This proves the product bound. The initial dimensions are $O(d)$, and the product is always at least one, giving the logarithmic bound on the number of resets. This bound concerns resets; the termination and work of intervening calls with unchanged reference dimensions are treated separately in Lemma 8.7. □

E.1.4 Step G4: comparisons on permuted summands

Step G4 compares individual pairs of summands. We must combine their answers using one element of H that matches all summands simultaneously. The following lemma gives that combination and its cost.

Lemma E.5 (Reduction using permuted summands)**.** *Suppose Step G4 applies to an input reached from* (8.1)*, with the computed decomposition $A_k = \bigoplus_{\nu=1}^{s} D_\nu$, $s > 1$. These inputs have $P = Z_{\mathrm{quad}} = 0$ by Lemma E.2. From the complete answers to at most s^2 pair comparisons* (8.8)*, each computed once, the step computes $g\,\mathrm{Trans}_{H,\alpha}(x, y)$ as emptiness or a coset with its full source stabilizer. It stores one transporter for each of the 2^s subsets of $\{1, \dots, s\}$ and performs at most $s2^{s-1} - s$ coset intersections. Apart from solving the pair comparisons, its work is $p^{O(d)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$, and each resulting input has bit length $\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$. Every pair comparison replaces A_k by a space of dimension at most $\dim A_k/2$, so it satisfies* (8.3)*.*

Proof. Since $P = Z_{\mathrm{quad}} = 0$, we have $x, y \in z + N \subseteq Z$; the translations $c_h \in Z$ may still be nonzero. Put $I = \{1, \dots, s\}$. Let Z_ν be the trilinear module with A_k replaced by D_ν, and let $q_\nu : Z \to Z_\nu$ be restriction. For $h \in H$, define $h\nu$ by $hD_\nu = D_{h\nu}$. The induced map $\rho_{h,\nu} : Z_\nu \to Z_{h\nu}$ is

$$[\rho_{h,\nu}(F)](a_1, a_2, a_3) = h_M\big(F(h_{A_1}^{-1}a_1, h_{A_2}^{-1}a_2, h_{A_3}^{-1}a_3)\big), \qquad a_k \in D_{h\nu},$$

where h_{A_i} and h_M are the given actions on A_i and M. The identity $q_{h\nu}\rho_h = \rho_{h,\nu}q_\nu$ and the cocycle identity for c give an H-action on the disjoint union of the spaces $\{\nu\} \times Z_\nu$:

$$h \cdot (\nu, t) = \big(h\nu, \rho_{h,\nu}(t) + q_{h\nu}(c_h)\big).$$

The single-pair transporter is therefore

$$C_{\nu\omega} := \{h \in H : hD_\nu = D_\omega, \quad q_\omega(\alpha_h(x)) = q_\omega(y)\}.$$

It is empty or a coset of the stabilizer of $(\nu, q_\nu(x))$. If $rD_\nu = D_\omega$ and $K = \mathrm{Stab}_H(D_\nu)$, then

$$C_{\nu\omega} = r\{h \in K : q_\nu(\alpha_h(x)) = q_\nu(\alpha_{r^{-1}}(y))\}.$$

This is exactly the complete answer to (8.8), whose kth argument is presented by $(\pi_k^{-1}(D_\nu), P_k)$. If no such r exists, or that answer is empty, set $C_{\nu\omega} = \varnothing$.

All summands must be matched by the same h. Since their restrictions determine a trilinear map on $A_k = \bigoplus_\nu D_\nu$,

$$\mathrm{Trans}_{H,\alpha}(x, y) = \bigcup_{\pi \in \mathrm{Sym}(I)} \bigcap_{\nu \in I} C_{\nu,\pi(\nu)}.$$

To compute this set from the stored pair answers, let $D(J)$, for $J \subseteq I$, consist of the elements matching the first $|J|$ source summands to exactly the target summands indexed by J. Under the action above, it is the transporter from

$$\{(\nu, q_\nu(x)) : 1 \le \nu \le |J|\} \quad \text{to} \quad \{(\omega, q_\omega(y)) : \omega \in J\}.$$

Choosing the target ω of the last source summand gives

$$\begin{aligned} D(\varnothing) &= H, \\ D(J) &= \bigcup_{\omega \in J} \big(C_{|J|,\omega} \cap D(J \setminus \{\omega\})\big) \qquad (J \neq \varnothing). \end{aligned}$$

The intersection requires one common h for all matched summands; the union covers every possible target for the last one. As a set transporter, each nonempty $D(J)$ is one coset of the source set's stabilizer, so (3.1) combines its branches into a complete coset. Finally, $D(I) = \mathrm{Trans}_{H,\alpha}(x, y)$, and returning $gD(I)$ gives the complete answer, including the full source stabilizer.

Each of the at most s^2 pair comparisons is computed once. There are 2^s stored transporters $D(J)$; initializing $D(\{\omega\}) = C_{1\omega}$ directly leaves

$$\sum_{j=2}^{s} j \binom{s}{j} = s2^{s-1} - s$$

coset intersections. Theorem 7.3 computes each in $p^{O(d)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$ time in the original matrix group. Taking unions and shortening generating sets have the same local bound. For elements returned by a child, constructive membership in the parent group supplies new words in its generators for evaluating c_h; the child's computation history need not be expanded. Since $s \le d$, the total local work has the claimed bound. Theorem 5.1 makes the summands a transitive H-set, so they have equal dimension and $\dim D_\nu = \dim A_k/s \le \dim A_k/2$. The restriction maps and group operations construct each pair input with bit length $\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$. □

The transporter identities for general P. For general $P \le N \le Z = Z_{\mathrm{quad}} \oplus Z_{\mathrm{tri}}$, extend q_ν by zero on Z_{quad} and choose representatives $\widetilde{x}, \widetilde{y} \in z + N$ of x, y. To use the same representative of c_h in all restrictions, work in

$$\widehat{H}_c = \{(h, v) \in H \times Z : v + P = c_h\}, \qquad (h, v)(h', v') = (hh', v + \rho_h(v')).$$

The cocycle identity makes this a group. It acts by $(h, v) \cdot (\nu, t) = (h\nu, \rho_{h,\nu}(t) + q_{h\nu}(v))$. The pair transporters are

$$\widehat{C}_{\nu\omega} = \{(h, v) \in \widehat{H}_c : hD_\nu = D_\omega, \quad q_\omega(\rho_h(\widetilde{x}) + v) = q_\omega(\widetilde{y})\}.$$

For $rD_\nu = D_\omega$, choose $v_r + P = c_r$ and put $K = \mathrm{Stab}_H(D_\nu)$. Removing the left factor r from the pair answer gives the comparison under K modulo $q_\nu(P)$, with target represented by $q_\nu(\rho_{r^{-1}}(\widetilde{y}-v_r))$. Lemma 3.6, applied to $\pi_{\mathrm{cov}}^{-1}(K) \to K$ with $\pi_{\mathrm{cov}}(h,v) = h$, recovers all lifts satisfying the restriction equation, including the kernel generators $(1,v)$ for a basis of $P \cap \ker q_\nu$. Left multiplication by (r, v_r) gives $\widehat{C}_{\nu\omega}$; an empty pair answer remains empty.

Apply the same subset recurrence in $\widehat{H}_c$, with these pair transporters and the representatives $\widetilde{x}, \widetilde{y}$, to obtain $\widehat{D}(I)$. Its elements use one common pair (h,v) for all restrictions. Consequently,

$$\pi_{\mathrm{cov}}(\widehat{D}(I)) = \mathrm{Trans}_H(x+V, y+V), \qquad V = ((N \cap Z_{\mathrm{quad}}) + P)/P.$$

Indeed, all restrictions match exactly when $\rho_h(\widetilde{x}) + v - \widetilde{y} \in Z_{\mathrm{quad}}$. Taking $y = x$ also gives the source stabilizer after projection. The remaining comparison is on $V \cong (N \cap Z_{\mathrm{quad}})/(P \cap Z_{\mathrm{quad}})$; Lemma 8.2 and Step G2 recover the complete original answer. For the recursive inputs of the lemma, $V = 0$, as used in the main proof.

E.1.5 Step G5: repeated copies of a natural module

When $A_k = B \otimes_{\mathbb{K}} \mathbb{K}^m$, the group H may mix the copies of B. This step compares the possible images of an ordered basis of $\mathbb{K}^m$, using the group action constructed below. For each resulting candidate coset, it replaces A_k by B and records the m tensor restrictions as a map into $M^{\oplus m}$.

Lemma E.6 (Reduction using repeated copies)**.** *Suppose Step G5 applies to an input reached from* (8.1)*, with $A_k = B \otimes_{\mathbb{K}} \mathbb{K}^m$, where B is a natural module for a large classical component and $m \geq 2$. In $p^{O(d)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$ local time one can compute a finite index set Ξ and cosets $u_\xi K \subseteq H$ such that*

$$H = \bigcup_{\xi \in \Xi} u_\xi K, \qquad \log_p |\Xi| \leq C\big(\dim_{\mathbb{F}_p} A_k - \dim_{\mathbb{F}_p} B\big)$$

for an absolute constant C. In each coset, the comparison is equivalent to (8.9)*: its new argument is B, its trilinear maps take values in $M^{\oplus m}$ with the old action on each coordinate, and its input has bit length* $\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$*. In particular, the new argument has at most half the previous dimension. The complete branch answers recover the original complete answer by* (8.6)*.*

Proof. Use the coordinates from Step G1 to write $A_k = B \otimes_{\mathbb{K}} \mathbb{K}^m$, and choose an ordered basis $e = (e_1, \ldots, e_m)$ of $\mathbb{K}^m$. We seek a subgroup $K \leq H$ such that, for each $h \in K$, one $\mathbb{F}_p$-linear map $h_B : B \to B$ satisfies

$$h(b \otimes e_j) = h_B(b) \otimes e_j \qquad (b \in B,\ 1 \leq j \leq m).$$

For $\mathtt{T} \in Z_{\mathrm{tri}}$, the tuple $(\mathtt{T}(\ldots, b \otimes e_j, \ldots))_{j=1}^m$ takes values in $M^{\oplus m}$, with the same h_B acting on its argument $b \in B$ in every coordinate. Thus, for candidates in $u_\xi K$, we replace the target by $\alpha_{u_\xi^{-1}}(y)$ and compare under K, using B as the kth argument and $M^{\oplus m}$ to record all m restrictions.

To obtain such a K, we need to fix the basis e under the maps on $\mathbb{K}^m$. The factorizations from Lemma 5.4 have the form

$$h(b \otimes t) = u(b) \otimes v(t), \qquad u : B \to B, \quad v : \mathbb{K}^m \to \mathbb{K}^m.$$

The pairs (u, v) and $(\lambda u, \lambda^{-1} v)$ give the same map on A_k for $\lambda \in \mathbb{K}^\times$, so h alone does not specify the image of e under v. We therefore record both factor maps in an auxiliary group before taking the stabilizer of e.

Write $\theta : H \twoheadrightarrow A$ for the action image on A_k. Lemma 5.4 supplies a group $\widehat{A}$ and a surjection $\pi_{\mathrm{fac}} : \widehat{A} \twoheadrightarrow A$ recording these factor maps. Form

$$\widehat{H} := \{(h, \widehat{a}) \in H \times \widehat{A} : \theta(h) = \pi_{\mathrm{fac}}(\widehat{a})\}, \qquad \pi(h, \widehat{a}) = h,$$

with componentwise multiplication. Each $\widehat{a}$ records invertible maps u, v with a common field automorphism σ of $\mathbb{K}$:

$$u(\lambda b) = \sigma(\lambda)u(b), \qquad v(\lambda t) = \sigma(\lambda)v(t) \quad (\lambda \in \mathbb{K}).$$

The group $\widehat{H}$ contains $(h, 1)$ for every $h \in \ker\theta$, including their possible actions on the other arguments and on M.

The group $\widehat{H}$ acts on ordered bases by its recorded map v. Define $\Xi := \widehat{H} \cdot e$. Choose $\widehat{u}_\xi$ sending e to each $\xi \in \Xi$, and put

$$\widehat{K} := \mathrm{Stab}_{\widehat{H}}(e), \qquad u_\xi := \pi(\widehat{u}_\xi), \qquad K := \pi(\widehat{K}).$$

Then

$$\widehat{H} = \bigsqcup_{\xi \in \Xi} \widehat{u}_\xi \widehat{K}, \qquad H = \bigcup_{\xi \in \Xi} u_\xi K.$$

The projected cosets may coincide, which does not affect the union.

If a lift in $\widehat{K}$ has field automorphism σ, its map v fixes all e_j and sends $\sum_j \lambda_j e_j$ to $\sum_j \sigma(\lambda_j)e_j$. Thus K has the required common action on the copies of B. The map h_B is obtained by restricting h to $B \otimes e_1$ and using $b \otimes e_1 \mapsto b$. A scalar λ^{-1} fixing all e_j is 1, so this map is independent of the lift. Field automorphisms that fix the basis remain included in the action.

Expansion in $A_k = \bigoplus_j (B \otimes e_j)$ gives the $\mathbb{F}_p$-linear bijection

$$\Phi : Z \longrightarrow Z' := Z_{\mathrm{quad}} \oplus \mathrm{Hom}(A_1' \otimes A_2' \otimes A_3', M^{\oplus m})$$

from Step G5, where $A_k' = B$ and $A_i' = A_i$ for $i \neq k$. Its inverse evaluates the tuple on the corresponding argument components and adds the results. The formula for $h(b \otimes e_j)$ shows that this is equivariant when K acts on every value coordinate as on M: the jth restriction is transformed by the inverse argument maps and then by the given action h_M on M, with the same h_B^{-1} in argument position k.

Let $\overline{\Phi} : Z/P \to Z'/\Phi(P)$ be the isomorphism induced by Φ and put $\rho_h' = \Phi\rho_h\Phi^{-1}$. For $h \in K$,

$$\overline{\Phi}\big(\alpha_h(x)\big) = \rho_h'\big(\overline{\Phi}(x)\big) + \overline{\Phi}(c_h).$$

Thus the equation for a candidate $u_\xi h$ is equivalent to

$$\rho_h'\big(\overline{\Phi}(x)\big) + \overline{\Phi}(c_h) = \overline{\Phi}\big(\alpha_{u_\xi^{-1}}(y)\big),$$

which proves (8.9), including the left factor gu_ξ. Since K preserves $B \otimes e_1$, the pair $(\pi_k^{-1}(B \otimes e_1), P_k)$ is a valid argument presentation. Taking the direct sum of m copies of the controlled presentation of M gives that of $M^{\oplus m}$. The catalogue modules and semilinear lines are unchanged, so their total dimension is unchanged as well. Formula (8.6) therefore recovers a coset representative and generators of the full source stabilizer.

Put $f = [\mathbb{K} : \mathbb{F}_p]$ and $a = \dim_{\mathbb{K}} B$. Every element of Ξ is an ordered $\mathbb{K}$-basis, even for semilinear maps, so

$$|\Xi| \leq |\mathrm{GL}(m, \mathbb{K})| \leq p^{fm^2}.$$

Orbit search and the stabilizer computation use this many points, with original lifts supplied by Lemma 5.4. Since A_k is a subquotient of an original argument, $fam = \dim A_k \le d$. The classical order estimate used in Lemma 5.5, together with the threshold $T_{\mathrm{cl}} = p^{C_* d}$, gives

$$C_* d < fa^2, \qquad fam \le d, \qquad \frac{m}{a} < \frac{1}{C_*}.$$

Since $m \ge 2$,

$$\log_p |\Xi| \le fm^2 \le \frac{2}{C_*}\, fa(m-1) = O\big(\dim_{\mathbb{F}_p} A_k - \dim_{\mathbb{F}_p} B\big).$$

The new argument dimension is $fa = \dim_{\mathbb{F}_p} A_k/m \le a_k/2$, and the other dimensions are unchanged. Step G3 therefore resets the reference dimensions in every branch. The isomorphism $\Phi : Z \to Z'$ gives $\dim Z' = \dim Z$. Storing the new coordinate maps as matrices therefore gives each child input bit length $\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$. Factorization, orbit computation, and operations in the original matrix group take $p^{O(d)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$ local time. □

E.1.6 Step G6: putting translations in the pairing submodule

The preceding structural reductions leave three single natural, dual, or twisted-natural arguments for large classical components. Step G6 changes the origin so that $c_h^b \in J(Z)/P$ for every $h \in H$. Thus the action on $Z/J(Z)$ is linear, while the action on Z/P may still have nonzero translations.

Lemma E.7 (Making the action on $Z/J(Z)$ linear)**.** *Consider a comparison reached from* (8.1) *at Step G6, with each A_i in the single natural, dual, or twisted-natural case just described. In time $p^{O(d)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$ one can construct an H-equivariant map*

$$\mathcal{J} : \mathcal{S} \longrightarrow Z_{\mathrm{tri}}, \qquad J(Z) := Z_{\mathrm{quad}} \oplus \operatorname{im} \mathcal{J},$$

where $\mathcal{S}$ has a controlled degree-2 presentation as in Definition 23, and an element $b \in Z$ such that $P \le J(Z)$ and, with $\overline{b} = b + P$,

$$c_h^b := c_h + \overline{b} - \rho_h(\overline{b}) \in J(Z)/P \qquad \text{for every } h \in H.$$

Thus the new action on $Z/J(Z)$ is linear. The update (8.10) *preserves the complete transporter and the source stabilizer, with the same group and argument spaces. When $c = 0$, one may take $b = 0$.*

Proof. Lemma E.2 gives $P = Z_{\mathrm{quad}} = 0$ and the trivial action of H on M. This does not imply $c_h = 0$; the following argument handles the actual affine translations.

The pairing map and its parameters. Use the composition flags in all three original spaces U_a^* and the group decompositions computed in Step G1. Write the flags and their successive quotients as

$$0 = F_{a,0} < \cdots < F_{a,\ell_a} = U_a^*, \qquad X_{a,j} := F_{a,j}/F_{a,j-1} \quad (a \in \{1,2,3\},\ 1 \le j \le \ell_a).$$

Let R be the kernel of the action of H on $\bigoplus_{a,j} X_{a,j}$. By Theorem 5.1, $R = \mathrm{O}_p(H)$, the largest normal p-subgroup of H. The current A_i are simple argument quotients after Step G1, so R acts trivially on them, and it acts trivially on M as well. Consequently the action on Z_{tri} factors through $G := H/R$.

Let $D \trianglelefteq G$ be the subgroup constructed in Theorem 5.2(3): the central product of the classical components whose central quotients have order greater than T_{cl}. On each A_i, its action is the specified natural, dual, or Frobenius-twisted natural action. The fixed threshold ensures the dimension and Witt-index bounds required by Theorems 6.2 and 6.3. Put $V_i = A_i^*$. Over $\mathbb{F}_p$, form

$$B_i = (V_j \otimes V_\ell)^D \quad (\{i,j,\ell\} = \{1,2,3\},\ j < \ell), \qquad \mathcal{S} = \bigoplus_{i=1}^{3} V_i \otimes B_i \otimes M.$$

Define $\mathcal{J} : \mathcal{S} \to Z_{\mathrm{tri}}$ by inserting the B_i coordinate in positions j, ℓ and the V_i coordinate in position i, leaving the value coordinate in M. Here B_i is the subspace fixed by every element of D in the other two dual argument positions. This is the prime-field form of the maps in Definition 16, with the specified defining fields, twists, and central-product quotient map. Theorem 6.2 applies componentwise after scalar extension and descent. Each basis coordinate of the trivial module M gives a separate copy with trivial D-action.

Since D is normal in G and H acts through G on the V_i, every B_i is preserved by H. The insertion maps commute with these induced actions, so $\mathcal{J}$ is H-equivariant. Write

$$J(Z) = \operatorname{im} \mathcal{J}, \qquad Q = Z/J(Z).$$

In particular, $P \le J(Z)$ because $P = 0$.

The parameters have the required controlled degree-2 presentation. Indeed, $B_i \cong \mathrm{Hom}_D(V_j^*, V_\ell)$ is zero unless the two indicated simple D-modules are isomorphic. In the nonzero case, put $\mathbb{K} = \mathrm{End}_D(V_j^*)$, the field of linear maps on V_j^* commuting with D. The space B_i is one-dimensional over this field. Conjugation by each $h \in H$ induces a field automorphism σ_h of $\mathbb{K}$, and

$$h(\lambda\eta) = \sigma_h(\lambda)\, h(\eta) \qquad (\lambda \in \mathbb{K},\ \eta \in B_i).$$

Hence B_i is a semilinear line in the sense of Definition 20. These fields, bases of $\mathrm{Hom}_D(V_j^*, V_\ell)$, and generator actions are obtained from the given natural coordinates by linear equations. The sum of their prime-field dimensions is $O(d)$, and their joint image is solvable. They therefore form a controlled catalogue as in Definition 21. Each summand of $\mathcal{S}$ has one active factor V_i and one semilinear line B_i, with a separate copy for each vector in the chosen basis of M. Taking the dual of the given presentation $A_i = N_i/P_i$ gives the required subquotient presentation of $V_i = A_i^*$. This gives a presentation as in Definition 23, with at most two positions per summand. The parameter space $\mathcal{S}$ is a subspace of a direct sum of three copies of Z_{tri}, so its explicit presentation has bit length $\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$. Bases and maps are obtained by linear algebra in these coordinates.

The restriction of c to R. Since $Z = Z_{\mathrm{tri}}$ and $P = 0$, the translations are vectors $c_h \in Z$ satisfying $c_{hh'} = c_h + \rho_h(c_{h'})$. The group R acts linearly as the identity on Z, so

$$c|_R : R \longrightarrow (Z, +), \qquad r \longmapsto c_r$$

is a homomorphism. Put $E := c(R)$. The cocycle identity shows that E is invariant under $G = H/R$. Lemma 3.7 says that every composition factor of E occurs in a space $\mathrm{Hom}_{\mathbb{F}_p}(X_{a,j}, X_{a,i})$, with $i < j$, between the original flag quotients defined above.

By Theorem 5.2, each $X_{a,j}$, restricted to D, is a direct sum of natural, dual, or Frobenius-twisted natural modules and trivial modules. Since

$$\mathrm{Hom}_{\mathbb{F}_p}(X_{a,j}, X_{a,i}) \cong X_{a,j}^* \otimes X_{a,i},$$

each of its composition factors occurs in a tensor product of at most two such natural modules, including their duals and twists. The same is true of every composition factor of E. Apply Theorem 6.2(ii) to $Z = V_1 \otimes V_2 \otimes V_3 \otimes M$, using its assertion over $\mathbb{F}_p$ and one direct-sum copy for each vector in a basis of M. It gives

$$E \le J(Z).$$

The linear action of R on Q is trivial, and the projected cocycle $\pi_J(c_r)$ vanishes for $r \in R$, where $\pi_J : Z \to Q$ is the quotient map. Both parts of the affine action on Q therefore descend to $G = H/R$. This verifies the descent for the actual cocycle; no complement to R in H is required.

A change of origin for H. Let ρ_Q, c_Q denote the linear action and cocycle on Q descended to G. Theorem 6.3 applies over $\mathbb{F}_p$, to the direct-sum copies indexed by a basis of M, and to the given central-product quotient map. It gives

$$Q^D := \{v \in Q : \rho_Q(d_0)v = v \text{ for all } d_0 \in D\} = 0,$$
$$H^1(D, Q) = 0.$$

The second assertion means that every cocycle on D has the form $d_0 \mapsto (\rho_Q(d_0) - I)b_Q$; the first says that this b_Q is unique. In particular, there is a unique $b_Q \in Q$ such that

$$c_Q(s) = (\rho_Q(s) - I)b_Q$$

on generators of D. It is obtained by solving these linear equations; the cocycle law extends them to every element of D. Values evaluated on original lifts are independent of the lifts because the projected cocycle and linear action are trivial on R.

Put $c_Q^b(h) = c_Q(h) + b_Q - \rho_Q(h)b_Q$. This cocycle vanishes on D. For $d_0 \in D$ and $h \in G$, normality gives

$$\rho_Q(d_0)c_Q^b(h) = c_Q^b(d_0h) = c_Q^b\big(h(h^{-1}d_0h)\big) = c_Q^b(h).$$

Hence $c_Q^b(h) \in Q^D = 0$ for every $h \in G$. This is the step that gives $c_Q^b(h) = 0$ for all $h \in G$, starting from its vanishing on D.

Choose a lift $b \in Z$ of b_Q, and let S_H be the given generating set of H. It follows that the linear system used in Step G6,

$$(\rho_s - I_Z)b + J(Z) = \pi_J(c_s) \qquad (s \in S_H),$$

has a solution. Solving it directly on the original generators returns such a lift, and the cocycle identity shows that $c_h^b = c_h + b - \rho_h(b)$ belongs to $J(Z)$ for every $h \in H$. When $c = 0$, the choice $b = 0$ is valid.

Finally, the coordinate map $v \mapsto v + b$ sends $z + N$ to $z + b + N$ and satisfies

$$\rho_h(v + b) + c_h^b = \rho_h(v) + c_h + b.$$

It follows that (8.10) has exactly the same transporter and source stabilizer as its input. More generally, for $P \le J(Z)$ the same coordinate-change identity holds in Z/P with $\overline{b} = b + P$; the subspace P is unchanged. The group computations above use the original lifts and kernels and cost $p^{O(d)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$; the remaining linear algebra has cost $\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$. This proves the construction and solution preservation asserted in the lemma. □

E.1.7 Step G7: handling the remaining affine comparison

Step G6 makes the action on $Z/J(Z)$ linear; it does not make the action on Z linear. For the recursive inputs under consideration, Lemma E.2 gives $P = Z_{\mathrm{quad}} = 0$, so the remaining equation is

$$\rho_h(x) + c_h = y, \qquad c_h \in J(Z) = \operatorname{im} \mathcal{J}, \qquad h \in H.$$

We use the same formula $\alpha_h(t) = \rho_h(t) + c_h$ on all of Z. We first prove two directly testable cases and the exact formula for separating a tensor and its pairing parameters. We then construct that separation or a proper-subgroup comparison by examining contractions modulo the pairing terms.

The large-component threshold ensures that the defining-field natural dimensions used below are at least 64. If such a module has dimension n over a field of degree f over $\mathbb{F}_p$, then $fn \le d$ and $|S/\mathrm{Z}(S)| \le p^{fn^2}$. Thus $|S/\mathrm{Z}(S)| > p^{C_* d}$ implies $n > C_*$, and the fixed constant may be chosen greater than 64.

Lemma E.8 (Two direct cases of Step G7)**.** *Let S_H be the given generating set of H, and put $\delta_h = \rho_h(x) + c_h - x$ and $w = y - x$.*

(i) *If $\delta_s \in J(Z)$ for every $s \in S_H$, the complete comparison is computable by the degree-2 algorithm, giving exit (a) of Step G7.*

(ii) *If the linear equations*

$$(\rho_s - I_Z)b = c_s \qquad (s \in S_H)$$

have a solution $b \in Z$, the original comparison is equivalent to the ordinary comparison

$$g \operatorname{Trans}_{H,\rho}(x + b, y + b),$$

giving exit (c) with unchanged group and argument spaces.

Both tests and the completion in (i) take $p^{O(d)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$ time.

Proof. For (i), the cocycle identity $\delta_{hk} = \delta_h + \rho_h(\delta_k)$ and the invariance of $J(Z)$ extend the generator test to all of H. Thus $\alpha_h(x) - x \in J(Z)$ for every h. If $w \notin J(Z)$, the answer is empty. Otherwise set

$$\mathcal{S}_N := \mathcal{J}^{-1}(N \cap J(Z)), \qquad \overline{\mathcal{J}} : \mathcal{S}_N/\ker \mathcal{J} \xrightarrow{\sim} N \cap J(Z).$$

The displayed isomorphism is induced by $\mathcal{J}$. Both δ_h and w belong to N, because $x, y \in z + N$ and α_h preserves $z + N$. Pulling the centered action back through $\overline{\mathcal{J}}$ gives the comparison with source 0, target $\overline{\mathcal{J}}^{-1}(w)$, and translations $\overline{\mathcal{J}}^{-1}(\delta_h)$ on $\mathcal{S}_N/\ker \mathcal{J}$. This is a subquotient of the controlled degree-2 presentation of $\mathcal{S}$ constructed in Lemma E.7. Theorem 7.9 therefore computes exactly $\{h \in H : \delta_h = w\}$ and its source stabilizer. Multiplication by g gives the required complete answer.

For (ii), the cocycle identity extends the equations from S_H to all of H. Hence

$$\rho_h(x) + c_h = y \quad \Longleftrightarrow \quad \rho_h(x + b) = y + b.$$

The same identity with $y = x$ proves preservation of the source stabilizer. The ordinary comparison may be placed in all of $Z = Z_{\mathrm{tri}}$, since only these two points and their transporter are being compared. Linear algebra gives the two tests; the degree-2 theorem gives the stated completion cost. □

The next lemma specifies what is needed to separate the two conditions in the remaining affine equation. Put

$$\overline{\mathcal{S}} := \mathcal{S}/\ker \mathcal{J}, \qquad \overline{\mathcal{J}} : \overline{\mathcal{S}} \xrightarrow{\sim} J(Z), \qquad \overline{\mathcal{J}}(r + \ker \mathcal{J}) = \mathcal{J}(r).$$

Write σ_h for the induced action on $\overline{\mathcal{S}}$, and let $\eta_h := \overline{\mathcal{J}}^{-1}(c_h)$. These translations define the affine action $\beta_h(v) = \sigma_h(v) + \eta_h$ on $\overline{\mathcal{S}}$.

Lemma E.9 (Separating a tensor and its pairing parameters)**.** *Suppose a set $\Omega \subseteq Z$ containing x, y is preserved by every ρ_h and by addition of every element of $J(Z)$. Suppose a map $s : \Omega \to \overline{\mathcal{S}}$ has been constructed such that*

$$\begin{aligned} s(\rho_h(t)) &= \sigma_h(s(t)), \quad h \in H,\ t \in \Omega, \\ s(t + \overline{\mathcal{J}}(r)) &= s(t) + r, \quad r \in \overline{\mathcal{S}},\ t \in \Omega. \end{aligned} \tag{E.3}$$

Define

$$\mathtt{T}_x := x - \overline{\mathcal{J}}(s(x)), \qquad \mathtt{T}_y := y - \overline{\mathcal{J}}(s(y)).$$

Then, for every $h \in H$,

$$\alpha_h(x) = y \quad \Longleftrightarrow \quad \rho_h(\mathtt{T}_x) = \mathtt{T}_y \text{ and } \beta_h(s(x)) = s(y). \tag{E.4}$$

The parameter comparison on $\overline{\mathcal{S}}$ is a controlled degree-2 comparison. If its complete answer is uK, where $K = \mathrm{Stab}_{H,\beta}(s(x))$, then

$$\begin{aligned} g\,\mathrm{Trans}_{H,\alpha}(x, y) &= gu\,\mathrm{Trans}_{K,\rho}(\mathtt{T}_x, \rho_{u^{-1}}(\mathtt{T}_y)), \\ \mathrm{Stab}_{H,\alpha}(x) &= \mathrm{Stab}_{K,\rho}(\mathtt{T}_x). \end{aligned} \tag{E.5}$$

An empty parameter answer gives an empty original answer. For $K < H$, the ordinary comparison on the right is a recursive comparison in exit (b) of Step G7; for $K = H$, it is the ordinary input in exit (c).

Once $s(x), s(y)$ and the hypotheses on s are available, the parameter comparison and construction of this ordinary input cost $p^{O(d)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$, excluding the solution of the ordinary comparison. There is at most one resulting candidate coset.

Proof. Equivariance of $\overline{\mathcal{J}}$ transfers the cocycle identity for c to η, proving the assertion about β. For $t \in \Omega$, write $\mathtt{T}_t = t - \overline{\mathcal{J}}(s(t))$. The two identities in (E.3) give

$$\begin{aligned} s(\alpha_h(t)) &= \sigma_h(s(t)) + \eta_h, \\ \mathtt{T}_{\alpha_h(t)} &= \rho_h(\mathtt{T}_t). \end{aligned}$$

The pair $(\mathtt{T}_t, s(t))$ determines t, since $t = \mathtt{T}_t + \overline{\mathcal{J}}(s(t))$. Matching both entries is therefore equivalent to matching the original points, which proves (E.4).

The quotient $\overline{\mathcal{S}}$ inherits the controlled degree-2 presentation of $\mathcal{S}$, including its actual H-action and the translations η_h. Apply Theorem 7.9 to compute $\mathrm{Trans}_{H,\beta}(s(x), s(y))$. For a nonempty answer uK, every original solution has the form uk with $k \in K$, and its remaining condition is

$$\rho_k(\mathtt{T}_x) = \rho_{u^{-1}}(\mathtt{T}_y).$$

This proves both identities in (E.5), the second by setting $y = x$. The child input is exactly (8.11), with the tensors defined above. Its argument spaces, value space, and linear action are unchanged, apart from restriction to K; its translations are zero. When $K = H$, the parameter transporter is all of H, so $u = 1$ may be chosen and this ordinary input goes directly to the tensor reductions. When $K < H$, it starts at Step G1. Thus this replacement alone introduces neither a family of candidate cosets nor a return to Step G1 with the same group and the same nonordinary input. The cost follows from the degree-2 theorem and linear algebra. □

One sufficient way to obtain the map in this lemma is to compute an H-equivariant linear map $s : Z \to \overline{\mathcal{S}}$ with $s\overline{\mathcal{J}} = I_{\overline{\mathcal{S}}}$. These are linear equations on s, and a solution satisfies (E.3) with $\Omega = Z$. Such a solution is not guaranteed: it would make $J(Z)$ a direct summand of Z as an H-module. The conclusion of Lemma E.7 does not assert this splitting, or that its remaining translations are coboundaries.

Writing the pairing maps in coordinates. We use the actual-input conclusion of Lemma E.2: $P = Z_{\mathrm{quad}} = 0$ and H fixes the value space M pointwise. Choose its ordered basis, write $m = \dim_{\mathbb{F}_p} M$, and write T_j for the j-th scalar coordinate of $T \in Z$. All m coordinates are processed together. The fields, intertwiners and forms below are computed from the representation data in Step G6. Identifications of dual or Frobenius-twisted occurrences use these intertwiners; the resulting maps are still maps over $\mathbb{F}_p$ on the original argument spaces. Choose the absolute large-component threshold so that every paired natural space used below has dimension at least 64 over its defining field.

Lemma E.10 (Coordinate formulas for the pairing maps)**.** *The pairing maps of Step G6 have the following coordinate forms. Put $f = [\mathbb{F} : \mathbb{F}_p]$ for the relevant defining field.*

(i) *Suppose that the three arguments are identified with $A = \mathbb{F}^a$ for a symplectic, orthogonal, or unitary natural action. Let b be its nondegenerate alternating, symmetric, or Hermitian form; the Hermitian form is linear in its first argument. Then the pairing parameters have the form*

$$[\mathcal{J}(\ell)]_j(x,y,z) = \mathrm{Tr}_{\mathbb{F}/\mathbb{F}_p}\big(\ell_{1,j}(x)b(y,z) + \ell_{2,j}(y)b(x,z) + \ell_{3,j}(z)b(x,y)\big), \tag{E.6}$$

where $\ell_{i,j} \in \mathrm{Hom}_{\mathbb{F}_p}(A,\mathbb{F})$ and $1 \le j \le m$. The formula includes all prime-field occurrences of the permitted pairings, rather than treating Frobenius twists as independent recursive inputs.

(ii) *For a mixed general-linear action, trace evaluation identifies the scalar trilinear map with a bilinear map $B_j : A \times A \to A$. Its pairing terms are*

$$[\mathcal{J}(\alpha,\beta)]_j(x,y) = \alpha_j(x)y + \beta_j(y)x, \qquad \alpha_j, \beta_j \in \mathrm{Hom}_{\mathbb{F}_p}(A,\mathbb{F}). \tag{E.7}$$

This is the case with two natural occurrences and one dual, or its dual after reversing the identification. The cases without a natural–dual pair have $J(Z) = 0$.

(iii) *Suppose that exactly one pair of arguments has an invariant pairing. Write those arguments as X, Y, with a nondegenerate $\mathbb{F}$-bilinear pairing $Q : X \times Y \to \mathbb{F}$, and put $C = A_k^*$ for the remaining argument. Currying gives maps $T_j : X \times Y \to C$. Their pairing terms are*

$$[\mathcal{J}(u)]_j(x,y) = u_j(Q(x,y)), \qquad u_j \in \mathrm{Hom}_{\mathbb{F}_p}(\mathbb{F},C). \tag{E.8}$$

Each displayed parameter space is identified with the corresponding part of $\mathcal{S}/\ker\mathcal{J}$ by explicit linear algebra. Its H-action is the action determined by $\rho_h\mathcal{J} = \mathcal{J}\sigma_h$; this includes field automorphisms and scalar multipliers on the pairings.

Proof. For a classical natural module the commuting algebra is its defining field. Thus, after identifying a paired space with the dual of the other, all invariant $\mathbb{F}_p$-bilinear forms are

$$(y,z) \longmapsto \mathrm{Tr}_{\mathbb{F}/\mathbb{F}_p}(t\,b(y,z)), \qquad t \in \mathbb{F},$$

or the corresponding trace-evaluation forms. These forms are nondegenerate when $t \neq 0$. Indeed, a nonzero value of b can be multiplied by every element of $\mathbb{F}$ in one argument, and the field trace

pairing is nondegenerate. Products of these invariant pairings with a prime-field linear functional on the remaining argument give (E.6). Trace evaluation instead gives (E.7). In the last case all such products are exactly the maps in (E.8).

Theorem 6.2 says that these are all the pairing terms. Distinct classical factors cannot pair. For a general-linear factor only a natural occurrence and its matching dual can pair. Self-duality supplied by the specified form gives the other displayed cases. Applying the statement to defining-field components and descending their maps gives the prime-field formulas above. In particular the functions $\ell_{i,j}$ need not be $\mathbb{F}$-linear. The actual parameter map and the displayed map have the same image in Z; comparing their matrix columns computes the identification modulo their kernels. Transporting the displayed formulas by the given matrices computes σ_h and preserves all relations between the original actions. □

What matching contraction operators forces. The group returned by a complete comparison need not centralize its operators: it may still move a defining field or a pairing multiplier. The following lemma keeps these transformations in the returned group and uses a subgroup only to bound its new large components.

Lemma E.11 (Contraction operators and new natural modules)**.** *At an input to Step G7, let W be the direct sum of one representative of each distinct natural argument module and required dual, with the defining-field identifications from Theorem 5.2. Let $B \le \mathrm{End}_{\mathbb{F}_p}(W)$ be the algebra generated by the defining fields, the projections onto these summands, and a matched tuple of contraction operators and their required adjoints. For a projected same-type contraction, include its projection P_v and match the operator modulo $\mathbb{K}P_v$.*

Let $H' \le H$ be the complete stabilizer of the marks and these observations. Write

$$C = \mathrm{End}_B(W), \qquad C/\operatorname{rad} C \cong \prod_t \mathrm{Mat}(m_t, \mathbb{F}_t), \qquad d_t = [\mathbb{F}_t : \mathbb{F}_p] m_t.$$

Every new large natural module obtained from the old argument spaces after restriction to H' has prime-field dimension at most $\max_t d_t$. The sum of the dimensions of its distinct natural types, including required duals once each, is at most $\sum_t d_t$. The same conclusions hold for ordinary contraction observations. The bounds also apply separately on an original represented summand, using the centralizer of its contraction operators. The original comparison is continued in the full group H', including every original kernel.

Proof. Apply the structural decomposition to H'. Let α be its action on all the recorded original argument grades, and let Q be the central product of the newly selected large components relevant to W. Its full preimage is used throughout. The kernel of the action on all these grades is a normal p-group.

Let A_{old} be the image of the preceding group on its three natural arguments and their formal duals, and let $D_{\mathrm{old}} \trianglelefteq A_{\mathrm{old}}$ be its selected classical product. Take β to be the restriction to H' of the homomorphism to $A_{\mathrm{old}}/D_{\mathrm{old}}$. Its image has order $p^{O(d)}$ by Lemma 5.5. In its kernel, the action belongs to the old classical product; it fixes the defining fields and pairing forms, and acts identically on the recorded equivalent copies. Thus it acts on the chosen space W, even when representatives of repeated types have been identified there. The old value space M is fixed by Lemma E.2. The image of β has no simple section of order greater than the fixed large-component threshold, once its absolute constant is chosen sufficiently large. Lemma 5.6 therefore shows that

$$H_0 = \alpha^{-1}(Q) \cap \ker \beta \quad \text{maps onto } Q.$$

This assertion is about the actual joint image, so it does not assume that the auxiliary actions descend through $\ker \alpha$.

The marks are fixed in H'. On H_0 the defining fields, forms, and projections are also fixed. An ordinary observation is then fixed as an operator. A projected observation has equations

$$hD_{j,v}h^{-1} = D_{j,v} + \lambda_{j,v}(h)P_v, \qquad \lambda_{j,v}(h) \in \mathbb{K}.$$

Here both $\mathbb{K}$ and P_v are fixed, so the simultaneous map λ to the additive group of these scalar coordinates is a homomorphism. Its kernel $H_1 \leq H_0$ centralizes every generator of B. Moreover $\alpha(H_1) \trianglelefteq Q$, and $Q/\alpha(H_1)$ is a quotient of the abelian group H_0/H_1. Since Q is perfect, $\alpha(H_1) = Q$. For ordinary observations one can take $H_1 = H_0$. Kernel computation and original lifts use Lemma 3.1; this subgroup is used for the proof and never replaces the complete returned group H'.

Filter W by powers of $\operatorname{rad} C$. Its factors are sums of the natural modules $\mathbb{F}_t^{m_t}$ for $C/\operatorname{rad} C$. Every simple H_1-constituent of W consequently occurs in the restriction of one of these modules. Since H_1 maps onto Q, the new natural types occur among these constituents; the normal p-kernel acts trivially on every simple constituent. Thus each type has dimension at most $\max_t d_t$. Taking one copy of each $\mathbb{F}_t^{m_t}$ already contains every possible distinct type among its composition factors, proving the sum bound. The old argument spaces are copies of the summands represented in W; this accounts for every relevant type and its dual without counting repeated occurrences. Further invariant-subspace or adjoint conditions restrict these modules and cannot increase either bound. For the separate bound on an operator argument A, note that B contains the old summand projections, so its centralizer preserves A. The subgroup H_1 on A commutes with the algebra generated by the tuple acting on A. Apply the same radical-filtration argument to that algebra's centralizer on A alone. Thus a large factor on another argument does not affect the bound for the new constituents of A. □

Lemma E.12 (A field quotient or smaller centralizer modules)**.** *Let $A = \mathbb{K}^n$, where $f = [\mathbb{K} : \mathbb{F}_p]$, and let $B \leq \operatorname{End}_{\mathbb{F}_p}(A)$ be a represented unital algebra containing the distinguished copy of $\mathbb{K}$. This copy need not be central in B. For $0 \leq r < n/2$, one can compute either*

(i) *a field quotient $B/I = F$ whose annihilator $A_I = \{a \in A : Ia = 0\}$ has $\mathbb{K}$-dimension at least $n - r$; this quotient is the unique field quotient with such a large annihilator; or*

(ii) *a decomposition of $\operatorname{End}_B(A)/\operatorname{rad}\operatorname{End}_B(A)$ all of whose natural modules have prime-field dimension at most $f(n - r)$.*

The construction takes $p^{O(d)}\operatorname{poly}(\operatorname{InSize}_{\mathrm{cur}}, d, \log p)$ time when $fn \leq d$. In the first case the quotient embeds $\mathbb{K}$ in F and $[F : \mathbb{K}] \leq n$. In the second case, matching the corresponding contraction tuple completely gives the same bound on every new large natural constituent of A.

Proof. First compute the field factors of $B/\operatorname{rad} B$ by Lemma 3.2, and compute the annihilator of each quotient kernel by linear algebra. Return the first alternative if an annihilator has the stated dimension. Its F-module structure implies $[F : \mathbb{K}] \leq n$. To prove the second alternative when this test fails, put $C = \operatorname{End}_B(A)$. Decompose A into indecomposable B-modules as $A = \bigoplus_t Q_t^{m_t}$, with the Q_t pairwise nonisomorphic. This can be computed from primitive idempotents of $C/\operatorname{rad} C$ and their lifts using Lemma 3.2. Write

$$\mathbb{F}_t = \operatorname{End}_B(Q_t)/\operatorname{rad}\operatorname{End}_B(Q_t), \qquad f_t = [\mathbb{F}_t : \mathbb{F}_p], \qquad \dim_{\mathbb{F}_p} Q_t = f_t j_t.$$

The last quotient is a finite field because the endomorphism algebra of an indecomposable finite-dimensional module is local. Its simple module has dimension f_t, so j_t is an integer. The semisimple quotient of C is $\prod_t \operatorname{Mat}(m_t, \mathbb{F}_t)$.

If $f_t m_t > f(n-r) > fn/2$, then $j_t = 1$ by the dimension of A. The radical of $\mathrm{End}_B(Q_t)$ therefore annihilates Q_t and is zero, since this endomorphism algebra acts faithfully. Its action is the field $\mathbb{F}_t$ on a one-dimensional space over that field. The image of B is a subfield of $\mathbb{F}_t$; if it were proper, Q_t would split into more than one copy of its one-dimensional natural module, contradicting indecomposability. Thus B would have the field quotient $\mathbb{F}_t$, and its kernel annihilates $Q_t^{m_t}$. The annihilator is $\mathbb{K}$-stable, and its dimension is greater than $n-r$. The quotient embeds $\mathbb{K}$ because a unital homomorphism from a field is injective. Two different field-quotient kernels are comaximal; their annihilators have zero intersection. They therefore cannot both have dimension greater than half that of A; this proves the uniqueness asserted in the first alternative.

When the initial field-quotient test fails, the preceding contradiction gives $f_t m_t \le f(n-r)$ for every t, which is the second alternative. Lemma E.11 passes this bound to the full stabilizer's new natural modules. All calculations are on spaces of dimension at most d^2 and use Lemma 3.2; its deterministic polynomial dependence on p is absorbed by $p^{O(d)}$. □

Lemma E.13 (Cost and progress of an off-diagonal contraction)**.** *Let ν be the sum of the prime-field dimensions of the distinct current natural argument types and required duals, each counted once. Suppose a matched contraction stack of rank ρ runs between two distinct such types. Let ν' be the corresponding sum for the new large natural types on the same old arguments after its complete stabilizer is computed. Then*

$$\nu' \le \nu - \rho/2.$$

In particular, in the one-pair case with paired arguments X, Y of dimension fn over $\mathbb{F}_p$, output argument C of dimension at least fn and of a different natural type from Y, and

$$P_{x,z}(y) = y - z\frac{Q(x,y)}{Q(x,z)}, \quad Q(x,z) \neq 0, \qquad D_{x,z} = \big(T_j(x,-)P_{x,z}\big)_j,$$

a stack of rank $\rho \ge fn/64$ has at most p^{2fn} target-mark choices, and every nonempty branch has the displayed loss in ν; its stabilizer is a proper subgroup when $\rho > 0$. Pairing corrections vanish in $D_{x,z}$. Each branch is matched by the degree-2 algorithm and continues with the original tensor equation in its complete original coset. All these branch constructions together take $p^{O(d)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$ time, excluding the recursive tensor comparisons.

Proof. Let W contain one representative of each current type and dual, so $\dim W = \nu$. Form the algebra B generated by the commutant of the old natural action, the component contractions, and their adjoints on the required dual spaces. The commutant contains the projections onto the distinct types. Hence each off-diagonal contraction and its adjoint belongs to the commutator ideal $I = B[B,B]B$: for its output projection e, an off-diagonal map D satisfies $eD - De = D$. The sum of the adjoint images has dimension equal to the rank ρ of the stack. Thus $\dim IW \ge \rho$.

Write $W = \bigoplus_t Q_t^{m_t}$ as a B-module and use f_t, j_t as in the preceding lemma. If $j_t = 1$, the image of B on Q_t is a field, so I annihilates Q_t. Therefore

$$\rho \le \dim IW \le \sum_{j_t \ge 2} f_t j_t m_t \le 2\sum_t f_t(j_t - 1)m_t.$$

The natural dimensions for the semisimple centralizer sum to $\sum_t f_t m_t$. Subtracting this from $\nu = \sum_t f_t j_t m_t$ proves the asserted loss for the centralizer, and Lemma E.11 proves it for the complete stabilizer's new large types.

For the displayed one-pair observation, $Q(x, P_{x,z}y) = 0$, so every correction $u_j(Q(x,y))$ disappears after projection. There are at most $|X||Y| = p^{2fn}$ target pairs; the denominator in the

projection also accommodates the actual pairing multiplier. First match the marks in the original group and then match the complete bilinear observation in that coset, keeping the full kernels. This is a controlled degree-2 comparison: the two moving arguments are Y and the output C, and M is fixed. Its denominator is zero. All correlations among the component maps are imposed in this one comparison. The two types are distinct by hypothesis. In the one-pair pattern this also follows from the absence of an additional invariant pairing: a nonzero equivariant map $Y \to C$ would give such a pairing between Y and the original argument C^*. No assertion that they belong to different classical factors is needed. Every solution of the original affine comparison carries the source marks to one of these target pairs and satisfies its projected observation, since all its translations lie in the pairing image. Conversely, every continued branch still imposes the original affine equation. Thus the complete branch answers have exactly the original solution set as their union, recovered by (8.6). Only the group, its restricted actions, and the pulled-back target change, as in (8.5); the reference dimensions and the invariants $P = Z_{\mathrm{quad}} = 0$ and trivial M are preserved. The strict loss shows that a surviving stabilizer cannot equal the preceding group. Since $fn \le d$, the degree-2 theorem and the list size give the stated local bound. □

Putting the field values in one field. The preceding algebra test supplies a distinguished field quotient for each tested tuple when it does not give smaller natural constituents. We need to compare the values for different tuples. The following lemma makes the necessary compatibility explicit.

Lemma E.14 (Compatible field quotients)**.** *Let $A = \mathbb{F}^a$, where $[\mathbb{F} : \mathbb{F}_p] = f$, and let $\mathcal{I}$ be a finite set of marks. For every subset $I \subseteq \mathcal{I}$ of size at most three, suppose that $B_I \le \mathrm{End}_{\mathbb{F}_p}(A)$ is generated by the scalar operators $\mathbb{F}$ and the operators belonging to the marks in I. Suppose B_I has a quotient field $\mathbb{K}_I = B_I/P_I$ whose annihilator*

$$A_I := \{v \in A : uv = 0 \text{ for all } u \in P_I\}$$

has $\mathbb{F}$-dimension at least $a - r$, with $2r < a$. Then these quotient fields admit compatible embeddings into one finite field $\mathbb{K} \supseteq \mathbb{F}$, unique up to a common $\mathbb{F}$-Frobenius automorphism, with

$$[\mathbb{K} : \mathbb{F}] = \mathrm{lcm}_{v \in \mathcal{I}}[\mathbb{K}_{\{v\}} : \mathbb{F}] \le \mathrm{lcm}(1, \dots, a) = \exp(O(a)).$$

Compatibility means that the images of $b + P_I$ and $b + P_{I'}$ agree whenever $b \in B_I$, $I \subseteq I'$, and $|I'| \le 3$. If $|\mathcal{I}| \le p^{O(d)}$ and $fa \le d$, the field and embeddings can be constructed in $p^{O(d)}$ times a polynomial in the explicit operator data. Conjugating the represented algebras transports this construction by one common field isomorphism.

Proof. The kernel of the quotient map on the scalar field $\mathbb{F}$ is zero, so $\mathbb{F}$ embeds in every $\mathbb{K}_I$. For $I \subseteq I'$, the action of B_I on $A_{I'}$ has field image in $\mathbb{K}_{I'}$. Both A_I and $A_{I'}$ have dimension greater than half of $\dim A$, so they intersect nontrivially. The kernels of their field actions of B_I must coincide: a nonzero element of one field image is invertible on every nonzero vector of its module. Consequently

$$P_{I'} \cap B_I = P_I, \qquad b + P_I \longmapsto b + P_{I'}$$

defines a field embedding. These embeddings compose under inclusions. In particular $A_{I'} \subseteq A_I$.

Put $e_v = [\mathbb{K}_{\{v\}} : \mathbb{F}]$. Since the field acts faithfully on a nonzero $\mathbb{F}$-subspace of A, $e_v \le a$. A pair algebra is generated by its singleton algebras, so its field image is their compositum, of degree $\mathrm{lcm}(e_v, e_w)$; the analogous statement holds for triples. Choose embeddings of all these fields into

a field of degree $E = \operatorname{lcm}_v e_v$ over $\mathbb{F}$. Two embeddings of a common subfield differ by a Frobenius power. Write the pair discrepancies as

$$d_{vw} \in \mathbb{Z}/\gcd(e_v, e_w)\mathbb{Z}.$$

The triple embeddings give $d_{vw} + d_{wu} = d_{vu}$ modulo $\gcd(e_v, e_w, e_u)$. For each prime $\ell \mid E$, choose a mark $v(\ell)$ for which $v_\ell(e_v)$ is maximal, where $v_\ell(e)$ denotes the exponent of ℓ in the integer e. Correct the embedding for v by the Frobenius exponent $d_{v,v(\ell)}$ modulo $\ell^{v_\ell(e_v)}$. The triple relation makes all pair embeddings agree at this prime. The Chinese remainder theorem combines the corrections. Any two corrected systems differ by one common exponent modulo E, proving uniqueness.

For completeness, $\log \operatorname{lcm}(1, \dots, a) = O(a)$: the quotient by $\operatorname{lcm}(1, \dots, \lceil a/2 \rceil)$ divides $\binom{a}{\lfloor a/2 \rfloor}$, and iteration gives the bound. To construct $\mathbb{K}$, construct a field extension of degree $\ell^{v_\ell(E)} \le a$ for each prime divisor ℓ; enumeration and irreducibility tests cost $|\mathbb{F}|^{O(a)}$. Their degrees are pairwise coprime, so their tensor product over $\mathbb{F}$ is a field of degree E. Its tensor basis permits arithmetic polynomial in E. A quotient field of degree $e \le a$ embeds in the subfield $\ker(\mathrm{Frob}_{\mathbb{F}}^e - I)$; enumerate that subfield to find a root of its defining polynomial, then make the Frobenius corrections above. The number of triples is $|\mathcal{I}|^{O(1)}$, giving the claimed cost. Uniqueness of the large annihilator and of the compatible embeddings proves the last assertion. □

Lemma E.15 (Using a nonzero field-value map)**.** *Suppose the source and target constructions give $\mathbb{F}_p$-linear maps λ_x, λ_y on a current irreducible argument A_i, with values in $\mathbb{K} \otimes_{\mathbb{F}_p} M$ or $(\mathbb{K}/\mathbb{F}) \otimes_{\mathbb{F}_p} M$. Suppose these maps are unchanged by adding pairing terms, and every solution satisfies*

$$\lambda_y \circ h_{A_i} = \kappa(\theta \otimes I_M) \circ \lambda_x$$

for one common Frobenius automorphism θ of $\mathbb{K}$ and one scalar $\kappa \in \mathbb{F}^\times$. The scalar accounts for the multiplier of the invariant pairing and is the same in all M-coordinates of this map. Assume $[\mathbb{K} : \mathbb{F}_p] \le p^{O(d)}$ and the maps have been computed in the same time bound, times a polynomial in the current input length. If $\lambda_x \neq 0$, one obtains either emptiness, one recursive comparison under a proper subgroup of H, or a complete degree-2 answer, with local cost $p^{O(d)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$. The recursive comparison keeps all three argument spaces and introduces no list of candidate choices.

Proof. The field and value actions are invertible, so every solution carries $\ker \lambda_x$ to $\ker \lambda_y$. A dimension mismatch rejects the comparison. If $0 < \ker \lambda_x < A_i$, compute their complete subspace transporter using Corollary 7.10. If it is $uK \neq \varnothing$, use (8.5) and return to Step G1. The subgroup $K = \mathrm{Stab}_H(\ker \lambda_x)$ is proper because A_i is irreducible. This is one coset, and the original affine equation is still imposed in the child comparison.

Otherwise λ_x is injective. After rejecting incompatible source and target field data, enumerate the at most $[\mathbb{K} : \mathbb{F}_p]$ Frobenius automorphisms θ and $|\mathbb{F}| - 1$ scalars κ. For each pair, the degree-2 algorithm computes the complete transporter

$$u_{\theta,\kappa} K_{\theta,\kappa} = \{h \in H : \lambda_y \circ h_{A_i} = \kappa(\theta \otimes I_M) \circ \lambda_x\}.$$

Indeed this is a linear comparison on $\mathrm{Hom}_{\mathbb{F}_p}(A_i, \mathbb{K} \otimes M)$, or on the indicated additive quotient, with a trivial action on the value coordinates and a single active argument factor. For a nonempty answer, its source stabilizer $K_{\theta,\kappa}$ fixes A_i pointwise: injectivity of λ_x gives $\lambda_x k_{A_i} = \lambda_x$ only when $k_{A_i} = I$. Under $K_{\theta,\kappa}$, the entire original tensor module therefore has at most two active factors. Give the fixed argument its trivial evaluation presentation, and apply Theorem 7.9 to the full affine comparison in (8.5). This includes its current subspaces and translations. Complete every (θ, κ)

case before combining the answers by (8.6). Thus no field-automorphism choices persist in recursive tensor calls.

The field coordinates may have $p^{O(d)}$ entries. They occur only in these completed local comparisons, so their polynomial description and arithmetic costs are absorbed in $p^{O(d)}$. The active argument spaces still have total dimension $O(d)$. The recursive subspace case stores only subspaces and original group matrices of polynomial size, proving the size and cost claims. □

Removing the pairing terms from marked contractions. First consider (E.6). Here $a \geq 64$; put $r = \lfloor a/32 \rfloor$. Write $q_0 = \mathrm{Tr}_{\mathbb{F}/\mathbb{F}_p} b$. For $x \neq 0$, choose a nondegenerate $\mathbb{F}$-subspace $E \leq A$ of dimension at most two containing x. If $b(x,x) \neq 0$, use $E = \mathbb{F}x$. Otherwise choose y with $b(x,y) \neq 0$ and use $E = \langle x, y \rangle_{\mathbb{F}}$. The Gram determinant is nonzero in the latter case. We use all such choices, recording the mark $v = (x, E)$; there are at most $|\mathbb{F}|^{2a}$ choices, since each mark is represented by a pair (x, y), using $y = x$ for the one-dimensional case. Let P_v be the orthogonal projection onto $E^{\perp}$. It is $\mathbb{F}$-linear and self-adjoint for q_0.

Define the prime-field linear operators $\mathcal{M}_{j,T}(x)$ by

$$q_0(\mathcal{M}_{j,T}(x)y, z) = T_j(x, y, z), \qquad D_{j,T,v} = P_v \mathcal{M}_{j,T}(x) P_v.$$

No commutation of $\mathcal{M}_{j,T}(x)$ with $\mathbb{F}$ is assumed. For a set $\mathcal{V}$ of marks put

$$B_{T,v} = \mathrm{alg}_{\mathbb{F}_p}(\mathbb{F}, I, P_v, D_{1,T,v}, \ldots, D_{m,T,v}), \qquad B_{T,\mathcal{V}} = \mathrm{alg}_{\mathbb{F}_p}\{B_{T,v} : v \in \mathcal{V}\}. \tag{E.9}$$

These are represented algebras on A, containing the specified scalar field $\mathbb{F}$, which need not be central in the whole algebra.

Lemma E.16 (Projected contractions and their quotient values)**.** *Adding* (E.6) *changes the displayed operators by*

$$D_{j,T,v} \longmapsto D_{j,T,v} + \ell_{1,j}(x) P_v. \tag{E.10}$$

Consequently all algebras in (E.9) *are unchanged by every translation in $J(Z)$.*

Suppose that, for every $\mathcal{V}$ of size at most three, there is a field quotient $\phi_{\mathcal{V}} : B_{T,\mathcal{V}} \twoheadrightarrow \mathbb{K}_{\mathcal{V}}$ whose kernel $I_{\mathcal{V}}$ satisfies

$$A_{\mathcal{V}} := \mathrm{Ann}_A(I_{\mathcal{V}}), \qquad \dim_{\mathbb{F}} A_{\mathcal{V}} \geq a - r.$$

Use the compatible field embeddings of Lemma E.14 into a common field $\mathbb{K} \supseteq \mathbb{F}$. The images of $D_{j,T,v}$ define a map

$$\lambda_T : A \longrightarrow \mathbb{K}^m$$

which is independent of the mark above x, is $\mathbb{F}_p$-linear, and satisfies

$$\lambda_{T+\mathcal{J}\ell,j}(x) = \lambda_{T,j}(x) + \ell_{1,j}(x). \tag{E.11}$$

The same quotient fields and compatible embeddings are used on both sides. Their images are generated by $\mathbb{F}$ and these values.

Proof. Self-adjointness of P_v gives $q_0(D_{j,T,v}y, z) = T_j(x, P_v y, P_v z)$. The two pairing terms containing $b(x, P_v y)$ or $b(x, P_v z)$ vanish; so do their reversed pairings. The remaining term is $q_0(\ell_{1,j}(x)P_v y, z)$, proving (E.10). The added scalar multiple of P_v already lies in the algebra, and subtraction gives the reverse inclusion. This proves equality of the algebras, jointly over every value coordinate.

The field quotient embeds $\mathbb{F}$ since its restriction to this unital subfield is injective. Thus $A_{\mathcal{V}}$ is $\mathbb{F}$-stable. On $A_{\mathcal{V}}$, every element with nonzero field image acts invertibly. The image of P_v is an

idempotent. It cannot be zero, since then $A_{\mathcal{V}} \subseteq \ker P_v$ would give $a - r \le 2$. Hence it is one. The field is therefore generated by the images of the $D_{j,T,v}$ and $\mathbb{F}$.

For every $\mathbb{F}_p$-linear operator D and every $\mathbb{F}$-linear projection P with kernel of $\mathbb{F}$-dimension at most two,

$$\operatorname{rank}_{\mathbb{F}_p}(D - PDP) \le \operatorname{rank}((I-P)D) + \operatorname{rank}(PD(I-P)) \le 4f.$$

Two marked operators at the same x differ by rank at most $8f$. Operators at $x, y, x+y$, with arbitrary marks, have additive defect of rank at most $12f$, because the uncompressed $\mathcal{M}_{j,T}(x)$ depend linearly on x. Each defect lies in the algebra of its at most three marks. A nonzero image in that algebra's field quotient would act invertibly on $A_{\mathcal{V}}$, and hence have rank at least $f(a-r) > 12f$. Every defect therefore has zero field image. This proves independence of the mark and additivity. Comparing the marks at x and tx, for $t \in \mathbb{F}_p$, proves homogeneity by the same argument. The value at zero is zero and needs no mark. Finally the image of P_v is one, so (E.10) gives (E.11). □

Constructing the extraction map when the quotient fields are scalar. The following is a finite test and a construction of the map used in the tensor/parameter separation, not an assumption that an equivariant projection of Z onto $J(Z)$ exists.

Lemma E.17 (Constructing the pairing parameters from scalar quotients)**.** *For each orientation of* (E.6)*, perform the following test on every set $\mathcal{V}$ of at most three marks: the represented algebra $B_{T,\mathcal{V}}$ must have a quotient onto the specified field $\mathbb{F}$, restricting to the identity on $\mathbb{F}$, with annihilator of $\mathbb{F}$-dimension at least $a - r$. Let $\Omega \subseteq Z$ be the set passing all these tests in all three orientations. Membership in Ω and an explicit map*

$$s : \Omega \longrightarrow \mathcal{S}/\ker \mathcal{J}$$

are computable in time $p^{O(fa)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, \log p)$. The set Ω is preserved by H and by every translation in $J(Z)$, and the computed map satisfies

$$s(\rho_h T) = \sigma_h s(T), \qquad s(T + \mathcal{J}t) = s(T) + (t + \ker \mathcal{J}) \quad (h \in H, T \in \Omega, t \in \mathcal{S}). \tag{E.12}$$

Proof. Lemma 3.2 computes the simple algebra quotients. A candidate quotient to $\mathbb{F}$ is a one-by-one matrix factor with field degree f; its restriction to the specified $\mathbb{F}$ is an isomorphism, which fixes its identification with $\mathbb{F}$. The common kernel on A of its ideal is computed by linear equations. This tests the required dimension.

There is at most one such quotient. Indeed, the annihilators of two different simple quotient types have zero intersection: a nonzero vector in the intersection would identify their simple scalar actions. Two subspaces of dimension greater than $\frac{1}{2}\dim_{\mathbb{F}_p} A$ cannot have zero intersection. For nested sets of at most three marks, restricting the larger quotient to the smaller algebra still fixes $\mathbb{F}$ and has an annihilator of the required dimension. Uniqueness therefore makes the quotient maps compatible. In this scalar case no choice of a common extension field is needed.

Take the unique quotient value of $D_{j,T,v}$ in the first orientation and denote it by $\ell_{1,j,T}(x)$. The rank-defect proof in Lemma E.16 proves that this is independent of v and is $\mathbb{F}_p$-linear in x. The other two orientations produce $\ell_{2,j,T}$ and $\ell_{3,j,T}$, using the order of the paired arguments in (E.6). These functions are the explicit parameter tuple defining $s(T)$, transferred to $\mathcal{S}/\ker \mathcal{J}$ through Lemma E.10.

Equation (E.10) leaves all tested algebras unchanged under $T \mapsto T + \mathcal{J}t$. It proves both translation invariance of Ω and the second identity in (E.12), for every parameter t, with no restriction to the actual values of the cocycle.

For completeness, the covariance uses the whole given group, including its scalar and field actions. Lemma 5.4 supplies these actions explicitly. After the computed identifications of the three natural occurrences, an element of H acts on them by semilinear maps g_1, g_2, g_3 with a common field automorphism τ, differing only by field scalars from one common natural-module transformation. In the first orientation there is a specified $\kappa_{23} \in \mathbb{F}^\times$ with

$$b(g_2 y, g_3 z) = \kappa_{23}\tau(b(y, z)).$$

The mark $v = (x, E)$ is carried to $(g_1 x, g_1 E)$, whose projector is $g_2 P_v g_2^{-1}$. The transformed compressed operator is

$$D_{j,\rho_h T,hv} = \kappa_{23}^{-1} g_2 D_{j,T,v} g_2^{-1}.$$

This conjugacy, together with the common field automorphism, carries the algebra and its unique quotient to those of the transformed tensor. Consequently its scalar quotient value is $\kappa_{23}^{-1}\tau(\ell_{1,j,T}(x))$. Substituting in (E.6) shows that this is exactly the first coordinate of the parameter action σ_h. The other orientations give the other coordinates. The same κ_{23} applies to every M-coordinate, since H fixes the chosen basis of M. It is computed by evaluating the two pairing forms on one pair with nonzero pairing value. Thus Ω is H-invariant and the first identity in (E.12) follows.

There are at most $p^{6fa+O(1)}$ tested triples of marks in each of three orientations. All algebras act on the original fa-dimensional space and contain all m scalar observations; matrix-span closure and the represented-algebra computations have polynomial dependence on $\mathrm{InSize}_{\mathrm{cur}}$ and on p. Values on a prime-field basis give the matrices of the extracted linear functions. This proves the stated bound. □

The mixed general-linear pattern. For (E.7), put $\mathcal{M}_{j,T}(x)y = B_j(x, y)$. A mark is (x, ϕ) with $x \neq 0$, $\phi \in \mathrm{Hom}_{\mathbb{F}}(A, \mathbb{F})$ and $\phi(x) = 1$. Use

$$P_{x,\phi} = I - x \otimes \phi, \qquad D_{j,T,(x,\phi)} = P_{x,\phi}\mathcal{M}_{j,T}(x)P_{x,\phi}.$$

There are at most $|\mathbb{F}|^{2a}$ marks. The added pairing term changes this operator by $\alpha_j(x)P_{x,\phi}$: the term $\beta_j(P_{x,\phi}y)x$ is killed by the left projection. The compression error, same-point difference and additive defect have ranks at most $2f$, $4f$ and $6f$, respectively. Thus the preceding construction and proofs apply with these smaller bounds. Apply the construction also to $B_j(y, x)$ to extract β_j. Naturality of $P_{x,\phi}$ under the given natural and dual actions replaces self-adjointness. More explicitly, if g_1, g_2, g_3 are the two input actions and the output action on B_j, write $g_3 = \kappa g_2$ after the fixed identifications. Then

$$D_{j,\rho_h T,hv} = \kappa g_2 D_{j,T,v} g_2^{-1},$$

with the field automorphism induced by g_2. The scalar $\kappa \in \mathbb{F}^\times$ is computed from these matrices and is common to all j. This gives an explicitly tested H- and $J(Z)$-invariant domain and the map s satisfying (E.12) for this pattern.

Lemma E.18 (Contractions for a smaller unpaired argument)**.** *In* (E.8)*, suppose* $\dim_{\mathbb{F}_p} C \leq fa$*, where* $a = \dim_{\mathbb{F}} X = \dim_{\mathbb{F}} Y$*. Put* $r = \lfloor a/32 \rfloor$*. Use the nondegenerate trace pairing* $q_0 = \mathrm{Tr}_{\mathbb{F}/\mathbb{F}_p} Q$*. For* $\xi \in C^*$ *define*

$$q_0(\mathcal{M}_{j,T}(\xi)x, y) = \xi(T_j(x, y)), \qquad \mathcal{M}_{j,T}(\xi) \in \mathrm{End}_{\mathbb{F}_p}(X).$$

Adding $\mathcal{J}(u)$ *changes this operator by* $s_{j,u}(\xi)I$*, where* $s_{j,u}(\xi) \in \mathbb{F}$ *is uniquely determined by*

$$\xi(u_j(t)) = \mathrm{Tr}_{\mathbb{F}/\mathbb{F}_p}(s_{j,u}(\xi)t) \qquad (t \in \mathbb{F}). \tag{E.13}$$

The algebras generated by $\mathbb{F}$ and the complete operator tuples are consequently invariant under all translations in $J(Z)$. Testing their tuples of size at most three gives the same field quotient alternatives. On the domain where all these quotients are $\mathbb{F}$ with annihilator dimension at least $a - r$, the quotient values explicitly construct the map s in (E.12)*. The computation costs $p^{O(fa)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, \log p)$.*

Proof. The nondegenerate field trace gives the unique value $s_{j,u}(\xi)$ in (E.13) and proves the scalar-shift formula. No projection is needed. The operator map is already $\mathbb{F}_p$-linear in ξ, so the compatible quotient values are $\mathbb{F}_p$-linear without a rank-defect argument. If they all lie in $\mathbb{F}$, call them $s_{j,T}(\xi)$. For each $t \in \mathbb{F}$, the rule

$$\xi \longmapsto \mathrm{Tr}_{\mathbb{F}/\mathbb{F}_p}(s_{j,T}(\xi)t)$$

is a linear functional on C^*, and hence determines a unique $u_{j,T}(t) \in C$. The resulting map $u_{j,T} : \mathbb{F} \to C$ is $\mathbb{F}_p$-linear and is computed by a trace-dual basis. Its tuple is the required $s(T)$ in the parameters of (E.8). For the covariance, let g_X, g_Y, g_C be the three given actions in these coordinates. There is a computed common field automorphism τ and $\kappa \in \mathbb{F}^\times$ such that $Q(g_X x, g_Y y) = \kappa\tau(Q(x, y))$. For $\xi' = \xi \circ g_C^{-1}$ one has

$$\mathcal{M}_{j,\rho_h T}(\xi') = \kappa^{-1} g_X \mathcal{M}_{j,T}(\xi) g_X^{-1}.$$

Again κ is common to all M-coordinates. This formula, uniqueness of the quotient maps, and the scalar-shift identity prove both identities in (E.12) for the entire H-action. There are at most $p^{3\dim_{\mathbb{F}_p} C}$ tested tuples, which is bounded by p^{3fa} under the dimension hypothesis. □

A common zero block chosen invariantly. For a matrix space $\mathcal{B} \le \mathrm{Hom}_{\mathbb{F}_q}(V, W)$ and $U \le V$, write $\mathcal{B}(U)$ for the linear span of all $B(u)$ with $B \in \mathcal{B}$ and $u \in U$.

Lemma E.19 (A canonical common zero block)**.** *Suppose every matrix in $\mathcal{B} \le \mathrm{Hom}_{\mathbb{F}_q}(V, W)$ has rank at most R. There is a uniquely specified pair $(U, \mathcal{B}(U))$ such that*

$$\mathrm{codim}_V U + \dim \mathcal{B}(U) \le 4R.$$

The pair is covariant under invertible changes in the domain and codomain and can be computed deterministically in $q^{O(1)}\mathrm{poly}(\dim V, \dim W, \dim \mathcal{B}, \log q)$ time. The assertion includes all finite fields, with no lower bound on q.

Proof. The rank bound is due to Haramaty–Shpilka [HS10]. We use its matrix-space formulation in [BCL$^+$25, Lemma 6.6], which applies over every field, including $\mathbb{F}_2$, and gives

$$\mathrm{ncrk}(\mathcal{B}) = \min_{U \le V} \big(\mathrm{codim}_V U + \dim \mathcal{B}(U)\big) \le 4 \max_{B \in \mathcal{B}} \mathrm{rank}\, B \le 4R.$$

Here [BCL$^+$25, Section 6.1] calls the middle quantity the minimal covering number; the formulation includes rectangular matrix spaces.

For the canonical choice, maximize

$$d_{\mathcal{B}}(U) := \dim U - \dim \mathcal{B}(U) \quad \text{over } U \le V,$$

and take the smallest maximizing subspace. It is unique: $d_{\mathcal{B}}$ is supermodular, since $\mathcal{B}(U + U') = \mathcal{B}(U) + \mathcal{B}(U')$ and $\mathcal{B}(U \cap U') \le \mathcal{B}(U) \cap \mathcal{B}(U')$. The intersection of two maximizers is therefore again a maximizer. This choice minimizes $\mathrm{codim}\, U + \dim \mathcal{B}(U)$, so it satisfies the cited bound. Its uniqueness gives covariance under every left-right equivalence of matrix spaces.

To compute it, pad the domain and codomain with zero summands to the same dimension n. The deterministic noncommutative-rank algorithm of [IQS18, Theorem 1.5 and Corollary 1.7(2)] computes a maximum-deficiency shrunk subspace. Its construction returns the smallest such subspace, as recorded in [IMQ22, Propositions 7–8 and the paragraph between them]. Every maximizing subspace contains the added zero domain summand; remove this summand from the computed subspace. If the padded maximum deficiency is zero, take the zero subspace instead. This gives precisely the choice above.

For small fields, Corollary 1.7(2) of [IQS18] and its proof in Section 5.2 apply: the noncommutative rank is unchanged by extension of the field, and the algorithm returns a shrunk subspace over the original field. Equivalently, compute the smallest maximizing subspace over a sufficiently large finite extension. Uniqueness makes it invariant under Galois automorphisms, so it descends to $\mathbb{F}_q$; its descended basis is computed by linear algebra. An extension of degree $O(\log_q(n+2))$ suffices. It can be constructed by enumerating monic polynomials in $q^{O(1)}\mathrm{poly}(n)$ time. The matrix algorithms and descent have polynomial cost, proving the stated conservative bound. □

A paired pair with a larger unpaired argument. We now treat the size case in which marking vectors in the unpaired argument would cost too much. By reorienting the trilinear map, write the tensor space as

$$Z \cong \mathrm{Hom}_{\mathbb{F}_p}(X \otimes_{\mathbb{F}_p} Y, C \otimes_{\mathbb{F}_p} M), \qquad C = A_k^*.$$

Here X and Y are the other two arguments, and C is dual to the unpaired argument. Let $\mathbb{F}/\mathbb{F}_p$ have degree f, let X, Y have dimension a over $\mathbb{F}$, and let $Q : X \times Y \to \mathbb{F}$ be the specified nondegenerate $\mathbb{F}$-bilinear pairing. Write $b = fa$, $c = \dim_{\mathbb{F}_p} C$, and $m = \dim_{\mathbb{F}_p} M$. The action on the pairing values is the given invertible $\mathbb{F}_p$-linear semilinear map τ_h satisfying

$$Q(h_X v, h_Y w) = \tau_h(Q(v, w)).$$

The only pairing terms in this case are

$$\kappa(u)(v, w) = u(Q(v, w)), \qquad u \in \mathrm{Hom}_{\mathbb{F}_p}(\mathbb{F}, C \otimes M), \qquad J(Z) = \mathrm{im}\,\kappa.$$

These formulas also cover a sesquilinear presentation after its recorded duality identification. The action on M is trivial by Lemma E.2.

Lemma E.20 (The larger unpaired argument)**.** *In the preceding one-pair case, suppose $a \geq 64$ and $c \geq b$. Let $\mathtt{F}, \mathtt{F}' \in Z$ be the current source and target, with affine translations in $J(Z)$. One can construct one of the following, in $p^{O(d)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$ local time:*

(i) *a complete degree-2 answer;*

(ii) *a single necessary transporter coset with a proper subgroup of H, under which the original comparison is continued;*

(iii) *at most p^{2b} necessary transporter cosets, each with a proper subgroup of H, obtained from a compressed contraction of rank $r \geq b/64$. The decrease of the distinct-natural-type dimension sum is at least $r/2$, as in Lemma E.13.*

All remaining comparisons keep the original tensor equation and have unchanged argument dimensions. Their complete answers recover the original transporter and its full source stabilizer. The new explicit data have polynomial length in the initial input.

Proof. For $v \in X \setminus \{0\}$ and $w \in Y$ with $Q(v,w) \neq 0$, set

$$P_{v,w}(y) = y - w\frac{Q(v,y)}{Q(v,w)}, \qquad D_{v,w}(\mathtt{F})(y) = \mathtt{F}(v, P_{v,w}y).$$

Thus $\operatorname{im} P_{v,w} = \ker Q(v,-)$ and $D_{v,w}(\kappa(u)) = 0$ for every u. The formula with the denominator makes the projector covariant also when a transported pair has pairing value different from 1. Changing the chosen nonzero scalar multiple of w does not change $P_{v,w}$.

Enumerate these pairs for both tensors and compute their ranks. There are at most p^{2b} pairs. The existence of a rank at least $b/64$ is preserved by every affine transporter, because pairing translations disappear and changes of arguments and values are invertible. If precisely one tensor has such a pair, return emptiness. Otherwise, if one exists, fix a source pair (v,w) and enumerate its orbit under H. For each orbit point ξ choose an original lift $r_\xi \in H$. Put $H_0 = \operatorname{Stab}_H(v,w)$ and pull back the target by $\alpha_{r_\xi^{-1}}$. Under H_0, the compressed observation is an ordinary simultaneous linear-map comparison

$$D_{v,w}(\rho_k \mathtt{F} + c_k) = k_C D_{v,w}(\mathtt{F}) k_Y^{-1} \quad (k \in H_0).$$

Its two active factors are Y^* and C, and its M coordinates are fixed copies. The degree-2 algorithm computes its complete answer uK or emptiness. In a nonempty branch, restrict the original affine comparison to $r_\xi uK$ using (8.5). Since $v \neq 0$ and the current X-action is a nontrivial irreducible natural action, $K \leq H_0 < H$. For the selected classical product D of Step G6, the operator source Y and output $C = A_k^*$ have different simple D-module types: a nonzero intertwiner $Y \to C$ would give an additional invariant pairing between Y and A_k, contrary to the one-pair hypothesis. Thus Lemma E.13 gives the claimed loss. In particular the exponent $2b$ is at most 256 times that loss. No equation from the original comparison is discarded. Taking the union of the completed branches by (8.6) gives its complete answer.

It remains to handle the case in which every compressed rank is less than $b/64$. Put $r_0 = \lfloor b/64 \rfloor$ and form the $\mathbb{F}_p$-linear pencil

$$\widetilde{D}_v : Y \longrightarrow (C \otimes M) \oplus \mathbb{F}, \qquad y \longmapsto (\mathtt{F}(v,y), Q(v,y)) \quad (v \in X).$$

For $v \neq 0$, the last coordinate has rank f and kernel $\operatorname{im} P_{v,w}$. Rank–nullity on that coordinate gives

$$\operatorname{rank} \widetilde{D}_v \leq r_0 + f.$$

This also holds for $v = 0$. Every member of the matrix space $\mathcal{B}_\mathtt{F} := \{\widetilde{D}_v : v \in X\}$ has this bound, since $v \mapsto \widetilde{D}_v$ is $\mathbb{F}_p$-linear. Apply Lemma E.19 over $\mathbb{F}_p$ to obtain its canonical pair (W,E), where

$$E = \mathcal{B}_\mathtt{F}(W), \qquad \operatorname{codim}_Y W + \dim E \leq 4(r_0 + f) \leq b/8 < b.$$

The penultimate inequality uses $a \geq 64$. Compute (W', E') in the same way for $\mathtt{F}'$.

We verify covariance for the actual affine equation. Write $c_h = \kappa(u_h)$. If $\alpha_h(\mathtt{F}) = \mathtt{F}'$, then the two pencils are related by the changes h_X and h_Y in their parameter and domain and by the invertible output map

$$(d,s) \longmapsto (h_C d + u_h(\tau_h(s)), \tau_h(s)) \quad (d \in C \otimes M,\ s \in \mathbb{F}).$$

Here h_C acts trivially on the M coordinate. By the uniqueness of the canonical pair, $h_Y W = W'$, and the displayed output map sends E to E'. In particular $W \neq 0$.

If $W \neq Y$, compare W and W' in the original Y-action using Corollary 7.10; if their dimensions differ, the answer is immediately empty. A nonempty answer is a single coset uH_W. Because

$0 < W < Y$ and H acts irreducibly on Y, $H_W < H$. Restrict the original comparison to this coset. This is an exhaustive necessary condition and introduces no family of candidate cosets.

Suppose next that $W = Y$. If $W' \neq Y$, return emptiness. Nondegeneracy of Q makes $E \to \mathbb{F}$ surjective. Put

$$E_0 = E \cap ((C \otimes M) \oplus 0), \qquad E_0' = E' \cap ((C \otimes M) \oplus 0),$$

and identify these subspaces with subspaces of $C \otimes M$. The output shear acts trivially on its last-coordinate kernel, so every affine solution satisfies $h_C E_0 = E_0'$. Fix the recorded basis of the trivial module M and its coordinate projections $\pi_j : C \otimes M \to C$. If $E_0 \neq 0$, choose the first j for which $W_C := \pi_j(E_0) \neq 0$, and put $W_C' := \pi_j(E_0')$. Every solution sends W_C to W_C'. Moreover

$$0 < \dim W_C \leq \dim E_0 \leq \dim E \leq b/8 < c.$$

Thus W_C is a proper nonzero subspace in the irreducible C-action. Equivalently use its annihilator in the original argument A_k. Corollary 7.10 computes the complete necessary subspace transporter as one coset with a proper source stabilizer, or proves emptiness. Restrict the original affine comparison to that coset. Fixing the basis of M is legitimate because its action is trivial; therefore the choice of the first nonzero coordinate is covariant.

Finally suppose $W = Y$ and $E_0 = 0$. The projection $E \to \mathbb{F}$ is an isomorphism, so E is the graph of a unique linear map $u : \mathbb{F} \to C \otimes M$. Since E contains all $(\mathtt{F}(v, y), Q(v, y))$, we have $\mathtt{F} = \kappa(u) \in J(Z)$. The entire affine orbit of $\mathtt{F}$ is therefore contained in $J(Z)$. If $\mathtt{F}' \notin J(Z)$, return emptiness. Otherwise pull the comparison back through the injective map κ. The parameter module $\mathrm{Hom}_{\mathbb{F}_p}(\mathbb{F}, C \otimes M)$ has one active natural factor and the semilinear evaluation line $\mathbb{F}$, with fixed copies indexed by the basis of M. Its action is $u \mapsto h_C u \tau_h^{-1}$ and its affine translations are the known u_h. This is a controlled degree-2 comparison and computes the full original transporter, including its kernel.

All cases have now been covered. The only enumeration used at this step is the list of at most p^{2b} marks. The canonical pencils have domain dimension $b \leq d$ and output dimension $cm + f$, which is polynomial in the explicit input and bounded by the recorded tensor coordinates. Their bases, intersections, coordinate projections, and graph maps are computed by linear algebra. Every subspace comparison adds a frame domain of dimension $O(d)$; the parameter comparison adds only the f-dimensional semilinear line. Thus every completed auxiliary call has the asserted controlled presentation and original dimension $O(d)$. None of the output comparisons enlarges a tensor argument or its value space. The recorded original coordinate maps give the asserted local time and uniform input length bounds. □

What a failed scalar test returns. Apply Lemma E.12 to every tested algebra. If it has no field quotient with the required large annihilator, the tuple is the input to Lemma E.11. In the fully paired cases, the images of its at most six marked vectors have at most p^{6fa} choices, and fa is the dimension of each argument. In the one-pair case, its at most three marks lie in C^*, giving at most $p^{3\dim_{\mathbb{F}_p} C}$ choices, where $\dim_{\mathbb{F}_p} C \leq fa$. Thus the logarithm of the number of choices is $O(\min_i \dim_{\mathbb{F}_p} A_i)$ in both cases. The complete comparison uses the operators modulo their scalar projection subspaces; the original tensor equation is still imposed in each returned comparison.

If every tested algebra has a large-annihilator field quotient, Lemma E.14 and Lemma E.16 give the common field $\mathbb{K}$ and a linear map λ_T. Its class

$$\overline{\lambda}_T : A \longrightarrow (\mathbb{K}/\mathbb{F})^m$$

is unchanged by all pairing translations. For the one-pair pattern its domain is C^* instead. Nonzero such maps are handled by Lemma E.15, with the field automorphism and the nonzero scalar multiplier of the pairing both included. If the map is zero, its entries all lie in $\mathbb{F}$; since these entries and

$\mathbb{F}$ generate $\mathbb{K}$, one has $\mathbb{K} = \mathbb{F}$. Performing the test in every permitted orientation therefore either gives one of those earlier exits or reaches the explicit domain of Lemma E.17 or Lemma E.18. The one-pair case with $\dim_{\mathbb{F}_p} C > fa$ is treated separately by Lemma E.20, which adjoins the pairing value to the output; it is not covered by the smaller-argument cost bound above. The pairing cases are selected from the computed invariant-pairing spaces, not solely from names of defining-field twists: after restriction to $\mathbb{F}_p$, such twists can be isomorphic. In particular, the single-pair case requires only that the third position has no additional invariant pairing with the other two, and does not assert that its action comes from a different classical factor. If there is no pairing at all, $J(Z) = 0$, so the normalized cocycle is zero and the comparison is already ordinary.

Lemma E.21 (Complete alternatives in Step G7)**.** *For every recursive input reaching Step G7, the three exits specified there are exhaustive and constructible. More precisely, one obtains emptiness, a complete degree-2 answer, an ordinary comparison under the same group and with the same argument spaces, or a family of comparisons under proper subgroups of H. Each comparison in the last family keeps the original argument spaces and reference dimensions and starts at Step G1. Its data have the form* (8.5) *or* (8.11)*. All outputs have $P = Z_{\mathrm{quad}} = 0$ and the same trivial value module M.*

The complete answers recover the original transporter and its full source stabilizer. Construction, degree-2 completion, and recovery take $p^{O(d)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$ time, excluding the internal work of resulting tensor comparisons. Only the marked contraction alternatives produce a family of recursive candidate cosets: each uses at most six marks in arguments of minimum current dimension. The other exits produce at most one recursive coset or complete all their cases locally.

Proof. The input has the normalized translations supplied by Lemma E.7. An ordinary input is passed on immediately. The direct tests of Lemma E.8 give the stated answers when they apply. We now cover every remaining input.

The threshold bound at the start of this subsubsection gives defining-field natural dimensions at least 64, so all the preceding contraction constructions apply.

Compute the pairing spaces and choose their coordinate form in Lemma E.10. If there is no pairing, $J(Z) = 0$ and the normalized cocycle vanishes, so the input is ordinary. If there is exactly one paired pair and the unpaired argument has greater prime-field dimension, apply Lemma E.20. This gives either a degree-2 completion, a single proper-subspace restriction, or the high-rank marked family, with all original tensor conditions still imposed. These are exits (a) and (b).

For the self-dual and mixed general-linear patterns, use the projected operators constructed above. For a single pair with a smaller or equal unpaired argument, use Lemma E.18. In each case, test every tuple of at most three marks for each tensor. The existence of a tuple without a sufficiently large field quotient is invariant under the group action and pairing translations. If precisely one tensor has such a tuple, return emptiness. If a source tuple has no field quotient with the required large annihilator, Lemma E.12 bounds all natural modules of its centralizer on the operator argument by the following number, where a is that argument's defining-field dimension and f is the field degree over $\mathbb{F}_p$:

$$f(a - \lfloor a/32 \rfloor) \le (63/64)fa.$$

Fix this source tuple and enumerate its possible images in H. First compute the mark transporter rH_0 in the original group. For each nonempty case, pull back the target by $\alpha_{r^{-1}}$ and match the complete operator tuple under H_0, modulo its scalar projection subspaces. Explicitly, the observation of a tensor t is

$$\varepsilon(t) = (D_{j,t,v} + \mathbb{F}P_v)_{j,v} \in \bigoplus_{j,v} \mathrm{End}_{\mathbb{F}_p}(A)/\mathbb{F}P_v,$$

where v runs over the chosen marks and $P_v = I$ in the unprojected one-pair case. Compare $\varepsilon(x)$ with $\varepsilon(\alpha_{r^{-1}}(y))$. This is a degree-2 comparison: fixing the contracted arguments leaves two active factors; the invariant scalar subspaces are its explicit relations. Pairing translations map into those relations, so the induced observation action is linear. The same element of H_0 is required to match every coordinate and every mark. A nonempty answer uK gives the child (8.5) for the coset ruK.

The marks include a nonzero vector of a nontrivial irreducible argument. Therefore $K \le H_0 < H$. Lemma E.11 passes the centralizer bound to every new large natural constituent, even when the full returned group still moves the field or the pairing multiplier. There are at most p^{6s} target tuples, where $s = \min_i \dim A_i$: in the fully paired cases all dimensions coincide; in the one-pair case it is the smaller unpaired argument that is marked. Every original solution has one of these images, and after the observation comparison the original affine equation is still solved in the resulting coset. Thus this is exit (b), preserving the complete solution set.

Suppose now that all the tested tuples have their large field quotients. Lemmas E.14 and E.16, or the unprojected version in Lemma E.18, construct the linear field-value maps. Reject unequal common field degrees; for equal degrees, use the same deterministically constructed prime-power extensions and tensor basis in the two constructions, embedding both sets of quotient fields in that one field as in Lemma E.14. Modulo the original defining field these maps are unchanged by all pairing translations. If any orientation gives a nonzero map, apply Lemma E.15: a proper nonzero kernel gives one proper-subgroup comparison, and an injective map allows the complete comparison to be solved by the degree-2 algorithm. All field-automorphism and multiplier cases are completed locally before their answers are combined.

If every orientation has zero quotient-value map, every field value lies in the original defining field. The compatible field is generated by these values and that field, so no extension remains. The source belongs to the explicitly tested domain Ω in Lemma E.17 or its mixed general-linear or one-pair version. Test the target as well. The domain is preserved by the whole H-action and by all $J(Z)$-translations, so failure of the target test proves emptiness. Otherwise compute $s(x)$ and $s(y)$ and apply Lemma E.9. An empty parameter answer rejects the comparison. A nonempty answer uK gives an ordinary comparison: if $K < H$ this is exit (b), and if $K = H$ this is exit (c). The exact identities (E.5) preserve both the transporter and its source stabilizer. This completes the exhaustive case analysis.

All comparisons under proper subgroups use the original ρ, c, N, z, x, y restricted and pulled back as in (8.5); the ordinary replacement uses the same tensor module and value space. Thus the input format, the three invariants, and the argument-dimension assertions are preserved. Empty branches are discarded and completed nonempty branches are combined by (8.6).

There are $p^{O(d)}$ tested marks and tuples. Each algebra, projection, subspace and pairing map uses matrices of polynomial dimension in the current explicit input. The represented-algebra and degree-2 theorems give the asserted local cost. A common field can have exponential dimension, but Lemma E.15 accounts for that dimension within the same $p^{O(d)}$ local bound and completes all its cases. No such field is added to a recursive input. The child inputs contain only subspaces, tensors, and group matrices in the preceding coordinate spaces, or the ordinary tensors obtained from scalar pairing parameters. Their uniform size is verified with the other recursive inputs below. □

Paying for repeated choices of marked images. An image choice is counted here only when its condition remains imposed while another tensor comparison is solved. Completed field cases, successive degree-2 equations, and the subset calculation of Step G4 contribute additive local work; they do not multiply the number of tensor descendants.

Lemma E.22 (Cumulative cost of marked images)**.** *Start at the initial call or a call made at Step G3, with reference dimensions $a_i = \dim A_i$. Follow one path until the next call made at Step G3, or until a local completion. Include the ordinary and projected contraction choices, as well as the multiplicity choices of Step G5. If their target-image lists are Ξ_e, then, for an absolute constant C,*

$$\sum_e \log_p |\Xi_e| \le \begin{cases} C \sum_i (a_i - a_i'), & \text{at the next reset with dimensions } a_i', \\ C \sum_i a_i, & \text{at a local completion or intermediate computation.} \end{cases}$$

Choices already imposed on the original coset are kept on every return to Step G1; such a return does not start a new image budget.

Proof. Put $c_0 = \min_i a_i$; a zero argument is completed without an image list. Multiplicity choices are paid directly by Lemma E.6, and their children immediately satisfy the reset test. Before the first contraction comparison no other persistent list has been opened.

At a contraction comparison let $c = \min_i \dim A_i \le c_0$. An ordinary high-rank contraction uses one mark and satisfies $\rho \ge c/64$. A projected off-diagonal contraction uses two marks on paired arguments of dimension $c = fn$ and satisfies $\rho \ge c/64$. Thus its exponent is at most $2c$ and at most $256(\nu - \nu')$, by Lemma E.13. A same-type tuple uses at most three marks in a smallest argument, or six marks on paired arguments of that dimension. When the unpaired argument is smaller, it is the contraction argument, so its at most three marks again cost at most $3c$. The bad-tuple alternative of Lemma E.12, with $r = \lfloor n/32 \rfloor$ and $n \ge 64$, bounds every new natural constituent of its operator argument by

$$f(n - r) \le (63/64) fn.$$

The ensuing group reductions either complete the comparison or produce reset calls: a flag quotient already this small meets the test; a nontrivial permuted decomposition or multiplicity reduction meets it by Steps G4 and G5. Consequently a tuple choice is a final charge of at most $6c_0$ before such an output.

It remains to sum high-rank choices which continue without a reset. At the first contraction comparison, tag the three natural arguments and their formal duals, giving at most six occurrences. Until the reset test holds, each occurrence has at most one continuing constituent of dimension greater than $127a_i/128$: two such constituents cannot fit in that original argument. Follow these unique constituents through restrictions. Two previously isomorphic occurrences remain isomorphic after restriction and selection of their unique large constituents. Here types mean types for the selected classical product: when that product changes, use the subgroup mapping onto the new product and lying in the old product's full preimage, constructed in Lemma E.11. The old intertwiners commute with this subgroup and pass to its simple constituents. Thus differing auxiliary characters in the full group do not split a previously identified type. Thus their partition into distinct types can only merge; at most five restriction steps merge classes.

Between mergers, the decrease in ν is at most twice the decrease in $\sum_i \dim A_i$, since each distinct type is represented by an argument or its formal dual. These losses telescope. Charge a step that merges classes, and the final step that completes or first splits off a reset comparison, by its bound $6c_0$ instead. There are at most six such exceptional charges. If a reset is reached, its total argument loss satisfies

$$\sum_i (a_i - a_i') \ge \frac{1}{128} a_j \ge \frac{1}{128} c_0$$

for some j, so this loss pays for all exceptional charges. All other charges telescope into that same argument loss. For a completion or an earlier local computation, the initial sum $\sum_i a_i$ pays the exceptional charges and the telescoping losses. This proves both bounds. Unbranched proper

subspace restrictions, field cases completed locally, and the single parameter-transporter coset add no persistent list. □

Every return from Step G7(b) to Step G1 strictly decreases the original subgroup H. Along a path with fixed reference dimensions there are therefore at most $\log_2 |\Gamma| = O(d^2 \log p)$ such returns. Exit (c) enters the tensor reductions directly, so it gives no same-group, same-dimension return to Step G1.

E.1.8 Proof of Proposition 8.4

We now combine the seven steps. For the local cost accounting, stop at a degree-2 completion, a child call made at Step G3, a proper-subgroup child produced by Step G7(b), or an ordinary input passed to the tensor reductions by Step G7. Include degree-2 completion costs, but exclude the internal work of the resulting tensor comparisons. Include construction of their inputs and recovery from their complete answers. In particular, a flag or summand comparison starting at Step G1 is still counted until it reaches one of these boundaries.

Lemma E.23 (Local work and input sizes for the group reductions)**.** *For inputs generated from* (8.1)*, the following hold, with an absolute constant C and a fixed polynomial P_0.*

(i) Fix one candidate coset whenever Step G5 or Step G7 branches. The part of the computation specified above has at most $P_0(d, \log p)$ applications of the structural theorems in Step G1 and recursive calls comparing successive flag quotients or pairs of summands. Each pair comparison in Step G4 is computed once. In a flag decomposition, at most one triple of argument quotients fails (8.3)*; every pair comparison in Step G4 satisfies that condition.*

(ii) For a multiplicity reduction $A_k = B \otimes_{\mathbb{K}} \mathbb{K}^m$, let Ξ index its candidate cosets $u_\xi K_\xi$. Then

$$H = \bigcup_{\xi \in \Xi} u_\xi K_\xi, \qquad \log_p |\Xi| \le C\big(\dim A_k - \dim B\big).$$

Among Steps G1–G6, only Step G5 requires solving an equivalent form of the whole current comparison recursively in every coset of an enumerated cover of H. The additional image lists in Step G7 are precisely the marked contraction lists in Lemma E.21; their cumulative cost, including later returns to Step G1, satisfies Lemma E.22.

(iii) If an ordinary input reaches Step G7 with unchanged group and argument dimensions, it is still ordinary: $Z = Z_{\mathrm{tri}}$, $P = 0$, and $c = 0$. It therefore enters the tensor reductions directly.

(iv) Every recursive input has bit length $\mathrm{poly}(\mathrm{InSize}, d, \log p)$*, with a fixed polynomial. Including all candidate cosets, the construction and recovery work specified above takes $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, d, \log p)$ time.*

Proof. For (i), consider a flag in A_i, whose successive quotient dimensions sum to $\dim A_i \le a_i$. At most one quotient has dimension greater than $127a_i/128$. Hence among the triples of successive argument quotients at most one fails (8.3). There are at most $(d+1)^3$ triples. If a triple consists of the current argument spaces, Step G1 passes its comparison to Step G2. Otherwise the argument dimension sum strictly decreases. Thus a sequence of successive recursive comparisons whose argument dimensions fail (8.3) has length at most d.

A child that already satisfies the dimension test still starts at Step G1, so we must count its possible further flag comparisons before it reaches Step G3. Give a trilinear comparison the integer

weight $\prod_i \dim A_i$. For one flag decomposition, the sum of these products over all triples of successive argument quotients equals the parent's weight. Omitting comparisons for which $Q_j/Q_{j-1} = 0$ only decreases this sum. Each recursive call on strictly smaller argument spaces decreases their dimension sum by at least one, so the depth is at most d. At each depth the total weight is at most d^3, and every comparison on a nonzero Q_j/Q_{j-1} has weight at least one. This gives $O(d^4)$ calls before a completion at Step G2 or a reset at Step G3. Since the dimension test continues to hold, none of these calls reaches Step G4 first.

By Lemma E.5, a summand decomposition introduces at most d^2 pair calls, each satisfying the dimension test, and reuses each answer. Combining this with the preceding bounds gives a fixed polynomial for (i). Each entry to Step G7 completes the comparison or produces output comparisons at the boundary just specified. It is therefore counted once, without expanding its proper-subgroup children. This adds at most a polynomial number of local constructions for fixed candidate choices. Lemmas E.6 and E.22 give (ii). Step G1 processes its quotient comparisons successively. Step G4 combines stored answers for pairs of summands. Neither requires solving the remaining tensor comparison separately for every permutation of the summands.

For (iii), recall that ordinary means $Z = Z_{\mathrm{tri}}$, $P = 0$, and $c = 0$. Any nontrivial flag, summand, or multiplicity reduction strictly decreases an argument dimension. If the group and dimensions are unchanged at entry to Step G7, none of these operations has changed the tensor comparison. Lemma E.7 allows $b = 0$ when $c = 0$, so Step G6 also leaves it ordinary. Step G7 passes it directly to the tensor reductions.

For (iv), first prove the uniform input-size bound directly from the data constructions, independently of the local running-time estimates. Lemma E.2 gives $P = Z_{\mathrm{quad}} = 0$ and a trivial action on M. Initially $\dim Z = \prod_i \dim_{\mathbb{F}_p} U_i^* \le d^3$. Passing to a quotient E_j in Step G1, or restricting to a summand in Step G4, does not increase $\dim Z$. The change of coordinates in Step G5 preserves it, since

$$(\dim M') \prod_i \dim A_i' = (\dim M) \prod_i \dim A_i.$$

Restricting H or changing the origin does not change Z. The restrictions in Step G7(b) keep Z and M. Its ordinary replacements use tensors in the same Z_{tri}, including the change of origin in Lemma E.8(ii). The ordinary replacement (8.11), and the tensor updates (8.13) and (8.14) keep M and replace the A_i by subquotients. Hence $\dim Z \le d^3$ in every continuing tensor comparison. Such a comparison has positive argument dimensions: otherwise $Z = 0$ and the algorithm returns immediately. Hence $\dim M \le \dim Z \le d^3$ as well.

Since H acts trivially on M, use $M \cong \mathbb{F}_p^{\oplus \dim M}$ as its controlled subquotient presentation. Store each flag subspace, inclusion, and quotient map in the original argument coordinates, replacing compositions of coordinate maps by their matrices. There are only $O(d)$ subspaces in the flags. A basis of N, the vector z, representatives of x, y, and the matrices ρ_h and vectors c_h for the generators h of H then require polynomially many field coordinates. A generating set in the original matrix group can be shortened to $O(d^2 \log p)$ elements. Each child input stores these generators as matrices in the original representation, together with their current action matrices and translation vectors. The straight-line programs recording their construction are shared in the computation history, rather than copied into each child input. A child's output programs refer to its input generators; linking these references recovers the programs in the original input generators required by Section 3.5. Thus the stored input does not include the child's entire computation history.

The auxiliary presentations in Steps G2 and G6, and in the direct G7 completions, have the same uniform size bound. The construction in Lemma E.3 uses at most three positions from spaces of total dimension $O(d)$, repeated $\dim M$ times. The parameter space $\mathcal{S}$ in Lemma E.7 is a subspace of a direct sum of three copies of Z_{tri}, so $\dim \mathcal{S} \le 3 \dim Z \le 3d^3$. Their bases, subspace maps,

and generator actions therefore require polynomially many field coordinates as well. The spaces $\mathcal{S}_N/\ker \mathcal{J}$ and $\overline{\mathcal{S}}$ in Lemmas E.8 and E.9 are subquotients of $\mathcal{S}$ and satisfy the same bound. This uses their explicit constructions, not their running-time estimates. Together these bounds give

$$\mathrm{InSize}_{\mathrm{cur}} \le \mathrm{poly}(\mathrm{InSize}, d, \log p),$$

with a fixed polynomial independent of the number of preceding calls. This proves the input-size assertion in (iv). The field arrays in Lemma E.14 may be larger, but they are temporary data in completed local comparisons. Their $p^{O(d)}$ cost is already included in Lemma E.15. They do not become part of any recursive input.

We can now express each local cost in terms of the initial input:

$$p^{O(d)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p) \le p^{O(d)}\mathrm{poly}(\mathrm{InSize}, d, \log p).$$

This applies to the structural computations, the subset calculation of Lemma E.5, and the constructions of Lemmas E.7 and E.21, including their completed degree-2 calls. It also bounds recovery from a child's answer: on receiving its matrices, constructive membership in the parent group H computes new programs in the parent's generators. Evaluating the parent's current affine action on these programs costs $p^{O(d)}\mathrm{poly}(\mathrm{InSize}_{\mathrm{cur}}, d, \log p)$, where $\mathrm{InSize}_{\mathrm{cur}}$ is the parent's input length, and hence $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, d, \log p)$ by the size bound just proved. This avoids expanding the child's computation history. Constructing the shared programs is charged to the group operations that produce them.

Part (i) bounds how many such local constructions are needed. A multiplicity choice is passed directly to Step G3. A marked choice in Step G7 produces a proper-subgroup output at which this local count stops. Thus the local construction does not expand a sequence of such choices. Each list has size $p^{O(d)}$, and all other G7 cases are either completed locally or produce one coset. Summing the costs over the polynomially many entries and these lists gives $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, d, \log p)$, including recovery from the output answers. When these outputs are expanded in the full algorithm, Lemma E.22 supplies the cumulative bound instead of charging $p^{O(d)}$ anew at each return. □

Proof of Proposition 8.4. We follow the steps until a completed comparison, a recursive output, or an ordinary input to the tensor reductions is obtained. Lemma E.1 constructs the inputs on Q_j/Q_{j-1} in Step G1 and recovers the original answer from their complete answers. A comparison with a valid degree-2 presentation is completed by Lemma E.3, giving outcome (i). Whenever (8.3) holds, Lemma E.4 justifies the unchanged comparison passed to a child with smaller reference dimensions, giving outcome (ii).

Lemma E.5 reduces Step G4 to pair comparisons satisfying the dimension test. Their complete answers recover the full transporter and its source stabilizer. If Step G5 applies instead, Lemma E.6 gives an exhaustive family of equivalent inputs, each with one argument at most half as large. These inputs also give outcome (ii).

Now suppose a comparison reaches Step G6. Its arguments are irreducible after Step G1, and Step G4 has found no nontrivial permuted decomposition. By Lemma E.3, failure of the degree-2 completion implies that all three arguments carry large classical components. Theorem 5.2 presents each as copies of a natural module, with the specified duals and twists. Since Step G5 does not apply, each multiplicity is one. Lemma E.7 therefore supplies the H-equivariant pairing map and a change of origin with

$$P \le J(Z), \qquad c_h^b \in J(Z)/P \quad (h \in H).$$

These are the entry conditions for Step G7, not yet the ordinary input required in outcome (iii).

Lemma E.21 now covers every input reaching Step G7. Its degree-2 completions give outcome (i). Its proper-subgroup comparisons give outcome (ii): their argument spaces are unchanged and their recursive calls start at Step G1 with the same reference dimensions. If subsequent group reductions satisfy (8.3), Step G3 resets those dimensions for the corresponding child call. Finally, a same-group ordinary replacement, or an input already ordinary, gives outcome (iii). The argument modules are the same single large natural, dual, or twisted-natural modules identified before Step G6.

In particular, separation of the tensor and its parameters first computes the complete degree-2 parameter transporter uK. If it is empty, the original comparison is empty. If $K < H$, (E.5) is a recursive ordinary comparison and belongs to outcome (ii). If $K = H$, it goes directly to the tensor reductions and belongs to outcome (iii). This distinguishes the two exits without invoking either reduction proposition in the construction of Step G7.

In Steps G1–G6, every new group is a subgroup of the preceding group and every new argument is a subquotient of its preceding argument; in Step G5 it is represented by the preimage of $B \otimes e_1$. The ordinary replacements in Step G7 keep the argument spaces. All other G7 outputs restrict the original comparison to a subgroup coset by (8.5), so they also keep the argument spaces. Thus $H' \leq H$ and $\dim A'_i \leq \dim A_i$. The preceding lemmas preserve the full solution sets. Lemma 8.2 recovers answers to successive comparisons, and (8.6) combines exhaustive candidate families. They return emptiness or guK with $K = \mathrm{Stab}_{H,\alpha}(x)$, as required. Finally, Lemma E.23(iv) gives the uniform input-size bound and, after substituting that bound into the local estimates, the total reduction-and-recovery cost $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, d, \log p)$. The internal work of the resulting recursive and ordinary tensor comparisons is excluded, as in the proposition. Lemma E.22 also accounts for the marked choices when these recursive outputs are later expanded. This proves all the assertions. □

E.2 Tensor reductions

We prove the ordinary tensor reductions of Algorithm 8.6. The input is a comparison $g\,\mathrm{Trans}_{H,\rho}(\mathtt{T}, \mathtt{T}')$ arising in case (iii) of Proposition 8.4 during the recursion from (8.1). The argument spaces are the three single large natural modules specified there, including their recorded duals and twists. Lemma E.2 gives the trivial action on the value space M for every input under consideration. We use this invariant throughout: a chosen basis of M gives fixed value coordinates. All tensor products, ranks, and dimensions are over $\mathbb{F}_p$ unless a different field is specified. The original parameter d and the threshold T_{cl} remain fixed.

For $v \in A_1$ and $\mathtt{F} \in Z_{\mathrm{tri}}$, contraction is

$$\mathcal{C}_v(\mathtt{F}) : A_2 \longrightarrow A_3^* \otimes M, \qquad \mathcal{C}_v(\mathtt{F})(b)(c) = \mathtt{F}(v, b, c).$$

For the progress estimate, let E be the represented semisimple algebra of the distinct selected natural types on the current arguments and their required duals, as in Subsection 5.4. Write

$$E \cong \prod_{\tau \in I} \mathrm{Mat}_{n_\tau}(\mathbb{K}_\tau), \qquad \nu(E) := \sum_{\tau \in I} [\mathbb{K}_\tau : \mathbb{F}_p] n_\tau.$$

Thus $\nu(E)$ is the dimension of one copy of each distinct selected natural module; repeated occurrences are counted once. For a subgroup $K \leq H$, write E_K for the corresponding algebra after applying the structural computation to the restricted action, with the same threshold and the same required duals.

E.2.1 Step T1: a contraction between different types

We use the group D and the action types from Step T0. By Lemma E.2, the current value module M is trivial. Thus, after choosing a basis of M, a contraction is a simultaneous tuple of bilinear

forms. This observation is used only to construct the degree-2 presentation; the rank is still that of the full map $A_2 \to A_3^* \otimes M$.

Lemma E.24 (Matching a contraction)**.** *Suppose A_2 and A_3^* have different types, with the argument positions ordered as in Step T0. If exactly one of $\mathtt{T}, \mathtt{T}'$ has a contraction of rank at least $\dim A_1/64$, their transporter is empty. If neither has such a contraction, both satisfy the rank bound at the entrance to Step T4.*

Otherwise choose $v \in A_1$ with $r = \operatorname{rank} \mathcal{C}_v(\mathtt{T}) \geq \dim A_1/64$ and put

$$H_v = \operatorname{Stab}_H(v), \qquad K = \operatorname{Stab}_{H_v}\big(\mathcal{C}_v(\mathtt{T})\big).$$

The degree-2 comparisons in Step T1 compute at most $p^{\dim A_1}$ candidate cosets, each with right subgroup $K < H$. The complete answers to the remaining tensor comparisons in these cosets recover the complete original transporter. The construction, excluding those remaining tensor comparisons, takes $p^{O(d)}\operatorname{poly}(\text{InSize}, d, \log p)$ time and produces inputs of polynomial length.

Let E and E_K be the algebras of the distinct selected natural types before and after restriction to K, including the required duals as above. Then

$$\nu(E_K) \leq \nu(E) - r.$$

Proof. Write h_i and h_M for the actions of h on A_i and M. If $\rho_h(\mathtt{T}) = \mathtt{T}'$, then for every $v \in A_1$,

$$\mathcal{C}_{h_1 v}(\mathtt{T}') = (h_3^{-T} \otimes h_M)\mathcal{C}_v(\mathtt{T})h_2^{-1}.$$

The maps on either side of the contraction are invertible. Hence the existence of a contraction of rank at least $\dim A_1/64$ is invariant under tensor isomorphism. This proves the first rejection. If neither tensor has one, every contraction has rank less than $\dim A_1/64$, which is at most $\min(\dim A_2, \dim A_3)/32$.

For a chosen v, compute its orbit and its stabilizer H_v by the matrix action on A_1, retaining a representative a with $av = v'$ for each $v' \in H \cdot v$. Every possible solution taking v to v' lies in aH_v. Contraction at v is H_v-equivariant, so Theorem 7.9 computes

$$\{k \in H_v : \mathcal{C}_v(\rho_k(\mathtt{T})) = \mathcal{C}_v(\rho_{a^{-1}}(\mathtt{T}'))\}.$$

A nonempty answer is a coset uK. Its complete tensor solutions are exactly the solutions retained by (8.5) in the candidate coset auK. The orbit covers every possible image of v, so (8.6) recovers the complete original answer. Since $r > 0$, we have $v \neq 0$. The natural action of the large component on A_1 has no nonzero fixed vector, giving $K \leq H_v < H$.

There are at most $p^{\dim A_1}$ vectors to inspect for the rank test, and at most $p^{\dim A_1}$ candidate images. A basis of the trivial space M expresses the bilinear comparison as fixed copies of $A_2^* \otimes A_3^*$. It therefore has a degree-2 presentation with the same original group and dimension budget. Fixed copies contribute only to input length in Definition 23. Orbit computation, the degree-2 calls, and coset recovery have the claimed local bound. Their matrices and original-group generators have polynomial length by the input conventions and the complete output guarantees of the cited algorithms.

It remains to prove the decrease in ν. We first retain the entire new selected product while restricting to the old product D. Let Γ be the joint image of H on the current arguments and their required duals, and identify D with its faithful image there. Then $D \trianglelefteq \Gamma$. Each of these spaces is a nontrivial simple D-module, so the joint space has no D-fixed summand and has dimension at most d. Thus Γ is a subgroup of the normalizer of D on that space. By Lemma 5.5, $[\Gamma : D] \leq p^{O(d)}$. The

fixed choice of C_* ensures that Γ/D has no simple section isomorphic to a simple central quotient selected at threshold T_{cl}. Apply the structural decomposition to K, let $\alpha : K \twoheadrightarrow G_K$ be its complete graded argument action, and let $D_K \trianglelefteq G_K$ be its selected product. Let $\beta : K \twoheadrightarrow \beta(K) \leq \Gamma/D$ be the induced quotient map, with its actual image as codomain. Apply Lemma 5.6 to α, β, and $D_K \leq G_K$. It gives a subgroup $L \leq K$ such that

$$\alpha(L) = D_K, \qquad \text{the old argument action of } L \text{ lies in } D.$$

Using the full product D_K in this application retains all new selected types simultaneously.

Let W be the direct sum of one representative of each distinct old natural type, including the required duals. Thus $\dim W = \nu(E)$. Every argument and required dual is D-isomorphic to a summand of W, and this identification remains valid on L. Use the composition flags for the K-action and restrict them to L. On their successive quotients, L acts through the full product D_K. Decomposing those quotients as D_K-modules shows that every new selected natural type is a composition factor of one of the argument spaces or duals restricted to L, and hence of $W|_L$. Distinct D_K-types remain distinct after pullback along the surjection $L \twoheadrightarrow D_K$. Consequently,

$$\nu(E_K) \leq \sum_{S \in \mathrm{cf}(W|_L)} \dim_{\mathbb{F}_p} S,$$

where $\mathrm{cf}(W|_L)$ denotes the set of isomorphism classes of simple composition factors, without multiplicities.

Put $A = A_2$, $B = A_3^*$, and $I = \operatorname{im} \mathcal{C}_v(\mathtt{T})$. Since $L \leq K$ and M is trivial, I is both a quotient of $A|_L$ and a submodule of a direct sum of copies of $B|_L$. For a simple L-module S, let $m_S(V)$ denote its composition multiplicity in V. Then

$$m_S(I) \leq m_S(A), \qquad m_S(I) > 0 \implies m_S(B) \geq 1.$$

The old D-types A and B are different, so W contains both as distinct summands. For every S occurring in I this gives

$$m_S(W) - 1 \geq m_S(A) + m_S(B) - 1 \geq m_S(I).$$

For all other composition factors of W, the left-hand side is nonnegative. Multiplying by $\dim S$ and summing yields

$$\begin{aligned} \nu(E) - \sum_{S \in \mathrm{cf}(W|_L)} \dim S &= \sum_{S \in \mathrm{cf}(W|_L)} (m_S(W) - 1) \dim S \\ &\geq \sum_S m_S(I) \dim S = \dim I = r. \end{aligned}$$

Together with the preceding bound on $\nu(E_K)$, this proves the claimed decrease. □

E.2.2 Step T2: short contraction tuples

Throughout this and the next subsubsection, put

$$A = A_2, \qquad \mathbb{F} = \mathrm{End}_{\mathbb{F}_p[D]}(A), \qquad f = [\mathbb{F} : \mathbb{F}_p], \qquad a = \dim_{\mathbb{F}} A, \qquad n = fa.$$

The current argument is one natural module for the selected classical product D. Thus $\mathbb{F}$ is a field and $a \geq 64$, by the stable-range convention in Subsection 6.1. The identification $\iota : A_2 \to A_3^*$ from Step T0 gives

$$L := \mathrm{Hom}_{\mathbb{F}_p[D]}(A_2, A_3^*) = \iota\mathbb{F}.$$

Normality of D makes L invariant under H, with action $\ell \mapsto h_{A_3^*}\ell h_{A_2}^{-1}$. Since h_{A_2} normalizes $\mathbb{F}$, this action lies in $\Gamma L_1(\mathbb{F})$ and has solvable image. Lemma E.2 also gives trivial H-action on M. Fix an $\mathbb{F}_p$-basis $m_1, \dots, m_b$ of M and write

$$\mathcal{C}_v(\mathtt{T}) = \sum_{j=1}^{b} C_{v,j} \otimes m_j, \qquad O_j(v) = \iota^{-1} C_{v,j} \in \mathrm{End}_{\mathbb{F}_p}(A).$$

The maps $O_j : A_1 \to \mathrm{End}_{\mathbb{F}_p}(A)$ are $\mathbb{F}_p$-linear; the operators $O_j(v)$ need not commute with $\mathbb{F}$.

The degree-2 comparison in Step T2 computes the complete joint stabilizer $K_{\mathbf{v}}(\mathtt{T})$, including the original group kernels. For any h with $\rho_h \mathtt{T} = \mathtt{T}'$, contraction equivariance gives

$$K_{h\mathbf{v}}(\mathtt{T}') = h K_{\mathbf{v}}(\mathtt{T}) h^{-1}.$$

The dimension test is independent of the structural choices. Put $n_k = \dim_{\mathbb{F}_p} A_k$. If a selected natural factor B has dimension greater than $63n_k/64$, the composition factor Q from which it is obtained is the unique composition factor exceeding that threshold, up to isomorphism, and occurs with multiplicity one. A nontrivial imprimitivity system or natural multiplicity at least two would give $\dim B \le n_k/2$. Hence Q is linearly primitive and its natural multiplicity is one. The corresponding normal quasisimple component acts irreducibly on Q and is unique: a distinct component would commute with it and lie in the multiplicative group of its endomorphism field, contradicting perfectness. This also follows from the conjugation-invariant local component selection in Appendix B.3. The constants c_0, C_* and the dimension budget d are fixed, so whether this component is selected by Theorem 5.2 is intrinsic as well. Thus the existence of a selected natural factor exceeding $63n_k/64$ depends only on the isomorphism type of the restricted argument representation. The displayed conjugacy therefore implies that the natural-module dimension test has the same answer on equivalent tensors. If a source tuple passes, its simultaneous contraction transporter in each image coset has right stabilizer $K_{\mathbf{v}}(\mathtt{T})$. The enumeration therefore covers every tensor solution. In particular this stabilizer is proper: equality with H would retain the current natural module A_k itself. There are at most p^{3a_1} tuple images and at most $p^{3\dim A_1}$ source tuples to test.

We verify the dimension progress using the local group reductions. Step G1 splits along argument flags by Lemma E.1. A nontrivial permuted-summand decomposition in any argument gives children with that argument at most half as large, by Lemma E.5. For a remaining layer with primitive argument actions, the absence of a large natural factor in its kth argument gives the degree-2 exit of Lemma E.3. Natural multiplicity at least two gives smaller children by Lemma E.6. If neither reduction applies to an unsolved layer, its kth argument is a single natural factor B from the tested decomposition. Its dimension satisfies

$$\dim B \le \frac{63}{64}\dim A_k \le \frac{63}{64}a_k < \frac{127}{128}a_k,$$

so Step G3 already applies. The inherited flags and decompositions remain available under the subgroups returned by successive layer comparisons. Applying these same alternatives to each resulting layer proves that the group reductions complete the comparison or reach a reference-dimension reset before Steps G6–G7.

It remains to prove what failure of all these tests supplies. For a subset $S \subseteq A_1$ of size at most three put

$$\mathscr{B}_S = \mathrm{alg}_{\mathbb{F}_p}(\mathbb{F}, I, \{O_j(v) : v \in S,\ 1 \le j \le b\}), \qquad \mathscr{E}_S = \mathrm{End}_{\mathscr{B}_S}(A).$$

All these represented algebras are computed by linear algebra and Lemma 3.2. Set $r = \lfloor a/32 \rfloor$.

Lemma E.25. *If every simple module of $\mathscr{E}_S/\operatorname{rad}\mathscr{E}_S$ has $\mathbb{F}_p$-dimension at most $n - fr$, then the tuple consisting of the elements of S passes the test in Step T2 with $k = 2$.*

Proof. Let $K = K_{\mathbf{v}}(\mathtt{T})$ and apply the structural computation to its joint action on the three current arguments. Let D_K be its full selected classical product in the resulting associated graded image, and let α be the homomorphism from K to that image. Let β be the action of K on L. The image of β is solvable, whereas the simple central quotients of the factors of D_K are nonabelian. Lemma 5.6 gives

$$K_0 = \alpha^{-1}(D_K) \cap \ker\beta, \qquad \alpha(K_0) = D_K.$$

Fixing every element of $L = \iota\mathbb{F}$ forces an element of K_0 to fix ι and centralize $\mathbb{F}$ on A. Since it fixes the marked contractions and M is trivial, it also centralizes every $O_j(v)$ for $v \in S$. Its image on A therefore lies in $\mathscr{E}_S^{\times}$.

A composition series for A as an $\mathscr{E}_S$-module has factors that are simple modules for $\mathscr{E}_S/\operatorname{rad}\mathscr{E}_S$. Restricting this series to K_0 shows that every K_0-composition factor of A has dimension at most $n - fr$. The graded pieces defining α on A_2 are K_0-subquotients of A. Since $\alpha(K_0) = D_K$, every selected natural factor arising from A_2 is a composition factor of one of these pieces, and hence of $A|_{K_0}$. Thus its dimension is at most $n - fr$. Finally, $a \geq 64$ gives $r \geq a/64$, so $n - fr \leq 63n/64$. □

Lemma E.26. *Suppose that $\mathscr{B} \leq \operatorname{End}_{\mathbb{F}_p}(A)$ is a unital algebra containing the specified copy of $\mathbb{F}$. If $\operatorname{End}_{\mathscr{B}}(A)/\operatorname{rad}\operatorname{End}_{\mathscr{B}}(A)$ has a simple module of dimension greater than $n - fr > n/2$, then $\mathscr{B}$ has a unique field quotient $\mathbb{K} = \mathscr{B}/P$ with*

$$U = \operatorname{Ann}_A(P), \qquad \dim_{\mathbb{F}} U > a - r.$$

The quotient embeds $\mathbb{F}$, and it and U are constructible by represented-algebra operations.

Proof. Write $\mathscr{E} = \operatorname{End}_{\mathscr{B}}(A)$ and choose a Wedderburn–Malcev complement to its radical. Suppose its factor $\operatorname{Mat}(m, \mathbb{K})$ has natural dimension $\delta = m[\mathbb{K} : \mathbb{F}_p] > n - fr > n/2$, and let e be the identity idempotent of this factor. Every factor of the complement occurs on the faithful module A. Hence the natural module of this factor occurs exactly once, and $U_0 = eA$ has dimension δ and is irreducible under $\operatorname{Mat}(m, \mathbb{K})$. It is also $\mathscr{B}$-invariant.

The radical of $e\mathscr{E}e = \operatorname{End}_{\mathscr{B}}(U_0)$ is nilpotent, so it has a nonzero annihilator on U_0. This annihilator is invariant under $\operatorname{Mat}(m, \mathbb{K})$ and hence equals U_0. The corner algebra acts faithfully on U_0, so its radical is zero and $\operatorname{End}_{\mathscr{B}}(U_0) = \operatorname{Mat}(m, \mathbb{K})$. The image of $\mathscr{B}$ on U_0 commutes with this matrix algebra and is therefore a subfield $\mathbb{L} \leq \mathbb{K}$. It must equal $\mathbb{K}$: otherwise $\operatorname{End}_{\mathbb{L}}(U_0)$ would strictly contain $\operatorname{Mat}(m, \mathbb{K})$. The kernel P of this field action has $U_0 \subseteq \operatorname{Ann}_A(P)$.

Nonzero elements of $\mathbb{F}$ are invertible on A, so the quotient map embeds $\mathbb{F}$. Since P is a two-sided ideal, its annihilator is $\mathbb{F}$-stable and has the required dimension. If two distinct field quotient kernels both had annihilators of dimension greater than $n/2$, their annihilators would meet nontrivially. A nonzero vector in the intersection would give an isomorphism between the two simple $\mathscr{B}$-modules, forcing their kernels to agree. This proves uniqueness. The complement, its matrix factors, the restricted action, and the annihilator are computed using Lemma 3.2 and linear equations. □

If no tuple passes Step T2, the contrapositive of Lemma E.25 and Lemma E.26 give, for every $|S| \leq 3$, a unique quotient

$$\mathbb{K}_S = \mathscr{B}_S/P_S, \qquad U_S = \operatorname{Ann}_A(P_S), \qquad \dim_{\mathbb{F}} U_S \geq a - r > a/2.$$

For $S = \varnothing$ these are simply $\mathbb{F}$ and A.

Lemma E.27. *For a reachable ordinary input of the same type, Step T2 either detects inequivalence, covers the complete answer by at most $p^{3\dim A_1}$ candidate cosets whose resulting comparisons complete or reset a reference dimension, or constructs the unique field quotients and annihilators above for every tuple of length at most three. Its local work costs $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$.*

Proof. The complete-stabilizer comparison, conjugacy identity, and tuple-image enumeration prove the first two alternatives and their asserted progress. In the remaining case, Lemmas E.25 and E.26 give the field quotients. There are at most $p^{3\dim A_1}$ tests; each uses the degree-2 and structural algorithms and represented-algebra computations on the current arguments, so has the asserted total local bound. □

E.2.3 Step T3: a common field-value map

Lemma E.28. *If the tuple test fails on a reachable ordinary input of the same type, one can construct a finite extension $\mathbb{K}/\mathbb{F}$ and an $\mathbb{F}_p$-linear map $\lambda_{\mathtt{T}} : A_1 \to \mathbb{K} \otimes M$ with $[\mathbb{K} : \mathbb{F}_p] \le n^2$. The construction has local cost $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$, is covariant under tensor transporters up to $z \mapsto \kappa z^{p^j}$ with $\kappa \in \mathbb{F}^\times$, and satisfies $\lambda_{\mathtt{T}}(v) = 0$ if and only if $\operatorname{rank} \mathcal{C}_v(\mathtt{T}) \le f\lfloor a/32 \rfloor$.*

Proof. We first construct the map used in the algorithm. The argument below uses only the field quotients just obtained, including those for pairs and triples; it does not assume that a short tuple generates the common field.

If $S \subseteq T$ and $|T| \le 3$, the image of $\mathscr{B}_S$ on U_T is a subfield of $\mathbb{K}_T$. Thus U_T is semisimple of one $\mathscr{B}_S$-type. Its intersection with U_S is nonzero by the dimension bound, so this type is the one defined by P_S. Therefore

$$P_T \cap \mathscr{B}_S = P_S, \qquad i_{ST} : \mathbb{K}_S \hookrightarrow \mathbb{K}_T, \quad u + P_S \longmapsto u + P_T, \qquad U_T \subseteq U_S.$$

These are $\mathbb{F}$-embeddings, and their formulas make successive inclusions commute. Write $\mathbb{K}_v$ for a singleton quotient and $e_v = [\mathbb{K}_v : \mathbb{F}] \le a$. A pair quotient is generated by its two singleton images, and a triple quotient by its three images. Their degrees are consequently the corresponding least common multiples.

Put $N = \operatorname{lcm}_{v \in A_1} e_v$ and take the abstract field $\mathbb{K}/\mathbb{F}$ of degree N. Initially choose arbitrary $\mathbb{F}$-embeddings of the singleton fields into $\mathbb{K}$. For each pair v, w, the prescribed embeddings into $\mathbb{K}_{vw}$ specify a relative Frobenius discrepancy

$$d_{vw} \in \mathbb{Z}/\gcd(e_v, e_w)\mathbb{Z}.$$

The embedding of the triple quotient gives

$$d_{uv} + d_{vw} = d_{uw} \pmod{\gcd(e_u, e_v, e_w)}.$$

For each prime $\ell \mid N$, choose w_ℓ for which $v_\ell(e_{w_\ell}) = v_\ell(N)$. Prescribe a Frobenius adjustment to the embedding of $\mathbb{K}_v$ by

$$s_v = d_{v,w_\ell} \pmod{\ell^{v_\ell(e_v)}}.$$

The triple relation verifies all pair constraints at this prime, and the Chinese remainder theorem combines these prescriptions modulo e_v. The adjusted embeddings thus extend compatibly to all pairs and triples. Any two such systems differ by one common $\mathbb{F}$-Frobenius: at each prime the differences are restrictions of the difference at w_ℓ, and another application of the Chinese remainder theorem combines them modulo N.

Define $\lambda_j(v)$ as the image in $\mathbb{K}$ of the class of $O_j(v)$ in $\mathbb{K}_v$, and set

$$\lambda_{\mathtt{T}}(v) = \sum_{j=1}^{b} \lambda_j(v) \otimes m_j \in \mathbb{K} \otimes M.$$

For $w = u + v$, the identity $O_j(w) = O_j(u) + O_j(v)$ holds in $\mathscr{B}_{uvw} = \mathscr{B}_{uv}$. Applying its quotient proves additivity of λ_j. The same argument for scalar multiples proves $\mathbb{F}_p$-homogeneity. The values $\lambda_j(v)$ generate $\mathbb{K}$ over $\mathbb{F}$, since they generate every singleton quotient.

Bounding and constructing the common field. The linearity just proved gives a stronger degree bound than the initial least-common-multiple bound. For each prime $\ell \mid N$, let $\mathbb{K}^{(\ell)}$ be the unique subfield of index ℓ over $\mathbb{F}$, and put

$$V_\ell = \{v \in A_1 : \lambda_j(v) \in \mathbb{K}^{(\ell)} \text{ for every } j\}.$$

This is a proper $\mathbb{F}_p$-subspace: otherwise all the values would lie in a proper subfield of the field they generate. For uniformly random $v \in A_1$, therefore, $\Pr(v \in V_\ell) \le 1/p$. If $v \notin V_\ell$, the field generated by its coordinates has degree e_v whose ℓ-part is the full ℓ-part of N. It follows that

$$\log a \;\ge\; \mathbb{E}_v \log e_v \;\ge\; (1 - 1/p) \sum_{\ell \mid N} v_\ell(N) \log \ell \;=\; (1 - 1/p) \log N.$$

Thus $N \le a^{p/(p-1)} \le a^2$, and

$$[\mathbb{K} : \mathbb{F}_p] = fN \le fa^2 \le n^2. \tag{E.14}$$

This also gives a direct construction with the required cost. For each prime $\ell \mid N$, its maximal prime-power divisor $d_\ell = \ell^{v_\ell(N)}$ occurs in some e_v and hence is at most a. Enumerate monic polynomials of degree d_ℓ over $\mathbb{F}$ until an irreducible one is found. The resulting fields have pairwise coprime degrees, so their tensor product over $\mathbb{F}$ is the field $\mathbb{K}$, with an explicit basis. Each search uses at most $|\mathbb{F}|^{d_\ell} \le p^n$ candidates. To embed a required quotient of degree $e \le a$, compute the subfield $\ker(\mathrm{Frob}_{\mathbb{F}}^e - I)$ of $\mathbb{K}$ by linear algebra, and enumerate its at most $p^{fe} \le p^n$ elements to find a root of a defining polynomial. Frobenius comparisons and the above congruences then align the embeddings. There are at most $p^{3 \dim A_1}$ tuple algebras; all algebra operations are on A, and (E.14) bounds the common-field arithmetic dimension polynomially. The total cost is $p^{O(\dim A_1 + n)} \mathrm{poly}(\mathrm{InSize}, \log p)$. Only the values on a basis of A_1 need be retained to specify $\lambda_{\mathtt{T}}$.

Ranks and covariance. If $\lambda_{\mathtt{T}}(v) = 0$, every $O_j(v)$ kills the same U_v, and consequently

$$\operatorname{rank} \mathcal{C}_v(\mathtt{T}) \le fr \le n/32. \tag{E.15}$$

If a coordinate value is nonzero, the corresponding operator is invertible on U_v and has rank at least $f(a - r) > fr$. In particular zero values are characterized by the low-rank alternative, even when the original operators do not commute with $\mathbb{F}$.

Now let h satisfy $\rho_h \mathtt{T} = \mathtt{T}'$. Put $q = h_{A_2}$ and $q' = h_{A_3^*}$. Since H normalizes D, there is a unique $\beta_h \in \mathbb{F}^\times$ such that

$$q' \iota q^{-1} = \iota \beta_h.$$

The operator q normalizes $\mathbb{F}$ and induces an $\mathbb{F}_p$-automorphism of it. Since M is trivial, contraction equivariance gives

$$O'_j(h_{A_1} v) = \beta_h q O_j(v) q^{-1}.$$

The scalar factor lies in $\mathbb{F}^\times$, so conjugation by q identifies the source tuple algebra with the corresponding target tuple algebra. It carries their unique large-annihilator quotient kernels to each other. Compatibility and uniqueness of the glued embeddings therefore give a single $\mathbb{F}_p$-field isomorphism $\theta_h : \mathbb{K}_{\mathrm{T}} \to \mathbb{K}_{\mathrm{T}'}$ and

$$\lambda_{\mathrm{T}'} h_{A_1} = (m_{\beta_h} \theta_h \otimes h_M) \lambda_{\mathrm{T}}, \qquad h_M = I_M, \tag{E.16}$$

where m_{β_h} denotes multiplication by β_h in the target field. Thus different field degrees exclude a tensor solution. After identifying both fields with a common extension of $\mathbb{F}$, every possible θ_h is a prime-field Frobenius. It suffices to enumerate

$$\psi_{\kappa,j} : z \longmapsto \kappa z^{p^j}, \quad \kappa \in \mathbb{F}^\times, \quad 0 \le j < [\mathbb{K} : \mathbb{F}_p].$$

The number of these maps is at most $p^f n^2 = p^{O(n)}$. Some maps need not be realized by H; the final tensor comparison removes their extraneous candidates. The coefficients $\mathbb{K}$ here are fixed $\mathbb{F}_p$-multiplicity coordinates, so their polynomially bounded dimension does not introduce another active argument. □

Lemma E.29. *With the field maps of Lemma E.28, Step T3 either detects inequivalence, gives a complete degree-2 solution, restricts to a proper subgroup with unchanged reference dimensions, or supplies the rank hypothesis of Step T4. All local work costs* $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$.

Proof. Equation (E.16) transports $W = \ker \lambda_{\mathrm{T}}$ to $W' = \ker \lambda_{\mathrm{T}'}$, which proves the unequal-dimension rejection. If $0 < W < A_1$, Corollary 7.10, applied to basis maps of W and W', computes their complete transporter. The current H-action on A_1 is irreducible, so $H_W < H$. Restricting to this coset therefore preserves all tensor solutions and gives the proper-subgroup exit asserted in Step T3(a).

If $W = 0$, both maps are injective. For a fixed $\psi = \psi_{\kappa,j}$, their comparison in Step T3(b) is a degree-2 comparison in $\mathrm{Hom}(A_1, \mathbb{K} \otimes M)$, with fixed $\mathbb{K}$-coordinates. Let its nonempty answer be uK. For $k \in K$, the source stabilizer identity, after cancelling the invertible fixed map ψ, is

$$\lambda_{\mathrm{T}} k_{A_1} = (I_{\mathbb{K}} \otimes k_M) \lambda_{\mathrm{T}}.$$

Since $k_M = I_M$, injectivity gives $k_{A_1} = I_{A_1}$. The original tensor comparison under K is therefore a simultaneous comparison of bilinear maps on $A_2 \times A_3$, indexed by bases of the now fixed spaces A_1 and M. The degree-2 algorithm computes its complete answer. Every ψ branch is completed before the answers are united; there is no further tensor recursion in this case.

Finally, if $W = A_1$, both maps vanish identically. Equation (E.15) holds for every contraction of both tensors. Since $A_2 \cong A_3^*$, their dimensions both equal n. This is precisely the rank hypothesis of Step T4. The subspace transport and each fixed-field-map comparison have the stated degree-2 cost, and there are at most $p^f n^2$ field-map cases. □

E.2.4 Step T4: a common zero block

We first construct the subspaces used in Step T4. For a matrix space $\mathcal{L} \le \mathrm{Hom}_{\mathbb{F}_p}(V, W)$ and $U \le V$, write

$$\mathcal{L}(U) := \sum_{A \in \mathcal{L}} A(U), \qquad b_{\mathcal{L}}(U) := \dim V - \dim U + \dim \mathcal{L}(U).$$

All dimensions in this subsubsection are over $\mathbb{F}_p$.

Lemma E.30 (Choosing a common zero block)**.** *Let $r = \max_{A\in\mathcal{L}} \operatorname{rank} A$. There is a unique smallest subspace $U \le V$ minimizing $b_{\mathcal{L}}(U)$, and it satisfies*

$$\mathcal{L}(U) \le W, \qquad b_{\mathcal{L}}(U) \le 4r.$$

The pair $(U, \mathcal{L}(U))$ can be computed deterministically in time $p^{O(1)}\mathrm{poly}(\mathrm{InSize}_{\mathcal{L}}, \log p)$, where $\mathrm{InSize}_{\mathcal{L}}$ is the explicit input length. If $\mathcal{L}' = t\mathcal{L}s^{-1}$ for linear bijections s, t, the corresponding pair is $(sU, t\mathcal{L}(U))$.

Proof. Apply Lemma E.19 with $q = p$, $\mathcal{B} = \mathcal{L}$, and $R = r$. Its canonical choice is precisely the unique smallest minimizer of $b_{\mathcal{L}}$, and it has the stated computation time and covariance. The inequality $\min_U b_{\mathcal{L}}(U) \le 4r$ is the Haramaty–Shpilka bound [HS10], in the matrix-space formulation of [BCL$^+$25, Lemma 6.6]; it holds over the original field, including $p = 2$. □

Lemma E.31 (The construction and progress in Step T4)**.** *Suppose an ordinary comparison reached in the recursion satisfies the rank bound at entry to Step T4. That step constructs its two pairs and their complete transporter in time $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, d, \log p)$. An empty pair transporter implies an empty tensor transporter. Otherwise its three successive comparisons have exactly the solution-recovery formulas* (8.13) *and* (8.14)*.*

For each of those three comparisons, Steps G1–G3 produce only degree-2 completions or calls satisfying (8.3)*. No argument dimension increases. The local construction and recovery have the same time bound, and all constructed data have bit length* $\mathrm{poly}(\mathrm{InSize}, d, \log p)$*.*

Proof. By Lemma E.2, H acts trivially on M. Form the matrix space

$$\mathcal{L}_{\mathtt{T}} := \{\mathcal{C}_v(\mathtt{T}) : v \in A_1\} \le \mathrm{Hom}(A_2, A_3^* \otimes M).$$

It is obtained by contracting on a basis of A_1 and taking the linear span. Apply Lemma E.30 to obtain X and $E = \mathcal{L}_{\mathtt{T}}(X)$, and do the same for $\mathtt{T}'$ to obtain X', E'. The assumed rank bound gives

$$\operatorname{codim}_{A_2} X + \dim E \le \tfrac{1}{8}\min(\dim A_2, \dim A_3),$$

and the same inequality for X', E'. These computations use the explicit tensor coordinates, with polynomial dependence on $\dim M$.

For a tensor solution h, equivariance of contraction gives

$$\mathcal{C}_{h_{A_1}v}(\mathtt{T}') = (h_{A_3}^{-T} \otimes h_M)\mathcal{C}_v(\mathtt{T})h_{A_2}^{-1}.$$

Thus the two matrix spaces are related by these domain and image maps. The last assertion of Lemma E.30 implies $hX = X'$ and $hE = E'$. If either pair of dimensions differs, the tensor transporter is empty. Otherwise apply Corollary 7.10 first to $X, X' \le A_2$ and then, under the resulting candidate coset, to $E, E' \le A_3^* \otimes M$. Use the original argument presentations for these actions. The auxiliary frame dimensions are at most $\dim A_2 + \dim E = O(d)$, and M is a fixed trivial multiplicity. These are degree-2 comparisons and give the complete transporter of the pair in $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, d, \log p)$ time. Pull back the target by its representative, as in (8.5). Henceforth both tensors use (X, E) and the current group preserves that pair.

Define the equivariant evaluation map

$$E \otimes M^* \longrightarrow A_3^*, \qquad e \otimes \ell \longmapsto \ell \circ e,$$

and denote its image by V. Then $Y = \bigcap_{e\in E} \ker e$ is the annihilator of V and $(A_3/Y)^* \cong V$. These spaces are invariant under the current group. Because M^* is trivial, $E \otimes M^*$ is a direct sum of copies

of E. Every simple factor of A_3/Y therefore has dimension at most $\dim E$. The same conclusion holds after restriction to any further subgroup: the displayed evaluation map remains equivariant, and its domain is still a direct sum of copies of the restricted module E.

Both tensors vanish on $A_1 \times X \times Y$. Let Z_0 be the space of all tensors with this property. The maps f_1, f_2 in Step T4 are well defined on Z_0: changing a representative adds an evaluation in this zero block. They are equivariant for the current group. Their common kernel consists of the tensors vanishing whenever the second argument is in X or the third argument is in Y. It is precisely the image of

$$\jmath : Z_3 = \mathrm{Hom}(A_1 \otimes (A_2/X) \otimes (A_3/Y), M) \hookrightarrow Z_0.$$

The first comparison returns gu_1H_1, so every remaining candidate has the form u_1h with $h \in H_1$. Equivariance gives the target $\rho^{(2)}_{u_1^{-1}} f_2(\mathtt{T}')$ for the second comparison. After its answer $gu_1u_2H_2$, put $u = u_1u_2$. The group H_2 stabilizes both source images, and both target images agree with the source after applying u^{-1}. Consequently the vectors defining w and μ_h in Step T4 belong to $\operatorname{im}\jmath$. The maps $v \mapsto \rho^{(3)}_h v + \mu_h$ form an affine action, and

$$\mathrm{Trans}_{H,\rho}(\mathtt{T}, \mathtt{T}') = u\,\mathrm{Trans}_{H_2,\rho^{(3)}+\mu}(0, w), \qquad \mathrm{Stab}_{H,\rho}(\mathtt{T}) = \mathrm{Stab}_{H_2,\rho^{(3)}+\mu}(0).$$

These are also the identities of Lemma 8.2 applied to the two observations in order. They prove the stated complete solution recovery, including all stabilizer generators.

It remains to justify the progress independently of Proposition 8.5. In the first and third comparisons,

$$\dim(A_2/X) \le \tfrac{1}{8}\dim A_2 \le \tfrac{1}{8}a_2.$$

Every argument quotient produced by Step G1 keeps this bound, so Step G2 completes the comparison or Step G3 makes the smaller recursive call. For the second comparison, the preceding evaluation argument shows that every simple quotient of A_3/Y has dimension at most

$$\dim E \le \tfrac{1}{8}\dim A_3 \le \tfrac{1}{8}a_3.$$

Step G1 uses composition flags, as specified in Lemma E.1, so every nonzero tensor layer has a third argument with this bound. The same conclusion then follows from Steps G2–G3. All other arguments are subquotients of their corresponding arguments at entry. Zero tensor spaces require no recursive comparison. In particular, none of these three inputs needs Step G7 before a degree-2 completion or a reset at Step G3.

Apart from solving the three resulting comparisons, the work consists of the canonical-subspace computations, the two frame comparisons, and linear algebra on the displayed maps. The maps, quotient presentations, and translations have polynomial explicit size. Their recovery uses a fixed number of group operations and the fibre identities. This proves the asserted local time and input-size bounds. No list of candidate images remains to be processed by subsequent tensor comparisons. □

Lemma E.32 (Bounds for contraction comparisons)**.** *For an applicable comparison in Step T1 or Step T2, let Ξ be its list of target images and let K be the right subgroup of any nonempty answer to that observation.*

(i) In Step T1, if the chosen contraction has rank r, then

$$|\Xi| \le p^{\dim A_1}, \qquad r \ge \frac{\dim A_1}{64}, \qquad \nu(E_K) \le \nu(E) - \frac{r}{2}.$$

(ii) In Step T2, there is an index k such that

$$|\Xi| \le p^{3\dim A_1}, \qquad \dim B \le \frac{63}{64}\dim A_k$$

for every selected natural factor B arising from A_k under K.

No other ordinary tensor reduction introduces a list of target images whose choices remain imposed while further tensor comparisons are solved.

Proof. Part (i) follows from Lemma E.24; its rank-decrease estimate is stronger than the displayed one. Part (ii) follows from Lemma E.27. The field and scalar choices in Step T3 are each completed by the degree-2 algorithm. The kernel comparison in that step and the pair comparison in Step T4 each return a single complete coset. The three subsequent comparisons in Step T4 are successive comparisons, not alternative target choices. This proves the last assertion. □

E.2.5 Proof of Proposition 8.5

Proof. Consider the ordinary input of the proposition, with its inherited reference dimensions a_i. If either tensor is zero, linearity gives the preliminary answer stated in the algorithm. A nonempty answer is also the degree-2 comparison on the zero module, so is covered by outcome (i). Now assume that both tensors are nonzero; then all three argument spaces are nonzero. Step T0 permutes their positions together with their quotient presentations and reference dimensions. This changes no equation or solution and does not reset a reference dimension. Its type test has the following two exhaustive outcomes.

If the types differ, Lemma E.24 justifies Step T1. A rank mismatch gives emptiness. If both tensors pass the test, the lemma gives a family of candidate cosets with unchanged argument spaces and right subgroups $K < H$. Since H is finite, $|K| \le |H|/2$. The unresolved comparisons start at Step G1 with the same reference dimensions, giving outcome (ii). If neither tensor passes, the rank hypothesis of Lemma E.31 holds, so the algorithm continues at Step T4.

If the types agree, Lemma E.27 justifies the search in Step T2, including rejection when only one tensor has an admissible tuple. When both have one, matching one source tuple gives candidate cosets with proper right subgroups, again giving outcome (ii). The natural-factor bound in that lemma also proves the stated progress when the group reductions are applied. When both tuple searches fail, the field construction and the kernel cases in Lemma E.29 apply. A failed compatibility test gives emptiness; a nonzero proper kernel gives outcome (ii); and an injective map completes the comparison by controlled degree-2 comparisons, giving outcome (i). If the maps vanish, that lemma gives the rank bound required by Step T4.

It remains to consider that common last step. By Lemma E.31, a failed pair comparison gives emptiness. A nonempty answer leaves at most three successive tensor comparisons. Every argument is a subquotient of its corresponding argument at entry, and Steps G1–G3 on each of these inputs give only degree-2 completions or comparisons satisfying (8.3). Step G3 sets the latter comparisons' reference dimensions to their current argument dimensions. This is outcome (iii). The last comparison may be affine; it has exactly the input format of Definition 24, as required by the proposition.

The local lemmas give the original-group representative and source stabilizer for every completed auxiliary comparison. Whenever a condition has been matched, (8.5) retains its entire candidate coset, and the remaining comparison is imposed inside that coset. Thus all earlier conditions remain satisfied. Lemma 8.2 gives the recovery identities for the successive comparisons in Step T4. For a family of candidates, the completed branches cover precisely the original transporter; (8.6) recovers

its representative and generators of its full source stabilizer. An empty answer at every branch gives emptiness. These facts prove the claimed complete output without assuming correctness of any unsolved recursive comparison: complete answers to those comparisons are exactly the data from which the proposition recovers its answer.

Finally, the local lemmas bound the construction and recovery work by $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, d, \log p)$, excluding the resulting recursive comparisons. The rank searches inspect at most $p^{\dim A_1}$ vectors, and the tuple searches at most $\sum_{t=1}^{3} p^{t \dim A_1}$ tuples, for each of the two tensors. Each test consists of linear algebra, degree-2 transport, and the structural group computations. The field construction and its scalar and Frobenius choices have the same local bound; all these choices are completed locally. The zero-block construction and its two auxiliary subspace transporters also have this bound. Summing a fixed number of such stages preserves $p^{O(d)}$.

All group computations retain the original lifts and kernels. The tensor spaces and their subquotients have polynomial explicit size. The common fields have polynomial degree, and their value coordinates are fixed multiplicities in the degree-2 presentations. The local lemmas retain the same value space in every resulting tensor input and give polynomial-size matrices for all quotient maps, restrictions and fibre translations. Consequently the size bound is uniform in the original $\mathrm{InSize}, d, \log p$, as asserted. □

E.2.6 Proof of Lemma 8.7

Proof. We use the local group and tensor bounds established above. At every call made at Step G3, count the call once and treat its complete answer as supplied, without expanding its computation. For the counts in (i) and (iii), retain one child at each choice of a candidate coset and retain all successive comparisons. This is only a way of counting the computation; it changes neither the inputs nor the answers used to construct later comparisons. The searches through contraction tuples and the field cases completed by the degree-2 algorithm belong to the local cost w. We do not expand those searches into recursive tensor calls.

First consider one application of the group reductions, with the boundaries used in Lemma E.23: stop at a completion, a call made at Step G3, a proper-subgroup output of Step G7(b), or an ordinary comparison passed to the tensor reductions. The portion counted by that lemma includes the flag and summand comparisons and the work performed after their answers return. By part (i) of that lemma, it has polynomially many such comparisons. Enlarge its polynomial bound to $P_0(d, \log p)$ to include the input updates and answer recoveries in this portion. This is possible because a flag has at most $d+1$ terms, each quotient is processed once, each pair of summands is computed once and reused, and every remaining group step makes a bounded number of updates for each retained candidate. Local searches and the subset calculation in Step G4 remain part of w.

There is at most one output of this portion that can continue beyond these boundaries without a reset of the reference dimensions. Indeed, in a flag on the current A_i, at most one successive quotient can have dimension greater than $127a_i/128$, since their dimensions sum to $\dim A_i \le a_i$. Thus at most one triple of flag quotients fails (8.3). Every other triple already satisfies that condition, which remains true upon further argument restriction. Its recursive flag comparisons therefore complete at Step G2 or reach Step G3 before any later group step. The same applies to every pair comparison in Step G4 and every retained child in Step G5, by Lemmas E.5 and E.6. Applying this observation through the nested flag calls leaves at most one output at Step G7. All other comparisons, including the work after their answers return, have already been counted in P_0. If an argument is zero, Lemma E.2 gives $Z = 0$, and the comparison returns immediately instead.

Now expand the possible continuing output. By Lemma E.21, a fixed candidate at Step G7(b) gives one child under a proper subgroup. An ordinary output enters Step T0 directly. Lemmas E.24,

E.27, and E.29 show that Steps T1–T3 either complete the comparison, continue to Step T4, or, for the fixed candidate choice, return to Step G1 under a proper subgroup. In the last case the subgroup is proper also in the group at the entrance to the preceding portion, since all intervening group restrictions only decrease the group. Passing from Step G7(c) to an ordinary comparison does not itself return to Step G1.

Step T4 gives at most three successive comparisons. By Lemma E.31, each is processed by Steps G1–G3 into completions or reset calls. Thus these comparisons add at most three further portions of the kind just counted, and none gives another continuing output. This statement also covers the affine final comparison in (8.14); its group reductions need not reach Step G7.

Consequently, after fixing candidate choices, the portions that can continue form one chain. Each transition to another such portion strictly decreases the original subgroup H. A proper subgroup of a finite group has at most half its order, so the chain has length at most

$$1 + \lfloor \log_2 |\Gamma| \rfloor.$$

The ambient group has a faithful prime-field representation of dimension at most d, and hence $\log_2 |\Gamma| = O(d^2 \log p)$. Absorbing the bounded number of tensor-step updates into P_0, the total number of comparisons, input updates, and recoveries counted here is at most

$$4P_0(d, \log p)\big(1 + \lfloor \log_2 |\Gamma| \rfloor\big) = \mathrm{poly}(d, \log p).$$

This proves (i) and (iii). In particular, the estimate counts all successive comparisons, including those constructed after a call made at Step G3 returns; it is not merely a bound on the length of a path. The simultaneous permutations in Step T0 only relabel the reference dimensions and do not affect the count.

The argument applies to every finite partial computation with the reset-call answers supplied. The local procedures terminate, and each candidate list is finite. An infinite tree of recursive calls would therefore have an infinite path, contradicting the uniform bounds on the continuing chain and its attached portions. Thus there is no infinite execution even when every candidate is processed. Assuming the calls made at Step G3 return, all calls counted here terminate. Each reset satisfies (8.3) by the test in Step G3, independently of this counting argument.

For (ii), the candidate lists whose choices remain imposed during further tensor comparisons occur only in Steps G5, G7, T1, and T2; see Lemma E.23(ii) and Lemma E.32. Apply Lemma E.22 along the path in the statement. It includes the ordinary contraction choices, whose rank and constituent-dimension bounds are supplied by Lemma E.32, as well as the projected and multiplicity choices. It gives

$$\sum_{\nu=1}^{t} \log_p |\Xi_\nu| \le \begin{cases} C \sum_i (a_i - a_i'), & \text{at the next call made at Step G3,} \\ C \sum_i a_i, & \text{at a local computation.} \end{cases}$$

Exponentiating proves (ii). A return to Step G1 with unchanged reference dimensions does not restart this budget. Choices internal to an unexpanded reset call belong to that child's computation; they are not appended to paths leading to its caller's later comparisons.

Finally, Lemma E.23(iv) gives the uniform input-size and local-work bounds for the group reductions, including recovery after a child's answer returns. The local cost bounds in Lemmas E.24, E.27, E.28, E.29, and E.31 give the same bounds for the tensor steps. The tensor-step constructions store the argument subquotients and group elements in the original coordinates; they retain the same value module in continuing tensor comparisons. The uniform size argument in Lemma E.23(iv) therefore applies after every such update. Auxiliary field coordinates occur only inside the local

comparisons and do not enlarge the recursive inputs. Writing $\mathrm{InSize}_{\mathrm{cur}}$ for a resulting input's bit length and increasing the absolute constant C if necessary gives

$$w \le p^{Cd}\mathrm{poly}(\mathrm{InSize}, d, \log p), \qquad \mathrm{InSize}_{\mathrm{cur}} \le \mathrm{poly}(\mathrm{InSize}, d, \log p)$$

with fixed polynomial bounds throughout the recursion, as claimed. □

E.3 Degree-3 quotient tensor module transporter

We give the proof and supporting results for the degree-3 quotient tensor module transporter of Subsection 8.3.

E.3.1 Proof of the main result

Theorem 8.10 (restated). *Given an instance of Definition 25 with original dimension d and input length* InSize*, its complete transporter can be computed deterministically in time $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$.*

Proof of Theorem 8.10. We use the recursive template of Proposition 7.11 with three positions. Its flag and group reductions are already available; we must supply its local step. The new ingredients are Lemma E.34, which detects the simple constituents not handled in degree 2, and Lemma E.33, which reduces their comparisons to Theorem 8.1. Both lemmas are proved below, independently of the present algorithm.

If $d = 0$, the linear action is trivial and $h \mapsto c_h$ is an additive homomorphism. Compute its image, lifts, and kernel as in Section 3.5; membership of $y - x$ in that image gives the complete answer in polynomial time. Assume henceforth that $d > 0$. At the local step, use H and $Q = N/P$ for the current group and space.

Center the source at 0, replacing the translations by $c_h := \alpha_h(x) - x$ and the target by $t := y - x$. The equation is now $c_h = t$. Lemma 7.5 supplies the joint action $\theta : H \twoheadrightarrow J$, its kernel R_0, and the normal classical subgroup $D \trianglelefteq J$. At this local step the number of selected auxiliary summands is bounded, so $[J : D] = p^{O(d)}$. The subgroup D fixes the selected auxiliary modules, multiplicity coordinates, and semilinear evaluation spaces pointwise. Consequently, restricted to D, the current tensor presentation is a direct sum of fixed copies of tensor products of at most three natural, dual, or recorded twisted-natural factors. The fixed threshold ensures the stable ranges in Lemma E.34.

Removing the translations of the action kernel. The linear action of R_0 on Q is trivial. By Lemma 3.6, compute $E := c(R_0) \le Q$, with the necessary additive kernels and lifts. The space E is invariant under the linear action of H, and the affine action on $\overline{Q} := Q/E$ factors through J. If $\overline{Q} = 0$, the same lemma completes the original transporter, including its full stabilizer. Otherwise let

$$A = \mathrm{span}_{\mathbb{F}_p} \rho_{\overline{Q}}(D) \le \mathrm{End}_{\mathbb{F}_p}(\overline{Q}), \qquad \mathfrak{r} = \mathrm{rad}\, A, \qquad M = \overline{Q}/\mathfrak{r}\overline{Q}.$$

Compute these data using Lemma 3.2. Nilpotence of $\mathfrak{r}$ gives $M \neq 0$, and M is semisimple as a D-module. Normality of D in J makes $\mathfrak{r}\overline{Q}$ invariant under J. Therefore the composite linear map

$$\pi_M : Q \longrightarrow \overline{Q} \longrightarrow M$$

is equivariant for the linear actions and induces an affine action on M. Write $U := \ker \pi_M$. Thus $E \le U$ and $\dim U = \dim Q - \dim M < \dim Q$.

Completing the equation on M. Enumerate $J = \bigsqcup_a aD$, with lifts of the representatives to H. On a coset aD, pull the target on M back by the induced affine action of a^{-1}. This leaves an equation

$$c_M(g) = t_a, \qquad g \in D,$$

where c_M is the restriction of the descended cocycle. It is independent of the chosen original lifts, since R_0 acts trivially on M.

Lemma E.34 gives the cubic targets $C_j = T_j/J(T_j)$. Compute bases of $\mathrm{Hom}_D(M, C_j)$ as in Lemma E.35. Stack all the maps in these bases to obtain

$$F : M \longrightarrow \bigoplus_j C_j^{\oplus n_j}, \qquad M_2 := \ker F.$$

Here $n_j = \dim_{\mathbb{F}_p} \mathrm{Hom}_D(M, C_j)$, and Lemma E.35 says that M_2 is precisely the sum of the simple summands whose types occur in degree at most two. In particular M_2 is semisimple.

For every map $f : M \to C_j$ in the computed bases, solve, under the full group D, the generator equations

$$f(c_M(s)) = (\rho_{C_j}(s) - I)b_f \qquad (s \text{ a generator of } D).$$

These equations are consistent by $H^1(D, C_j) = 0$. Compute all b_f before imposing any transporter condition. The cocycle identity then gives $f(c_M(g)) = gb_f - b_f$ for every $g \in D$. Thus the detected part of the equation is exactly the simultaneous system

$$gb_f = b_f + f(t_a) \qquad \text{for every basis map } f.$$

Lemma E.33 computes each equation in $C_j = T_j/J(T_j)$ by an ordinary 3-tensor comparison with total argument dimension $O(d)$. Impose these equations successively, using the full transporter and source stabilizer returned by each preceding equation, as in the successive-equation construction (3.2). Every f is D-equivariant, so its coordinate is fixed under all these subgroups. The precomputed identities for b_f also remain valid. No vanishing-cohomology assertion for a returned subgroup is needed. This computes exactly the transporter on M/M_2.

If that transporter is nonempty, Lemma 3.5 leaves one affine comparison on M_2 under its full returned stabilizer. Let W_2 be the direct sum of all tensor products of at most two current natural or dual modules, with twists in their finite field periods. Lemma E.36 presents M_2 as a subquotient of fixed copies of W_2 on those same natural spaces. Its maps are D-equivariant and remain equivariant under the returned subgroup. Transfer the two points and the actual generator translations through this presentation, and apply Theorem 7.1. This completes the entire equation on M in the coset aD. Empty systems are discarded. If $D = 1$, the coset comparisons are simply equality tests.

When the natural representations require a central-product cover, perform the preceding target constructions and comparisons on the specified product cover, pulling back the action on M and its cocycle. The covering kernel acts trivially on M and has zero cocycle. Equivariance makes every $f(M)$ fixed by this kernel; the displayed coboundary identity also makes every b_f fixed by it. Hence each target equation, and the completed equation on M, contains the full covering kernel in its source stabilizer. Project the complete answers to D, with their original lifts, as in the central-product convention for Proposition 7.2. This avoids treating a covering natural module as a representation of the quotient when it does not descend.

One remaining fibre. Combine the completed aD cases by the coset-union identity (3.1). This gives the complete transporter on M in J, or proves emptiness. Take its full inverse image in H, including every generator of R_0. This is exactly the original transporter on Q/U, because the affine

action on that quotient factors through J. Only after all coset cases have been combined, apply Lemma 3.5 to $Q \to Q/U$. It returns one affine comparison on U under the full source stabilizer. In the original presentation this is

$$U = N_U/P, \qquad N_U := \{n \in N : n + P \in U\}.$$

Use the fibre formula for its target and generator translations. Thus no lift of the original cocycle to N and no equivariant section of $N \to Q$ is required. Its dimension is $\dim Q - \dim M < \dim Q$.

Cost and data size. The algebra on $\overline{Q}$, its radical, the Hom spaces, and all changes of origin are computed on explicit spaces of dimension polynomial in the current explicit input length. Lemmas E.34 and E.36 give polynomially many targets, equations, and fixed copies. Each completed target comparison has argument dimension $O(d)$, and the quadratic calls use the same original natural spaces, including their recorded duals and twists. The semisimple head M, the subspace M_2, and their multiplicities are not charged as new moving representations. Thus each coset costs $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$; the $p^{O(d)}$ coset cases are all completed before the single remaining fibre is returned. Here the local estimates first use the current explicit input length. The uniform input-size bounds in the recursive template express it as a fixed polynomial in the initial $\mathrm{InSize}, d, \log p$, giving the displayed bounds in terms of the initial input. The enlarged target spaces and the quadratic presentation are temporary data for these completed calls. The recursive fibre uses the caller's tensor presentation and only replaces the subspace N, subgroup, points, and generator translations. The uniform coordinate and input-size conventions of the common recursion therefore apply. All kernels, preimages, and coset operations use the original group encoding.

The basic input has at most three positions and no auxiliary evaluation representations. The structural preparation and exact subquotient updates of Proposition 7.11 preserve its hypotheses with $r = 3$. The local construction above either completes the comparison or completes a positive-dimensional quotient, combines all its coset answers, and returns one strictly smaller affine fibre in the original presentation. It uses only the established degree-2 and ordinary 3-tensor algorithms, with local cost $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$. Thus all hypotheses of that proposition hold. It gives termination, the stated total running time, and the complete output in the original group, including every source-stabilizer generator and every failed case. □

E.3.2 Reduction of pairing quotients to 3-tensor isomorphism

A comparison modulo $J(T)$ permits adding invariant bilinear pairings multiplied by linear forms in the remaining argument. The following construction realizes precisely these additions as linear changes of coordinates on a larger ordinary tensor.

Lemma E.33 (Transport modulo pairing insertions)**.** *Let H be given in the original group encoding, with actions on based $\mathbb{F}_p$-spaces X_1, X_2, X_3, and put*

$$T = \mathrm{Hom}_{\mathbb{F}_p}(X_1 \otimes X_2 \otimes X_3, \mathbb{F}_p).$$

For $\{i, j, k\} = \{1, 2, 3\}$ with $j < k$, let $B_i \le \mathrm{Hom}_{\mathbb{F}_p}(X_j \otimes X_k, \mathbb{F}_p)$ be an explicitly given H-invariant space, with its induced action on bilinear forms. Set

$$S = \bigoplus_{i=1}^{3} \mathrm{Hom}_{\mathbb{F}_p}(X_i, B_i), \qquad \phi : S \longrightarrow T, \qquad \phi(r)(x_1, x_2, x_3) = \sum_{i=1}^{3} r_i(x_i)(x_j, x_k).$$

Let $N \le \operatorname{im}\phi$ be an explicitly given H-invariant subspace, and let $t, t' \in T$. Suppose that the sum of the original representation dimensions, including those needed for the group encoding, and $\sum_i \dim X_i + \sum_i \dim B_i$ is $O(d)$. Then the complete transporter

$$\{h \in H : h \cdot t - t' \in N\}$$

can be computed in time $p^{O(d)}\mathrm{poly}(\mathrm{InSize}, \log p)$, where InSize *includes all explicit input data. Its witness and full source stabilizer are returned in the original group H.*

Proof. All actions on forms are contragredient. The induced action on S is $(h \cdot r)_i = h_{B_i} r_i h_{X_i}^{-1}$, and ϕ is H-equivariant. Thus $R = \phi^{-1}(N)$ is invariant and is computed by linear algebra; it includes $\ker\phi$. Put $\widetilde{X}_i = X_i \oplus B_i$ and define

$$\widetilde{t}(x_1 + \beta_1, x_2 + \beta_2, x_3 + \beta_3) = t(x_1, x_2, x_3) + \sum_{i=1}^{3} \beta_i(x_j, x_k).$$

Define $\widetilde{t'}$ by the same formula with t' in place of t. Every coefficient involving two or more B_i coordinates is zero. Let d_h act on $\widetilde{X}_i$ as $h_{X_i} \oplus h_{B_i}$. For $r \in R$, let s_r act by

$$s_r(x_i, \beta_i) = (x_i, \beta_i + r_i(x_i)) \quad (i = 1, 2, 3).$$

Direct substitution, remembering the inverse in the action on forms, gives

$$d_h \cdot \widetilde{t} = \widetilde{h \cdot t}, \qquad s_r \cdot \widetilde{t} = \widetilde{t - \phi(r)}, \qquad d_h s_r d_h^{-1} = s_{h \cdot r}.$$

There are no mixed shear terms: a pairing term uses just one added coordinate and two original coordinates, and every shear fixes all original coordinates.

Let $\theta : H \twoheadrightarrow A$ be the joint image on the X_i. Its kernel also fixes the B_i, since their actions are induced from the X_i. Consequently the explicit matrix group

$$\widetilde{A} = \langle d_h,\ s_r : h \text{ is an input generator},\ r \text{ is a basis vector of } R \rangle \le \prod_i \mathrm{GL}(\widetilde{X}_i)$$

is $R \rtimes A$. Restriction to the original-coordinate quotients defines its surjection $\pi : \widetilde{A} \twoheadrightarrow A$. Its augmented transporter consists of the elements $s_r d_h$ with $h \cdot t - t' = \phi(r)$. Since $\phi(R) = N$, its image under π is exactly the transporter of $t + N$ to $t' + N$ in A. The same assertion with $t' = t$ identifies the image of its full source stabilizer with the full stabilizer of $t + N$ in A.

Apply Theorem 8.1 on the dual spaces $\widetilde{X}_i^*$, using the contragredient image of $\widetilde{A}$ as the prescribed subgroup. Project the returned witness and stabilizer generators through π. Lift the resulting witness and every stabilizer generator through θ, and adjoin generators of $\ker\theta$ to the lifted stabilizer. These preimages are the complete answer in H, because its linear action on T/N factors through θ. Thus no faithfulness of the displayed tensor action is required. In particular, elements of $\ker\phi$ are retained during the augmented comparison, and elements lost by the original argument action are restored by the full preimage operation.

The total augmented argument dimension is $\sum_i(\dim X_i + \dim B_i) = O(d)$, although $\dim R = O(d^2)$. An additive basis of R suffices for its shear generators; its vectors are not enumerated. All coordinate constructions are polynomial in the explicit input size. The kernel and lifting operations use Section 3.5 in the original encoding, with the stated dimension budget. This proves the running-time bound. □

Defining fields and prime-field pairing coordinates. The preceding lemma uses the actual prime-field spaces of pairings. For the classical module T in Definition 16, take $B_i = (X_j^* \otimes_{\mathbb{F}_p} X_k^*)^D$ and use the insertion map of Theorem 6.2. The resulting $\operatorname{im}\phi$ is exactly its prime-field $J(T)$. For three single natural arguments, each nonzero B_i is a one-dimensional space over the relevant intertwiner field $\mathbb{K}_i$, so $\sum_i \dim_{\mathbb{F}_p} B_i = O(d)$. More explicitly, an invariant $\mathbb{K}$-valued pairing b supplies the prime-field forms

$$b_\lambda(x_j, x_k) = \operatorname{Tr}_{\mathbb{K}/\mathbb{F}_p}(\lambda b(x_j, x_k)), \qquad \lambda \in \mathbb{K}.$$

The trace pairing is nondegenerate. In the nonzero natural-pairing case these forms identify the added space B_i with $\mathbb{K}$ over $\mathbb{F}_p$, including its actual induced scalar or semilinear action. Thus one adds $[\mathbb{K} : \mathbb{F}_p]$ coordinates, not one coordinate for each element of $\mathbb{K}$. The same construction applies to specified dual and Frobenius-twisted pairings; all maps used by the algorithm are their prime-field matrices.

If a target was first written as a defining-field cubic tensor, compose its associated trilinear form with the field trace and retain its defining-field linearity relations. This is an injective prime-field realization: a nonzero defining-field multilinear form assumes every value of its field by scaling an argument, and the trace is a nonzero functional. Use the exact trace image of its denominator as N in the lemma. Even when this is a proper subspace of all prime-field pairing insertions, $R = \phi^{-1}(N)$ imposes exactly that denominator. One may instead use the original prime-field component presentations throughout; no compositum of the defining fields is required.

E.3.3 Cubic targets for the semisimple comparison

Let D be a product of the selected classical derived groups, in the stable range of Section 6.1. Retain the given natural representations, their duals, and their specified field and twist identifications. All algorithmic modules and Hom spaces in this subsubsection are over $\mathbb{F}_p$. Let $\mathscr{L}_2$ be the set of simple composition-factor types occurring in tensor products of at most two of these representations. In forming this set, allow all Frobenius twists in the finite field periods. These twists are coordinate identifications on the original prime-field spaces; their repetitions do not create new original representations.

Lemma E.34 (Cubic targets detecting the remaining simple types)**.** *There is a polynomial-size, explicitly constructible family*

$$C_j = T_j / J(T_j)$$

with the following properties. Each T_j is a tensor product over $\mathbb{F}_p$ of three of the specified natural or dual representations. Each target therefore has an ordinary trilinear presentation whose three argument dimensions have sum $O(d)$. Every simple type outside $\mathscr{L}_2$ occurring in the given degree-three tensor modules embeds in at least one C_j, whereas no type in $\mathscr{L}_2$ embeds in any C_j. Moreover

$$C_j^D = 0, \qquad H^1(D, C_j) = 0.$$

The target matrices, the pairing maps, and bases of all Hom spaces into these targets are computable by linear algebra in polynomial explicit dimension. The same assertions hold for a specified central-product image through which all the representations factor.

Proof. Take all triples from the natural and dual catalogue needed by the input, allowing repetitions. One may include the polynomially many specified twists as well. This is a polynomial-size list, and every target has polynomial coordinate dimension. Theorems 6.2 and 6.3 construct $J(T_j)$, exclude

the types in $\mathscr{L}_2$ from the socle, and give the two vanishing assertions. We prove the remaining assertion, namely that all other simple types occur in these socles.

We use the following consequence of the insertion theorem. If $P \leq T$ has at least one composition factor outside $\mathscr{L}_2$ and exactly one such isomorphism type S, then S embeds in $T/J(T)$. Indeed, $J(T)$ has only lower-degree composition factors, so $P/(P \cap J(T))$ is nonzero. It embeds in $T/J(T)$, whose socle has no lower-degree simple type. Any simple submodule of $P/(P \cap J(T))$ therefore has type S. Thus it suffices to exhibit such an actual submodule for each remaining simple type. A subquotient with that property would not suffice for this argument.

We first make this calculation over an algebraic closure, for a single connected classical group and equal twists. Write V for its natural module, ϵ_i for its coordinate weights, and ω_i for its fundamental weights. In the stable ranks in use, the possible highest weights in a pure cube outside the degree-at-most-two list are among

$$3\omega_1, \quad \omega_1 + \omega_2, \quad \omega_3. \tag{*}$$

For a mixed type-A product $V \otimes V \otimes V^*$ the corresponding list is

$$2\omega_1 + \omega_{n-1}, \quad \omega_2 + \omega_{n-1}. \tag{**}$$

These lists can also be read directly from the tensor weights: a dominant weight involving three natural positions is, away from the spin nodes, one of the three partitions of three in the pure case, or one of the two signed partitions in the mixed case; all remaining dominant weights are on the quadratic or natural list. The square constituents used here are those in the degree-two representation calculation; see also [GT05, Section 3.2]. A self-dual natural module reduces a mixed product to the pure case. Applying the same position constructions with V^* in place of V and reversing the type-A root labels treats the remaining choices; we do not dualize a submodule into a quotient.

Here are the weight multiplicities needed to separate the types. For the first three positive coordinates, use the weights

$$a = 3\epsilon_1, \qquad b = 2\epsilon_1 + \epsilon_2, \qquad c = \epsilon_1 + \epsilon_2 + \epsilon_3.$$

The position action of S_3 commutes with the classical group. For $p > 3$, its symmetric, standard-isotypic, and alternating summands have the following multiplicities:

	a	b	c
$\operatorname{Sym}^3 V$	1	1	1
standard isotypic summand	0	2	4
$\bigwedge^3 V$	0	0	1.

(***)

They follow by permuting the basis tensors with index patterns 111, 112, and 123.

For completeness, the simple highest-weight modules contribute the stated middle weights for all the classical types. In a simple highest-weight module $L(\lambda)$, the sum of the weight spaces whose weights differ from λ only by the first two simple roots is the simple highest-weight module for the corresponding SL_3 Levi subgroup. To see simplicity, a nonzero Levi submodule has a vector fixed by its positive root groups. Positive root groups outside this Levi also fix that vector, since increasing an outside root coefficient leaves the weight set below λ. The highest-weight line of the simple ambient module is its only line of such fixed vectors. Ordered divided-power negative-root operators generate a highest-weight module from that line. An operator involving any root outside the Levi has a nonzero outside coefficient in its total weight decrease, so cannot contribute to this

face. Hence the negative Levi root operators alone generate the entire face from the highest-weight line. Thus no proper nonzero Levi submodule exists. For $3\omega_1$ and $p > 3$, this Levi module is the simple symmetric cube, and it has multiplicity one at each of a, b, c. For $\omega_1 + \omega_2$ it is the simple adjoint quotient of $\mathfrak{sl}_3$. Its middle weight has dimension two if $p \neq 3$, and dimension one if $p = 3$, when the scalar center has been factored out. The same argument on the first two coordinates shows that, for odd p, the simple mixed highest type $2\omega_1 + \omega_{n-1}$ contains the weight $\epsilon_1 + \epsilon_2 - \epsilon_n$ with multiplicity one.

For $p > 3$, the top simple in the symmetric summand of $(***)$ uses all three displayed one-dimensional weight spaces, so that summand has only type $3\omega_1$ outside the quadratic list. In the standard summand, the weight b forces two copies of $L(\omega_1 + \omega_2)$; their two-dimensional middle weights exhaust its weight-c space. Hence no $L(\omega_3)$ occurs there. The alternating summand has only type ω_3 outside the quadratic list. The preceding socle argument now applies to these three actual submodules.

For $p = 3$, $L(3\omega_1) = V^{[1]}$ is already on the lower-degree list. Use the actual invariant-tensor submodule $\Gamma^3 V = (V^{\otimes 3})^{S_3}$ and the image of the alternating insertion $\bigwedge^3 V \hookrightarrow V^{\otimes 3}$. The latter map is injective even in characteristic three: different basis triples have disjoint six-term supports, with coefficients 1 and -1. It has only type ω_3 outside the quadratic list. In $\Gamma^3 V$, the multiplicities at b, c are both one. The weight b forces $\omega_1 + \omega_2$; its one-dimensional Levi middle weight exhausts weight c. Thus $\Gamma^3 V$ has exactly that type outside the quadratic list. These submodules again give both required embeddings.

For $p = 2$, put $e = (123) + (132)$ in the position algebra. Then $e^2 = e$, and direct inspection of the same three index patterns gives

	a	b	c
$eV^{\otimes 3}$	0	2	4
$(1-e)V^{\otimes 3}$	1	1	2.

$(****)$

Steinberg's tensor-product theorem gives $L(3\omega_1) = V^{[1]} \otimes V$, a two-position module; its multiplicities at a, b, c are $1, 1, 0$. In the first row of $(***)$, the two copies forced at weight b have type $\omega_1 + \omega_2$. Each has a two-dimensional Levi middle weight, since $\mathfrak{sl}_3$ is simple in characteristic two. They exhaust weight c, excluding ω_3 from this summand. In the second row, the copy of $L(3\omega_1)$ forced at weight a exhausts weight b, excluding $\omega_1 + \omega_2$. Its zero weight-c multiplicity leaves type ω_3, which is therefore the only type outside the quadratic list in this summand. This argument applies to type A, symplectic type, and even orthogonal type alike; any additional natural constituents have none of the three displayed weights. The socle argument gives the embeddings in characteristic two without a semisimplicity assumption on either position summand.

For the mixed type-A list $(**)$ in odd characteristic, split the two natural positions by $(1 \pm (12))/2$. In the symmetric summand, the top simple uses the one-dimensional weight $\epsilon_1 + \epsilon_2 - \epsilon_n$, excluding the other listed type. The alternating summand has only that other type. For $p = 2$, the first listed simple is $V^{[1]} \otimes V^*$ by Steinberg's theorem and is lower degree. The entire mixed product therefore has at most one type outside the quadratic list, namely $\omega_2 + \omega_{n-1}$. The socle argument applies directly.

It remains to distribute the occurrences between twists and classical factors. If exactly two occurrences on one factor have the same exponent, use the following actual square submodules: in odd characteristic, the symmetric and alternating summands of a pure square; in characteristic two, the whole pure square; and, for a natural–dual pair, its whole mixed square. Each has at most one restricted quadratic simple type apart from trivial and Frobenius-natural types. Tensor it with the remaining occurrence at a different exponent. Steinberg's tensor-product theorem makes

the restricted quadratic type times that occurrence simple. A Frobenius-natural type times the occurrence still has a two-position presentation, and a trivial type leaves one position. Thus each chosen submodule has at most one type outside $\mathscr{L}_2$. Three distinct exponents give a simple product directly. For occurrences on independent factors, the same argument uses simple external tensor products: the partitions of three occurrences are $2+1$ and $1+1+1$. This exhausts the cases.

We check finite restriction before descending. For a group with field parameter $q = p^f$, normalize exponents to $0, \dots, f-1$. For unitary groups, whose natural matrices are over $\mathbb{F}_{q^2}$, record the identity $V^{[a+f]} \cong (V^{[a]})^*$ when making this normalization. Each remaining genuinely cubic simple has a p-restricted weight at each occupied exponent, hence a q-restricted total weight. Steinberg's finite restriction theorem, including its twisted form [Ste67, Theorems 41 and 43], therefore preserves its simplicity. The two exceptional carries already used, $3\omega_1$ at $p = 3$ and $3\omega_1$ or $2\omega_1 + \omega_{n-1}$ at $p = 2$, leave at most two natural positions. They remain on the lower-degree list even when the new exponent wraps around its field period. The other cubic weights are restricted at $q = 2$ or 3 and are distinct from the quadratic list in the stable ranks. The stable type-D graph action fixes the low-index weights used above. Thus each actual submodule just constructed still has at most one nonquadratic simple type for the finite group.

Finally, scalar extension of an actual prime-field tensor T_j decomposes into the tensor products of its defining-field constituents with their Frobenius-conjugate twists. Invariants and the insertion maps commute with scalar extension, since they are kernels and images of generator-matrix linear systems. Consequently its quotient $T_j/J(T_j)$ contains all the corresponding quotient components used in the calculation. No common extension field is constructed by the algorithm. If a simple $\mathbb{F}_pD$-module S outside $\mathscr{L}_2$ occurs in the input, one of its absolutely simple constituents therefore embeds in a scalar extension of a target. Since $\mathbb{F}_p$ is perfect, the scalar extension of S is semisimple. Projection onto that constituent followed by its embedding gives a nonzero map from the scalar extension of S. Scalar extension also commutes with the equations for $\mathrm{Hom}_D(S, C_j)$, so this Hom space is nonzero already over $\mathbb{F}_p$. Any nonzero map from S is injective. This proves detection over the actual input field.

All constructions required by the algorithm are the original prime-field tensors, their pairing maps, and intertwining linear systems; highest-weight coordinates are used only in the preceding proof. They have polynomial explicit length. Pulling back through a specified central-product quotient changes neither invariant subspaces nor Hom spaces, since its kernel acts trivially throughout. This proves the last assertion. □

Lemma E.35 (Detection of a semisimple quotient)**.** *Let D be a finite group, let M be an explicit semisimple $\mathbb{F}_pD$-module, and let $\mathcal{L}$ be a class of simple module types. Suppose a finite list of explicit $\mathbb{F}_pD$-modules C_j has no type in $\mathcal{L}$ in its socles, and every simple type of M outside $\mathcal{L}$ embeds in some C_j. For each j, compute a basis $\mathcal{F}_j$ of $\mathrm{Hom}_D(M, C_j)$ and form*

$$F : M \longrightarrow \bigoplus_j C_j^{\oplus|\mathcal{F}_j|}, \qquad m \longmapsto (f(m))_{j,f\in\mathcal{F}_j}.$$

Then $\ker F$ is precisely the sum of the simple summands of M whose types lie in $\mathcal{L}$. The maps and this kernel are computed by linear algebra in polynomial time in their explicit sizes.

Proof. A nonzero map from a simple module to C_j is injective and has image in its socle. Hence all maps kill the indicated summands whose types lie in $\mathcal{L}$. Conversely, project a nonzero vector in the complementary semisimple summand to a simple direct summand on which it is nonzero, and follow this projection by a specified embedding into one C_j. Some map in $\mathrm{Hom}_D(M, C_j)$ therefore detects that vector, and its expression in the computed basis shows that F detects it. The equations

$f\rho_M(s) = \rho_{C_j}(s)f$ on group generators compute the required Hom spaces; Gaussian elimination computes $\ker F$. □

Lemma E.36 (A quadratic presentation of semisimple low-degree modules)**.** *Let W_2 be an explicitly given degree-at-most-two tensor module as in Definition 17, on original spaces of total dimension $O(d)$, and let M be an explicit semisimple $\mathbb{F}_pD$-module. Suppose every simple type of M occurs as a composition factor of W_2. One can construct, in polynomial time in the explicit module dimensions and input length, an equivariant isomorphism*

$$M \cong N_2/P_2, \qquad P_2 \le N_2 \le W_2^{\oplus k},$$

where k is polynomial in $\dim W_2 + \dim M$ and the copy labels are fixed by D. This presentation uses the same underlying representation spaces. It remains equivariant under every subgroup of D and transfers any given quotient-valued affine action on M to the displayed quotient.

Proof. Put $A_2 = \operatorname{span}_{\mathbb{F}_p} \rho_{W_2}(D)$. An element of the group algebra annihilating W_2 annihilates all its composition factors, and hence annihilates the semisimple module M. Therefore the action on M factors through A_2. Compute the simultaneous enveloping algebra on $W_2 \oplus M$ by matrix-span closure. Projection to its W_2 block is an isomorphism onto A_2; linear algebra thus computes the action of every basis element of A_2 on M.

Let $w_1, \dots, w_n$ be a basis of W_2 and $m_1, \dots, m_m$ a basis of M. The maps

$$A_2 \longrightarrow W_2^{\oplus n}, \quad a \longmapsto (aw_1, \dots, aw_n), \qquad \psi : A_2^{\oplus m} \longrightarrow M, \quad (a_1, \dots, a_m) \longmapsto \sum_i a_i m_i$$

are respectively an injective and a surjective map of left A_2-modules. The first is injective because A_2 is an actual matrix algebra on W_2; the second is surjective because $1 \in A_2$. Apply the first map in each copy, obtaining an embedding $\iota : A_2^{\oplus m} \hookrightarrow W_2^{\oplus nm}$, and set $N_2 = \operatorname{im} \iota$, $P_2 = \iota(\ker \psi)$. The map $\iota(a) + P_2 \mapsto \psi(a)$ is the required equivariant isomorphism. Its coordinate matrices and the subspace bases are obtained by linear algebra. All spaces have polynomial dimension. The copies carry no additional moving action, so they increase only the explicit length, not the original representation budget. Transfer points and generator translations through the isomorphism; the cocycle identity is needed only in the quotient. The case $M = 0$ is handled by the zero presentation. □

F Technical results for group isomorphism

The subsections follow the corresponding topics of Section 9, starting with first-cohomology transport. The short proofs of the cohomology theorems follow their analyses in Section 9, and the group-isomorphism theorem is proved in §9.6. Here the local constructions precede the correctness and running-time conclusions that use them. Results stated in the main section are restated before their proofs; the remaining technical statements appear here with their proofs. Numbered formulas are stated only once and are cited where needed.

F.1 First-cohomology transport

We restate and prove the technical claims used in §9.3. The module-filtration and restriction constructions come first, followed by correctness of the recursion. The proof of Theorem 9.2 and its application to module extensions follow the running-time analysis in §9.3.3.

F.1.1 Module filtrations and inflation

Proposition 9.6 (restated). *The formulas for module grades* (9.10) *hold in degrees* $j = 1, 2$. *The restriction and inflation formulas* (9.12)–(9.13) *hold in degree one. All these identifications commute with compatible transformations of* R_i *and the modules involved, and are computable by linear algebra on normalised cochains.*

Proof. The long exact cohomology sequence of a short exact sequence of modules gives (9.9); see [Bro82, Proposition III.6.1]. Apply its maps to normalised cochains and pass to cycles modulo boundaries to compute their matrices.

For a fixed module grade, put $A_k = \phi_k^{-1}(N)$ and $E_k = \phi_k^{-1}(P)$. Exactness gives $\phi_k(\ker \lambda_k) = F_{k-1}$. The surjection $A_k \to \lambda_k(A_k)/\lambda_k(E_k)$ has kernel $E_k + (A_k \cap \ker \lambda_k)$. Since $\ker \phi_k \leq E_k$, applying ϕ_k identifies this quotient with

$$\frac{N \cap F_k}{(P \cap F_k) + (N \cap F_{k-1})} \cong W_k/W_{k-1}.$$

This is exactly (9.10). In particular, the denominator includes the kernels introduced by the maps $H^j(R_i, B_k) \to H^j(R_i, B)$.

For restriction and inflation, a 1-cocycle that vanishes on S_i takes values in $\mathrm{Fix}_{S_i}(B)$ and factors through R_{i+1}. If its restriction is a coboundary, subtract that coboundary first. This proves exactness at $H^1(R_i, B)$ in (9.11). Inflation is injective: if an inflated cocycle is the coboundary of $b \in B$, evaluation on S_i forces $b \in \mathrm{Fix}_{S_i}(B)$, so the original quotient cocycle was already a coboundary. This is the usual inflation–restriction sequence; see [DHW12, Section 1]. The first isomorphism theorem applied to the induced restriction map $N/P \to \mathrm{res}(N)/\mathrm{res}(P)$ gives (9.12). Taking inverse images under the inflation isomorphism onto F then gives (9.13).

All maps induced by module homomorphisms, as well as restriction and inflation, commute with compatible transformations on cochains. Their kernels, images, and quotient identifications are therefore equivariant, and their matrices are obtained by elimination. No equivariant choice of cocycle representatives is required. □

F.1.2 Elementary normal subgroups

Let R be a finite group, let $C \trianglelefteq R$ be an elementary abelian subgroup with $C \cong \mathbb{F}_p^r$, and let B be an $\mathbb{F}_p R$-module. The Γ-action on R preserves C and is compatible with its action on B.

Lemma 9.8 (restated). *Let* I_C *be the augmentation ideal of* $\mathbb{F}_p[C]$ *from Definition 28. Set* $B_k = I_C^k B$ *for* $k \geq 0$ *and* $D_k = B_k/B_{k+1}$. *Then* $B_0 = B$, *and the* B_k *form a descending filtration by* R*-submodules preserved by* Γ. *Every nonzero* B_k *properly contains* B_{k+1}, *and* C *acts trivially on every grade* D_k. *There are at most* $\dim_{\mathbb{F}_p} B$ *nonzero grades, and their dimensions sum to* $\dim_{\mathbb{F}_p} B$. *The filtration and its induced actions are computable by linear algebra on the given action matrices.*

Proof. Choose a vector-space basis $x_1, \dots, x_r$ of C. The ideal I_C is generated by $x_1 - 1, \dots, x_r - 1$ in $\mathbb{F}_p[C]$. These elements commute and have pth power zero, so $I_C^{r(p-1)+1} = 0$. In B,

$$B_{k+1} = \mathrm{span}_{\mathbb{F}_p}\{cb - b : c \in C,\ b \in B_k\} = \sum_{i=1}^{r} (x_i - 1) B_k.$$

Here $x_i - 1$ acts on B_k through the given representation of C; the ideal I_C itself lies in the group algebra. Normality of C makes each B_k an R-submodule. Compatibility gives $\alpha_g(cb - b) =$

$\beta_g(c)\alpha_g(b) - \alpha_g(b)$ for $g \in \Gamma$, and $\beta_g(C) = C$, so induction proves preservation by Γ. By definition, $cb - b \in B_{k+1}$ for $b \in B_k$, which proves that C acts trivially on D_k. Its R-action therefore factors through R/C; its Γ-action need not be trivial.

Nilpotence of I_C makes the filtration end at zero. If $B_k = B_{k+1}$, repeated multiplication by I_C gives $B_k = I_C^m B_k = 0$ for sufficiently large m. Thus all inclusions before zero are strict, proving both dimension assertions. The displayed sum computes each next subspace from the images of r linear maps; restriction and passage to quotients compute the induced actions. □

Lemma 9.9 (restated). *Let D be a filtration grade of B on which C acts trivially. There is an equivariant identification $H^1(C, D) = \mathrm{Hom}_{\mathbb{F}_p}(C, D) = D \otimes_{\mathbb{F}_p} C^*$, where $C^* = \mathrm{Hom}_{\mathbb{F}_p}(C, \mathbb{F}_p)$. Under this identification, $g \in \Gamma$ acts by $f \mapsto \alpha_g \circ f \circ \beta_g^{-1}$. Consequently the restriction comparison adds one tensor occurrence, with new moving argument of dimension $r = \log_p |C|$. For $P \le N \le H^1(R, D)$, the quotient compared in Step R2 is $\mathrm{res}(N)/\mathrm{res}(P)$, as a subquotient of $H^1(C, D)$. The remaining fibre is a subquotient of $H^1(R/C, D)$ and is the only continuing cohomology task for this grade.*

Proof. Since C acts trivially on D, its 1-cocycles are additive homomorphisms and its 1-coboundaries are zero. Additive maps between these elementary abelian groups are $\mathbb{F}_p$-linear. The cochain action of Γ gives the stated formula on $\mathrm{Hom}_{\mathbb{F}_p}(C, D)$, hence the equivariant tensor identification.

Write $F = \ker \mathrm{res}$ and $W = N/P$. The kernel of the induced restriction map on W is $W_F = ((N \cap F) + P)/P$, and the first isomorphism theorem gives $W/W_F \cong \mathrm{res}(N)/\mathrm{res}(P)$. Exactness of (9.11), with $\mathrm{Fix}_C(D) = D$, identifies F with the inflation image of $H^1(R/C, D)$. Thus W_F is a subquotient of that cohomology space, as used in Step R3. □

F.1.3 Restriction to products of simple groups

Proposition 9.10 (restated). *Let $S \trianglelefteq R$ be a direct product of nonabelian finite simple groups, and let B be an $\mathbb{F}_p R$-module. Let Γ act compatibly on R and B, preserving S. Suppose B has an explicit tensor subquotient presentation of degree at most e on argument spaces of total dimension d, with compatible actions of an initial group Q and a quotient map $Q \twoheadrightarrow R$ as in §9.3.1. Let $j \in \{1, 2\}$ and $e + j \le 3$. Every affine transport problem on an invariant subquotient of $H^j(S, B)$ reduces to at most $|S|$ cubic affine transporter calls. Each has moving argument dimension at most $C(d + \log_p |S|)$ and explicit input length polynomial in the original data, apart from any faithful vector-set realization of these moving arguments.*

The complete answers are combined and returned in the original group Γ before any subsequent quotient step. Temporary covering groups and corrected actions are not passed to that step. The only additional moving argument is $Y = \mathbb{F}_p^{\tau_p}$, indexed by the simple factors whose orders are divisible by p, with $\dim Y = \tau_p \le \log_p |S|$. It occurs at most j times in degree j, and its dimension is counted once.

Proof. Let the current finite group be R, let $S \trianglelefteq R$ be a product of nonabelian simple groups, and let B be the current $\mathbb{F}_p R$-module. For $j \in \{1, 2\}$, the restriction term in $H^j(S, B)$ is handled in four steps. Find its simple direct factors; enumerate the bounded set of cosets in their automorphism group modulo inner automorphisms and permutations of equal factors; correct inner automorphisms using the covering group $\widehat{\Gamma}_0 \twoheadrightarrow \Gamma_0 \le \Gamma$ defined below; and express degree-j cochains using at most j moving factor indices. Those indices fit into the remaining tensor degree. We describe the correction first, then give the arrays and linear relations presenting the cochains as quotients N/P of subspaces of tensor arrays.

Finding the factors and limiting the outer actions. If the current quotient group $R \neq 1$ has $\operatorname{srad}(R) = 1$, its socle $S = \operatorname{Soc}(R)$ is a characteristic product $T_1 \times \cdots \times T_\tau$ of nonabelian simple groups, with $\tau \in \mathbb{N}$. This is also the structural starting point of the radical-free calculation in Cannon and Holt [CH03, Sections 3.1–3.2]. Algorithmically, compute the normal closure of every element and generate the product of the inclusion-minimal nontrivial ones. Repeating the normal-closure calculation inside that product gives its simple direct factors. All these operations use the multiplication tables of R and its subgroups.

Every nonabelian finite simple group is generated by two elements [GK00, Theorem I and its Corollary]. A generating pair is found by enumeration. Enumerating its possible images finds all isomorphisms to another simple factor and all its automorphisms in polynomial table time. Fix a common table model and an explicit identification for each isomorphism type. The pure permutations of isomorphic factors now form a subgroup $\Pi \leq \operatorname{Aut}(S)$. Its elements permute the factors $T_i \leq S$ through the fixed identifications. Since $S = T_1 \times \cdots \times T_\tau$ is centerless,

$$J = \operatorname{Inn}(S) \rtimes \Pi \leq \operatorname{Aut}(S), \qquad [\operatorname{Aut}(S) : J] = \prod_{i=1}^{\tau} |\operatorname{Out}(T_i)| \leq |S|. \tag{F.1}$$

Here $\operatorname{Inn}(S) \leq \operatorname{Aut}(S)$ consists of the conjugation maps $x \mapsto nxn^{-1}$ for $n \in S$ and is called the *inner automorphism group*. The *outer automorphism group* $\operatorname{Out}(T_i) = \operatorname{Aut}(T_i)/\operatorname{Inn}(T_i)$ identifies automorphisms that differ by such a conjugation. For each $1 \leq i \leq \tau$, the inequality follows from $|\operatorname{Aut}(T_i)| \leq |T_i|^2$ and $|\operatorname{Inn}(T_i)| = |T_i|$. We enumerate cosets modulo J using these outer automorphism tables.

Removing inner actions without losing original elements. After restricting the outer action, each remaining $g \in \Gamma$ acts on S by a factor permutation followed by conjugation. Inner conjugation acts trivially on cohomology when accompanied by the corresponding action on B. We remove it so that element labels inside each simple factor become fixed. The removal must be made on the original factor representations as well as on $B = N_B/P_B$; the covering group below records all choices needed for that operation.

The construction uses the fixed initial group Q, even when the action on B factors through the current quotient R. For a second-cohomology input, use the presentation with one argument $V_1 = B$, numerator $N_B = B$, and denominator $P_B = 0$, with its Q-action pulled back from R. For a first-cohomology input, use its input presentation. Write $\nu : Q \twoheadrightarrow R$ for the quotient map, with kernel $K \trianglelefteq Q$; in the recursive algorithms these are ν_i and N_i. The presentation includes representations $\rho_i : Q \to \mathrm{GL}(V_i)$ on $\mathbb{F}_p$-vector spaces V_i, for $1 \leq i \leq s$ and $s \in \mathbb{Z}_{\geq 0}$. The group $\Gamma \leq \operatorname{Sym}(\Omega)$ has homomorphisms $\beta^Q : \Gamma \to \operatorname{Aut}(Q)$ and $\alpha_i : \Gamma \to \mathrm{GL}(V_i)$ satisfying $\alpha_i(g)\rho_i(q)\alpha_i(g)^{-1} = \rho_i(\beta_g^Q(q))$ for $g \in \Gamma$, $q \in Q$, and $1 \leq i \leq s$, where $\beta_g^Q = \beta^Q(g) \in \operatorname{Aut}(Q)$. The input automorphisms β_g^Q preserve $K \trianglelefteq Q$ and induce the given action β_g on R: $\nu\beta_g^Q = \beta_g\nu$. Write $B = N_B/P_B$, with $P_B \leq N_B \leq Z_B$, where Z_B consists of the specified tensor arrays on the spaces V_i. Both subspaces are preserved by Q and Γ. The input requires that every $k \in K$ act as the identity modulo P_B, so the action on B depends only on $qK \in R$. It does not require k to act trivially on each V_i. Keeping the matrices ρ_i on V_i makes the correction below explicit.

Restrict Γ to $\Gamma_0 = \{g \in \Gamma : \beta_g|_S \in J\}$, the preimage of J. Its index $[\Gamma : \Gamma_0]$ is at most $|S|$, and generators of $\Gamma_0 \leq \Gamma$ and all coset representatives are constructible. Indeed, form the finite outer automorphism tables of the simple factors. The product of their outer-coordinate sets has size at most $|S|$; Γ acts on these tuples, one outer-automorphism class per factor, by permuting isomorphic factors and multiplying by its outer-automorphism coordinates. The stabilizer of the

identity tuple is $\Gamma_0 \le \Gamma$. Orbit and Schreier calculations on at most $|S|$ outer-coordinate tuples give its full generators and representatives.

For each $g \in \Gamma_0$, there are unique $n_g \in S$ and $\sigma_g \in \Pi$ such that $\beta_g|_S = \mathrm{Inn}(n_g)\sigma_g$. Compute n_g by enumeration in S. Define the group

$$\begin{aligned} \widehat{\Gamma}_0 &= \{(g,q) \in \Gamma_0 \times Q : \nu(q) = n_g\}, \\ (g,q)(h,r) &= (gh, \beta_g^Q(r)q). \end{aligned} \tag{F.2}$$

For $(g,q),(h,r) \in \widehat{\Gamma}_0$, the identity $n_{gh} = \beta_g(n_h)n_g$ proves closure under this product. Associativity follows from the identity $\beta_{gh}^Q = \beta_g^Q \beta_h^Q$ in $\mathrm{Aut}(Q)$. The inverse of $(g,q) \in \widehat{\Gamma}_0$ is $(g^{-1}, (\beta_g^Q)^{-1}(q^{-1})) \in \widehat{\Gamma}_0$. The projection $\widehat{\Gamma}_0 \to \Gamma_0$, $(g,q) \mapsto g$, is surjective with kernel $\{(1,k) : k \in K\}$. For each generator $g \in \Gamma_0$, enumerate Q to choose q with $\nu(q) = n_g$. Surjectivity of ν guarantees such a choice. Use the pairs (g,q) and add generators of this kernel. Their images generate Γ_0 and the added elements generate the projection kernel, so together they generate $\widehat{\Gamma}_0$.

For $1 \le i \le s$ and $(g,q) \in \widehat{\Gamma}_0$, define corrected maps on V_i and on Q by

$$\alpha_i'(g,q) = \rho_i(q)^{-1}\alpha_i(g), \qquad \beta^{Q\prime}(g,q) = \mathrm{Inn}(q)^{-1}\beta_g^Q. \tag{F.3}$$

These define homomorphisms $\alpha_i' : \widehat{\Gamma}_0 \to \mathrm{GL}(V_i)$ and $\beta^{Q\prime} : \widehat{\Gamma}_0 \to \mathrm{Aut}(Q)$: for example, for $(g,q),(h,r) \in \widehat{\Gamma}_0$,

$$\rho_i(q)^{-1}\alpha_i(g)\rho_i(r)^{-1}\alpha_i(h) = \rho_i(\beta_g^Q(r)q)^{-1}\alpha_i(gh).$$

They satisfy the same compatibility equation as α_i, β^Q, with $\alpha_i', \beta^{Q\prime}$ in their places. Since the original Q- and Γ-actions preserve N_B and P_B, the corrected maps preserve them as well. On $S \le R$, the corrected automorphism is the pure factor permutation $\sigma_g \in \Pi$. The group $\widehat{\Gamma}_0$ is represented faithfully on the disjoint union $\Omega \sqcup Q$: its action on Ω is the original permutation $g \in \Gamma_0$, and its action on Q is $x \mapsto q^{-1}\beta_g^Q(x)$. For $(g,q),(h,r) \in \widehat{\Gamma}_0$, composition gives (F.2). Identity on Ω forces $g = 1$ in Γ_0, and identity on Q then forces $q = 1$ in Q. Thus this construction adds $|Q|$ points to the permutation domain.

For $j \in \{1,2\}$, the corrected action on $H^j(S,B)$ agrees with the original action. They differ by the simultaneous inner action of $n_g \in S$ on the group S and on the module B. Such an action is trivial on cohomology [Bro82, Proposition III.8.3], even when S acts nontrivially on B. A kernel element $(1,q) \in \widehat{\Gamma}_0$ has $q \in K$; it acts trivially on B and on R, hence on $H^j(S,B)$. On the current quotient of $H^j(S,B)$, give $(g,q) \in \widehat{\Gamma}_0$ the affine map of its projection $g \in \Gamma_0$, including the same translation. This kernel therefore has zero affine displacement as well. A further kernel of the corrected matrices can have a nonzero displacement and is handled by Lemma 3.6.

Presenting cochains using the moving factor indices. In degree j, a tensor product of resolutions uses at most j nonzero-degree factors. Consequently it has at most j moving simple-factor indices, even when each fixed table contributes many cochain coordinates. This separates the dimension charged in the exponent from the number of explicitly stored coordinates. We now give the tensor spaces, subspaces, and quotients for this corrected action. First let $S_{p'} \trianglelefteq S$ be the product of the simple factors whose orders are prime to p. For $j \in \{1,2\}$, replace $H^j(S,B)$ by $H^j(S/S_{p'}, \mathrm{Fix}_{S_{p'}}(B))$ by Lemma 9.7. The subgroup $S_{p'} \le S$ is characteristic. If $\pi_B : N_B \to B$ is the quotient map, put $N_B' = \pi_B^{-1}(\mathrm{Fix}_{S_{p'}}(B))$. Then $\mathrm{Fix}_{S_{p'}}(B) = N_B'/P_B$, using the same array space Z_B, and the corrected action of $\widehat{\Gamma}_0$ preserves both subspaces. Relabel the remaining factors as $T_1, \dots, T_{\tau_p}$, where $\tau_p \in \mathbb{Z}_{\ge 0}$, and let $Y = \mathbb{F}_p^{\tau_p}$ be the permutation module on these factor indices. Then $\dim_{\mathbb{F}_p} Y = \tau_p$ and $p^{\tau_p} \le |S|$, as asserted in (9.14). Indeed, every remaining T_i has order

divisible by p, so their product has order at least p^{τ_p}. The moving index space therefore has dimension at most $\log_p |S|$. The number of elements inside each simple factor can be much larger; their table labels will index fixed copies of these moving spaces and enter only the explicit coordinate size.

For $1 \le i \le \tau_p$, use the normalised bar resolution over $\mathbb{F}_p[T_i]$. A free resolution of the trivial module is an exact augmented chain complex of free $\mathbb{F}_p[T_i]$-modules ending in $\mathbb{F}_p$. In this resolution the degree-k free generators are symbols $[x_1 \mid \cdots \mid x_k]$ with $x_h \in T_i \setminus \{1\}$. In the first two degrees the boundary is $\partial[x] = x[\,] - [\,]$ and $\partial[x \mid y] = x[y] - [xy] + [x]$, where a symbol containing the identity is zero. Applying $\mathrm{Hom}_{\mathbb{F}_p[T_i]}(-, B)$ gives the normalised cochain differential used earlier. Tensor these resolutions over $\mathbb{F}_p$, with differential $\sum_i (-1)^{\nu_1+\cdots+\nu_{i-1}}\partial_i$ on multidegree $(\nu_1, \ldots, \nu_{\tau_p}) \in \mathbb{Z}_{\ge 0}^{\tau_p}$, where ∂_i is the boundary map on the ith resolution. If $\tau_p = 0$, the quotient $S/S_{p'}$ is trivial and its positive-degree cohomology is zero. The tensor complex is a free $\mathbb{F}_p[S/S_{p'}]$-resolution of the trivial module: tensor products of the free group-algebra factors are free over the product group, and exactness follows by splitting the augmented complexes as vector-space complexes and tensoring their contracting homotopies; these are the standard resolution and comparison constructions [Bro82, Chapters I–III]. Applying $\mathrm{Hom}_{\mathbb{F}_p[S/S_{p'}]}(-, \mathrm{Fix}_{S_{p'}}(B))$ gives a cochain complex $\mathcal{C}^\bullet$ in which its degree-j term, for $j \in \{1, 2\}$, has at most j factor indices of positive cochain degree. Explicitly,

$$\mathcal{C}^1 = \bigoplus_{i=1}^{\tau_p} \mathrm{Map}(T_i \setminus \{1\}, \mathrm{Fix}_{S_{p'}}(B)),$$

$$\mathcal{C}^2 = \bigoplus_{i=1}^{\tau_p} \mathrm{Map}((T_i \setminus \{1\})^2, \mathrm{Fix}_{S_{p'}}(B)) \;\oplus \bigoplus_{1 \le i < k \le \tau_p} \mathrm{Map}((T_i \setminus \{1\}) \times (T_k \setminus \{1\}), \mathrm{Fix}_{S_{p'}}(B)).$$

To construct the tensor arrays, we separate the index of a simple factor from the element label within its fixed table model. Let Λ be the disjoint union of the nonidentity elements of the chosen table models, one model for each occurring simple-group type. This is a fixed label set. In $\bigoplus_{u \in \Lambda} Y$, let E_1 be spanned by symbols (i, u) for which u belongs to the model of T_i. In $\bigoplus_{(u,v) \in \Lambda^2} Y$, let $E_{2,\mathrm{same}}$ be spanned by (i, u, v) for which both labels belong to that model. Finally, in $\bigoplus_{(u,v) \in \Lambda^2} (Y \otimes_{\mathbb{F}_p} Y)$, take the subspace spanned by $(i, u; k, v)$ with $i \ne k$ and the corresponding type conditions, and quotient it by the relations $(i, u; k, v) + (k, v; i, u) = 0$. Call the resulting space $E_{2,\mathrm{mix}}$. These linear relations define the quotient also when $p = 2$. Sending the ordered symbol with $i < k$ to the corresponding 2-factor cochain coordinate gives

$$\mathcal{C}^1 = \mathrm{Fix}_{S_{p'}}(B) \otimes_{\mathbb{F}_p} E_1, \qquad \mathcal{C}^2 = \mathrm{Fix}_{S_{p'}}(B) \otimes_{\mathbb{F}_p} (E_{2,\mathrm{same}} \oplus E_{2,\mathrm{mix}}).$$

For $h = (g, q) \in \widehat{\Gamma}_0$, these identifications send its action to

$$b \otimes (i, u) \longmapsto \alpha'_B(h)b \otimes (\sigma_g i, u), \qquad b \otimes (i, u; k, v) \longmapsto \alpha'_B(h)b \otimes (\sigma_g i, u; \sigma_g k, v),$$

with the analogous formula on (i, u, v). Here α'_B is computed by applying the corrected tensor matrices to a representative in N'_B and reducing modulo P_B. The displayed relations insert a minus sign when the two factors change order; in characteristic 2 that sign is $+1$. The type conditions are preserved. Thus u, v are fixed labels after correction, while the transformations act on Y and the tensor arguments V_i. Tensoring the specified numerator spaces and quotienting by all the specified denominator relations gives the displayed cochain spaces.

If the arrays defining B have at most $b \in \mathbb{Z}_{\ge 0}$ tensor occurrences, the arrays just constructed for $\mathcal{C}^j$ have at most $b + j$ occurrences, for $j \in \{1, 2\}$. The corrected $\widehat{\Gamma}_0$-action commutes with the differential. Write $\kappa_j : N^{[j]} \to \mathcal{C}^j = N^{[j]}/P^{[j]}$ for the constructed array quotient. Set

$$N_H = \kappa_j^{-1}(\ker(\delta : \mathcal{C}^j \to \mathcal{C}^{j+1})), \qquad P_H = \kappa_j^{-1}(\mathrm{im}(\delta : \mathcal{C}^{j-1} \to \mathcal{C}^j)).$$

Then $H^j(S,B)$ is represented by N_H/P_H, and $P^{[j]} \le P_H \le N_H \le N^{[j]}$ are all preserved by the corrected action. The cycle condition is the linear system defined by the matrix of the differential on the degree-j coordinates, so it adds no tensor factors.

Comparing the resolution with ordinary cochains. The input classes use ordinary cochain coordinates. The comparison theorem for projective resolutions identifies these with the preceding presentation, canonically on cohomology [Bro82, Theorem I.7.5 and the following remarks]. In particular, this identification commutes with compatible actions, although the chosen chain maps need not do so. Let $\mathcal{P}_\bullet$ and $\mathcal{Q}_\bullet$ be the tensor and bar resolutions over $\mathbb{F}_p[S/S_{p'}]$. Construct comparison maps in both directions through degree $j+1$, lifting the identity on the trivial module: for each free generator, solve for a preimage of its mapped boundary in the other resolution, then extend by the module action. Exactness guarantees a solution. Passing to cocycles modulo coboundaries gives the mutually inverse coordinate maps. For the first-cohomology call, degree two suffices. Degree-three cochains occur only when this shared construction is used for second cohomology; they give the equations defining the 2-cocycles, not a separate transport problem in H^3. For example, in degree 2 a mixed generator indexed by $u_i \in T_i$ and $v_k \in T_k$, with $i < k$, maps to $[u_i \mid v_k] - [v_k \mid u_i]$ in the ordinary bar resolution, with the same sign convention as in the quotient above.

Since $j+1 \le 3$, the free ranks through the required degree are bounded by a fixed multiple of $(\sum_i |T_i|)^3 \le |S|^3$; as vector spaces the resolutions have dimension at most a fixed polynomial in $|S|$. Hence these maps, their matrices on cocycle classes modulo coboundaries, all corrected actions, and all quotient coordinates are obtained by polynomial-size linear algebra.

Returning the complete answer. Finally solve every coset of $\Gamma_0 \le \Gamma$. If the current space is N/P, with $P \le N \le H^j(S,B)$ preserved by Γ and points $x, y \in N/P$, then in a coset $g\Gamma_0$, for $g \in \Gamma$, replace the target y by its image under the inverse affine action of g and perform the calculation in $\widehat{\Gamma}_0$, pulled back from Γ_0. Project the complete answer through $\widehat{\Gamma}_0 \to \Gamma_0$. Combine the projected complete cosets by (3.1). Return the complete coset in Γ before continuing with another cohomology factor. The extension group $\widehat{\Gamma}_0$ and its corrected module actions are used only for this restriction calculation; subsequent quotient problems use the original Q and its original module actions.

The construction gives degree at most $e+j$. The only new moving space is Y, whose dimension is at most $\log_p |S|$ by (9.14); its repeated occurrences are counted once. The coordinate calculations are polynomial in the input tables, tensor presentation of B, and action matrices. There are at most $|S|$ outer-action cosets by (F.1). Together with the cubic transport theorem and Lemma 3.6, these observations prove the result. □

F.1.4 Correctness of the recursion

Proposition 9.11 (restated). *Assuming the local constructions in §9.3.2, Algorithm 9.5 returns the complete coset specified in Definition 29, and every call terminates.*

Proof. Use lexicographic induction on $(\ell - i, \dim B)$. A call in Step R1 decreases the second component, and a call in Step R3 decreases the first. The zero cases are immediate. Exactness of (9.9) identifies each induced cohomology grade with the specified subquotient over a smaller module; there is no assumption that the filtration of B splits. Exactness of (9.11) identifies the remaining two pieces with the restriction image and the inflation term.

The restriction solvers are complete cubic affine transporter computations. In the nonabelian case, every outer-action coset is processed and every temporary cover is projected back, as required

by Proposition 9.10. The other comparisons are complete by induction. Formula (9.8) computes the full stabilizer after each partial match, and its cocycle expresses precisely the unsatisfied fibre equation. Hence multiplication of the returned cosets gives every solution and only solutions in the input Γ, including elements in kernels of any auxiliary action. This proves correctness, the full-stabilizer output, and termination. □

F.2 Module-action compatibility

These proofs justify the semisimple comparison and reconstruction steps in Section 9.4. The standard module computations are cited in Proposition 9.15. We use the input notation of Definition 27.

F.2.1 The semisimple terminal case

Proposition 9.17 (restated). *On semisimple inputs, Algorithm 9.14 returns the complete answer of Definition 27. It runs deterministically in time $p^{O(a)} \cdot \mathrm{poly}(\mathrm{InSize})$, where $n = |Q|$ and* InSize *is the total explicit input length. It closes the occurring simple types under $\beta(\Gamma)$ to a set $\mathcal{S}$ of size at most n, compares their multiplicities, and lifts the complete resulting coset to compatible pairs. The output includes the full kernel of projection to the allowed quotient automorphisms.*

Proof. The zero-dimensional case returns the unique linear map together with every element of Γ, as required. For positive dimension, close the simple isomorphism types occurring in M, M' under the generators of $\beta(\Gamma)$, using the simple-module test of Proposition 9.15. There are at most $|Q|$ simple $\mathbb{F}_pQ$-module types: every simple module is a quotient of the regular module, by sending 1 to a nonzero vector, and hence is a composition factor of that $|Q|$-dimensional module. Thus this closure is found by polynomially many steps. Its representations have dimension at most a, since twisting preserves dimension. Write $\mathcal{S}$ for this finite set of simple-module isomorphism types, with $|\mathcal{S}| \le |Q|$. Compute its permutation action $\lambda : \Gamma \to \mathrm{Sym}(\mathcal{S})$ and record each generator as a permutation in the original group $\Gamma \le \mathrm{Sym}(\Omega)$. Explicitly, if ρ_S is a representative of a simple type $[S]$, define $\lambda_g[S] = [\rho_S \circ \beta_g^{-1}]$. This is a left action. Condition (9.2) means that λ_g sends the simple types of M, with their multiplicities, to those of M'.

For each multiplicity $e \in \{1, \dots, a\}$, apply Lemma 9.16 to the two subsets of $\mathcal{S}$ whose multiplicity in M and M' is e. At most a members of $\mathcal{S}$ have positive multiplicity in either module. Applying (3.2) to these subset actions gives the simultaneous transporter in $\Gamma \le \mathrm{Sym}(\Omega)$. For its witness $g \in \Gamma$ and each source-stabilizer generator $h \in \Gamma$, match the simple summands to construct the corresponding $\mathbb{F}_p$-linear intertwiner. Add generators of $\mathrm{Aut}_Q(M) \le \mathrm{GL}(M)$, the kernel of projection from compatible pairs to Γ. The result is exactly all pairs satisfying (9.2). In particular, the answer includes every element of $\ker \beta \le \Gamma$, together with its original permutation and matrix actions.

The closure uses polynomially many simple-module tests because $|\mathcal{S}| \le n$. There are at most a multiplicity classes, and each subset has size at most a, so the small-set comparisons cost $2^{O(a)}$ times a polynomial in the explicit permutation records. Proposition 9.15 bounds the decompositions, intertwiners, and kernel generators by $p^{O(a)} \cdot \mathrm{poly}(\mathrm{InSize})$. Preimage computations and reduction to succinct generating sets take polynomial time in their explicit records. This proves the stated bound and the complete output guarantee. □

F.2.2 The diagonal comparison and extension-class test

Let $U = \mathrm{rad}(M)$, $U' = \mathrm{rad}(M')$, $V = M/U$, and $V' = M'/U'$, with $0 < \dim U = \dim U' < \dim M$. The symbols θ_U, θ_V and their primed versions denote the induced actions. For the class comparison,

choose linear sections and use the cocycles z, z' from (9.19), with $X = \mathrm{Hom}_{\mathbb{F}_p}(V, U)$ and $X' = \mathrm{Hom}_{\mathbb{F}_p}(V', U')$.

Proposition 9.18 (restated). *Assume that the recursive call in step R1 returns the complete coset of compatible radical pairs. Then step R2 returns exactly all triples (u, v, g) with $u : U \to U'$ and $v : V \to V'$ invertible and $g \in \Gamma$ satisfying*

$$\begin{aligned} u\theta_U(q) &= \theta'_U(\beta_g q)u, \\ v\theta_V(q) &= \theta'_V(\beta_g q)v \qquad (q \in Q). \end{aligned}$$

If this set is nonempty, it is the coset $h_0\Delta$ used in step R3, where $h_0 = (u_0, v_0, g_0)$ is one such triple and Δ is its full source stabilizer. Thus $\Delta \leq \mathrm{GL}(U) \times \mathrm{GL}(V) \times \Gamma$ satisfies the same identities with the primed actions replaced by the unprimed actions. Triples compose componentwise.

Proof. Write the radical answer as $(u_{\mathrm{rad}}, g_{\mathrm{rad}})K_U$. Every possible radical pair has the form $(u_{\mathrm{rad}}u, g_{\mathrm{rad}}g)$ for $(u, g) \in K_U$. After twisting the target quotient action by $\beta_{g_{\mathrm{rad}}}$, the semisimple call in step R2 tests exactly which of these pairs admit a compatible map $v : V \to V'$. The semisimple call ranges over $(u, g) \in K_U$, so the same g controls the radical and quotient equations. Its complete output, composed with the radical witness, is therefore exactly the stated coset.

For the source stabilizer, the projection $\Delta \to K_U$, $(u, v, g) \mapsto (u, g)$, has kernel $\mathrm{Aut}_Q(V)$. Its image consists of the radical pairs whose quotient automorphism also admits a compatible map on V; it need not be all of K_U. The semisimple call computes this image and includes the full kernel. Each triple still records its original permutation in Γ and both linear maps. □

Lemma 9.19 (restated). *The maps z, z' are cocycles, and their cohomology classes do not depend on the chosen sections. For a diagonal witness $h_0 = (u_0, v_0, g_0)$, the formula $z'_0(q) = u_0^{-1}z'(\beta_{g_0}q)v_0$ defines a cocycle in the source module X. The source diagonal group Δ acts on $H^1(Q, X)$ by the cochain formula $((u, v, g)z)(q) = u\, z(\beta_g^{-1}q)\, v^{-1}$. For a diagonal triple $h = (u, v, g) \in h_0\Delta$, a map lifting it has the form $\alpha = \left(\begin{smallmatrix} u & t \\ 0 & v \end{smallmatrix}\right)$ with $t \in \mathrm{Hom}_{\mathbb{F}_p}(V, U')$. It is invertible because u, v are invertible. Its remaining compatibility condition is the linear system*

$$uc(q) + t\theta_V(q) = \theta'_U(\beta_g q)t + c'(\beta_g q)v \qquad (q \in Q).$$

Equivalently, with $b = u^{-1}t \in \mathrm{Hom}_{\mathbb{F}_p}(V, U)$, this condition is

$$z(q) - u^{-1}z'(\beta_g q)v = q \cdot b - b \qquad (q \in Q).$$

The triples in $h_0\Delta$ that lift to compatible maps $M \to M'$ are exactly $h_0\,\mathrm{Trans}_\Delta([z], [z'_0])$.

Proof. Multiplying the block matrices in (9.19) gives $z(qr) = z(q) + q \cdot z(r)$. Changing a section by a linear map $V \to U$ changes z by a coboundary; the same calculation applies to z'. The diagonal identities for h_0 show that its pullback z'_0 satisfies the source cocycle identity. In particular, the composition defining $z'_0(q)$ has types

$$V \xrightarrow{v_0} V' \xrightarrow{z'(\beta_{g_0}q)} U' \xrightarrow{u_0^{-1}} U.$$

The diagonal identities for elements of Δ also show that their cochain action commutes with the differential, so it induces the stated action on $H^1(Q, X)$.

Fix a diagonal triple $h = (u, v, g) \in h_0\Delta$. A map lifting it has the form $\alpha = \left(\begin{smallmatrix} u & t \\ 0 & v \end{smallmatrix}\right)$ with $t \in \mathrm{Hom}_{\mathbb{F}_p}(V, U')$. It is invertible because u, v are invertible. The two diagonal compatibility conditions already hold; the remaining condition is the linear system

$$uc(q) + t\theta_V(q) = \theta'_U(\beta_g q)t + c'(\beta_g q)v \qquad (q \in Q).$$

Putting $b = u^{-1}t \in \mathrm{Hom}_{\mathbb{F}_p}(V, U)$ and using the diagonal identities rewrites this system as

$$z(q) - u^{-1}z'(\beta_g q)v = q \cdot b - b \qquad (q \in Q).$$

Thus h lifts exactly when the two cocycles on the left have the same class. If $h = h_0 k$ with $k \in \Delta$, then $u^{-1}z'(\beta_g q)v = (k^{-1}z'_0)(q)$. The lifting condition is consequently $[z] = k^{-1}[z'_0]$, or equivalently $k[z] = [z'_0]$, proving the transporter formula. □

F.2.3 Complete reconstruction and the full algorithm

Proposition 9.20 (restated). *If step R3 returns the nonempty transporter h_1K, with $K = \mathrm{Stab}_\Delta([z])$, restriction to U and passage to V give the exact sequence*

$$1 \longrightarrow \mathrm{Hom}_Q(V, U)_{\mathrm{add}} \longrightarrow K_\theta \longrightarrow K \longrightarrow 1.$$

Step R4 therefore returns the complete coset of compatible pairs for M, M'.

Proof. Lemma 9.19 shows that (9.21) has a solution for the witness h_0h_1. Its source-to-source version has a solution for every element of K, so the homomorphism $K_\theta \to K$ is surjective. Its kernel consists exactly of the pairs

$$\left(\begin{pmatrix} I & b \\ 0 & I \end{pmatrix}, 1_\Gamma\right), \qquad b \in \mathrm{Hom}_Q(V, U),$$

because the off-diagonal compatibility equation for such a pair is precisely $b\theta_V(q) = \theta_U(q)b$. Composition of these shears adds their parameters, proving exactness.

For a fixed liftable triple, equation (9.22) also shows that the solutions for $b = u^{-1}t$ form an affine coset of $\mathrm{Hom}_Q(V, U)$; the differences between solutions for t lie in $u\,\mathrm{Hom}_Q(V, U)$. Hence the linear systems in step R4(a)–(b) produce one compatible witness and lifts of all generators of K. The basis computed in step R4(c) generates the additive group $\mathrm{Hom}_Q(V, U)$, since the field is $\mathbb{F}_p$. The resulting shears generate the kernel, while the chosen lifts generate the image. Together they generate all of K_θ. Composing with the witness gives every compatible pair required by Definition 27. □

Theorem 9.3 (restated). *Given an instance of Definition 27, with $n = |Q|$ and total explicit input length* InSize*, the complete set of compatible pairs can be computed deterministically in time $p^{O(a)} \cdot \mathrm{poly}(\mathrm{InSize})$.*

Proof of Theorem 9.3. We prove correctness by induction on the module dimension. If $a = 0$, every $g \in \Gamma$ is compatible with the unique linear map. For a nonzero module, the radical is a proper submodule and is preserved by every module isomorphism and every twist by an automorphism of Q. Unequal radical dimensions therefore imply emptiness. If both radicals are zero, the modules are semisimple. Their simple types and multiplicities determine their isomorphism classes. The simultaneous small-set comparisons in Section 9.4.3 find precisely the quotient transformations

allowed by these data. The intertwiners above a witness and stabilizer generators, together with the kernel $\mathrm{Aut}_Q(M)$, give the required complete coset of pairs.

Otherwise, the recursive call on U, U' is on smaller modules, so the induction hypothesis gives every compatible radical pair. Proposition 9.18 then shows that step R2 finds all compatible diagonal triples. Lemma 9.19 identifies the liftable triples with the class transporter computed in step R3 by Corollary 9.12. Finally, Proposition 9.20 proves that step R4 gives a compatible witness and generators of the full K_θ. Every step records the original permutation in Γ, including elements in the kernel of its action on Q. This proves correctness and completeness of Algorithm 9.13.

The data representation and running-time analysis in Section 9.4.2 apply at every node: the only recursive module call is on the proper radical, the semisimple routine satisfies Proposition 9.17, and each cohomology root has tensor-argument parameter at most a. The explicit sizes and the sum over at most a levels therefore have the bound stated in the theorem. □

F.3 Second-cohomology transport

This subsection gives the constructions used by Algorithm 9.21. We first establish the filtration from restriction and inflation, then give the tensor presentations for restriction and the modules used by the middle comparison. The final two propositions prove degree closure and correctness.

F.3.1 Restriction, conjugation defect, and inflation

Let $S \trianglelefteq R$, let B be an $\mathbb{F}_pR$-module, and put

$$F_1 = \ker\big(H^2(R,B) \longrightarrow H^2(S,B)\big), \qquad F_2 = \mathrm{im}\big(H^2(R/S, \mathrm{Fix}_S(B)) \longrightarrow H^2(R,B)\big).$$

A class $[f] \in F_1$ has a splitting over S: choose a normalised $s \in C^1(S,B)$ satisfying $s(xy) = s(x) + xs(y) + f(x,y)$ for $x, y \in S$. The map $x \mapsto (s(x), x)$ is then a homomorphism into the extension E_f with multiplication (9.4). For $q \in R$ and $x \in S$, define its conjugation defect by

$$\zeta_q(x) = qs(q^{-1}xq) - s(x) + f(q, q^{-1}xq) - f(x,q). \tag{F.4}$$

The induced R/S-action on $H^1(S,B)$ sends $[z]$ to the class of $x \mapsto qz(q^{-1}xq)$. The obstruction map is

$$\omega_S : F_1 \longrightarrow H^1(R/S, H^1(S,B)), \qquad [f] \longmapsto [qS \mapsto [\zeta_q]]. \tag{F.5}$$

It measures whether a splitting over S is preserved by conjugation from R. The following lemma identifies the exact filtration used in R3; the chosen splitting need not commute with compatible automorphisms of R and B preserving S.

Lemma F.1 (Exact degree-2 middle term)**.** *Let p be prime, R a finite group, $S \trianglelefteq R$, and B a finite-dimensional $\mathbb{F}_pR$-module. With $F_1 = \ker(H^2(R,B) \to H^2(S,B))$ and $F_2 = \mathrm{im}(H^2(R/S, \mathrm{Fix}_S(B)) \to H^2(R,B))$, the map $\omega_S : F_1 \to H^1(R/S, H^1(S,B))$ of* (F.5) *has kernel $F_2 \le F_1$. In particular, $0 \le F_2 \le F_1 \le H^2(R,B)$ is a filtration preserved by every compatible automorphism preserving S. Its three consecutive quotients are, respectively, quotients of subspaces of*

$$H^2(R/S, \mathrm{Fix}_S(B)), \qquad H^1(R/S, H^1(S,B)), \qquad H^2(S,B).$$

The map ω_S is linear and commutes with every such compatible automorphism. The filtration spaces, obstruction map, and all induced maps on these subquotients are computable by polynomial-size linear systems.

Proof. Take $[f] \in F_1$ and choose a splitting cochain s as in the construction preceding (F.4). The coboundary map $\delta_S : B \to Z^1(S, B)$ is given by $(\delta_S b)(x) = xb - b$ for $b \in B$ and $x \in S$. The conjugation action is $(qz)(x) = qz(q^{-1}xq)$ for $q \in R$, $z \in Z^1(S, B)$, and $x \in S$. The actions on B use its given $\mathbb{F}_p R$-module structure. The formula for qz preserves cocycles and coboundaries, and thus sends $[z]$ to $[qz]$ in $H^1(S, B)$. Every $x \in S$ acts trivially on these classes, so this class action depends only on $qS \in R/S$. Extend s to a normalised cochain $\widetilde{s} \in C^1(R, B)$ and set $f_0 = f + \delta\widetilde{s}$. Since $\delta s = -f|_{S\times S}$, we have $f_0|_{S\times S} = 0$. Direct substitution gives

$$f_0(q, q^{-1}xq) - f_0(x, q) = \zeta_q(x) - (\delta_S \widetilde{s}(q))(x) \qquad (q \in R,\ x \in S).$$

Thus [DHW12, Section 10.3], with ambient group R, normal subgroup S, quotient R/S, and module B, identifies $qS \mapsto [\zeta_q]$ with its conjugation cocycle. It therefore gives the map (F.5). This map is well defined and additive, hence $\mathbb{F}_p$-linear, by [DHW12, Corollary 6.5 and Theorem 6.7]. By [DHW12, Corollary 6.6 and Section 8], changing the coordinates in B and group labels by a compatible automorphism g preserving S gives $\omega_S(g[f]) = g\omega_S([f])$. Thus ω_S commutes with every such compatible automorphism on classes, even though the chosen splitting cochains need not do so.

Exactness is [DHW12, Lemma 6.8]; the three factors identify with the inflation image, $\operatorname{im}\omega_S$, and the restriction image, respectively. Form the cochain spaces, restrictions, inflations, and F_1, F_2 by elimination. For each basis class $[f] \in F_1$, solve for the splitting cochain s, evaluate (F.4), and project modulo $B^1(S, B)$ and then $B^1(R/S, H^1(S, B))$. This computes ω_S and the identification $F_1/F_2 \cong \operatorname{im}\omega_S$ by polynomial-size linear systems. □

The lemma identifies exact images and quotients, rather than a direct sum decomposition of $H^2(R, B)$. Inflation need not be injective, and $\operatorname{im}\omega_S$ need not be the entire first-cohomology space. R3(a) includes their kernels in the displayed numerator–denominator presentations. Intersecting the filtration with N and adding P gives the filtration of $W = N/P$ used by R3.

F.3.2 Restriction presentations and modules for the middle term

When R2 applies, it replaces B by grades with trivial S_i-action. The following statement gives the tensor presentation for restriction to an elementary abelian S_i in R3(b), and the module used by the middle call in R3(c).

Proposition F.2 (Elementary restriction and the module for the middle term)**.** *Let $C \cong \mathbb{F}_p^r$ be normal in a finite group R, and let D be an $\mathbb{F}_p R$-module on which C acts trivially. There are natural identifications*

$$H^2(C, \mathbb{F}_p) \cong \begin{cases} \operatorname{Sym}^2(C^*), & p = 2, \\ \bigwedge^2(C^*) \oplus C^*, & p \text{ odd}, \end{cases} \qquad H^2(C, D) \cong D \otimes_{\mathbb{F}_p} H^2(C, \mathbb{F}_p). \tag{F.6}$$

Consequently, the three factors in Lemma F.1 are subquotients of

$$H^2(R/C, D), \qquad H^1(R/C, D \otimes_{\mathbb{F}_p} C^*), \qquad D \otimes_{\mathbb{F}_p} H^2(C, \mathbb{F}_p), \tag{F.7}$$

respectively. All identifications and the actions induced by compatible automorphisms of R and D preserving C are computable by polynomial-size linear systems. The last term has a cubic tensor presentation on the argument spaces D, C; the middle term uses $D \otimes C^*$, with a quadratic tensor presentation on the same two spaces. In both cases their total argument dimension is $\dim D + r$, with repeated occurrences of the C-action counted once.

Proof. With scalar coefficients, (F.6) gives the natural degree-two component of the elementary abelian cohomology ring; see [Bel20, Introduction]. Here $\mathrm{Sym}^2(C^*)$ means the quotient of $C^* \otimes C^*$ by the span of $u \otimes v - v \otimes u$, also in characteristic two. Use the cup-product convention $(a \smile b)(u, v) = a(u)b(v)$. Cup products give the quadratic summand, and for odd p the Bockstein of $0 \to \mathbb{F}_p \to \mathbb{Z}/p^2\mathbb{Z} \to \mathbb{F}_p \to 0$ gives the linear summand.

For a dual basis $x_1, \dots, x_r \in C^*$, write $u_i = x_i(u)$ and $v_i = x_i(v)$. Normalized representatives are

$$f_{ij}(u, v) = u_i v_j, \qquad b_i(u, v) = \left\lfloor \frac{\widetilde{u}_i + \widetilde{v}_i}{p} \right\rfloor \pmod p,$$

where tildes are representatives in $\{0, \dots, p-1\}$; see [Bel20, Lemma 2.2]. Use f_{ij} for $i \le j$ when $p = 2$, and f_{ij} for $i < j$ together with b_i for odd p. Passing to classes modulo $B^2(C, \mathbb{F}_p)$ computes the displayed isomorphism; elimination computes its inverse. Naturality of cup products and the Bockstein gives $\mathrm{GL}(C)$-equivariance. Since C acts trivially on D, tensoring its scalar cochain complex with D proves the second identification in (F.6).

The same trivial action gives $H^1(C, D) = D \otimes C^*$. Both D and C are specified R/C-modules: the action on D descends by assumption, and conjugation on C factors through R/C because C is abelian. Applying Lemma F.1 proves (F.7). Its restriction factor uses at most three tensor occurrences, namely one of D and at most two of C^*. The same matrix on C^* acts in both occurrences. The odd-characteristic linear summand uses only one such occurrence. Thus the distinct argument dimensions sum to $\dim D + r$. The middle term uses the module $D \otimes C^*$, with the corresponding quadratic presentation. □

Products of nonabelian simple groups. For $S_i = T_1 \times \cdots \times T_\tau$, the restriction comparison is exactly the $j = 2$ case of Proposition 9.10 with B represented by one argument. After the local inner-action correction, its tensor presentation has at most three occurrences and adds only the permutation module on the p-relevant simple-factor indices. Its dimension is at most $\log_p |S_i|$ by (9.14). All local outer-action cases are reunited, and all local covering groups are projected away, before the algorithm continues to the middle term.

For this nonabelian step put $E = H^1(S_i, B)$. The original R_i-action on cochains induces an action on E on which S_i acts trivially, so E is an $\mathbb{F}_p[R_{i+1}]$-module. Compute its basis, the R_{i+1}-action, and the matrices induced by the generators of H on cocycles modulo coboundaries. The following bound permits E itself to be used as a single representation argument in Theorem 9.2.

Lemma F.3 (First cohomology of direct products)**.** *Let p be a prime, let $S = T_1 \times \cdots \times T_\tau$ be a product of nonabelian finite simple groups with $\tau \in \mathbb{N}$, and let A be a finite-dimensional $\mathbb{F}_p S$-module. Then $\dim_{\mathbb{F}_p} H^1(S, A) \le 2 \dim_{\mathbb{F}_p} A$.*

Proof. We charge the contribution of one simple factor only to the quotient on which that factor acts nontrivially. The fixed subspace is then available for induction on the remaining factors, so the same module dimensions are not charged repeatedly. All dimensions are over $\mathbb{F}_p$. Induct on the number of factors, with the empty product as the trivial base case. Write $S = T \times S'$, where T is one nonabelian simple direct factor and S' is the product of the remaining factors, and put $A_0 = \mathrm{Fix}_T(A) \le A$. The factors commute, so A_0 is an $\mathbb{F}_p S'$-module. Inflation–restriction (9.11) gives $\dim H^1(S, A) \le \dim H^1(S', A_0) + \dim H^1(T, A)$. Since T equals its commutator subgroup and acts trivially on A_0, every homomorphism from T to the abelian group $(A_0, +)$ is zero. Thus $H^1(T, A_0) = \mathrm{Hom}(T, (A_0, +)) = 0$. Applying (9.9) to $0 \to A_0 \to A \to A/A_0 \to 0$ therefore gives an injection $H^1(T, A) \hookrightarrow H^1(T, A/A_0)$. The group T is generated by two elements [GK00, Theorem I and its Corollary], and a 1-cocycle is determined by its values on these generators. Hence

$\dim H^1(T, A/A_0) \le 2\dim(A/A_0)$. By induction $\dim H^1(S', A_0) \le 2\dim A_0$, so $\dim H^1(S, A) \le 2\dim A_0 + 2\dim(A/A_0) = 2\dim A$. □

The first-cohomology input. The call in R3(c) uses the same initial group Q, subgroup series, and position $i+1$, with ambient group Γ_0 and current subgroup K. Its tensor presentation $\mathcal{P}_E$ uses $\nu_{i+1} : Q \to R_{i+1}$. In the elementary case, $E = H^1(S_i, B) = B \otimes S_i^*$, with argument spaces B, S_i; in the nonabelian case, use the single argument E. These argument spaces are R_{i+1}-modules: in the elementary case S_i acts trivially on B and on itself by conjugation, and in the nonabelian case its induced action on E is trivial. Pull their actions back along ν_{i+1} to obtain the Q-actions used by the first-cohomology algorithm. In each case the numerator is the whole displayed tensor space and the denominator is zero, with the computed identification with E and all action matrices included in the input. Both presentations meet the input requirements of Theorem 9.2.

F.3.3 Degree closure and correctness

Proposition F.4 (Degree closure of the second-cohomology calls)**.** *Every module in a continuing second-cohomology call of Algorithm 9.21 is a subquotient of the initial module. At position i, with current module B and selected subgroup S_i, each restriction comparison has a tensor presentation of degree at most three. Each middle comparison is a valid input to Theorem 9.2: its module $E = H^1(S_i, B)$ has a tensor presentation on B, S_i in the elementary case, and uses the single argument E of dimension at most $2\dim B$ in the nonabelian case. The induced numerators, denominators, and affine actions satisfy the corresponding call contracts throughout the recursion.*

Proof. In R1 and R3, the next module is a fixed subspace of the current module; in R2 it is a filtration grade. Each is therefore a subquotient of the current module and has dimension at most the initial d, and it can be treated as one argument. Each restriction comparison adds at most two tensor occurrences, on the same elementary section or simple-factor index space, and is therefore an instance of cubic affine transport.

For the middle term, the elementary case uses $B \otimes S_i^*$, with a quadratic tensor presentation on the two modules B, S_i. The nonabelian case gives the single module $E = H^1(S_i, B)$ of dimension at most $2\dim B$ by Lemma F.3. Both meet the input requirements of Theorem 9.2. In its later calls, module subquotients use these original Q-actions and the quotient maps ν_i. They are not replaced by a new vector-set argument of their possibly larger dimension. In particular, a branch uses a positive-degree cohomology space as its module only once, when passing from degree two to degree one.

All subspaces and maps above are preserved by the current subgroup H. Projecting the current affine action along these equivariant maps gives the corresponding induced cocycle; normalizing a nonempty quotient match gives the cocycle on its fiber. Therefore arbitrary affine subquotient inputs, not only linear comparisons of whole cohomology spaces, are supported at every step. □

Proposition F.5 (Correctness of the second-cohomology recursion)**.** *Algorithm 9.21 returns exactly the affine transporter specified in Theorem 9.4, including the full source stabilizer in Γ_0.*

Proof. Use lexicographic induction on $(\ell - i, \dim B)$ as specified after the algorithm. At $N_\ell = Q$, the current group is $R_\ell = 1$, so positive-degree cohomology is zero. Module filtrations are processed through (9.9), which identifies the required factors with explicit invariant subquotients of the grade cohomologies. These subquotients include the kernels arising from mapping back to the original module.

For each grade, Lemma F.1 identifies the restriction image, middle obstruction, and inflation image, with all kernels included. Intersecting this filtration with the given numerator and passing modulo the denominator gives the exact factors of the current affine subquotient. The restriction presentation and the cubic affine transporter solve the restriction comparison in Step R3. Theorem 9.2 solves its middle comparison, by Proposition F.4. The induction hypothesis solves its inflation comparison. The coprime shortcut is valid by Lemma 9.7.

After each match, Lemma 3.5 uses its complete source stabilizer to compare the remaining affine fiber and lifts the complete answer. Hence the resulting transporter contains exactly those original transformations satisfying all three conditions. The same argument combines the filtration grades. In a nonabelian restriction, the local covering group is projected back to Γ_0 and all outer-action cases are combined before the next comparison, as required by Proposition 9.10. No auxiliary transformation is returned in place of an original one. These facts prove the inductive claim, including completeness of the returned stabilizer. □

G Detailed AI methodology statement

Summary of AI methodology.

The main results in this paper were generated by ChatGPT during sustained conversations between the authors and ChatGPT. The AI-generated results have clear traces back to previous results on group isomorphism, including several in the past 20 years. The authors put in their best efforts in a limited time to understand and rewrite the results in the presentation in the current form, and take full responsibility for the results presented in this paper. A more detailed AI methodology statement can be found at the end of the paper.

Intermediate results from ChatGPT leading to this paper.

Below is a sequence of intermediate results that led to the results in the current paper. These happened from August to early October.

- We experimented with ChatGPT to tackle Tensor Isomorphism, with little progress.
- We asked ChatGPT about Tensor Isomorphism for tensors of low slice rank. Such tensors were viewed as a hard case to the strategy of Xiaorui Sun's algorithm (STOC'23); Xiaorui explicitly mentioned this in a talk at a Shonan Meeting in late April, in response to a question of James B. Wilson. One day, GhatGPT produced a faster algorithm testing Tensor Isomorphism for tensors of low analytical rank.
- Then ChatGPT generated an algorithm for testing $n \times n \times n$ Tensor Isomorphism over $\mathbb{F}_q$ in time $q^{\tilde{O}(n^{4/3})}$.
- We used ChatGPT Astra to get an algorithm for Tensor Isomorphism in time $q^{\tilde{O}(n)}$.
- We used ChatGPT to improve the running time further to $q^{O(n)}$ for odd q. This implies an algorithm for testing isomorphism of p-groups of class 2 and exponent p in time polynomial in the group order.
- We asked ChatGPT to imrpove the above result to 2-groups of Frattini class 2. ChatGPT produced an algorithm for group isomorphism overall, and suggested that what we asked is a special case of its new algorithm.

Future communications with the community.

We maintain a website `https://group-iso.github.io/` for this paper (with archived versions), materials produced by AI, and supplementary materials such as lecture notes and recordings. We plan two online seminars, one in November and one in December, 2026.

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